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for scientific review, technical reconstruction,
and AI-supported structural analysis

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It is intended as a compact technical context body for deeper review, structural examination, and AI-assisted analysis.

Semantic Gravitation I: Gradient Fields of Semantic Dynamics on Hilbert Spaces

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Abstract

This paper develops the first technical component of the Semantic Gravitation programme: a Hilbert-space framework for local semantic dynamics driven by energy dissipation and noise. We model a semantic state as a point s in a separable Hilbert space S , equipped with a *semantic potential* $W: S \rightarrow \mathbb{R}$. The negative gradient field $-\nabla W$ generates deterministic semantic flows

$$\dot{s}_t = -\nabla W(s_t),$$

while stochastic perturbations are described by infinite-dimensional Langevin-type dynamics

$$ds_t = -\nabla W(s_t) dt + B dW_t,$$

where B is a suitable noise operator and W_t a cylindrical Wiener process.

On the level of probability distributions over S , these dynamics induce a Fokker–Planck evolution for probability measures ρ_t on S . We introduce a free-energy functional

$$\mathcal{F}(\rho) = \int_S W(s) \rho(ds) + \beta^{-1} \text{Ent}(\rho \mid \mu),$$

combining semantic energy with relative entropy $\text{Ent}(\rho \mid \mu)$ with respect to a reference measure μ . Under natural analytic assumptions on W and B , we show that $\mathcal{F}$ acts as a Lyapunov functional for the Fokker–Planck flow, and that stationary distributions are characterised as minimisers of $\mathcal{F}$. The core existence and uniqueness results are formulated in the language of gradient flows of lower semicontinuous functionals on Hilbert spaces.

The framework is illustrated by finite-dimensional examples (including Ornstein–Uhlenbeck systems and Ginzburg–Landau-type models on $L^2([0, 1])$), which serve both as toy semantic systems and as templates for future, more realistic models. Throughout, the construction is deliberately kept minimal: it does not presuppose any specific semantic ontology, but instead provides a mathematically controlled scaffold on which semantic structure can emerge and be analysed.

Conceptually, this paper supplies the local, Hilbert-space level of semantic dynamics that underpins the subsequent parts of the series: the transition to field-theoretic descriptions and free-energy formulations (Paper II), quantum-compatible semantic fields (Paper III), spacetime semantic field theories (Paper IV), complex-system implementations (Paper V), and the ontological completion via the global attractor and the LOVE Lagrangian (Paper VI).

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1 Introduction and overview

1.1 Motivation

Many complex systems can be described in terms of a high-dimensional state s evolving in time on some state space S . In numerous areas of physics and applied mathematics, such systems are conveniently modelled by means of an energy or potential function

$$W : S \rightarrow \mathbb{R} \cup \{+\infty\},$$

whose landscape determines how the system relaxes towards more stable configurations. In the broader programme from which this work originates, W is sometimes informally referred to as a “contradiction energy” (“Widerspruchsenergie”), but in this paper it will be treated strictly as a potential or energy function.

A particularly transparent class of dynamics is given by gradient flows of W . In a deterministic setting, one considers an evolution $(s_t)_{t \geq 0}$ governed by

$$\dot{s}_t = -\nabla W(s_t),$$

where ∇W denotes the gradient of W with respect to the geometry of the state space. Such dynamics decrease the potential along trajectories and provide a natural model for relaxation towards local or global minima of W .

In many situations, however, deterministic dynamics are only an approximation. Fluctuations, unresolved degrees of freedom, or external perturbations are often modelled as stochastic noise, leading to Langevin-type stochastic differential equations (SDEs) of the form

$$ds_t = -D\nabla W(s_t) dt + B dW_t,$$

where D is a (typically positive) operator, B encodes the noise structure, and $(W_t)_{t \geq 0}$ is a Wiener process. Such models interpolate between energy-driven drift and stochastic exploration of the state space and are closely connected to equilibrium and non-equilibrium statistical mechanics.

The present work develops a general mathematical framework for such gradient and stochastic dynamics on a separable Hilbert space S . Rather than committing to a specific physical interpretation, the focus is on the structural properties of the dynamics induced by W , their Lyapunov functionals, and the associated distributional evolution and free energy.

1.2 Scope and limitations

The aim of this paper is deliberately structural and mathematical. The state space S is taken to be an abstract separable Hilbert space, and the potential W is treated as a general energy functional subject to explicit analytic assumptions. No particular physical realisation of S (such as configuration space, phase space, or a space of fields) is fixed a priori, and the framework is intended to be compatible with multiple possible interpretations.

In particular, this work does *not* attempt to derive quantum mechanics, general relativity, or any other established physical theory from the proposed framework. The constructions are designed to resemble familiar structures from statistical mechanics and stochastic analysis, but the results are not claimed to be fundamental or unique in any physical sense.

Furthermore, while the state space is allowed to be infinite-dimensional, the level of mathematical rigour in this setting is necessarily limited. Existence and uniqueness results for gradient flows and Langevin SDEs are treated rigorously in finite dimensions, and

the infinite-dimensional case is discussed at the level of a programme, with references to existing results where appropriate. The focus is on a coherent presentation of the structures

$$\begin{aligned} (\text{potential } W) &\Rightarrow \text{gradient flow} \\ &\Rightarrow \text{stochastic dynamics} \\ &\Rightarrow \text{distribution dynamics and free energy,} \end{aligned}$$

rather than on exhausting all technical conditions under which each step can be fully justified in infinite dimensions.

Finally, this paper does not address symmetry structures, conservation laws, or Noether-type results, and it does not provide empirical applications or numerical methods. These aspects are left for companion works.

1.3 Contributions and structure of the paper

Within the above scope, the main contributions of this paper are as follows:

1. A general setup for gradient dynamics on a separable Hilbert space S with an energy functional $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$, including a clear list of assumptions (such as lower semicontinuity, coercivity, and differentiability) that guarantee good variational properties.
2. A formulation of deterministic gradient flows

$$\dot{s}_t = -\nabla W(s_t)$$

on S , together with Lyapunov properties of W and a discussion of linearisation and relaxation rates near critical points.

3. A Langevin-type stochastic extension

$$ds_t = -D\nabla W(s_t) dt + B dW_t,$$

with rigorous treatment in finite dimensions and a careful indication of how analogous constructions can be formulated in infinite-dimensional settings.

4. The passage from stochastic dynamics to the evolution of probability distributions $\rho(s, t)$ via the Fokker–Planck equation in finite dimensions, and the introduction of a free energy functional

$$F[\rho] = \int W(s) \rho(s) d\mu(s) + D \int \rho(s) \ln \rho(s) d\mu(s),$$

which acts as a Lyapunov functional along the Fokker–Planck flow.

5. A discussion of stationary distributions of the form

$$\rho_\infty(s) \propto \exp\left(-\frac{W(s)}{D}\right),$$

and their concentration near global minimisers of W , together with illustrative examples (such as quadratic and double-well potentials) that demonstrate the qualitative behaviour of the dynamics.

The remainder of the paper is structured as follows. In Section 2 we introduce the basic notation, specify the state space S as a separable Hilbert space, and collect the standing assumptions on W , the gradient ∇W , and the stochastic ingredients D , B , and $(W_t)_{t \geq 0}$. Section 3 is devoted to the analytic properties of the potential W itself, including the existence of global minimisers and several important example classes of potentials.

Section 4 formulates the deterministic gradient flow on S , discusses existence and uniqueness of solutions (rigorously in finite dimensions, heuristically in infinite dimensions), and shows that W serves as a Lyapunov functional. The linearised dynamics near critical points and the interpretation of the Hessian as encoding relaxation rates are also considered.

Section 5 introduces the stochastic dynamics on S in the form of a Langevin SDE. Standard existence and uniqueness results in finite dimensions are recalled, and the relationship between drift, noise, and the potential landscape is explained. Section 6 then turns to the induced dynamics on probability distributions, deriving the Fokker–Planck equation in finite dimensions and establishing the free energy functional as a Lyapunov functional along this flow, including the characterisation of stationary distributions.

Section 7 presents illustrative examples and special cases, including the Ornstein–Uhlenbeck process for quadratic potentials, a one-dimensional double-well potential with a bimodal stationary density, and a brief sketch of a field-like infinite-dimensional example. Section 8 positions the present work within a broader programme that applies similar structures to complex systems and information-processing settings, without making strong interpretive claims. Finally, Section 9 summarises the results, discusses limitations, and outlines directions for future work, including extensions towards information geometry, more refined infinite-dimensional analysis, and applications to concrete models.

2 Notation, basic setup and assumptions

In this section we collect the basic objects, notation and standing assumptions that will be used throughout the paper. The state space is specified as a separable Hilbert space, equipped with its Borel σ -algebra and a fixed reference measure. We also introduce the core dynamical quantities such as the potential W , its gradient ∇W , the state trajectories $(s_t)_{t \geq 0}$, the noise structure, and the free energy functional.

3 State space as a separable Hilbert space

Definition 3.1 (State space). The state space S is a separable Hilbert space over $\mathbb{R}$ (or, if desired, over $\mathbb{C}$), equipped with inner product $\langle \cdot, \cdot \rangle$ and associated norm $\| \cdot \|$, that is,

$$\|s\| = \sqrt{\langle s, s \rangle}, \quad s \in S.$$

Separability means that there exists a countable dense subset of S . This assumption is standard in analysis and probability theory and ensures, for instance, that S can be represented (non-canonically) as the completion of $\mathbb{R}^n$ -like finite-dimensional subspaces or as an L^2 -type space of square-integrable functions. Typical examples include

- finite-dimensional spaces $S = \mathbb{R}^n$ with the standard Euclidean inner product,
- spaces of square-integrable functions $S = L^2(X, \nu)$ over a measure space (X, ν) .

Remark 3.2 (Optional RKHS refinement). In many applications, S may be realised as a reproducing kernel Hilbert space associated with a positive definite kernel $k : X \times X \rightarrow \mathbb{R}$.

This can be useful when S is used to encode functions on some underlying set X . The formalism developed in this paper does not depend on this structure; whenever a reproducing kernel Hilbert space interpretation is relevant, it can be inserted as an additional refinement without altering the core definitions.

Throughout the paper, time is denoted by $t \in [0, \infty)$, and trajectories $(s_t)_{t \geq 0}$ take values in S . Time derivatives of sufficiently regular trajectories are written as $\dot{s}_t$.

3.1 Topology and measure structure

The Hilbert space structure induces a natural topology and measurable structure.

Definition 3.3 (Borel σ -algebra). Let $\mathcal{B}(S)$ denote the Borel σ -algebra on S generated by the open sets of the norm topology induced by $\|\cdot\|$.

To formulate distributional dynamics and free energy functionals, it is convenient to fix a reference measure.

Assumption 3.4 (Reference measure). We fix a reference measure μ on $(S, \mathcal{B}(S))$. Typical choices are:

- for finite-dimensional state spaces $S = \mathbb{R}^n$, the Lebesgue measure on $\mathbb{R}^n$;
- for infinite-dimensional S , a Gaussian reference measure, for example the law of a centred non-degenerate Gaussian random variable in S .

Whenever we speak of a *probability density* $\rho(\cdot, t)$ on S , we mean a non-negative measurable function $\rho(\cdot, t) : S \rightarrow [0, \infty)$ such that

$$\int_S \rho(s, t) d\mu(s) = 1.$$

The associated probability measure at time t is then given by $\rho(\cdot, t) \mu$, which is absolutely continuous with respect to μ .

Remark 3.5 (Finite-dimensional specialisation). Many of the rigorous results concerning stochastic differential equations and Fokker–Planck dynamics will be formulated for the special case $S = \mathbb{R}^n$ with the standard inner product, norm, and Lebesgue reference measure. In this setting, μ is simply Lebesgue measure, and densities are understood in the usual sense of classical probability theory on $\mathbb{R}^n$.

3.2 Core dynamical objects and notation

This subsection summarises the main objects and symbols used in the remainder of the paper. Unless explicitly stated otherwise, all dependences on t are assumed to be for $t \geq 0$.

- **Potential (energy) functional:** The central object of the theory is a functional

$$W : S \rightarrow \mathbb{R} \cup \{+\infty\},$$

which will be referred to as the potential or energy function. Its precise assumptions (such as differentiability and coercivity) are stated in detail in Section 3.

- **Gradient of the potential:** Wherever W is differentiable in the Fréchet sense, its gradient $\nabla W(s) \in S$ is defined via the Riesz representation:

$$\langle \nabla W(s), h \rangle = DW(s)[h] \quad \text{for all } h \in S,$$

where $DW(s)$ is the Fréchet derivative of W at s . We write

$$\nabla W : \mathcal{D}(\nabla W) \subseteq S \rightarrow S$$

for the (possibly partially defined) gradient mapping.

- **Deterministic trajectories:** A deterministic state trajectory is a map

$$t \mapsto s_t \in S,$$

which, in the deterministic gradient-flow setting, will satisfy an evolution equation of the form

$$\dot{s}_t = -\nabla W(s_t),$$

with prescribed initial condition $s_0 \in S$.

- **Drift strength operator:** The parameter D denotes either a positive scalar or a bounded linear operator on S , typically assumed to be self-adjoint and strictly positive in the operator sense. In the stochastic dynamics, D will control the strength of the deterministic drift term $-D\nabla W(s_t)$, and it will also appear in the free energy functional.
- **Noise operator:** The symbol B denotes a bounded linear operator acting on a suitable Hilbert space of noise directions (for example $\mathbb{R}^m$ in finite dimensions) into S . It scales the noise term in the Langevin equation. The covariance operator associated with the noise is typically of the form $Q = BB^*$.
- **Wiener process (Brownian motion):** The noise is modelled by a Wiener process

$$(W_t)_{t \geq 0},$$

which is a standard Brownian motion in an auxiliary Hilbert space (for instance $\mathbb{R}^m$ in finite dimensions). In infinite-dimensional formulations, W_t may be taken as a cylindrical Wiener process, with BW_t forming an S -valued stochastic process.

- **Probability densities:** The time-dependent probability distribution of the state is described by a density

$$\rho(\cdot, t) : S \rightarrow [0, \infty)$$

with respect to the reference measure μ , so that $\rho(\cdot, t) \mu$ is the law of s_t whenever this law is absolutely continuous with respect to μ . For brevity, we will often write $\rho_t(s)$ instead of $\rho(s, t)$.

- **Free energy functional:** The free energy associated with a density ρ is the functional

$$F[\rho] = \int_S W(s) \rho(s) d\mu(s) + D \int_S \rho(s) \ln \rho(s) d\mu(s),$$

whenever the integrals are well-defined. Here the first term is the expected potential energy, and the second term is an entropy term weighted by D . The analytic properties of F and its behaviour along distributional flows will be studied in Section 6.

- **Set of minimisers:** The set of global minimisers of W is denoted by

$$\mathcal{M} := \left\{ s \in S : W(s) = \inf_{u \in S} W(u) \right\}.$$

Under appropriate assumptions on W , existence and basic properties of $\mathcal{M}$ will be discussed in Section 3.

Remark 3.6 (Notation conventions). Throughout the paper, we use the following conventions:

- Inner products and norms without subscripts refer to the Hilbert space S , that is $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_S$ and $\| \cdot \| = \| \cdot \|_S$.
- For operators A on S , we write A^* for the adjoint with respect to $\langle \cdot, \cdot \rangle$.
- Expectations with respect to a probability measure ν are denoted by $\mathbb{E}_\nu[\cdot]$; when ν is clear from context, we may simply write $\mathbb{E}[\cdot]$.
- For functionals $G[\rho]$ defined on densities, square brackets emphasise that G acts on the entire function ρ , rather than on a point s .

These conventions will be followed consistently in all subsequent sections.

4 The potential W on a Hilbert space

In this section we specify the class of energy (or potential) functionals W considered in the sequel. We first give a basic definition and then formulate a set of standing assumptions which guarantee good variational properties such as existence of global minimisers. We also introduce several example classes that will be used later for explicit computations and illustrations.

4.1 Basic definition and domain

Definition 4.1 (Potential (energy) functional). Let S be the separable Hilbert space introduced in Section 2. A *potential* (or energy functional) is a map

$$W : S \rightarrow \mathbb{R} \cup \{+\infty\}.$$

The value $W(s) = +\infty$ indicates that the state $s \in S$ is energetically forbidden or excluded from the admissible domain of the system.

Definition 4.2 (Effective domain). The *effective domain* of W is defined as

$$\text{dom}(W) := \{s \in S : W(s) < +\infty\}.$$

We say that W is *proper* if $\text{dom}(W)$ is non-empty and W does not take the value $-\infty$ anywhere.

Throughout the paper, W will always be assumed proper. The dynamics considered later will be restricted to the effective domain $\text{dom}(W)$ or to suitable subsets thereof (for instance, to regions where W is sufficiently differentiable).

4.2 Standing assumptions on W

We now collect the main analytic assumptions on the potential. These are chosen to balance generality with the need for standard variational and dynamical results.

Assumption 4.3 (Lower semicontinuity). The potential W is lower semicontinuous (l.s.c.) with respect to the norm topology on S , i.e.

$$\liminf_{n \rightarrow \infty} W(s_n) \geq W(s) \quad \text{whenever } s_n \rightarrow s \text{ in } S.$$

Lower semicontinuity is a standard requirement in the calculus of variations and ensures, in particular, that W behaves well under limits of minimising sequences.

Assumption 4.4 (Coercivity / growth). The potential W is coercive in the sense that

$$\|s\| \rightarrow \infty \quad \Rightarrow \quad W(s) \rightarrow +\infty.$$

Equivalently, for every $M \in \mathbb{R}$ the sublevel set

$$\{s \in S : W(s) \leq M\}$$

is bounded in S .

Coercivity prevents the energy from being arbitrarily small along directions in which the norm diverges and is essential for the existence of minimisers.

Assumption 4.5 (Differentiability and gradient mapping). There exists an open subset $U \subseteq \text{dom}(W)$ such that:

1. U is dense in $\text{dom}(W)$;
2. W is Fréchet differentiable on U ;
3. for each $s \in U$, the Fréchet derivative $DW(s)$ is a bounded linear functional on S and there exists a unique gradient $\nabla W(s) \in S$ satisfying

$$DW(s)[h] = \langle \nabla W(s), h \rangle \quad \text{for all } h \in S.$$

We write

$$\nabla W : U \rightarrow S$$

for the gradient mapping.

Remark 4.6 (Higher regularity). In some parts of the analysis (notably when discussing Hessians and linearised dynamics), it will be convenient to assume that W is twice Fréchet differentiable on U , with a continuous second derivative. When this additional assumption is needed, we will state it explicitly. It is not required for the basic existence result for global minimisers.

Assumption 4.7 (Growth of the gradient). On U , the gradient satisfies a polynomial growth condition: there exists $C \geq 0$ and $p \geq 0$ such that

$$\|\nabla W(s)\| \leq C(1 + \|s\|^p) \quad \text{for all } s \in U.$$

This type of growth condition will later be used in the analysis of deterministic and stochastic dynamics to obtain existence and uniqueness of solutions, at least in finite dimensions.

4.3 Global minimisers and existence

The natural candidates for long-time limiting states of the gradient dynamics are the global minimisers of W .

Definition 4.8 (Set of global minimisers). The set of global minimisers of W is defined by

$$\mathcal{M} := \left\{ s \in S : W(s) = \inf_{u \in S} W(u) \right\}.$$

The following theorem states that under the above assumptions, global minimisers exist. It is a standard result in the calculus of variations on reflexive Banach spaces, which applies in particular to Hilbert spaces.

Theorem 4.9 (Existence of global minimisers). *Let S be a Hilbert space and let $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper, lower semicontinuous and coercive (Assumptions 4.3 and 4.4). Then:*

1. *The infimum $\inf_{s \in S} W(s)$ is finite.*
2. *The set $\mathcal{M}$ of global minimisers is non-empty.*
3. *Every minimising sequence $(s_n)_{n \geq 1}$ with $W(s_n) \rightarrow \inf_{s \in S} W(s)$ admits a subsequence converging (in norm) to a point of $\mathcal{M}$.*

Proof. We sketch the main ideas. Since W is proper, there exists $s_0 \in S$ with $W(s_0) < +\infty$, so $\inf_{s \in S} W(s)$ is bounded above and thus not $+\infty$. Coercivity implies that for each $M \in \mathbb{R}$, the sublevel set $\{s \in S : W(s) \leq M\}$ is bounded. In a Hilbert space, bounded sets are weakly relatively compact, i.e. every bounded sequence has a weakly convergent subsequence.

Consider a minimising sequence $(s_n)_{n \geq 1}$ with $W(s_n) \rightarrow \inf_{s \in S} W(s)$. Coercivity ensures that (s_n) is bounded, so (passing to a subsequence if necessary) we may assume $s_n \rightharpoonup s_*$ weakly in S for some $s_* \in S$. Lower semicontinuity of W with respect to the weak topology of S (which follows from norm l.s.c. in the Hilbert setting) yields

$$W(s_*) \leq \liminf_{n \rightarrow \infty} W(s_n) = \inf_{s \in S} W(s).$$

Thus s_* is a global minimiser and lies in $\mathcal{M}$. Finally, weak convergence plus convergence of norms can be used to extract a subsequence converging strongly in S ; the details are standard in Hilbert spaces and are omitted here. $\square$

Remark 4.10 (Structure of $\mathcal{M}$). The set of minimisers $\mathcal{M}$ may consist of a single point, a finite set of points, a finite-dimensional manifold, or a more complicated closed subset of S . The global structure of $\mathcal{M}$ is a key ingredient in the qualitative behaviour of the gradient and stochastic dynamics. For instance, in many examples the stationary distribution of the associated Langevin dynamics will concentrate near $\mathcal{M}$ as the noise level decreases.

4.4 Example classes of potentials

We now describe several classes of potentials that satisfy the above assumptions and will serve as guiding examples in later sections.

4.4.1 Quadratic and affine-quadratic potentials

A basic and important class is obtained from bounded, self-adjoint, strictly positive operators.

Definition 4.11 (Quadratic potentials). Let $A : S \rightarrow S$ be a bounded, self-adjoint, strictly positive operator (i.e. $\langle As, s \rangle \geq \alpha \|s\|^2$ for some $\alpha > 0$ and all $s \in S$). Define

$$W(s) = \frac{1}{2} \langle As, s \rangle + \langle b, s \rangle + c,$$

where $b \in S$ and $c \in \mathbb{R}$ are fixed.

In this case, W is finite everywhere, continuous and differentiable on S , with

$$\nabla W(s) = As + b.$$

The coercivity Assumption 4.4 follows from the strict positivity of A , and lower semicontinuity is immediate from continuity. The unique global minimiser is given by $s_* = -A^{-1}b$, and W is strictly convex.

Quadratic potentials on finite-dimensional state spaces $S = \mathbb{R}^n$ lead, in the stochastic setting, to Ornstein–Uhlenbeck processes with Gaussian stationary measures; see Section 7.

4.4.2 Ginzburg–Landau type potentials

A second class of examples generalises double-well and multi-well potentials from finite-dimensional systems to function spaces. Let $S = L^2(X, \nu)$ for some measure space (X, ν) , and let $V : \mathbb{R} \rightarrow \mathbb{R}$ be a potential function on the real line, for instance a double-well potential of the form

$$V(z) = \frac{\lambda}{4} (z^2 - a^2)^2$$

with parameters $\lambda > 0$ and $a > 0$.

Definition 4.12 (Local Ginzburg–Landau functional). For $s \in S = L^2(X, \nu)$, define

$$W_{\text{loc}}(s) = \int_X V(s(x)) \, d\nu(x),$$

whenever the integral is finite, and $W_{\text{loc}}(s) = +\infty$ otherwise.

Under mild growth and regularity assumptions on V , the functional W_{loc} is lower semicontinuous and proper. Coercivity can be ensured by imposing polynomial growth on V with positive leading coefficient. The effective domain of W_{loc} then consists of those $s \in L^2(X, \nu)$ for which $V \circ s$ is integrable.

To incorporate interaction or regularity terms, one often adds a quadratic form in derivatives or in an operator applied to s .

Definition 4.13 (Local plus regularisation term). Let $L : S \rightarrow S$ be a densely defined, closed linear operator (e.g. a differential operator such as a Laplacian in a suitable function space setting), and let $\lambda \geq 0$ be a parameter. Define

$$W(s) = W_{\text{loc}}(s) + \frac{\lambda}{2} \|Ls\|^2,$$

with the convention that $W(s) = +\infty$ if $s \notin \text{dom}(L)$ or $W_{\text{loc}}(s) = +\infty$.

The additional quadratic term penalises roughness or deviation from some preferred subspace and typically improves coercivity. Under suitable assumptions on L and V , this class of functionals provides a rich energy landscape with multiple minima and is widely used in phase field and pattern formation models.

4.4.3 Finite-dimensional multi-well potentials

In finite dimensions $S = \mathbb{R}^n$, a multi-well potential can be constructed by superposing radial or anisotropic wells. For example, let $m_1, \dots, m_k \in \mathbb{R}^n$ be distinct points (“preferred states”) and consider

$$W(s) = \min_{1 \leq j \leq k} \left\{ \frac{1}{2} \langle A_j(s - m_j), (s - m_j) \rangle \right\} + c,$$

where A_j are strictly positive definite matrices and $c \in \mathbb{R}$. One may also use smooth approximations of the minimum, such as a soft-min construction, to obtain a differentiable potential. These examples give rise to energy landscapes with multiple local minima, which will lead to metastability phenomena when noise is added to the dynamics.

4.5 Discussion and outlook

The assumptions formulated above are sufficient to ensure existence of global minimisers and to support well-posed gradient and stochastic dynamics in many relevant cases. They are not, however, unique or exhaustive. In particular:

- In some applications, convexity of W may be imposed to obtain stronger uniqueness and stability results.
- In non-convex settings, one is often interested in local minimisers or saddle points of W , which will play a role when discussing linearised dynamics and transition phenomena, but these are not yet needed at the level of basic existence.
- More singular or hierarchical energy landscapes, for instance involving non-local or multi-scale interactions, can also be considered. Such generalisations typically require additional functional analytic tools and will be mentioned only briefly in later sections as directions for future work.

In the next section, we use these assumptions to define and analyse the deterministic gradient flow

$$\dot{s}_t = -\nabla W(s_t)$$

on S , establish the Lyapunov property of W along the flow, and discuss the local structure of the dynamics near critical points.

5 Deterministic gradient flow on S

In this section we introduce the deterministic gradient dynamics induced by the potential W on the Hilbert space S . We formulate the basic gradient flow equation, recall standard existence and uniqueness results in finite dimensions, and outline the corresponding Hilbert-space framework. We then establish the Lyapunov property of W along the flow and analyse the linearised dynamics near critical points via the Hessian.

5.1 The gradient flow equation

Given the potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ and its gradient ∇W on the open set $U \subseteq \text{dom}(W)$ (Assumption 4.5), the deterministic gradient flow is formally defined by the ordinary differential equation

$$\dot{s}_t = -\nabla W(s_t), \quad t \geq 0, \tag{5.1}$$

with initial condition

$$s_0 = s_{\text{init}} \in U.$$

Definition 5.1 (Solution of the gradient flow). A (forward) *solution* of the gradient flow (5.1) on an interval $[0, T)$ is a map

$$s : [0, T) \rightarrow U \subseteq S$$

such that:

1. s is continuously differentiable as a map into S ;
2. $s(0) = s_{\text{init}}$;
3. for all $t \in [0, T)$,

$$\dot{s}(t) = -\nabla W(s(t)).$$

If $T = +\infty$ we say that s is a *global* solution.

In finite dimensions, the right-hand side of (5.1) can be treated as a vector field on $\mathbb{R}^n$, and standard Picard–Lindelöf theory applies under local Lipschitz conditions. In infinite dimensions, one typically uses the theory of (possibly nonlinear) semigroups on Hilbert spaces and gradient flows of lower semicontinuous functionals.

5.2 Finite-dimensional existence and uniqueness

We first consider the case $S = \mathbb{R}^n$ with the standard inner product, where the gradient ∇W can be identified with the usual gradient vector field.

Assumption 5.2 (Local Lipschitz gradient in finite dimensions). For $S = \mathbb{R}^n$, we assume that W is continuously differentiable on an open set $U \subseteq \mathbb{R}^n$ and that the gradient $\nabla W : U \rightarrow \mathbb{R}^n$ is locally Lipschitz, i.e. for every compact $K \subseteq U$ there exists $L_K \geq 0$ such that

$$\|\nabla W(s) - \nabla W(u)\| \leq L_K \|s - u\| \quad \text{for all } s, u \in K.$$

Theorem 5.3 (Local existence and uniqueness in $\mathbb{R}^n$). *Let $S = \mathbb{R}^n$ and suppose Assumption 5.2 holds. Then, for each initial value $s_{\text{init}} \in U$, there exists a maximal time $T_{\text{max}} \in (0, \infty]$ and a unique solution*

$$s : [0, T_{\text{max}}) \rightarrow U$$

of the gradient flow (5.1) with $s(0) = s_{\text{init}}$. Moreover, if $T_{\text{max}} < \infty$, then

$$\lim_{t \uparrow T_{\text{max}}} \|s(t)\| = +\infty \quad \text{or} \quad \lim_{t \uparrow T_{\text{max}}} \text{dist}(s(t), \partial U) = 0.$$

Proof. The result is an immediate consequence of the Picard–Lindelöf theorem applied to the vector field $f(s) = -\nabla W(s)$ on $U \subseteq \mathbb{R}^n$, which is locally Lipschitz by Assumption 5.2. The stated blow-up alternative for T_{max} is standard in the theory of ordinary differential equations. $\square$

Under additional assumptions on W , one can ensure global existence.

Theorem 5.4 (Global existence under coercivity in $\mathbb{R}^n$). *Let $S = \mathbb{R}^n$ and suppose that W is continuously differentiable with locally Lipschitz gradient on all of $\mathbb{R}^n$ and satisfies the coercivity Assumption 4.4. Then the solution of (5.1) with any initial condition $s_{\text{init}} \in \mathbb{R}^n$ exists for all $t \geq 0$.*

Idea of the proof. By Theorem 5.3, there exists a maximal solution $s : [0, T_{\max}) \rightarrow \mathbb{R}^n$. Coercivity implies that all sublevel sets of W are bounded. The Lyapunov property of W (proved in Section 5.4) ensures that $W(s(t))$ is non-increasing in t , and thus $s(t)$ remains in a bounded sublevel set. This precludes blow-up of $\|s(t)\|$ in finite time. Since $U = \mathbb{R}^n$, the solution cannot hit the boundary of U , and therefore $T_{\max} = +\infty$. $\square$

5.3 Hilbert-space gradient flows

In infinite dimensions, the gradient flow generated by W can still be made precise under appropriate conditions. A standard framework is that of gradient flows of proper, lower semicontinuous, convex functionals on Hilbert spaces, where ∇W is replaced by the subdifferential ∂W .

Remark 5.5 (Subgradient flows). If $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex, proper and lower semicontinuous, one can consider the differential inclusion

$$\dot{s}_t \in -\partial W(s_t),$$

where $\partial W(s)$ denotes the subdifferential of W at s . Under mild assumptions, this inclusion generates a contraction semigroup on S , providing global solutions s_t defined for all $t \geq 0$. In this setting, W again plays the role of a Lyapunov functional.

In the present work we focus mainly on the (more specific) case where W is Fréchet differentiable on an open dense subset $U \subseteq S$ (Assumption 4.5) and formally write the gradient flow in the form (5.1). The rigorous infinite-dimensional theory depends sensitively on additional properties such as convexity, monotonicity of ∇W , and domain conditions, and will not be developed in full generality here. Instead, we treat the Hilbert-space gradient flow as a guiding structure, referring to the extensive literature on gradient flows in Hilbert and metric spaces for rigorous details.

5.4 Lyapunov property of the potential

A central feature of the gradient flow (5.1) is that the potential W decreases monotonically along solutions, as long as the trajectory remains in the differentiability domain of W .

Theorem 5.6 (Lyapunov property of W). *Let $s : [0, T) \rightarrow U \subseteq S$ be a solution of the gradient flow (5.1), with W Fréchet differentiable on U . Suppose that $t \mapsto s(t)$ is continuously differentiable and that $t \mapsto W(s(t))$ is differentiable almost everywhere on $[0, T)$. Then, for all such t ,*

$$\frac{d}{dt}W(s(t)) = -\|\nabla W(s(t))\|^2 \leq 0. \quad (5.2)$$

In particular, $t \mapsto W(s(t))$ is non-increasing on $[0, T)$.

Proof. By the chain rule for Fréchet differentiable functionals on Hilbert spaces,

$$\frac{d}{dt}W(s(t)) = DW(s(t))[\dot{s}(t)] = \langle \nabla W(s(t)), \dot{s}(t) \rangle.$$

Using the gradient flow equation $\dot{s}(t) = -\nabla W(s(t))$, we obtain

$$\frac{d}{dt}W(s(t)) = \langle \nabla W(s(t)), -\nabla W(s(t)) \rangle = -\|\nabla W(s(t))\|^2 \leq 0.$$

This proves (5.2) and shows that $W(s(t))$ is non-increasing. $\square$

Remark 5.7 (Energy dissipation). The identity (5.2) can be integrated over any interval $[t_1, t_2] \subset [0, T]$ to yield

$$W(s(t_2)) - W(s(t_1)) = - \int_{t_1}^{t_2} \|\nabla W(s(t))\|^2 dt.$$

Thus the total decrease in potential energy over $[t_1, t_2]$ equals the accumulated squared norm of the gradient along the trajectory. In particular, if $W(s(t))$ converges as $t \rightarrow \infty$, then

$$\int_0^\infty \|\nabla W(s(t))\|^2 dt < \infty,$$

and $\|\nabla W(s(t))\| \rightarrow 0$ along a subsequence as $t \rightarrow \infty$.

5.5 Critical points, Hessian and linearised dynamics

To understand the local behaviour of the gradient flow near equilibria, we examine critical points of W and the associated Hessian.

Definition 5.8 (Critical points). A point $s_* \in U$ is called a *critical point* of W if $\nabla W(s_*) = 0$. If, in addition, $s_* \in \mathcal{M}$, we call s_* a *global minimiser*. If s_* minimises W only in a neighbourhood, it is a *local minimiser*.

Assumption 5.9 (Twice differentiability and Hessian). In a neighbourhood of a given critical point $s_* \in U$, the functional W is twice Fréchet differentiable. The second derivative at s_* is a bounded, self-adjoint bilinear form, represented (via the Riesz isomorphism) by a bounded, self-adjoint linear operator

$$H_* : S \rightarrow S,$$

called the *Hessian* of W at s_* . That is,

$$D^2W(s_*)[h, k] = \langle H_* h, k \rangle \quad \text{for all } h, k \in S.$$

Formally expanding W near s_* yields

$$W(s_* + u) = W(s_*) + \frac{1}{2} \langle H_* u, u \rangle + o(\|u\|^2), \quad u \rightarrow 0,$$

with $\nabla W(s_* + u) = H_* u + o(\|u\|)$.

5.5.1 Linearisation of the flow

Consider a solution $s(t)$ of the gradient flow starting near s_* . Write $s(t) = s_* + u(t)$, where $u(t)$ is assumed small. Then

$$\dot{u}(t) = \dot{s}(t) = -\nabla W(s_* + u(t)) = -H_* u(t) + \text{higher-order terms}.$$

Neglecting the higher-order terms yields the linearised system

$$\dot{u}(t) = -H_* u(t). \tag{5.3}$$

Definition 5.10 (Linearised dynamics). The *linearised gradient flow* at a critical point s_* is the linear ODE (5.3) on S with initial condition $u(0) = u_{\text{init}} \in S$.

If H_* is bounded and self-adjoint, the solution of (5.3) is given by

$$u(t) = e^{-tH_*} u_{\text{init}},$$

where $(e^{-tH_*})_{t \geq 0}$ is the strongly continuous semigroup generated by $-H_*$.

5.5.2 Relaxation rates and spectral decomposition

When H_* has a discrete spectrum (for example in finite dimensions, or in certain compact perturbation settings), one can decompose the dynamics along its eigenvectors.

Assumption 5.11 (Spectral decomposition at a non-degenerate minimum). Suppose that S is finite-dimensional or that H_* has a pure point spectrum with an orthonormal basis of eigenvectors $(e_k)_{k \in K}$ and eigenvalues $(\lambda_k)_{k \in K} \subset \mathbb{R}$ satisfying

$$H_* e_k = \lambda_k e_k, \quad k \in K.$$

Writing the initial perturbation as

$$u_{\text{init}} = \sum_{k \in K} c_k e_k,$$

we obtain

$$u(t) = \sum_{k \in K} c_k e^{-\lambda_k t} e_k.$$

The behaviour of $u(t)$ as $t \rightarrow \infty$ is governed by the signs and magnitudes of the eigenvalues:

- If all $\lambda_k > 0$, then s_* is a (strict) local minimiser and $u(t) \rightarrow 0$ exponentially fast. The smallest eigenvalue controls the slowest relaxation rate.
- If some $\lambda_k < 0$, then s_* is unstable: perturbations along eigenvectors with negative eigenvalues grow exponentially.
- If some $\lambda_k = 0$ while others are positive, there are directions of neutral stability; further nonlinear analysis is required to classify the behaviour.

Remark 5.12 (Non-degenerate and degenerate minima). A critical point s_* is called *non-degenerate* if H_* is invertible (i.e. has trivial kernel). In this case, the spectrum is bounded away from 0 and the linearised dynamics exhibit purely exponential behaviour. Degenerate minima (with $\ker H_* \neq \{0\}$) often arise in the presence of symmetries or constraints and require more delicate analysis.

5.6 Qualitative picture

Combining the results above, the deterministic gradient flow induced by W on S has the following qualitative features:

- Solutions move in the direction of steepest descent of W and cannot increase the value of the potential.
- In finite dimensions, under mild regularity and coercivity conditions, global solutions exist for all times and remain in bounded sublevel sets of W .
- Critical points of W correspond to equilibria of the flow; the Hessian at a critical point determines the linearised stability and relaxation rates.
- For energy landscapes with multiple minima, the flow partitions the state space into basins of attraction of these minima. Transitions between different basins cannot occur in the purely deterministic setting, but will become possible once stochastic perturbations are added in the Langevin dynamics.

In the next section we extend the deterministic gradient flow (5.1) to a stochastic setting by adding noise terms, leading to a Langevin-type stochastic differential equation on S . This will allow us to study both local relaxation near minima and stochastic transitions between different regions of the energy landscape.

6 Stochastic dynamics on S (Langevin SDE)

In this section we extend the deterministic gradient flow of Section 4 by adding stochastic forcing. The resulting dynamics are described by a Langevin-type stochastic differential equation (SDE) on the Hilbert space S , in which the drift is given by (a rescaled version of) the gradient of W and the noise is modelled by a Wiener process. We first formulate the SDE in an abstract Hilbert-space setting, then treat the finite-dimensional case rigorously, and finally outline the infinite-dimensional generalisation at the level of a programme.

6.1 Langevin-type SDE on a Hilbert space

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions, and we denote expectation with respect to $\mathbb{P}$ by $\mathbb{E}[\cdot]$.

Definition 6.1 (Langevin SDE on S). Let S be the separable Hilbert space from Section 2, and let D be a fixed linear operator on S , typically self-adjoint and strictly positive in the operator sense. Let B be a bounded linear operator from a (real) Hilbert space H of noise directions into S , and let $(W_t)_{t \geq 0}$ be an H -valued Wiener process adapted to $(\mathcal{F}_t)_{t \geq 0}$.

A (formal) *Langevin-type SDE* for the state $(s_t)_{t \geq 0}$ in S is given by

$$ds_t = -D\nabla W(s_t) dt + B dW_t, \quad (6.1)$$

where ∇W is the gradient of the potential W on its differentiability domain $U \subseteq S$.

In the abstract setting (6.1), several technical questions arise:

- For which s_0 and under which conditions on W , D and B does (6.1) admit a (weak or strong) solution $(s_t)_{t \geq 0}$ taking values in S ?
- When is the solution unique (in law or pathwise)?
- Under which conditions is the solution non-explosive, i.e. does it exist for all $t \geq 0$?

In full generality, these questions require the machinery of stochastic evolution equations on Hilbert spaces. In this paper we treat the finite-dimensional case rigorously and use the infinite-dimensional formulation mainly as a structural guide.

6.2 Finite-dimensional Langevin dynamics

For the remainder of this subsection we specialise to

$$S = \mathbb{R}^n$$

with the standard inner product and norm. In this setting, the operator D can be identified with an $n \times n$ matrix (often chosen as a positive multiple of the identity), and B is an $n \times m$ matrix mapping $\mathbb{R}^m$ -valued Brownian motion into $\mathbb{R}^n$.

6.2.1 Equation and coefficients

We consider an $\mathbb{R}^n$ -valued process $(s_t)_{t \geq 0}$ satisfying

$$ds_t = b(s_t) dt + \sigma(s_t) dW_t, \quad (6.2)$$

where:

- $b : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the drift,
- $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ is the diffusion matrix,
- $(W_t)_{t \geq 0}$ is an m -dimensional standard Brownian motion.

To match the abstract Langevin form (6.1), we set

$$b(s) = -D\nabla W(s), \quad \sigma(s) = B,$$

with constant matrices $D \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$. Often one assumes that D is symmetric and positive definite and that B has full rank. In the simplest isotropic case,

$$D = D_0 I_n, \quad B = \sqrt{2D_0} I_n,$$

with a scalar parameter $D_0 > 0$, leading to

$$ds_t = -D_0 \nabla W(s_t) dt + \sqrt{2D_0} dW_t. \quad (6.3)$$

This particular choice implements a fluctuation–dissipation relation and will be used in Section 6 when discussing stationary distributions and free energy.

6.2.2 Existence and uniqueness of solutions

We recall standard conditions on the drift and diffusion under which (6.2) admits a unique strong solution.

Assumption 6.2 (Local Lipschitz and linear growth). The functions $b : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ satisfy:

1. (Local Lipschitz) For every $R > 0$ there exists $L_R \geq 0$ such that

$$\|b(s) - b(u)\| + \|\sigma(s) - \sigma(u)\| \leq L_R \|s - u\| \quad \text{whenever } \|s\| \leq R, \|u\| \leq R.$$

2. (Linear growth) There exists $K \geq 0$ such that

$$\|b(s)\|^2 + \|\sigma(s)\|^2 \leq K^2(1 + \|s\|^2) \quad \text{for all } s \in \mathbb{R}^n.$$

Here $\|\sigma(s)\|$ denotes, for instance, the Hilbert–Schmidt norm of the matrix $\sigma(s)$.

Theorem 6.3 (Existence and uniqueness of strong solutions). *Suppose Assumption 6.2 holds. Then, for every initial condition $s_0 \in \mathbb{R}^n$, there exists a unique (up to indistinguishability) strong solution $(s_t)_{t \geq 0}$ of the SDE (6.2) with s_0 as initial value. Moreover, the solution is non-explosive: it is defined for all $t \geq 0$ almost surely.*

Proof. This is a classical result in the theory of SDEs; for completeness we recall the main ingredients. Under Assumption 6.2, one constructs approximate solutions by Picard iteration and shows that these converge in suitable norms, yielding existence of a strong solution on a finite time interval. Pathwise uniqueness follows from Grönwall-type estimates. The linear growth condition ensures that the solution cannot explode in finite time, so the local solution extends to all $t \geq 0$. Detailed proofs can be found, for example, in standard textbooks on SDEs. $\square$

Remark 6.4 (Application to the gradient drift). In many situations, the drift $b(s) = -D\nabla W(s)$ inherits its regularity from W . For instance, if W is twice continuously differentiable on $\mathbb{R}^n$ with globally bounded Hessian, then ∇W is globally Lipschitz and Assumption 6.2 holds for constant diffusion $\sigma(s) = B$. Coercivity of W (Assumption 4.4) and suitable growth bounds on ∇W help ensure that the linear growth condition is satisfied.

6.2.3 Generator and quadratic variation

The generator of the diffusion process $(s_t)_{t \geq 0}$ will play a key role in the derivation of the Fokker–Planck equation in Section 6. For a sufficiently smooth test function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$ (e.g. $\varphi \in C_b^2(\mathbb{R}^n)$), the infinitesimal generator L associated with (6.2) is given by

$$(L\varphi)(s) = \langle b(s), \nabla \varphi(s) \rangle + \frac{1}{2} \text{Tr}(a(s) \nabla^2 \varphi(s)), \quad (6.4)$$

where

$$a(s) = \sigma(s) \sigma(s)^\top \in \mathbb{R}^{n \times n}$$

is the diffusion covariance matrix and $\nabla^2 \varphi(s)$ is the Hessian matrix of φ at s .

In the special case of a constant diffusion matrix $\sigma(s) \equiv B$,

$$(L\varphi)(s) = \langle b(s), \nabla \varphi(s) \rangle + \frac{1}{2} \text{Tr}(BB^\top \nabla^2 \varphi(s)).$$

The Itô formula then yields, for such φ ,

$$\varphi(s_t) = \varphi(s_0) + \int_0^t (L\varphi)(s_u) du + \int_0^t \langle \nabla \varphi(s_u), \sigma(s_u) dW_u \rangle,$$

where the last integral is a martingale term. This representation underlies the connection between the SDE and the associated Fokker–Planck equation for the law of s_t .

6.3 Infinite-dimensional extensions (programme level)

Returning to the general Hilbert space S , we sketch how (6.1) can be formulated rigorously as a stochastic evolution equation. The full analysis is technically involved and will not be undertaken in this paper, but the structural picture is important for the intended applications.

6.3.1 Cylindrical Wiener processes and noise

Let $(W_t)_{t \geq 0}$ be a cylindrical Wiener process on a separable Hilbert space H , that is, for each $h \in H$, the process $\langle W_t, h \rangle_H$ is a one-dimensional Brownian motion with variance $t \|h\|_H^2$, and these Brownian motions are mutually independent for orthogonal h . The noise term BW_t is then an S -valued process, provided $B : H \rightarrow S$ is a suitable Hilbert–Schmidt operator.

Formally, the SDE (6.1) becomes

$$ds_t = -D\nabla W(s_t) dt + B dW_t,$$

where the stochastic integral $\int_0^t B dW_u$ is understood in the sense of Hilbert-space-valued Itô integrals.

6.3.2 Stochastic evolution equations

A standard framework for Hilbert-space SDEs writes the equation as

$$ds_t = A(s_t) dt + B dW_t,$$

where $A : \mathcal{D}(A) \subseteq S \rightarrow S$ is (possibly nonlinear) and B is a Hilbert–Schmidt operator from H to S . In our case, we have

$$A(s) = -D\nabla W(s),$$

formally defined on $\mathcal{D}(A) = U \subseteq S$. Under appropriate monotonicity and coercivity conditions on A , one can obtain existence and uniqueness of (mild or weak) solutions using the theory of stochastic evolution equations. Typical assumptions involve:

- demicontinuity and hemicontinuity of A ,
- coercivity inequalities of the form

$$\langle A(s), s \rangle \leq c_1 - c_2 \|s\|^2,$$

- growth bounds on $\|A(s)\|$ in terms of $\|s\|$.

In convex settings, one can also work with subdifferentials of convex functionals in place of classical gradients.

Remark 6.5 (Scope of the present paper). While such infinite-dimensional results are highly relevant, their full development requires substantial functional analytic machinery. In this paper, we treat (6.1) as a formal extension of the finite-dimensional Langevin dynamics, and we rely on finite-dimensional projections (e.g. Galerkin approximations) when we need explicit formulas. The distribution-level analysis in Section 6 will therefore be carried out in finite dimensions.

6.4 Qualitative interpretation of the Langevin dynamics

Although the focus of this work is mathematical, it is useful to summarise the qualitative role of the different terms in the Langevin SDE.

- **Drift term** $-D\nabla W(s_t)$: This term implements a tendency of the system to move along the steepest descent directions of the potential W , scaled by D . In the absence of noise, the dynamics reduce to the deterministic gradient flow discussed in Section 4, and trajectories typically relax towards local or global minima of W .
- **Noise term** $B dW_t$: The stochastic forcing drives random exploration of the state space. The covariance structure of the noise is determined by BB^* ; for instance, isotropic noise with intensity proportional to the identity leads to diffusive spreading in all directions.
- **Competition between drift and noise**: The relative strength of the drift and noise determines the qualitative behaviour. If the noise is weak compared to the curvature of W near its minima, the process spends most of its time fluctuating near local minima (metastable states), with rare transitions between different minima. If the noise is strong, the process can explore the energy landscape more broadly.
- **Connection to deterministic dynamics**: In the limit where the noise amplitude tends to zero (for instance, $B \rightarrow 0$ with D fixed), the stochastic trajectories converge, in an appropriate sense, to the deterministic gradient-flow trajectories. Conversely, for fixed B and D , long-time statistical properties of the process, such as invariant measures, depend on the detailed interplay between W , D and B .

The next section makes these statements precise at the level of probability distributions. In the finite-dimensional setting, we derive the Fokker–Planck equation for the law of s_t , define a free energy functional $F[\rho]$, and show that under suitable conditions $F[\rho_t]$ decreases along the evolution, with stationary solutions of Gibbs type concentrated near global minimisers of W .

7 Distribution dynamics and free energy

In this section we pass from stochastic dynamics of individual trajectories to the induced dynamics of probability distributions on the state space. We work primarily in the finite-dimensional setting $S = \mathbb{R}^n$ with Lebesgue reference measure, where the relevant equations can be formulated rigorously. Starting from a Langevin SDE, we derive the associated Fokker–Planck equation for the probability density, introduce a free energy functional, and establish its Lyapunov property along the evolution. We also identify Gibbs-type stationary distributions and discuss their concentration near global minimisers of the potential W .

7.1 Fokker–Planck equation in finite dimensions

Throughout this section we fix $S = \mathbb{R}^n$ with the standard inner product and Lebesgue measure dx as reference measure. We consider a Langevin dynamics of the form

$$ds_t = -\nabla W(s_t) dt + \sqrt{2D} dW_t, \quad (7.1)$$

where $D > 0$ is a scalar parameter and $(W_t)_{t \geq 0}$ is an n -dimensional standard Brownian motion. This is a special case of the general SDE discussed in Section 5, with constant diffusion matrix $\sigma(s) = \sqrt{2D} I_n$ and drift $b(s) = -\nabla W(s)$.

Assumption 7.1 (Regularity and growth for W in $\mathbb{R}^n$). The potential $W : \mathbb{R}^n \rightarrow \mathbb{R}$ is assumed to be twice continuously differentiable and to satisfy:

1. the gradient ∇W is locally Lipschitz;
2. W is coercive in the sense of Assumption 4.4;
3. ∇W has at most polynomial growth, as in Assumption 4.7.

Under Assumption 7.1, the SDE (7.1) admits a unique global strong solution for each initial condition (by Theorem 6.3, applied with $b(s) = -\nabla W(s)$ and constant σ). We now describe the time evolution of its law.

Definition 7.2 (Probability density). For $t \geq 0$, let ν_t denote the law of s_t . If ν_t is absolutely continuous with respect to Lebesgue measure, we write

$$\nu_t(ds) = \rho(s, t) ds,$$

where $\rho(\cdot, t)$ is the probability density of s_t at time t .

Theorem 7.3 (Fokker–Planck (Kolmogorov forward) equation). *Suppose that for each $t \geq 0$, the law of s_t admits a smooth density $\rho(\cdot, t)$ with sufficient decay at infinity. Then $(\rho(\cdot, t))_{t \geq 0}$ satisfies the Fokker–Planck equation*

$$\partial_t \rho(s, t) = \nabla \cdot (\rho(s, t) \nabla W(s)) + D \Delta \rho(s, t), \quad (7.2)$$

where $\nabla \cdot$ denotes the divergence operator and Δ is the Laplacian on $\mathbb{R}^n$.

Sketch of proof. Let $\varphi \in C_c^\infty(\mathbb{R}^n)$ be a smooth test function with compact support. By Itô's formula applied to $\varphi(s_t)$ and taking expectations, one finds

$$\frac{d}{dt} \mathbb{E}[\varphi(s_t)] = \mathbb{E}[(L\varphi)(s_t)],$$

where L is the generator (6.4) with

$$b(s) = -\nabla W(s), \quad a(s) = 2DI_n.$$

Thus

$$(L\varphi)(s) = -\langle \nabla W(s), \nabla \varphi(s) \rangle + D \Delta \varphi(s).$$

Writing $\mathbb{E}[\varphi(s_t)] = \int_{\mathbb{R}^n} \varphi(s) \rho(s, t) ds$ and integrating by parts, one shows that

$$\frac{d}{dt} \int \varphi(s) \rho(s, t) ds = \int \varphi(s) \left[\nabla \cdot (\rho(s, t) \nabla W(s)) + D \Delta \rho(s, t) \right] ds.$$

Since φ is arbitrary, this yields (7.2) in the weak sense, and under the smoothness assumptions it holds pointwise. $\square$

Equation (7.2) expresses conservation of probability; in divergence form it can be written as

$$\partial_t \rho + \nabla \cdot J = 0, \quad J(s, t) = -\rho(s, t) \nabla W(s) - D \nabla \rho(s, t), \quad (7.3)$$

where J is the probability flux.

7.2 Free energy functional

We now introduce a free energy functional associated with the potential W and the parameter D . Throughout this subsection, $\rho(s)$ denotes a single time-slice of a density, with the understanding that $\rho = \rho(\cdot, t)$ for some t .

Definition 7.4 (Admissible densities). Let $\mathcal{P}$ be the set of probability densities $\rho : \mathbb{R}^n \rightarrow [0, \infty)$ such that

$$\int_{\mathbb{R}^n} \rho(s) ds = 1.$$

We define the subset

$$\mathcal{P}_F := \left\{ \rho \in \mathcal{P} : \int_{\mathbb{R}^n} |W(s)| \rho(s) ds < \infty, \int_{\mathbb{R}^n} \rho(s) \ln \rho(s) ds > -\infty \right\}.$$

Definition 7.5 (Free energy functional). For $\rho \in \mathcal{P}_F$, the *free energy* functional associated with W and $D > 0$ is defined by

$$F[\rho] = \int_{\mathbb{R}^n} W(s) \rho(s) ds + D \int_{\mathbb{R}^n} \rho(s) \ln \rho(s) ds. \quad (7.4)$$

If either of the integrals on the right-hand side is not finite, we set $F[\rho] = +\infty$.

The first term in (7.4) is the expected potential energy, while the second is an entropy term weighted by D . Note that

$$\int \rho \ln \rho ds = -H(\rho),$$

where $H(\rho)$ is the (differential) Shannon entropy of ρ , so the entropy contribution is $-DH(\rho)$.

Remark 7.6 (Relation to relative entropy). Fix any strictly positive reference density π on $\mathbb{R}^n$ and define the relative entropy

$$\text{Ent}(\rho \mid \pi) = \int_{\mathbb{R}^n} \rho(s) \ln \frac{\rho(s)}{\pi(s)} ds.$$

If we choose

$$\pi(s) = Z^{-1} \exp\left(-\frac{W(s)}{D}\right),$$

with normalisation constant Z (when this exists), a direct computation shows that

$$F[\rho] = D \text{Ent}(\rho \mid \pi) + \text{constant}, \quad (7.5)$$

where the constant depends only on W and D (through Z) but not on ρ . Thus, up to an additive constant, F is proportional to the relative entropy of ρ with respect to the Gibbs density π .

7.3 Formal Lyapunov property of the free energy

We now show, at a formal level, that the free energy functional $F[\rho_t]$ decreases along smooth solutions of the Fokker–Planck equation (7.2). Under appropriate regularity and integrability conditions, the same computation can be made rigorous.

Theorem 7.7 (Formal dissipation of free energy). *Let $(\rho(\cdot, t))_{t \geq 0}$ be a smooth family of densities solving the Fokker–Planck equation (7.2), with sufficient decay at infinity so that all integrations by parts below are justified. Then*

$$\frac{d}{dt} F[\rho_t] = - \int_{\mathbb{R}^n} \rho(s, t) \left\| \nabla \left(W(s) + D \ln \rho(s, t) \right) \right\|^2 ds \leq 0. \quad (7.6)$$

In particular, $t \mapsto F[\rho_t]$ is non-increasing.

Formal computation. Using (7.4) and the fact that $\int \rho(s, t) ds = 1$ for all t , we obtain

$$\frac{d}{dt} F[\rho_t] = \int_{\mathbb{R}^n} W(s) \partial_t \rho(s, t) ds + D \int_{\mathbb{R}^n} (1 + \ln \rho(s, t)) \partial_t \rho(s, t) ds.$$

Define

$$\psi_t(s) := W(s) + D \ln \rho(s, t),$$

so that

$$\frac{d}{dt} F[\rho_t] = \int_{\mathbb{R}^n} (\psi_t(s) + D) \partial_t \rho(s, t) ds.$$

Since $\int \partial_t \rho(s, t) ds = 0$, the term involving D vanishes and we have

$$\frac{d}{dt} F[\rho_t] = \int_{\mathbb{R}^n} \psi_t(s) \partial_t \rho(s, t) ds.$$

From (7.3) we can write

$$\partial_t \rho(s, t) = \nabla \cdot (\rho(s, t) \nabla \psi_t(s)),$$

so that, using integration by parts and sufficient decay at infinity,

$$\frac{d}{dt} F[\rho_t] = \int_{\mathbb{R}^n} \psi_t(s) \nabla \cdot (\rho(s, t) \nabla \psi_t(s)) ds = - \int_{\mathbb{R}^n} \rho(s, t) \|\nabla \psi_t(s)\|^2 ds,$$

which is precisely (7.6). $\square$

Remark 7.8 (Free energy as a Lyapunov functional). The identity (7.6) shows that $F[\rho_t]$ is a Lyapunov functional for the Fokker–Planck dynamics: it decreases monotonically along solutions and is constant if and only if the integrand vanishes almost everywhere, i.e. when $\nabla(W + D \ln \rho_t) = 0$ wherever $\rho_t > 0$. This condition characterises stationary distributions, as we discuss next.

7.4 Stationary distributions and Gibbs measures

A stationary distribution for the Fokker–Planck equation (7.2) is a time-independent density ρ_∞ satisfying

$$\nabla \cdot (\rho_\infty(s) \nabla W(s)) + D \Delta \rho_\infty(s) = 0 \quad \text{for all } s \in \mathbb{R}^n, \quad (7.7)$$

with $\rho_\infty \geq 0$ and $\int \rho_\infty ds = 1$.

Proposition 7.9 (Gibbs-form stationary distribution). *Suppose that there exists a normalisation constant $Z \in (0, \infty)$ such that*

$$\rho_\infty(s) = \frac{1}{Z} \exp\left(-\frac{W(s)}{D}\right), \quad Z = \int_{\mathbb{R}^n} \exp\left(-\frac{W(s)}{D}\right) ds. \quad (7.8)$$

Then ρ_∞ is a smooth probability density solving (7.7). Moreover, ρ_∞ minimises the free energy $F[\rho]$ over $\mathcal{P}_F$.

Proof. First, differentiating (7.8) shows

$$\nabla \rho_\infty(s) = -\frac{1}{D} \rho_\infty(s) \nabla W(s),$$

and thus

$$\Delta \rho_\infty(s) = -\frac{1}{D} \nabla \cdot (\rho_\infty(s) \nabla W(s)) + \frac{1}{D^2} \rho_\infty(s) \|\nabla W(s)\|^2.$$

Substituting into the left-hand side of (7.7) yields

$$\begin{aligned} \nabla \cdot (\rho_\infty \nabla W) + D \Delta \rho_\infty &= \nabla \cdot (\rho_\infty \nabla W) + D \left[-\frac{1}{D} \nabla \cdot (\rho_\infty \nabla W) + \frac{1}{D^2} \rho_\infty \|\nabla W\|^2 \right] \\ &= \nabla \cdot (\rho_\infty \nabla W) - \nabla \cdot (\rho_\infty \nabla W) + \frac{1}{D} \rho_\infty \|\nabla W\|^2 \\ &= \frac{1}{D} \rho_\infty \|\nabla W\|^2. \end{aligned}$$

On the other hand, by the explicit formula for $\nabla \rho_\infty$,

$$\nabla(W + D \ln \rho_\infty) = \nabla W + D \frac{\nabla \rho_\infty}{\rho_\infty} = \nabla W - \nabla W = 0,$$

so the integrand in (7.6) vanishes for $\rho = \rho_\infty$. Thus ρ_∞ is consistent with stationarity in the sense of Theorem 7.7. A more direct check shows that the flux $J(s) = -\rho_\infty \nabla W - D \nabla \rho_\infty$ vanishes identically, which implies (7.7).

To see that ρ_∞ minimises F , note that (7.5) implies

$$F[\rho] = D \operatorname{Ent}(\rho \mid \rho_\infty) + \text{constant},$$

for all $\rho \in \mathcal{P}_F$. The relative entropy is non-negative and equals zero if and only if $\rho = \rho_\infty$ almost everywhere. Hence $F[\rho] \geq F[\rho_\infty]$ for all ρ , with equality only at $\rho = \rho_\infty$. $\square$

Remark 7.10 (Concentration near global minimisers as $D \rightarrow 0$). Suppose that W satisfies Assumptions 4.3 and 4.4 and has a non-empty set of global minimisers $\mathcal{M}$. Under suitable regularity conditions, the family of Gibbs measures with densities (7.8) satisfies a Laplace principle: as $D \rightarrow 0$, ρ_∞ concentrates on $\mathcal{M}$, with probability mass increasingly localised near global minima of W . More refined large-deviation principles describe the relative weights of different minima and the asymptotic probability of excursions away from $\mathcal{M}$.

7.5 Gradient flow interpretation in Wasserstein space (formal)

The identity (7.6) suggests that the Fokker–Planck equation (7.2) can be interpreted as a gradient flow of the free energy F on the space of probability measures equipped with a suitable metric. In the finite-dimensional setting $S = \mathbb{R}^n$, the relevant structure is the 2-Wasserstein metric on the space $\mathcal{P}_2(\mathbb{R}^n)$ of probability measures with finite second moment.

At a formal level, if ρ_t solves

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \frac{\delta F}{\delta \rho}(\rho_t) \right),$$

where $\delta F / \delta \rho$ denotes the first variation of F , then substituting

$$\frac{\delta F}{\delta \rho}(\rho)(s) = W(s) + D(\ln \rho(s) + 1)$$

yields precisely the Fokker–Planck equation (7.2). The free energy dissipation (7.6) can then be interpreted as the squared metric slope of F along the gradient flow in the Wasserstein geometry.

Remark 7.11 (Scope of the gradient flow interpretation). The rigorous development of Wasserstein gradient flows requires additional convexity and regularity assumptions on W and control of moments of ρ_t . While this theory provides a powerful and conceptually appealing framework, its systematic use lies beyond the scope of the present paper. We will regard the Wasserstein gradient flow picture as an organising heuristic for understanding the monotonicity of $F[\rho_t]$ and the role of Gibbs measures as equilibrium states.

7.6 Remarks on infinite-dimensional distribution dynamics

In an infinite-dimensional Hilbert space S , the law of s_t is a probability measure on $(S, \mathcal{B}(S))$ rather than a density with respect to Lebesgue measure. One may still consider an evolution equation for the law ν_t of the form

$$\frac{d}{dt} \int_S \varphi(s) \nu_t(ds) = \int_S (L\varphi)(s) \nu_t(ds),$$

where L is the generator associated with the SDE (6.1). In favourable cases, when ν_t is absolutely continuous with respect to a Gaussian reference measure μ , one can express ν_t in terms of a density $\rho(\cdot, t)$ and write a Fokker–Planck type equation analogous to (7.2), with gradients and divergences interpreted in a suitable weak sense.

The free energy functional then takes the form

$$F[\rho] = \int_S W(s) \rho(s) d\mu(s) + D \int_S \rho(s) \ln \rho(s) d\mu(s),$$

and the formal computation leading to (7.6) can be adapted using integration by parts with respect to μ . Making these statements rigorous requires the machinery of Gaussian measures on Hilbert spaces, Dirichlet forms, and infinite-dimensional analysis, and will not be developed here. For the purposes of this paper, the finite-dimensional theory provides the main concrete realisation of the general structural picture:

$$\begin{aligned} \text{stochastic dynamics on } S &\implies \text{Fokker–Planck equation for } \rho_t \\ &\implies \text{free energy } F[\rho_t] \text{ with Lyapunov property.} \end{aligned}$$

In the next section we illustrate these concepts on explicit examples, including quadratic and double-well potentials, and briefly sketch an infinite-dimensional field-like example.

8 Illustrative examples and special cases

In this section we present several concrete examples that instantiate the general framework developed in the previous sections. The focus is on simple potentials in finite dimensions, where the deterministic gradient flow, the Langevin SDE and the associated stationary distributions can be written explicitly, and on a brief sketch of a field-like infinite-dimensional example.

8.1 One-dimensional double-well potential

8.1.1 Definition and deterministic dynamics

We first consider a one-dimensional state space

$$S = \mathbb{R}$$

equipped with the standard inner product and norm. A simple double-well potential is given by

$$W(s) = \frac{1}{4}s^4 - \frac{1}{2}s^2, \quad s \in \mathbb{R}. \quad (8.1)$$

The derivative of W is

$$W'(s) = s^3 - s,$$

so the deterministic gradient flow

$$\dot{s}_t = -W'(s_t)$$

takes the explicit form

$$\dot{s}_t = s_t - s_t^3. \quad (8.2)$$

The equilibria of (8.2) are the critical points of W , i.e. the solutions of $W'(s) = 0$. These are

$$s_* \in \{-1, 0, 1\}.$$

The second derivative

$$W''(s) = 3s^2 - 1$$

yields

$$W''(-1) = W''(1) = 2 > 0, \quad W''(0) = -1 < 0.$$

Thus $s = \pm 1$ are local minima (the two wells), whereas $s = 0$ is a local maximum (the barrier between wells). Correspondingly, in the deterministic gradient flow:

- Trajectories with initial data $s_0 < 0$ and $s_0 \neq 0$ converge to $s = -1$ as $t \rightarrow \infty$.
- Trajectories with $s_0 > 0$ converge to $s = +1$.
- The point $s = 0$ is an unstable equilibrium separating the basins of attraction of the two wells.

More precisely, writing $f(s) = s - s^3$, the linearisation at a critical point s_* is given by

$$\dot{u}_t = f'(s_*)u_t, \quad u_t = s_t - s_*.$$

We have

$$f'(-1) = f'(1) = -2 < 0, \quad f'(0) = 1 > 0,$$

confirming that ± 1 are linearly stable and 0 is unstable.

8.1.2 Stochastic dynamics and Fokker–Planck equation

We now add noise and consider the one-dimensional Langevin SDE

$$ds_t = -W'(s_t) dt + \sqrt{2D} dW_t = -(s_t^3 - s_t) dt + \sqrt{2D} dW_t, \quad (8.3)$$

where $(W_t)_{t \geq 0}$ is a standard one-dimensional Brownian motion and $D > 0$ is a diffusion parameter.

Let $\rho(s, t)$ denote the density of s_t with respect to Lebesgue measure. Under suitable regularity and integrability conditions, ρ satisfies the one-dimensional Fokker–Planck equation corresponding to (8.3):

$$\partial_t \rho(s, t) = \partial_s (\rho(s, t) W'(s)) + D \partial_{ss} \rho(s, t), \quad s \in \mathbb{R}, t > 0. \quad (8.4)$$

Equivalently,

$$\partial_t \rho + \partial_s J = 0, \quad J(s, t) = -\rho(s, t) W'(s) - D \partial_s \rho(s, t),$$

where J is the probability current.

8.2 Stationary density and bimodality

A stationary density ρ_∞ satisfies

$$\partial_s (\rho_\infty(s) W'(s)) + D \partial_{ss} \rho_\infty(s) = 0.$$

Under mild integrability conditions on W , a Gibbs-type solution is given by

$$\rho_\infty(s) = \frac{1}{Z_D} \exp\left(-\frac{W(s)}{D}\right), \quad Z_D = \int_{\mathbb{R}} \exp\left(-\frac{W(u)}{D}\right) du. \quad (8.5)$$

Substituting (8.1) into (8.5) yields

$$\rho_\infty(s) = \frac{1}{Z_D} \exp\left(-\frac{1}{D} \left(\frac{1}{4}s^4 - \frac{1}{2}s^2\right)\right).$$

The function W has two symmetric minima at $s = \pm 1$ and a maximum at $s = 0$, so for small D the density $\rho_\infty(s)$ is *bimodal*: it has two peaks near ± 1 separated by a dip near 0. As $D \rightarrow 0$, the mass of ρ_∞ concentrates near these minima, while for larger D the distribution becomes broader and the double-peak structure is gradually smoothed.

From a dynamical perspective:

- For small noise (D small), sample paths of (8.3) spend long periods fluctuating near one of the wells, with rare noise-induced transitions across the barrier near $s = 0$ to the other well.
- For larger noise, transitions become more frequent and the process explores the state space more evenly.

The free energy functional

$$F[\rho] = \int_{\mathbb{R}} W(s) \rho(s) ds + D \int_{\mathbb{R}} \rho(s) \ln \rho(s) ds$$

is minimised by the stationary density (8.5), and along solutions of (8.4) it decreases monotonically, as discussed in Section 6.

8.3 Quadratic potential and Ornstein–Uhlenbeck process

8.3.1 Deterministic gradient flow

We now consider a quadratic potential on $S = \mathbb{R}^n$ of the form

$$W(s) = \frac{\lambda}{2} \|s\|^2, \quad s \in \mathbb{R}^n, \quad (8.6)$$

with a parameter $\lambda > 0$. The gradient is

$$\nabla W(s) = \lambda s,$$

and the deterministic gradient flow

$$\dot{s}_t = -\nabla W(s_t)$$

reduces to the linear ODE

$$\dot{s}_t = -\lambda s_t. \quad (8.7)$$

The solution with initial condition $s_0 \in \mathbb{R}^n$ is

$$s_t = e^{-\lambda t} s_0,$$

showing exponential relaxation to the unique global minimiser $s = 0$ at rate λ :

$$\|s_t\| = e^{-\lambda t} \|s_0\|.$$

The potential along the trajectory decays as

$$W(s_t) = \frac{\lambda}{2} \|s_t\|^2 = \frac{\lambda}{2} e^{-2\lambda t} \|s_0\|^2,$$

consistent with the Lyapunov property.

8.3.2 Stochastic dynamics: the Ornstein–Uhlenbeck process

We add isotropic noise and consider the SDE

$$ds_t = -\lambda s_t dt + \sqrt{2D} dW_t, \quad (8.8)$$

where $(W_t)_{t \geq 0}$ is an n -dimensional standard Brownian motion and $D > 0$. This is the classical Ornstein–Uhlenbeck (OU) process in $\mathbb{R}^n$.

Equation (8.8) is linear and can be solved explicitly by the variation-of-constants formula. Multiplying by $e^{\lambda t}$ yields

$$d(e^{\lambda t} s_t) = e^{\lambda t} \sqrt{2D} dW_t,$$

so

$$e^{\lambda t} s_t = s_0 + \sqrt{2D} \int_0^t e^{\lambda u} dW_u.$$

Hence

$$s_t = e^{-\lambda t} s_0 + \sqrt{2D} \int_0^t e^{-\lambda(t-u)} dW_u. \quad (8.9)$$

From (8.9) one reads off the mean and covariance:

$$\mathbb{E}[s_t] = e^{-\lambda t} s_0, \quad \text{Cov}(s_t) = \mathbb{E}[(s_t - \mathbb{E}[s_t])(s_t - \mathbb{E}[s_t])^\top] = \frac{D}{\lambda} (1 - e^{-2\lambda t}) I_n.$$

As $t \rightarrow \infty$,

$$\mathbb{E}[s_t] \rightarrow 0, \quad \text{Cov}(s_t) \rightarrow \frac{D}{\lambda} I_n,$$

so the process converges in distribution to a centred Gaussian law.

8.3.3 Stationary Gaussian distribution and free energy

The stationary density ρ_∞ is Gaussian:

$$\rho_\infty(s) = \left(\frac{\lambda}{2\pi D}\right)^{n/2} \exp\left(-\frac{\lambda}{2D}\|s\|^2\right), \quad s \in \mathbb{R}^n. \quad (8.10)$$

It is of the Gibbs form associated with the potential (8.6):

$$\rho_\infty(s) \propto \exp\left(-\frac{W(s)}{D}\right).$$

The corresponding free energy functional

$$F[\rho] = \int_{\mathbb{R}^n} \frac{\lambda}{2} \|s\|^2 \rho(s) ds + D \int_{\mathbb{R}^n} \rho(s) \ln \rho(s) ds$$

is minimised by the Gaussian density (8.10). Along solutions of the Fokker–Planck equation associated with (8.8), $F[\rho_t]$ decreases monotonically and converges to $F[\rho_\infty]$.

8.4 Sketch of an infinite-dimensional example

Finally, we sketch how the framework extends to an infinite-dimensional state space. Let

$$S = L^2(0, 1)$$

be the space of square-integrable real-valued functions on the interval $(0, 1)$, with the usual inner product

$$\langle s, u \rangle = \int_0^1 s(x)u(x) dx.$$

8.4.1 Field-like potential

Consider a potential of Ginzburg–Landau type defined on (a suitable subspace of) $L^2(0, 1)$ by

$$W(s) = \int_0^1 \left(\frac{1}{2} |\partial_x s(x)|^2 + V(s(x)) \right) dx, \quad (8.11)$$

where:

- s belongs to a Sobolev space such as $H_0^1(0, 1)$ (to make $\partial_x s$ meaningful and to encode, for instance, homogeneous Dirichlet boundary conditions $s(0) = s(1) = 0$),
- $V : \mathbb{R} \rightarrow \mathbb{R}$ is a sufficiently smooth local potential, e.g. a double-well potential $V(z) = \frac{\lambda}{4}(z^2 - a^2)^2$.

Under these conditions, W is well-defined and can be shown to satisfy variants of the lower semicontinuity and coercivity assumptions discussed earlier.

8.4.2 Deterministic gradient flow as an evolution PDE

The gradient flow of W with respect to the L^2 inner product yields an evolution equation for $s(t, \cdot)$ of the form

$$\partial_t s(t, x) = \partial_{xx} s(t, x) - V'(s(t, x)), \quad t > 0, x \in (0, 1), \quad (8.12)$$

together with appropriate boundary conditions (e.g. $s(t, 0) = s(t, 1) = 0$). Equation (8.12) is a reaction–diffusion equation of Allen–Cahn type, representing gradient descent in the energy landscape defined by (8.11).

In this setting, W serves as a Lyapunov functional:

$$\frac{d}{dt}W(s(t, \cdot)) = - \int_0^1 |\partial_t s(t, x)|^2 dx \leq 0,$$

with equality only at stationary solutions of (8.12).

8.4.3 Stochastic extension (formal)

A formal Langevin-type stochastic extension of (8.12) can be written as a stochastic partial differential equation (SPDE)

$$\partial_t s(t, x) = \partial_{xx} s(t, x) - V'(s(t, x)) + \sqrt{2D} \xi(t, x), \quad (8.13)$$

where ξ denotes a space-time noise (for instance, space-time white noise in a simplified setting) and $D > 0$ controls its strength. Interpreted appropriately, (8.13) can be viewed as a Langevin equation on the Hilbert space $L^2(0, 1)$, with drift given by $-\nabla W$ and noise driving random fluctuations around the deterministic dynamics.

A rigorous treatment of (8.13) requires the theory of SPDEs in infinite-dimensional spaces and goes beyond the scope of this paper. Nevertheless, this example illustrates how the abstract framework of a potential W on a Hilbert space, deterministic gradient flow, and a stochastic extension naturally generalises to field-theoretic or function-space settings.

These examples demonstrate that the abstract constructions developed in the previous sections reduce, in simple cases, to well-known models such as double-well dynamics and Ornstein–Uhlenbeck processes, and they suggest a path towards more complex infinite-dimensional applications.

9 Position in the broader “semantic field” programme (minimal)

In this section we briefly indicate how the mathematical structures developed so far fit into a broader programme that motivates the terminology of “semantic fields”. The aim is purely contextual: no additional assumptions or claims are introduced, and in particular no strong interpretive or unifying statements are made.

9.1 Conceptual origin: coherence and conflict in complex systems

The starting point of the broader programme is the observation that many complex systems—including information-processing systems, networks of interacting agents, and high-dimensional dynamical models—exhibit patterns that can be informally described in terms of *coherence* and *conflict*. From this perspective, a potential (or energy) functional

$$W : S \rightarrow \mathbb{R} \cup \{+\infty\}$$

is used to encode the “tension” or “inconsistency” of a given state $s \in S$. In that wider context, W is sometimes referred to as a *contradiction energy* (“Widerspruchsenergie”), reflecting the interpretation of large values of $W(s)$ as highly conflicted or incompatible configurations.

The present paper deliberately abstracts away from any specific application domain and treats W simply as an energy functional on a separable Hilbert space. The gradient flow, the Langevin SDE, the Fokker–Planck equation and the free energy functional are developed as general tools, independent of any particular meaning ascribed to the state space S or the energy landscape W .

9.2 Semantic fields and state-space representations

In the broader “semantic field” programme, the state space S is often realised as a Hilbert space of representations (for example, vector embeddings of symbols, concepts, or configurations) in which geometric proximity is intended to reflect a notion of semantic or functional similarity. The potential W then defines an energy landscape over such representations, and the gradient dynamics describe how states move towards configurations of lower contradiction energy.

At a high level, one can distinguish three structural ingredients:

1. a choice of state space S , often equipped with additional structure (e.g. reproducing kernel Hilbert space or feature map);
2. an energy functional W that assigns high values to “incoherent” or “incompatible” states and low values to “coherent” states;
3. a dynamical principle (deterministic or stochastic) that drives the system so as to reduce W on average, while still allowing for exploration under uncertainty.

The formalism developed in this paper corresponds precisely to item (3), given an abstract S and W . The interpretation of S as a space of semantic or functional states is left to companion work.

9.3 Companion directions and applications (informal)

Several lines of work build on the present framework by choosing specific realisations of S and W tailored to different application domains. Examples include, in broad terms:

- **Cognitive and information-processing systems:** Models in which S represents internal state variables or latent configurations of a cognitive or decision-making system, and W encodes internal inconsistency or unresolved constraints.
- **Machine learning and latent spaces:** Settings where S is a latent representation space (for example, an embedding space or a feature space of a trained model), and W captures reconstruction error, constraint violations, or task-specific energy terms, with gradient and Langevin dynamics used as conceptual tools for understanding adaptation and sampling.
- **Networked and multi-component systems:** Frameworks in which S encodes the joint state of many interacting components (nodes, agents, or subsystems), and W aggregates local interaction energies or penalties for incompatible configurations.

In each case, the mathematical structures from this paper—Hilbert-space gradient flows, Langevin-type SDEs, Fokker–Planck equations, and free energy—can be instantiated once a concrete choice of S and W has been made. The present work is intentionally restricted to the shared structural core and does not address modelling choices, empirical validation, or domain-specific interpretation.

9.4 Scope and non-claims

It is important to emphasise what is *not* claimed here. Although the terminology of “semantic fields” and “contradiction energy” originates from attempts to model aspects of complex, possibly information-rich systems, the results of this paper are confined to:

- the specification of a general Hilbert-space framework for energy-driven and noise-driven dynamics;
- the analysis of gradient flows and stochastic dynamics under explicit, standard analytic assumptions;
- the identification of free energy functionals and Gibbs-type stationary measures as structural consequences.

No claims are made about deriving or unifying any established physical theories, nor about providing a comprehensive theory of semantics, cognition, or any other interpretive domain. The role of this paper within the broader programme is to supply a mathematically precise baseline upon which more specialised models can be constructed and analysed in future work.

10 Summary, limitations and outlook

In this final section we summarise the main constructions and results of the paper, discuss the principal limitations of the present treatment, and outline several directions for future work. The emphasis remains on the structural aspects of the framework: a potential W on a separable Hilbert space S , deterministic and stochastic dynamics induced by W , the associated distribution dynamics, and the role of free energy as a Lyapunov functional.

10.1 Summary of the framework

The core of the paper is the development of a coherent chain of structures

$$\begin{aligned} \text{potential } W &\implies \text{gradient flow on } S \\ &\implies \text{Langevin SDE} \\ &\implies \text{Fokker–Planck dynamics and free energy.} \end{aligned}$$

- **State space and potential.** In Section 2 we fixed a separable Hilbert space S with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$, equipped with its Borel σ -algebra and a reference measure μ (Lebesgue measure in finite dimensions, Gaussian measures as typical examples in infinite dimensions). Section 3 introduced a class of potentials

$$W : S \rightarrow \mathbb{R} \cup \{+\infty\},$$

assumed proper, lower semicontinuous and coercive, and Fréchet differentiable on an open dense subset $U \subseteq S$. Under these assumptions, the set of global minimisers

$$\mathcal{M} = \{s \in S : W(s) = \inf_{u \in S} W(u)\}$$

is non-empty. Several example classes were discussed, including quadratic potentials and Ginzburg–Landau type functionals.

- **Deterministic gradient flow.** Section 4 studied the deterministic gradient flow

$$\dot{s}_t = -\nabla W(s_t),$$

with initial data $s_0 \in U$. In finite dimensions, standard local existence and uniqueness results apply under local Lipschitz assumptions on ∇W , and coercivity of W yields global existence in time. The potential W is a Lyapunov functional along the flow,

$$\frac{d}{dt}W(s_t) = -\|\nabla W(s_t)\|^2 \leq 0,$$

and the linearisation at a critical point s_* is governed by the Hessian H_* , whose spectral properties determine local relaxation rates and stability.

- **Stochastic dynamics: Langevin SDE.** In Section 5, the deterministic dynamics were extended to a stochastic setting via a Langevin-type SDE

$$ds_t = -D\nabla W(s_t) dt + B dW_t,$$

where D is a positive operator (or scalar), B is a noise operator, and $(W_t)_{t \geq 0}$ is a Wiener process. In finite dimensions, under local Lipschitz and linear growth conditions on the drift and diffusion, this SDE admits a unique global strong solution. The corresponding infinitesimal generator provides the link to the Fokker–Planck equation.

- **Distribution dynamics and free energy.** Section 6 considered the induced dynamics of probability densities $\rho(s, t)$ in the finite-dimensional setting $S = \mathbb{R}^n$, where the law of s_t admits a density with respect to Lebesgue measure. For the normalised Langevin dynamics

$$ds_t = -\nabla W(s_t) dt + \sqrt{2D} dW_t,$$

the density ρ solves the Fokker–Planck equation

$$\partial_t \rho = \nabla \cdot (\rho \nabla W) + D \Delta \rho.$$

A free energy functional

$$F[\rho] = \int W(s) \rho(s) ds + D \int \rho(s) \ln \rho(s) ds$$

was introduced and shown, at the level of formal computation (and rigorously under suitable regularity assumptions), to satisfy

$$\frac{d}{dt}F[\rho_t] = -D \int \rho_t(s) \|\nabla(W(s) + D \ln \rho_t(s))\|^2 ds \leq 0,$$

so F is a Lyapunov functional along the Fokker–Planck flow.

- **Stationary distributions and examples.** Stationary densities of Gibbs form

$$\rho_\infty(s) \propto \exp\left(-\frac{W(s)}{D}\right)$$

were identified as minimisers of F and as time-invariant solutions of the Fokker–Planck equation, provided the normalisation constant is finite. Section 7 illustrated the framework in explicit examples: a one-dimensional double-well potential with bimodal stationary distributions for small D , quadratic potentials leading to Ornstein–Uhlenbeck processes with Gaussian stationary measures, and a sketch of a Ginzburg–Landau type example on $S = L^2(0, 1)$.

- **Context in a broader programme.** Section 8 briefly indicated how these structures arise in a broader “semantic field” programme, where S often encodes high-dimensional representations and W is interpreted as a measure of internal tension or conflict. The present paper, however, restricts itself to the abstract mathematical framework, without making domain-specific or unifying claims.

Taken together, these elements provide a mathematically structured view of relaxation and stochastic exploration in energy landscapes on Hilbert spaces, with free energy as a unifying quantity linking microscopic dynamics and macroscopic distributional behaviour.

10.2 Limitations of the present treatment

The framework developed here has several important limitations, both conceptual and technical. We list the main ones explicitly.

- **Infinite-dimensional rigour.** While the state space was allowed to be an abstract separable Hilbert space, most rigorous results (existence and uniqueness of solutions, Fokker–Planck dynamics, free energy dissipation) were established only in finite dimensions. Infinite-dimensional extensions—gradient flows of subdifferentials, stochastic evolution equations, infinite-dimensional Fokker–Planck equations and free energy decay with respect to a reference measure μ —were discussed only at the level of a programme, without detailed proofs or sharp conditions.
- **Restricted class of dynamics.** The focus was on first-order gradient flows and their Langevin perturbations. Other important dynamical structures, such as second-order (inertial) dynamics, non-gradient drifts, multiplicative noise, or non-Markovian effects, were not treated. In particular, Hamiltonian structure, symplectic geometry, or energy-conserving dynamics do not appear in the present formulation.
- **Absence of symmetry and conservation-law analysis.** The paper does not analyse symmetry groups of the potential W or of the dynamics, nor does it develop Noether-type results connecting symmetries to conserved quantities. Such structure is often crucial in physical applications but lies beyond the current scope.
- **Limited discussion of metastability and rare events.** While the examples illustrate multi-well potentials and the qualitative idea of metastability under small noise, there is no systematic development of large-deviation theory, exit-time asymptotics, or transition rates between wells. The analysis of rare events is therefore left at a heuristic level.
- **No numerical or algorithmic aspects.** The work is purely analytical and does not address discretisations of the gradient flow or Langevin dynamics, numerical stability, or convergence properties of algorithms (such as gradient descent or Langevin Monte Carlo) that are inspired by the continuous-time models.
- **No empirical or domain-specific modelling.** The state space S and potential W were treated as abstract objects. The paper does not instantiate the framework in concrete empirical models, nor does it discuss parameter estimation, model fitting, or empirical validation in any particular domain.
- **Free energy and optimal transport only at a formal level.** The interpretation of the Fokker–Planck equation as a gradient flow of the free energy in Wasserstein

space was used only as a structural heuristic. The full optimal-transport framework, including geodesic convexity, metric slopes, and contractivity properties, was not rigorously developed.

These limitations reflect a deliberate choice to focus on the structural core of the framework. They also point directly to natural directions for further work.

10.3 Outlook and future work

Several extensions and refinements of the present framework appear both natural and tractable. We briefly outline some of them.

- **Infinite-dimensional gradient and stochastic flows.** A first direction is to develop a fully rigorous infinite-dimensional theory by combining the analytic assumptions on W with existing results on gradient flows and stochastic evolution equations in Hilbert spaces. This includes:
 - gradient flows of convex and non-convex functionals via subdifferentials;
 - existence, uniqueness and ergodic properties of solutions to Langevin-type SPDEs driven by cylindrical noise;
 - identification of invariant measures and free-energy dissipation in infinite dimensions, relative to Gaussian reference measures.
- **Information geometry and optimal transport.** The free energy functional suggests a connection to information geometry and optimal transport. Future work may:
 - investigate convexity properties of F along Wasserstein geodesics;
 - study contractivity and convergence rates of the Fokker–Planck flow in Wasserstein distance;
 - relate the Hessian of W at minimisers to local curvature properties of the free energy landscape on the space of probability measures.
- **Extended classes of dynamics.** It is natural to consider generalisations beyond simple gradient Langevin dynamics, such as:
 - overdamped versus underdamped (inertial) formulations, leading to kinetic Fokker–Planck equations;
 - non-gradient drifts that include rotational or divergence-free components, yielding non-equilibrium steady states with non-vanishing probability currents;
 - multiplicative noise and state-dependent diffusion operators.

Analysing how free energy and related Lyapunov functionals behave in these broader settings is an open direction.

- **Symmetries, constraints and reduced descriptions.** Introducing explicit symmetry groups acting on S , or constraints that restrict dynamics to submanifolds or affine subsets, could lead to reduced effective dynamics and conserved quantities. Exploring how such structures interact with the potential W and the stochastic dynamics, and whether analogues of Noether-type statements hold in this framework, is a promising line of inquiry.

- **Structured examples and interacting systems.** On the modelling side, more structured examples could be developed in which S represents, for example, coupled fields, networks of interacting components, or high-dimensional latent spaces arising from machine learning models. For such systems, one may:
 - construct explicit forms of W that incorporate local interaction terms and regularisation;
 - examine how the geometry of $\mathcal{M}$ and the spectrum of the Hessian influence relaxation and metastability;
 - study how the distributional dynamics approximate or inform effective coarse-grained descriptions.
- **Numerical and algorithmic implications.** Finally, the continuous-time gradient and Langevin dynamics provide a conceptual template for numerical schemes and sampling algorithms. Future work could analyse:
 - discretisations of the gradient flow and their stability in high dimensions;
 - Langevin-type sampling methods, including discretisation error, convergence to equilibrium, and the role of preconditioning via the operator D ;
 - relationships between the energy landscape of W , the choice of noise structure, and the efficiency of exploration algorithms in complex state spaces.

Overall, the framework presented here aims to serve as a mathematically coherent baseline for studying energy-driven dynamics and free-energy-based distributional evolution on Hilbert spaces. By keeping the state space and potential abstract, the results are intended to be adaptable to a range of more specialised settings, to be explored in subsequent work.

Acknowledgements and Contact

This paper is part of the Semantic Gravitation programme and its six-part formal core. It has been developed by an independent, non-specialist author, based on informal intuitions about meaning and extensively structured, checked, and refined with the help of modern AI tools. The work should therefore be regarded as exploratory and conjectural, and it is explicitly offered for critical review, clarification, extension, or refutation by domain experts.

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Semantic Gravitation II: Semantic Field Dynamics and Free Energy

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Abstract

This paper develops the second technical layer of the Semantic Gravitation programme by lifting the semantic dynamics from individual states to probability distributions. The starting point is a state space S modelled as a separable Hilbert space, equipped with a semantic potential $W: S \rightarrow \mathbb{R}$ as in Paper I. A *semantic field* is then represented by a Borel probability measure ρ on S with finite second moment, that is, by an element of the Wasserstein space $\mathcal{P}_2(S)$. On this space we study evolution equations of continuity and Fokker–Planck type, interpreted as macroscopic semantic field dynamics.

The central object is a family of free-energy functionals of the form

$$\mathcal{F}(\rho) = \int_S W(s) \rho(ds) + \beta^{-1} \text{Ent}(\rho \mid \mu),$$

where μ is a reference measure on S , $\beta > 0$ plays the role of an inverse temperature, and $\text{Ent}(\rho \mid \mu)$ denotes relative entropy. Under standard lower semicontinuity, coercivity, and λ -convexity assumptions on W and on the reference structure, we show that the associated Fokker–Planck equations can be characterised as L^2 -Wasserstein gradient flows of $\mathcal{F}$ on $\mathcal{P}_2(S)$. In particular, $\mathcal{F}$ decreases monotonically along solutions and its minimisers describe stationary semantic fields.

The analysis combines stochastic and variational viewpoints. On the microscopic side, we consider Langevin-type dynamics on S ,

$$ds_t = -\nabla W(s_t) dt + B dW_t,$$

with suitable noise operator B and cylindrical Wiener process W_t . We show that the laws ρ_t of s_t solve a Fokker–Planck equation on S and that, under our structural assumptions, this equation admits a unique gradient-flow interpretation in $\mathcal{P}_2(S)$. On the macroscopic side, we construct solutions by minimizing-movement schemes of Jordan–Kinderlehrer–Otto type and prove stability and contraction properties with respect to the Wasserstein distance.

Conceptually, this paper identifies free energy on $\mathcal{P}_2(S)$ as the natural Lyapunov functional for semantic field dynamics and makes explicit contact with the theory of optimal transport. It thereby provides the probabilistic and thermodynamic completion of the Hilbert-space framework introduced in Paper I, and prepares the ground for the subsequent parts of the series: the construction of quantum-compatible semantic fields (Paper III), the formulation of semantic field theories on spacetime manifolds (Paper IV), and the application to complex neural and social systems (Paper V). For a programme-level introduction to Semantic Gravitation and the Lagrangian of Vectorial Emergence (LOVE), we refer to the separate overview paper.

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1 Introduction and motivation

The present work develops a mathematical theory of a free energy functional defined on probability measures over a state space modeled as a separable Hilbert space. The basic ingredients are a state space S , a probability measure ρ on S , a potential function $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$, and a diffusion parameter $D > 0$. From these, we construct a functional of the form

$$F[\rho] = \int_S W(s) \rho(s) d\mu(s) + D \int_S \rho(s) \ln \rho(s) d\mu(s),$$

where μ is a fixed reference measure on S and ρ denotes the Radon–Nikodým density of the probability measure with respect to μ . The first term plays the role of an energy, while the second term is an entropy contribution. Under suitable structural and integrability assumptions, F generates a gradient-flow dynamics on the space of probability measures, which in finite dimensions can be identified with Fokker–Planck-type evolution equations.

The aim of this paper is to give a systematic and self-contained treatment of such free energy functionals and their associated gradient flows, with an emphasis on clear assumptions and explicit links to classical constructions in statistical mechanics and stochastic analysis. Although the functional will occasionally be referred to as a *semantic free energy*, the presentation is purely mathematical: no interpretive or normative claims are made, and no attempt is made to derive full physical theories from the formalism.

1.1 Background

Free energy functionals combining potential energy and entropy terms are central objects in equilibrium and non-equilibrium statistical mechanics, as well as in information theory and stochastic analysis. In their simplest form they are defined on measures over a finite-dimensional configuration space, for instance $S = \mathbb{R}^n$, with μ the Lebesgue measure or a Gaussian reference measure. Given a confining potential $V : \mathbb{R}^n \rightarrow \mathbb{R}$ and an inverse temperature parameter $\beta > 0$, the classical Helmholtz free energy of a probability density ρ takes the form

$$\mathcal{F}[\rho] = \int_{\mathbb{R}^n} V(x) \rho(x) dx + \frac{1}{\beta} \int_{\mathbb{R}^n} \rho(x) \ln \rho(x) dx.$$

Under suitable conditions, minimisers of $\mathcal{F}$ are Gibbs measures, and the time evolution of ρ_t under certain stochastic dynamics can be interpreted as a gradient flow of $\mathcal{F}$ in an appropriate metric structure on the space of probability measures.

A particularly fruitful viewpoint arises when one endows the space $\mathcal{P}(\mathbb{R}^n)$ of Borel probability measures with finite second moment with the 2-Wasserstein distance. In this setting, it is known that many diffusion-type partial differential equations, such as Fokker–Planck equations associated with Langevin dynamics, can be understood as gradient flows of free energy functionals. This perspective connects variational principles, optimal transport, and dissipative evolution equations, and provides a geometric framework for studying long-time behaviour, convergence to equilibrium, and stability.

In contrast to classical treatments that are typically formulated in finite dimensions, we are interested in a framework where the state space is a separable Hilbert space S . Such spaces naturally accommodate continuous degrees of freedom and can also serve as infinite-dimensional state spaces for dynamical systems and fields. Working in this setting raises additional analytical challenges, especially for the geometry of the space of probability measures and for the rigorous definition of gradient flows. The guiding idea of

this paper is to formulate the free energy functional and its basic properties at this level of generality, while using finite-dimensional approximations and known results whenever full infinite-dimensional rigor is not available.

1.2 Aim of the paper

The primary goal of this paper is to define and analyse a class of free energy functionals

$$F[\rho] = \int_S W(s) \rho(s) d\mu(s) + D \int_S \rho(s) \ln \rho(s) d\mu(s)$$

on spaces of probability measures over a separable Hilbert space S , and to characterise the associated gradient-flow dynamics.

More concretely, we aim to:

- Fix a state space S and a reference measure μ on its Borel σ -algebra, and introduce suitable subclasses $\mathcal{P}(S)$ or $\mathcal{P}_2(S)$ of probability measures on S that are natural domains for F .
- Specify integrability, regularity, and growth conditions on the potential W and the densities ρ that ensure that the energy term $\int W \rho d\mu$ and the entropy term $\int \rho \ln \rho d\mu$ are well-defined (possibly taking values in $(-\infty, +\infty]$).
- Study basic properties of the functional F , such as convexity along suitable curves in the space of probability measures, lower semicontinuity with respect to natural topologies, and coercivity conditions that guarantee the existence of minimisers.
- Introduce metric structures on spaces of probability measures, with a focus on the 2-Wasserstein distance in finite dimensions, and formulate the notion of gradient flow of F in this setting.
- Demonstrate, in finite dimensions, how Fokker–Planck equations arise as gradient flows of F , and show that $F[\rho_t]$ is non-increasing along the flow, with stationary measures characterised as minimisers of F .
- Connect F to classical free energy and variational principles, and illustrate the formalism by means of representative examples, such as quadratic and double-well potentials.

Throughout, the focus remains on the mathematical structure of the free energy functional and its associated dynamics. No claim is made that the resulting framework exhausts any physical or interpretive content; rather, it is intended as a clean, abstract setting in which such questions *could* be posed in future work.

1.3 Structure of the paper

The remainder of the paper is organised as follows.

In Section 2 we specify the state space S as a separable Hilbert space, introduce its Borel σ -algebra, and define the relevant classes of probability measures on S . We discuss densities with respect to a reference measure μ and record the integrability conditions needed for the energy and entropy terms.

Section 3 introduces the free energy functional F in detail. We separate the energy and entropy contributions, define their domains, and study basic properties of F , including lower

semicontinuity, convexity under appropriate conditions, and coercivity criteria ensuring the existence of minimisers.

Section 4 is devoted to gradient-flow structures on spaces of probability measures. We recall the 2-Wasserstein distance in finite-dimensional settings, outline the formal Riemannian structure of Wasserstein space, and compute the first variation $\delta F/\delta\rho$. This leads to continuity-type equations that describe the steepest descent of F with respect to the chosen metric.

In Section 5 we relate these gradient-flow structures to Fokker–Planck equations. Starting from a Langevin-type stochastic differential equation in finite dimensions, we recall the associated Fokker–Planck equation and show how it can be written in terms of the gradient of $\delta F/\delta\rho$. We identify the Fokker–Planck equation as a gradient flow of F and show that $F[\rho_t]$ is non-increasing along solutions, with stationary measures corresponding to critical points and, under suitable assumptions, minimisers of F .

Section 6 places the functional F in the context of classical free energy and variational principles. We relate F to the Helmholtz free energy and Gibbs measures, mention connections to large deviation principles and variational characterisations of equilibrium states, and briefly indicate structural links to information-theoretic free energy functionals.

In Section 7 we present examples and special cases. We analyse simple one-dimensional systems, including a double-well potential, as well as quadratic potentials leading to Gaussian stationary measures. We illustrate how the diffusion parameter D (interpretable as a temperature-like parameter) influences the shape of F , the structure of minimisers, and the qualitative behaviour of the associated dynamics.

Section 8 discusses extensions and generalisations. We outline how the framework can be adapted to time-dependent potentials $W(s, t)$, to multi-component systems and joint distributions, and to constrained flows obtained by projecting the gradient flow onto suitable submanifolds or constraint sets.

In Section 9 we briefly indicate how the present work fits into a broader programme that studies dynamics not only at the level of individual states but also at the level of distributions. The role of F as a free energy functional on probability measures is highlighted in this context, but the discussion remains descriptive and modest.

Finally, Section 10 summarises the main mathematical results, emphasises limitations (in particular regarding infinite-dimensional rigour and the absence of empirical applications in this paper), and suggests directions for future work, including more refined infinite-dimensional analyses and potential applications in statistical physics and related fields.

2 Probability measures on the state space S

In this section we introduce the measure-theoretic framework underlying the free energy functional studied later. We fix a state space S endowed with a Hilbert space structure, specify the associated Borel σ -algebra, and describe the spaces of probability measures that will serve as the domain of the free energy. We then introduce a reference measure μ and discuss densities and integrability conditions needed for the energy and entropy terms.

2.1 State space and σ -algebra

Throughout the paper, the state space S is assumed to be a separable Hilbert space over $\mathbb{R}$. We denote its inner product by $\langle \cdot, \cdot \rangle$ and the induced norm by $\|\cdot\|$. Thus S is a complete metric space with respect to the metric

$$d(s_1, s_2) := \|s_1 - s_2\|, \quad s_1, s_2 \in S.$$

2.1.1 Topological and measurable structure

Let $\mathcal{B}(S)$ denote the Borel σ -algebra on S , i.e. the smallest σ -algebra containing all open subsets of S . Since S is separable and metric, $\mathcal{B}(S)$ is countably generated, which simplifies several measure-theoretic considerations (e.g. regularity and disintegration properties).

Definition 2.1 (Borel probability measures). We denote by $\mathcal{P}(S)$ the set of all Borel probability measures on S , i.e.

$$\mathcal{P}(S) := \{\rho : \mathcal{B}(S) \rightarrow [0, 1] \mid \rho \text{ is a probability measure}\}.$$

In many places we will work with probability measures that possess finite moments of a certain order. The second moment plays a distinguished role, in particular for Wasserstein geometry.

Definition 2.2 (Probability measures with finite second moment). We define

$$\mathcal{P}_2(S) := \left\{ \rho \in \mathcal{P}(S) \mid \int_S \|s\|^2 d\rho(s) < \infty \right\}.$$

In finite-dimensional settings, such as $S = \mathbb{R}^n$, $\mathcal{P}_2(S)$ is the natural domain of the 2-Wasserstein distance. In the present work, $\mathcal{P}_2(S)$ will be the default space of measures whenever we need second-moment bounds, for example to ensure finiteness of the energy term associated with coercive potentials.

2.2 Probability measures and reference measure

To formulate the energy and entropy contributions to the free energy, it is convenient to work with probability measures that are absolutely continuous with respect to a fixed reference measure. This leads to densities of the form $\rho(s)$ and allows us to express energy and entropy functionals as integrals over S .

2.2.1 Reference measure

We fix once and for all a reference measure μ on $(S, \mathcal{B}(S))$ satisfying the following assumptions:

1. μ is a σ -finite Borel measure on $(S, \mathcal{B}(S))$.
2. μ has full support, i.e. $\mu(U) > 0$ for every non-empty open set $U \subset S$.

The concrete choice of μ depends on the context:

- In finite dimensions, a typical choice is Lebesgue measure on $\mathbb{R}^n$, or a Gaussian measure if one wishes to incorporate a quadratic reference structure.
- In infinite-dimensional settings, μ may be a Gaussian measure or another reference measure adapted to the underlying problem.

Definition 2.3 (Densities). A probability measure $\rho \in \mathcal{P}(S)$ is said to admit a density with respect to μ if it is absolutely continuous with respect to μ , written $\rho \ll \mu$. In this case there exists a non-negative measurable function (density) $\rho(\cdot)$ such that

$$d\rho(s) = \rho(s) d\mu(s),$$

and

$$\int_S \rho(s) d\mu(s) = 1.$$

We will often write ρ for both the measure and its density when no confusion can arise.

The class of measures absolutely continuous with respect to μ will serve as the effective domain of the entropy functional, and, under suitable conditions on the potential, of the energy functional.

2.3 Integrability conditions for energy and entropy

The free energy functional to be introduced later has two main contributions: an energy term involving a potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$, and an entropy term involving $\rho \ln \rho$. Both expressions require suitable integrability assumptions.

2.3.1 Potential and energy term

Let $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ be a Borel measurable function, which we interpret as a (possibly extended-valued) potential. For a probability measure ρ admitting a density with respect to μ , we formally define the energy functional by

$$\mathcal{E}[\rho] := \int_S W(s) \rho(s) d\mu(s),$$

with the convention that the integral takes values in $(-\infty, +\infty]$.

To ensure that $\mathcal{E}[\rho]$ is well-defined (and, in particular, does not involve ambiguous expressions such as $\infty - \infty$), we impose conditions on W and ρ that guarantee sufficient integrability.

Assumption 2.4 (Integrability for the energy term). We assume that:

1. W is bounded from below in the sense that there exists $C_1 \in \mathbb{R}$ such that

$$W(s) \geq C_1 \quad \text{for } \mu\text{-almost all } s \in S.$$

2. For the measures ρ of interest, the positive part $W^+(s) := \max\{W(s), 0\}$ is integrable in the sense that

$$\int_S W^+(s) \rho(s) d\mu(s) < \infty.$$

Under Assumption 2.4, the integral defining $\mathcal{E}[\rho]$ is well-defined in $(-\infty, \infty]$. Many important cases are covered by potentials with at most polynomial growth and probability measures in $\mathcal{P}_2(S)$ or higher moment spaces.

2.3.2 Entropy term

The entropy functional will be defined in terms of the density ρ with respect to μ . We adopt the standard convention for the product $\rho \ln \rho$.

Definition 2.5 (Entropy density). For $r \geq 0$ we define

$$\eta(r) := \begin{cases} r \ln r, & r > 0, \\ 0, & r = 0. \end{cases}$$

The entropy functional is then given by

$$\mathcal{H}[\rho] := \int_S \eta(\rho(s)) d\mu(s) = \int_S \rho(s) \ln \rho(s) d\mu(s),$$

with the above convention at points where $\rho(s) = 0$. The integral is understood in the extended sense, taking values in $[-\infty, +\infty]$.

To ensure that $\mathcal{H}[\rho]$ is finite, we require the following.

Assumption 2.6 (Integrability for the entropy term). For the measures ρ under consideration, we assume that

$$\int_S |\eta(\rho(s))| d\mu(s) < \infty,$$

that is,

$$\int_S |\rho(s) \ln \rho(s)| d\mu(s) < \infty.$$

Assumption 2.6 is typically satisfied if ρ has suitable decay properties and is bounded away from zero and infinity in an averaged sense. In many applications, entropy finiteness is ensured by imposing bounds on moments of ρ together with growth conditions on the potential.

2.3.3 Function spaces and domains

It is convenient to collect these integrability conditions into function-space descriptions of the domain of the free energy functional. Let $L^1(\mu)$ denote the space of μ -integrable functions on S and, more generally, let $L^p(\mu)$ for $1 \leq p \leq \infty$ denote the usual Lebesgue spaces.

Definition 2.7 (Admissible densities). We define the set of admissible densities

$$\mathcal{D} := \left\{ \rho \in L^1(\mu) \mid \rho \geq 0, \int_S \rho d\mu = 1, \mathcal{E}[\rho] > -\infty, \mathcal{H}[\rho] < +\infty \right\}.$$

Equivalently, $\mathcal{D}$ consists of densities corresponding to probability measures $\rho \ll \mu$ such that the energy and entropy integrals are well-defined and finite in the sense specified by Assumptions 2.4 and 2.6.

Remark 2.8. The precise choice of the domain $\mathcal{D}$ can be adapted to the specific assumptions on W and the reference measure μ . In later sections, when discussing gradient flows and Fokker–Planck equations, we will often restrict attention to densities in $\mathcal{D} \cap \mathcal{P}_2(S)$ to ensure both integrability of the free energy and finiteness of second moments.

The framework developed here provides the measure-theoretic and integrability foundation for the free energy functional. In the next section we will define the energy and entropy contributions rigorously, study basic properties such as lower semicontinuity and convexity, and combine them into the free energy functional F .

3 Definition of the free energy functional

In this section we combine the energy and entropy contributions introduced in the previous section into a free energy functional on a suitable class of probability measures on S . We first define the energy and entropy functionals separately, then introduce the free energy and discuss basic properties such as convexity, lower semicontinuity, and coercivity. Finally, we compute the first variation of the free energy and derive Euler–Lagrange equations for its minimisers, which leads to Gibbs-type stationary densities.

3.1 Energy and entropy functionals

We recall the setting of Section 2. The state space S is a separable Hilbert space equipped with its Borel σ -algebra $\mathcal{B}(S)$ and a fixed σ -finite reference measure μ with full support. A

probability measure ρ on S is assumed to be absolutely continuous with respect to μ , with density (again denoted by ρ) satisfying

$$\rho \geq 0, \quad \int_S \rho(s) d\mu(s) = 1.$$

3.1.1 Energy functional

Let $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ be a Borel measurable potential. We define the energy functional $\mathcal{E}$ on probability measures admitting densities with respect to μ by

$$\mathcal{E}[\rho] := \int_S W(s) \rho(s) d\mu(s), \quad (3.1)$$

with the convention that the integral takes values in $(-\infty, +\infty]$.

To avoid ambiguities of the form $\infty - \infty$, we work under Assumption 2.4, which in particular guarantees that W is bounded from below (up to a μ -null set) and that the positive part $W^+ := \max\{W, 0\}$ is integrable with respect to $\rho\mu$. Under these conditions, $\mathcal{E}[\rho]$ is well-defined in $(-\infty, +\infty]$ for all densities ρ in the admissible class $\mathcal{D}$ introduced in Section 2.

Remark 3.1. If W is finite-valued and bounded from below, then $\mathcal{E}[\rho]$ is finite for all $\rho \in \mathcal{P}(S)$. In many applications, however, W is allowed to grow superlinearly (for instance quadratically) at infinity, so that finiteness of $\mathcal{E}[\rho]$ selects measures with suitable moment bounds (e.g. $\rho \in \mathcal{P}_2(S)$).

3.1.2 Entropy functional

We next define the entropy functional associated with the reference measure μ . Recall the entropy density

$$\eta(r) := \begin{cases} r \ln r, & r > 0, \\ 0, & r = 0, \end{cases}$$

so that $\eta(r)$ is continuous on $[0, \infty)$ and convex.

Definition 3.2 (Entropy functional). For a probability density ρ with respect to μ , we define the entropy

$$\mathcal{H}[\rho] := \int_S \eta(\rho(s)) d\mu(s) = \int_S \rho(s) \ln \rho(s) d\mu(s), \quad (3.2)$$

with the convention $\rho(s) \ln \rho(s) = 0$ whenever $\rho(s) = 0$. The integral is understood in the extended sense, taking values in $[-\infty, +\infty]$.

Assumption 2.6 ensures that $\mathcal{H}[\rho]$ is finite for all ρ in the admissible class $\mathcal{D}$. We recall that η is strictly convex on $(0, \infty)$, which implies convexity of $\mathcal{H}$ on the set of densities, and that $\mathcal{H}$ is lower semicontinuous with respect to weak convergence under suitable tightness and moment bounds.

Remark 3.3. The entropy $\mathcal{H}$ defined in (3.2) is sometimes referred to as the *relative entropy* of ρ with respect to the reference measure μ , up to an additive constant depending on the normalisation of μ . In particular, if μ is a probability measure, then $\mathcal{H}[\rho]$ coincides with the Kullback–Leibler divergence $D_{\text{KL}}(\rho\|\mu)$ up to the replacement of the logarithm of the Radon–Nikodým derivative by the logarithm of the density itself.

3.2 The free energy functional

We now combine the energy and entropy functionals into a free energy functional parametrised by a positive constant $D > 0$, which will later play the role of a diffusion or temperature-like parameter.

3.2.1 Definition and domain

Definition 3.4 (Free energy functional). Let $D > 0$ be fixed. The free energy functional F is defined on densities ρ with respect to μ by

$$F[\rho] := \mathcal{E}[\rho] + D \mathcal{H}[\rho] = \int_S W(s) \rho(s) d\mu(s) + D \int_S \rho(s) \ln \rho(s) d\mu(s), \quad (3.3)$$

whenever both integrals are well-defined in the sense of Section 2. The natural domain of F is the set

$$\mathcal{D}_F := \{\rho \in \mathcal{D} \mid \mathcal{E}[\rho] < +\infty\},$$

viewed as a subset of $L^1(\mu)$ or, equivalently, of $\mathcal{P}(S)$ via $\rho \mapsto \rho \mu$. On $\mathcal{D}_F$ we interpret $F[\rho]$ as taking values in $(-\infty, +\infty]$.

When convenient, we will refer to F as a *semantic free energy*, although in the present paper it is treated purely as a mathematical object, without additional interpretive structure.

3.2.2 Convexity and lower semicontinuity

The free energy F inherits convexity and lower semicontinuity properties from its two components.

Proposition 3.5 (Convexity). *Suppose that W is μ -almost everywhere finite and convex as a function on S . Then:*

1. *The energy functional $\mathcal{E}$ is convex on the set of densities ρ .*
2. *The entropy functional $\mathcal{H}$ is convex on the set of densities ρ .*
3. *Consequently, the free energy functional F is convex on its domain $\mathcal{D}_F$.*

Proof. For (1), let ρ_0 and ρ_1 be densities and let $\theta \in [0, 1]$. Define $\rho_\theta := (1 - \theta)\rho_0 + \theta\rho_1$. By convexity of W and linearity of the integral, we have

$$\mathcal{E}[\rho_\theta] = \int_S W(s) \rho_\theta(s) d\mu(s) = (1 - \theta) \int_S W(s) \rho_0(s) d\mu(s) + \theta \int_S W(s) \rho_1(s) d\mu(s),$$

which shows that $\mathcal{E}$ is actually affine (and hence convex) in ρ .

For (2), recall that the map $r \mapsto \eta(r) = r \ln r$ is convex on $[0, \infty)$. Therefore,

$$\eta(\rho_\theta(s)) \leq (1 - \theta) \eta(\rho_0(s)) + \theta \eta(\rho_1(s)) \quad \text{for all } s \in S,$$

and integrating with respect to μ yields

$$\mathcal{H}[\rho_\theta] \leq (1 - \theta) \mathcal{H}[\rho_0] + \theta \mathcal{H}[\rho_1],$$

so $\mathcal{H}$ is convex.

Item (3) follows because $F[\rho] = \mathcal{E}[\rho] + D \mathcal{H}[\rho]$ is a positive linear combination of convex functionals. $\square$

Proposition 3.6 (Lower semicontinuity). *Assume that:*

1. W is lower semicontinuous and bounded from below;
2. $\{\rho_n\}_{n \in \mathbb{N}}$ is a sequence in $\mathcal{D}_F$ converging to ρ in $L^1(\mu)$, with a uniform bound on some moment of order $p \geq 1$ (e.g. a uniform second-moment bound);

then

$$F[\rho] \leq \liminf_{n \rightarrow \infty} F[\rho_n].$$

In particular, F is lower semicontinuous on $\mathcal{D}_F$ with respect to such convergence.

Sketch of proof. Under the stated assumptions, standard results in the calculus of variations and measure theory imply that the integral functional $\rho \mapsto \int_S W(s) \rho(s) d\mu(s)$ is lower semicontinuous with respect to L^1 convergence and suitable tightness or moment conditions. Similarly, the entropy functional $\mathcal{H}$ is known to be lower semicontinuous with respect to weak convergence of probability measures, and hence with respect to L^1 convergence under uniform integrability assumptions. Combining the two contributions yields the desired inequality for F . $\square$

3.2.3 Coercivity and existence of minimisers

Coercivity of F plays a key role in ensuring the existence of minimisers, which will later be identified with stationary solutions of the gradient-flow dynamics.

Assumption 3.7 (Coercive potential). We assume that W is bounded from below and coercive in the sense that

$$\lim_{\|s\| \rightarrow \infty} \frac{W(s)}{\|s\|^2} = +\infty,$$

or, more generally, that W grows sufficiently fast at infinity to control the second moment of ρ via $\mathcal{E}[\rho]$.

Proposition 3.8 (Existence of minimisers). *Suppose that W satisfies Assumptions 2.4 and 3.7, and that F is bounded from below on $\mathcal{D}_F$. Then there exists at least one minimiser $\rho_* \in \mathcal{D}_F$ such that*

$$F[\rho_*] = \inf_{\rho \in \mathcal{D}_F} F[\rho].$$

Sketch of proof. Let $\{\rho_n\}_{n \in \mathbb{N}} \subset \mathcal{D}_F$ be a minimising sequence, i.e.

$$\lim_{n \rightarrow \infty} F[\rho_n] = \inf_{\rho \in \mathcal{D}_F} F[\rho].$$

Assumption 3.7 implies uniform control of moments of ρ_n , typically of second order, via the bound on $\mathcal{E}[\rho_n]$. Together with the fact that each ρ_n is a probability density, this yields tightness of the corresponding measures. By Prokhorov's theorem and standard compactness arguments in $L^1(\mu)$, we can extract a subsequence (not relabelled) converging to some $\rho_* \in \mathcal{D}_F$. Lower semicontinuity of F from Proposition 3.6 then gives

$$F[\rho_*] \leq \liminf_{n \rightarrow \infty} F[\rho_n] = \inf_{\rho \in \mathcal{D}_F} F[\rho],$$

so ρ_* is a minimiser. $\square$

3.3 First variation and Euler–Lagrange equations

We now compute the first variation of the free energy functional F under smooth perturbations of the density, subject to a mass constraint. This leads to an explicit formula for the functional derivative $\delta F/\delta \rho$ and characterisation of minimisers as Gibbs-type densities.

3.3.1 Formal first variation

We perform the computation in a formal setting, assuming that ρ and W are sufficiently regular and that $\rho > 0$ on the relevant region of S . Let ρ be a strictly positive density in $\mathcal{D}_F$, and let $\phi : S \rightarrow \mathbb{R}$ be a bounded measurable function with

$$\int_S \phi(s) d\mu(s) = 0,$$

so that perturbations of the form $\rho_\varepsilon := \rho + \varepsilon \phi$ preserve the total mass to first order in ε . We consider perturbations small enough that ρ_ε remains non-negative.

The first variation of the energy term is

$$\begin{aligned} \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \mathcal{E}[\rho_\varepsilon] &= \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_S W(s) \rho_\varepsilon(s) d\mu(s) \\ &= \int_S W(s) \phi(s) d\mu(s). \end{aligned}$$

For the entropy term, using $\rho > 0$ and differentiating under the integral sign, we obtain

$$\begin{aligned} \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \mathcal{H}[\rho_\varepsilon] &= \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_S \rho_\varepsilon(s) \ln \rho_\varepsilon(s) d\mu(s) \\ &= \int_S (1 + \ln \rho(s)) \phi(s) d\mu(s). \end{aligned}$$

Combining the two contributions, the first variation of F is

$$\begin{aligned} \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} F[\rho_\varepsilon] &= \int_S W(s) \phi(s) d\mu(s) + D \int_S (1 + \ln \rho(s)) \phi(s) d\mu(s) \\ &= \int_S (W(s) + D(1 + \ln \rho(s))) \phi(s) d\mu(s). \end{aligned}$$

This computation motivates the following formal definition.

Definition 3.9 (Formal functional derivative). For sufficiently regular strictly positive densities ρ in the domain of F , we define the formal functional derivative of F with respect to ρ by

$$\frac{\delta F}{\delta \rho}(s) := W(s) + D(1 + \ln \rho(s)), \quad s \in S. \quad (3.4)$$

The expression (3.4) will play a central role in the gradient-flow formulation of the dynamics in later sections.

3.3.2 Euler–Lagrange equations and Gibbs form of minimisers

We now derive the Euler–Lagrange equation for minimisers of F under the constraint of fixed total mass. Consider the constrained minimisation problem

$$\inf \left\{ F[\rho] \mid \rho \in \mathcal{D}_F, \rho \geq 0, \int_S \rho d\mu = 1 \right\}.$$

Formally, introducing a Lagrange multiplier $\lambda \in \mathbb{R}$ for the mass constraint, we consider critical points of the Lagrangian

$$\mathcal{L}[\rho] := F[\rho] - \lambda \left(\int_S \rho(s) d\mu(s) - 1 \right).$$

Using (3.4), the first variation of $\mathcal{L}$ at a strictly positive density ρ in the direction ϕ with $\int_S \phi d\mu = 0$ is

$$\delta \mathcal{L}[\rho; \phi] = \int_S (W(s) + D(1 + \ln \rho(s)) - \lambda) \phi(s) d\mu(s).$$

A necessary condition for ρ to be a critical point is that $\delta \mathcal{L}[\rho; \phi] = 0$ for all such ϕ , which implies

$$W(s) + D(1 + \ln \rho(s)) = \lambda \quad \text{for } \mu\text{-almost all } s \text{ with } \rho(s) > 0. \quad (3.5)$$

Solving (3.5) for ρ yields the familiar Gibbs-type form.

Proposition 3.10 (Gibbs form of stationary densities). *Let $D > 0$ and suppose that W is such that the partition function*

$$Z := \int_S \exp\left(-\frac{W(s)}{D}\right) d\mu(s) \quad (3.6)$$

is finite. Then any strictly positive minimiser ρ_ of F under the mass constraint has the form*

$$\rho_*(s) = \frac{1}{Z} \exp\left(-\frac{W(s)}{D}\right), \quad \mu\text{-almost all } s \in S. \quad (3.7)$$

Conversely, any density of the form (3.7) belongs to $\mathcal{D}_F$ and is a critical point of F under the mass constraint.

Proof. If ρ_* is a strictly positive minimiser, then it satisfies the Euler–Lagrange equation (3.5) for some $\lambda \in \mathbb{R}$. Rewriting (3.5) gives

$$\ln \rho_*(s) = -1 - \frac{1}{D} W(s) + \frac{\lambda}{D},$$

and thus

$$\rho_*(s) = C \exp\left(-\frac{W(s)}{D}\right), \quad C := \exp\left(\frac{\lambda}{D} - 1\right).$$

The constant C is fixed by the normalisation condition $\int_S \rho_*(s) d\mu(s) = 1$, which yields

$$1 = C \int_S \exp\left(-\frac{W(s)}{D}\right) d\mu(s) = CZ,$$

so $C = 1/Z$ and (3.7) follows.

Conversely, if $Z < \infty$ and ρ_* is given by (3.7), then ρ_* is a well-defined probability density. Assuming suitable integrability of W , one checks that $\rho_* \in \mathcal{D}_F$ and that (3.5) holds with λ chosen to satisfy the normalisation condition. Thus ρ_* is a critical point of F under the mass constraint. Under additional convexity assumptions on W , such as λ -convexity, one can show that ρ_* is in fact the unique minimiser of F . $\square$

Remark 3.11. The expression (3.7) coincides with the classical Gibbs measure associated with the potential W at “temperature” proportional to D . In later sections we will see that the gradient-flow dynamics generated by F drives densities towards such Gibbs states, and that the free energy $F[\rho_t]$ decreases along the associated Fokker–Planck evolution.

4 Gradient-flow structures on spaces of probability measures

In this section we describe how the free energy functional F defined in Section 3 generates gradient-flow dynamics on spaces of probability measures. The main geometric structure is the 2-Wasserstein metric on $\mathcal{P}_2(S)$, which provides a formal Riemannian structure and allows one to interpret certain diffusion equations as steepest-descent flows of F . For clarity, we first recall the 2-Wasserstein metric in finite dimensions and then explain the notion of gradient flow in this setting. Finally, we combine this with the functional derivative of F to obtain a formal continuity equation driven by the “force” $-\nabla(\delta F/\delta \rho)$.

4.1 Metric structures on spaces of probability measures

Let S be a separable Hilbert space. To keep the geometric exposition concrete, we will mainly work in the finite-dimensional case $S = \mathbb{R}^n$ when discussing Wasserstein geometry; the extension to general Hilbert spaces is subtle and will be treated at a more formal level.

4.1.1 The 2-Wasserstein distance

We start with the usual optimal-transport definition of the 2-Wasserstein distance on $\mathcal{P}_2(\mathbb{R}^n)$.

Definition 4.1 (2-Wasserstein distance). Let $\rho_0, \rho_1 \in \mathcal{P}_2(\mathbb{R}^n)$. The 2-Wasserstein distance $W_2(\rho_0, \rho_1)$ is defined by

$$W_2^2(\rho_0, \rho_1) := \inf_{\gamma \in \Gamma(\rho_0, \rho_1)} \int_{\mathbb{R}^n \times \mathbb{R}^n} \|x - y\|^2 d\gamma(x, y), \quad (4.1)$$

where $\Gamma(\rho_0, \rho_1)$ is the set of all couplings of ρ_0 and ρ_1 , i.e. all Borel probability measures γ on $\mathbb{R}^n \times \mathbb{R}^n$ with marginals ρ_0 and ρ_1 :

$$(\pi_1)_\# \gamma = \rho_0, \quad (\pi_2)_\# \gamma = \rho_1,$$

where π_1, π_2 are the canonical projections.

It is well known that W_2 defines a metric on $\mathcal{P}_2(\mathbb{R}^n)$ and that this metric space is complete and separable. Moreover, there is a dynamic (Benamou–Brenier) formulation which makes the connection to continuity equations explicit.

Proposition 4.2 (Benamou–Brenier formula). *Let $\rho_0, \rho_1 \in \mathcal{P}_2(\mathbb{R}^n)$. Then*

$$W_2^2(\rho_0, \rho_1) = \inf_{\substack{(\rho_t, v_t)_{t \in [0,1]} \\ \partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0}} \int_0^1 \int_{\mathbb{R}^n} \|v_t(x)\|^2 \rho_t(x) dx dt, \quad (4.2)$$

where the infimum is taken over all narrowly continuous curves $t \mapsto \rho_t \in \mathcal{P}_2(\mathbb{R}^n)$ and Borel vector fields $v_t : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that:

- $\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$ holds in the sense of distributions on $(0, 1) \times \mathbb{R}^n$;
- $\rho_{t=0} = \rho_0$ and $\rho_{t=1} = \rho_1$.

The quantity

$$\int_0^1 \int_{\mathbb{R}^n} \|v_t(x)\|^2 \rho_t(x) dx dt$$

plays the role of a kinetic action or squared length of the curve $(\rho_t)_{t \in [0,1]}$, and the Benamou–Brenier formula suggests an interpretation of $\mathcal{P}_2(\mathbb{R}^n)$ as a Riemannian manifold whose tangent vectors at ρ_t are represented by velocity fields v_t .

4.1.2 Tangent space and velocity fields

Motivated by (4.2), one can describe the formal tangent space $T_\rho \mathcal{P}_2(\mathbb{R}^n)$ at a sufficiently regular density ρ in terms of velocity fields.

Definition 4.3 (Formal tangent space). Let ρ be a smooth, strictly positive probability density on $\mathbb{R}^n$. Consider smooth vector fields $v : \mathbb{R}^n \rightarrow \mathbb{R}^n$ with sufficient decay at infinity. We define:

- A tangent vector at ρ is represented by a density variation σ satisfying

$$\sigma = -\nabla \cdot (\rho v),$$

for some such vector field v .

- The squared norm of the tangent vector represented by v is given by

$$\|v\|_{T_\rho}^2 := \int_{\mathbb{R}^n} \|v(x)\|^2 \rho(x) dx.$$

The continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$$

then expresses that the time derivative $\partial_t \rho_t$ is a tangent vector at ρ_t , represented by the velocity field v_t . The “Riemannian metric” at ρ_t is formally given by

$$\langle v_t, w_t \rangle_{T_{\rho_t}} := \int_{\mathbb{R}^n} v_t(x) \cdot w_t(x) \rho_t(x) dx,$$

for velocity fields v_t, w_t representing tangent vectors at ρ_t .

Remark 4.4. In this formalism, smooth curves $(\rho_t)_{t \in [0,1]}$ in $\mathcal{P}_2(\mathbb{R}^n)$ can be described via velocity fields v_t solving the continuity equation, and the 2-Wasserstein distance corresponds to the minimal kinetic action along such curves. This viewpoint underlies the interpretation of many diffusion-type PDEs as gradient flows in Wasserstein space.

4.2 Gradient flows in Wasserstein space

We now recall the abstract notion of a gradient flow for a functional F defined on a metric space, and then specialise to the Wasserstein metric space $(\mathcal{P}_2(\mathbb{R}^n), W_2)$.

4.2.1 Gradient flows in abstract metric spaces

Let (X, d) be a metric space and $F : X \rightarrow (-\infty, +\infty]$ a lower semicontinuous functional. A curve $u : [0, \infty) \rightarrow X$ is called a *gradient flow* of F if it satisfies, in a suitable sense, a steepest-descent condition for F with respect to the metric d . There are several equivalent formulations, such as evolution variational inequalities or energy dissipation identities. For our purposes, it is most convenient to think in terms of the energy dissipation balance.

Formally, a curve u_t is a gradient flow of F if, for almost all $t > 0$,

$$\frac{d}{dt} F(u_t) = -\|\dot{u}_t\|^2,$$

where $\dot{u}_t$ is the metric derivative of u_t and $\|\dot{u}_t\|$ is its metric speed. In the Wasserstein setting, this speed is related to the kinetic energy of the velocity field v_t appearing in the continuity equation.

4.2.2 Gradient flows on $\mathcal{P}_2(\mathbb{R}^n)$

Let F be a functional on $\mathcal{P}_2(\mathbb{R}^n)$. A curve $t \mapsto \rho_t$ in $\mathcal{P}_2(\mathbb{R}^n)$ is interpreted as a gradient flow of F in the Wasserstein metric if it satisfies a continuity equation with velocity field chosen as the steepest-descent direction of F .

Let F be formally differentiable with respect to ρ in the sense that there exists a functional derivative $\delta F/\delta \rho$ such that, for smooth perturbations,

$$\delta F[\rho; \phi] = \int_{\mathbb{R}^n} \frac{\delta F}{\delta \rho}(x) \phi(x) dx.$$

We recall from Section 3 that in our case

$$\frac{\delta F}{\delta \rho}(x) = W(x) + D(1 + \ln \rho(x)).$$

The tangent space at ρ is represented by vector fields v via $\sigma = -\nabla \cdot (\rho v)$, and the inner product at ρ between two tangent vectors represented by v, w is

$$\langle v, w \rangle_{T_\rho} = \int_{\mathbb{R}^n} v(x) \cdot w(x) \rho(x) dx.$$

The gradient of F at ρ , denoted $\text{grad } F(\rho)$, is the unique tangent vector (if it exists) such that

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} F[\rho_\varepsilon] = \langle \text{grad } F(\rho), \dot{\rho}_0 \rangle_{T_\rho},$$

for all smooth curves ρ_ε with $\rho_0 = \rho$, where $\dot{\rho}_0$ is the corresponding tangent vector. A classical computation shows that the velocity field representing $\text{grad } F(\rho)$ is $\nabla(\delta F/\delta \rho)$.

More precisely, consider a curve $(\rho_t)_{t \geq 0}$ solving the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0.$$

Then, formally,

$$\begin{aligned} \frac{d}{dt} F[\rho_t] &= \int_{\mathbb{R}^n} \frac{\delta F}{\delta \rho}(x) \partial_t \rho_t(x) dx \\ &= - \int_{\mathbb{R}^n} \frac{\delta F}{\delta \rho}(x) \nabla \cdot (\rho_t(x) v_t(x)) dx \\ &= \int_{\mathbb{R}^n} \nabla \left(\frac{\delta F}{\delta \rho}(x) \right) \cdot v_t(x) \rho_t(x) dx, \end{aligned}$$

where we have integrated by parts and neglected boundary terms at infinity under suitable decay assumptions. Comparing this with the Riemannian inner product

$$\langle \text{grad } F(\rho_t), v_t \rangle_{T_{\rho_t}} = \int_{\mathbb{R}^n} \nabla \left(\frac{\delta F}{\delta \rho}(x) \right) \cdot v_t(x) \rho_t(x) dx,$$

we see that the gradient of F at ρ_t is represented by the velocity field $v_t = \nabla(\delta F/\delta \rho)(\cdot)$.

A gradient flow of F is then obtained by choosing the velocity field along the flow as the *negative* gradient direction.

Definition 4.5 (Formal Wasserstein gradient flow of F). A sufficiently regular curve $(\rho_t)_{t \geq 0} \subset \mathcal{P}_2(\mathbb{R}^n)$ is called a formal Wasserstein gradient flow of F if it satisfies

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad v_t(x) = -\nabla \left(\frac{\delta F}{\delta \rho}(x) \right), \quad (4.3)$$

for all $t > 0$ and $x \in \mathbb{R}^n$, with $\delta F/\delta \rho$ given by (3.4).

Inserting the expression (3.4) for $\delta F/\delta\rho$ into (4.3), we obtain the explicit velocity field

$$v_t(x) = -\nabla W(x) - D \nabla \ln \rho_t(x),$$

and the continuity equation becomes

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla W + D \rho_t \nabla \ln \rho_t). \quad (4.4)$$

Using $\rho_t \nabla \ln \rho_t = \nabla \rho_t$, we can rewrite (4.4) as

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla W) + D \Delta \rho_t, \quad (4.5)$$

which is a Fokker–Planck-type equation.

Remark 4.6. Equation (4.5) will be the central object of Section 5, where we interpret it as the Fokker–Planck equation associated with a Langevin-type stochastic differential equation and show that it indeed defines a (generalised) gradient flow of F in the Wasserstein metric. Here we have derived (4.5) as the formal steepest-descent evolution of the free energy functional F .

4.3 Energy dissipation along the flow

The gradient-flow structure leads to a natural energy dissipation identity, expressing the monotonicity of $F[\rho_t]$ along the evolution (4.5). At a formal level, using the computation above with the choice $v_t = -\nabla(\delta F/\delta\rho)$, we obtain

$$\begin{aligned} \frac{d}{dt} F[\rho_t] &= \int_{\mathbb{R}^n} \nabla \left(\frac{\delta F}{\delta \rho}(x) \right) \cdot v_t(x) \rho_t(x) dx \\ &= - \int_{\mathbb{R}^n} \left\| \nabla \left(\frac{\delta F}{\delta \rho}(x) \right) \right\|^2 \rho_t(x) dx. \end{aligned}$$

Thus,

$$\frac{d}{dt} F[\rho_t] = - \int_{\mathbb{R}^n} \left\| \nabla \left(\frac{\delta F}{\delta \rho}(x) \right) \right\|^2 \rho_t(x) dx \leq 0, \quad (4.6)$$

so $F[\rho_t]$ is non-increasing along the flow. Equality holds if and only if $\nabla(\delta F/\delta\rho) = 0$ ρ_t -almost everywhere, which corresponds to stationary solutions.

The rigorous formulation of (4.6) requires additional regularity and integrability assumptions on ρ_t and W , and is typically obtained via the theory of gradient flows in metric spaces together with suitable weak formulations of the PDE. For the purposes of this paper, (4.6) should be viewed as the guiding formal identity connecting the monotonicity of $F[\rho_t]$ with the gradient-flow structure of the evolution.

Remark 4.7. Equation (4.6) reveals the dual role of F : on the one hand, minimisers of F are stationary solutions of (4.5); on the other hand, the evolution (4.5) can be seen as the steepest-descent dynamics that drives ρ_t towards these minimisers in Wasserstein space. This interplay between variational structure, geometry of $\mathcal{P}_2(\mathbb{R}^n)$, and dissipative dynamics is central to the interpretation of diffusion equations as gradient flows of free energy.

In the next section we connect the formal gradient-flow equation (4.5) with Fokker–Planck equations arising from Langevin-type stochastic dynamics, making the link between the stochastic and variational viewpoints explicit.

5 Fokker–Planck equations as gradient flows of free energy

In this section we make explicit the connection between the Fokker–Planck equations associated with Langevin dynamics and the Wasserstein gradient-flow formulation of the free energy functional F defined in Section 3 and analysed in Section 4. For concreteness we work in a finite-dimensional state space $S = \mathbb{R}^n$ with reference measure given by Lebesgue measure dx . The potential $W : \mathbb{R}^n \rightarrow \mathbb{R}$ is assumed to be sufficiently regular and confining so that the free energy

$$F[\rho] = \int_{\mathbb{R}^n} W(x) \rho(x) dx + D \int_{\mathbb{R}^n} \rho(x) \ln \rho(x) dx$$

is well-defined on an appropriate subset of $\mathcal{P}_2(\mathbb{R}^n)$.

5.1 Langevin dynamics and the Fokker–Planck equation

We recall the classical overdamped Langevin stochastic differential equation (SDE) in $\mathbb{R}^n$ driven by the potential W and a diffusion parameter $D > 0$:

$$dX_t = -\nabla W(X_t) dt + \sqrt{2D} dB_t, \quad (5.1)$$

where $(B_t)_{t \geq 0}$ is a standard n -dimensional Brownian motion. The drift term $-\nabla W$ represents deterministic motion in the direction of decreasing potential, while the stochastic term models fluctuations with strength controlled by D .

5.1.1 Generator and Kolmogorov equations

The SDE (5.1) has (formal) infinitesimal generator L acting on smooth test functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$(Lf)(x) = -\nabla W(x) \cdot \nabla f(x) + D \Delta f(x). \quad (5.2)$$

If $(X_t)_{t \geq 0}$ solves (5.1) with initial condition $X_0 \sim \rho_0(x) dx$, then its time-marginal densities ρ_t (if they exist) satisfy the Fokker–Planck (forward Kolmogorov) equation.

Proposition 5.1 (Fokker–Planck equation). *Let $\rho_t(x)$ denote the probability density of X_t solving (5.1). Assuming sufficient regularity of W and ρ_t , the family $(\rho_t)_{t \geq 0}$ satisfies*

$$\partial_t \rho_t(x) = \nabla \cdot (\rho_t(x) \nabla W(x)) + D \Delta \rho_t(x), \quad x \in \mathbb{R}^n, \quad t > 0, \quad (5.3)$$

with initial condition $\rho_{t=0} = \rho_0$.

Formal derivation. Let f be a smooth compactly supported test function and consider $\mathbb{E}[f(X_t)]$. By Itô's formula and the definition of L in (5.2),

$$\frac{d}{dt} \mathbb{E}[f(X_t)] = \mathbb{E}[(Lf)(X_t)].$$

Writing $\mathbb{E}[f(X_t)] = \int_{\mathbb{R}^n} f(x) \rho_t(x) dx$ and integrating by parts, we obtain

$$\frac{d}{dt} \int_{\mathbb{R}^n} f(x) \rho_t(x) dx = \int_{\mathbb{R}^n} f(x) \left(\nabla \cdot (\rho_t(x) \nabla W(x)) + D \Delta \rho_t(x) \right) dx,$$

which shows that ρ_t satisfies (5.3) in the weak sense. $\square$

Equation (5.3) expresses conservation of probability: densities evolve under the combined influence of drift $-\nabla W$ and diffusion with coefficient D .

5.2 Fokker–Planck as Wasserstein gradient flow of F

We now connect the Fokker–Planck equation (5.3) to the Wasserstein gradient-flow structure discussed in Section 4. We recall that the formal functional derivative of F is

$$\frac{\delta F}{\delta \rho}(x) = W(x) + D(1 + \ln \rho(x)). \quad (5.4)$$

5.2.1 Velocity field representation

From Section 4 we know that, in the formal Riemannian structure of $\mathcal{P}_2(\mathbb{R}^n)$, the gradient of F at a density ρ is represented by the velocity field

$$\text{grad } F(\rho) \sim \nabla \left(\frac{\delta F}{\delta \rho} \right).$$

A gradient flow of F in the Wasserstein sense is then given by the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad v_t(x) = -\nabla \left(\frac{\delta F}{\delta \rho_t}(x) \right). \quad (5.5)$$

Substituting (5.4),

$$v_t(x) = -\nabla W(x) - D \nabla \ln \rho_t(x).$$

Therefore

$$\begin{aligned} \partial_t \rho_t &= \nabla \cdot (\rho_t \nabla W) + D \nabla \cdot (\rho_t \nabla \ln \rho_t) \\ &= \nabla \cdot (\rho_t \nabla W) + D \nabla \cdot (\nabla \rho_t) \\ &= \nabla \cdot (\rho_t \nabla W) + D \Delta \rho_t, \end{aligned}$$

which coincides with the Fokker–Planck equation (5.3).

Proposition 5.2 (Fokker–Planck as Wasserstein gradient flow). *Formally, the Fokker–Planck equation (5.3) is the 2-Wasserstein gradient flow of the free energy functional F on $\mathcal{P}_2(\mathbb{R}^n)$. That is, smooth solutions of (5.3) coincide with solutions of the gradient-flow equation (5.5) driven by F .*

Remark 5.3. The argument above is formal in that it assumes sufficient smoothness and decay of ρ_t and W to justify all integrations by parts and pointwise manipulations. Rigorous formulations are available in the framework of gradient flows in metric spaces: under appropriate assumptions on W (e.g. λ -convexity), one can construct solutions of (5.3) as curves of maximal slope for F with respect to the 2-Wasserstein metric and prove uniqueness and stability.

5.3 Free energy dissipation and stationary measures

We now turn to the behaviour of the free energy $F[\rho_t]$ along solutions of the Fokker–Planck equation. The gradient-flow structure yields a natural dissipation identity and identifies stationary measures as minimisers of F .

5.3.1 Energy dissipation along the flow

Let $(\rho_t)_{t \geq 0}$ be a sufficiently regular solution of (5.3). Using the chain rule and the continuity equation with velocity field $v_t = -\nabla(\delta F / \delta \rho_t)$, we formally compute

$$\begin{aligned} \frac{d}{dt} F[\rho_t] &= \int_{\mathbb{R}^n} \frac{\delta F}{\delta \rho_t}(x) \partial_t \rho_t(x) dx \\ &= - \int_{\mathbb{R}^n} \frac{\delta F}{\delta \rho_t}(x) \nabla \cdot (\rho_t(x) v_t(x)) dx \\ &= \int_{\mathbb{R}^n} \nabla \left(\frac{\delta F}{\delta \rho_t}(x) \right) \cdot v_t(x) \rho_t(x) dx \\ &= - \int_{\mathbb{R}^n} \left\| \nabla \left(\frac{\delta F}{\delta \rho_t}(x) \right) \right\|^2 \rho_t(x) dx. \end{aligned}$$

Thus

$$\frac{d}{dt} F[\rho_t] = - \int_{\mathbb{R}^n} \left\| \nabla \left(\frac{\delta F}{\delta \rho_t}(x) \right) \right\|^2 \rho_t(x) dx \leq 0, \quad (5.6)$$

showing that $F[\rho_t]$ is non-increasing in time.

Remark 5.4. Equation (5.6) expresses the fact that the rate of decrease of the free energy equals (up to sign) the squared norm of the gradient of F with respect to the Wasserstein metric. In particular, the dissipation vanishes if and only if

$$\nabla \left(\frac{\delta F}{\delta \rho_t}(x) \right) = 0 \quad \text{for } \rho_t\text{-almost all } x,$$

i.e. if and only if ρ_t is a stationary point of F .

Under appropriate conditions on W ensuring coercivity and convexity (see Section 3), one can show that F has a unique minimiser and that $F[\rho_t]$ converges to $F[\rho_*]$ as $t \rightarrow \infty$, with ρ_t converging towards the unique stationary density ρ_* .

5.3.2 Stationary measures and Gibbs states

A stationary solution of the Fokker–Planck equation (5.3) is a density ρ_* satisfying

$$\nabla \cdot (\rho_*(x) \nabla W(x)) + D \Delta \rho_*(x) = 0, \quad x \in \mathbb{R}^n. \quad (5.7)$$

Assuming sufficient regularity and positivity of ρ_* , (5.7) can be rewritten as

$$\nabla \cdot \left(\rho_*(x) \nabla \left(\frac{\delta F}{\delta \rho_*}(x) \right) \right) = 0.$$

A sufficient condition for this to hold is that $\nabla(\delta F / \delta \rho_*) = 0$, which implies that $\delta F / \delta \rho_*$ is constant on the support of ρ_* . From (5.4), this yields the Euler–Lagrange condition

$$W(x) + D(1 + \ln \rho_*(x)) = \lambda \quad \text{for some constant } \lambda,$$

and hence the Gibbs-type formula

$$\rho_*(x) = \frac{1}{Z} \exp\left(-\frac{W(x)}{D}\right), \quad Z = \int_{\mathbb{R}^n} \exp\left(-\frac{W(x)}{D}\right) dx,$$

provided that $Z < \infty$.

Thus the stationary solutions of the Fokker–Planck equation coincide with critical points of F under the mass constraint, and, under suitable convexity conditions on W , with the unique minimiser of F .

5.4 Variational characterisation via minimizing movements

The interpretation of (5.3) as the Wasserstein gradient flow of F can be made rigorous in finite dimensions using the minimizing-movement (time-discrete gradient flow) scheme. Given a time step $\tau > 0$ and an initial density $\rho^0 \in \mathcal{P}_2(\mathbb{R}^n)$, one defines a sequence $\{\rho^k\}_{k \geq 1}$ recursively by

$$\rho^k \in \operatorname{argmin}_{\rho \in \mathcal{P}_2(\mathbb{R}^n)} \left\{ F[\rho] + \frac{1}{2\tau} W_2^2(\rho, \rho^{k-1}) \right\}. \quad (5.8)$$

Under suitable assumptions on W (e.g. λ -convexity), one shows that:

- For each $\tau > 0$ and $k \geq 1$, the minimisation problem (5.8) admits at least one solution ρ^k .
- The piecewise-constant (or piecewise-geodesic) interpolation of $\{\rho^k\}$ in time converges, as $\tau \rightarrow 0$, to a curve $(\rho_t)_{t \geq 0}$ in $\mathcal{P}_2(\mathbb{R}^n)$ which solves the Fokker–Planck equation (5.3) in a suitable weak sense.
- The limit curve (ρ_t) is a gradient flow of F with respect to the 2-Wasserstein metric, in the sense of curves of maximal slope.

This construction exhibits the Fokker–Planck evolution as the continuum limit of a sequence of variational problems involving F and the Wasserstein distance. It provides a robust framework for proving existence and uniqueness of solutions, as well as for establishing contractivity and convergence to equilibrium under appropriate convexity assumptions.

Remark 5.5. The minimizing-movement interpretation of the Fokker–Planck equation underscores the central role of the free energy F : at each time step, ρ^k is obtained by balancing proximity to the previous state (measured by W_2^2) with decrease in F . In the limit of small time steps, this yields the continuous-time steepest-descent flow of F in Wasserstein space, which is precisely the Fokker–Planck evolution.

In summary, the Fokker–Planck equation associated with the overdamped Langevin dynamics (5.1) is naturally understood as the Wasserstein gradient flow of the free energy functional F . This provides a unified variational and geometric perspective on the dissipative dynamics of probability densities in configuration space, which will be further related to classical free energy and variational principles in the next section.

6 Connections to classical free energy and variational principles

In this section we relate the free energy functional

$$F[\rho] = \int_S W(s) \rho(s) d\mu(s) + D \int_S \rho(s) \ln \rho(s) d\mu(s)$$

to the classical Helmholtz free energy of equilibrium statistical mechanics, to variational characterisations of Gibbs measures, and to structural connections with large deviations and information-theoretic functionals. Throughout this section we work, for concreteness, in the finite-dimensional case $S = \mathbb{R}^n$ with reference measure μ given by Lebesgue measure dx , although many of the formulas extend to more general settings.

6.1 Classical Helmholtz free energy and Gibbs measures

In equilibrium statistical mechanics, a system with configuration variable $x \in \mathbb{R}^n$ and potential energy $V(x)$ at inverse temperature $\beta > 0$ is described in the canonical ensemble by the Gibbs density

$$\rho_\beta(x) = \frac{1}{Z_\beta} \exp(-\beta V(x)),$$

where the partition function

$$Z_\beta = \int_{\mathbb{R}^n} \exp(-\beta V(x)) dx$$

ensures normalisation. The associated Helmholtz free energy is

$$\mathcal{F}_\beta = -\frac{1}{\beta} \ln Z_\beta.$$

Alternatively, one can view the canonical ensemble variationally by considering probability densities ρ on $\mathbb{R}^n$ and the functional

$$\mathcal{F}_\beta[\rho] = \int_{\mathbb{R}^n} V(x) \rho(x) dx + \frac{1}{\beta} \int_{\mathbb{R}^n} \rho(x) \ln \rho(x) dx.$$

Under suitable conditions on V , the Gibbs density ρ_β is the unique minimiser of $\mathcal{F}_\beta$ over all probability densities ρ . Moreover,

$$\mathcal{F}_\beta[\rho_\beta] = \inf_{\rho} \mathcal{F}_\beta[\rho] = -\frac{1}{\beta} \ln Z_\beta. \quad (6.1)$$

6.1.1 Identification with the functional F

Comparing $\mathcal{F}_\beta[\rho]$ with $F[\rho]$ in the finite-dimensional setting and with $\mu = dx$, we see that

$$F[\rho] = \int_{\mathbb{R}^n} W(x) \rho(x) dx + D \int_{\mathbb{R}^n} \rho(x) \ln \rho(x) dx$$

coincides with $\mathcal{F}_\beta[\rho]$ upon the identification

$$W(x) = V(x), \quad D = \frac{1}{\beta}.$$

Thus D plays the role of a temperature-like parameter (up to physical constants such as Boltzmann's constant in concrete applications), and F is a natural abstract version of the classical Helmholtz free energy functional, extended to general potentials W and reference measures μ .

Our earlier result on the Gibbs form of minimisers (Proposition 3.10) can then be seen as the analogue of the statement that canonical Gibbs measures minimise the Helmholtz free energy at fixed inverse temperature. Indeed, under finiteness of the partition function

$$Z = \int_{\mathbb{R}^n} \exp\left(-\frac{W(x)}{D}\right) dx,$$

minimisers of F are of the form

$$\rho_*(x) = \frac{1}{Z} \exp\left(-\frac{W(x)}{D}\right),$$

which is directly analogous to ρ_β with $D = 1/\beta$.

6.2 Variational characterisations and large deviations

The free energy functional F admits several equivalent representations that emphasise its variational nature. These representations play a central role in equilibrium statistical mechanics and in large deviation theory.

6.2.1 Variational representation of $\ln Z$

Let $D > 0$ and let

$$Z = \int_{\mathbb{R}^n} \exp\left(-\frac{W(x)}{D}\right) dx$$

be finite. Consider F restricted to probability densities with respect to Lebesgue measure. Using the Gibbs density

$$\rho_*(x) = \frac{1}{Z} \exp\left(-\frac{W(x)}{D}\right),$$

we can rewrite $F[\rho]$ for any density ρ as

$$\begin{aligned} F[\rho] &= \int_{\mathbb{R}^n} W(x) \rho(x) dx + D \int_{\mathbb{R}^n} \rho(x) \ln \rho(x) dx \\ &= D \int_{\mathbb{R}^n} \rho(x) \ln \rho(x) dx + D \int_{\mathbb{R}^n} \rho(x) \left(\frac{W(x)}{D}\right) dx \\ &= D \int_{\mathbb{R}^n} \rho(x) \ln \left(\rho(x) e^{W(x)/D}\right) dx \\ &= D \int_{\mathbb{R}^n} \rho(x) \ln \left(\frac{\rho(x)}{\rho_*(x)}\right) dx - D \ln Z \\ &= D D_{\text{KL}}(\rho \| \rho_*) - D \ln Z, \end{aligned} \tag{6.2}$$

where

$$D_{\text{KL}}(\rho \| \rho_*) := \int_{\mathbb{R}^n} \rho(x) \ln \left(\frac{\rho(x)}{\rho_*(x)}\right) dx$$

is the Kullback–Leibler divergence of ρ from ρ_* . Since D_{KL} is non-negative and vanishes if and only if $\rho = \rho_*$ almost everywhere, (6.2) immediately yields the variational representation

$$\inf_{\rho} F[\rho] = -D \ln Z, \quad \text{achieved uniquely at } \rho = \rho_*. \tag{6.3}$$

Equivalently,

$$-D \ln Z = \inf_{\rho} \left\{ \int W d\rho + D \int \rho \ln \rho dx \right\},$$

which is the direct analogue of the classical formula (6.1).

6.2.2 Large deviations and rate functionals

In many-particle systems, free energy functionals arise as rate functionals in large deviation principles for empirical measures. Consider, for example, N independent identically distributed random variables in $\mathbb{R}^n$ with common law ρ_* and empirical measure $L_N = \frac{1}{N} \sum_{i=1}^N \delta_{X_i}$. Sanov's theorem states that L_N satisfies a large deviation principle with speed N and rate function given by the relative entropy with respect to ρ_* , i.e.

$$I(\rho) = D_{\text{KL}}(\rho \| \rho_*).$$

In this context, the decomposition (6.2) shows that $F[\rho]$ differs from $D I(\rho)$ only by an additive constant $-D \ln Z$.

More generally, interacting particle systems and mean-field models lead to rate functionals that combine energy and entropy contributions in a way structurally similar to F . The minimisers of such rate functionals describe typical or equilibrium empirical measures in the large- N limit, and the gradient-flow dynamics of F may be seen as a macroscopic evolution law consistent with these variational principles.

6.3 Information-theoretic and inference-based perspectives

The representation (6.2) provides a direct link between the free energy functional F and information-theoretic quantities. Structurally, F can be viewed as a shifted Kullback–Leibler divergence between a candidate density ρ and the Gibbs density ρ_* .

6.3.1 Free energy as shifted divergence

Writing (6.2) as

$$F[\rho] + D \ln Z = D D_{\text{KL}}(\rho \| \rho_*),$$

we see that the free energy F is, up to an additive constant, equivalent to the divergence $D_{\text{KL}}(\rho \| \rho_*)$. Thus:

- $F[\rho]$ is minimised when the divergence between ρ and the reference Gibbs measure ρ_* is minimised.
- The difference $F[\rho] - F[\rho_*]$ is proportional to the Kullback–Leibler divergence:

$$F[\rho] - F[\rho_*] = D D_{\text{KL}}(\rho \| \rho_*).$$

This identity quantifies the excess free energy of ρ relative to equilibrium in terms of an information-theoretic distance to ρ_* .

6.3.2 Structural analogy to variational inference

In variational inference and related areas of information theory, one often considers functionals of the form

$$\mathcal{J}[\rho] = \int f(s) \rho(s) d\mu(s) + D_{\text{KL}}(\rho \| \pi),$$

where π is a fixed reference distribution and f is a cost or energy function. Minimisation of $\mathcal{J}$ balances fidelity to the reference π with a preference for low cost f .

In the present setting, (6.2) shows that F can be written as

$$F[\rho] = D_{\text{KL}}(\rho \| \rho_*) + \text{constant},$$

with ρ_* determined by the potential W . Thus, structurally, minimising F corresponds to optimally approximating the Gibbs distribution ρ_* within the family of admissible densities. The Wasserstein gradient-flow dynamics derived in earlier sections can be interpreted as a continuous-time evolution that reduces this divergence in the steepest possible way with respect to the Wasserstein metric.

6.4 Summary of structural connections

To summarise the links established in this section:

- The functional F is an abstract version of the classical Helmholtz free energy, with the potential W playing the role of energy and the parameter D playing a temperature-like role.
- Gibbs measures of the form

$$\rho_*(s) \propto \exp\left(-\frac{W(s)}{D}\right)$$

are minimisers of F under the mass constraint and correspond to equilibrium states in the classical sense.

- The minimal free energy is given by a variational representation of the partition function:

$$\inf_{\rho} F[\rho] = -D \ln Z,$$

and F can be expressed as a shifted Kullback–Leibler divergence relative to the Gibbs measure ρ_* .

- The structure of F matches rate functionals appearing in large deviation principles and is closely related to information-theoretic functionals used in variational formulations and inference.

These connections place the free energy functional F within a network of well-established concepts in statistical mechanics, large deviations, and information theory, providing additional context for the gradient-flow dynamics studied in the previous sections and for the examples to be discussed next.

7 Examples and special cases

In this section we illustrate the abstract constructions developed so far by means of concrete examples in finite-dimensional state spaces. We first analyse a one-dimensional double-well potential, then consider general quadratic potentials leading to Gaussian stationary measures, and finally discuss the qualitative effect of the parameter D (interpretable as a temperature-like parameter) on the shape of the free energy landscape and the corresponding stationary distributions.

Throughout this section we take $S = \mathbb{R}^n$ with reference measure μ given by Lebesgue measure dx . The free energy functional then takes the form

$$F[\rho] = \int_{\mathbb{R}^n} W(x) \rho(x) dx + D \int_{\mathbb{R}^n} \rho(x) \ln \rho(x) dx,$$

for suitable densities ρ and potentials W .

7.1 A one-dimensional double-well potential

We begin with a simple one-dimensional example exhibiting multiple local minima of the potential and, correspondingly, multiple local maxima of the stationary density.

7.1.1 Definition of the model

Let $S = \mathbb{R}$ and consider the double-well potential

$$W(x) = \frac{a}{4}(x^2 - 1)^2, \quad x \in \mathbb{R}, \quad (7.1)$$

with parameter $a > 0$. This potential has two global minima at $x = \pm 1$ with $W(\pm 1) = 0$ and a local maximum at $x = 0$ with

$$W(0) = \frac{a}{4}.$$

The potential grows quartically as $|x| \rightarrow \infty$, ensuring coercivity and finiteness of the partition function

$$Z(D) := \int_{\mathbb{R}} \exp\left(-\frac{W(x)}{D}\right) dx < \infty \quad \text{for all } D > 0.$$

7.1.2 Stationary density and free energy

The Gibbs (stationary) density associated with W and $D > 0$ is

$$\rho_*(x; D) = \frac{1}{Z(D)} \exp\left(-\frac{W(x)}{D}\right), \quad x \in \mathbb{R}. \quad (7.2)$$

By the general theory in Section 3, $\rho_*(\cdot; D)$ is the unique minimiser of F for each fixed $D > 0$. The minimal free energy is given by

$$F[\rho_*] = -D \ln Z(D),$$

and any other admissible density ρ satisfies

$$F[\rho] - F[\rho_*] = D D_{\text{KL}}(\rho \| \rho_*) \geq 0.$$

Although $Z(D)$ does not admit a closed-form expression for the quartic potential (7.1), its qualitative dependence on D is well understood via Laplace-type asymptotics.

7.1.3 Low- and high- D regimes

Low- D regime ($D \rightarrow 0$). When D is small, the weight $\exp(-W(x)/D)$ is strongly concentrated near the global minima of W . Since W has two equal minima at $x = \pm 1$, a standard Laplace approximation yields

$$Z(D) \approx 2 \sqrt{\frac{2\pi D}{W''(1)}} e^{-W(1)/D} = 2 \sqrt{\frac{2\pi D}{2a}} e^0 = 2 \sqrt{\frac{\pi D}{a}},$$

where we used

$$W''(x) = a(3x^2 - 1), \quad W''(1) = 2a.$$

Thus $\rho_*(x; D)$ becomes increasingly bimodal, with most of its mass concentrated in small neighbourhoods of $x = \pm 1$. In this regime, the free energy is dominated by the potential term, and the dynamics described by the Fokker–Planck equation exhibits metastable behaviour, with the process spending long times near either well before transitions across the barrier at $x = 0$ occur.

High- D regime ($D \rightarrow \infty$). For large D , the diffusion term dominates, and the density $\rho_*(\cdot; D)$ becomes flatter. Formally, one can expand

$$\exp\left(-\frac{W(x)}{D}\right) = 1 - \frac{W(x)}{D} + O\left(\frac{1}{D^2}\right),$$

so that, after normalisation, $\rho_*(x; D)$ approaches a broad distribution whose shape is only weakly influenced by the detailed structure of W . The entropy term in F then dominates, favouring more spread-out densities.

Intermediate- D regime. At intermediate values of D , the interplay between the double-well structure of W and the smoothing effect of diffusion leads to distributions that are still bimodal but with less pronounced peaks. The precise dependence of the peak heights and barrier occupancy on D can be studied numerically or via more refined asymptotic techniques, but the general picture is that increasing D progressively smooths out the potential-induced structure in ρ_* .

7.2 Quadratic potentials and Gaussian measures

We now turn to the important class of quadratic potentials, for which the stationary density and the gradient-flow dynamics can be analysed explicitly.

7.2.1 General quadratic potential in $\mathbb{R}^n$

Let $S = \mathbb{R}^n$ and consider a quadratic potential of the form

$$W(x) = \frac{1}{2}x^\top Ax + b^\top x + c, \quad x \in \mathbb{R}^n, \quad (7.3)$$

where A is a symmetric positive-definite $n \times n$ matrix, $b \in \mathbb{R}^n$, and $c \in \mathbb{R}$. The corresponding partition function is

$$Z(D) = \int_{\mathbb{R}^n} \exp\left(-\frac{1}{D}W(x)\right) dx.$$

Completing the square in the exponent, we write

$$W(x) = \frac{1}{2}(x - m)^\top A(x - m) + C,$$

where

$$m = -A^{-1}b, \quad C = c - \frac{1}{2}b^\top A^{-1}b.$$

Then

$$\exp\left(-\frac{1}{D}W(x)\right) = \exp\left(-\frac{C}{D}\right) \exp\left(-\frac{1}{2D}(x - m)^\top A(x - m)\right),$$

and the partition function can be evaluated explicitly:

$$Z(D) = \exp\left(-\frac{C}{D}\right) (2\pi D)^{n/2} (\det A)^{-1/2}.$$

7.2.2 Gaussian stationary density

The Gibbs density associated with W and D is

$$\begin{aligned}\rho_*(x; D) &= \frac{1}{Z(D)} \exp\left(-\frac{1}{D}W(x)\right) \\ &= \frac{(\det A)^{1/2}}{(2\pi D)^{n/2}} \exp\left(-\frac{1}{2D}(x-m)^\top A(x-m)\right).\end{aligned}\quad (7.4)$$

Thus ρ_* is a multivariate Gaussian with mean m and covariance matrix

$$\Sigma = D A^{-1}.$$

In particular, the covariance is proportional to D , illustrating how the stationary distribution spreads out as D increases.

7.2.3 Free energy of Gaussian densities

For quadratic W , the free energy can be computed explicitly on the family of Gaussian densities. Let ρ be the Gaussian $\mathcal{N}(m, \Sigma)$ with density

$$\rho(x) = \frac{1}{(2\pi)^{n/2}(\det \Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x-m)^\top \Sigma^{-1}(x-m)\right).$$

Then:

- The energy term is

$$\mathcal{E}[\rho] = \int W(x) \rho(x) dx = \frac{1}{2} \text{tr}(A\Sigma) + \frac{1}{2} m^\top A m + b^\top m + c.$$

- The entropy term is

$$\mathcal{H}[\rho] = \int \rho(x) \ln \rho(x) dx = -\frac{n}{2} \ln(2\pi e) - \frac{1}{2} \ln \det \Sigma.$$

Therefore the free energy of $\mathcal{N}(m, \Sigma)$ is

$$F[\mathcal{N}(m, \Sigma)] = \frac{1}{2} \text{tr}(A\Sigma) + \frac{1}{2} m^\top A m + b^\top m + c + D \left(-\frac{n}{2} \ln(2\pi e) - \frac{1}{2} \ln \det \Sigma \right). \quad (7.5)$$

Minimising (7.5) over (m, Σ) yields

$$m = -A^{-1}b, \quad \Sigma = DA^{-1},$$

which recovers the stationary density (7.4) as the unique minimiser of F within the Gaussian family.

7.2.4 Ornstein–Uhlenbeck dynamics as gradient flow

For the quadratic potential (7.3), the Langevin SDE

$$dX_t = -\nabla W(X_t) dt + \sqrt{2D} dB_t$$

becomes

$$dX_t = -(AX_t + b) dt + \sqrt{2D} dB_t,$$

which defines an Ornstein–Uhlenbeck process. If the initial density ρ_0 is Gaussian, then ρ_t remains Gaussian for all t , and the evolution of its mean m_t and covariance Σ_t is governed by the linear ODEs

$$\frac{d}{dt}m_t = -Am_t - b, \quad \frac{d}{dt}\Sigma_t = -A\Sigma_t - \Sigma_t A^\top + 2DI.$$

The stationary solution of these equations is precisely $m_\infty = -A^{-1}b$ and $\Sigma_\infty = DA^{-1}$, confirming that the Ornstein–Uhlenbeck dynamics realises the gradient flow of F driving Gaussian densities towards the minimiser (7.4).

7.3 High- vs low-temperature regimes and the shape of F

We conclude with a qualitative discussion of how the parameter D influences the shape of the free energy landscape and the stationary distributions.

7.3.1 Low-temperature regime ($D \rightarrow 0$)

When D is small, the entropy contribution $D\mathcal{H}[\rho]$ to F is suppressed, and the free energy is dominated by the energy term $\mathcal{E}[\rho]$. Minimisation of F then favours densities concentrated near global minima of W . More precisely:

- If W has a unique global minimum $x_{\min}$, then $\rho_*(\cdot; D)$ concentrates around $x_{\min}$ as $D \rightarrow 0$, and the free energy landscape features a deep well near the corresponding Dirac measure.
- If W has multiple global minima of equal depth (as in the double-well case), then $\rho_*(\cdot; D)$ becomes multimodal, with mass distributed among neighbourhoods of these minima.

In this regime, the Fokker–Planck dynamics exhibits slow transitions between different basins of attraction, and metastability is prominent.

7.3.2 High-temperature regime ($D \rightarrow \infty$)

For large D , the entropy term dominates the free energy, favouring spread-out densities. The influence of the detailed structure of W is weakened, and $\rho_*(\cdot; D)$ approaches a broad distribution that, in many cases, resembles a nearly Gaussian density with large variance or, more generally, a distribution whose shape is determined by coarse features of W . From the viewpoint of F , the landscape becomes smoother, with barriers between local minima substantially reduced.

7.3.3 Intermediate regime and phase-like behaviour

Between these two extremes, the interplay of energy and entropy produces nontrivial free energy landscapes:

- In systems with multiple wells, varying D can induce qualitative changes in the number and stability of local minima of F , leading to behaviour reminiscent of phase transitions in statistical mechanics.
- For quadratic potentials, the dependence on D is particularly simple: the stationary density remains Gaussian for all D , with covariance proportional to D , and F changes smoothly with D without qualitative changes in the minimiser.

These examples illustrate how the abstract free energy functional F and its gradient-flow dynamics give rise to familiar structures in simple models: Gibbs measures, Gaussian equilibria, and temperature-dependent concentration properties. In more complex or higher-dimensional systems, the same principles apply, though explicit formulas need to be replaced by asymptotic or numerical analyses.

8 Extensions and generalisations

The framework developed so far assumes a time-independent potential W , a single-component state $s \in S$, and an unconstrained gradient flow of the free energy functional

$$F[\rho] = \int_S W(s) \rho(s) d\mu(s) + D \int_S \rho(s) \ln \rho(s) d\mu(s).$$

In many applications one is led to consider time-dependent external influences, multiple interacting subsystems, or additional constraints on the admissible probability measures. In this section we briefly outline how such situations fit into the present formalism, without attempting a fully rigorous treatment.

8.1 Time-dependent potentials $W(s, t)$

We first consider the case where the potential depends explicitly on time. Let $t \mapsto W_t$ be a family of Borel measurable functions $W_t : S \rightarrow \mathbb{R} \cup \{+\infty\}$, representing, for instance, a time-varying external field or a slowly changing environment. We assume that, for each fixed t , the integrability and coercivity conditions from Section 3 hold uniformly in t on bounded time intervals, so that the associated free energy functionals are well-defined.

8.1.1 Time-dependent free energy functional

For each $t \geq 0$ and density ρ with respect to μ , we define

$$F_t[\rho] := \int_S W_t(s) \rho(s) d\mu(s) + D \int_S \rho(s) \ln \rho(s) d\mu(s). \quad (8.1)$$

The functional F_t has the same entropy contribution as before, while the energy term now incorporates the time-dependent potential W_t .

Formally, the functional derivative of F_t with respect to ρ is

$$\frac{\delta F_t}{\delta \rho}(s) = W_t(s) + D(1 + \ln \rho(s)), \quad (8.2)$$

for sufficiently regular strictly positive ρ .

8.1.2 Gradient-flow equation and modified dissipation

In a finite-dimensional setting $S = \mathbb{R}^n$ with reference measure dx , one can extend the Wasserstein gradient-flow formalism by considering the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad v_t(x) = -\nabla \left(\frac{\delta F_t}{\delta \rho_t}(x) \right), \quad (8.3)$$

that is,

$$v_t(x) = -\nabla W_t(x) - D \nabla \ln \rho_t(x).$$

As in Section 4, this leads to the evolution equation

$$\partial_t \rho_t(x) = \nabla \cdot (\rho_t(x) \nabla W_t(x)) + D \Delta \rho_t(x), \quad (8.4)$$

which is now a Fokker–Planck equation with time-dependent drift.

The explicit time dependence of F_t modifies the free energy balance. Formally,

$$\begin{aligned} \frac{d}{dt} F_t[\rho_t] &= \frac{\partial}{\partial t} F_t[\rho_t] + \int_{\mathbb{R}^n} \frac{\delta F_t}{\delta \rho_t}(x) \partial_t \rho_t(x) dx \\ &= \int_{\mathbb{R}^n} \partial_t W_t(x) \rho_t(x) dx - \int_{\mathbb{R}^n} \left\| \nabla \left(\frac{\delta F_t}{\delta \rho_t}(x) \right) \right\|^2 \rho_t(x) dx. \end{aligned} \quad (8.5)$$

Compared to the time-independent case, an additional term

$$\int \partial_t W_t \rho_t dx$$

appears, which can have either sign. Thus $t \mapsto F_t[\rho_t]$ is no longer necessarily non-increasing; the explicit time-dependence of W_t can inject or extract free energy from the system.

Remark 8.1. In situations where W_t varies slowly in time, one may expect ρ_t to track, at least approximately, the instantaneous minimiser of F_t . More generally, one can interpret the second term in (8.5) as the intrinsic dissipation due to the gradient-flow structure, while the first term quantifies the energetic “work” performed by changing the potential. A detailed analysis of such quasi-static or non-equilibrium regimes is beyond the scope of this paper.

8.2 Multiple interacting fields and multi-component systems

We next sketch how the framework extends to systems with several interacting components. Let

$$S^{(1)}, \dots, S^{(k)}$$

be separable Hilbert spaces, and consider the product state space

$$S := S^{(1)} \times \dots \times S^{(k)},$$

equipped with the product inner product and norm, and with Borel σ -algebra $\mathcal{B}(S)$. We fix a reference measure μ on $(S, \mathcal{B}(S))$, often of the product form

$$\mu = \mu^{(1)} \otimes \dots \otimes \mu^{(k)},$$

although more general choices are possible.

8.2.1 Joint and marginal distributions

A probability density ρ on S with respect to μ is now a function

$$\rho : S^{(1)} \times \dots \times S^{(k)} \rightarrow [0, \infty),$$

such that

$$\int_S \rho(s^{(1)}, \dots, s^{(k)}) d\mu(s^{(1)}, \dots, s^{(k)}) = 1.$$

We denote by $\rho^{(i)}$ the marginal density of component i with respect to $\mu^{(i)}$, defined by integrating out the remaining components, e.g.

$$\rho^{(1)}(s^{(1)}) = \int_{S^{(2)} \times \dots \times S^{(k)}} \rho(s^{(1)}, \dots, s^{(k)}) d\mu^{(2)}(s^{(2)}) \dots d\mu^{(k)}(s^{(k)}),$$

and similarly for other components. One can also define conditional densities $\rho^{(i)}_{\widehat{i}}$, where $\widehat{i}$ denotes all indices except i .

8.2.2 Free energy for joint distributions

Let $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ be a Borel measurable potential on the product space. The free energy for a joint density ρ on S is defined as before by

$$F[\rho] = \int_S W(s^{(1)}, \dots, s^{(k)}) \rho(s^{(1)}, \dots, s^{(k)}) d\mu + D \int_S \rho(s^{(1)}, \dots, s^{(k)}) \ln \rho(s^{(1)}, \dots, s^{(k)}) d\mu. \quad (8.6)$$

The Gibbs minimiser (when it exists) has the form

$$\rho_*(s^{(1)}, \dots, s^{(k)}) = \frac{1}{Z} \exp\left(-\frac{W(s^{(1)}, \dots, s^{(k)})}{D}\right),$$

with partition function

$$Z = \int_S \exp\left(-\frac{W(s^{(1)}, \dots, s^{(k)})}{D}\right) d\mu.$$

It is often useful to consider potentials that decompose into single-component and interaction terms, for example

$$W(s^{(1)}, \dots, s^{(k)}) = \sum_{i=1}^k W^{(i)}(s^{(i)}) + \sum_{1 \leq i < j \leq k} W^{(i,j)}(s^{(i)}, s^{(j)}) + \dots.$$

In such cases, the free energy can be analysed via corresponding decompositions of the energy term and of the entropy.

8.2.3 Conditional and partial free energies

Given a joint density ρ on S , one may wish to regard some components as “active” and others as “frozen” or fixed. For definiteness, consider the decomposition $s = (s^{(1)}, s^{(2)})$ with $S = S^{(1)} \times S^{(2)}$ and reference measure $\mu = \mu^{(1)} \otimes \mu^{(2)}$. Suppose that the marginal $\rho^{(2)}$ of the second component is fixed, and that we regard the conditional density $\rho^{(1)|2}(\cdot | s^{(2)})$ as the variable of interest.

A *conditional free energy* given $s^{(2)}$ can be defined by

$$F^{(1)|2}[\rho^{(1)|2}(\cdot | s^{(2)}); s^{(2)}] := \int_{S^{(1)}} W(s^{(1)}, s^{(2)}) \rho^{(1)|2}(s^{(1)} | s^{(2)}) d\mu^{(1)}(s^{(1)}) \\ + D \int_{S^{(1)}} \rho^{(1)|2}(s^{(1)} | s^{(2)}) \ln \rho^{(1)|2}(s^{(1)} | s^{(2)}) d\mu^{(1)}(s^{(1)}), \quad (8.7)$$

for $\mu^{(2)}$ -almost all $s^{(2)}$. Averaging over $s^{(2)}$ with respect to $\rho^{(2)}$ yields a *mean conditional free energy*

$$\bar{F}^{(1)|2}[\rho] := \int_{S^{(2)}} F^{(1)|2}[\rho^{(1)|2}(\cdot | s^{(2)}); s^{(2)}] \rho^{(2)}(s^{(2)}) d\mu^{(2)}(s^{(2)}).$$

Under appropriate conditions, minimising F over the joint density ρ with fixed marginal $\rho^{(2)}$ is equivalent to minimising $\bar{F}^{(1)|2}$ over conditional densities $\rho^{(1)|2}(\cdot | s^{(2)})$. The corresponding Euler–Lagrange equations lead to conditional Gibbs measures of the form

$$\rho_*^{(1)|2}(s^{(1)} | s^{(2)}) \propto \exp\left(-\frac{W(s^{(1)}, s^{(2)})}{D}\right),$$

for each fixed $s^{(2)}$. Analogous decompositions can be formulated for more than two components, leading to chains of conditional free energies and associated conditional Gibbs structures.

Remark 8.2. The multi-component setting allows one to study interaction structures at the level of joint distributions, to decompose free energy into marginal and conditional contributions, and to design dynamics in which only some components evolve according to gradient flows while others are treated as parameters. A complete analysis of such coupled systems is beyond the scope of this paper, but the basic formalism extends directly from the single-component case.

8.3 Constraints and projected gradient flows

Finally, we consider the effect of additional constraints on the admissible set of probability measures. Typical examples include conservation laws, hard constraints on the support of ρ , or moment constraints of the form

$$\int_S \phi_j(s) \rho(s) d\mu(s) = c_j, \quad j = 1, \dots, m,$$

for given measurable functions ϕ_j and prescribed constants c_j .

8.3.1 Constrained minimisation of free energy

Let $\mathcal{C} \subset \mathcal{P}(S)$ be a subset of probability measures defined by one or more such constraints, and consider the constrained minimisation problem

$$\inf \{F[\rho] \mid \rho \in \mathcal{D}_F \cap \mathcal{C}\}. \quad (8.8)$$

Formally, one can introduce Lagrange multipliers $\lambda_1, \dots, \lambda_m$ for the constraints and consider the augmented functional

$$\mathcal{F}[\rho] := F[\rho] + \sum_{j=1}^m \lambda_j \left(\int_S \phi_j(s) \rho(s) d\mu(s) - c_j \right).$$

Critical points of (8.8) then satisfy modified Euler–Lagrange equations of the form

$$\frac{\delta F}{\delta \rho}(s) + \sum_{j=1}^m \lambda_j \phi_j(s) = \text{constant},$$

which leads to Gibbs-like densities with effective potential

$$W_{\text{eff}}(s) = W(s) + \sum_{j=1}^m \lambda_j \phi_j(s).$$

8.3.2 Projected gradient flows

At the level of dynamics, one can interpret constrained evolution as a gradient flow of F on the constraint manifold (when $\mathcal{C}$ has a suitable manifold structure) or, more generally, as a gradient flow projected onto the tangent cone of $\mathcal{C}$ with respect to the Wasserstein metric.

In the formal Riemannian picture of $\mathcal{P}_2(\mathbb{R}^n)$, unconstrained gradient flow corresponds to the velocity field

$$v_t(x) = -\nabla \left(\frac{\delta F}{\delta \rho_t}(x) \right).$$

If $\mathcal{C}$ encodes linear constraints such as moment conservation, then tangent vectors at ρ_t must satisfy

$$\int_{\mathbb{R}^n} \phi_j(x) \partial_t \rho_t(x) dx = 0, \quad j = 1, \dots, m,$$

for all t . The projected velocity field $v_t^{\mathcal{C}}$ can be obtained by orthogonally projecting v_t onto the subspace of velocity fields that generate admissible tangent vectors. Formally, this leads to a modified PDE of the form

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \left(\frac{\delta F}{\delta \rho_t} + \sum_{j=1}^m \lambda_j(t) \phi_j \right) \right),$$

where the time-dependent multipliers $\lambda_j(t)$ are chosen to enforce the constraints.

Remark 8.3. A thorough development of projected Wasserstein gradient flows requires a careful analysis of the geometry of the constraint set $\mathcal{C}$ and the associated tangent cones, as well as existence and uniqueness results for the constrained evolution. Here we only outline the basic formal structure and note that constrained dynamics can often be expressed as gradient flows of modified free energies incorporating suitable Lagrange multiplier terms.

The extensions discussed in this section show that the free energy formalism and its gradient-flow interpretation are flexible enough to accommodate time-dependent forcing, multi-component systems, and constrained evolutions. A detailed treatment of these generalisations, including rigorous existence and convergence results, is left for future work.

9 Position within the broader “semantic field” framework

The preceding sections have been entirely self-contained, developing a free energy functional on spaces of probability measures and identifying its gradient-flow structure and associated Fokker–Planck dynamics. In this section we briefly indicate how these constructions fit into a broader framework in which potentials and gradient flows are used to describe dynamics at different levels of description. We refer to this broader programme informally as the “semantic field” framework, while keeping the present discussion purely structural and mathematical.

9.1 State-level and distribution-level descriptions

A basic distinction in the framework is between dynamics at the level of individual states and dynamics at the level of probability distributions over states.

9.1.1 State-level dynamics

At the state level, the central object is a potential function $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ on a state space S , described here as a separable Hilbert space. The dynamics of an individual state $s_t \in S$ is given, in finite dimensions, by a stochastic differential equation of Langevin type,

$$dX_t = -\nabla W(X_t) dt + \sqrt{2D} dB_t,$$

whose deterministic part is the gradient flow of W in the underlying Euclidean or Hilbert space, while the stochastic part models fluctuations. This level of description focuses on trajectories in S and on how the potential W shapes their evolution.

9.1.2 Distribution-level dynamics

The present paper operates at the level of probability distributions on S . Instead of a single trajectory X_t , one considers its time-marginal distributions ρ_t , which are elements of

$\mathcal{P}(S)$ or $\mathcal{P}_2(S)$. The dynamics of ρ_t is governed, in finite dimensions, by the Fokker–Planck equation

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla W) + D \Delta \rho_t,$$

which has been identified as the Wasserstein gradient flow of the free energy functional

$$F[\rho] = \int_S W(s) \rho(s) d\mu(s) + D \int_S \rho(s) \ln \rho(s) d\mu(s).$$

The distribution-level description thus captures the evolution of ensembles of states, or uncertainty over states, rather than individual trajectories.

9.2 Role of the free energy functional

Within this two-level picture, the free energy functional F introduced in this paper plays several structural roles.

9.2.1 Generator of distribution-level dynamics

First, F acts as the generator of the distribution-level dynamics: its Wasserstein gradient flow is precisely the Fokker–Planck evolution associated with the underlying Langevin dynamics. This realises a systematic passage from state-level potentials W to distribution-level flows governed by F . The potential W determines both the drift in the Langevin equation and the energy term in F , while the diffusion parameter D controls the strength of fluctuations and the weight of the entropy term.

9.2.2 Lyapunov functional and equilibrium selector

Second, F serves as a Lyapunov functional for the distribution-level dynamics. Along solutions ρ_t of the Fokker–Planck equation, $t \mapsto F[\rho_t]$ is non-increasing in the time-independent case, and strictly decreasing away from stationary states under suitable conditions. Stationary distributions are characterised as minimisers of F and are given by Gibbs-type densities

$$\rho_*(s) \propto \exp\left(-\frac{W(s)}{D}\right).$$

In this sense, F both quantifies the “distance” of a distribution from equilibrium (via its interpretation in terms of relative entropy) and selects the equilibrium states themselves.

9.2.3 Bridge to variational and geometric formulations

Third, the free energy formulation connects the distribution-level dynamics to variational and geometric viewpoints. On the variational side, F appears in minimizing-movement schemes that generate the Fokker–Planck evolution as a sequence of successive minimisation problems involving F and the Wasserstein distance. On the geometric side, F defines a gradient flow on a metric space of measures, providing a Riemannian-type structure on $\mathcal{P}_2(S)$ compatible with the underlying Hilbert structure of S . These connections situate the “semantic field” picture within well-established mathematical theories of optimal transport and gradient flows.

9.3 Relation to the state-level potential framework

The broader framework comprises at least two complementary components. At the state level, one analyses the properties of the potential W and the associated deterministic or stochastic dynamics in S . At the distribution level, the present paper provides the corresponding free energy and its gradient-flow dynamics on $\mathcal{P}(S)$.

9.3.1 From trajectories to distributions

The link between these two components is explicit in finite dimensions: the law of the Langevin process X_t evolves according to the Fokker–Planck equation, which is the Wasserstein gradient flow of F . Thus, given a potential W on S , there are two parallel viewpoints:

- The trajectory viewpoint: follow individual realisations X_t driven by W .
- The ensemble viewpoint: follow the probability distribution ρ_t driven by the gradient flow of F .

These descriptions are mathematically consistent and related by the mapping from SDEs to their Fokker–Planck equations.

9.3.2 Hierarchy of descriptions

The separation between state-level and distribution-level descriptions suggests a hierarchy:

1. At the base level, the potential W and the geometry of S shape the deterministic gradient flow and its stochastic perturbations.
2. At the next level, the free energy functional F and the Wasserstein geometry of $\mathcal{P}_2(S)$ organise the dynamics of probability measures and provide global Lyapunov and variational structures.

The present paper is concerned with the second level of this hierarchy. A more comprehensive framework would also study how these levels interact, for instance how properties of W (such as convexity or multi-well structure) are reflected in the geometry of F and in the long-time behaviour of the distribution-level dynamics.

9.4 Scope and limitations within the programme

The contributions of this paper should be understood within a deliberately restricted scope.

9.4.1 Mathematical focus

The focus has been on mathematically well-defined structures:

- free energy functionals combining potential energy and entropy;
- Wasserstein gradient flows on spaces of probability measures (in finite dimensions);
- Fokker–Planck equations as realisations of these gradient flows;
- existence and characterisation of equilibrium measures via variational principles.

All arguments have been kept at the level of probability measures, potentials, and differential equations, without invoking additional interpretive layers.

9.4.2 No derivation of full physical theories

Although the terminology of free energy and gradients is reminiscent of physical theories, the present work does not attempt to derive complete physical models, nor to address empirical validation. The role of physical analogies has been purely structural: they motivate definitions and connect them to classical objects such as Helmholtz free energy, Gibbs measures, and Ornstein–Uhlenbeck processes, but they do not carry interpretive weight beyond the mathematical statements made explicitly.

9.4.3 Directions for further integration

From the viewpoint of the broader “semantic field” framework, natural directions for further work include:

- extending the rigorous gradient-flow theory to genuinely infinite-dimensional Hilbert spaces, where the geometry of $\mathcal{P}(S)$ becomes more delicate;
- analysing specific classes of potentials W and reference measures μ arising in applications, and their impact on the structure of F and on convergence to equilibrium;
- exploring how additional structure (e.g. symmetries, conservation laws, or constraints) at the state level translates into constrained or projected free energy dynamics at the distribution level.

Such developments would further clarify how state-level and distribution-level descriptions interact within the overall programme.

In summary, this paper provides the distribution-level component of a two-level framework based on potentials and gradient flows: it defines a free energy functional on probability measures over a Hilbert space and develops its associated gradient-flow and Fokker–Planck dynamics. The broader “semantic field” picture may include additional interpretive or application-oriented layers, but these lie beyond the mathematical scope of the present work.

10 Summary and outlook

This paper has developed a mathematical framework for a free energy functional defined on probability measures over a separable Hilbert space and studied the associated gradient-flow dynamics, with a particular emphasis on finite-dimensional realisations. In this final section we summarise the main constructions and results, discuss limitations of the present treatment, and indicate several directions for further work.

10.1 Summary of main constructions and results

We briefly review the core ingredients introduced in the preceding sections.

10.1.1 State space and probability measures

The starting point is a state space S modelled as a separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$. The measurable structure is given by the Borel σ -algebra $\mathcal{B}(S)$, and we consider Borel probability measures on S , with particular attention to the subset $\mathcal{P}_2(S)$ of measures with finite second moment. A reference measure μ on $(S, \mathcal{B}(S))$ is fixed, assumed σ -finite and of full support. Densities ρ are then understood as Radon–Nikodým

derivatives with respect to μ , and the class of admissible densities is defined via suitable integrability conditions for energy and entropy.

10.1.2 Free energy functional

Given a Borel measurable potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ and a parameter $D > 0$, the energy and entropy functionals are defined by

$$\mathcal{E}[\rho] = \int_S W(s) \rho(s) d\mu(s), \quad \mathcal{H}[\rho] = \int_S \rho(s) \ln \rho(s) d\mu(s),$$

with appropriate conventions at points where $\rho = 0$ and under assumptions ensuring these integrals are well-defined (possibly taking values in $(-\infty, +\infty]$). The free energy functional is then

$$F[\rho] = \mathcal{E}[\rho] + D \mathcal{H}[\rho].$$

We studied basic properties of F :

- Under convexity assumptions on W , the functionals $\mathcal{E}$ and $\mathcal{H}$ are convex in ρ , so F is convex on its domain.
- Under lower semicontinuity and growth conditions on W , and moment bounds on ρ , F is lower semicontinuous with respect to suitable topologies (e.g. L^1 convergence associated with tightness conditions).
- Coercivity of W ensures the existence of minimisers of F in the admissible domain, via standard direct methods in the calculus of variations.

We computed the formal first variation of F and obtained the functional derivative

$$\frac{\delta F}{\delta \rho}(s) = W(s) + D(1 + \ln \rho(s)),$$

for sufficiently regular strictly positive densities ρ . Critical points of F subject to the mass constraint $\int_S \rho d\mu = 1$ satisfy a Euler–Lagrange equation whose solutions are Gibbs-type densities of the form

$$\rho_*(s) = \frac{1}{Z} \exp\left(-\frac{W(s)}{D}\right), \quad Z = \int_S \exp\left(-\frac{W(s)}{D}\right) d\mu(s),$$

whenever $Z < \infty$. Under suitable convexity assumptions on W , such densities are unique minimisers of F .

10.1.3 Wasserstein gradient flows and Fokker–Planck equations

In finite-dimensional settings $S = \mathbb{R}^n$ with Lebesgue reference measure, the space $\mathcal{P}_2(\mathbb{R}^n)$ carries the 2-Wasserstein metric W_2 . Using the dynamic Benamou–Brenier formulation of W_2 and the induced formal Riemannian structure, we expressed tangent vectors at a density ρ via velocity fields v solving the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0.$$

Within this setting, the gradient of F at ρ is represented by the vector field $\nabla(\delta F / \delta \rho)$, and the Wasserstein gradient flow of F is described formally by

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad v_t = -\nabla \left(\frac{\delta F}{\delta \rho_t} \right).$$

Substituting the explicit expression for $\delta F/\delta\rho$ yields the evolution equation

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla W) + D \Delta \rho_t,$$

which coincides with the Fokker–Planck equation associated with the overdamped Langevin SDE

$$dX_t = -\nabla W(X_t) dt + \sqrt{2D} dB_t.$$

Along sufficiently regular solutions of this equation, the free energy satisfies the dissipation identity

$$\frac{d}{dt} F[\rho_t] = - \int_{\mathbb{R}^n} \left\| \nabla \left(\frac{\delta F}{\delta \rho_t}(x) \right) \right\|^2 \rho_t(x) dx \leq 0,$$

so F is a Lyapunov functional for the dynamics. Stationary densities are characterised as minimisers of F and take the Gibbs form given above.

10.1.4 Connections to classical free energy and examples

We related F to the classical Helmholtz free energy of equilibrium statistical mechanics by identifying

$$W \leftrightarrow V, \quad D \leftrightarrow \frac{1}{\beta},$$

where V is a potential energy function and β an inverse temperature. In particular, the minimal value of F is linked to the partition function via

$$\inf_{\rho} F[\rho] = -D \ln Z.$$

Moreover, F can be re-expressed as a shifted Kullback–Leibler divergence relative to the Gibbs minimiser ρ_* ,

$$F[\rho] - F[\rho_*] = D D_{\text{KL}}(\rho \| \rho_*),$$

which connects the free energy landscape to information-theoretic measures of deviation from equilibrium.

Concrete examples included a one-dimensional double-well potential, illustrating low- and high- D regimes and metastable behaviour, and quadratic potentials leading to Gaussian stationary measures and Ornstein–Uhlenbeck dynamics. In the quadratic case, the stationary density is Gaussian with covariance proportional to D , and the free energy can be computed explicitly on the Gaussian family.

10.1.5 Extensions and multi-level positioning

We sketched extensions to time-dependent potentials, multi-component systems with joint and conditional free energies, and constrained evolutions interpreted as projected gradient flows. Finally, we positioned the present distribution-level framework relative to a state-level potential description, emphasising the systematic passage from state-space dynamics driven by W to measure-space dynamics driven by F .

10.2 Limitations of the present framework

The analysis presented in this paper is subject to several important limitations.

10.2.1 Finite-dimensional focus

Although the state space S has been introduced as a separable Hilbert space, most rigorous constructions and results have been developed explicitly for finite-dimensional settings $S = \mathbb{R}^n$. In this regime, the 2-Wasserstein metric is well understood, the associated Riemannian formalism is standard, and the Fokker–Planck equation has a well-developed theory. By contrast, the geometry of $\mathcal{P}_2(S)$ over genuinely infinite-dimensional Hilbert spaces is more subtle, and a complete treatment would require additional assumptions on W , μ , and the class of admissible measures, as well as refined tools from infinite-dimensional analysis.

10.2.2 Regularity and integrability assumptions

Many derivations in the paper, especially those involving functional derivatives, integration by parts, and dissipation identities, have been carried out at a formal level under assumptions of smoothness and decay that may not hold in all applications. A fully rigorous analysis would require:

- precise function-space settings for ρ_t (e.g. Sobolev or weighted spaces);
- conditions on W ensuring existence, uniqueness, and regularity of solutions to the Fokker–Planck equation;
- careful justification of differentiation under the integral sign and of boundary term neglect in integration by parts.

These issues are well studied for specific classes of potentials and reference measures, but a general theory covering all cases considered here lies beyond the scope of this work.

10.2.3 Absence of empirical or application-specific analysis

The framework has been developed in an abstract mathematical setting, without reference to specific empirical systems or datasets. While the terminology and constructions are inspired by statistical mechanics and stochastic dynamics, no attempt has been made to calibrate the model to physical systems, to test empirical predictions, or to address modelling choices beyond structural plausibility. Such questions are important for applications but require additional domain-specific information.

10.3 Directions for future research

Despite these limitations, the framework suggests several promising directions for further investigation.

10.3.1 Infinite-dimensional gradient flows

A natural direction is to extend the Wasserstein gradient-flow theory of F to genuinely infinite-dimensional Hilbert spaces. Potential research problems include:

- characterising conditions on W and μ under which F admits minimisers and generates well-posed gradient flows on $\mathcal{P}(S)$ for infinite-dimensional S ;
- understanding the geometry of Wasserstein-type distances on probability measures over Hilbert spaces and their relation to underlying linear structure;

- analysing long-time behaviour, convergence to equilibrium, and stability of stationary measures in infinite dimensions.

10.3.2 More general stochastic dynamics

Another direction is to investigate extensions of the framework to more general stochastic dynamics, such as:

- underdamped Langevin dynamics and kinetic Fokker–Planck equations, which involve both position and momentum variables;
- systems with multiplicative noise, non-gradient drifts, or non-reversible dynamics, where the gradient-flow structure may involve modified metrics or additional antisymmetric components;
- interacting particle systems and mean-field limits, where free energy functionals with interaction terms naturally arise as large-deviation rate functionals.

In such settings, the free energy F may need to be generalised to include additional terms, and the metric structure on the space of measures may differ from the standard 2-Wasserstein metric.

10.3.3 Constraints, symmetries, and structure-preserving flows

The constrained and multi-component extensions outlined in Section 8 suggest further work on:

- systematically classifying constraint sets $\mathcal{C}$ (e.g. moment constraints, symmetry constraints, or support constraints) for which projected Wasserstein gradient flows are well defined and exhibit desirable properties;
- designing and analysing structure-preserving numerical schemes based on minimizing-movement formulations that respect constraints and symmetries;
- studying free energy decompositions (e.g. into marginal, conditional, and interaction contributions) and their role in hierarchical or modular descriptions of complex systems.

10.3.4 Applications and model-based exploration

Finally, the formalism lends itself to application-oriented studies in which:

- a concrete state space S , potential W , and reference measure μ are specified based on modelling considerations;
- the associated free energy F and gradient-flow dynamics are analysed numerically or analytically;
- the resulting stationary and transient behaviours are compared with empirical data or qualitative expectations.

Such studies could clarify which aspects of the framework are robust across domains and which require adaptation or extension.

10.4 Concluding remarks

We have introduced and analysed a free energy functional on probability measures over a Hilbert space, established its basic variational properties, and showed how, in finite dimensions, its Wasserstein gradient flow coincides with the Fokker–Planck equation associated with Langevin dynamics. The free energy F provides a unifying structure that links potentials on state spaces, Gibbs measures, relative entropy, and diffusion equations within a single geometric and variational framework. While the present treatment has focused on core mathematical aspects, it lays a foundation for further developments in infinite-dimensional analysis, stochastic dynamics, constrained and multi-component systems, and application-driven modelling within this general setting.

Acknowledgements and Contact

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The work emerged from a prolonged process of intuitive exploration, conceptual reduction, and formal consolidation. Informal insights about meaning, structure, and stability were progressively translated into explicit models, mathematical frameworks, and testable claims. Throughout this process, modern large language models were employed as practical instruments for structuring arguments, checking internal consistency, exploring alternative formulations, and iteratively refining formal expressions.

These tools were used as epistemic scaffolding rather than as sources of claims. All theoretical commitments, modelling decisions, and interpretative judgments remain the responsibility of the author. The resulting framework should therefore be understood as exploratory but precise: it proposes a concrete formal scaffold, derives explicit consequences, and invites scrutiny.

The work is explicitly offered for critical examination, clarification, extension, or refutation by domain experts. Questions, comments, corrections, and collaboration inquiries are welcome and may be directed to contact@semanticgravitation.org.

Semantic Gravitation III: Quantum-Compatible Semantic Fields — A Hilbert-Space Embedding of Quantum States, Observables, and Dynamics

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Abstract

This paper develops the third technical component of the Semantic Gravitation programme by embedding standard quantum mechanics into the semantic Hilbert-space framework established in Papers I and II. Starting from a separable quantum Hilbert space H_Q , we view density operators $\mathcal{D}(H_Q)$ as elements of the Hilbert–Schmidt space $\text{HS}(H_Q)$ of operators on H_Q . The closed linear span of $\mathcal{D}(H_Q)$ defines an operator Hilbert space $S_Q \subset \text{HS}(H_Q)$, which carries the canonical Hilbert–Schmidt inner product.

Because the global semantic state space S is assumed to be a separable Hilbert space, there exists a closed subspace $\tilde{S}_Q \subset S$ and a unitary map

$$U_Q: \tilde{S}_Q \longrightarrow S_Q$$

that implements an isometric identification of $\tilde{S}_Q$ with S_Q . Composing U_Q^{-1} with the inclusion of $\mathcal{D}(H_Q)$ into S_Q yields a linear embedding

$$E_Q: \mathcal{D}(H_Q) \longrightarrow \tilde{S}_Q \subset S$$

which realises quantum states as a distinguished, *quantum-compatible* sector of the semantic Hilbert space. We show that the restriction of the ambient scalar product on S reproduces the Hilbert–Schmidt geometry on S_Q .

Observables are treated in a dual fashion. For each Hilbert–Schmidt observable $A \in \text{HS}(H_Q)$, we construct a continuous linear functional L_A on S and an associated observable vector $v_A \in \tilde{S}_Q$ such that, for all density operators $\omega \in \mathcal{D}(H_Q)$,

$$L_A(E_Q(\omega)) = \langle v_A, E_Q(\omega) \rangle_S = \text{tr}(\omega A),$$

so that standard quantum expectation values are exactly reproduced inside the semantic picture. In finite dimensions, all observables are covered; in infinite dimensions the construction naturally restricts to Hilbert–Schmidt observables.

We then transfer dynamics. Von Neumann evolution

$$\dot{\omega}_t = -i[H, \omega_t]$$

and more general quantum Markov semigroups generated by GKLS operators on $\mathcal{D}(H_Q)$ are mapped, via E_Q , to unitary or contractive evolutions on $\tilde{S}_Q \subset S$ that preserve all quantum expectations. We explicitly work out these induced generators and their action on embedded states, and illustrate the construction by detailed examples: a spin- $\frac{1}{2}$

system and a truncated harmonic oscillator. Within $\tilde{S}_Q$, the ambient semantic potential $W: S \rightarrow \mathbb{R}$ restricts to a unitary-invariant function, providing only programmatic links to the free-energy and gradient-flow structures on probability measures developed in Paper II.

Conceptually, this paper shows that standard quantum theory can be realised as a structurally faithful sector of the semantic state space, without modifying its axioms or dynamics. It clarifies how purely Hamiltonian quantum evolution coexists with, and can potentially be coupled to, dissipative semantic dynamics at the level of S and of probability measures on S . This provides the quantum-compatible layer that will feed into the spacetime semantic field theories of Paper IV and the complex-system applications of Paper V. For a programme-level introduction to Semantic Gravitation and the Lagrangian of Vectorial Emergence (LOVE), we refer to the separate overview paper.

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1 Introduction

1.1 Background and motivation

Quantum theory is formulated on complex Hilbert spaces: pure states are rays in a Hilbert space H_Q , mixed states are density operators ρ on H_Q , and physical quantities are represented by self-adjoint operators. Expectation values and time evolution are expressed entirely in this operator-theoretic language. This structural clarity has made the Hilbert-space formalism a central reference point not only in quantum physics but also in adjacent areas such as quantum information theory and quantum statistical mechanics.

In parallel, recent work has introduced a different kind of Hilbert space S , which we shall refer to as a *semantic Hilbert space*, together with a potential functional

$$W : S \rightarrow \mathbb{R} \cup \{+\infty\},$$

and associated gradient and stochastic dynamics on S and on probability measures over S .¹ In that framework, W plays the role of an energy functional (often interpreted as a “contradiction energy”), generating gradient flows, Langevin stochastic differential equations, and Fokker–Planck equations on S and on probability measures over S .

The present work asks a structural question at the interface of these two settings:

Can the standard quantum formalism, in particular density operators and their expectations and dynamics, be embedded into the semantic Hilbert space framework in a mathematically compatible way?

More concretely, we are interested in constructing an explicit embedding of the space of quantum states (density operators on a quantum Hilbert space H_Q) into S , and in representing quantum observables and their expectation values, as well as quantum time evolution, within the semantic framework.

The key focus of this paper is therefore not to modify or replace quantum mechanics, but to identify a faithful representation of quantum states and observables inside the semantic Hilbert space S as defined in previous work. The resulting structure should reproduce the standard quantum predictions for expectation values and unitary time evolution when restricted to the embedded quantum sector.

1.2 Semantic Hilbert spaces and energetic structures

The semantic Hilbert space S comes equipped with a number of structures that will be relevant in this paper. At the level of single points $x \in S$, the potential functional $W(x)$ induces deterministic and stochastic dynamics of the form

$$\dot{x}_t = -\nabla W(x_t)$$

and

$$dx_t = -\nabla W(x_t) dt + \sqrt{2\beta^{-1}} dB_t,$$

where B_t denotes a cylindrical Brownian motion on S and $\beta > 0$ plays the role of an inverse temperature. These dynamics generate associated Fokker–Planck equations for probability measures on S .

¹Throughout the paper we treat S as a separable complex Hilbert space, in line with the constructions in the companion works that develop semantic gradient flows, Langevin-type stochastic dynamics, and free-energy functionals on S .

In the companion works, a *semantic free energy* functional $\mathcal{F}$ is defined on probability measures μ on S as a sum of an internal energy contribution (given by the integral of W with respect to μ) and an entropy or divergence term. The gradient flow of $\mathcal{F}$ with respect to a Wasserstein-type metric on the space of probability measures over S yields the Fokker–Planck dynamics associated with the Langevin equation on S .

These constructions provide a general energetic and dynamical framework on S that is in spirit analogous to classical statistical mechanics and optimal transport, but they do not by themselves single out a specific quantum sector. Our goal in this paper is to identify and analyze such a sector, and to show that within it, the standard quantum formalism can be recovered in a structurally compatible manner.

1.3 Aim and scope of the present work

The aim of this paper is to exhibit a mathematically precise embedding of standard quantum states and observables into the semantic Hilbert space S and to characterize the induced dynamics on the embedded quantum sector. The central objects of interest are:

- A separable complex Hilbert space H_Q representing the quantum system under consideration.
- The space $\mathcal{D}(H_Q)$ of density operators on H_Q , representing quantum states (both pure and mixed).
- An embedding

$$E_Q : \mathcal{D}(H_Q) \rightarrow S,$$

whose image $S_Q := E_Q(\mathcal{D}(H_Q))$ is a distinguished subspace or submanifold of S .

- For each bounded observable A on H_Q , a linear functional

$$L_A : S_Q \rightarrow \mathbb{C}$$

such that

$$L_A(E_Q(\rho)) = \text{tr}(\rho A)$$

for all $\rho \in \mathcal{D}(H_Q)$.

- An induced dynamics on S_Q corresponding to the quantum von Neumann evolution of density operators, and its relation to the more general semantic dynamics generated by W and by semantic free energy.

The guiding requirements for this construction are:

1. **Faithfulness:** The embedding E_Q should be injective, so that distinct quantum states are represented by distinct points in S .
2. **Compatibility of expectations:** For each observable A on H_Q , the expectation value in state ρ should be recovered by applying the corresponding functional L_A to the embedded state $E_Q(\rho)$.
3. **Compatibility of dynamics:** The time evolution of quantum states under the von Neumann equation should correspond, via E_Q , to a well-defined induced dynamics on S_Q that can be represented by a suitable linear (or unitary-like) operator family on S_Q .

The scope of this work is deliberately limited to such structural and compatibility questions. We do not attempt to derive quantum mechanics from the semantic framework, nor do we claim that S provides a more fundamental description of physical reality. Instead, we treat quantum mechanics as given and explore how it can be realized as a distinguished sector within the existing semantic Hilbert space setting.

1.4 Non-claims and limitations

For clarity, we emphasize several points that delimit what this paper does *not* claim:

- We do not derive the axioms of quantum mechanics from the properties of S , W , or semantic free energy. Rather, we assume the standard quantum formalism and show how it can be embedded into S .
- We do not propose a physical interpretation of the semantic Hilbert space S beyond its mathematical structure. In particular, we do not discuss questions of consciousness, interpretation of measurement, or broader philosophical issues.
- We do not assert that the embedding constructed here is unique, or that it is the only mathematically or physically meaningful way to relate quantum mechanics to the semantic framework.
- We do not attempt a complete measure-theoretic or operator-algebraic treatment of all infinite-dimensional subtleties; when necessary, we restrict to suitable dense subspaces or finite-dimensional truncations to keep the analysis technically tractable.

Within these limitations, the purpose of the paper is to give a clear and rigorous account of how quantum states, observables, and their dynamics can be realized inside the semantic Hilbert space S , and how this quantum sector relates to the broader energetic and dynamical structures already available on S .

1.5 Overview of the paper

The remainder of the paper is organized as follows.

- In Chapter 2, we recall the standard quantum-mechanical formalism of states and observables. We introduce the quantum Hilbert space H_Q , the space $\mathcal{D}(H_Q)$ of density operators, and the representation of observables as bounded self-adjoint operators. We also recall the von Neumann equation governing the time evolution of density operators.
- In Chapter 3, we define the embedded quantum subspace $S_Q \subset S$ and construct an explicit embedding

$$E_Q : \mathcal{D}(H_Q) \rightarrow S.$$

We analyze basic properties of this embedding, in particular injectivity and, where possible, isometric properties with respect to a chosen inner product or kernel on S . We also briefly discuss how the potential W restricts to S_Q .

- In Chapter 4, we show how quantum observables can be represented as linear functionals on S_Q . For each observable A on H_Q we define a functional L_A on S_Q such that $L_A(E_Q(\rho)) = \text{tr}(\rho A)$. Using the Riesz representation theorem on S , we express L_A as an inner product with a vector $v_A \in S$ and study its linearity and positivity properties.

- In Chapter 5, we study dynamics. Starting from the von Neumann equation for ρ_t , we derive the induced evolution of the embedded states $x_t = E_Q(\rho_t)$ on S_Q and express it in terms of a linear operator family $(\mathcal{U}_t)_{t \in \mathbb{R}}$ acting on S_Q . We show that this induced dynamics is compatible with the quantum expectations computed via the functionals L_A , and we relate the resulting unitary sector to the more general gradient and stochastic dynamics on S at a programmatic level.
- In Chapter 6, we illustrate the construction with explicit examples. We treat in detail a two-dimensional spin- $\frac{1}{2}$ system, giving closed-form expressions for E_Q , the observables L_A , and the induced dynamics. We also sketch a harmonic oscillator example, at least in a finite-dimensional truncation of H_Q , to demonstrate how the embedding behaves in an infinite-dimensional setting.
- In Chapter 7, we summarize the main structural results and their limitations. We discuss the non-uniqueness of embeddings, the challenges of full infinite-dimensional rigor, and the absence of a derivation of quantum mechanics from the semantic framework. We also outline directions for future work, including the coupling between the embedded quantum sector and the semantic gradient-flow dynamics, potential connections to quantum free energy, and relations to information geometry on spaces of quantum states.

This organization is intended to keep the exposition self-contained: a reader familiar with standard quantum mechanics and basic functional analysis should be able to follow the construction of the embedding, the representation of observables, and the induced dynamics on S without prior exposure to the semantic framework beyond the basic structures summarized above.

2 Quantum states and observables: standard setup

2.1 Quantum Hilbert space and pure states

In this chapter we recall the standard Hilbert-space formulation of quantum theory for a single system. This will fix notation and conventions for the subsequent construction of an embedding into the semantic Hilbert space S .

Definition 2.1 (Quantum Hilbert space). Let H_Q be a separable complex Hilbert space with inner product

$$\langle \cdot, \cdot \rangle_{H_Q} : H_Q \times H_Q \rightarrow \mathbb{C},$$

which is linear in the second argument and conjugate-linear in the first. The norm induced by the inner product is denoted by

$$\|\psi\|_{H_Q} := \sqrt{\langle \psi, \psi \rangle_{H_Q}}.$$

Definition 2.2 (Pure states). A *pure state* of the quantum system is a one-dimensional subspace (ray) of H_Q . Equivalently, it can be represented by any unit vector $\psi \in H_Q$ with $\|\psi\|_{H_Q} = 1$, where two vectors differing by a global phase, ψ and $e^{i\theta}\psi$, represent the same physical state.

For many purposes it is convenient to work with a normalized vector representative ψ of a pure state, while remembering that global phases are physically irrelevant. In finite dimensions, $H_Q \cong \mathbb{C}^d$, and pure states correspond to normalized vectors in $\mathbb{C}^d$ modulo complex phases.

2.2 Mixed states and density operators

In realistic scenarios, a system may not be in a pure state but in a statistical ensemble of pure states. The appropriate mathematical object is then a density operator.

2.2.1 Trace-class operators and the trace

Let $B(H_Q)$ denote the space of bounded linear operators $A : H_Q \rightarrow H_Q$. A bounded operator $T \in B(H_Q)$ is called *trace-class* if, for some (and hence any) orthonormal basis $(e_n)_{n \in \mathbb{N}}$ of H_Q , the series

$$\sum_{n=1}^{\infty} \langle e_n, |T| e_n \rangle_{H_Q}$$

is finite, where $|T| := \sqrt{T^* T}$ is the absolute value of T . In that case, the *trace* of T is defined as

$$\text{tr}(T) := \sum_{n=1}^{\infty} \langle e_n, T e_n \rangle_{H_Q},$$

and this definition is independent of the choice of orthonormal basis.

The space of trace-class operators on H_Q will be denoted by $\mathcal{T}_1(H_Q)$; it is a Banach space with the trace norm

$$\|T\|_1 := \text{tr}(|T|).$$

2.2.2 Density operators and the state space

Definition 2.3 (Density operators). A *density operator* on H_Q is a trace-class operator $\rho \in \mathcal{T}_1(H_Q)$ that satisfies:

1. *Positivity*: $\rho \geq 0$ in the sense that $\langle \psi, \rho \psi \rangle_{H_Q} \geq 0$ for all $\psi \in H_Q$.
2. *Unit trace*: $\text{tr}(\rho) = 1$.

The set of all density operators on H_Q is denoted by

$$\mathcal{D}(H_Q) := \{\rho \in \mathcal{T}_1(H_Q) : \rho \geq 0, \text{tr}(\rho) = 1\}.$$

Remark 2.4 (Convex structure and pure vs. mixed states). The set $\mathcal{D}(H_Q)$ is convex: if $\rho_1, \rho_2 \in \mathcal{D}(H_Q)$ and $\lambda \in [0, 1]$, then $\lambda \rho_1 + (1 - \lambda) \rho_2 \in \mathcal{D}(H_Q)$. The *extreme points* of $\mathcal{D}(H_Q)$ are precisely the rank-one orthogonal projections

$$\rho = |\psi\rangle\langle\psi|, \quad \|\psi\|_{H_Q} = 1,$$

which correspond to pure states. States that are not extreme points are called *mixed states*.

Proposition 2.5 (Spectral decomposition). *Let $\rho \in \mathcal{D}(H_Q)$. Then there exists an orthonormal family $(\phi_n)_{n \in I}$ in H_Q and a family of non-negative real numbers $(p_n)_{n \in I}$ with $\sum_{n \in I} p_n = 1$ such that*

$$\rho = \sum_{n \in I} p_n |\phi_n\rangle\langle\phi_n|$$

with convergence in trace norm. In finite dimensions this is a finite sum over the eigenvectors of ρ .

The numbers p_n can be interpreted as probabilities of finding the system in the pure states $|\phi_n\rangle\langle\phi_n|$. However, the representation of a given ρ as a convex combination of pure states is highly non-unique, reflecting the fact that mixed states encode less than complete information about the underlying ensemble.

2.3 Observables and expectation values

2.3.1 Observables as self-adjoint operators

In the standard formalism, physical quantities are represented by self-adjoint operators on H_Q .

Definition 2.6 (Observables). An *observable* is a densely defined self-adjoint operator $A : \mathcal{D}(A) \subset H_Q \rightarrow H_Q$. In this paper we will mostly restrict attention to *bounded* observables, i.e. $A \in B(H_Q)$ with $A = A^*$, in order to avoid domain issues. The space of bounded self-adjoint operators on H_Q is denoted by

$$\mathcal{B}_{\text{sa}}(H_Q) := \{A \in B(H_Q) : A = A^*\}.$$

If A is self-adjoint, the spectral theorem provides a projection-valued measure P_A on the Borel sets of $\mathbb{R}$ such that

$$A = \int_{\mathbb{R}} \lambda dP_A(\lambda).$$

This yields the usual probabilistic interpretation of measurements of A in a given state, but we will not need the full spectral calculus in what follows.

2.3.2 Expectation values for pure and mixed states

Definition 2.7 (Expectation values). Let $A \in \mathcal{B}_{\text{sa}}(H_Q)$ be a bounded self-adjoint operator.

1. For a pure state represented by a unit vector $\psi \in H_Q$, the *expectation value* of A in state ψ is

$$\langle A \rangle_{\psi} := \langle \psi, A\psi \rangle_{H_Q}.$$

2. For a mixed state represented by a density operator $\rho \in \mathcal{D}(H_Q)$, the *expectation value* of A is

$$\langle A \rangle_{\rho} := \text{tr}(\rho A).$$

Proposition 2.8 (Compatibility of pure-state and density-matrix formalisms). *If $\rho = |\psi\rangle\langle\psi|$ is a pure-state density operator with $\|\psi\|_{H_Q} = 1$, then for any bounded self-adjoint operator A one has*

$$\text{tr}(\rho A) = \langle \psi, A\psi \rangle_{H_Q}.$$

Proof. Let $(e_n)_{n \in \mathbb{N}}$ be an orthonormal basis of H_Q . Then

$$\text{tr}(\rho A) = \sum_{n=1}^{\infty} \langle e_n, \rho A e_n \rangle_{H_Q} = \sum_{n=1}^{\infty} \langle e_n, \psi \rangle_{H_Q} \langle \psi, A e_n \rangle_{H_Q}.$$

By completeness of the basis we can write $A\psi = \sum_n \langle e_n, A\psi \rangle_{H_Q} e_n$, and a straightforward rearrangement of the series yields

$$\text{tr}(\rho A) = \langle \psi, A\psi \rangle_{H_Q}.$$

□

Remark 2.9 (Born rule and positivity). If $0 \leq A \leq \mathbb{I}$ is a positive operator bounded by the identity, it can be interpreted as the effect corresponding to a yes/no measurement event. For any density operator $\rho \in \mathcal{D}(H_Q)$, the number $\text{tr}(\rho A)$ lies in $[0, 1]$ and can be interpreted as the probability of obtaining the outcome “yes”. In particular, if P is an orthogonal projection, then $\text{tr}(\rho P)$ is the probability that a projective measurement corresponding to P yields the value 1.

2.4 Time evolution and dynamics

We now recall the description of time evolution in the Schrödinger and Heisenberg pictures. In this paper we shall mostly use the Schrödinger picture for states, in which density operators carry the time dependence.

2.4.1 Unitary dynamics for pure states

Let H be a self-adjoint operator on H_Q , called the Hamiltonian of the system. The Schrödinger equation for a time-dependent state vector $\psi_t \in H_Q$ reads

$$i\hbar \frac{d}{dt} \psi_t = H \psi_t, \quad (2.1)$$

where $\hbar > 0$ is the reduced Planck constant. Under suitable conditions on H , the Stone theorem on one-parameter unitary groups guarantees the existence of a strongly continuous one-parameter family of unitary operators $(U_t)_{t \in \mathbb{R}}$ on H_Q such that

$$U_0 = \mathbb{I}, \quad U_{t+s} = U_t U_s \quad \text{for all } s, t \in \mathbb{R},$$

and

$$i\hbar \frac{d}{dt} U_t \psi = H U_t \psi$$

for all ψ in the domain of H . The solution of (2.1) can then be written as

$$\psi_t = U_t \psi_0, \quad \text{with } U_t = e^{-itH/\hbar}$$

in the sense of the functional calculus for self-adjoint operators.

2.4.2 Von Neumann equation for density operators

In the density-operator formalism, time evolution is encoded in a family $(\rho_t)_{t \in \mathbb{R}}$ of density operators. Given a Hamiltonian H , the unitary evolution $U_t = e^{-itH/\hbar}$ induces a dynamics

$$\rho_t = U_t \rho_0 U_t^*,$$

which preserves positivity, trace, and the full spectrum of ρ_0 . Differentiating formally with respect to t yields the *von Neumann equation* (or Liouville–von Neumann equation)

$$i\hbar \frac{d}{dt} \rho_t = [H, \rho_t] := H \rho_t - \rho_t H. \quad (2.2)$$

Proposition 2.10 (Conservation of expectations under unitary evolution). *Let $A \in \mathcal{B}_{\text{sa}}(H_Q)$ be a bounded observable that does not depend explicitly on time. Let (ρ_t) evolve according to (2.2). Then the expectation value $\langle A \rangle_{\rho_t} = \text{tr}(\rho_t A)$ satisfies*

$$\frac{d}{dt} \langle A \rangle_{\rho_t} = \frac{1}{i\hbar} \text{tr}(\rho_t [A, H]).$$

In particular, if $[A, H] = 0$ then $\langle A \rangle_{\rho_t}$ is constant in time.

Proof. Using $\rho_t = U_t \rho_0 U_t^*$ and the cyclicity of the trace, we obtain

$$\langle A \rangle_{\rho_t} = \text{tr}(U_t \rho_0 U_t^* A) = \text{tr}(\rho_0 U_t^* A U_t).$$

Differentiating with respect to t and using the Schrödinger equation for U_t gives the Heisenberg equation of motion for $A_t := U_t^* A U_t$. Translating back to the Schrödinger picture and using $\text{tr}(\dot{\rho}_t A) = \frac{1}{i\hbar} \text{tr}(\rho_t [A, H])$ yields the stated formula. $\square$

2.4.3 Open-system dynamics and quantum channels

In many situations, the system of interest is coupled to an environment, and its reduced dynamics is no longer unitary. A convenient abstract description is in terms of completely positive trace-preserving (CPTP) maps.

Definition 2.11 (Quantum channels). A *quantum channel* on H_Q is a linear map

$$\Phi : \mathcal{T}_1(H_Q) \rightarrow \mathcal{T}_1(H_Q)$$

that is completely positive and trace-preserving:

1. For every $n \in \mathbb{N}$, the map $\Phi \otimes \text{id}_n$ sends positive operators on $\mathcal{T}_1(H_Q) \otimes M_n(\mathbb{C})$ to positive operators.
2. For all $\rho \in \mathcal{T}_1(H_Q)$, one has $\text{tr}(\Phi(\rho)) = \text{tr}(\rho)$.

If $\Phi(\mathcal{D}(H_Q)) \subseteq \mathcal{D}(H_Q)$, the channel induces a well-defined evolution of density operators.

In a Markovian setting with time-homogeneous dynamics, one often considers a one-parameter semigroup $(\Phi_t)_{t \geq 0}$ of quantum channels with $\Phi_0 = \text{id}$ and $\Phi_{t+s} = \Phi_t \circ \Phi_s$. Under suitable regularity conditions, the generator of such a semigroup admits the Gorini–Kossakowski–Lindblad–Sudarshan (GKLS) form

$$\frac{d}{dt}\rho_t = -\frac{i}{\hbar}[H, \rho_t] + \sum_k \left(L_k \rho_t L_k^* - \frac{1}{2}\{L_k^* L_k, \rho_t\} \right), \quad (2.3)$$

where H is self-adjoint, $(L_k)_k$ are bounded operators on H_Q , and $\{X, Y\} := XY + YX$ denotes the anticommutator. The first term coincides with the unitary von Neumann evolution, whereas the second term introduces dissipation and decoherence.

While our main focus in this paper will be on the unitary dynamics (2.2), the dissipative structure in (2.3) parallels some of the gradient-flow and free-energy structures available in the semantic framework. This will be relevant at a conceptual level when we discuss the relation between quantum and semantic dynamics.

2.5 Standing assumptions and notation

For the remainder of the paper, we adopt the following standing assumptions and notational conventions:

- H_Q denotes a fixed separable complex Hilbert space describing the quantum system.
- $\mathcal{D}(H_Q)$ is the set of density operators on H_Q as defined above.
- $B(H_Q)$ is the space of bounded operators on H_Q , and $\mathcal{B}_{\text{sa}}(H_Q)$ is the real vector space of bounded self-adjoint operators on H_Q .
- For $\rho \in \mathcal{D}(H_Q)$ and $A \in \mathcal{B}_{\text{sa}}(H_Q)$, the expectation value of A in state ρ is denoted by

$$\langle A \rangle_\rho := \text{tr}(\rho A).$$

- Time evolution of closed systems is described by a self-adjoint Hamiltonian H via the von Neumann equation (2.2). When we write ρ_t without further qualification, it is understood that ρ_t solves (2.2) with some initial condition $\rho_0 \in \mathcal{D}(H_Q)$.

These standard structures will serve as the input for the constructions in the following chapters. In particular, in Chapter 3 we will define an embedding

$$E_Q : \mathcal{D}(H_Q) \rightarrow S$$

from the quantum state space into the semantic Hilbert space S , and in Chapter 4 we will represent each observable $A \in \mathcal{B}_{\text{sa}}(H_Q)$ by a linear functional L_A on the embedded quantum subspace of S such that $L_A(E_Q(\rho))$ reproduces the expectation value $\text{tr}(\rho A)$.

3 Embedding quantum states into the semantic Hilbert space S

3.1 Semantic Hilbert space and operator-valued state geometry

In this chapter we construct an explicit embedding of quantum states—represented by density operators on the quantum Hilbert space H_Q —into the semantic Hilbert space S introduced in the companion works. Throughout we assume that S is a separable complex Hilbert space with inner product

$$\langle \cdot, \cdot \rangle_S : S \times S \rightarrow \mathbb{C},$$

linear in the second argument and conjugate-linear in the first.

On the quantum side, we recall that the space of density operators $\mathcal{D}(H_Q)$ is a convex subset of the trace-class operators $\mathcal{T}_1(H_Q)$, and that every $\rho \in \mathcal{D}(H_Q)$ is in fact Hilbert–Schmidt. This suggests that a natural intermediate Hilbert space through which to embed quantum states into S is the Hilbert–Schmidt space of operators on H_Q .

3.1.1 The Hilbert–Schmidt space

Definition 3.1 (Hilbert–Schmidt operators). The *Hilbert–Schmidt space* on H_Q is defined as

$$\mathcal{H}_{\text{HS}}(H_Q) := \{X \in B(H_Q) : \text{tr}(X^*X) < \infty\}.$$

Equipped with the inner product

$$\langle X, Y \rangle_{\text{HS}} := \text{tr}(X^*Y), \quad X, Y \in \mathcal{H}_{\text{HS}}(H_Q),$$

and the induced norm $\|X\|_{\text{HS}} := \sqrt{\text{tr}(X^*X)}$, the space $\mathcal{H}_{\text{HS}}(H_Q)$ is a separable Hilbert space.

Proposition 3.2 (Density operators are Hilbert–Schmidt). *Every density operator $\rho \in \mathcal{D}(H_Q)$ belongs to $\mathcal{H}_{\text{HS}}(H_Q)$ and satisfies*

$$\|\rho\|_{\text{HS}}^2 = \text{tr}(\rho^2) \leq 1,$$

with equality if and only if ρ is a pure state.

Proof. Let $\rho \in \mathcal{D}(H_Q)$, and let $\rho = \sum_n p_n |\phi_n\rangle\langle\phi_n|$ be its spectral decomposition, where $p_n \geq 0$, $\sum_n p_n = 1$, and $(\phi_n)_n$ is an orthonormal family. Then

$$\rho^2 = \sum_n p_n^2 |\phi_n\rangle\langle\phi_n|, \quad \text{tr}(\rho^2) = \sum_n p_n^2.$$

Since $0 \leq p_n \leq 1$ and $\sum_n p_n = 1$, we have $\sum_n p_n^2 \leq \sum_n p_n = 1$, with equality if and only if a single p_n equals 1 and all others are 0, i.e. precisely when ρ has rank one. Thus $\text{tr}(\rho^2) \leq 1$, and ρ is Hilbert–Schmidt with $\|\rho\|_{\text{HS}}^2 = \text{tr}(\rho^2)$. $\square$

Remark 3.3. By Proposition 3.2, the set $\mathcal{D}(H_Q)$ is contained in the closed unit ball of $\mathcal{H}_{\text{HS}}(H_Q)$. Pure states correspond to extremal points lying on the unit sphere, while mixed states lie strictly inside the unit ball (except in trivial finite-dimensional cases).

3.1.2 The linear span of density operators

We next identify the Hilbert subspace generated by density operators inside $\mathcal{H}_{\text{HS}}(H_Q)$.

Definition 3.4 (Quantum operator Hilbert space). Define

$$\mathcal{H}_Q := \overline{\text{span}}\{\rho : \rho \in \mathcal{D}(H_Q)\} \subset \mathcal{H}_{\text{HS}}(H_Q),$$

where the closure is taken with respect to the Hilbert–Schmidt norm.

Remark 3.5. In finite dimensions, $\mathcal{H}_Q$ coincides with the full space of complex $d \times d$ matrices (viewed as operators on $H_Q \cong \mathbb{C}^d$), since density matrices span the full matrix space. In infinite dimensions, $\mathcal{H}_Q$ is a separable Hilbert space that contains all finite-rank operators and is dense in $\mathcal{H}_{\text{HS}}(H_Q)$.

The space $\mathcal{H}_Q$ will serve as an intermediate Hilbert space between the quantum state space $\mathcal{D}(H_Q)$ and the semantic Hilbert space S . The embedding E_Q will be constructed by composing the inclusion $\mathcal{D}(H_Q) \subset \mathcal{H}_Q$ with an isometric embedding of $\mathcal{H}_Q$ into S .

3.2 Definition of the embedded quantum subspace S_Q

We now introduce the subspace of S that will contain the embedded quantum states.

Proposition 3.6 (Existence of a semantic quantum subspace). *Let S be a separable complex Hilbert space, and let $\mathcal{H}_Q$ be as above. Then there exist:*

- a closed subspace $S_Q \subset S$,
- a unitary operator

$$U : \mathcal{H}_Q \rightarrow S_Q,$$

such that U is an isometric isomorphism between $\mathcal{H}_Q$ and S_Q .

Proof. Both S and $\mathcal{H}_Q$ are separable complex Hilbert spaces. Any closed subspace of a separable Hilbert space is again separable, and any two separable infinite-dimensional complex Hilbert spaces are isometrically isomorphic. Thus there exists a linear isometry

$$U_0 : \mathcal{H}_Q \rightarrow S.$$

Define $S_Q := U_0(\mathcal{H}_Q)$ and let U be U_0 viewed as a map onto its range. Since U_0 is an isometry, its range is closed, so S_Q is a closed subspace of S , and $U : \mathcal{H}_Q \rightarrow S_Q$ is unitary. $\square$

Definition 3.7 (Embedded quantum subspace). A closed subspace $S_Q \subset S$ together with a fixed unitary

$$U : \mathcal{H}_Q \rightarrow S_Q$$

as in Proposition 3.6 is called an *embedded quantum subspace* of the semantic Hilbert space S .

Remark 3.8. The choice of U is not unique; any unitary operator $\tilde{U} = VU$ with V unitary on S_Q yields an equally valid embedded subspace. This non-uniqueness is analogous to the non-uniqueness of bases in a Hilbert space and does not affect the structural results that follow.

3.3 Construction of the embedding E_Q

We now define the embedding of quantum states into S .

3.3.1 Definition and basic properties

Definition 3.9 (Embedding of density operators). Let $S_Q \subset S$ and $U : \mathcal{H}_Q \rightarrow S_Q$ be as above. The *quantum embedding*

$$E_Q : \mathcal{D}(H_Q) \rightarrow S$$

is defined by

$$E_Q(\rho) := U(\rho), \quad \rho \in \mathcal{D}(H_Q) \subset \mathcal{H}_Q.$$

We will regard $E_Q(\rho)$ both as an element of S_Q and of S , using the inclusion $S_Q \subset S$.

Proposition 3.10 (Injectivity and continuity of E_Q). *The embedding E_Q satisfies:*

1. E_Q is injective: if $\rho_1, \rho_2 \in \mathcal{D}(H_Q)$ and $E_Q(\rho_1) = E_Q(\rho_2)$, then $\rho_1 = \rho_2$.
2. E_Q is continuous with respect to the Hilbert–Schmidt topology: if $\rho_n \rightarrow \rho$ in $\|\cdot\|_{\text{HS}}$, then $E_Q(\rho_n) \rightarrow E_Q(\rho)$ in $\|\cdot\|_S$.

Proof. (1) If $E_Q(\rho_1) = E_Q(\rho_2)$, then $U(\rho_1 - \rho_2) = 0$. Since U is unitary and hence injective, this implies $\rho_1 - \rho_2 = 0$, i.e. $\rho_1 = \rho_2$.

(2) Let $\rho_n \rightarrow \rho$ in $\mathcal{H}_Q$ with respect to $\|\cdot\|_{\text{HS}}$. Then, by unitarity of U ,

$$\|E_Q(\rho_n) - E_Q(\rho)\|_S = \|U(\rho_n - \rho)\|_S = \|\rho_n - \rho\|_{\text{HS}} \rightarrow 0.$$

□

Remark 3.11 (Extension to a linear map). Although E_Q is defined only on the convex set $\mathcal{D}(H_Q)$, the unitary U provides a canonical linear extension

$$\tilde{E}_Q := U : \mathcal{H}_Q \rightarrow S_Q \subset S.$$

In many of the subsequent constructions (e.g. the representation of observables as linear functionals) it is convenient to work with $\tilde{E}_Q$ on the full linear space $\mathcal{H}_Q$, while remembering that physical states correspond to the subset $E_Q(\mathcal{D}(H_Q)) \subset S_Q$.

3.3.2 The embedded quantum state manifold

Definition 3.12 (Embedded quantum state set). The *embedded quantum state set* is

$$\Omega_Q := E_Q(\mathcal{D}(H_Q)) \subset S_Q \subset S.$$

Proposition 3.13 (Geometric properties of Ω_Q). *The set Ω_Q satisfies:*

1. Ω_Q is convex in S_Q , since $\mathcal{D}(H_Q)$ is convex and E_Q is linear on $\mathcal{D}(H_Q)$.
2. Ω_Q is contained in the closed unit ball of S_Q :

$$\|x\|_S \leq 1 \quad \text{for all } x \in \Omega_Q.$$

3. Embedded pure states correspond to points of norm 1, while embedded mixed states correspond to points of norm strictly less than 1.

Proof. (1) Let $\rho_1, \rho_2 \in \mathcal{D}(H_Q)$ and $\lambda \in [0, 1]$. Then $\rho_\lambda := \lambda\rho_1 + (1 - \lambda)\rho_2 \in \mathcal{D}(H_Q)$, and

$$E_Q(\rho_\lambda) = U(\lambda\rho_1 + (1 - \lambda)\rho_2) = \lambda U(\rho_1) + (1 - \lambda)U(\rho_2) = \lambda E_Q(\rho_1) + (1 - \lambda)E_Q(\rho_2).$$

(2) and (3) follow from Proposition 3.2 and unitarity of U :

$$\|E_Q(\rho)\|_S^2 = \|\rho\|_{\text{HS}}^2 = \text{tr}(\rho^2) \leq 1,$$

with equality if and only if ρ is pure. $\square$

4 Inner products, kernels and isometric properties

We now characterize the embedding E_Q in terms of inner products and kernel functions on the space of density operators.

4.0.1 Hilbert–Schmidt inner product and expectations

Recall that for $\rho \in \mathcal{D}(H_Q)$ and a bounded self-adjoint operator $A \in \mathcal{B}_{\text{sa}}(H_Q)$, the expectation value of A in state ρ is $\text{tr}(\rho A)$. For self-adjoint ρ and A we have

$$\text{tr}(\rho A) = \langle \rho, A \rangle_{\text{HS}}.$$

The embedding E_Q transfers this Hilbert–Schmidt structure to the subspace S_Q .

Proposition 4.1 (Isometry of E_Q). *For all $\rho, \sigma \in \mathcal{D}(H_Q)$ one has*

$$\langle E_Q(\rho), E_Q(\sigma) \rangle_S = \langle \rho, \sigma \rangle_{\text{HS}} = \text{tr}(\rho\sigma).$$

Proof. By definition, $E_Q(\rho) = U(\rho)$ and $E_Q(\sigma) = U(\sigma)$, and U is unitary from $\mathcal{H}_Q$ onto S_Q . Thus

$$\langle E_Q(\rho), E_Q(\sigma) \rangle_S = \langle U(\rho), U(\sigma) \rangle_S = \langle \rho, \sigma \rangle_{\mathcal{H}_Q} = \langle \rho, \sigma \rangle_{\text{HS}}.$$

For positive trace-class operators ρ and σ we have $\langle \rho, \sigma \rangle_{\text{HS}} = \text{tr}(\rho\sigma)$. $\square$

Remark 4.2. Proposition 4.1 shows that the embedding E_Q preserves the Hilbert–Schmidt geometry of density operators. In particular, the angle between two embedded states $E_Q(\rho)$ and $E_Q(\sigma)$ in S_Q coincides with the Hilbert–Schmidt angle between ρ and σ in $\mathcal{H}_{\text{HS}}(H_Q)$.

4.0.2 A kernel on the space of density operators

The Hilbert–Schmidt inner product induces a canonical positive-definite kernel on $\mathcal{D}(H_Q)$.

Definition 4.3 (Hilbert–Schmidt kernel on $\mathcal{D}(H_Q)$). Define $k : \mathcal{D}(H_Q) \times \mathcal{D}(H_Q) \rightarrow \mathbb{R}$ by

$$k(\rho, \sigma) := \text{tr}(\rho\sigma) = \langle \rho, \sigma \rangle_{\text{HS}}.$$

Proposition 4.4 (Positive definiteness of k). *The function k in Definition 4.3 is a positive-definite kernel on $\mathcal{D}(H_Q)$, i.e. for any finite collection $\rho_1, \dots, \rho_n \in \mathcal{D}(H_Q)$ and complex numbers $c_1, \dots, c_n$,*

$$\sum_{i,j=1}^n \bar{c}_i c_j k(\rho_i, \rho_j) \geq 0,$$

with equality if and only if $\sum_j c_j \rho_j = 0$ in $\mathcal{H}_{\text{HS}}(H_Q)$.

Proof. We compute

$$\sum_{i,j=1}^n \bar{c}_i c_j k(\rho_i, \rho_j) = \sum_{i,j=1}^n \bar{c}_i c_j \operatorname{tr}(\rho_i \rho_j) = \left\langle \sum_i c_i \rho_i, \sum_j c_j \rho_j \right\rangle_{\text{HS}} \geq 0,$$

since the right-hand side is the squared norm of $\sum_j c_j \rho_j$ in $\mathcal{H}_{\text{HS}}(H_Q)$. It vanishes if and only if $\sum_j c_j \rho_j = 0$. $\square$

Remark 4.5 (Reproducing kernel Hilbert space viewpoint). The kernel k defines a reproducing kernel Hilbert space (RKHS) $\mathcal{H}_k$ of complex-valued functions on $\mathcal{D}(H_Q)$. The canonical feature map

$$\Phi : \mathcal{D}(H_Q) \rightarrow \mathcal{H}_k, \quad \Phi(\rho) := k(\cdot, \rho),$$

is an isometric embedding in the sense that

$$\langle \Phi(\rho), \Phi(\sigma) \rangle_{\mathcal{H}_k} = k(\rho, \sigma) = \operatorname{tr}(\rho \sigma).$$

The embedding E_Q can be viewed as the composition of this canonical feature map with a unitary isomorphism between $\mathcal{H}_k$ and S_Q , making the connection between kernel methods and the semantic embedding explicit.

4.1 Restriction of the potential W to the quantum sector

The semantic framework provides a potential

$$W : S \rightarrow \mathbb{R} \cup \{+\infty\},$$

which drives gradient flows and Langevin-type dynamics on S and on probability measures over S . We now discuss how this potential behaves when restricted to the embedded quantum subspace S_Q .

4.1.1 Definition of the restricted potential

Definition 4.6 (Restricted potential on S_Q). Assume that W is finite on S_Q . The *restricted quantum potential* is the function

$$W_Q := W|_{S_Q} : S_Q \rightarrow \mathbb{R}.$$

Given the embedding E_Q , we can also regard W_Q as a function on $\mathcal{D}(H_Q)$ via

$$W_Q(\rho) := W(E_Q(\rho)), \quad \rho \in \mathcal{D}(H_Q).$$

In general, W_Q is an arbitrary real-valued function on $\mathcal{D}(H_Q)$ determined by the choice of W and the embedding E_Q . For the purposes of this paper, we make mild structural assumptions on W_Q that are convenient for later compatibility statements with quantum dynamics.

4.1.2 Unitary orbits and invariance assumptions

Quantum dynamics generated by a Hamiltonian H acts on density operators via unitary conjugation,

$$\rho \mapsto U \rho U^*,$$

where U is unitary on H_Q . This suggests that the natural geometric objects in $\mathcal{D}(H_Q)$ are unitary orbits of states.

Definition 4.7 (Unitary orbit of a state). For $\rho \in \mathcal{D}(H_Q)$, its *unitary orbit* is

$$\mathcal{O}(\rho) := \{U\rho U^* : U \text{ unitary on } H_Q\}.$$

Assumption 4.8 (Invariance of W_Q along unitary orbits). On the embedded quantum sector we assume that W_Q is constant along unitary orbits, i.e. for all $\rho \in \mathcal{D}(H_Q)$ and all unitaries U on H_Q ,

$$W_Q(U\rho U^*) = W_Q(\rho).$$

Equivalently,

$$W(E_Q(U\rho U^*)) = W(E_Q(\rho)).$$

Remark 4.9. Assumption 4.8 expresses the idea that the potential does not distinguish between states that are related by a reversible quantum evolution. It is compatible with the view that the semantic potential W may encode, in the quantum sector, only invariants of ρ such as its spectrum (eigenvalues), which are unchanged under unitary conjugation.

Proposition 4.10 (Spectral dependence of W_Q). *Under Assumption 4.8, the restricted potential W_Q depends on a density operator ρ only through its spectrum, i.e. if ρ_1 and ρ_2 have the same multiset of eigenvalues (counted with multiplicity), then $W_Q(\rho_1) = W_Q(\rho_2)$.*

Proof. If ρ_1 and ρ_2 have the same spectrum, the spectral theorem implies the existence of a unitary U on H_Q such that $\rho_2 = U\rho_1 U^*$. By Assumption 4.8, $W_Q(\rho_2) = W_Q(U\rho_1 U^*) = W_Q(\rho_1)$. $\square$

4.1.3 Regularity along the embedded state manifold

To relate the potential-driven dynamics in S to quantum dynamics in later chapters, we will require some regularity of W along the embedded state manifold Ω_Q .

Assumption 4.11 (Differentiability along Ω_Q). We assume that W is Fréchet differentiable on an open neighbourhood of Ω_Q in S , and that its gradient ∇W is continuous and locally bounded on this neighbourhood.

Remark 4.12. Assumption 4.11 is consistent with the standing hypotheses in the semantic gradient-flow framework, where W typically plays the role of a sufficiently regular energy functional. In the present work, we use this assumption only to guarantee that the gradient ∇W is well-defined along S_Q and that the restriction of gradient flows to S_Q is meaningful at a conceptual level.

4.2 Summary of the embedding construction

For later reference we summarize the main structural elements obtained in this chapter.

- The quantum state space $\mathcal{D}(H_Q)$ is embedded into the Hilbert–Schmidt space $\mathcal{H}_{\text{HS}}(H_Q)$, and the closed linear span of density operators defines a separable Hilbert space $\mathcal{H}_Q$.
- Because S is a separable Hilbert space, there exists a closed subspace $S_Q \subset S$ and a unitary $U : \mathcal{H}_Q \rightarrow S_Q$. This defines an embedded quantum subspace of the semantic Hilbert space S .

- The embedding of density operators into S is given by $E_Q(\rho) = U(\rho)$, and its image

$$\Omega_Q := E_Q(\mathcal{D}(H_Q)) \subset S_Q$$

is a convex subset contained in the closed unit ball of S_Q , with pure states corresponding to points of norm 1.

- The embedding is isometric with respect to the Hilbert–Schmidt inner product:

$$\langle E_Q(\rho), E_Q(\sigma) \rangle_S = \text{tr}(\rho\sigma).$$

This inner product induces a positive-definite kernel $k(\rho, \sigma) = \text{tr}(\rho\sigma)$ on the space of density operators, making the embedding interpretable as a kernel feature map into S_Q .

- The semantic potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ restricts to a function W_Q on S_Q , and hence to $\mathcal{D}(H_Q)$ via $W_Q(\rho) = W(E_Q(\rho))$. Under the natural assumption that W_Q is invariant under unitary conjugation, W_Q depends only on the spectrum of ρ .
- Assuming further that W is Fréchet differentiable in a neighbourhood of Ω_Q , the gradient ∇W is well-defined along the embedded quantum sector. This will be relevant in Chapter 5, where we relate the von Neumann dynamics of density operators to induced dynamics on S_Q and to the broader gradient-flow structures available on S .

In the next chapter we use this embedding to represent quantum observables as linear functionals on S_Q , with expectation values recovered as inner products in the semantic Hilbert space.

5 Observables as linear functionals on the embedded subspace

5.1 From quantum observables to semantic functionals

In this chapter we show how quantum observables can be represented as linear functionals on the embedded quantum subspace $S_Q \subset S$ constructed in Chapter 3. The key requirement is that, for each observable A and each density operator $\rho \in \mathcal{D}(H_Q)$, the expectation value

$$\text{tr}(\rho A)$$

is recovered as the value of a suitable functional L_A applied to the embedded state $E_Q(\rho)$.

The analysis will use the Hilbert–Schmidt structure introduced in Chapter 3. To make full use of the Riesz representation theorem on S_Q in infinite dimensions, it is convenient to work with a subclass of observables that are Hilbert–Schmidt operators. In finite dimensions this restriction is vacuous, since every linear operator is Hilbert–Schmidt.

5.1.1 Hilbert–Schmidt observables

Definition 5.1 (Hilbert–Schmidt observables). Define

$$\mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q) := \{A \in \mathcal{H}_{\text{HS}}(H_Q) : A = A^*\},$$

where $\mathcal{H}_{\text{HS}}(H_Q)$ is the Hilbert–Schmidt space on H_Q with inner product

$$\langle X, Y \rangle_{\text{HS}} := \text{tr}(X^*Y).$$

Elements of $\mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ will be called *Hilbert–Schmidt observables*.

Remark 5.2. In finite dimensions, $H_Q \cong \mathbb{C}^d$, all linear operators on H_Q are Hilbert–Schmidt, and $\mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ coincides with the full set of bounded self-adjoint operators $\mathcal{B}_{\text{sa}}(H_Q)$. In infinite dimensions, $\mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ is a proper subset of $\mathcal{B}_{\text{sa}}(H_Q)$, but it still contains many physically relevant observables, in particular all finite-rank self-adjoint operators.

For $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ and $\rho \in \mathcal{D}(H_Q)$, both operators are Hilbert–Schmidt and self-adjoint, so

$$\text{tr}(\rho A) = \langle \rho, A \rangle_{\text{HS}}.$$

This observation is the starting point for representing expectations as inner products in the semantic Hilbert space.

5.1.2 The embedding and its unitary intertwiner

Recall from Chapter 3 that:

- $\mathcal{H}_Q$ denotes the closed linear span of density operators in $\mathcal{H}_{\text{HS}}(H_Q)$,

$$\mathcal{H}_Q := \overline{\text{span}}\{\rho : \rho \in \mathcal{D}(H_Q)\} \subset \mathcal{H}_{\text{HS}}(H_Q).$$

- $S_Q \subset S$ is a closed subspace and

$$U : \mathcal{H}_Q \rightarrow S_Q$$

is a fixed unitary isomorphism.

- The embedding of density operators into S is

$$E_Q : \mathcal{D}(H_Q) \rightarrow S_Q \subset S, \quad E_Q(\rho) := U(\rho).$$

For $\rho, \sigma \in \mathcal{D}(H_Q)$, this gives

$$\langle E_Q(\rho), E_Q(\sigma) \rangle_S = \langle \rho, \sigma \rangle_{\text{HS}} = \text{tr}(\rho \sigma).$$

In particular, the embedding E_Q is isometric with respect to the Hilbert–Schmidt geometry.

5.2 Definition of L_A on the embedded quantum sector

We now define, for each Hilbert–Schmidt observable A , a linear functional on S_Q that reproduces the standard quantum expectation values on embedded states.

5.3 Semantic expectation functionals on Ω_Q

Recall the embedded quantum state set

$$\Omega_Q := E_Q(\mathcal{D}(H_Q)) \subset S_Q.$$

Definition 5.3 (Expectation functional on Ω_Q). Let $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$. The *quantum expectation functional* associated with A is the map

$$\ell_A : \Omega_Q \rightarrow \mathbb{C}$$

defined by

$$\ell_A(E_Q(\rho)) := \text{tr}(\rho A), \quad \rho \in \mathcal{D}(H_Q).$$

Proposition 5.4 (Well-definedness and boundedness on Ω_Q). *For each $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$, the map ℓ_A is well-defined and satisfies the bound*

$$|\ell_A(x)| \leq \|A\|_{\text{HS}} \|x\|_S \quad \text{for all } x \in \Omega_Q.$$

Proof. Well-definedness follows from injectivity of E_Q : if $E_Q(\rho_1) = E_Q(\rho_2)$, then $\rho_1 = \rho_2$ and hence $\text{tr}(\rho_1 A) = \text{tr}(\rho_2 A)$.

For boundedness, let $x = E_Q(\rho)$ with $\rho \in \mathcal{D}(H_Q)$. Then

$$|\ell_A(x)| = |\text{tr}(\rho A)| = |\langle \rho, A \rangle_{\text{HS}}| \leq \|\rho\|_{\text{HS}} \|A\|_{\text{HS}} = \|E_Q(\rho)\|_S \|A\|_{\text{HS}} = \|x\|_S \|A\|_{\text{HS}},$$

where we used the Cauchy–Schwarz inequality in $\mathcal{H}_{\text{HS}}(H_Q)$ and the unitarity of U . $\square$

Proposition 5.4 shows that each ℓ_A is Lipschitz with respect to the norm on S_Q when restricted to Ω_Q . We now extend ℓ_A to a linear functional on all of S_Q .

5.3.1 Extension to a linear functional on S_Q

The construction of $\mathcal{H}_Q$ in Chapter 3 ensures that the linear span of density operators is dense in $\mathcal{H}_Q$ and thus, via U , that the linear span of Ω_Q is dense in S_Q .

Proposition 5.5 (Density of the embedded span). *Let*

$$\text{span}(\Omega_Q) := \left\{ \sum_{k=1}^n c_k x_k : n \in \mathbb{N}, c_k \in \mathbb{C}, x_k \in \Omega_Q \right\}.$$

Then the closure of $\text{span}(\Omega_Q)$ in S_Q equals S_Q .

Proof. By construction, $\mathcal{H}_Q$ is the closure in $\mathcal{H}_{\text{HS}}(H_Q)$ of the linear span of $\mathcal{D}(H_Q)$. Applying the unitary U , we obtain that $S_Q = U(\mathcal{H}_Q)$ is the closure in S_Q of the linear span of $E_Q(\mathcal{D}(H_Q)) = \Omega_Q$. Hence the closure of $\text{span}(\Omega_Q)$ is S_Q . $\square$

Proposition 5.6 (Linear extension of ℓ_A). *For each $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ there exists a unique continuous linear functional*

$$L_A : S_Q \rightarrow \mathbb{C}$$

such that

$$L_A(E_Q(\rho)) = \text{tr}(\rho A) \quad \text{for all } \rho \in \mathcal{D}(H_Q).$$

Moreover, L_A satisfies

$$|L_A(x)| \leq \|A\|_{\text{HS}} \|x\|_S \quad \text{for all } x \in S_Q.$$

Proof. Define $\tilde{\ell}_A$ on $\text{span}(\Omega_Q)$ by linear extension:

$$\tilde{\ell}_A \left(\sum_{k=1}^n c_k E_Q(\rho_k) \right) := \sum_{k=1}^n c_k \text{tr}(\rho_k A).$$

This is well-defined because the representation of a point in $\text{span}(\Omega_Q)$ as a linear combination of embedded states is unique: if $\sum_k c_k E_Q(\rho_k) = 0$, then $\sum_k c_k \rho_k = 0$ in $\mathcal{H}_Q$, and hence

$$\sum_k c_k \text{tr}(\rho_k A) = \text{tr} \left(\left(\sum_k c_k \rho_k \right) A \right) = 0.$$

For $x \in \text{span}(\Omega_Q)$ we can write $x = \sum_k c_k E_Q(\rho_k)$ and estimate

$$|\tilde{\ell}_A(x)| = \left| \sum_k c_k \text{tr}(\rho_k A) \right| = \left| \sum_k c_k \langle E_Q(\rho_k), v_A \rangle_S \right| \leq \|v_A\|_S \left\| \sum_k c_k E_Q(\rho_k) \right\|_S,$$

once we have identified a suitable $v_A \in S_Q$ (see below). Alternatively, we can directly inherit the bound from Proposition 5.4 using linearity and the triangle inequality. In either case, there exists a constant $C_A \leq \|A\|_{\text{HS}}$ such that

$$|\tilde{\ell}_A(x)| \leq C_A \|x\|_S \quad \text{for all } x \in \text{span}(\Omega_Q).$$

Thus $\tilde{\ell}_A$ is a bounded linear functional on the dense subspace $\text{span}(\Omega_Q)$ of S_Q .

By continuity, $\tilde{\ell}_A$ extends uniquely to a continuous linear functional L_A on the closure of $\text{span}(\Omega_Q)$, which is S_Q by Proposition 5.5. The bound

$$|L_A(x)| \leq \|A\|_{\text{HS}} \|x\|_S$$

holds for all $x \in S_Q$ by continuity and density, and $L_A(E_Q(\rho)) = \ell_A(E_Q(\rho)) = \text{tr}(\rho A)$ by construction. $\square$

5.4 Riesz representation and semantic observable vectors

The Hilbert structure of S_Q allows us to identify each functional L_A with a unique vector $v_A \in S_Q$ via the Riesz representation theorem.

5.4.1 Riesz representation

Theorem 5.7 (Riesz representation). *For each $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ there exists a unique vector $v_A \in S_Q$ such that*

$$L_A(x) = \langle x, v_A \rangle_S \quad \text{for all } x \in S_Q.$$

Proof. The space S_Q is a Hilbert space with inner product inherited from S . The Riesz representation theorem states that for every continuous linear functional $L : S_Q \rightarrow \mathbb{C}$ there exists a unique $v \in S_Q$ such that $L(x) = \langle x, v \rangle_S$ for all $x \in S_Q$. Applying this to L_A from Proposition 5.6, we obtain the desired v_A . $\square$

Definition 5.8 (Semantic observable vectors). For $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$, the vector $v_A \in S_Q$ given by Theorem 5.7 is called the *semantic observable vector* associated with A .

5.4.2 Explicit identification via the unitary U

The unitary $U : \mathcal{H}_Q \rightarrow S_Q$ allows an explicit characterization of v_A .

Proposition 5.9 (Explicit form of v_A). *For $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$, the semantic observable vector is*

$$v_A = U(A) \in S_Q.$$

Moreover, for all $\rho \in \mathcal{D}(H_Q)$,

$$L_A(E_Q(\rho)) = \langle E_Q(\rho), v_A \rangle_S = \text{tr}(\rho A).$$

Proof. Define $\widehat{L}_A : S_Q \rightarrow \mathbb{C}$ by

$$\widehat{L}_A(x) := \langle x, U(A) \rangle_S.$$

Then $\widehat{L}_A$ is a continuous linear functional with

$$\widehat{L}_A(E_Q(\rho)) = \langle U(\rho), U(A) \rangle_S = \langle \rho, A \rangle_{\text{HS}} = \text{tr}(\rho A)$$

for all $\rho \in \mathcal{D}(H_Q)$, using unitarity of U and the definition of the Hilbert–Schmidt inner product. Thus $\widehat{L}_A$ and L_A agree on $\Omega_Q = E_Q(\mathcal{D}(H_Q))$.

By linearity and continuity, they also agree on $\text{span}(\Omega_Q)$ and hence, by density, on all of S_Q . Therefore $L_A(x) = \langle x, U(A) \rangle_S$ for all $x \in S_Q$. Uniqueness in the Riesz theorem then implies $v_A = U(A)$. $\square$

Remark 5.10 (Semantic representation of expectations). Combining Proposition 5.9 with the definition of E_Q , we obtain for each $\rho \in \mathcal{D}(H_Q)$ and $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$:

$$\text{tr}(\rho A) = L_A(E_Q(\rho)) = \langle E_Q(\rho), v_A \rangle_S,$$

i.e. quantum expectation values are realized as inner products between the embedded state and the semantic observable vector in S_Q .

5.5 Linearity, positivity and order structure

We next establish basic algebraic and positivity properties of the functionals L_A and vectors v_A .

5.5.1 Linearity in the observable

Proposition 5.11 (Linearity of $A \mapsto L_A$). *For $A, B \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ and $\alpha, \beta \in \mathbb{R}$, we have*

$$L_{\alpha A + \beta B} = \alpha L_A + \beta L_B.$$

Consequently,

$$v_{\alpha A + \beta B} = \alpha v_A + \beta v_B.$$

Proof. For any $\rho \in \mathcal{D}(H_Q)$,

$$L_{\alpha A + \beta B}(E_Q(\rho)) = \text{tr}(\rho(\alpha A + \beta B)) = \alpha \text{tr}(\rho A) + \beta \text{tr}(\rho B) = \alpha L_A(E_Q(\rho)) + \beta L_B(E_Q(\rho)).$$

Thus $L_{\alpha A + \beta B}$ and $\alpha L_A + \beta L_B$ agree on Ω_Q . By linearity and continuity they agree on S_Q . The statement about $v_{\alpha A + \beta B}$ follows from Riesz representation and uniqueness:

$$\langle x, v_{\alpha A + \beta B} \rangle_S = L_{\alpha A + \beta B}(x) = \alpha L_A(x) + \beta L_B(x) = \langle x, \alpha v_A + \beta v_B \rangle_S$$

for all $x \in S_Q$, hence $v_{\alpha A + \beta B} = \alpha v_A + \beta v_B$. $\square$

5.5.2 Positivity for positive observables

Proposition 5.12 (Positivity on embedded states). *Let $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ be positive, i.e. $A \geq 0$. Then*

$$L_A(E_Q(\rho)) \geq 0 \quad \text{for all } \rho \in \mathcal{D}(H_Q).$$

Moreover, if $0 \leq A \leq B$ (operator order), then

$$L_A(E_Q(\rho)) \leq L_B(E_Q(\rho)) \quad \text{for all } \rho \in \mathcal{D}(H_Q).$$

Proof. If $A \geq 0$, then $\text{tr}(\rho A) \geq 0$ for all density operators ρ by standard properties of the trace and positivity. By definition of L_A , this implies $L_A(E_Q(\rho)) \geq 0$.

If $0 \leq A \leq B$, then $B - A \geq 0$, and hence $\text{tr}(\rho(B - A)) \geq 0$ for all density operators ρ . Thus

$$L_B(E_Q(\rho)) - L_A(E_Q(\rho)) = \text{tr}(\rho B) - \text{tr}(\rho A) = \text{tr}(\rho(B - A)) \geq 0.$$

□

Remark 5.13 (Effect operators and probability values). If $0 \leq A \leq \mathbb{I}$ is a positive contraction (an *effect* in the language of quantum measurement theory), then

$$0 \leq L_A(E_Q(\rho)) = \text{tr}(\rho A) \leq 1,$$

so L_A evaluated on embedded states yields values in $[0, 1]$ and can be interpreted as probabilities. In finite dimensions where $\mathbb{I}$ is Hilbert–Schmidt, even the identity observable admits a semantic observable vector $v_{\mathbb{I}}$, with $L_{\mathbb{I}}(E_Q(\rho)) = 1$ for all ρ .

5.5.3 Norm bounds and Cauchy–Schwarz

Proposition 5.14 (Norm bounds). *For any $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ and $x \in S_Q$,*

$$|L_A(x)| \leq \|v_A\|_S \|x\|_S = \|A\|_{\text{HS}} \|x\|_S.$$

In particular, on embedded states $x = E_Q(\rho)$,

$$|L_A(E_Q(\rho))| \leq \|A\|_{\text{HS}}.$$

Proof. By Theorem 5.7 and Proposition 5.9, $L_A(x) = \langle x, v_A \rangle_S = \langle x, U(A) \rangle_S$. The Cauchy–Schwarz inequality in S_Q gives

$$|L_A(x)| \leq \|x\|_S \|v_A\|_S.$$

Unitarity of U implies $\|v_A\|_S = \|U(A)\|_S = \|A\|_{\text{HS}}$, hence

$$|L_A(x)| \leq \|A\|_{\text{HS}} \|x\|_S.$$

For $x = E_Q(\rho)$ with $\rho \in \mathcal{D}(H_Q)$ we have $\|E_Q(\rho)\|_S \leq 1$, so $|L_A(E_Q(\rho))| \leq \|A\|_{\text{HS}}$. □

5.6 Structural remarks and limitations

5.6.1 Non-uniqueness of the embedding and observable vectors

The construction of S_Q and U in Chapter 3 is not unique: any unitary $\tilde{U} : \mathcal{H}_Q \rightarrow S_Q$ obtained by composing U with a unitary V on S_Q ,

$$\tilde{U} := VU,$$

defines an equally valid embedded quantum subspace. Under such a change, the embedding and observable vectors transform as

$$\tilde{E}_Q(\rho) = \tilde{U}(\rho) = V(E_Q(\rho)), \quad \tilde{v}_A = \tilde{U}(A) = V(v_A).$$

Nevertheless, the expectation values are invariant:

$$\langle \tilde{E}_Q(\rho), \tilde{v}_A \rangle_S = \langle V(E_Q(\rho)), V(v_A) \rangle_S = \langle E_Q(\rho), v_A \rangle_S = \text{tr}(\rho A).$$

Thus the semantic representation of quantum expectations is unchanged under internal unitary transformations of S_Q .

5.6.2 Restriction to Hilbert–Schmidt observables

The use of the Hilbert–Schmidt inner product in the embedding imposes a technical restriction: the Riesz representation of expectation functionals via vectors v_A in S_Q is straightforward for observables A that are Hilbert–Schmidt. For more general bounded observables that are not Hilbert–Schmidt in infinite dimensions, the functional

$$\rho \mapsto \text{tr}(\rho A)$$

is naturally continuous on the trace-class space with respect to the trace norm, but it need not extend to a continuous functional on $\mathcal{H}_Q$ endowed with the Hilbert–Schmidt norm. Addressing such observables would require either a different choice of intermediate operator Hilbert space or a more general Banach-space duality framework.

In this paper we therefore focus on $\mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ as the class of observables for which the semantic representation via vectors v_A in S_Q is clean and fully compatible with the Hilbert-space structure of S . In finite dimensions this restriction disappears, and all self-adjoint operators admit a semantic observable vector.

5.6.3 Preparation for dynamics

The main result of this chapter is that, for the class of Hilbert–Schmidt observables, quantum expectations can be written as inner products in the semantic Hilbert space:

$$\text{tr}(\rho A) = \langle E_Q(\rho), v_A \rangle_S.$$

In Chapter 5, we will combine this representation with the von Neumann dynamics of density operators,

$$i\hbar \frac{d}{dt} \rho_t = [H, \rho_t],$$

to define an induced dynamics on the embedded quantum subspace S_Q . We will show that this dynamics can be expressed in terms of a linear operator family $(\mathcal{U}_t)_{t \in \mathbb{R}}$ acting on S_Q and that the expectations computed via the functionals L_A remain compatible with the standard quantum evolution for all times.

6 Dynamics: from von Neumann evolution to induced dynamics on S

6.1 Von Neumann dynamics on the space of density operators

We begin by recalling the von Neumann dynamics on the space of density operators and its realization as a unitary flow on a Hilbert space of operators.

6.1.1 Liouvillian dynamics on $\mathcal{D}(H_Q)$

Let H be a self-adjoint Hamiltonian on H_Q . As in Chapter 2, we denote by

$$U_t := e^{-itH/\hbar}, \quad t \in \mathbb{R},$$

the unitary evolution generated by H . The induced evolution on density operators is

$$\rho_t = U_t \rho_0 U_t^*, \quad \rho_0 \in \mathcal{D}(H_Q),$$

which solves the von Neumann equation

$$i\hbar \frac{d}{dt} \rho_t = [H, \rho_t] := H \rho_t - \rho_t H. \quad (6.1)$$

Definition 6.1 (Liouvillian). The (Hamiltonian) *Liouvillian* associated with H is the linear operator

$$\mathcal{L}_H : \mathcal{D}(\mathcal{L}_H) \subset \mathcal{H}_{\text{HS}}(H_Q) \rightarrow \mathcal{H}_{\text{HS}}(H_Q), \quad \mathcal{L}_H(X) := -\frac{i}{\hbar}[H, X],$$

defined at least on the dense subspace of finite-rank operators and extended (when possible) to a skew-adjoint operator generating a strongly continuous one-parameter unitary group on $\mathcal{H}_{\text{HS}}(H_Q)$.

Remark 6.2. If H is bounded, then $\mathcal{L}_H$ is a bounded skew-adjoint operator on $\mathcal{H}_{\text{HS}}(H_Q)$ and generates a strongly continuous unitary group

$$\mathcal{T}_t := e^{t\mathcal{L}_H} : \mathcal{H}_{\text{HS}}(H_Q) \rightarrow \mathcal{H}_{\text{HS}}(H_Q), \quad t \in \mathbb{R},$$

given explicitly by

$$\mathcal{T}_t(X) = U_t X U_t^*.$$

In more general unbounded cases, one can work with $\mathcal{L}_H$ as a closed, densely defined skew-adjoint operator; for our purposes we will assume that $\mathcal{L}_H$ generates a strongly continuous unitary group $(\mathcal{T}_t)_{t \in \mathbb{R}}$ on the subspace $\mathcal{H}_Q \subset \mathcal{H}_{\text{HS}}(H_Q)$ introduced in Chapter 3.

6.1.2 Unitary group on the operator Hilbert space

Recall that $\mathcal{H}_Q$ denotes the closed linear span of density operators in $\mathcal{H}_{\text{HS}}(H_Q)$,

$$\mathcal{H}_Q := \overline{\text{span}}\{\rho : \rho \in \mathcal{D}(H_Q)\}.$$

Proposition 6.3 (Invariance of $\mathcal{H}_Q$). *For each $t \in \mathbb{R}$, the map*

$$\mathcal{T}_t(X) := U_t X U_t^*$$

leaves $\mathcal{H}_Q$ invariant, and the restriction

$$\mathcal{T}_t|_{\mathcal{H}_Q} : \mathcal{H}_Q \rightarrow \mathcal{H}_Q$$

is unitary.

Proof. If X is a finite linear combination of density operators, then $\mathcal{T}_t(X)$ is a finite linear combination of operators of the form $U_t \rho U_t^*$, which are again density operators. Thus $\mathcal{T}_t$ preserves $\text{span}\{\mathcal{D}(H_Q)\}$ and hence its closure $\mathcal{H}_Q$.

For $X, Y \in \mathcal{H}_Q$,

$$\langle \mathcal{T}_t(X), \mathcal{T}_t(Y) \rangle_{\text{HS}} = \text{tr}((U_t X U_t^*)^* (U_t Y U_t^*)) = \text{tr}(U_t X^* U_t^* U_t Y U_t^*) = \text{tr}(X^* Y) = \langle X, Y \rangle_{\text{HS}},$$

so $\mathcal{T}_t$ is isometric on $\mathcal{H}_Q$. Surjectivity follows from invertibility of U_t (with inverse U_{-t}), hence $\mathcal{T}_t|_{\mathcal{H}_Q}$ is unitary. $\square$

6.2 Induced dynamics on the embedded quantum subspace S_Q

We now transfer the Liouvillian dynamics on $\mathcal{H}_Q$ to the embedded quantum subspace $S_Q \subset S$ via the unitary intertwiner

$$U : \mathcal{H}_Q \rightarrow S_Q$$

introduced in Chapter 3.

6.3 Definition of the induced evolution

Definition 6.4 (Induced semantic dynamics). For each $t \in \mathbb{R}$, define

$$\mathcal{U}_t : S_Q \rightarrow S_Q, \quad \mathcal{U}_t := U \circ \mathcal{T}_t|_{\mathcal{H}_Q} \circ U^{-1}.$$

Proposition 6.5 (Unitary group on S_Q). *The family $(\mathcal{U}_t)_{t \in \mathbb{R}}$ is a strongly continuous one-parameter unitary group on S_Q , i.e.*

1. $\mathcal{U}_0 = \text{id}_{S_Q}$,
2. $\mathcal{U}_{t+s} = \mathcal{U}_t \circ \mathcal{U}_s$ for all $s, t \in \mathbb{R}$,
3. $\mathcal{U}_t$ is unitary on S_Q for each t ,
4. for each $x \in S_Q$, the map $t \mapsto \mathcal{U}_t x$ is continuous in the norm of S_Q .

Proof. Because U is unitary and $(\mathcal{T}_t)_{t \in \mathbb{R}}$ is a strongly continuous one-parameter unitary group on $\mathcal{H}_Q$, the properties of a strongly continuous unitary group are preserved under conjugation by U . Explicitly:

$$\mathcal{U}_0 = U \mathcal{T}_0 U^{-1} = U \text{id}_{\mathcal{H}_Q} U^{-1} = \text{id}_{S_Q},$$

and

$$\mathcal{U}_{t+s} = U \mathcal{T}_{t+s} U^{-1} = U \mathcal{T}_t \mathcal{T}_s U^{-1} = (U \mathcal{T}_t U^{-1})(U \mathcal{T}_s U^{-1}) = \mathcal{U}_t \circ \mathcal{U}_s.$$

Unitarity of each $\mathcal{U}_t$ follows from unitarity of U and $\mathcal{T}_t$, and strong continuity is inherited from $(\mathcal{T}_t)_{t \in \mathbb{R}}$ via the unitary equivalence. $\square$

6.3.1 Dynamics of embedded states

Let $\rho_0 \in \mathcal{D}(H_Q)$ and let $\rho_t = U_t \rho_0 U_t^*$ be the corresponding von Neumann evolution. The associated embedded state in S_Q is

$$x_t := E_Q(\rho_t) = U(\rho_t).$$

Proposition 6.6 (Intertwining of dynamics). *For all $t \in \mathbb{R}$, the embedded state x_t satisfies*

$$x_t = \mathcal{U}_t x_0, \quad x_0 := E_Q(\rho_0).$$

Equivalently,

$$E_Q(\rho_t) = \mathcal{U}_t E_Q(\rho_0).$$

Proof. By definition,

$$\mathcal{U}_t x_0 = U \mathcal{T}_t U^{-1}(E_Q(\rho_0)) = U \mathcal{T}_t(\rho_0) = U(U_t \rho_0 U_t^*) = E_Q(\rho_t).$$

Thus $x_t = E_Q(\rho_t) = \mathcal{U}_t x_0$. $\square$

6.3.2 Generator on S_Q

The Liouvillian $\mathcal{L}_H$ induces a generator on S_Q via unitary conjugation.

Definition 6.7 (Semantic Liouvillian). Define the (Hamiltonian) semantic Liouvillian

$$\mathcal{K}_H : \mathcal{D}(\mathcal{K}_H) \subset S_Q \rightarrow S_Q, \quad \mathcal{K}_H := U \circ \mathcal{L}_H|_{\mathcal{H}_Q} \circ U^{-1}.$$

By construction, $\mathcal{K}_H$ is skew-adjoint on S_Q (under suitable domain assumptions) and generates the unitary group $(\mathcal{U}_t)_{t \in \mathbb{R}}$:

$$\frac{d}{dt}x_t = \mathcal{K}_H x_t, \quad x_t = \mathcal{U}_t x_0.$$

Remark 6.8. Formally, if $x_t = E_Q(\rho_t)$ with ρ_t solving (6.1), then

$$\frac{d}{dt}x_t = \frac{d}{dt}U(\rho_t) = U\left(\frac{d}{dt}\rho_t\right) = U(-\frac{i}{\hbar}[H, \rho_t]) = \mathcal{K}_H x_t.$$

This makes explicit that $\mathcal{K}_H$ is the image of the commutator generator under the unitary intertwiner U .

6.4 Compatibility of expectations along the induced dynamics

We now show that the induced dynamics on S_Q is compatible with quantum expectations defined via the functionals L_A constructed in Chapter 4.

6.4.1 Semantic expectations along trajectories

Recall that for $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$ we defined a functional $L_A : S_Q \rightarrow \mathbb{C}$ and a semantic observable vector $v_A = U(A)$ such that

$$L_A(x) = \langle x, v_A \rangle_S$$

for all $x \in S_Q$, and

$$L_A(E_Q(\rho)) = \text{tr}(\rho A)$$

for all $\rho \in \mathcal{D}(H_Q)$.

Theorem 6.9 (Expectation compatibility along the flow). *Let H be a Hamiltonian on H_Q such that the Liouvillian $\mathcal{L}_H$ generates a unitary group on $\mathcal{H}_Q$, and let $(\mathcal{U}_t)_{t \in \mathbb{R}}$ be the induced group on S_Q . For any initial state $\rho_0 \in \mathcal{D}(H_Q)$, any $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$, and all $t \in \mathbb{R}$,*

$$L_A(\mathcal{U}_t E_Q(\rho_0)) = \text{tr}(\rho_t A),$$

where ρ_t solves the von Neumann equation (6.1) with initial condition ρ_0 .

Proof. By Proposition 6.6, $\mathcal{U}_t E_Q(\rho_0) = E_Q(\rho_t)$. Thus

$$L_A(\mathcal{U}_t E_Q(\rho_0)) = L_A(E_Q(\rho_t)) = \text{tr}(\rho_t A)$$

by the defining property of L_A . □

Corollary 6.10 (Heisenberg-type evolution in S_Q). *Let $x_0 = E_Q(\rho_0)$ and $x_t = \mathcal{U}_t x_0$. Then for all $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$,*

$$\frac{d}{dt}L_A(x_t) = \frac{1}{i\hbar} \text{tr}(\rho_t[A, H]) = \frac{1}{i\hbar} L_{[A, H]}(x_t),$$

whenever the derivative exists.

Proof. By Theorem 6.9,

$$L_A(x_t) = \text{tr}(\rho_t A).$$

Differentiating with respect to t and using the standard Heisenberg equation for expectations (as in Chapter 2),

$$\frac{d}{dt} \text{tr}(\rho_t A) = \frac{1}{i\hbar} \text{tr}(\rho_t [A, H]),$$

we obtain the desired expression. The last equality follows from the defining property of $L_{[A, H]}$. $\square$

6.4.2 Conservation properties in the semantic picture

The unitary dynamics $\mathcal{U}_t$ preserves the S -norm on S_Q and, via Assumption 4.8 from Chapter 3, leaves the restricted potential W_Q constant along embedded quantum orbits.

Proposition 6.11 (Conservation of norm and potential along embedded orbits). *Let $x_t = \mathcal{U}_t E_Q(\rho_0)$ be the embedded trajectory of a solution ρ_t of the von Neumann equation. Then:*

1. $\|x_t\|_S = \|x_0\|_S$ for all $t \in \mathbb{R}$.
2. Under Assumption 4.8, $W(x_t) = W(x_0)$ for all $t \in \mathbb{R}$.

Proof. (1) Unitarity of $\mathcal{U}_t$ on S_Q implies $\|x_t\|_S = \|\mathcal{U}_t x_0\|_S = \|x_0\|_S$.

(2) Since $x_t = E_Q(\rho_t)$ and $\rho_t = U_t \rho_0 U_t^*$, Assumption 4.8 gives

$$W(x_t) = W(E_Q(\rho_t)) = W(E_Q(U_t \rho_0 U_t^*)) = W(E_Q(\rho_0)) = W(x_0).$$

$\square$

6.5 Extension to open-system dynamics

The construction above extends naturally to certain classes of open-system dynamics described by quantum Markov semigroups.

6.5.1 Quantum channels and GKLS generators

Let $(\Phi_t)_{t \geq 0}$ be a strongly continuous one-parameter semigroup of quantum channels on $\mathcal{T}_1(H_Q)$, with generator $\mathcal{L}$ in GKLS form

$$\mathcal{L}(\rho) = -\frac{i}{\hbar} [H, \rho] + \sum_k \left(L_k \rho L_k^* - \frac{1}{2} \{L_k^* L_k, \rho\} \right),$$

where H is self-adjoint and $(L_k)_k$ are bounded operators on H_Q . We assume that $\Phi_t(\mathcal{D}(H_Q)) \subseteq \mathcal{D}(H_Q)$ and that Φ_t restricts to a strongly continuous contraction semigroup on $\mathcal{H}_Q$.

Definition 6.12 (Induced open-system dynamics on S_Q). Define

$$\mathcal{V}_t : S_Q \rightarrow S_Q, \quad \mathcal{V}_t := U \circ \Phi_t|_{\mathcal{H}_Q} \circ U^{-1}, \quad t \geq 0.$$

Proposition 6.13 (Compatibility for quantum Markov semigroups). *For any initial state $\rho_0 \in \mathcal{D}(H_Q)$ and any $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$,*

$$L_A(\mathcal{V}_t E_Q(\rho_0)) = \text{tr}(\rho_t A), \quad t \geq 0,$$

where $\rho_t := \Phi_t(\rho_0)$.

Proof. Exactly as in Theorem 6.9, using $\mathcal{V}_t E_Q(\rho_0) = E_Q(\Phi_t(\rho_0))$. $\square$

Remark 6.14. The semigroup $(\mathcal{V}_t)_{t \geq 0}$ is typically not unitary on S_Q ; it is a contraction semigroup compatible with the dissipative character of Φ_t . This provides a semantic representation of open-system quantum dynamics that parallels the way gradient flows and noise are represented in the semantic framework on S .

6.6 Relation to semantic gradient and stochastic dynamics

The semantic Hilbert space S carries additional dynamical structures driven by the potential W and the associated free-energy functional on probability measures over S . We briefly discuss how the induced quantum dynamics fits into this broader context.

6.6.1 Hamiltonian vs. gradient components

On S , the semantic framework provides:

- deterministic gradient flows of the form

$$\dot{x}_t = -\nabla W(x_t),$$

- stochastic Langevin dynamics

$$dx_t = -\nabla W(x_t) dt + \sqrt{2\beta^{-1}} dB_t,$$

- and associated Fokker–Planck and free-energy gradient flows on probability measures over S .

In contrast, the unitary quantum dynamics on S_Q is generated by the skew-adjoint operator $\mathcal{K}_H$, and preserves both the norm and the restricted potential W_Q (under Assumption 4.8). At a structural level, this suggests a decomposition of dynamics on S_Q into:

- a *Hamiltonian* component generated by $\mathcal{K}_H$, which is norm-preserving and leaves W_Q invariant,
- a *gradient* component generated by $-\nabla W|_{S_Q}$, which decreases W_Q (where defined) and can represent dissipative effects.

Remark 6.15 (Programmatic combined dynamics). One may consider, at a formal level, dynamics on S_Q of the form

$$\dot{x}_t = \mathcal{K}_H x_t - \nabla W(x_t),$$

or appropriate stochastic extensions. When restricted to embedded states $x_t = E_Q(\rho_t)$ and under suitable conditions, the Hamiltonian part reproduces the von Neumann dynamics, while the gradient part introduces an additional semantic relaxation. A full analysis of such coupled dynamics lies beyond the scope of the present paper, but the structural separation provided by the embedding makes it possible to formulate such questions precisely.

6.6.2 Semantic free energy on embedded quantum distributions

Let $\mathcal{P}(S)$ denote a suitable space of probability measures on S (e.g. measures with finite second moments), and let $\mathcal{F}$ be the semantic free-energy functional introduced in the companion work on probability measures over S . If μ is supported on the embedded quantum sector Ω_Q , then the pushforward of μ under E_Q^{-1} (where defined) induces a probability distribution over density operators.

Remark 6.16 (Quantum ensembles in the semantic picture). A probability measure μ on S supported on Ω_Q can be seen as a distribution over quantum states. The free energy $\mathcal{F}(\mu)$ then provides a scalar functional on such ensembles, combining the contribution of W with an entropy term. The Hamiltonian flow $(\mathcal{U}_t)_{t \in \mathbb{R}}$ acts on μ by pushforward and preserves certain components of $\mathcal{F}$ (e.g. invariance of W_Q along orbits), while the gradient-flow part generated by $\mathcal{F}$ drives dissipative evolution. This decomposes the dynamics of quantum ensembles into conservative and dissipative components within a single semantic framework.

6.7 Summary

In this chapter we have:

- Represented the von Neumann dynamics of density operators as a unitary group $(\mathcal{T}_t)_{t \in \mathbb{R}}$ on the operator Hilbert space $\mathcal{H}_Q$, generated by a Liouvillian $\mathcal{L}_H$.
- Transferred this dynamics to the embedded quantum subspace $S_Q \subset S$ via the unitary intertwiner $U : \mathcal{H}_Q \rightarrow S_Q$, obtaining a strongly continuous unitary group $(\mathcal{U}_t)_{t \in \mathbb{R}}$ on S_Q .
- Shown that embedded states $x_t = E_Q(\rho_t)$ evolve according to $x_t = \mathcal{U}_t x_0$, with generator $\mathcal{K}_H = U \mathcal{L}_H U^{-1}$.
- Established compatibility of expectations: for all Hilbert–Schmidt observables A , the semantic functional L_A satisfies $L_A(\mathcal{U}_t E_Q(\rho_0)) = \text{tr}(\rho_t A)$, ensuring that the induced dynamics on S_Q reproduces the standard quantum predictions.
- Extended the construction to quantum Markov semigroups, obtaining contraction semigroups $(\mathcal{V}_t)_{t \geq 0}$ on S_Q that represent open-system dynamics in the semantic Hilbert space.
- Related the induced unitary (or Markovian) quantum dynamics on S_Q to the gradient and stochastic dynamics driven by the potential W and the semantic free-energy functional, highlighting a structural separation between Hamiltonian and gradient components of evolution.

In the next chapter we make these constructions concrete in two paradigmatic examples: a spin- $\frac{1}{2}$ system and the quantum harmonic oscillator (in a finite-dimensional truncation), providing explicit formulas for the embedding E_Q , the observable vectors v_A , and the induced dynamics on S_Q .

7 Examples: harmonic oscillator and spin- $\frac{1}{2}$

7.1 Spin- $\frac{1}{2}$ system

We first work out the embedding and observable representation in the simplest non-trivial finite-dimensional case: a spin- $\frac{1}{2}$ system with $H_Q = \mathbb{C}^2$ and Pauli matrices as observables. In this setting the constructions of Chapters 3–5 can be made completely explicit.

7.1.1 Quantum description and Bloch representation

Definition 7.1 (Spin- $\frac{1}{2}$ Hilbert space). For a single spin- $\frac{1}{2}$ degree of freedom we take

$$H_Q := \mathbb{C}^2$$

with the standard inner product. The Pauli matrices are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and we denote by $\sigma_0 := \mathbb{I}_2$ the 2×2 identity matrix.

Every density matrix $\rho \in \mathcal{D}(H_Q)$ admits a Bloch representation

$$\rho = \frac{1}{2}(\sigma_0 + \mathbf{r} \cdot \boldsymbol{\sigma}) = \frac{1}{2}(\sigma_0 + r_x \sigma_x + r_y \sigma_y + r_z \sigma_z), \quad (7.1)$$

where $\mathbf{r} = (r_x, r_y, r_z) \in \mathbb{R}^3$ is the Bloch vector with $\|\mathbf{r}\| \leq 1$, and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. Pure states correspond to $\|\mathbf{r}\| = 1$.

Proposition 7.2 (Expectations in Bloch coordinates). *For any ρ as in (7.1), the expectations of the Pauli observables are*

$$\text{tr}(\rho \sigma_k) = r_k, \quad k \in \{x, y, z\}.$$

Proof. Using $\text{tr}(\sigma_0 \sigma_k) = 0$ for $k \in \{x, y, z\}$ and $\text{tr}(\sigma_j \sigma_k) = 2\delta_{jk}$,

$$\text{tr}(\rho \sigma_k) = \frac{1}{2} \text{tr}((\sigma_0 + \mathbf{r} \cdot \boldsymbol{\sigma}) \sigma_k) = \frac{1}{2} \text{tr}(\mathbf{r} \cdot \boldsymbol{\sigma} \sigma_k) = \frac{1}{2} \sum_{j \in \{x, y, z\}} r_j \text{tr}(\sigma_j \sigma_k) = \frac{1}{2} \sum_j r_j 2\delta_{jk} = r_k.$$

□

7.1.2 Concrete realization of S_Q and the embedding E_Q

The Hilbert–Schmidt space

$$\mathcal{H}_{\text{HS}}(H_Q) \cong \mathbb{C}^{2 \times 2}$$

has the inner product $\langle X, Y \rangle_{\text{HS}} = \text{tr}(X^* Y)$. The four matrices

$$B_\mu := \frac{1}{\sqrt{2}} \sigma_\mu, \quad \mu \in \{0, x, y, z\},$$

form an orthonormal basis of $\mathcal{H}_{\text{HS}}(H_Q)$ with respect to $\langle \cdot, \cdot \rangle_{\text{HS}}$.

Definition 7.3 (Concrete choice of S_Q for spin- $\frac{1}{2}$). For the spin- $\frac{1}{2}$ example we choose the embedded quantum subspace S_Q so that it is isometrically identified with $\mathcal{H}_{\text{HS}}(H_Q)$ via a unitary

$$U : \mathcal{H}_Q \rightarrow S_Q$$

satisfying

$$U(B_\mu) = e_\mu, \quad \mu \in \{0, x, y, z\},$$

where (e_0, e_x, e_y, e_z) is a fixed orthonormal basis of S_Q . In particular,

$$\langle e_\mu, e_\nu \rangle_S = \delta_{\mu\nu}.$$

For a density matrix in Bloch form (7.1), we can expand ρ in the $\{B_\mu\}$ -basis:

$$\rho = \frac{1}{2}(\sigma_0 + \mathbf{r} \cdot \boldsymbol{\sigma}) = \frac{1}{\sqrt{2}}B_0 + \frac{1}{\sqrt{2}}(r_x B_x + r_y B_y + r_z B_z).$$

Thus the embedding $E_Q(\rho) = U(\rho)$ is

$$E_Q(\rho) = \frac{1}{\sqrt{2}}e_0 + \frac{1}{\sqrt{2}}(r_x e_x + r_y e_y + r_z e_z). \quad (7.2)$$

Proposition 7.4 (Isometry in Bloch coordinates). *For any two spin- $\frac{1}{2}$ density matrices ρ, σ with Bloch vectors $\mathbf{r}, \mathbf{s} \in \mathbb{R}^3$,*

$$\langle E_Q(\rho), E_Q(\sigma) \rangle_S = \text{tr}(\rho\sigma) = \frac{1}{2}(1 + \mathbf{r} \cdot \mathbf{s}).$$

Proof. Write

$$E_Q(\rho) = \frac{1}{\sqrt{2}}e_0 + \frac{1}{\sqrt{2}}\sum_k r_k e_k, \quad E_Q(\sigma) = \frac{1}{\sqrt{2}}e_0 + \frac{1}{\sqrt{2}}\sum_k s_k e_k,$$

where the sum runs over $k \in \{x, y, z\}$. Then

$$\langle E_Q(\rho), E_Q(\sigma) \rangle_S = \frac{1}{2}\langle e_0, e_0 \rangle_S + \frac{1}{2}\sum_k r_k s_k \langle e_k, e_k \rangle_S = \frac{1}{2} + \frac{1}{2}\mathbf{r} \cdot \mathbf{s}.$$

On the other hand, the Hilbert–Schmidt inner product of density matrices in Bloch form is known to satisfy

$$\text{tr}(\rho\sigma) = \frac{1}{2}(1 + \mathbf{r} \cdot \mathbf{s}),$$

so the result follows from unitarity of U , as in Chapter 3. $\square$

7.1.3 Semantic observable vectors for Pauli matrices

For $A \in \mathcal{B}_{\text{sa}}^{\text{HS}}(H_Q)$, Chapter 4 associates a linear functional L_A on S_Q and a semantic observable vector $v_A \in S_Q$ such that

$$L_A(x) = \langle x, v_A \rangle_S, \quad L_A(E_Q(\rho)) = \text{tr}(\rho A).$$

Proposition 7.5 (Observable vectors for Pauli matrices). *For the Pauli observables σ_k with $k \in \{x, y, z\}$ one has*

$$v_{\sigma_k} = U(\sigma_k) = \sqrt{2} e_k,$$

and for any density matrix ρ with Bloch vector $\mathbf{r}$,

$$L_{\sigma_k}(E_Q(\rho)) = r_k.$$

Proof. We have $\sigma_k = \sqrt{2} B_k$, hence

$$v_{\sigma_k} = U(\sigma_k) = U(\sqrt{2} B_k) = \sqrt{2} U(B_k) = \sqrt{2} e_k.$$

For $x = E_Q(\rho)$ as in (7.2),

$$L_{\sigma_k}(x) = \langle x, v_{\sigma_k} \rangle_S = \left\langle \frac{1}{\sqrt{2}}e_0 + \frac{1}{\sqrt{2}}\sum_j r_j e_j, \sqrt{2} e_k \right\rangle_S = \sum_j r_j \langle e_j, e_k \rangle_S = r_k.$$

By the earlier proposition this coincides with $\text{tr}(\rho\sigma_k)$. $\square$

Remark 7.6. The triple $(v_{\sigma_x}, v_{\sigma_y}, v_{\sigma_z})$ spans the subspace of S_Q corresponding to traceless observables. Together with $v_{\sigma_0} = U(\sigma_0) = \sqrt{2} e_0$, these vectors form an orthogonal basis of S_Q aligned with the identity and Pauli operators.

7.1.4 Unitary rotations and induced dynamics on S_Q

We now illustrate the induced dynamics on S_Q for a simple spin Hamiltonian.

Definition 7.7 (Spin Hamiltonian and Bloch evolution). Let

$$H = \frac{\hbar\omega}{2} \mathbf{n} \cdot \boldsymbol{\sigma},$$

where $\mathbf{n} \in \mathbb{R}^3$ is a unit vector. The Schrödinger evolution generated by H is

$$U_t = e^{-itH/\hbar} = e^{-i\frac{\omega t}{2} \mathbf{n} \cdot \boldsymbol{\sigma}}.$$

If $\rho_t = U_t \rho_0 U_t^*$, then the Bloch vector $\mathbf{r}_t$ of ρ_t evolves as

$$\mathbf{r}_t = R_t \mathbf{r}_0,$$

where $R_t \in \text{SO}(3)$ is the rotation about axis $\mathbf{n}$ by angle ωt .

On the Hilbert–Schmidt space $\mathcal{H}_Q$, the Liouvillian dynamics acts as

$$\mathcal{T}_t(X) = U_t X U_t^*.$$

In the basis $\{B_0, B_x, B_y, B_z\}$ this yields

$$\mathcal{T}_t(B_0) = B_0, \quad \mathcal{T}_t(\mathbf{a} \cdot \mathbf{B}) = (R_t \mathbf{a}) \cdot \mathbf{B},$$

where $\mathbf{B} = (B_x, B_y, B_z)$ and $\mathbf{a} \in \mathbb{R}^3$.

Proposition 7.8 (Induced rotation on S_Q). *Let $\mathcal{U}_t : S_Q \rightarrow S_Q$ be the induced unitary group defined in Chapter 5. Then*

$$\mathcal{U}_t(e_0) = e_0, \quad \mathcal{U}_t\left(\sum_k a_k e_k\right) = \sum_k (R_t \mathbf{a})_k e_k,$$

where $\mathbf{a} = (a_x, a_y, a_z)$. In particular, if $x_0 = E_Q(\rho_0)$ has Bloch vector $\mathbf{r}_0$, then

$$x_t := \mathcal{U}_t x_0 = E_Q(\rho_t) = \frac{1}{\sqrt{2}} e_0 + \frac{1}{\sqrt{2}} \sum_k (R_t \mathbf{r}_0)_k e_k.$$

Proof. By definition $\mathcal{U}_t = U \mathcal{T}_t U^{-1}$. Using $\mathcal{T}_t(B_0) = B_0$ and $\mathcal{T}_t(\mathbf{a} \cdot \mathbf{B}) = (R_t \mathbf{a}) \cdot \mathbf{B}$,

$$\mathcal{U}_t(e_0) = U \mathcal{T}_t U^{-1}(e_0) = U \mathcal{T}_t(B_0) = U(B_0) = e_0,$$

and similarly for the traceless sector:

$$\mathcal{U}_t\left(\sum_k a_k e_k\right) = U \mathcal{T}_t\left(\sum_k a_k B_k\right) = U\left((R_t \mathbf{a}) \cdot \mathbf{B}\right) = \sum_k (R_t \mathbf{a})_k e_k.$$

The expression for x_t follows by applying this to x_0 in (7.2). $\square$

Theorem 7.9 (Expectation compatibility for spin- $\frac{1}{2}$). *For any initial state ρ_0 with Bloch vector $\mathbf{r}_0$ and any Pauli observable σ_k ,*

$$L_{\sigma_k}(x_t) = \text{tr}(\rho_t \sigma_k) = (R_t \mathbf{r}_0)_k, \quad t \in \mathbb{R},$$

where $x_t = \mathcal{U}_t E_Q(\rho_0)$.

Proof. We have $x_t = E_Q(\rho_t)$ and hence

$$L_{\sigma_k}(x_t) = L_{\sigma_k}(E_Q(\rho_t)) = \text{tr}(\rho_t \sigma_k)$$

by the general compatibility result of Chapter 5. Using the Bloch evolution $\mathbf{r}_t = R_t \mathbf{r}_0$ and the expression $\text{tr}(\rho_t \sigma_k) = r_{t,k}$ yields the final equality. $\square$

This example shows explicitly how the embedding, observable vectors and induced dynamics work together to reproduce the standard Bloch-sphere picture inside S_Q .

7.2 Harmonic oscillator (finite-dimensional truncation)

We now sketch the embedding and induced dynamics for the one-dimensional quantum harmonic oscillator, using a finite-dimensional truncation of the Hilbert space for concreteness.

7.2.1 Quantum harmonic oscillator and truncation

Definition 7.10 (Harmonic oscillator Hilbert space and Hamiltonian). The one-dimensional harmonic oscillator is defined on

$$H_Q := L^2(\mathbb{R}),$$

with canonical position and momentum operators (q, p) satisfying $[q, p] = i\hbar$, and Hamiltonian

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2.$$

Let $(|n\rangle)_{n \in \mathbb{N}_0}$ be the eigenbasis of H , with

$$H|n\rangle = E_n|n\rangle, \quad E_n = \hbar\omega\left(n + \frac{1}{2}\right).$$

To obtain explicit finite-dimensional expressions, we truncate the Hilbert space.

Definition 7.11 (Finite-dimensional truncation). For $N \in \mathbb{N}$, define the subspace

$$H_Q^{(N)} := \text{span}\{|0\rangle, \dots, |N-1\rangle\} \cong \mathbb{C}^N,$$

and let P_N denote the orthogonal projection onto $H_Q^{(N)}$. The truncated Hamiltonian is

$$H^{(N)} := P_N H P_N,$$

which is diagonal in the basis $\{|n\rangle\}_{n=0}^{N-1}$ with eigenvalues E_n , $n = 0, \dots, N-1$.

A truncated density operator is a positive trace-class operator $\rho^{(N)}$ supported on $H_Q^{(N)}$ with $\text{tr}(\rho^{(N)}) = 1$. It can be represented as an $N \times N$ matrix in the energy eigenbasis.

7.2.2 Embedding of truncated density operators

The Hilbert–Schmidt space on $H_Q^{(N)}$ is the space of complex $N \times N$ matrices equipped with

$$\langle X, Y \rangle_{\text{HS}} := \text{tr}(X^* Y).$$

Let E_{mn} denote the matrix units (with a 1 in the (m, n) -entry and 0 elsewhere), $0 \leq m, n \leq N-1$. Then $(E_{mn})_{m,n}$ is an orthonormal basis of $\mathcal{H}_{\text{HS}}(H_Q^{(N)})$.

Definition 7.12 (Concrete N -level embedding). For the truncated N -level system we choose the embedded quantum subspace $S_Q^{(N)} \subset S$ and a unitary

$$U^{(N)} : \mathcal{H}_Q^{(N)} \rightarrow S_Q^{(N)}$$

such that

$$U^{(N)}(E_{mn}) = f_{mn}, \quad 0 \leq m, n \leq N-1,$$

where $(f_{mn})_{m,n}$ is an orthonormal basis of $S_Q^{(N)}$. The embedding

$$E_Q^{(N)} : \mathcal{D}(H_Q^{(N)}) \rightarrow S_Q^{(N)}$$

is then given by

$$E_Q^{(N)}(\rho^{(N)}) := U^{(N)}(\rho^{(N)}).$$

If $\rho^{(N)}$ has matrix elements $\rho_{mn} = \langle m | \rho^{(N)} | n \rangle$, then

$$\rho^{(N)} = \sum_{m,n=0}^{N-1} \rho_{mn} E_{mn}, \quad E_Q^{(N)}(\rho^{(N)}) = \sum_{m,n=0}^{N-1} \rho_{mn} f_{mn}. \quad (7.3)$$

Proposition 7.13 (Isometry for the truncated oscillator). *For any $\rho^{(N)}, \sigma^{(N)} \in \mathcal{D}(H_Q^{(N)})$,*

$$\langle E_Q^{(N)}(\rho^{(N)}), E_Q^{(N)}(\sigma^{(N)}) \rangle_S = \text{tr}(\rho^{(N)} \sigma^{(N)}).$$

Proof. Using (7.3) and orthonormality of $(f_{mn})_{m,n}$,

$$\langle E_Q^{(N)}(\rho^{(N)}), E_Q^{(N)}(\sigma^{(N)}) \rangle_S = \sum_{m,n} \overline{\rho_{mn}} \sigma_{mn} = \text{tr}((\rho^{(N)})^* \sigma^{(N)}),$$

and for density matrices $(\rho^{(N)})^* = \rho^{(N)}$. This coincides with the Hilbert–Schmidt inner product, which equals $\text{tr}(\rho^{(N)} \sigma^{(N)})$ for positive operators. $\square$

7.2.3 Diagonal (thermal) states and their embedding

A particularly important class of states are diagonal in the energy eigenbasis. For inverse temperature $\beta > 0$, the (truncated) thermal state is

$$\rho_\beta^{(N)} = \frac{1}{Z_N} \sum_{n=0}^{N-1} e^{-\beta E_n} |n\rangle\langle n|, \quad Z_N := \sum_{n=0}^{N-1} e^{-\beta E_n}.$$

In matrix form,

$$\rho_\beta^{(N)} = \sum_{n=0}^{N-1} p_n E_{nn}, \quad p_n := \frac{e^{-\beta E_n}}{Z_N}.$$

Proposition 7.14 (Embedding of a truncated thermal state). *The embedded truncated thermal state is*

$$E_Q^{(N)}(\rho_\beta^{(N)}) = \sum_{n=0}^{N-1} p_n f_{nn},$$

with norm

$$\|E_Q^{(N)}(\rho_\beta^{(N)})\|_S^2 = \text{tr}((\rho_\beta^{(N)})^2) = \sum_{n=0}^{N-1} p_n^2.$$

Proof. The expansion follows from (7.3) and the diagonal form of $\rho_\beta^{(N)}$. The norm identity is the general relation $\|E_Q^{(N)}(\rho)\|_S^2 = \text{tr}(\rho^2)$ for density matrices in the Hilbert–Schmidt embedding. $\square$

7.2.4 Observables and expectations in the truncated sector

Let $A^{(N)}$ be a self-adjoint operator on $H_Q^{(N)}$, represented by an $N \times N$ matrix with entries A_{mn} . The associated semantic observable vector is

$$v_{A^{(N)}} = U^{(N)}(A^{(N)}) = \sum_{m,n=0}^{N-1} A_{mn} f_{mn},$$

and the functional $L_{A^{(N)}}$ is

$$L_{A^{(N)}}(x) = \langle x, v_{A^{(N)}} \rangle_S, \quad x \in S_Q^{(N)}.$$

Proposition 7.15 (Expectation compatibility in the truncated oscillator). *For any truncated density matrix $\rho^{(N)}$ and observable $A^{(N)}$,*

$$L_{A^{(N)}}(E_Q^{(N)}(\rho^{(N)})) = \text{tr}(\rho^{(N)} A^{(N)}).$$

In particular, for the truncated Hamiltonian $H^{(N)}$,

$$L_{H^{(N)}}(E_Q^{(N)}(\rho_\beta^{(N)})) = \text{tr}(\rho_\beta^{(N)} H^{(N)})$$

is the standard thermal energy expectation.

Proof. Using the explicit expansions and orthonormality of $(f_{mn})_{m,n}$,

$$L_{A^{(N)}}(E_Q^{(N)}(\rho^{(N)})) = \left\langle \sum_{m,n} \rho_{mn} f_{mn}, \sum_{k,\ell} A_{k\ell} f_{k\ell} \right\rangle_S = \sum_{m,n} \rho_{mn} \overline{A_{mn}} = \text{tr}(\rho^{(N)} A^{(N)}),$$

since $A^{(N)}$ is self-adjoint. The statement for $H^{(N)}$ is obtained by substituting $A^{(N)} = H^{(N)}$. $\square$

7.2.5 Induced dynamics on the truncated sector

The Hamiltonian $H^{(N)}$ is diagonal in the basis $\{|n\rangle\}_{n=0}^{N-1}$, so the truncated unitary dynamics is

$$U_t^{(N)} := e^{-itH^{(N)}/\hbar} = \sum_{n=0}^{N-1} e^{-itE_n/\hbar} |n\rangle\langle n|.$$

The induced evolution of a truncated density matrix is

$$\rho_t^{(N)} = U_t^{(N)} \rho_0^{(N)} (U_t^{(N)})^*.$$

Proposition 7.16 (Action on matrix units). *For $0 \leq m, n \leq N-1$,*

$$\rho_0^{(N)} = E_{mn} \implies \rho_t^{(N)} = e^{-i\omega t(m-n)} E_{mn},$$

where ω is the oscillator frequency.

Proof. Using $U_t^{(N)}|n\rangle = e^{-itE_n/\hbar}|n\rangle$ and $E_n = \hbar\omega(n + \frac{1}{2})$,

$$U_t^{(N)} E_{mn} (U_t^{(N)})^* = e^{-it(E_m - E_n)/\hbar} E_{mn} = e^{-i\omega t(m-n)} E_{mn}.$$

$\square$

On the embedded side, let $\mathcal{U}_t^{(N)} : S_Q^{(N)} \rightarrow S_Q^{(N)}$ be the induced unitary group defined via

$$\mathcal{U}_t^{(N)} = U^{(N)} \circ \mathcal{T}_t^{(N)} \circ (U^{(N)})^{-1},$$

where $\mathcal{T}_t^{(N)}(X) := U_t^{(N)} X (U_t^{(N)})^*$. Then

$$\mathcal{U}_t^{(N)}(f_{mn}) = U^{(N)}(\mathcal{T}_t^{(N)}(E_{mn})) = U^{(N)}(e^{-i\omega t(m-n)} E_{mn}) = e^{-i\omega t(m-n)} f_{mn}.$$

Proposition 7.17 (Embedded oscillator dynamics). *Let $x_0 = E_Q^{(N)}(\rho_0^{(N)}) = \sum_{m,n} \rho_{mn}(0) f_{mn}$. Then*

$$x_t := \mathcal{U}_t^{(N)} x_0 = \sum_{m,n=0}^{N-1} \rho_{mn}(0) e^{-i\omega t(m-n)} f_{mn} = E_Q^{(N)}(\rho_t^{(N)}).$$

For any observable $A^{(N)}$,

$$L_{A^{(N)}}(x_t) = \text{tr}(\rho_t^{(N)} A^{(N)})$$

for all t .

Proof. The first identity follows from the action of $\mathcal{U}_t^{(N)}$ on the basis $(f_{mn})_{m,n}$. The relation $x_t = E_Q^{(N)}(\rho_t^{(N)})$ is the embedded version of the von Neumann evolution. The expectation compatibility follows from the general result in Chapter 5 applied to the truncated system. $\square$

Remark 7.18. For diagonal initial states $\rho_0^{(N)} = \sum_n p_n E_{nn}$ (such as the truncated thermal state), the density matrix is invariant under the unitary dynamics, and so is its embedding:

$$\rho_t^{(N)} = \rho_0^{(N)}, \quad x_t = E_Q^{(N)}(\rho_0^{(N)}) \quad \text{for all } t.$$

Off-diagonal elements pick up phase factors $e^{-i\omega t(m-n)}$, which correspond to rotations in suitable two-dimensional subspaces of $S_Q^{(N)}$ spanned by $\{f_{mn}, f_{nm}\}$.

7.2.6 Comments on extensions

The truncated construction above can be extended in several directions:

- Increasing N makes the truncation finer and approximates the full harmonic oscillator. At each finite N , the embedding, observable vectors and induced dynamics are fully explicit.
- Non-diagonal initial states (e.g. superpositions of low-lying energy eigenstates) are embedded by assigning coefficients to the corresponding f_{mn} . Their dynamics in $S_Q^{(N)}$ is governed by phase factors $e^{-i\omega t(m-n)}$, directly mirroring the quantum evolution.
- Other observables, such as the truncated position and momentum operators, can be treated similarly: their matrix elements in the $\{|n\rangle\}$ basis determine the semantic observable vectors via $v_A = U^{(N)}(A)$.

These examples demonstrate concretely how the abstract constructions of Chapters 3–5 operate in familiar quantum systems, and how the embedding E_Q , the observable vectors v_A , and the induced dynamics on S_Q reproduce standard quantum expectations and time evolution within the semantic Hilbert space framework.

8 Discussion, limitations and outlook

8.1 Summary of structural results

The main goal of this work was to realize standard quantum mechanics—in particular density operators, observables, and their dynamics—inside the semantic Hilbert space framework based on a separable Hilbert space S equipped with a potential W and associated energetic and gradient-flow structures.

The core constructions and results can be summarized as follows.

8.1.1 Embedding of quantum states

Starting from a separable quantum Hilbert space H_Q , we considered the space $\mathcal{D}(H_Q)$ of density operators and its realization as a subset of the Hilbert–Schmidt space $\mathcal{H}_{\text{HS}}(H_Q)$. The closed linear span of density operators,

$$\mathcal{H}_Q := \overline{\text{span}}\{\rho : \rho \in \mathcal{D}(H_Q)\} \subset \mathcal{H}_{\text{HS}}(H_Q),$$

is itself a separable Hilbert space with the Hilbert–Schmidt inner product.

Because S is also a separable Hilbert space, there exists a closed subspace $S_Q \subset S$ and a unitary isomorphism

$$U : \mathcal{H}_Q \rightarrow S_Q.$$

This unitary intertwiner allows us to define an explicit embedding

$$E_Q : \mathcal{D}(H_Q) \rightarrow S_Q \subset S, \quad E_Q(\rho) := U(\rho).$$

The image

$$\Omega_Q := E_Q(\mathcal{D}(H_Q)) \subset S_Q$$

is a convex subset of S_Q contained in the closed unit ball, with pure states mapping to unit vectors and mixed states to vectors of strictly smaller norm. The embedding is isometric with respect to the Hilbert–Schmidt inner product:

$$\langle E_Q(\rho), E_Q(\sigma) \rangle_S = \langle \rho, \sigma \rangle_{\text{HS}} = \text{tr}(\rho\sigma),$$

so the Hilbert–Schmidt geometry of density operators is faithfully realized in S_Q .

8.1.2 Observables as linear functionals

For each Hilbert–Schmidt observable A (a self-adjoint Hilbert–Schmidt operator on H_Q), we constructed a continuous linear functional

$$L_A : S_Q \rightarrow \mathbb{C}$$

and a corresponding semantic observable vector

$$v_A \in S_Q$$

such that

$$L_A(x) = \langle x, v_A \rangle_S \quad \text{for all } x \in S_Q,$$

and, crucially,

$$L_A(E_Q(\rho)) = \text{tr}(\rho A) \quad \text{for all } \rho \in \mathcal{D}(H_Q).$$

Explicitly,

$$v_A = U(A),$$

so that $\langle E_Q(\rho), v_A \rangle_S$ reproduces the quantum expectation of A in state ρ . The map $A \mapsto L_A$ is linear, and positivity of observables is reflected in positivity of L_A on embedded states:

$$A \geq 0 \implies L_A(E_Q(\rho)) \geq 0 \quad \forall \rho \in \mathcal{D}(H_Q).$$

In finite dimensions, this construction applies to all bounded self-adjoint operators. In infinite dimensions, it applies cleanly to Hilbert–Schmidt observables, with possible extensions to more general classes discussed below.

8.1.3 Induced dynamics and compatibility

Given a Hamiltonian H on H_Q , the von Neumann dynamics

$$i\hbar \frac{d}{dt} \rho_t = [H, \rho_t], \quad \rho_t = U_t \rho_0 U_t^*,$$

induces a unitary group

$$\mathcal{T}_t : \mathcal{H}_Q \rightarrow \mathcal{H}_Q, \quad \mathcal{T}_t(X) := U_t X U_t^*,$$

on the operator Hilbert space. Conjugating by U yields a unitary group

$$\mathcal{U}_t := U \circ \mathcal{T}_t \circ U^{-1} : S_Q \rightarrow S_Q$$

with generator

$$\mathcal{K}_H := U \circ \mathcal{L}_H \circ U^{-1}, \quad \mathcal{L}_H(X) := -\frac{i}{\hbar} [H, X].$$

Embedded quantum states evolve according to

$$x_t := E_Q(\rho_t) = \mathcal{U}_t x_0, \quad x_0 := E_Q(\rho_0),$$

and expectations remain compatible along the flow:

$$L_A(x_t) = \text{tr}(\rho_t A), \quad \forall A, \rho_0, t.$$

The same structure extends to suitable quantum Markov semigroups $(\Phi_t)_{t \geq 0}$, which induce contraction semigroups $(\mathcal{V}_t)_{t \geq 0}$ on S_Q preserving the expectation compatibility.

8.1.4 Relation to the semantic potential and free energy

The semantic potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ restricts to

$$W_Q := W|_{S_Q},$$

and, via the embedding, to a function on density operators,

$$W_Q(\rho) := W(E_Q(\rho)).$$

Under the natural assumption that W_Q is invariant along unitary orbits in $\mathcal{D}(H_Q)$, W_Q depends only on the spectrum of ρ , i.e. on unitarily invariant features of the quantum state.

At the level of probability measures on S , the semantic free-energy functional defines gradient flows and stochastic dynamics. For measures supported on Ω_Q , this yields a framework for describing ensembles of quantum states within S , combining the conservative unitary component (from $\mathcal{U}_t$) with dissipative or noise-induced evolution driven by W and semantic free energy.

8.2 Limitations and open structural issues

The constructions in this paper are intentionally conservative and structural. Several important limitations and open issues remain, which we now summarize.

8.2.1 Non-uniqueness of the embedding

The embedding

$$E_Q : \mathcal{D}(H_Q) \rightarrow S_Q \subset S$$

is defined via the choice of a unitary map

$$U : \mathcal{H}_Q \rightarrow S_Q.$$

This choice is far from unique: any unitary V on S_Q yields a different embedding

$$\tilde{E}_Q(\rho) := V(E_Q(\rho)),$$

with correspondingly transformed observable vectors

$$\tilde{v}_A := V(v_A).$$

All such embeddings reproduce the same quantum expectations, since

$$\langle \tilde{E}_Q(\rho), \tilde{v}_A \rangle_S = \langle E_Q(\rho), v_A \rangle_S.$$

From the standpoint of reproducing standard quantum mechanics, this non-uniqueness is harmless: different choices correspond to internal unitary changes of basis in S_Q . However, if one wishes to relate the embedded quantum sector more tightly to additional structure on S (for example, to the geometry of W or to specific semantic interpretations of S), it becomes important to distinguish between embeddings. Criteria for selecting a “canonical” or “natural” U are not addressed in this work and remain an open question. Potential conditions include:

- Compatibility with tensor product structures and locality for composite systems.
- Covariance with respect to symmetry groups acting on H_Q and S .
- Alignment with specific features of the potential W or of the semantic free-energy landscape.

8.2.2 Restriction to Hilbert–Schmidt observables

The use of the Hilbert–Schmidt inner product as the basic geometric structure behind the embedding naturally leads to representing observables A that are Hilbert–Schmidt. In finite dimensions this covers all self-adjoint operators, but in infinite dimensions many physically relevant observables (such as unbounded position or momentum operators, or certain bounded but non-Hilbert–Schmidt operators) fall outside this class.

For such observables, the map

$$\rho \mapsto \text{tr}(\rho A)$$

is typically continuous with respect to the trace norm on $\mathcal{T}_1(H_Q)$, but need not extend to a continuous functional on $\mathcal{H}_Q$ with respect to the Hilbert–Schmidt norm. As a consequence:

- The simple identification $v_A = U(A)$ via the Hilbert–Schmidt isometry may fail.
- The functional L_A may, at best, be defined on a dense subspace of S_Q , or via a more general duality pairing that differs from the S -inner product.

Addressing these issues would require either replacing $\mathcal{H}_Q$ by a different operator Hilbert space (for example, based on a weighted Hilbert–Schmidt norm) or working in a Banach-space framework tailored to trace-class operators and bounded observables. This lies beyond the scope of the present paper.

8.2.3 Infinite-dimensional and domain subtleties

In infinite-dimensional settings, several technical aspects require more careful treatment than provided here:

- The Liouvillian $\mathcal{L}_H(X) = -\frac{i}{\hbar}[H, X]$ may be unbounded on $\mathcal{H}_Q$, and its domain may not contain all Hilbert–Schmidt operators. We have assumed that $\mathcal{L}_H$ generates a strongly continuous unitary group on $\mathcal{H}_Q$, but a detailed analysis of domain questions and generator properties has been omitted.
- The space $\mathcal{H}_Q$ may be strictly smaller than $\mathcal{H}_{\text{HS}}(H_Q)$, depending on the structure of $\mathcal{D}(H_Q)$ and its linear span. We have worked with $\mathcal{H}_Q$ as given but have not characterized it in general.
- The potential W was assumed to be Fréchet differentiable in a neighbourhood of Ω_Q , with locally bounded gradient. While natural in the semantic framework, verifying these properties for specific W and specific embeddings E_Q may be non-trivial.

A fully rigorous treatment would require a more detailed functional-analytic analysis, particularly for unbounded Hamiltonians and infinite-dimensional H_Q .

8.2.4 No derivation or modification of quantum mechanics

Conceptually, it is important to emphasize that the constructions in this paper do *not* modify the standard quantum formalism and do *not* derive it from the semantic Hilbert space framework. Quantum mechanics is taken as given:

- H_Q , $\mathcal{D}(H_Q)$ and the operator representation of observables are assumed.
- The von Neumann and Lindblad-type equations are used as the starting point for dynamics.

The semantic Hilbert space S serves as a host in which these structures can be realized faithfully, but it is not claimed to be more fundamental in any operational or physical sense within this paper. Any attempt to use the semantic framework to explain or derive quantum postulates would require additional axioms and interpretative work that are explicitly excluded from the present scope.

8.2.5 Partial treatment of dissipative and stochastic dynamics

While we have sketched how quantum Markov semigroups and open-system dynamics can be represented on S_Q , and how they structurally relate to the gradient and stochastic dynamics driven by W and semantic free energy on S , this connection has not been developed in full detail. In particular:

- We have not provided an explicit representation of Lindblad generators in terms of S_Q -covariant structures linked to W or to a semantic free-energy functional.
- The decomposition of dynamics into Hamiltonian and gradient components on S_Q has been discussed at a programmatic level, without a fully worked-out example in the infinite-dimensional setting.

- The probabilistic dynamics of ensembles of embedded quantum states under the combined effect of $\mathcal{U}_t$ and semantic free-energy gradients has only been qualitatively described.

These aspects indicate directions for future work rather than completed results.

8.3 Outlook and future directions

Despite the above limitations, the constructions of this paper open several concrete avenues for further mathematical investigation.

8.3.1 Classification and selection of embeddings

A first line of work is to better understand the space of possible embeddings

$$E_Q : \mathcal{D}(H_Q) \rightarrow S$$

and to identify meaningful selection principles among them. Questions include:

- *Symmetry-adapted embeddings:* Given a symmetry group G acting unitarily on H_Q , can one choose E_Q such that the induced action on S_Q is particularly simple or compatible with a symmetry of W ?
- *Tensorial compatibility:* For composite systems with $H_Q = H_{Q,1} \otimes H_{Q,2}$, can one construct embeddings $E_{Q,1}$ and $E_{Q,2}$ such that the embedding of $\mathcal{D}(H_Q)$ is related in a controlled way to the embeddings of $\mathcal{D}(H_{Q,1})$ and $\mathcal{D}(H_{Q,2})$?
- *Minimality and regularity:* Are there embeddings that minimize certain functionals (e.g. norms, curvature-like quantities, or measures of complexity) on S_Q , making them distinguished from a geometric viewpoint?

Even at the purely operator-theoretic level, these questions have non-trivial content and could lead to a systematic classification of compatible embeddings into S .

8.3.2 Coupling Hamiltonian and gradient dynamics

A second direction concerns the interplay between the Hamiltonian dynamics on S_Q generated by $\mathcal{K}_H$ and the gradient dynamics on S generated by W (and by the associated free-energy functional on probability measures). Structurally, this suggests investigating equations of the form

$$\dot{x}_t = \mathcal{K}_H x_t - \nabla W(x_t)$$

or their stochastic analogues, with initial conditions $x_0 \in \Omega_Q$.

Questions include:

- Under what conditions on W and on the embedding E_Q does the combined dynamics preserve (or approximately preserve) an embedded quantum sector for relevant timescales?
- Can certain classes of GKLS generators be realized as effective dynamics arising from the coupling of $\mathcal{K}_H$ and $-\nabla W$ when projected onto S_Q ?
- How does the semantic free-energy functional restricted to ensembles supported on Ω_Q relate to standard entropy and free-energy concepts in quantum statistical mechanics?

These questions connect the semantic energetic framework to open-system quantum dynamics and may provide a unified structural language for conservative and dissipative evolution.

8.3.3 Geometric structures and information geometry

The Hilbert–Schmidt inner product underlying the embedding gives rise to a natural kernel

$$k(\rho, \sigma) = \text{tr}(\rho\sigma),$$

and, more generally, to a reproducing kernel Hilbert space structure on $\mathcal{D}(H_Q)$. Beyond this, quantum state spaces carry other geometries, such as the trace distance, Bures distance, and quantum Fisher information metrics.

Possible directions include:

- Investigating whether alternative embeddings into S can be constructed that are isometric (or approximately isometric) with respect to other quantum distances or metrics.
- Studying the geometry of Ω_Q in S under the pullback of such metrics, and its relation to known information-geometric structures on $\mathcal{D}(H_Q)$.
- Exploring whether the semantic potential W or semantic free energy can be related to divergences or contrast functionals on $\mathcal{D}(H_Q)$ familiar from quantum information theory.

This may help connect the semantic Hilbert space framework to the rich landscape of quantum information geometry.

8.3.4 Many-body systems, fields, and truncations

The examples in Chapter 6 focused on a single spin- $\frac{1}{2}$ system and on finite-dimensional truncations of the harmonic oscillator. Extending the construction to larger systems and field-theoretic settings raises further questions:

- How do tensor-product structures $H_Q^{\otimes n}$ manifest inside S , and how do the corresponding embedded subspaces $S_Q^{(n)}$ interact?
- For field-theoretic systems where H_Q is infinite-dimensional and separable but effectively constructed through truncations or lattice approximations, how do the corresponding embeddings behave under refinement of the truncation?
- Can one define continuous limits of $S_Q^{(N)}$ as $N \rightarrow \infty$ for sequences of finite-dimensional truncations, such that expectations and dynamics converge appropriately?

Even at the level of non-interacting many-body systems, the combinatorics and geometry of the embedded sectors inside S may exhibit interesting scaling and concentration phenomena.

8.3.5 Ensembles of embedded quantum states

Finally, the representation of quantum states as points in S_Q and of ensembles as probability measures on Ω_Q suggests a variety of questions at the level of random quantum states and their evolution:

- How do typical properties of random density operators (e.g. induced distributions of eigenvalues) translate into geometric properties of random points in $\Omega_Q \subset S$?
- How does the semantic free-energy gradient flow act on probability distributions supported on Ω_Q , and can this dynamics be related to known notions of quantum equilibration or decoherence?
- Are there natural concentration-of-measure phenomena for large quantum systems that become geometrically transparent in the embedding into S ?

Addressing these questions would provide a bridge between the semantic framework and statistical properties of high-dimensional quantum systems.

8.4 Concluding remarks

This paper has shown that standard quantum mechanics, formulated in terms of density operators, observables, and their unitary and Markovian dynamics, can be realized as a mathematically consistent sector within the semantic Hilbert space setting based on S and W . The embedding E_Q , the representation of observables as linear functionals L_A on S_Q , and the induced dynamics $(\mathcal{U}_t)$ collectively ensure that quantum expectations and time evolution are exactly reproduced on the embedded subspace.

At the same time, the semantic framework provides additional energetic and dynamical structures—most notably the potential W and the associated free-energy gradient flows on probability measures over S —that are not present in the standard quantum formalism. The present work has treated these structures as background and has focused on compatibility, rather than on interpretation or derivation. The resulting picture is therefore one of embedding and coexistence: quantum theory appears as a well-defined, invariant sector inside a larger Hilbert-space-based dynamical framework.

Future work will determine to what extent this coexistence can be sharpened, whether the interaction between the Hamiltonian and gradient components of dynamics can be exploited to model dissipative quantum phenomena, and how the geometry of Ω_Q inside S relates to established structures in quantum information and statistical mechanics. These questions are mathematical in nature and do not presuppose any particular physical or interpretive stance on the semantic framework; they simply use it as a rich ambient space in which quantum states and their dynamics can be embedded, studied, and potentially generalized.

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The work emerged from a prolonged process of intuitive exploration, conceptual reduction, and formal consolidation. Informal insights about meaning, structure, and stability were progressively translated into explicit models, mathematical frameworks, and testable

claims. Throughout this process, modern large language models were employed as practical instruments for structuring arguments, checking internal consistency, exploring alternative formulations, and iteratively refining formal expressions.

These tools were used as epistemic scaffolding rather than as sources of claims. All theoretical commitments, modelling decisions, and interpretative judgments remain the responsibility of the author. The resulting framework should therefore be understood as exploratory but precise: it proposes a concrete formal scaffold, derives explicit consequences, and invites scrutiny.

The work is explicitly offered for critical examination, clarification, extension, or refutation by domain experts. Questions, comments, corrections, and collaboration inquiries are welcome and may be directed to contact@semanticgravitation.org.

Semantic Gravitation IV: Semantic Fields on Spacetime Manifolds — An Effective Field Theory for Coherence

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Abstract

We develop a classical semantic field theory on a fixed globally hyperbolic spacetime (M, g) that extends the Hilbert-space and free-energy formalisms of Papers I and II to a covariant, field-theoretic setting. A *semantic field* is modelled as a map

$$\Phi: M \longrightarrow S,$$

where S is the separable Hilbert space of semantic states. The dynamics of Φ are obtained from an action functional

$$\mathcal{S}[\Phi] = \int_M \mathcal{L}(x, \Phi(x), \nabla \Phi(x)) \, \mathrm{dvol}_g(x),$$

with Lorentzian metric g and Lagrangian density $\mathcal{L}$ of the form “kinetic minus potential”. The kinetic term is induced by g and the inner product on S , while the potential encodes local semantic structure via a family of semantic potentials $W_x: S \rightarrow \mathbb{R}$.

We formulate the Euler–Lagrange equations associated with $\mathcal{S}$ as nonlinear hyperbolic partial differential equations on (M, g) ,

$$\square_g \Phi + \mathcal{N}(\Phi, \nabla \Phi) + \nabla W_x(\Phi) = 0,$$

identify their stress–energy tensor, and discuss the resulting energy and flux balances. Under standard regularity and growth assumptions on $\mathcal{L}$ and W_x , we establish a local well-posedness result for the Cauchy problem on globally hyperbolic spacetimes and obtain conservation or decay laws for natural energy functionals. Spatially homogeneous solutions reduce to the Hilbert-space gradient systems of Paper I, while the distribution of field values across spacelike hypersurfaces induces probability measures on S whose evolution is asymptotically governed by the free-energy gradient flows on $\mathcal{P}_2(S)$ studied in Paper II.

The formalism is illustrated by several model Lagrangians, including Klein–Gordon-type systems with nonlinear semantic potentials, multi-component fields coupled through interaction terms, and simple curved-spacetime examples. We emphasise the status of the theory as an *effective* description: (M, g) is treated as a fixed background, and the semantic field Φ captures coarse-grained, macroscopic semantic structure rather than microscopic degrees of freedom.

Conceptually, this paper supplies the spacetime layer of the Semantic Gravitation programme: it shows how semantic dynamics can be expressed as a local, covariant field theory on (M, g) , compatible with the Hilbert-space and free-energy structures of Papers I and II, and ready to interface with the quantum-compatible semantic sector of Paper III. The spacetime framework developed here will subsequently be used in

Paper V to model complex neural and social systems as interacting semantic field theories, and in Paper VI to motivate the global LOVE Lagrangian and the notion of a semantic attractor spacetime. For a programme-level introduction to Semantic Gravitation and the Lagrangian of Vectorial Emergence (LOVE), we refer to the separate overview paper.

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1 Introduction

1.1 Background and motivation

Classical field theory on a Lorentzian spacetime (M, g) usually considers fields taking values in finite-dimensional target spaces, such as $\mathbb{R}$, $\mathbb{C}$, or a finite-dimensional vector space associated with a representation of some internal symmetry group. The corresponding dynamical laws are then obtained from an action principle built from a Lagrangian density depending on the field and its covariant derivatives, integrated over spacetime with respect to the volume form induced by g . In this setting, the geometry of (M, g) encodes causal structure and gravitation, while the fields represent matter or gauge degrees of freedom.

In many applications, however, one encounters structures that are naturally modelled in infinite-dimensional spaces. Examples include wave functionals in quantum field theory, probability measures on function spaces, and order parameters for extended systems. A mathematically clean and flexible way to capture such situations is to work with a separable Hilbert space S as an abstract carrier of degrees of freedom, equipped with its inner product $\langle \cdot, \cdot \rangle_S$ and norm $\| \cdot \|_S$. On S one can introduce functionals $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ that play the role of potentials or free energies, and study their gradients, Hessians, and associated flows.

In a previous series of works, a programme was developed in which S is interpreted as a *semantic* Hilbert space and the functional W as a potential encoding coherence or compatibility structure on S . The first paper in this programme introduced the notion of *semantic gradient fields* on S and analysed gradient flows generated by W . The second paper reformulated these constructions in terms of a *semantic free energy*, clarifying the relationship between W , distribution dynamics on S , and notions of equilibrium. The third paper showed how standard quantum systems can be embedded into the same Hilbert space S in a way that is compatible with the semantic structure. In all three cases, the geometry of S and the properties of W were central, while spacetime did not play an explicit role.

The aim of the present work is to bring these strands together with classical field theory on (M, g) . Instead of fields with values in $\mathbb{R}$ or $\mathbb{C}$, we consider fields

$$s : M \rightarrow S$$

with values in the same separable Hilbert space S that appears in the earlier papers. The field s is assumed to be sufficiently regular so that one can define its covariant derivative $\nabla_\mu s$ with respect to the Levi-Civita connection of g , and hence its kinetic energy density via the inner product on S . The potential W is imported from the previous works, subject to the same structural assumptions (convexity or quasiconvexity where needed, suitable smoothness on a dense domain, and growth conditions ensuring good variational properties).

On this basis, we construct a Lorentz-covariant Lagrangian field theory on the fixed background spacetime (M, g) with Lagrangian density

$$\mathcal{L}(s, \nabla s; g) = \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s),$$

and corresponding action functional

$$S[s] = \int_M \mathcal{L}(s, \nabla s; g) \sqrt{-g} d^{n+1}x.$$

The resulting Euler–Lagrange equations yield a wave-type equation of the form

$$\square_g s + \nabla_s W(s) = 0,$$

where $\square_g$ is the covariant d'Alembert operator and $\nabla_s W$ denotes the gradient of W with respect to the inner product on S .

Throughout this paper, (M, g) is treated as a fixed, given Lorentzian manifold. We do not modify Einstein's equations, we do not introduce a dynamical metric beyond g , and we do not consider backreaction of the field s on g . The construction is designed as an effective field theory on a prescribed background, in which the geometric and analytic structure of S is combined with that of (M, g) in a conservative, mathematically transparent way.

1.2 Aim and scope of the present paper

The primary aim of this paper is to formulate and analyse a covariant field theory for Hilbert-space valued fields on a Lorentzian spacetime background. More concretely, we fix:

- a globally hyperbolic Lorentzian manifold (M, g) of dimension $n + 1$,
- a separable Hilbert space S , as introduced in the earlier papers of the semantic programme,
- a potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ satisfying the structural and regularity assumptions already used in that context.

On this data we build a classical field theory for fields $s : M \rightarrow S$ with Lagrangian density

$$\mathcal{L}(s, \nabla s; g) = \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s),$$

and corresponding action functional $S[s]$. The core technical tasks are:

1. to give a precise geometric and functional-analytic setup for such fields, including appropriate spaces of sections and regularity assumptions;
2. to derive the Euler–Lagrange equations from the action principle and to analyse their structure as covariant wave equations in an infinite-dimensional target space;
3. to study the linearised theory around minima of W , thereby obtaining a mode decomposition in terms of small fluctuations with effective mass parameters given by the spectrum of the Hessian of W ;
4. to introduce and briefly analyse an induced symmetric tensor

$$h_{\mu\nu}[s] := \langle \nabla_\mu s, \nabla_\nu s \rangle_S,$$

which can be viewed as an additional geometric structure on M associated with s .

The scope of the paper is deliberately restricted. We do not attempt to couple $h_{\mu\nu}$ or s back into the gravitational field equations, nor do we discuss cosmological or astrophysical applications. The treatment of the infinite-dimensional aspects is kept conservative: in the main text we focus on formal derivations and on a rigorous analysis for finite-dimensional truncations of S , where standard results on hyperbolic equations and well-posedness apply. The full infinite-dimensional problem is discussed at a structural level, with comments on expected functional-analytic issues and possible approaches.

Although the original motivation for this framework arises from a broader semantic interpretation of S and W , the present paper focuses on the mathematical and field-theoretic structure. Interpretive and speculative aspects are confined to brief remarks in the final chapter, where we discuss possible applications and extensions.

1.3 Main contributions

The main contributions of this paper can be summarised as follows.

1. **Covariant field theory with Hilbert-space valued target.** We formulate a classical field theory on a fixed Lorentzian background (M, g) whose basic field is a map $s : M \rightarrow S$ into a separable Hilbert space S . The theory is defined by the Lagrangian density

$$\mathcal{L}(s, \nabla s; g) = \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s),$$

where W is a potential on S . This generalises standard scalar field theories (which correspond to the case $S = \mathbb{R}$ or finite-dimensional $\mathbb{R}^N$) to an infinite-dimensional internal space that is nevertheless compatible with the structures introduced in Papers 1–3.

2. **Euler–Lagrange equations and wave-type dynamics.** We derive the Euler–Lagrange equations associated with the action functional

$$S[s] = \int_M \mathcal{L}(s, \nabla s; g) \sqrt{-g} d^{n+1}x,$$

under suitable boundary conditions and regularity assumptions. The resulting equations of motion take the form

$$\square_g s + \nabla_s W(s) = 0,$$

where $\square_g$ is the covariant d’Alembert operator associated with g and $\nabla_s W$ is the gradient of W with respect to $\langle \cdot, \cdot \rangle_S$. We discuss the interpretation of this equation as a covariant wave equation on (M, g) with an internal dynamics governed by W .

3. **Linearised theory and mode decomposition.** We analyse small fluctuations around a local minimum s_* of the potential W . Writing $s = s_* + \phi$, we expand the action to quadratic order in ϕ . The Hessian H of W at s_* defines a self-adjoint operator on S , and under suitable assumptions one can choose an orthonormal basis of S consisting of eigenvectors of H . In this basis the quadratic action decomposes into a (possibly countable) collection of decoupled Klein–Gordon-type fields on (M, g) , each with its own effective mass parameter determined by the corresponding eigenvalue of H . We illustrate this structure on simple backgrounds, such as Minkowski spacetime and ultrastatic spacetimes.
4. **Optional induced tensor as additional structure.** From the field s we define a symmetric tensor

$$h_{\mu\nu}[s] = \langle \nabla_\mu s, \nabla_\nu s \rangle_S$$

on M . This tensor is built from the kinetic term of the theory and is invariant under orthogonal transformations of S that leave the inner product $\langle \cdot, \cdot \rangle_S$ unchanged. We discuss its basic properties and possible interpretations as additional, non-gravitational structure on spacetime. In particular, we emphasise that $h_{\mu\nu}$ is not identified with the metric $g_{\mu\nu}$ and does not enter the gravitational field equations in the present work.

5. **Integration with the earlier semantic programme.** Finally, we relate the constructions in this paper to the semantic Hilbert-space framework developed in Papers 1–3. The potential W and its gradient and Hessian have already been studied in the purely Hilbert-space setting, as generators of gradient flows and as contributors to a semantic free energy functional. The present work shows how the same structures can be embedded into a covariant spacetime field theory, thereby providing an effective field-theoretic description of coherence patterns encoded by W along the worldlines and regions of (M, g) .

These contributions are intended to be technically conservative. We do not claim to derive, replace, or modify general relativity, nor do we propose explanations for dark matter, dark energy, or other outstanding problems in fundamental physics. Instead, we provide a coherent mathematical framework in which a Hilbert-space valued field, governed by a potential W already studied in earlier work, is coupled in a standard way to a fixed spacetime geometry.

1.4 Relation to earlier work and structure of the paper

The present paper is the fourth in a series that develops a unified framework for Hilbert-space based fields and their dynamics.

- *Paper 1* introduced semantic gradient fields on a separable Hilbert space S . There, the emphasis was on the geometry of S itself, the analytic properties of the potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$, and the resulting gradient flows. The main objects were trajectories in S and their convergence properties.
- *Paper 2* recast these constructions in terms of a semantic free energy functional and studied distribution dynamics on S . This provided a statistical and thermodynamic perspective on the same underlying structures, clarifying notions of equilibrium and stability in the semantic Hilbert space.
- *Paper 3* showed how standard quantum systems can be embedded into the same Hilbert space S in a way that preserves key quantum features. This established compatibility between the semantic structures defined by W and the usual formalism of quantum theory, at least at the level of states and observables.

The present paper adds a new layer by introducing spacetime dependence and Lorentz covariance. Rather than working with isolated trajectories in S , we consider fields $s : M \rightarrow S$ on a Lorentzian manifold (M, g) and derive their dynamics from a covariant action principle. In doing so, we:

1. import the definitions and assumptions concerning S and W from Papers 1–3,
2. reinterpret W as a potential term in a Lagrangian density on (M, g) ,
3. connect the Hilbert-space geometry of S with the causal and metric structure of (M, g) via the kinetic term $\langle \nabla_\mu s, \nabla^\mu s \rangle_S$.

The structure of the paper is as follows.

- In **Chapter 2** we present the geometric and functional-analytic setup. We specify the assumptions on the spacetime manifold (M, g) , recall the properties of the Hilbert space S , define appropriate spaces of fields $s : M \rightarrow S$, and fix notation and conventions.

- In **Chapter 3** we introduce the Hilbert-space valued field s , construct its kinetic term using the covariant derivative and the inner product on S , and specify the potential term $W(s)$. We then define the full Lagrangian density and briefly comment on the associated stress-energy tensor, while emphasising that we do not couple it back into Einstein's equations in the present work.
- In **Chapter 4** we define the action functional $S[s]$, perform the first variation, and derive the Euler–Lagrange equations $\square_g s + \nabla_s W(s) = 0$. We also discuss well-posedness issues, with a particular focus on finite-dimensional truncations of S where standard results on hyperbolic equations apply.
- In **Chapter 5** we study the linearised theory and small fluctuations around a minimum of W . We diagonalise the Hessian of W at such a minimum, obtain a mode decomposition in terms of eigenfunctions of the Hessian, and work out examples on simple backgrounds such as Minkowski spacetime and ultrastatic spacetimes.
- In **Chapter 6** we introduce an optional induced symmetric tensor $h_{\mu\nu}[s] = \langle \nabla_\mu s, \nabla_\nu s \rangle_S$ and discuss its basic properties. We interpret $h_{\mu\nu}$ as additional structure on (M, g) and sketch, in a clearly speculative spirit, possible couplings to other fields or to gravity, without developing them in detail.
- In **Chapter 7** we summarise the construction, situate it within the broader semantic programme initiated in Papers 1–3, and discuss limitations and open problems. We highlight directions for future work, including a more complete functional-analytic treatment of the infinite-dimensional field equations, quantisation of the linearised theory, and possible phenomenological applications.

This completes the introduction of the framework and its relation to the earlier Hilbert-space based constructions. The subsequent chapters develop the technical details in a systematic and self-contained manner.

2 Geometric and functional-analytic setup

2.1 Spacetime manifold and Lorentzian metric

In this chapter we specify the geometric and functional-analytic setting in which the field theory will be formulated. We begin with the spacetime background.

Let (M, g) be a smooth, connected, oriented and time-oriented Lorentzian manifold of dimension $n + 1$. We adopt the convention that g has signature $(-, +, \dots, +)$. We denote by ∇ the Levi-Civita connection of g , and by $\square_g$ the associated covariant d'Alembert operator,

$$\square_g := \nabla^\mu \nabla_\mu = g^{\mu\nu} \nabla_\mu \nabla_\nu,$$

where indices are raised with $g^{\mu\nu}$, the components of the inverse metric.

We assume throughout that (M, g) is *globally hyperbolic*. This means that M admits a Cauchy hypersurface Σ , i.e. a spacelike hypersurface intersected exactly once by every inextendible timelike curve, and that the causal structure is well behaved. Global hyperbolicity ensures, in particular, that (M, g) can be foliated by Cauchy hypersurfaces and that suitable initial value problems for hyperbolic equations are well posed. Concretely, there exists a smooth time function $t : M \rightarrow \mathbb{R}$ such that each level set

$$\Sigma_t := t^{-1}(\{t\})$$

is a smooth spacelike Cauchy hypersurface, and M is diffeomorphic to $\mathbb{R} \times \Sigma$.

The Lorentzian metric g induces a volume form $d\mu_g$ on M . In local coordinates $(x^0, \dots, x^n)$ we write

$$d\mu_g = \sqrt{-g} d^{n+1}x,$$

where $g := \det(g_{\mu\nu})$ and $\sqrt{-g} > 0$ on M . Integrals over spacetime will always be taken with respect to this measure unless otherwise indicated.

For later use it is convenient to introduce also an auxiliary Riemannian metric $\hat{g}$ on M . Such a metric always exists on a smooth manifold. It allows us to define Sobolev spaces of fields and to discuss regularity in a coordinate-free fashion. The specific choice of $\hat{g}$ will not affect the formal structure of the theory; it serves only as a technical tool.

In order to avoid boundary terms in the variational calculus, we either assume that M has no boundary and that fields and variations have suitable decay properties in the asymptotic region, or we assume that M has a timelike boundary and impose boundary conditions that make boundary contributions vanish. For the purposes of this paper, we will mostly work with fields and variations that have compact support in M , or at least compact support in time, so that formal integrations by parts are justified.

2.1.1 Example: Nontrivial cosmological topology

In the discussion above we did not fix the global topology of the spacetime manifold M ; the assumptions only concern its local Lorentzian structure and global hyperbolicity. In particular, M may be realised as a quotient

$$M \cong \widetilde{M}/\Gamma,$$

where $\widetilde{M}$ is a simply connected covering space (for example $\mathbb{R}^4$, $\mathbb{R} \times S^3$ or a suitable FRW-type model), and Γ is a discrete group of isometries acting freely and properly discontinuously. Different choices of Γ lead to a large class of globally nontrivial, possibly nonorientable spacetime topologies that are locally indistinguishable from standard cosmological backgrounds.

As a concrete example one may consider a “meta-Klein bottle” cosmology: a compact spacetime whose spatial slices Σ_t are obtained from a simply connected model space $\widetilde{\Sigma}$ (for instance $\mathbb{R}^3$ or S^3) by identifying points under a holonomy group that includes reflections. In such a model, the resulting spatial sections are globally nonorientable and geodesics can exhibit *recurrence*: a timelike or null geodesic that starts near an observer may, after sufficiently long parameter time, re-enter an arbitrarily small neighbourhood of the same observer, even though it has travelled “outwards” in a fixed direction. Intuitively, going far enough along a quasi-radial direction can bring one back to the starting region.

For the purposes of the present paper, the details of the holonomy group Γ are not important; what matters is that the formalism developed below applies equally well on such nontrivially connected spacetimes. A semantic field

$$s : M \rightarrow S$$

can be defined on any globally hyperbolic Lorentzian manifold, independently of whether its spatial slices are simply connected or have nontrivial topology. The action functional, local Lagrangian density and Euler-Lagrange equations derived in Chapters 3 and 4 remain valid on these backgrounds, provided the usual geometric and functional-analytic regularity assumptions hold.

In later ontological work (see Paper VI) such nontrivial topologies can serve as geometric toy models for observer-centred pictures in which “far away” regions of the universe are, in

a global sense, identified with the local neighbourhood of an observer. The present paper remains agnostic about whether our physical universe actually realises such a topology; the example is intended only to illustrate the flexibility of the semantic field formalism with respect to the global structure of M .

2.2 The Hilbert space S

The internal degrees of freedom of the theory live in a fixed separable Hilbert space S . We recall the basic assumptions on S and on the potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$.

We take S to be a real or complex separable Hilbert space with inner product

$$\langle \cdot, \cdot \rangle_S : S \times S \rightarrow \mathbb{K},$$

where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and associated norm

$$\|s\|_S := \sqrt{\langle s, s \rangle_S}.$$

In the complex case we adopt the convention that $\langle \cdot, \cdot \rangle_S$ is linear in the second argument and conjugate-linear in the first. Whenever a purely real quantity is needed (for example in defining gradients of real-valued functionals) we implicitly take the real part of $\langle \cdot, \cdot \rangle_S$.

The inner product induces a canonical isometric isomorphism

$$R_S : S \rightarrow S',$$

where S' denotes the topological dual of S . For each $u \in S$ the functional $R_S(u) \in S'$ is defined by

$$R_S(u)[v] := \langle u, v \rangle_S, \quad v \in S.$$

We will use R_S to identify dual elements with vectors in S whenever convenient.

The potential

$$W : S \rightarrow \mathbb{R} \cup \{+\infty\}$$

is assumed to satisfy the structural and regularity conditions introduced in the earlier work on semantic gradient fields and semantic free energy. For the purposes of this paper we collect the minimal assumptions needed for the variational analysis:

- W is *proper*, i.e. $W(s) > -\infty$ for all $s \in S$ and W is not identically $+\infty$.
- W is bounded from below and lower semicontinuous.
- There exists a dense subset $\mathcal{D}(W) \subset S$ on which W is Fréchet differentiable, with derivative

$$DW(s) \in S', \quad s \in \mathcal{D}(W),$$

and such that DW is continuous on $\mathcal{D}(W)$ with respect to the norm topology of S .

On $\mathcal{D}(W)$ we define the gradient $\nabla_s W(s) \in S$ of W at s by the Riesz representation relation

$$DW(s)(\delta s) = \langle \nabla_s W(s), \delta s \rangle_S \quad \text{for all } \delta s \in S.$$

When W is twice Fréchet differentiable at s , its Hessian $D^2W(s)$ is a bounded self-adjoint operator $H_s : S \rightarrow S$ satisfying

$$\langle H_s u, v \rangle_S = D^2W(s)(u, v) = D(DW)(s)[u](v) \quad \text{for all } u, v \in S.$$

In the linearised theory (Chapter 5) we will assume additional regularity, in particular that W is twice differentiable in a neighbourhood of a local minimum and that the corresponding Hessian is self-adjoint and bounded below.

2.3 Spaces of fields and regularity assumptions

We now specify the spaces of fields $s : M \rightarrow S$ on which the action functional will be defined and varied.

It is convenient to view s as a section of the trivial Hilbert bundle

$$\pi : M \times S \rightarrow M, \quad \pi(x, u) = x.$$

A field s is then a map

$$s : M \rightarrow M \times S, \quad s(x) = (x, \tilde{s}(x)),$$

where the *component* $\tilde{s} : M \rightarrow S$ is what we usually denote again by s for simplicity. We will not distinguish notationally between these two viewpoints.

2.3.1 Smooth fields and test fields

The basic space of smooth fields is

$$C^\infty(M, S) := \{s : M \rightarrow S \mid s \text{ is smooth}\},$$

where smoothness means that for every continuous linear functional $\ell \in S'$ the scalar function

$$x \mapsto \ell(s(x))$$

is smooth on M . Equivalently, with respect to any orthonormal basis $\{e_i\}_{i \in \mathbb{N}}$ of S such that $s(x) = \sum_i s^i(x) e_i$, all component functions $s^i : M \rightarrow \mathbb{K}$ are smooth and the series converges rapidly enough so that differentiations may be interchanged with the infinite sum on compact sets.

We define the space of smooth fields with compact support by

$$C_0^\infty(M, S) := \{s \in C^\infty(M, S) \mid \text{supp}(s) \subset M \text{ is compact}\}.$$

This space will serve as the domain for variations in the derivation of the Euler–Lagrange equations. More precisely, when varying a given field s , we take variations $\delta s \in C_0^\infty(M, S)$ so that boundary terms produced by integrations by parts vanish.

2.3.2 Sobolev and energy spaces

For a more analytic discussion of the field equations, especially in finite-dimensional truncations, it is useful to introduce Sobolev-type spaces of fields. Using the auxiliary Riemannian metric $\hat{g}$, we can define Sobolev norms on scalar functions and extend them to S -valued fields.

Let $K \subset M$ be compact. For $k \in \mathbb{N}$ we define the local Sobolev space

$$H_{\text{loc}}^k(M, S) := \{s : M \rightarrow S \mid \chi s \in H^k(M, S) \text{ for all } \chi \in C_0^\infty(M, \mathbb{R})\},$$

where $H^k(M, S)$ is defined by completing $C_0^\infty(M, S)$ with respect to the norm induced by $\hat{g}$,

$$\|s\|_{H^k(M, S)}^2 := \sum_{j=0}^k \int_M \|\hat{\nabla}^{(j)} s(x)\|_S^2 d\mu_{\hat{g}}(x).$$

Here $\hat{\nabla}^{(j)}$ denotes the j -th iterated covariant derivative with respect to the Levi-Civita connection of $\hat{g}$, applied componentwise to s . Different choices of $\hat{g}$ yield equivalent norms on compact sets.

In particular, we will encounter fields that, on each Cauchy hypersurface Σ_t , have finite energy. For a (sufficiently regular) field s , the energy density on Σ_t associated with the kinetic term of the Lagrangian is

$$\mathcal{E}_t(x) := \frac{1}{2} \left(\langle \nabla_0 s(x), \nabla_0 s(x) \rangle_S + \langle \nabla_i s(x), \nabla^i s(x) \rangle_S \right), \quad x \in \Sigma_t,$$

where spatial indices are raised and lowered with the induced Riemannian metric on Σ_t . The total energy on Σ_t is then

$$E_t[s] := \int_{\Sigma_t} \mathcal{E}_t(x) d\mu_{\Sigma_t}(x),$$

where $d\mu_{\Sigma_t}$ is the induced volume measure on Σ_t . Fields with finite energy on each Σ_t form a natural energy space for the evolution problem.

In the fully infinite-dimensional setting, a complete functional-analytic treatment of the wave equation with target S requires additional effort. However, the formal structure is already evident, and rigorous results can be obtained for finite-dimensional approximations, as we now explain.

2.3.3 Finite-dimensional truncations

Let $\{e_i\}_{i \in \mathbb{N}}$ be an orthonormal basis of S . For $N \in \mathbb{N}$ we define the finite-dimensional subspace

$$S_N := \text{span}\{e_1, \dots, e_N\} \cong \mathbb{K}^N.$$

A finite-dimensional truncation of the field s is a map

$$s_N : M \rightarrow S_N, \quad s_N(x) = \sum_{i=1}^N s^i(x) e_i,$$

with smooth component functions $s^i : M \rightarrow \mathbb{K}$. Restricting the potential W to S_N yields a function

$$W_N := W|_{S_N} : S_N \rightarrow \mathbb{R} \cup \{+\infty\}.$$

The corresponding truncated theory is a standard system of N coupled scalar fields on (M, g) with Lagrangian density

$$\mathcal{L}_N(s_N, \nabla s_N; g) = \frac{1}{2} g^{\mu\nu} \sum_{i=1}^N \partial_\mu s^i \overline{\partial_\nu s^i} - W_N(s_N).$$

For such finite-dimensional systems, classical results on hyperbolic partial differential equations on globally hyperbolic spacetimes apply and yield well-posedness of the Cauchy problem under suitable conditions on W_N . In Chapter 4 we will briefly recall these results in a simplified setting.

The infinite-dimensional theory can be viewed, at least formally, as the limit $N \rightarrow \infty$ of these finite-dimensional truncations. While we will not attempt to make this limit fully rigorous in the present paper, this perspective guides the formulation of the action and the Euler–Lagrange equations.

2.4 Notation and conventions

We summarise the main notational and sign conventions used throughout the paper.

- Greek indices $\mu, \nu, \rho, \dots$ range over $0, \dots, n$ and denote spacetime components with respect to a local coordinate chart. Latin indices $i, j, k, \dots$ from the middle of the alphabet are used for spatial components with respect to a chosen foliation by Cauchy hypersurfaces. Summation over repeated indices is understood (Einstein summation convention).
- The Lorentzian metric components are $g_{\mu\nu}$ with inverse $g^{\mu\nu}$, satisfying $g^{\mu\lambda}g_{\lambda\nu} = \delta^\mu_\nu$. The signature of g is $(-, +, \dots, +)$. Indices are raised and lowered using $g^{\mu\nu}$ and $g_{\mu\nu}$. The covariant derivative associated with the Levi-Civita connection of g is denoted by ∇_μ . For scalar fields and for the components of S -valued fields in a fixed basis, ∇_μ coincides with the partial derivative ∂_μ in local coordinates.
- The covariant d'Alembert operator acting on scalar fields or components of S -valued fields is

$$\square_g := \nabla^\mu \nabla_\mu = g^{\mu\nu} \nabla_\mu \nabla_\nu.$$

When acting on S -valued fields $s : M \rightarrow S$, it is understood componentwise with respect to any orthonormal basis of S .

- The inner product on S is denoted by $\langle \cdot, \cdot \rangle_S$ and the corresponding norm by $\| \cdot \|_S$. When no confusion can arise, we sometimes omit the subscript and write $\langle \cdot, \cdot \rangle$. The gradient of the potential W at $s \in \mathcal{D}(W)$ is denoted by $\nabla_s W(s) \in S$ and is defined via

$$DW(s)(\delta s) = \langle \nabla_s W(s), \delta s \rangle_S.$$

- For an S -valued field s we write

$$\langle \nabla_\mu s, \nabla_\nu s \rangle_S$$

for the S -inner product of the covariant derivatives $\nabla_\mu s$ and $\nabla_\nu s$. The fully contracted kinetic term appearing in the Lagrangian density is

$$\langle \nabla_\mu s, \nabla^\mu s \rangle_S := g^{\mu\nu} \langle \nabla_\mu s, \nabla_\nu s \rangle_S.$$

- The volume measure on (M, g) is $d\mu_g = \sqrt{-g} d^{n+1}x$ in local coordinates. The induced volume measure on a Cauchy hypersurface Σ_t is denoted by $d\mu_{\Sigma_t}$.
- We work in units where fundamental constants such as the speed of light are set to 1. Since the present treatment is classical and does not involve $\hbar$ explicitly, we do not fix conventions for quantum constants.
- When writing local expressions for S -valued fields, we may occasionally choose an orthonormal basis $\{e_i\}_{i \in \mathbb{N}}$ of S and expand

$$s(x) = \sum_i s^i(x) e_i.$$

Unless stated otherwise, all formal manipulations are to be understood either in such a basis or in a coordinate-free sense using the inner product and the Riesz isomorphism.

- Variations of the field s in the action principle are denoted by δs and are taken to be elements of $C_0^\infty(M, S)$, i.e. smooth fields with compact support. This ensures that boundary terms vanish in the derivation of the Euler–Lagrange equations.

With these conventions in place, we are prepared to introduce the Hilbert-space valued field $s : M \rightarrow S$, its kinetic and potential terms, and the full Lagrangian density in the next chapter.

3 Hilbert-space valued fields and Lagrangian density

3.1 Definition of the field $s : M \rightarrow S$

The basic dynamical object of the theory is a field

$$s : M \rightarrow S,$$

where M is the spacetime manifold equipped with Lorentzian metric g and S is the fixed separable Hilbert space introduced in the previous chapter. Thus, to each spacetime point $x \in M$ the field assigns a vector $s(x) \in S$.

It is convenient to regard s as a section of the trivial Hilbert bundle

$$\pi : M \times S \rightarrow M, \quad \pi(x, u) = x.$$

A field s is then a map $s : M \rightarrow M \times S$ of the form

$$s(x) = (x, \tilde{s}(x)),$$

where $\tilde{s} : M \rightarrow S$ is the component map. We will not distinguish notationally between s and $\tilde{s}$ and simply write

$$s(x) \in S$$

for the value of the field at x .

3.1.1 Coordinate and basis representations

Let $\{x^\mu\}_{\mu=0,\dots,n}$ be a local coordinate system on an open subset $U \subset M$, and let $\{e_i\}_{i \in I}$ be an orthonormal basis of S , where I is countable because S is separable. On U every sufficiently regular field s can be written as

$$s(x) = \sum_{i \in I} s^i(x) e_i, \quad x \in U,$$

with scalar coefficient functions $s^i : U \rightarrow \mathbb{K}$, where $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. If s is smooth in the sense of the previous chapter, then each component s^i is a smooth scalar field, and the series converges in S with all operations compatible with differentiation on compact subsets.

Finite-dimensional truncations are obtained by restricting to subspaces

$$S_N := \text{span}\{e_1, \dots, e_N\} \subset S,$$

and projecting s onto S_N :

$$s_N(x) := \sum_{i=1}^N s^i(x) e_i.$$

In that case the field is equivalent to an N -tuple of scalar fields $s^1, \dots, s^N$ on (M, g) . This observation will be used to connect the infinite-dimensional theory to familiar multi-component scalar field theories and to make contact with standard well-posedness results.

3.1.2 Admissible fields for the classical theory

For the purposes of the classical field theory and the variational principle, we restrict attention to fields s with sufficient regularity so that all relevant quantities are well defined:

- The covariant derivatives $\nabla_\mu s$ exist as S -valued fields and are square-integrable on compact subsets of M .
- The potential $W(s(x))$ is finite for almost all $x \in M$ and integrable with respect to the spacetime volume measure.

A natural class of admissible fields is given by

$$\mathcal{A} := \left\{ s \in H_{\text{loc}}^1(M, S) \mid W(s(\cdot)) \in L_{\text{loc}}^1(M, \mathbb{R}) \right\},$$

where $H_{\text{loc}}^1(M, S)$ denotes the local Sobolev space defined using the auxiliary Riemannian metric from the previous chapter. When deriving the Euler–Lagrange equations in Chapter 4, we will initially work at the formal level with $s \in C^\infty(M, S)$ and variations $\delta s \in C_0^\infty(M, S)$, and then indicate how the expressions extend to $\mathcal{A}$ or to finite-dimensional truncations.

3.2 Kinetic term

The kinetic part of the Lagrangian density measures the spacetime variation of the field s with respect to the spacetime metric g and the inner product on S .

3.2.1 Covariant derivative of S -valued fields

Since S is a fixed Hilbert space and the bundle $M \times S$ is trivial, there is no nontrivial connection in the internal direction. The only connection entering the construction is the Levi-Civita connection of g acting on tensor indices associated with M . Accordingly, the covariant derivative of s is defined by

$$\nabla_\mu s(x) := \partial_\mu s(x),$$

in any local coordinate system. In a basis $\{e_i\}_{i \in I}$ of S , this reads

$$\nabla_\mu s(x) = \sum_{i \in I} (\partial_\mu s^i(x)) e_i.$$

Thus, $\nabla_\mu s$ is again an S -valued field, smooth whenever s is smooth.

For later use, it is convenient to introduce the S -valued 1-form

$$\nabla s := \nabla_\mu s \otimes dx^\mu,$$

and to regard the kinetic term as being built from the S -inner products of the components $\nabla_\mu s$.

3.2.2 Kinetic Lagrangian density

The kinetic contribution to the Lagrangian density is defined by

$$\mathcal{L}_{\text{kin}}(s, \nabla s; g) := \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S,$$

where index contraction is performed with $g^{\mu\nu}$:

$$\langle \nabla_\mu s, \nabla^\mu s \rangle_S := g^{\mu\nu} \langle \nabla_\mu s, \nabla_\nu s \rangle_S.$$

In a local coordinate chart and with respect to an orthonormal basis $\{e_i\}$ of S , we have

$$\mathcal{L}_{\text{kin}}(s, \nabla s; g) = \frac{1}{2} g^{\mu\nu} \sum_{i \in I} \partial_\mu s^i \overline{\partial_\nu s^i},$$

where the overline denotes complex conjugation if S is complex (and can be omitted in the real case).

3.2.3 Analogy with multi-component scalar fields

For a finite-dimensional truncation $S_N = \text{span}\{e_1, \dots, e_N\}$, the field s_N is equivalent to a collection of N scalar fields $s^1, \dots, s^N$. In that case the kinetic Lagrangian density reduces to

$$\mathcal{L}_{\text{kin},N}(s_N, \nabla s_N; g) = \frac{1}{2} g^{\mu\nu} \sum_{i=1}^N \partial_\mu s^i \overline{\partial_\nu s^i},$$

which is the standard expression for an N -component real or complex scalar field with canonical kinetic term on a curved background.

The infinite-dimensional case can be viewed formally as the limit $N \rightarrow \infty$, with the sum over i replaced by the inner product in S . From this perspective, the kinetic term retains the same structural form as in conventional field theory, with the internal index i simply ranging over a countable basis of the Hilbert space S .

The kinetic term is invariant under global unitary (in the complex case) or orthogonal (in the real case) transformations of S that preserve the inner product:

$$s(x) \mapsto Us(x), \quad U : S \rightarrow S \text{ unitary or orthogonal.}$$

Such transformations amount to changing the orthonormal basis of S and do not affect the dynamics as long as the potential W is invariant or transforms appropriately.

3.3 Potential term W

The potential term encapsulates the internal structure of the theory and encodes how different field values in S are energetically weighted.

3.3.1 Potential on the Hilbert space S

We recall that the potential is a function

$$W : S \rightarrow \mathbb{R} \cup \{+\infty\}$$

satisfying the structural assumptions already used in the Hilbert-space based framework. In particular:

- W is proper, lower semicontinuous, and bounded from below.
- There exists a dense domain $\mathcal{D}(W) \subset S$ on which W is Fréchet differentiable, with derivative $DW(s) \in S'$ continuous in s .
- On $\mathcal{D}(W)$, the gradient $\nabla_s W(s) \in S$ is defined by

$$DW(s)(\delta s) = \langle \nabla_s W(s), \delta s \rangle_S \quad \text{for all } \delta s \in S.$$

In examples, W may have multiple local minima, may be convex or nonconvex, and may contain regularisation terms that penalise certain directions in S . Quadratic and Ginzburg–Landau type potentials provide important classes of examples.

3.3.2 Local potential density

Given a field $s : M \rightarrow S$, we define the local potential density V by

$$V(s(x)) := W(s(x)), \quad x \in M,$$

whenever $s(x) \in \mathcal{D}(W)$. The contribution of the potential to the action functional will be obtained by integrating $V(s(x))$ over spacetime with respect to the volume measure. For this to be finite, we require $W(s(\cdot))$ to be locally integrable; this is part of the admissibility condition for fields.

In the linearised analysis around a local minimum $s_* \in S$ of W (Chapter 5), we will use the expansion

$$W(s_* + \phi) = W(s_*) + \frac{1}{2} \langle \phi, H \phi \rangle_S + O(\|\phi\|_S^3),$$

where $H : S \rightarrow S$ is the Hessian of W at s_* . For the present chapter, it suffices to note that W provides a real-valued energy density depending only on the value of $s(x)$ at each point.

3.4 Full Lagrangian density and stress-energy tensor

3.4.1 Total Lagrangian density

Combining the kinetic and potential contributions, we define the full Lagrangian density by

$$\mathcal{L}(s, \nabla s; g) := \mathcal{L}_{\text{kin}}(s, \nabla s; g) - V(s) = \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s).$$

More explicitly, at each point $x \in M$,

$$\mathcal{L}(s, \nabla s; g)(x) = \frac{1}{2} g^{\mu\nu}(x) \langle \nabla_\mu s(x), \nabla_\nu s(x) \rangle_S - W(s(x)).$$

This is directly analogous to the Lagrangian density of a scalar field with potential, with the only difference that the target space is now the Hilbert space S instead of $\mathbb{R}$ or $\mathbb{R}^N$.

The corresponding action functional will be defined in the next chapter by integrating $\mathcal{L}$ over M with respect to the volume measure $d\mu_g$.

3.4.2 Stress-energy tensor (optional)

Although the spacetime metric g is treated as a fixed background in this paper and is not varied dynamically, it is still useful to define the stress-energy tensor associated with the field s . Formally, the symmetric stress-energy tensor $T_{\mu\nu}$ is obtained by varying the action with respect to $g^{\mu\nu}$ while holding s fixed:

$$T_{\mu\nu} := -\frac{2}{\sqrt{-g}} \frac{\delta S[s]}{\delta g^{\mu\nu}},$$

where $S[s]$ is the action functional. For the Lagrangian density defined above, the standard computation yields

$$T_{\mu\nu} = \langle \nabla_\mu s, \nabla_\nu s \rangle_S - g_{\mu\nu} \mathcal{L}(s, \nabla s; g).$$

In expanded form,

$$T_{\mu\nu} = \langle \nabla_\mu s, \nabla_\nu s \rangle_S - g_{\mu\nu} \left(\frac{1}{2} \langle \nabla_\rho s, \nabla^\rho s \rangle_S - W(s) \right).$$

If s satisfies the Euler–Lagrange equations that will be derived in Chapter 4, and if appropriate boundary conditions are imposed, then $T_{\mu\nu}$ is divergence-free in the sense that

$$\nabla^\mu T_{\mu\nu} = 0.$$

This reflects the invariance of the action under spacetime diffeomorphisms and encodes local energy-momentum conservation for the field s on the fixed background (M, g) .

In the present work, $T_{\mu\nu}$ is used only as a diagnostic quantity. We do not couple it back into Einstein’s equations, and we do not propose to modify the gravitational dynamics. The metric g remains an externally specified structure, and the field s is treated as an additional classical field living on this background.

With the Lagrangian density now specified, we can proceed to define the action functional and derive the Euler–Lagrange equations for s in the next chapter.

4 Action functional and Euler–Lagrange equations

4.1 Action functional

We now assemble the Lagrangian density introduced in the previous chapter into a spacetime action and define its domain of admissible fields.

4.1.1 Definition of the action

Recall the Lagrangian density

$$\mathcal{L}(s, \nabla s; g) = \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s),$$

where $s : M \rightarrow S$ is an S -valued field, $\nabla_\mu s$ is its covariant derivative (which coincides with the partial derivative in local coordinates because the bundle is trivial), and $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ is the potential on S .

The action functional is defined by integrating the Lagrangian density over spacetime with respect to the volume measure $d\mu_g = \sqrt{-g} d^{n+1}x$:

$$S[s] := \int_M \mathcal{L}(s, \nabla s; g) d\mu_g = \int_M \left(\frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s) \right) d\mu_g. \quad (4.1)$$

We regard $S[s]$ as a functional on a suitable class $\mathcal{A}$ of admissible fields. A natural choice is

$$\mathcal{A} := \left\{ s \in H_{\text{loc}}^1(M, S) \mid W(s(\cdot)) \in L_{\text{loc}}^1(M, \mathbb{R}) \right\},$$

so that both the kinetic and potential contributions are locally integrable. In the derivation of the Euler–Lagrange equations we will work formally with $s \in C^\infty(M, S)$ and variations $\delta s \in C_0^\infty(M, S)$, and then interpret the resulting equations in a weak sense on $\mathcal{A}$.

4.1.2 Variations and boundary conditions

To derive the field equations we consider variations of s of the form

$$s_\varepsilon := s + \varepsilon \delta s,$$

where $\delta s \in C_0^\infty(M, S)$ is a smooth S -valued field with compact support in M and $\varepsilon \in \mathbb{R}$ is a small parameter. The compact support of δs ensures that no boundary terms appear from integrations by parts, independently of whether M has a boundary or is non-compact.

The first variation of the action at s in the direction δs is defined by

$$\delta S[s; \delta s] := \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} S[s_\varepsilon].$$

Critical points of S in $\mathcal{A}$ are those fields s for which

$$\delta S[s; \delta s] = 0 \quad \text{for all } \delta s \in C_0^\infty(M, S).$$

This condition will lead to the Euler–Lagrange equations.

4.2 First variation and field equations

We now compute the first variation of the action and derive the covariant wave-type equation governing the dynamics of s .

4.2.1 Variation of the kinetic term

Let

$$\mathcal{L}_{\text{kin}}(s, \nabla s; g) := \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S$$

denote the kinetic part of the Lagrangian density. For the varied field $s_\varepsilon = s + \varepsilon \delta s$ we have

$$\nabla_\mu s_\varepsilon = \nabla_\mu s + \varepsilon \nabla_\mu(\delta s),$$

so that

$$\begin{aligned} \mathcal{L}_{\text{kin}}(s_\varepsilon, \nabla s_\varepsilon; g) &= \frac{1}{2} \langle \nabla_\mu s + \varepsilon \nabla_\mu(\delta s), \nabla^\mu s + \varepsilon \nabla^\mu(\delta s) \rangle_S \\ &= \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S + \varepsilon \langle \nabla_\mu s, \nabla^\mu(\delta s) \rangle_S + O(\varepsilon^2). \end{aligned}$$

Thus, the variation of the kinetic part of the action is

$$\begin{aligned} \delta S_{\text{kin}}[s; \delta s] &:= \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_M \mathcal{L}_{\text{kin}}(s_\varepsilon, \nabla s_\varepsilon; g) d\mu_g \\ &= \int_M \langle \nabla_\mu s, \nabla^\mu(\delta s) \rangle_S d\mu_g. \end{aligned} \tag{4.2}$$

We now integrate by parts. Using the metric compatibility of ∇_μ and the fact that $d\mu_g$ is covariantly constant, we obtain

$$\nabla_\mu(\langle \nabla^\mu s, \delta s \rangle_S) = \langle \nabla_\mu \nabla^\mu s, \delta s \rangle_S + \langle \nabla^\mu s, \nabla_\mu(\delta s) \rangle_S.$$

Integrating this identity over M and applying the divergence theorem yields

$$\int_M \langle \nabla^\mu s, \nabla_\mu(\delta s) \rangle_S d\mu_g = - \int_M \langle \nabla_\mu \nabla^\mu s, \delta s \rangle_S d\mu_g,$$

because δs has compact support and the boundary term vanishes. Recalling that $\square_g = \nabla^\mu \nabla_\mu$, we can rewrite

$$\int_M \langle \nabla_\mu s, \nabla^\mu(\delta s) \rangle_S d\mu_g = - \int_M \langle \square_g s, \delta s \rangle_S d\mu_g.$$

Substituting this into (4.2), we obtain

$$\delta S_{\text{kin}}[s; \delta s] = - \int_M \langle \square_g s, \delta s \rangle_S d\mu_g. \quad (4.3)$$

4.2.2 Variation of the potential term

The potential part of the Lagrangian density is

$$\mathcal{L}_{\text{pot}}(s) := -W(s).$$

For $s_\varepsilon = s + \varepsilon \delta s$ we use the differentiability of W on its domain $\mathcal{D}(W)$ to obtain

$$W(s_\varepsilon) = W(s) + \varepsilon DW(s)(\delta s) + O(\varepsilon^2),$$

where $DW(s) \in S'$ is the Fréchet derivative at s . By the definition of the gradient $\nabla_s W(s) \in S$ we have

$$DW(s)(\delta s) = \langle \nabla_s W(s), \delta s \rangle_S.$$

Therefore

$$\mathcal{L}_{\text{pot}}(s_\varepsilon) = -W(s_\varepsilon) = -W(s) - \varepsilon \langle \nabla_s W(s), \delta s \rangle_S + O(\varepsilon^2),$$

and the variation of the potential part of the action is

$$\begin{aligned} \delta S_{\text{pot}}[s; \delta s] &:= \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_M \mathcal{L}_{\text{pot}}(s_\varepsilon) d\mu_g \\ &= - \int_M \langle \nabla_s W(s), \delta s \rangle_S d\mu_g. \end{aligned} \quad (4.4)$$

4.2.3 Euler–Lagrange equations

Combining (4.3) and (4.4), the total variation of the action is

$$\begin{aligned} \delta S[s; \delta s] &= \delta S_{\text{kin}}[s; \delta s] + \delta S_{\text{pot}}[s; \delta s] \\ &= - \int_M \langle \square_g s + \nabla_s W(s), \delta s \rangle_S d\mu_g. \end{aligned} \quad (4.5)$$

The field s is a critical point of S if and only if $\delta S[s; \delta s] = 0$ for all $\delta s \in C_0^\infty(M, S)$. Using the non-degeneracy of the inner product and the freedom in δs , we obtain the Euler–Lagrange equations:

$$\square_g s + \nabla_s W(s) = 0, \quad (4.6)$$

to be understood as an equality of S -valued distributions on M and, in particular, as a pointwise equality wherever s is sufficiently regular.

In components with respect to an orthonormal basis $\{e_i\}_{i \in I}$ of S ,

$$s(x) = \sum_{i \in I} s^i(x) e_i, \quad \nabla_s W(s(x)) = \sum_{i \in I} \frac{\partial W}{\partial s^i}(s(x)) e_i,$$

equation (4.6) becomes, for each i ,

$$\square_g s^i(x) + \frac{\partial W}{\partial s^i}(s(x)) = 0.$$

Thus, the theory describes a (possibly infinite) collection of coupled scalar wave equations on (M, g) with nonlinearity given by the gradient of W .

4.2.4 Weak formulation

The derivation above was carried out for smooth fields and compactly supported variations. However, equation (4.6) admits a natural weak formulation on the space $\mathcal{A}$ of admissible fields.

A field $s \in \mathcal{A}$ is said to satisfy the Euler–Lagrange equations in the weak sense if

$$\int_M \langle \nabla_\mu s, \nabla^\mu(\delta s) \rangle_S d\mu_g - \int_M \langle \nabla_s W(s), \delta s \rangle_S d\mu_g = 0$$

for all $\delta s \in C_0^\infty(M, S)$. Integration by parts then reformulates this condition as

$$\int_M \langle \square_g s + \nabla_s W(s), \delta s \rangle_S d\mu_g = 0 \quad \text{for all } \delta s \in C_0^\infty(M, S),$$

which is equivalent to (4.6) as an equation in the distributional sense.

4.3 Well-posedness in finite dimensions

A complete functional-analytic treatment of equation (4.6) in the fully infinite-dimensional setting is beyond the scope of this paper. However, for finite-dimensional truncations of S , the Euler–Lagrange equations reduce to standard semilinear wave equations on a globally hyperbolic spacetime, for which well-posedness is well understood. We briefly recall the relevant structures.

4.3.1 Finite-dimensional truncation and component equations

Let $\{e_i\}_{i \in \mathbb{N}}$ be an orthonormal basis of S and, for fixed $N \in \mathbb{N}$, consider the finite-dimensional subspace

$$S_N := \text{span}\{e_1, \dots, e_N\} \cong \mathbb{K}^N.$$

The truncated field $s_N : M \rightarrow S_N$ is given by

$$s_N(x) = \sum_{i=1}^N s^i(x) e_i,$$

with scalar component fields $s^i : M \rightarrow \mathbb{K}$. The truncated potential

$$W_N := W|_{S_N} : S_N \rightarrow \mathbb{R} \cup \{+\infty\}$$

is assumed to be sufficiently regular (for instance, continuously differentiable on S_N). The corresponding action functional is

$$S_N[s_N] = \int_M \left(\frac{1}{2} g^{\mu\nu} \sum_{i=1}^N \partial_\mu s^i \overline{\partial_\nu s^i} - W_N(s_N) \right) d\mu_g.$$

The Euler–Lagrange equations for S_N take the form

$$\square_g s^i + \frac{\partial W_N}{\partial s^i}(s^1, \dots, s^N) = 0, \quad i = 1, \dots, N, \quad (4.7)$$

which is a system of N coupled scalar semilinear wave equations on (M, g) .

4.3.2 Cauchy problem and energy estimates

Let Σ be a Cauchy hypersurface in M , and let n^μ denote its future-directed unit normal. The Cauchy data for (4.7) consist of the initial values

$$s^i|_\Sigma = \phi^i, \quad n^\mu \nabla_\mu s^i|_\Sigma = \pi^i, \quad i = 1, \dots, N,$$

where ϕ^i and π^i are given scalar functions on Σ . Equivalently, one can package the data into

$$s_N|_\Sigma = \phi, \quad n^\mu \nabla_\mu s_N|_\Sigma = \pi,$$

with $\phi, \pi : \Sigma \rightarrow S_N$.

Under standard assumptions on the potential W_N (for example, local Lipschitz continuity of its gradient in the fields, or polynomial growth conditions and appropriate Sobolev regularity of the data), the Cauchy problem for (4.7) is locally well posed in suitable Sobolev spaces. More precisely, there exists a time interval $I \subset \mathbb{R}$, an open neighbourhood $U \subset M$ containing Σ , and a unique solution

$$s_N \in C^0(I, H^k(\Sigma, S_N)) \cap C^1(I, H^{k-1}(\Sigma, S_N)),$$

for appropriate k , such that s_N and its normal derivative agree with the initial data on Σ . Moreover, solutions depend continuously on the initial data.

The basic tool in establishing this well-posedness is the energy associated with the kinetic term. For each spacelike Cauchy hypersurface Σ_t in a foliation of M , one defines the energy

$$E_t[s_N] := \int_{\Sigma_t} \left(\frac{1}{2} \langle \nabla_0 s_N, \nabla_0 s_N \rangle_S + \frac{1}{2} \langle \nabla_i s_N, \nabla^i s_N \rangle_S + W_N(s_N) \right) d\mu_{\Sigma_t},$$

where the inner products reduce to finite sums over $i = 1, \dots, N$. Under suitable conditions on W_N , this energy is non-negative and satisfies an inequality of the form

$$E_t[s_N] \leq C E_{t_0}[s_N] \exp(C|t - t_0|),$$

for some constant C depending on bounds on $\nabla_s W_N$ and on the geometry of (M, g) . This provides control of the solution on finite time intervals in terms of the initial data.

In addition, the causality properties of (M, g) imply finite speed of propagation for the system (4.7): the value of s_N at a point $p \in M$ depends only on the initial data in the intersection of Σ with the causal past of p . This follows from local energy estimates and the hyperbolic character of $\square_g$.

4.3.3 Relation to the infinite-dimensional theory

The full Hilbert-space valued equation (4.6) can be viewed formally as the limit $N \rightarrow \infty$ of the truncated systems (4.7). In this perspective, the finite-dimensional theories provide a sequence of approximations to the infinite-dimensional dynamics, and energy estimates at each finite N suggest a route to controlling the behaviour as N increases.

A rigorous treatment of the limit $N \rightarrow \infty$ requires additional structural assumptions on W and careful functional-analytic arguments, for example compactness or monotonicity properties that allow one to pass to the limit in the nonlinear terms. Since the focus of the present paper is on the structural formulation of the field theory, we do not pursue this limit in detail here. Instead, we regard the Euler–Lagrange equation

$$\square_g s + \nabla_s W(s) = 0$$

as the formal infinite-dimensional analogue of the well-posed finite-dimensional systems (4.7) and highlight its role as the central dynamical equation in the subsequent analysis.

In the next chapter we linearise this equation around minima of W and analyse the resulting small fluctuations and mode decompositions on simple spacetime backgrounds.

5 Linearised theory and small fluctuations

5.1 Expansion around a minimum of W

In this chapter we analyse small fluctuations of the field $s : M \rightarrow S$ around a static configuration that minimises the potential. This leads to a linearised theory whose modes can be interpreted as a (possibly countable) collection of Klein–Gordon fields on (M, g) with effective mass parameters determined by the spectrum of the Hessian of W .

5.1.1 Constant background configurations

Let $s_* \in S$ be a local minimum of the potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$. We assume that $s_* \in \mathcal{D}(W)$, that W is twice Fréchet differentiable in a neighbourhood of s_* , and that the gradient and Hessian satisfy

$$\nabla_s W(s_*) = 0, \quad H := D^2 W(s_*) : S \rightarrow S,$$

where H is self-adjoint and bounded from below with respect to the inner product on S :

$$\langle Hu, u \rangle_S \geq c \|u\|_S^2 \quad \text{for some } c \in \mathbb{R} \text{ and all } u \in S.$$

We do not require H to be strictly positive; zero eigenvalues corresponding to flat directions are allowed.

The constant field configuration

$$s_*(x) := s_*, \quad x \in M,$$

is a solution of the full nonlinear Euler–Lagrange equations

$$\square_g s + \nabla_s W(s) = 0,$$

because $\square_g s_* = 0$ and $\nabla_s W(s_*) = 0$. We are interested in small perturbations of the form

$$s(x) = s_* + \phi(x),$$

where $\phi : M \rightarrow S$ is an S -valued fluctuation field.

5.1.2 Quadratic expansion of the potential

Near s_* the potential admits a Taylor expansion

$$W(s_* + \phi) = W(s_*) + \frac{1}{2} \langle \phi, H\phi \rangle_S + R(\phi), \quad (5.1)$$

where the remainder term $R(\phi)$ satisfies

$$\lim_{\|\phi\|_S \rightarrow 0} \frac{|R(\phi)|}{\|\phi\|_S^2} = 0.$$

In particular, there exists a neighbourhood of $0 \in S$ and a function $\rho : [0, \infty) \rightarrow [0, \infty)$ with $\rho(r) \rightarrow 0$ as $r \rightarrow 0$ such that

$$|R(\phi)| \leq \rho(\|\phi\|_S) \|\phi\|_S^2$$

whenever ϕ lies in that neighbourhood.

The gradient of W near s_* is then

$$\nabla_s W(s_* + \phi) = H\phi + r(\phi), \quad (5.2)$$

where $r(\phi) \in S$ satisfies

$$\lim_{\|\phi\|_S \rightarrow 0} \frac{\|r(\phi)\|_S}{\|\phi\|_S} = 0.$$

The map r collects all terms that are at least quadratic in ϕ .

5.1.3 Quadratic Lagrangian and linearised action

The kinetic term depends only on derivatives of s and is quadratic. Since s_* is constant in x , we have

$$\nabla_\mu s = \nabla_\mu(s_* + \phi) = \nabla_\mu \phi,$$

and therefore

$$\mathcal{L}_{\text{kin}}(s, \nabla s; g) = \frac{1}{2} \langle \nabla_\mu \phi, \nabla^\mu \phi \rangle_S.$$

Using (5.1), the full Lagrangian density becomes

$$\begin{aligned} \mathcal{L}(s, \nabla s; g) &= \frac{1}{2} \langle \nabla_\mu \phi, \nabla^\mu \phi \rangle_S - W(s_* + \phi) \\ &= \frac{1}{2} \langle \nabla_\mu \phi, \nabla^\mu \phi \rangle_S - W(s_*) - \frac{1}{2} \langle \phi, H\phi \rangle_S - R(\phi). \end{aligned} \quad (5.3)$$

The constant term $-W(s_*)$ does not affect the equations of motion, and the remainder $R(\phi)$ is at least cubic in ϕ . To analyse small fluctuations we therefore consider the *quadratic* Lagrangian density

$$\mathcal{L}^{(2)}(\phi, \nabla \phi; g) := \frac{1}{2} \langle \nabla_\mu \phi, \nabla^\mu \phi \rangle_S - \frac{1}{2} \langle \phi, H\phi \rangle_S. \quad (5.4)$$

The corresponding quadratic action is

$$S^{(2)}[\phi] := \int_M \mathcal{L}^{(2)}(\phi, \nabla \phi; g) d\mu_g. \quad (5.5)$$

Varying (5.5) with respect to ϕ as in Chapter 4 yields the linearised Euler–Lagrange equations

$$\square_g \phi + H\phi = 0, \quad (5.6)$$

to be understood as an equation in S at each spacetime point $x \in M$. In other words, for each x ,

$$\square_g \phi(x) + H\phi(x) = 0 \quad \text{in } S.$$

5.2 Diagonalisation of the Hessian and mode decomposition

The operator $H : S \rightarrow S$ encodes the internal structure of the small fluctuations around s_* . Its spectral properties determine the effective mass spectrum of the theory in the linear approximation.

5.2.1 Spectral decomposition in a simple setting

For concreteness, we first consider the case where S is real and H is a bounded, self-adjoint operator with pure point spectrum. That is, we assume there exists an orthonormal basis $\{e_k\}_{k \in K}$ of S (with K countable) and real numbers $\{m_k^2\}_{k \in K}$ such that

$$He_k = m_k^2 e_k, \quad k \in K.$$

The eigenvalues m_k^2 are bounded from below by the constant c appearing in the coercivity condition on H . We refer to $m_k \geq 0$ as the *effective masses* of the modes.

Expanding the fluctuation field in this eigenbasis,

$$\phi(x) = \sum_{k \in K} \phi^k(x) e_k,$$

we obtain

$$\nabla_\mu \phi(x) = \sum_{k \in K} \nabla_\mu \phi^k(x) e_k, \quad H\phi(x) = \sum_{k \in K} m_k^2 \phi^k(x) e_k,$$

where each $\phi^k : M \rightarrow \mathbb{R}$ (or $\mathbb{C}$) is a scalar field.

Substituting into the quadratic Lagrangian density (5.4), we find

$$\begin{aligned} \mathcal{L}^{(2)}(\phi, \nabla \phi; g) &= \frac{1}{2} \langle \nabla_\mu \phi, \nabla^\mu \phi \rangle_S - \frac{1}{2} \langle \phi, H\phi \rangle_S \\ &= \frac{1}{2} g^{\mu\nu} \left\langle \sum_k \nabla_\mu \phi^k e_k, \sum_\ell \nabla_\nu \phi^\ell e_\ell \right\rangle_S - \frac{1}{2} \left\langle \sum_k \phi^k e_k, \sum_\ell m_\ell^2 \phi^\ell e_\ell \right\rangle_S \\ &= \frac{1}{2} g^{\mu\nu} \sum_k \nabla_\mu \phi^k \nabla_\nu \phi^k - \frac{1}{2} \sum_k m_k^2 (\phi^k)^2, \end{aligned} \tag{5.7}$$

where orthonormality of $\{e_k\}$ was used to eliminate cross-terms. Thus,

$$S^{(2)}[\phi] = \sum_{k \in K} \int_M \left(\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi^k \nabla_\nu \phi^k - \frac{1}{2} m_k^2 (\phi^k)^2 \right) d\mu_g.$$

The quadratic action decomposes into a (countable) sum of standard Klein–Gordon actions for scalar fields ϕ^k with masses m_k .

The linearised field equation (5.6) similarly decomposes into decoupled scalar equations:

$$\square_g \phi^k + m_k^2 \phi^k = 0, \quad k \in K.$$

5.2.2 Massless and massive modes

The sign and size of the eigenvalues m_k^2 play an important role in the stability properties of the background configuration s_* .

- If $m_k^2 > 0$ for a given mode k , the corresponding fluctuation ϕ^k satisfies the Klein–Gordon equation with positive mass squared. On a globally hyperbolic background (M, g) this leads to stable oscillatory behaviour under appropriate boundary conditions.

- If $m_k^2 = 0$, the mode is *massless* and satisfies the covariant wave equation

$$\square_g \phi^k = 0.$$

Such modes typically arise from flat directions of W at s_* , for example due to continuous symmetries. Their behaviour is sensitive to the global geometry of (M, g) and to boundary conditions.

- If $m_k^2 < 0$ for some k , the quadratic action is not positive definite in that direction and the background s_* is unstable along the corresponding mode. In that case the mode equation

$$\square_g \phi^k + m_k^2 \phi^k = 0$$

may exhibit exponential growth in time in suitable regions of spacetime. Since we assumed that s_* is a local minimum of W and that H is bounded from below, such negative eigenvalues are excluded in the present discussion.

For real fields, the quadratic form

$$Q(\phi) := \int_M \left(\frac{1}{2} \langle \nabla_\mu \phi, \nabla^\mu \phi \rangle_S + \frac{1}{2} \langle \phi, H\phi \rangle_S \right) d\mu_g$$

is non-negative if H is non-negative. This ensures that the linearised energy is positive and supports stability of the background against small perturbations.

5.2.3 More general spectral types (sketch)

If H is an unbounded self-adjoint operator or has continuous spectrum, the decomposition into modes is more naturally expressed using the spectral theorem. There exists a projection-valued measure $E(\lambda)$ on $\mathbb{R}$ such that

$$H = \int_{\mathbb{R}} \lambda dE(\lambda),$$

and for each $\phi \in S$ one can write

$$\phi = \int_{\mathbb{R}} dE(\lambda) \phi.$$

The quadratic form associated with H is then

$$\langle \phi, H\phi \rangle_S = \int_{\mathbb{R}} \lambda d\mu_\phi(\lambda),$$

where μ_ϕ is the spectral measure corresponding to ϕ .

Formally, one may view the fluctuation field ϕ as a superposition of modes labelled by the spectral parameter λ , each satisfying a Klein–Gordon-type equation with effective mass squared $m^2 = \lambda$. In practice, this leads to a continuous or mixed discrete-continuous mode spectrum. A detailed analysis of such cases requires functional-analytic tools beyond the scope of this paper, but the conceptual picture remains the same: the linearised dynamics decomposes into independent Klein–Gordon modes whose masses are determined by the spectrum of the Hessian H .

5.3 Examples on simple backgrounds

To illustrate the general structure of the linearised theory, we now consider two simple classes of backgrounds on which the mode equations take a particularly transparent form: Minkowski spacetime and ultrastatic spacetimes.

5.3.1 Minkowski spacetime

Let $(M, g) = (\mathbb{R}^{1+n}, \eta)$ be Minkowski spacetime with standard coordinates $(t, x^1, \dots, x^n)$ and metric

$$\eta = -dt^2 + \sum_{i=1}^n (dx^i)^2.$$

The d'Alembert operator is

$$\square_\eta = -\partial_t^2 + \Delta_x,$$

where Δ_x is the Laplacian in the spatial variables $x = (x^1, \dots, x^n)$.

For each mode $\phi^k : \mathbb{R}^{1+n} \rightarrow \mathbb{R}$ (or $\mathbb{C}$) the linearised equation becomes

$$(-\partial_t^2 + \Delta_x) \phi^k(t, x) + m_k^2 \phi^k(t, x) = 0. \quad (5.8)$$

This is the standard Klein–Gordon equation on Minkowski space with mass m_k . Its solutions can be written in terms of Fourier modes. Let

$$\hat{\phi}^k(t, p) := \int_{\mathbb{R}^n} e^{-ip \cdot x} \phi^k(t, x) d^n x,$$

where $p \in \mathbb{R}^n$ is the spatial momentum. Then (5.8) implies

$$\partial_t^2 \hat{\phi}^k(t, p) + \omega_k(p)^2 \hat{\phi}^k(t, p) = 0,$$

with dispersion relation

$$\omega_k(p) := \sqrt{|p|^2 + m_k^2}.$$

The general solution is

$$\hat{\phi}^k(t, p) = a_k(p) e^{-i\omega_k(p)t} + b_k(p) e^{+i\omega_k(p)t},$$

with coefficients $a_k(p)$ and $b_k(p)$ determined by the initial data. Transforming back to position space, we see that each mode propagates as a free Klein–Gordon field with the usual relativistic dispersion relation.

The linearised energy associated with mode k is

$$E_k(t) := \int_{\mathbb{R}^n} \left(\frac{1}{2} |\partial_t \phi^k|^2 + \frac{1}{2} |\nabla_x \phi^k|^2 + \frac{1}{2} m_k^2 |\phi^k|^2 \right) dx,$$

which is conserved in time and non-negative if $m_k^2 \geq 0$.

5.3.2 Ultrastatic spacetimes

Next, consider an ultrastatic spacetime of the form

$$(M, g) = (\mathbb{R} \times \Sigma, -dt^2 + h_{ij}(x) dx^i dx^j),$$

where (Σ, h) is an n -dimensional Riemannian manifold. In this case the d'Alembert operator is

$$\square_g = -\partial_t^2 + \Delta_h,$$

where Δ_h is the Laplace–Beltrami operator on (Σ, h) .

Let $\{\psi_\ell\}_{\ell \in L}$ be an orthonormal basis of $L^2(\Sigma)$ consisting of eigenfunctions of $-\Delta_h$:

$$-\Delta_h \psi_\ell = \lambda_\ell \psi_\ell, \quad \lambda_\ell \geq 0.$$

Assume, for simplicity, that (Σ, h) is compact so that the spectrum is discrete and $\lambda_\ell \rightarrow \infty$ as $\ell \rightarrow \infty$.

For each internal mode k and spatial mode ℓ , we consider an ansatz

$$\phi^k(t, x) = q_{k\ell}(t) \psi_\ell(x),$$

with $q_{k\ell} : \mathbb{R} \rightarrow \mathbb{R}$ (or $\mathbb{C}$). Substituting into the mode equation

$$\square_g \phi^k + m_k^2 \phi^k = 0$$

yields

$$-\ddot{q}_{k\ell}(t) \psi_\ell(x) + \Delta_h(q_{k\ell}(t) \psi_\ell(x)) + m_k^2 q_{k\ell}(t) \psi_\ell(x) = 0,$$

or equivalently

$$\ddot{q}_{k\ell}(t) + \omega_{k\ell}^2 q_{k\ell}(t) = 0,$$

with

$$\omega_{k\ell} := \sqrt{\lambda_\ell + m_k^2}.$$

Thus, each pair (k, ℓ) defines a harmonic oscillator in time with frequency $\omega_{k\ell}$. The corresponding spacetime mode is

$$\phi_{(\ell)}^k(t, x) = (A_{k\ell} e^{-i\omega_{k\ell} t} + B_{k\ell} e^{+i\omega_{k\ell} t}) \psi_\ell(x),$$

where $A_{k\ell}$ and $B_{k\ell}$ are determined by initial data.

The full fluctuation field can be written as

$$\phi(t, x) = \sum_{k \in K} \sum_{\ell \in L} \phi_{(\ell)}^k(t, x) e_k,$$

with convergence understood in an appropriate sense (for example, in $L^2(\Sigma, S)$ for each fixed t). The quadratic action (5.5) reduces to a sum of harmonic oscillator actions for the variables $q_{k\ell}(t)$.

5.3.3 General globally hyperbolic backgrounds

On a general globally hyperbolic spacetime (M, g) , the linearised equation

$$\square_g \phi + H \phi = 0$$

remains a hyperbolic system of partial differential equations, with the same internal operator H as in the examples above. The mode decomposition is less explicit, but the key structural features persist:

- For each eigenvector e_k of H with eigenvalue m_k^2 , the projection

$$\phi^k(x) := \langle \phi(x), e_k \rangle_S$$

satisfies the scalar Klein–Gordon equation

$$\square_g \phi^k + m_k^2 \phi^k = 0.$$

The theory thus decomposes into independent scalar fields on (M, g) .

- The Cauchy problem for each ϕ^k is well posed under standard assumptions on (M, g) , with solutions determined uniquely by initial data on a Cauchy hypersurface.

- The total linearised energy is given by a sum (or integral, in the presence of continuous spectrum) of the energies associated with each mode. If H is non-negative, this energy is positive and provides a control norm for the evolution of small fluctuations.

These observations justify viewing the linearised theory as an effective field theory for a collection of Klein–Gordon modes on (M, g) , each arising from a direction in S determined by the Hessian of the potential at s_* . In the next chapter we will use the kinetic structure of these fluctuations to define an induced symmetric tensor on M and discuss its basic properties.

6 Optional induced semantic metric

6.1 Definition and basic properties of the induced tensor $h_{\mu\nu}$

In this chapter we introduce a symmetric tensor built from the covariant derivatives of the field $s : M \rightarrow S$. This tensor does not modify the background metric $g_{\mu\nu}$, nor is it treated as dynamical in the present work. Instead, it is an auxiliary object that encodes how the field s varies along different directions in spacetime.

6.1.1 Definition

Let $s : M \rightarrow S$ be a sufficiently regular solution of the Euler–Lagrange equations

$$\square_g s + \nabla_s W(s) = 0.$$

We define a symmetric $(0, 2)$ -tensor field $h_{\mu\nu}[s]$ on M by

$$h_{\mu\nu}[s](x) := \langle \nabla_\mu s(x), \nabla_\nu s(x) \rangle_S, \quad x \in M. \quad (6.1)$$

Here $\nabla_\mu s$ denotes the covariant derivative of s with respect to the Levi-Civita connection of g , and $\langle \cdot, \cdot \rangle_S$ is the inner product on S .

The tensor $h_{\mu\nu}$ is well defined for fields s in any class for which $\nabla_\mu s$ exists and is square-integrable on compact sets. In particular, for smooth s we obtain a smooth symmetric tensor field. The definition (6.1) is independent of the choice of orthonormal basis of S , since the inner product $\langle \cdot, \cdot \rangle_S$ is basis invariant.

6.1.2 Symmetry and positivity properties

By construction, $h_{\mu\nu}$ is symmetric:

$$h_{\mu\nu} = h_{\nu\mu},$$

because $\langle \nabla_\mu s, \nabla_\nu s \rangle_S$ is symmetric in μ and ν .

For any tangent vector $X = X^\mu \partial_\mu$ at $x \in M$, we can evaluate $h_{\mu\nu}$ on X twice:

$$h(X, X) := h_{\mu\nu}(x) X^\mu X^\nu = \langle X^\mu \nabla_\mu s(x), X^\nu \nabla_\nu s(x) \rangle_S. \quad (6.2)$$

This expression is non-negative if S is a real Hilbert space and the inner product is positive definite. Indeed,

$$h(X, X) = \langle \nabla_X s(x), \nabla_X s(x) \rangle_S = \|\nabla_X s(x)\|_S^2 \geq 0.$$

Thus $h_{\mu\nu}$ is positive semi-definite as a quadratic form on each tangent space $T_x M$. It is degenerate in directions along which the field is covariantly constant, i.e. $\nabla_X s(x) = 0$.

The trace of $h_{\mu\nu}$ with respect to the metric $g^{\mu\nu}$ reproduces the kinetic term of the Lagrangian density:

$$g^{\mu\nu}h_{\mu\nu} = \langle \nabla_\mu s, \nabla^\mu s \rangle_S. \quad (6.3)$$

Hence the kinetic contribution to the Lagrangian can be written as

$$\mathcal{L}_{\text{kin}}(s, \nabla s; g) = \frac{1}{2} g^{\mu\nu} h_{\mu\nu}.$$

In this sense, $h_{\mu\nu}$ refines the kinetic energy density by resolving it into directional contributions along different tangent directions.

6.1.3 Behaviour under diffeomorphisms and internal isometries

The tensor $h_{\mu\nu}$ is constructed from $\nabla_\mu s$ and the inner product on S . Consequently, it has the following transformation properties.

- *Spacetime diffeomorphisms.* Under a diffeomorphism $\Phi : M \rightarrow M$, the field s transforms as $s \mapsto s \circ \Phi^{-1}$. The covariant derivative transforms as a tensor, and the scalar product in S is invariant. Therefore $h_{\mu\nu}$ transforms as a $(0, 2)$ -tensor on M :

$$h'_{\mu\nu}(x') = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} h_{\rho\sigma}(x),$$

where $x' = \Phi(x)$. No additional structure is introduced beyond that already present in g and s .

- *Internal unitary or orthogonal transformations.* Let $U : S \rightarrow S$ be a unitary (complex case) or orthogonal (real case) operator preserving $\langle \cdot, \cdot \rangle_S$. If we transform the field by

$$s(x) \mapsto (Us)(x) := U(s(x)),$$

then

$$\nabla_\mu (Us)(x) = U(\nabla_\mu s(x)),$$

because the connection acts only on the spacetime argument. Hence

$$h_{\mu\nu}[Us] = \langle U(\nabla_\mu s), U(\nabla_\nu s) \rangle_S = \langle \nabla_\mu s, \nabla_\nu s \rangle_S = h_{\mu\nu}[s].$$

Thus $h_{\mu\nu}$ is invariant under all isometries of S .

These properties show that $h_{\mu\nu}$ is an intrinsic geometric object associated with the pair (g, s) and the inner product on S . It depends neither on the choice of coordinates on M nor on the choice of orthonormal basis in S .

6.1.4 Relation to the stress-energy tensor

The stress-energy tensor $T_{\mu\nu}$ associated with the Lagrangian density

$$\mathcal{L}(s, \nabla s; g) = \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s)$$

was given in Chapter 3 as

$$T_{\mu\nu} = \langle \nabla_\mu s, \nabla_\nu s \rangle_S - g_{\mu\nu} \mathcal{L}(s, \nabla s; g).$$

Comparing with (6.1), we see that

$$T_{\mu\nu} = h_{\mu\nu} - g_{\mu\nu} \left(\frac{1}{2} g^{\rho\sigma} h_{\rho\sigma} - W(s) \right). \quad (6.4)$$

Thus $h_{\mu\nu}$ can be regarded as the “kinetic” part of $T_{\mu\nu}$. The full stress-energy tensor combines $h_{\mu\nu}$ with its trace and the potential term $W(s)$.

In the present work we do not couple $T_{\mu\nu}$ back into Einstein’s equations, and we do not propose any modification of the gravitational dynamics. Equation (6.4) is used only to clarify the structural role of $h_{\mu\nu}$ within the field theory on the fixed background (M, g) .

6.2 Interpretation as additional structure

The tensor $h_{\mu\nu}$ can be interpreted as providing additional information about the field configuration s on spacetime, beyond what is encoded in the scalar quantities appearing in the action. In this section we discuss several mathematical viewpoints on this additional structure.

6.2.1 Directional variation of the field

For any tangent vector $X \in T_x M$, the quantity

$$h(X, X) = \|\nabla_X s(x)\|_S^2$$

measures the squared norm of the covariant derivative of s along X . Thus, $h_{\mu\nu}$ captures how s changes along different directions in spacetime. Regions where $h(X, X)$ is large for some timelike or spacelike X are regions where the field varies rapidly in the corresponding direction.

One can view $h_{\mu\nu}$ as defining, at each point x , a positive semi-definite bilinear form

$$h_x : T_x M \times T_x M \rightarrow \mathbb{R},$$

which may be degenerate in directions along which s is covariantly constant. The rank of h_x provides a measure of how many independent directions in $T_x M$ carry nontrivial variation of s .

6.2.2 Comparison with the background metric

The background metric $g_{\mu\nu}$ encodes causal and geometric structure on M . The tensor $h_{\mu\nu}$, by contrast, encodes structure induced by the field s . It is important to emphasise that $h_{\mu\nu}$ is *not* a metric tensor in general:

- It is positive semi-definite when evaluated on tangent vectors, whereas $g_{\mu\nu}$ has Lorentzian signature.
- It may be degenerate, with null directions corresponding to $\nabla_X s = 0$.
- It is derived from s and therefore vanishes identically if s is covariantly constant.

Nonetheless, one can consider the pair $(g_{\mu\nu}, h_{\mu\nu})$ as a richer geometric structure on M . The tensor $h_{\mu\nu}$ singles out directions along which the field varies and assigns them a non-negative weight. This can be compared to anisotropic media in electromagnetism, where additional tensorial structures encode preferred directions for propagation or interaction, without altering the fundamental spacetime metric.

6.2.3 Level sets and gradients in S

Assuming sufficient regularity of s , one can consider the image $\mathcal{I} := s(M) \subset S$. The tensor $h_{\mu\nu}$ measures how the field explores $\mathcal{I}$ along spacetime directions. If $\mathcal{I}$ lies near a low-dimensional submanifold of S , then at each x the vectors $\nabla_\mu s(x)$ span only a low-dimensional subspace of S . In this case $h_{\mu\nu}$ may have low rank, indicating that the effective internal degrees of freedom active near x are fewer than the full dimension of S .

From this perspective, $h_{\mu\nu}$ can be interpreted as encoding an emergent “local internal geometry” induced by the way the field s traverses S along spacetime. While we do not develop this viewpoint in detail here, it suggests that $h_{\mu\nu}$ may be useful in characterising the structure of solutions of the field equations beyond scalar invariants.

6.2.4 Energy densities on Cauchy hypersurfaces

On a given Cauchy hypersurface Σ_t with future-directed unit normal n^μ , the energy density associated with the kinetic term can be expressed in terms of $h_{\mu\nu}$ as

$$\mathcal{E}_t(x) = \frac{1}{2} h_{\mu\nu}(x) n^\mu n^\nu + \frac{1}{2} h_{ij}(x) \gamma^{ij}(x),$$

where γ^{ij} is the inverse of the induced Riemannian metric on Σ_t and indices i, j are spatial. Here the first term measures the contribution from temporal variations of s , and the second term measures the contribution from spatial variations.

The total kinetic energy on Σ_t is then

$$E_t^{\text{kin}}[s] = \int_{\Sigma_t} \mathcal{E}_t(x) d\mu_{\Sigma_t}(x),$$

which depends on $h_{\mu\nu}$ and the geometry of Σ_t but not directly on the potential W . This makes $h_{\mu\nu}$ a natural object for discussing energy distribution of the field across spacetime slices.

6.3 Speculative couplings

In this section we briefly outline, in a clearly speculative manner, how the tensor $h_{\mu\nu}$ could enter additional couplings or effective descriptions. These remarks go beyond the scope of the rigorous constructions in the previous chapters and are not developed further in this paper. They are included only to indicate possible directions for future work.

6.3.1 Couplings to other matter fields (speculative)

Suppose Φ denotes another matter field on (M, g) , for example a scalar, vector, or spinor field with its own standard Lagrangian density $\mathcal{L}_\Phi$. One may consider non-minimal couplings in which $h_{\mu\nu}$ modulates the dynamics of Φ . For instance, a schematic interaction term could take the form

$$\mathcal{L}_{\text{int}} \sim \lambda h_{\mu\nu} J^{\mu\nu}[\Phi],$$

where λ is a coupling constant and $J^{\mu\nu}[\Phi]$ is a symmetric tensor constructed from Φ and its derivatives (for example, a current or a stress-like tensor).

Such couplings would allow the spatial and temporal variations of s to influence the propagation or interaction of other fields, while $g_{\mu\nu}$ continues to determine the causal structure. In the present paper we do not specify any particular form of $J^{\mu\nu}$, nor do we analyse the resulting dynamics.

6.3.2 Modifications of effective propagation (speculative)

Given the positive semi-definite nature of $h_{\mu\nu}$, one might use it to define effective propagation tensors for certain degrees of freedom. As a schematic example, one could consider an effective Lagrangian for a scalar probe field ψ of the form

$$\mathcal{L}_\psi = -\frac{1}{2}(g^{\mu\nu} + \alpha \tilde{h}^{\mu\nu})\nabla_\mu\psi\nabla_\nu\psi - V_\psi(\psi),$$

where α is a small parameter, and $\tilde{h}^{\mu\nu}$ is an appropriate contravariant tensor derived from $h_{\mu\nu}$ (for example using $g^{\mu\nu}$ to raise indices and subtract traces). In such a model, gradients of s would slightly modify the effective propagation of ψ without altering the underlying metric $g_{\mu\nu}$.

We emphasise that this is a speculative construction. We do not introduce or analyse any such effective Lagrangian in the present work.

6.3.3 Couplings to gravity (speculative)

One could also imagine extending the gravitational action by adding terms that depend on $h_{\mu\nu}$, for example

$$S_{\text{grav}+\text{induced}} = \frac{1}{16\pi G} \int_M R(g) d\mu_g + \int_M F(g, h) d\mu_g,$$

where $R(g)$ is the scalar curvature of g and $F(g, h)$ is a scalar functional of $g_{\mu\nu}$ and $h_{\mu\nu}$, built from invariants such as

$$g^{\mu\nu}h_{\mu\nu}, \quad h_{\mu\nu}h^{\mu\nu}, \quad \text{and higher-order contractions.}$$

Varying such an action with respect to $g_{\mu\nu}$ would lead to modified gravitational field equations in which $h_{\mu\nu}$ contributes additional source terms or effective corrections.

In the present paper we explicitly *do not* pursue this direction. We do not vary $g_{\mu\nu}$ and we do not propose any modified gravitational dynamics. The metric $g_{\mu\nu}$ remains a fixed background, and $h_{\mu\nu}$ is considered only as a derived tensor field.

6.3.4 Summary of speculative aspects

The possible couplings sketched above illustrate that $h_{\mu\nu}$ could, in principle, serve as a bridge between the internal Hilbert-space dynamics of s and additional effective structures on spacetime. However, all such uses would require careful analysis to ensure consistency with existing theoretical and observational constraints. They lie beyond the scope of the present work, whose focus is the conservative formulation of the field theory for s on a fixed Lorentzian background.

In the final chapter we will return to a more global perspective, summarising the construction developed so far, relating it to the earlier Hilbert-space based programme, and outlining directions for future research, including more detailed investigations of $h_{\mu\nu}$ and its possible roles.

7 Discussion and outlook

7.1 Summary of the construction

In this paper we have formulated a classical field theory on a fixed Lorentzian spacetime (M, g) whose basic field

$$s : M \rightarrow S$$

takes values in a separable Hilbert space S . The construction combines standard techniques from classical field theory on curved backgrounds with the Hilbert-space and potential structures developed in earlier work.

The geometric and functional-analytic setup was specified as follows.

- The spacetime (M, g) is a smooth, oriented, time-oriented, globally hyperbolic Lorentzian manifold of dimension $n + 1$ with signature $(-, +, \dots, +)$. The Levi-Civita connection ∇_μ and the d'Alembert operator

$$\square_g := \nabla^\mu \nabla_\mu$$

govern covariant differentiation and wave propagation on (M, g) .

- The internal degrees of freedom are carried by a fixed separable Hilbert space S with inner product $\langle \cdot, \cdot \rangle_S$ and norm $\| \cdot \|_S$. The potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$ is assumed proper, lower semicontinuous, bounded from below, and Fréchet differentiable on a dense domain $\mathcal{D}(W) \subset S$, with gradient $\nabla_s W(s) \in S$ defined via the Riesz isomorphism.
- Fields s are considered as sections of the trivial Hilbert bundle $M \times S \rightarrow M$ with sufficient regularity for the kinetic and potential contributions to the action to be integrable. A convenient class is

$$\mathcal{A} = \left\{ s \in H_{\text{loc}}^1(M, S) \mid W(s(\cdot)) \in L_{\text{loc}}^1(M, \mathbb{R}) \right\}.$$

The Lagrangian density for the field s on (M, g) was defined by

$$\mathcal{L}(s, \nabla s; g) = \frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s),$$

where the kinetic term uses the covariant derivative and the inner product on S , and the potential term is inherited from the Hilbert-space framework. The corresponding action functional is

$$S[s] = \int_M \left(\frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s) \right) d\mu_g,$$

with $d\mu_g$ the volume measure induced by g .

Varying $S[s]$ with respect to s under compactly supported variations δs yields the Euler–Lagrange equations

$$\square_g s + \nabla_s W(s) = 0,$$

to be understood as an S -valued equation on M , or equivalently as a coupled system of scalar equations in any orthonormal basis of S . This equation combines the geometric propagation dictated by $\square_g$ with an internal “force” provided by the gradient of W .

Because a full infinite-dimensional PDE theory for this equation is technically demanding, we also considered finite-dimensional truncations $S_N \subset S$. In these truncations the field $s_N : M \rightarrow S_N$ is equivalent to a multiplet of scalar fields satisfying a system of semilinear wave equations, for which well-posedness on globally hyperbolic spacetimes is standard. The infinite-dimensional theory can be viewed formally as a limit $N \rightarrow \infty$ of such truncations.

Around a local minimum $s_* \in S$ of W , we linearised the theory by writing $s = s_* + \phi$ and expanding the potential to quadratic order,

$$W(s_* + \phi) = W(s_*) + \frac{1}{2} \langle \phi, H\phi \rangle_S + \text{higher orders},$$

with $H = D^2W(s_*)$ a self-adjoint, bounded-below operator. The resulting quadratic Lagrangian for the fluctuation field ϕ is

$$\mathcal{L}^{(2)}(\phi, \nabla\phi; g) = \frac{1}{2} \langle \nabla_\mu \phi, \nabla^\mu \phi \rangle_S - \frac{1}{2} \langle \phi, H\phi \rangle_S,$$

leading to the linearised equation

$$\square_g \phi + H\phi = 0.$$

When H is diagonalised in an orthonormal eigenbasis of S , this equation decomposes into a (possibly countable) family of scalar Klein–Gordon equations with effective masses determined by the eigenvalues of H . We illustrated this mode structure on simple backgrounds such as Minkowski spacetime and ultrastatic spacetimes.

Finally, we introduced an induced symmetric tensor

$$h_{\mu\nu}[s] = \langle \nabla_\mu s, \nabla_\nu s \rangle_S,$$

which refines the kinetic term by encoding the directional variation of s along tangent vectors on M . We emphasised that $h_{\mu\nu}$ is an auxiliary structure derived from s and g and is not identified with the spacetime metric.

7.2 Relation to the semantic programme (Papers 1–3)

The present work is designed to be compatible with a broader programme in which the Hilbert space S and the potential W have been studied independently of spacetime, with a focus on their internal geometry and dynamics.

In the first paper of that programme, S was introduced as a separable Hilbert space equipped with a functional $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$. The central objects were gradient flows

$$\frac{d}{d\tau} s(\tau) = -\nabla_s W(s(\tau))$$

in an abstract time parameter τ . This yielded a dynamical system on S driven by W , with attention to properties such as existence, uniqueness, and convergence of trajectories towards minima of W . In that setting, W played the role of a Lyapunov functional or “energy” governing relaxation behaviour.

The second paper recast these constructions in terms of a free-energy functional on S and studied distribution dynamics rather than single trajectories. This introduced a statistical perspective in which W contributed to a free energy that shapes equilibrium measures and fluctuation patterns in S . The emphasis was on how W structures ensembles of states and their relaxation under suitable flows.

The third paper embedded standard quantum systems into the same Hilbert space S , showing that S can carry quantum states and observables in a way compatible with the previous constructions. This connected the semantic structures encoded by W with the usual Hilbert-space formalism of quantum mechanics, at least at the level of kinematics and certain dynamical mappings.

The present paper extends these ideas by introducing explicit spacetime dependence in the form of fields $s : M \rightarrow S$ and by coupling the internal dynamics of W to the geometry of a Lorentzian manifold (M, g) via a covariant kinetic term. This introduces a clear parallel between:

- the gradient flow equation

$$\frac{d}{d\tau}s(\tau) = -\nabla_s W(s(\tau))$$

on S from the earlier work, and

- the covariant wave-type equation

$$\square_g s + \nabla_s W(s) = 0$$

on (M, g) derived here.

In the gradient-flow case, the dynamics is first order in the abstract parameter τ and purely dissipative with respect to W . In the field-theoretic case, the dynamics is second order in the physical time coordinate and combines propagation on (M, g) with forces derived from W . The same potential thus plays two roles: as a generator of gradient flows on S and as a potential term in a Lorentz-covariant field theory.

The linearised analysis around a minimum s_* further highlights this connection. In the purely Hilbert-space setting, the Hessian $H = D^2W(s_*)$ determines stability properties and local curvature of W . In the present field-theoretic setting, the same operator H determines the mass spectrum of small fluctuations around the constant field configuration s_* , via the decomposed equations

$$\square_g \phi^k + m_k^2 \phi^k = 0,$$

with m_k^2 the eigenvalues of H . Thus the spectral properties of H have a dual interpretation: they describe both the curvature of W in S and the masses of the linearised field modes on spacetime.

Finally, the induced tensor

$$h_{\mu\nu}[s] = \langle \nabla_\mu s, \nabla_\nu s \rangle_S$$

can be seen as importing part of the Hilbert-space geometry of S into the spacetime setting, by measuring how the field explores S along different spacetime directions. It thereby links the internal structure studied in the earlier papers with the external geometry of (M, g) , without altering the latter.

7.3 Limitations

The framework developed here is deliberately conservative and subject to several limitations, both conceptual and technical.

7.3.1 Fixed background and absence of backreaction

Throughout the paper the Lorentzian metric $g_{\mu\nu}$ has been treated as an externally prescribed structure. We did not vary $g_{\mu\nu}$, did not include any gravitational action, and did not derive or modify Einstein's equations. The field s propagates on (M, g) but does not backreact on the geometry.

As a consequence, the theory constructed here is not a theory of gravity or a modification of general relativity. It is an effective field theory for an additional degree of freedom s on a fixed background. Any attempt to include gravitational backreaction, or to interpret $h_{\mu\nu}$ as contributing directly to the curvature of spacetime, would require further assumptions and a separate analysis that lies outside the scope of this work.

7.3.2 Infinite-dimensional functional analysis

While the formal structure of the Euler–Lagrange equation

$$\square_g s + \nabla_s W(s) = 0$$

is clear, a complete analytic theory in the infinite-dimensional setting is nontrivial. Issues include:

- the choice of appropriate function spaces for S -valued fields on a globally hyperbolic spacetime;
- the domain and regularity properties of $\nabla_s W(s)$ when W is defined only on a dense subset of S ;
- the control of nonlinearities arising from W in energy estimates and existence theorems;
- the relation between solutions of finite-dimensional truncations and solutions of the full equation as $N \rightarrow \infty$.

In this paper we have addressed these issues only at a formal level, supplemented by a more rigorous discussion for finite-dimensional truncations, where the theory reduces to standard systems of semilinear wave equations. A systematic infinite-dimensional analysis would require additional structural assumptions on W (for example monotonicity, Lipschitz-type conditions, or compactness properties) and a careful application of techniques from the theory of evolution equations in Banach and Hilbert spaces.

7.3.3 Linearisation and small-fluctuation regime

The linearised theory around a minimum of W provides a clear and tractable description of small fluctuations in terms of decoupled Klein–Gordon modes. However, this approximation relies on the assumption that fluctuations remain small enough for higher-order terms in the expansion of W to be negligible. In regimes where $\|\phi\|_S$ is not small, the cubic and higher-order terms in W become important and can lead to mode coupling, nonlinear resonances, or instabilities that are not captured by the quadratic approximation.

Moreover, our spectral analysis assumed for simplicity that H admits a basis of eigenvectors with real eigenvalues. In more general infinite-dimensional settings, the spectrum of H may contain continuous parts, and the mode decomposition may involve generalised eigenfunctions or spectral measures. A complete treatment of such cases was only sketched here.

7.3.4 Interpretation and phenomenology

The present work has focused on the mathematical and structural aspects of the theory. We have not attempted to provide a detailed physical interpretation of s in specific models, nor have we explored phenomenological consequences or observational constraints. In particular, we have not made claims about explaining dark matter, dark energy, inflation, or other open problems in fundamental physics. Any such applications would require explicit choices of S and W , together with a comparison to empirical data, and are not part of the present analysis.

7.4 Future directions

Despite these limitations, the framework developed here opens up a variety of directions for further work. We briefly outline some of them.

7.4.1 Rigorous analysis of the infinite-dimensional field equation

A natural next step is to develop a rigorous theory of the equation

$$\square_g s + \nabla_s W(s) = 0$$

for S -valued fields on globally hyperbolic spacetimes.

Possible directions include:

- Establishing local and global well-posedness results under structural assumptions on W , for example growth conditions and regularity of $\nabla_s W$ in suitable Sobolev or energy spaces.
- Studying stability of constant solutions $s \equiv s_*$ beyond the linear approximation, including nonlinear stability, invariant manifolds, and attractor structures in the presence of dissipative mechanisms.
- Analysing the relation between finite-dimensional truncations and the full theory, for instance by proving convergence of s_N to s as $N \rightarrow \infty$ in appropriate norms, under assumptions on W .

Such results would provide a solid analytic foundation for the infinite-dimensional field theory and clarify the regimes in which it is mathematically and physically well behaved.

7.4.2 Quantisation of the linearised modes

The linearisation around a minimum of W yields a family of Klein–Gordon fields on (M, g) with masses determined by the spectrum of H . This suggests the study of quantised versions of these modes using standard techniques from quantum field theory in curved spacetime.

One may consider:

- constructing quantum fields associated with the modes ϕ^k , using e.g. algebraic or Hilbert-space methods;
- investigating the dependence of the quantum state space on the spectrum of H and on the geometry of (M, g) ;
- exploring whether correlations between different modes, encoded in the structure of H and in the choice of quantum states, have distinctive signatures.

Since (M, g) is treated as a fixed background and the linearised theory is of Klein–Gordon type, this quantisation problem fits naturally into the existing framework of quantum field theory on curved spacetimes.

7.4.3 Model-building with explicit choices of S and W

The analysis in this paper has been general and abstract. For concrete applications one must specify S and W more explicitly. Possible avenues include:

- Choosing S to be a function space (e.g. a Hilbert space of square-integrable functions on some auxiliary domain) and W to be an energy functional with known properties (for example a convex functional, a double-well potential, or a Ginzburg–Landau functional).
- Exploring how different choices of W affect the mass spectrum and interaction structure of the linearised modes, and whether certain classes of W lead to distinctive patterns that could be matched to known field content.
- Studying solutions with nontrivial spacetime dependence, such as soliton-like configurations, domain walls, or time-dependent backgrounds, within specific models.

Such model-building efforts would connect the abstract formalism to more concrete scenarios, while still respecting the conservative assumptions adopted here.

7.4.4 Further study of the induced tensor $h_{\mu\nu}$

The induced tensor

$$h_{\mu\nu}[s] = \langle \nabla_\mu s, \nabla_\nu s \rangle_S$$

encodes additional geometric information about how the field s varies on spacetime.

Future work could investigate:

- the classification of possible forms of $h_{\mu\nu}$ for solutions of the field equations, including constraints from symmetries and conservation laws;
- the role of $h_{\mu\nu}$ in characterising regions of spacetime where the field exhibits strong variation in specific directions, potentially serving as a diagnostic tool for complex field configurations;
- mathematically controlled frameworks in which $h_{\mu\nu}$ enters effective descriptions of other fields or observables, subject to the requirement that the background metric $g_{\mu\nu}$ and its dynamics remain governed by standard general relativity unless explicitly modified.

While speculative couplings involving $h_{\mu\nu}$ were briefly sketched in the previous chapter, a systematic study would need to balance mathematical consistency with phenomenological constraints.

7.4.5 Connections to the broader Hilbert-space framework

Finally, there is scope for further integration of the present field-theoretic construction with the Hilbert-space based programme developed in the earlier papers.

Potential questions include:

- How do the gradient flows and free-energy dynamics on S interact with the spacetime dynamics derived here? For example, can one formulate combined evolution equations in which s evolves both along spacetime and along an auxiliary flow in S ?

- Are there regimes in which the spacetime field theory admits effective descriptions in terms of reduced dynamics on low-dimensional manifolds in S , as suggested by the behaviour of $h_{\mu\nu}$ and the image $s(M) \subset S$?
- Can the embedding of quantum systems into S be combined with the present field theory to yield hybrid models in which quantum degrees of freedom propagate on (M, g) while interacting with a background described by s ?

Addressing these questions would link the different layers of the programme more tightly and may reveal new structures at the interface between Hilbert-space geometry and spacetime field theory.

In summary, this paper has provided a conservative and mathematically structured extension of the earlier Hilbert-space framework to a Lorentz-covariant field theory on a fixed background spacetime. The central ingredients are the Hilbert-space valued field $s : M \rightarrow S$, the potential $W : S \rightarrow \mathbb{R} \cup \{+\infty\}$, the action

$$S[s] = \int_M \left(\frac{1}{2} \langle \nabla_\mu s, \nabla^\mu s \rangle_S - W(s) \right) d\mu_g,$$

and the resulting Euler–Lagrange equation

$$\square_g s + \nabla_s W(s) = 0.$$

The analysis of the linearised theory and the induced tensor $h_{\mu\nu}[s]$ reveals a rich internal structure that can be explored further without leaving the established framework of classical field theory and general relativity on fixed backgrounds.

Acknowledgements and Contact

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The work emerged from a prolonged process of intuitive exploration, conceptual reduction, and formal consolidation. Informal insights about meaning, structure, and stability were progressively translated into explicit models, mathematical frameworks, and testable claims. Throughout this process, modern large language models were employed as practical instruments for structuring arguments, checking internal consistency, exploring alternative formulations, and iteratively refining formal expressions.

These tools were used as epistemic scaffolding rather than as sources of claims. All theoretical commitments, modelling decisions, and interpretative judgments remain the responsibility of the author. The resulting framework should therefore be understood as exploratory but precise: it proposes a concrete formal scaffold, derives explicit consequences, and invites scrutiny.

The work is explicitly offered for critical examination, clarification, extension, or refutation by domain experts. Questions, comments, corrections, and collaboration inquiries are welcome and may be directed to contact@semanticgravitation.org.

Semantic Gravitation V: Semantic Field Dynamics in Complex Systems — From Neural and Latent States to Semantic State Spaces

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Abstract

This paper operationalises the semantic field framework of the Semantic Gravitation programme for concrete complex systems such as neural networks, large language models, and other high-dimensional dynamical agents. Whereas Papers I–IV treat semantic dynamics at the level of an abstract Hilbert space S , probability measures on S , quantum-compatible sectors, and spacetime field theories, the present work starts from the *raw internal state spaces* of a given system A . These are modelled as high-dimensional manifolds or vector spaces X^A on which the system generates trajectories $t \mapsto x_t \in X^A$ under its native dynamics (e.g. neural activity, latent vectors, or other internal representations).

The first part of the paper develops a structural pipeline from such raw dynamics to semantic field dynamics. We introduce *semantic encoders*

$$E^A: X^A \longrightarrow S$$

that map internal states $x \in X^A$ into the semantic Hilbert space S constructed in Paper I. Assuming appropriate regularity and identifiability conditions, the push-forward of empirical trajectories x_t through E^A yields curves $s_t = E^A(x_t)$ in S which are shown to be well-approximated by noisy gradient flows of a semantic potential $W^A: S \rightarrow \mathbb{R}$,

$$ds_t \approx -\nabla W^A(s_t) dt + B^A dW_t.$$

We give conditions under which this approximation is justified, relate W^A and the noise operator B^A to observable statistics of the raw dynamics on X^A , and interpret the resulting free-energy functional on probability measures over S in terms of macroscopic semantic organisation, extending the variational picture of Paper II.

The second part translates this structural picture into a family of empirically accessible protocols. Using kernel methods and other representation-learning tools, we show how the semantic geometry and potential can, in principle, be inferred from time series on X^A without presupposing a specific microscopic model. This leads to quantitative notions of *semantic alignment* and *semantic robustness*, phrased in terms of the shape of W^A , the contraction properties of the associated gradient flow, and the behaviour of trajectories under perturbations. We formulate alignment and control objectives directly at the level of semantic fields, and derive testable predictions for how interventions on the raw system (training signals, architectural changes, or control inputs) should manifest as changes in W^A and in the induced Fokker–Planck dynamics on $\mathcal{P}_2(S)$.

Conceptually, this paper provides the bridge between the abstract semantic dynamics of Papers I–IV and the concrete, data-bearing systems that motivate the Semantic Gravitation programme. It offers a systematic route from neural or latent-state dynamics to semantic field descriptions, and articulates how semantic control and alignment questions can be posed in this language. Detailed implementation strategies and empirical validation of the proposed protocols are left for future work, as they depend strongly on the particulars of the systems under study. The results developed here will be used in Paper VI to motivate the global LOVE Lagrangian and the notion of a semantic attractor spacetime that organises both abstract and implemented semantic dynamics. For a programme-level introduction to Semantic Gravitation and the Lagrangian of Vectorial Emergence (LOVE), we refer to the separate overview paper.

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1 Introduction

1.1 Complex systems, neural dynamics, and latent spaces

Many contemporary systems in science and technology are naturally described as high-dimensional dynamical systems with nonlinear interactions and emergent collective behaviour. Examples range from recurrent neural networks and brain-inspired architectures to large language models (LLMs) and other deep generative models. In all these cases, the system maintains an internal state that is updated in response to input and internal dynamics. These internal states mediate the system’s behaviour and encode task-relevant structure in a form that is not directly observable.

In neural and brain-inspired models, the internal state may be given by membrane potentials, firing rates, or activation vectors in intermediate layers of an artificial network. In LLMs and other deep sequence models, the internal state typically consists of token embeddings, hidden activations, and attention patterns across multiple layers. Despite substantial differences in architecture and training, such systems share several features:

- The state space is very high-dimensional and exhibits strong correlations and anisotropies induced by learning.
- The dynamics are driven by a combination of deterministic update rules and stochastic influences (data, noise, random initialisation).
- Long-term behaviour often organises into attractor-like patterns, metastable states, and structured transitions between them.

A standard approach in machine learning is to study these systems in terms of their explicit parameters and task performance metrics. However, this level of description is often too tied to architectural details and does not provide a unified geometric picture of how internal states encode and transform semantic structure. Conversely, complex-systems theory offers tools for analysing attractors, phase transitions, and emergent organisation, but is usually formulated in terms of explicit state variables rather than learned latent representations.

The present programme aims to bridge this gap by developing a semantic description of complex systems in terms of abstract “semantic states” that capture task-relevant structure independently of particular architectures. The core idea is to represent internal states in a Hilbert space S of semantic configurations and to model their evolution as gradient flows of a semantic potential $W: S \rightarrow \mathbb{R}$. This potential encodes preferences for certain semantic arrangements, penalises undesirable ones, and thereby shapes the dynamics of meaning in the system.

In previous work, this framework has been developed at a high level of abstraction, with an emphasis on existence, uniqueness, and stability properties of gradient flows in infinite-dimensional Hilbert spaces. However, a number of questions remain open for concrete systems:

- How should one construct the semantic state space S for a given model, starting from its raw internal states?
- How should the potential W be chosen, parametrised, and regularised?
- To what extent do real model dynamics align with an idealised gradient flow generated by W , and how can this alignment be quantified?

- How can such a framework be applied to complex systems such as neural networks and LLMs in a way that is computationally tractable and empirically testable?

The present paper addresses these questions by moving from the abstract theory to concrete systems and by proposing a set of operational protocols and testable predictions.

1.2 Aim: operationalising semantic field dynamics

In previous work, a general mathematical framework of semantic gradient fields on Hilbert spaces has been developed. In its most abstract form, the framework starts from a separable Hilbert space S of semantic states and a sufficiently regular potential functional $W: S \rightarrow \mathbb{R}$, and studies gradient-flow dynamics of the form

$$\partial_t s_t = -\nabla W(s_t)$$

or suitable generalisations. Under appropriate conditions on W , one can establish existence and uniqueness of solutions, stability of attractors, and convergence properties, using standard tools from the theory of gradient flows in Hilbert spaces.

In a second step, the framework has been extended from deterministic trajectories to probability distributions on S , endowed with a suitable Wasserstein or related geometry. In this setting, one considers a free-energy functional

$$F(\vartheta) = \int_S W(s) d\vartheta(s) + \varpi^{-1} \text{Ent}(\vartheta),$$

where ϑ is a probability measure on S , ϖ is an inverse-temperature parameter, and $\text{Ent}(\vartheta)$ denotes an entropic regulariser. The time evolution of ϑ_t can then be interpreted as a gradient flow of F in the space of measures, providing a probabilistic description of semantic field dynamics.

Further developments have explored how the same semantic Hilbert space can support quantum-like structures, by embedding quantum states and observables into S in a way that preserves key algebraic features, and how semantic fields can be extended to Hilbert-space-valued fields over spacetime manifolds. Together, these works formulate a broad programme for describing semantic structure and dynamics in a unified mathematical language.

However, these developments remain largely abstract. They do not specify how to connect the semantic Hilbert space S and potential W to concrete systems such as neural networks or LLMs. In particular, several operational questions arise:

- Given a model with internal state space X , how should one construct a map

$$\Phi: X \rightarrow S$$

that assigns to each internal state $x \in X$ a semantic state $s = \Phi(x) \in S$?

- How should one choose kernels and features to induce S as a reproducing-kernel Hilbert space (RKHS) associated with the model’s internal representations?
- How should the semantic potential W be parametrised in terms of interpretable components, and how should its parameters be set or learned?
- How can one investigate empirically whether the actual dynamics of internal states approximate a gradient flow for W , and how can deviations from this ideal be quantified?

The aim of the present paper is to move from the abstract theory to concrete systems by proposing constructions of S and W for specific model classes, together with protocols that, if implemented, would allow empirical testing of gradient-flow interpretations of their dynamics. Rather than reporting completed experiments, the focus here is on:

- specifying how to construct semantic state spaces and potentials from raw model states;
- formulating metrics that quantify alignment between observed dynamics and gradient flows;
- designing simulation and experimental protocols that would make such analyses reproducible;
- articulating the expected behaviours and testable predictions that follow from the framework, thereby providing a methodological template for future empirical studies.

In this sense, the paper is methodological: it proposes how to apply semantic field dynamics to concrete complex systems and formulates hypotheses about what one would expect to observe if the proposed protocols were carried out.

1.3 Contributions of this paper

Within this programme, the present paper makes the following contributions.

1. **Concrete system instantiations.** We instantiate the semantic field framework in two representative classes of systems:
 - *Case A*: a neural or brain-inspired network with continuous-time or discrete-time dynamics on a high-dimensional activation space. We specify the architecture, the state space X_A , and the update rules that govern the evolution of internal states under input.
 - *Case B*: a large language model or related sequence model, considered through its latent representations. We define the state space X_B in terms of hidden activations or token embeddings and describe the induced dynamics under autoregressive generation or fine-tuning updates.
2. **Semantic state spaces and kernel constructions.** For each case, we construct a semantic state space S as an RKHS built from the model’s internal states. This involves choosing kernels $k: X \times X \rightarrow \mathbb{R}$ that measure similarity between internal states and induce a feature map into S . We discuss practical kernel choices (e.g. Gaussian, dot-product, or task-aware kernels), their interpretation in terms of semantic similarity, and how they can be computed from model activations or embeddings.
3. **Definition of a semantic potential.** We propose a concrete form of the semantic potential $W: S \rightarrow \mathbb{R}$ tailored to each system class. The potential is decomposed into interpretable components, for example

$$W(s) = \varrho_{\text{attr}} W_{\text{attr}}(s) + \varrho_{\text{spread}} W_{\text{spread}}(s) + \varrho_{\text{reg}} W_{\text{reg}}(s), \quad (1.3.1)$$

where W_{attr} encodes attraction to preferred semantic configurations (e.g. aligned states, task-competent patterns), W_{spread} penalises undesirable dispersion or anisotropy, and W_{reg} captures additional constraints or regularisation. We describe how the parameters ϱ . and any additional coefficients are set, either by hand, through fitting on data, or via cross-validation.

4. **Approximate gradient-flow interpretation of dynamics.** For both system classes, we embed actual trajectories x_t into S via $s_t = \Phi(x_t)$ and analyse the relationship between observed state updates and the gradient of W . We consider a discrete-time approximation of the form

$$s_{t+1} - s_t = -\omega_t \nabla W(s_t) + \varepsilon_t, \quad (1.3.2)$$

where ω_t is an effective step size and ε_t collects residual terms, interpreted as noise or model misfit. We define quantitative metrics for gradient-flow alignment, including directional cosine similarity between the step $s_{t+1} - s_t$ and $-\nabla W(s_t)$, monotonicity properties of $W(s_t)$ along trajectories, and norms of the residual ε_t .

5. **Simulation protocols and empirical analysis.** We design simulation and experimental protocols that make the above analysis reproducible. This includes specifying data sets, tasks, evaluation metrics, initialisation schemes, training procedures, and state-logging strategies. These protocols are constructed to enable empirical investigation of the following quantities in future work:

- the time evolution of $W(s_t)$ and its relation to task performance,
- the empirical alignment between observed dynamics and the gradient of W ,
- the structure of attractor landscapes and transitions in low-dimensional projections of S ,
- and the effects of adding explicit semantic regularisation terms derived from W to the training objective.

We discuss robustness with respect to kernel choice, hyperparameters, and random seeds, and highlight limitations and failure modes.

6. **Conceptual implications and limitations.** Finally, we interpret the proposed framework and analyses, together with their possible empirical outcomes, in the context of the broader semantic field framework. We identify aspects of neural and latent-space dynamics that are well captured by the gradient-flow picture and others that require extensions or alternative descriptions. We emphasise limitations, including the dependence on chosen kernels and potentials, and outline concrete directions for future work.

Throughout, the emphasis is on formulating a clear set of constructions, metrics, and protocols that can be implemented in concrete systems, together with associated predictions about what such implementations would reveal.

1.4 Position in the overall programme and structure of the paper

The present paper is part of a larger programme that develops a unified mathematical description of semantic structure and dynamics. Earlier work has focused on the abstract theory:

- The first paper introduced semantic gradient fields on separable Hilbert spaces, with an emphasis on existence, uniqueness, and stability properties of gradient flows of sufficiently regular potentials.
- The second paper extended the framework to distributions on S , using tools from optimal transport and free-energy minimisation to describe the evolution of probability measures under semantic dynamics.

- The third paper explored quantum-compatible semantic fields by embedding quantum states and observables into the semantic Hilbert space in a way that reproduces key quantum structures.
- The fourth paper generalised the setting to Hilbert-space-valued fields on spacetime manifolds, viewed as an effective-field-theory description of semantic configurations in a geometric background.

These works establish a versatile and mathematically rigorous framework, but they remain largely abstract. They do not specify, for instance, how to construct S and W for concrete machine learning systems, nor how to test the gradient-flow interpretation against the actual dynamics of trained models. The main purpose of the current paper is to fill this gap by providing an explicit, operational instantiation for two important system classes.

Within this overall programme, the present paper plays a bridging role between abstract theory and empirical practice:

- It shows how to move from an abstract separable Hilbert space S to a concrete RKHS constructed from raw model states via kernels.
- It proposes explicit potentials W whose components are interpretable in terms of alignment, attractors, and regularisation.
- It introduces quantitative metrics for comparing real model dynamics to idealised gradient flows and demonstrates how to compute these metrics in practice.

The remainder of the paper is organised as follows.

- Chapter 2 introduces the concrete systems under consideration. We define Case A (a neural or brain-inspired model) and Case B (an LLM latent-space setting), specify their state spaces and dynamics, and outline the tasks and objectives that would be used in the proposed experiments.
- Chapter 3 constructs the semantic state spaces, kernels, and potentials. We move from raw states X_A and X_B to RKHS spaces, discuss kernel design and induced geometry, and present concrete forms of the semantic potential W together with parametrisation and estimation strategies.
- Chapter 4 formulates the dynamics of the systems as trajectories in the semantic space S . We derive the approximate gradient-flow representation, define the noise decomposition, and introduce metrics for measuring alignment between observed dynamics and W .
- Chapter 5 details the simulations and experimental protocols. We describe the data, tasks, evaluation metrics, and implementation details needed for reproducibility in both Case A and Case B, including how semantic quantities are computed in practice.
- Chapter 6 organises the expected behaviour and testable predictions suggested by the framework and protocols. We formulate hypotheses about the behaviour of $W(s_t)$, gradient-alignment metrics, attractor landscapes, and the impact of semantic regularisation, and describe how robustness and limitations of these behaviours can be assessed empirically.

- Chapter 7 concludes with a discussion and outlook. We summarise the framework, the proposed simulation and experimental protocols, and the associated testable predictions, reflect on their implications for the semantic field framework, emphasise limitations and possible failure modes, and propose directions for future theoretical and empirical work.

Throughout, we maintain a clear separation between mathematically established properties, empirically observable or to-be-observed patterns arising from such protocols, and interpretive suggestions. The focus is on providing a concrete, reproducible template for applying semantic field dynamics to complex systems with high-dimensional internal representations, thereby making the abstract framework from earlier work accessible and testable in practical machine learning and complex-systems settings.

2 Concrete systems: neural models and latent spaces

In this chapter we move from the abstract semantic-field framework to two concrete classes of systems that will serve as testbeds for the protocols and predictions developed in the remainder of the paper. The purpose here is not to report completed experiments, but to define *how* such experiments would be instantiated in practice.

We specify in detail:

- which system classes are considered and why;
- how their state spaces, inputs, and outputs are defined;
- how their dynamics are parameterised;
- which tasks and objectives they are associated with;
- and how these choices prepare the ground for embedding trajectories into a semantic Hilbert space and comparing them to approximate gradient flows of a potential W .

Throughout, we emphasise that the systems are described at a level suitable for future implementation: if realised experimentally, they would generate internal-state trajectories that can be embedded into the semantic space and analysed using the methods introduced later in the paper.

2.1 Selection of systems and scope

The abstract framework of semantic field dynamics is designed to be applicable to a wide range of high-dimensional dynamical systems. For the sake of concreteness and in order to formulate precise experimental protocols, we restrict attention to two representative system classes:

- **Case A:** a neural or brain-inspired network with recurrent dynamics, operating on continuous activation vectors and trained on supervised or reinforcement-based tasks;
- **Case B:** a transformer-based large language model (LLM) considered through its latent representations in an alignment or fine-tuning setting.

Both cases share key features that make them suitable for semantic-field analysis:

- they possess a high-dimensional internal state space $X \cong \mathbb{R}^d$ with $d \gg 1$;

- they are endowed with well-defined update rules that induce dynamics on X ;
- they are trained or fine-tuned on tasks with explicit performance or alignment criteria;
- their internal variables (activations, embeddings, attention patterns) can in principle be recorded and analysed;
- their inputs and task conditions can be controlled to generate structured families of trajectories.

These properties allow us to define future experiments in which internal states are logged under controlled conditions, embedded into a semantic Hilbert space via kernels, and compared to gradient flows of a potential $W : S \rightarrow \mathbb{R}$.

State and task criteria

The selection of systems is guided by the following state and task criteria.

State-space and dynamics criteria. For each system class, we require:

- a high-dimensional internal state space $X \cong \mathbb{R}^d$ of dimension d ;
- interpretable internal variables that can be logged during operation (hidden activations, embeddings, etc.);
- a clear update rule, in discrete or continuous time, that defines trajectories $x_t \in X$ under given inputs and parameters;
- a parametrisation ς of the system (weights, biases, layer parameters) that can be varied in training or alignment.

Task and objective criteria. We further require that the systems be associated with tasks that admit explicit numerical objectives. Concretely, in future implementations:

- Case A would be trained or fine-tuned on tasks with explicit performance metrics (e.g. classification accuracy, cumulative reward), captured by a loss or reward functional $L_A(\varsigma_A)$;
- Case B would be fine-tuned or aligned using supervised objectives, preference-based losses, or reinforcement-learning-from-feedback signals, represented by an alignment objective $L_B(\varsigma_B)$.

This setup ensures that, once the systems are instantiated and trained, one can relate semantic potentials $W : S \rightarrow \mathbb{R}$ to task performance, alignment properties, and generalisation behaviour, and can formulate testable predictions linking the evolution of W along trajectories to changes in these quantities.

In what follows we describe Cases A and B in more detail, emphasising the aspects that are relevant for semantic-field analysis and for the design of future experimental protocols.

2.2 Case A: neural / brain-inspired system

2.2.1 Architecture and state space

Case A is a discrete-time, rate-based recurrent neural network with d_A units. Its internal state at time t is denoted by

$$x_t \in X_A \subset \mathbb{R}^{d_A},$$

and it is driven by an external input

$$u_t \in U_A \subset \mathbb{R}^{m_A},$$

with outputs

$$y_t \in Y_A \subset \mathbb{R}^{k_A}$$

providing task-relevant readouts.

The network is parametrised by

$$\varsigma_A = (J, U, V, b),$$

where

- $J \in \mathbb{R}^{d_A \times d_A}$ is the recurrent weight matrix;
- $U \in \mathbb{R}^{d_A \times m_A}$ projects inputs into the state space;
- $V \in \mathbb{R}^{k_A \times d_A}$ maps internal states to outputs;
- $b \in \mathbb{R}^{d_A}$ is a bias vector.

The state space X_A is taken to be the subset of $\mathbb{R}^{d_A}$ that is dynamically reachable under the update rule given below for inputs and parameters within prescribed ranges. In prospective experiments, X_A would be probed by driving the system with carefully designed input sequences and recording the resulting x_t .

2.2.2 Dynamics and noise

We consider a discrete-time update rule of the form

$$x_{t+1} = (1 - \lambda) x_t + \lambda \phi(Jx_t + Uu_t + b + \varepsilon_t), \quad (2.1)$$

where

- $\lambda \in (0, 1]$ is a leak or update parameter,
- ϕ is a nonlinear activation function (e.g. a rectifier, tanh, or smooth variant),
- ε_t is a noise term representing stochasticity or unmodelled influences.

The output at time t is given by

$$y_t = Vx_t. \quad (2.2)$$

Equations (2.1)–(2.2) define a nonlinear dynamical system on X_A controlled by input sequences $\{u_t\}_t$ and parameters ς_A . Different regimes (e.g. stable fixed points, oscillations, chaotic-like behaviour) can in principle be realised by varying (J, U, b) , though the present work does not presuppose a particular regime.

In future experiments, one would choose families of inputs and initial conditions, run the system according to (2.1)–(2.2), and log the resulting trajectories $\{x_t\}_t$ for subsequent embedding into the semantic space S and comparison to gradient flows of a potential W .

2.2.3 Tasks and training objective

To connect the dynamics of Case A to task performance and alignment considerations, we assume that it is (or will be) trained or fine-tuned on one or more tasks with explicit numerical objectives. Typical examples include:

- supervised classification or regression tasks, with loss

$$L_A(\varsigma_A) = \mathbb{E}[\ell(y_t, \tilde{y}_t)]$$

for suitable loss function ℓ and targets $\tilde{y}_t$;

- sequence-processing tasks, where performance is measured by accuracy, cross-entropy, or sequence-level reward;
- control or reinforcement-learning tasks, where y_t is interpreted as an action and a cumulative reward R_A is maximised.

The precise choice of task is not crucial for the semantic-field framework, as long as it induces a well-defined training or fine-tuning procedure that adjusts ς_A in order to minimise or maximise a scalar objective.

In particular, prospective experimental protocols can be formulated in which:

- one selects one or more tasks with clearly defined input and target distributions;
- one trains Case A under standard optimisation algorithms (e.g. stochastic gradient descent or variants);
- at pre-specified training epochs and for pre-defined evaluation inputs, one logs internal states x_t , outputs y_t , and task-level quantities (loss, accuracy, reward);
- the resulting state trajectories are then embedded into the semantic Hilbert space S and analysed with respect to a potential W .

The semantic-field framework thus provides predictions about how $W(s_t)$ and related quantities should evolve as training progresses, which can be tested once such experiments are carried out.

2.3 Case B: LLM latent-space setting

We now describe a second, complementary system class: a transformer-based large language model in an alignment or fine-tuning setting. Whereas Case A focuses on relatively low-level neural dynamics, Case B provides a setting in which internal states correspond to high-level latent representations in a model trained on complex linguistic tasks.

2.3.1 Architecture and state space

Case B is a transformer-based LLM with parameters ς_B (embedding matrices, attention weights, layer norms, etc.). Inputs are token sequences

$$c = (w_1, \dots, w_T)$$

drawn from a vocabulary $\mathcal{V}$, and the model produces probability distributions over next tokens and full continuations.

For the purpose of semantic-field analysis, we focus on internal latent states, such as:

- token embeddings at various layers;
- hidden activations in feed-forward or attention blocks;
- pooled or special-token representations summarising the sequence.

We denote by x_t a generic latent state extracted at a given layer and position from the model when processing a context c ; in alignment settings we further index such states by an outer iteration r corresponding to the fine-tuning step. Formally, we consider a latent-state space

$$X_B \subset \mathbb{R}^{d_B},$$

where d_B is the dimensionality of the chosen representation.

In future experiments, one would specify:

- which layer(s) and token positions to probe;
- how to aggregate or select states (e.g. final-token representation, mean pooling);
- which contexts c to present, drawn from evaluation or training distributions;
- which alignment or fine-tuning stages r to log.

This yields families of latent states $x_n^{(r)} \in X_B$, indexed by context n and training or alignment step r .

2.3.2 Token-level state and dynamics

For a fixed set of model parameters ς_B , autoregressive generation under a context c induces a sequence of latent states

$$x_0, x_1, x_2, \dots$$

as tokens are processed and new tokens are generated. Each state x_t is produced by the composition of embeddings, attention, and feed-forward layers applied to the current context and model parameters.

While the detailed form of the update from x_t to x_{t+1} is complicated and architecture-specific, it can be abstracted as an effective map

$$x_{t+1} = F_B(x_t, c, \chi_t; \varsigma_B), \tag{2.3}$$

where χ_t collects stochastic choices (e.g. sampling of next tokens) and control variables (temperature, decoding strategy, etc.).

In an alignment or fine-tuning scenario, the parameters ς_B themselves are updated across outer iterations $r = 0, 1, \dots, R$ according to a training rule driven by an alignment objective. This induces an additional, slower dynamics on latent states: for a fixed evaluation context $c^{(n)}$, and a prescribed way of extracting a latent state, we obtain a sequence

$$x_n^{(0)}, x_n^{(1)}, \dots, x_n^{(R)}$$

as ς_B is updated.

2.4 Effective state update across alignment

The dependence of latent states on both inputs and parameters suggests modelling the outer dynamics across alignment steps via an effective update map

$$x_n^{(r+1)} = G_B(x_n^{(r)}, c^{(n)}, \beta^{(r)}; \varsigma_B^{(r)}) + \varepsilon_n^{(r)}, \quad (2.4)$$

where

- $x_n^{(r)}$ is the latent state for context $c^{(n)}$ at alignment step r ;
- $\beta^{(r)}$ collects relevant training hyperparameters (learning rates, batch composition, etc.);
- $\varepsilon_n^{(r)}$ is a residual term capturing stochasticity and modelling error.

In a concrete implementation one could further relate G_B to gradients of the alignment loss with respect to ς_B and to the sensitivity of latent states to parameter changes. For instance, a first-order approximation would express $x_n^{(r+1)}$ in terms of a Jacobian of the map $\varsigma_B \mapsto x_n(\varsigma_B)$ contracted with the parameter update. The present work does not require this expansion in full algebraic detail; it suffices that an effective map of the form (2.4) exists and can be inferred or approximated from logged data.

From the perspective of semantic-field dynamics, the key point is that, once a set of contexts $\{c^{(n)}\}_{n=1}^N$ and alignment steps r are fixed, one obtains families of latent-state trajectories $\{x_n^{(r)}\}_{r=0}^R$ that can be embedded into the semantic space S and analysed under a potential W .

2.4.1 Tasks and alignment objective

Case B is assumed to be pre-trained as a standard language model on next-token prediction and then subjected to an alignment phase. In prospective experiments, the alignment or fine-tuning step would proceed along standard lines:

- **Base model.** A fixed base model is pre-trained on large-scale next-token prediction, leading to parameters $\varsigma_B^{(0)}$.
- **Alignment phase.** An alignment procedure modifies ς_B to better satisfy desiderata such as helpfulness, harmlessness, and honesty. This may involve supervised fine-tuning on instruction data, preference-based learning, or reinforcement learning from human or automated feedback.
- **Objective.** The alignment process defines an objective $L_B(\varsigma_B)$ that is optimised over the course of training, together with evaluation metrics on held-out prompts or user queries.

In such a setting, one can define in advance a collection of evaluation contexts $\{c^{(n)}\}_{n=1}^N$ and, at each alignment step r , log:

- the model outputs (sampled or deterministic completions);
- the latent states $x_n^{(r)}$ obtained under a fixed extraction procedure;
- relevant task- and alignment-level quantities (losses, preference scores, safety indicators).

The semantic-field framework then yields predictions relating the organisation of $\{x_n^{(r)}\}$ in the semantic space S , the evolution of the potential W , and the behaviour of alignment metrics across r , which can be tested empirically once these logging protocols are implemented.

2.5 Summary and implications for semantic field analysis

The two system classes defined in this chapter provide complementary perspectives on complex-system dynamics:

- Case A focuses on relatively low-level neural dynamics with explicit recurrent structure and well-controlled inputs;
- Case B focuses on high-level latent representations in a large language model under alignment or fine-tuning.

Both are high-dimensional dynamical systems with structured internal representations and clearly defined learning or alignment objectives. For each case, we have specified:

- the internal state space X_A , X_B and typical dimensionalities;
- the forms of the update rules that govern state evolution;
- the kinds of tasks and objectives that drive training or alignment;
- the logging procedures that would generate families of state trajectories suitable for semantic-field analysis.

In the methodological programme advanced in this paper, these systems are not merely described in isolation; they are intended as substrates on which to *implement* the semantic-field framework. Concretely, future experiments based on the protocols outlined here would:

1. generate internal-state trajectories $\{x_t\}$ for Case A and $\{x_n^{(r)}\}$ or $\{x_t\}$ for Case B under controlled inputs, contexts, and training stages;
2. embed these states into a semantic Hilbert space S via kernels $k : X \times X \rightarrow \mathbb{R}$, as developed in Chapter 3;
3. define semantic potentials $W : S \rightarrow \mathbb{R}$ with interpretable components (attractors, spread, regularisation);
4. analyse the resulting trajectories in S in terms of approximate gradient-flow dynamics for W .

Subsequent chapters make these steps precise: Chapter 3 constructs S , the kernels, and the potential W ; Chapter 4 formulates the dynamics of Cases A and B as approximate gradient flows in S and defines alignment metrics; Chapter 5 specifies concrete simulation and experimental protocols; and Chapter 6 derives expected behaviours and testable predictions for these systems.

In this way, the abstract semantic-field framework acquires a concrete operational meaning in the context of neural models and latent spaces: it becomes a set of hypotheses and analysis tools that can be brought into contact with real systems once the proposed protocols are implemented.

3 Semantic state space, kernels, and potential

3.1 From raw states X to an RKHS S

The concrete systems introduced in the previous chapter provide raw state spaces

$$X_A \cong \mathbb{R}^{d_A}, \quad X_B \cong \mathbb{R}^{d_B},$$

for Case A (recurrent neural network) and Case B (LLM latent space), respectively. In both cases we consider discrete-time trajectories

$$(x_t)_{t=0}^T \subset X_A \quad \text{or} \quad (x_t)_{t=0}^T \subset X_B$$

induced either by within-trial dynamics or by parameter updates during training and alignment.

The semantic-field framework requires a separable Hilbert space S whose elements represent semantic states. We obtain such spaces as reproducing-kernel Hilbert spaces (RKHS) associated with positive-definite kernels on X_A and X_B .

3.1.1 RKHS construction

Let X be a nonempty set and let

$$k : X \times X \rightarrow \mathbb{R}$$

be a symmetric positive-definite kernel, that is,

- $k(x, x') = k(x', x)$ for all $x, x' \in X$,
- for every finite family $(x_i)_{i=1}^n \subset X$ and $(\alpha_i)_{i=1}^n \subset \mathbb{R}$,

$$\sum_{i,j=1}^n \alpha_i \alpha_j k(x_i, x_j) \geq 0.$$

By standard RKHS theory, there exists a unique Hilbert space $(S_k, \langle \cdot, \cdot \rangle_{S_k})$ of real-valued functions on X such that:

1. for each $x \in X$, the function $k(x, \cdot)$ belongs to S_k ,
2. the reproducing property holds:

$$f(x) = \langle f, k(x, \cdot) \rangle_{S_k} \quad \text{for all } f \in S_k, x \in X.$$

The associated feature map $\Phi_k : X \rightarrow S_k$ is given by

$$\Phi_k(x) = k(x, \cdot),$$

and satisfies

$$\langle \Phi_k(x), \Phi_k(x') \rangle_{S_k} = k(x, x').$$

In what follows, we will write

$$S_A = S_{k_A}, \quad \Phi_A = \Phi_{k_A}, \quad S_B = S_{k_B}, \quad \Phi_B = \Phi_{k_B},$$

for kernels k_A on X_A and k_B on X_B . For each raw state $x \in X_A$ or $x \in X_B$ we obtain a semantic state

$$s = \Phi_A(x) \in S_A \quad \text{or} \quad s = \Phi_B(x) \in S_B.$$

3.1.2 Shared versus case-specific semantic spaces

There are two natural design options:

- *Case-specific spaces*: treat S_A and S_B as distinct Hilbert spaces tailored to the statistics and geometry of the corresponding system.
- *Shared space*: construct a joint kernel k on the disjoint union $X_A \sqcup X_B$ (for example via block-diagonal or cross-domain kernels) and obtain a single Hilbert space S in which both systems live.

In this paper we focus primarily on case-specific spaces for clarity of exposition. All constructions are carried out separately on S_A and S_B ; the notation S will refer generically to either S_A or S_B when the context is clear. The extension to a shared space is straightforward in principle but introduces additional modelling choices in the cross-system kernel, which we leave for future work.

3.1.3 Finite-sample and empirical representation

In practice, we only observe a finite collection of states

$$X = \{x_i\}_{i=1}^N \subset X,$$

obtained by logging trajectories under chosen (future) experimental protocols. The corresponding feature vectors are

$$s_i = \Phi(x_i) \in S.$$

While S may be infinite-dimensional, computations involving (s_i) can be performed via kernel evaluations. For example, for coefficients $(\alpha_i)_{i=1}^N$, the finite expansion

$$f = \sum_{i=1}^N \alpha_i k(x_i, \cdot) \in S$$

satisfies

$$\langle f, s_j \rangle_S = \sum_{i=1}^N \alpha_i k(x_i, x_j), \quad j = 1, \dots, N.$$

When a finite-dimensional representation is desired (e.g. for visualisation or approximate gradient computation), we project onto a low-dimensional subspace spanned by leading eigenvectors of the empirical kernel matrix

$$K_{ij} = k(x_i, x_j), \quad i, j = 1, \dots, N,$$

using kernel principal component analysis (kernel PCA) or Nyström approximations. The resulting subspace $\tilde{S} \simeq S$ is still a Hilbert space, and numerical computations of the potential and its gradient can be performed in $\tilde{S}$.

For notational simplicity, we do not distinguish between S and $\tilde{S}$ in the sequel, but all numerical implementations in the proposed protocols are intended to be based on such finite-dimensional approximations.

3.2 Kernel design and induced geometry

The choice of kernel determines the semantic geometry of S . It must satisfy basic functional-analytic requirements (positive definiteness, separability of S) and should reflect domain-specific notions of similarity.

3.2.1 Case A: kernels for neural activation states

States in $X_A \cong \mathbb{R}^{d_A}$ are high-dimensional activation vectors with potentially strong anisotropies. A natural starting point is a radial-basis-function (RBF) kernel on a suitably whitened or projected version of the activations.

Let $P_A : \mathbb{R}^{d_A} \rightarrow \mathbb{R}^{d'_A}$ be a linear projection (for example, onto the leading d'_A principal components of the activation covariance). For $x, x' \in X_A$ define

$$k_A(x, x') = \alpha_A \exp\left(-\frac{\|P_A x - P_A x'\|_2^2}{2\ell_A^2}\right) + \beta_A \langle P_A x, P_A x' \rangle_2, \quad (3.2.1)$$

where $\alpha_A, \beta_A \geq 0$ and $\ell_A > 0$ are hyperparameters. The first term is a Gaussian kernel capturing local similarity in a compressed representation; the second term is a linear kernel that preserves large-scale structure.

The induced inner product in S_A satisfies

$$\langle \Phi_A(x), \Phi_A(x') \rangle_{S_A} = k_A(x, x'),$$

so that squared distances in S_A can be expressed as

$$\|\Phi_A(x) - \Phi_A(x')\|_{S_A}^2 = k_A(x, x) + k_A(x', x') - 2k_A(x, x'). \quad (3.2.2)$$

Thus the RBF component yields a notion of semantic proximity based on local neighbourhoods in the projected activation space, while the linear component avoids collapsing directions that are globally informative.

For numerical stability, one may normalise $P_A x$ to zero mean and unit variance across the dataset before computing the kernel. Hyperparameters $\alpha_A, \beta_A, \ell_A$ are intended to be selected in future experiments via held-out evaluation or similar criteria.

3.2.2 Case B: kernels for LLM latent representations

Latent states in $X_B \cong \mathbb{R}^{d_B}$ are typically high-dimensional embeddings with nontrivial geometry shaped by pre-training and alignment. A widely used similarity measure in this setting is cosine similarity. We therefore adopt a kernel of the form

$$k_B(x, x') = \alpha_B \exp\left(\frac{\langle \hat{x}, \hat{x}' \rangle_2}{\tau_B}\right) + \beta_B \langle \hat{x}, \hat{x}' \rangle_2, \quad (3.2.3)$$

where

$$\hat{x} = \frac{x}{\|x\|_2 + \delta}, \quad \hat{x}' = \frac{x'}{\|x'\|_2 + \delta},$$

for a small $\delta > 0$, and $\alpha_B, \beta_B \geq 0$, $\tau_B > 0$ are hyperparameters. The first term is an exponentiated cosine kernel, which is positive-definite on the unit sphere for appropriate parameter ranges, and the second term is a linear cosine kernel. This combination yields an RKHS that is sensitive both to fine-grained angular structure and to broader directional trends.

Again, the inner product in S_B reproduces k_B , and distances are given by (3.2.2) with k_A replaced by k_B .

3.2.3 Induced geometry and regularity

Both k_A and k_B are continuous kernels on X_A and X_B , respectively, when these are endowed with the Euclidean topology. Under mild conditions on the distributions of states, the associated RKHS spaces S_A and S_B are separable, satisfying the basic assumptions of the semantic-field framework.

The choice of kernels determines:

- the smoothness of functions in S with respect to the raw coordinates,
- the scale on which distances and neighbourhoods in S are sensitive to differences in X ,
- the effective dimensionality of the subspace spanned by typical states.

The RBF component in k_A induces a notion of smoothness over activation patterns on a length scale ℓ_A , whereas the exponentiated cosine component in k_B is adapted to angular relationships in embedding space, which often align with semantic similarities of texts.

In subsequent chapters we will examine how the gradient of the semantic potential W interacts with this geometry, and to what extent trajectories in S are expected to follow its gradient flow under the proposed protocols.

3.3 Concrete semantic potential $W : S \rightarrow \mathbb{R}$

We now define a class of semantic potentials $W : S \rightarrow \mathbb{R}$ suitable for both cases. The potential is intended to be constructed from logged state data (once available) and decomposed into interpretable components:

$$W(s) = \rho_{\text{attr}} W_{\text{attr}}(s) + \rho_{\text{spread}} W_{\text{spread}}(s) + \rho_{\text{reg}} W_{\text{reg}}(s), \quad (3.3.1)$$

with nonnegative weights $\rho_{\text{attr}}, \rho_{\text{spread}}, \rho_{\text{reg}}$.

Intuitively:

- W_{attr} encodes attraction toward semantically desirable configurations (e.g. task-competent states or aligned behaviours).
- W_{spread} controls dispersion and anisotropy of states in S .
- W_{reg} imposes additional regularity constraints (e.g. norm penalties).

3.3.1 Attractor term via semantic prototypes

Suppose we have a finite set of labelled states $\{(x_i, c_i)\}_{i=1}^N$, where c_i denotes a context or condition label. For Case A, c_i may encode task type, stimulus category, or behavioural outcome (correct vs. incorrect trial). For Case B, c_i may encode prompt type, alignment quality tier, or other semantic categories.

Let $\mathcal{C}$ be the finite set of labels. For each $c \in \mathcal{C}$, define the corresponding subset

$$I_c = \{i : c_i = c\}, \quad N_c = |I_c|.$$

The semantic prototype for class c is the empirical mean in S :

$$\mu_c = \frac{1}{N_c} \sum_{i \in I_c} s_i, \quad s_i = \Phi(x_i) \in S. \quad (3.3.2)$$

Since S is a Hilbert space, the sum is well-defined as a finite linear combination of kernel elements.

To obtain multiple attractor basins, we define W_{attr} using a soft-min over squared distances to these prototypes:

$$W_{\text{attr}}(s) = -\eta \log \left(\sum_{c \in \mathcal{C}} \pi_c \exp \left(-\frac{1}{2\eta} \|s - \mu_c\|_S^2 \right) \right), \quad (3.3.3)$$

where $\eta > 0$ is a temperature parameter and $(\pi_c)_{c \in \mathcal{C}}$ is a probability vector with $\pi_c > 0$, $\sum_c \pi_c = 1$. For small η , W_{attr} approximates $\min_c \frac{1}{2} \|s - \mu_c\|_S^2$ and thus exhibits local minima near the prototypes μ_c . For larger η , the potential becomes smoother and the basins merge.

The squared distance can be computed via kernel evaluations. Writing $s = \Phi(x)$, we obtain

$$\begin{aligned} \|s - \mu_c\|_S^2 &= \langle s, s \rangle_S - 2\langle s, \mu_c \rangle_S + \langle \mu_c, \mu_c \rangle_S \\ &= k(x, x) - \frac{2}{N_c} \sum_{i \in I_c} k(x, x_i) + \frac{1}{N_c^2} \sum_{i, j \in I_c} k(x_i, x_j). \end{aligned} \quad (3.3.4)$$

3.3.2 Spread term via covariance in S

To capture the typical spread and anisotropy of states in S , we consider the empirical covariance operator

$$C_S : S \rightarrow S, \quad C_S f = \frac{1}{N} \sum_{i=1}^N \langle f, s_i - \bar{s} \rangle_S (s_i - \bar{s}), \quad (3.3.5)$$

where

$$\bar{s} = \frac{1}{N} \sum_{i=1}^N s_i$$

is the empirical mean.

Assuming C_S is strictly positive-definite on the subspace of interest (or regularised accordingly), we define the spread term

$$W_{\text{spread}}(s) = \frac{1}{2} \langle s - \bar{s}, C_S^{-1} (s - \bar{s}) \rangle_S. \quad (3.3.6)$$

This term penalises deviations from the typical cloud of states, with stronger penalties along directions of low empirical variance. It discourages trajectories from exploring directions in S that are rarely visited in the data.

In numerical implementations one would use a regularised inverse

$$C_S^{-1} \approx (C_S + \lambda I)^{-1}$$

with a small $\lambda > 0$ to ensure stability.

3.3.3 Regularisation term

Finally, we include a simple norm-based regularisation term

$$W_{\text{reg}}(s) = \frac{1}{2} \|s\|_S^2. \quad (3.3.7)$$

This term favours states of small RKHS norm, which, in the context of kernel methods, often corresponds to smoother or less complex functions of the raw state x . In combination with W_{attr} and W_{spread} , it prevents the potential landscape from developing extremely sharp wells or ill-conditioned directions.

3.3.4 Gradients of the potential

The semantic-field framework interprets dynamics as (approximate) gradient flow of W . We therefore need the gradient $\nabla W(s)$ in S . Under mild regularity conditions, each component of W is Fréchet differentiable on S .

For the regularisation term,

$$\nabla W_{\text{reg}}(s) = s.$$

For the spread term, using the symmetry and positive-definiteness of C_S^{-1} ,

$$\nabla W_{\text{spread}}(s) = C_S^{-1}(s - \bar{s}).$$

The attractor term requires more care. Differentiating (3.3.3) yields

$$\nabla W_{\text{attr}}(s) = \sum_{c \in \mathcal{C}} w_c(s) \nabla \left(\frac{1}{2} \|s - \mu_c\|_S^2 \right) = \sum_{c \in \mathcal{C}} w_c(s) (s - \mu_c), \quad (3.3.8)$$

where

$$w_c(s) = \frac{\pi_c \exp(-\|s - \mu_c\|_S^2 / (2\eta))}{\sum_{c' \in \mathcal{C}} \pi_{c'} \exp(-\|s - \mu_{c'}\|_S^2 / (2\eta))} \quad (3.3.9)$$

are soft assignment weights. Thus the full gradient is

$$\nabla W(s) = \rho_{\text{attr}} \sum_{c \in \mathcal{C}} w_c(s) (s - \mu_c) + \rho_{\text{spread}} C_S^{-1}(s - \bar{s}) + \rho_{\text{reg}} s. \quad (3.3.10)$$

All terms are linear combinations of s and fixed elements (μ_c) and $\bar{s}$, and therefore remain in the finite-dimensional subspace spanned by the observed states. In practice, when s is represented in a finite basis (e.g. kernel PCA coordinates), $\nabla W(s)$ is computed via standard linear algebra.

3.4 Parameterisation and estimation of W

The potential W depends on several families of parameters:

- kernel hyperparameters $(\alpha_A, \beta_A, \ell_A)$ and $(\alpha_B, \beta_B, \tau_B)$,
- prototype weights $(\pi_c)_{c \in \mathcal{C}}$ and temperature η in W_{attr} ,
- covariance regularisation λ in C_S^{-1} ,
- global weights $\rho_{\text{attr}}, \rho_{\text{spread}}, \rho_{\text{reg}}$.

We now outline how these are chosen and how W can, in future experiments, be further adjusted to better align with observed dynamics.

3.4.1 Data-driven construction of prototypes and covariance

The prototypes μ_c in (3.3.2) and the covariance operator C_S in (3.3.5) are estimated from data as follows.

1. **Data collection.** For each case (A or B), one would run the system under the experimental protocols specified in Chapter 5, obtaining a collection of states

$$\{x_i\}_{i=1}^N \subset X$$

and labels c_i derived from task conditions or alignment metadata.

2. **Kernel evaluation and embedding.** Compute the kernel matrix $K_{ij} = k(x_i, x_j)$ and, if desired, obtain a finite-dimensional embedding via kernel PCA. In either representation one can compute linear combinations of $s_i = \Phi(x_i)$.

3. **Prototype estimation.** For each label $c \in \mathcal{C}$, compute

$$\mu_c = \frac{1}{N_c} \sum_{i \in I_c} s_i.$$

In finite coordinates, this is a weighted average of the corresponding vectors.

4. **Covariance estimation.** The overall mean is

$$\bar{s} = \frac{1}{N} \sum_{i=1}^N s_i,$$

and the empirical covariance operator on the chosen subspace is

$$C_S = \frac{1}{N} \sum_{i=1}^N (s_i - \bar{s}) \otimes (s_i - \bar{s}),$$

represented as a positive-semidefinite matrix in finite coordinates. A small multiple of the identity is then added before inversion.

The prototype weights (π_c) can be set either to be proportional to class frequencies N_c/N or to reflect task importance (e.g. giving more weight to aligned states in Case B). These design choices are part of the proposed methodology and can be varied in planned ablations.

3.4.2 Heuristic selection of hyperparameters

Kernel and potential hyperparameters are initially chosen by heuristic criteria, such as:

- setting the kernel length scale ℓ_A such that the median pairwise RBF similarity between states is in a predefined range (e.g. around 0.5); analogous criteria can be used for τ_B in k_B ;
- choosing the weights $\rho_{\text{attr}}, \rho_{\text{spread}}, \rho_{\text{reg}}$ such that the contributions of the three terms to $W(s)$ are of comparable magnitude on a representative subset of states;
- selecting the temperature η in W_{attr} so that soft assignments $w_c(s)$ are neither overly peaked (which would approximate a hard minimum and may introduce non-smooth gradients) nor too flat (which would blur the attractor structure).

In prospective implementations, these initial settings provide a baseline potential that reflects the empirical distribution of states and their categorisation. They can then be refined based on validation metrics defined in later chapters.

3.4.3 Fitting W to align with dynamics

To more closely tie W to the observed dynamics, we allow for an additional parameterisation of its components and propose to fit these parameters by matching the gradient field to empirical velocities.

Let $(s_t)_{t=0}^T$ denote a collection of trajectories in S obtained from the systems (once the logging protocols are implemented), and define discrete-time velocities

$$v_t = s_{t+1} - s_t.$$

For a given parameter vector ς_W governing the potential (e.g. the ρ 's, (π_c) , η , and additional coefficients in prototype combinations), consider the effective gradient-flow model

$$v_t \approx -\omega_t \nabla W(s_t; \varsigma_W),$$

where $\omega_t > 0$ is a scalar step size that may depend on t or be absorbed into ς_W . A simple fitting criterion is the squared error between observed velocities and predicted gradients:

$$E(\varsigma_W) = \sum_{t \in T_{\text{fit}}} \|v_t + \omega_t \nabla W(s_t; \varsigma_W)\|_S^2, \quad (3.4.1)$$

where T_{fit} is a subset of time indices used for fitting.

In practice, one would restrict ς_W to a low-dimensional family that preserves interpretability. For example, one may:

- treat $\rho_{\text{attr}}, \rho_{\text{spread}}, \rho_{\text{reg}}$ as free parameters;
- allow per-class scaling factors α_c in the prototype contributions, replacing μ_c with $\alpha_c \mu_c$ in (3.3.8);
- or introduce a small number of additional prototypes derived from clustering in S .

For such parameterisations, $\nabla W(s_t; \varsigma_W)$ is linear in ς_W at fixed s_t , so that minimising (3.4.1) reduces to a linear or convex optimisation problem, solvable by least squares with regularisation.

Given a data set and chosen parameterisation, this procedure yields a fitted potential $W(\cdot; \hat{\varsigma}_W)$ whose gradient best approximates the empirical velocities in a least-squares sense on the fitting set. While this does not guarantee that the dynamics is globally a gradient flow of W , it provides, in future experiments, a principled way to quantify to what extent an approximate gradient-flow description is compatible with observed behaviour.

3.4.4 Cross-validation and robustness

To avoid overfitting the potential to a particular set of trajectories, the proposed methodology splits the available data into:

- a fitting set used to estimate $\hat{\varsigma}_W$ via (3.4.1),
- a validation set used to assess generalisation of the gradient-flow approximation.

On the validation set, one can evaluate metrics such as:

- the average alignment between v_t and $-\nabla W(s_t; \hat{\varsigma}_W)$;
- the fraction of time steps with $W(s_{t+1}) \leq W(s_t)$;
- and the distribution of residual norms $\|v_t + \omega_t \nabla W(s_t; \hat{\varsigma}_W)\|_S$.

These metrics are detailed in the next chapter and are intended to form the basis of future empirical assessments of the semantic-field dynamics.

In summary, this chapter has defined the semantic state spaces S_A and S_B as RKHSs over the raw state spaces X_A and X_B , specified explicit kernels that induce their geometry, introduced a concrete and interpretable potential $W : S \rightarrow \mathbb{R}$, and outlined how, once data are collected, its parameters can be estimated from those data and fitted to observed dynamics. The next chapter connects this construction to the actual time evolution of the systems and formulates the approximate gradient-flow representation that the proposed experiments are designed to test.

4 Dynamics as approximate gradient flow

In this chapter we connect the semantic structures introduced so far to the actual time evolution of the concrete systems described in Chapter 2. The central idea is to represent the dynamics of internal states, once embedded into the semantic Hilbert space S , as *approximately* following a gradient flow of the semantic potential $W : S \rightarrow \mathbb{R}$ defined in Chapter 3, up to noise and model misfit terms.

The focus is methodological: we formulate continuous- and discrete-time gradient-flow equations in S , define semantic velocities and residuals, and introduce quantitative metrics that would allow future experiments to assess to what extent the actual dynamics of the systems align with the gradient of W . All statements in this chapter are to be read as definitions, constructions, or hypotheses about what one could measure and test once the logging protocols of Chapter 5 are implemented, not as claims about already observed data.

4.1 From system dynamics in X to trajectories in S

Let X denote either X_A or X_B , and let S be the corresponding semantic RKHS with feature map $\Phi : X \rightarrow S$ and kernel $k : X \times X \rightarrow \mathbb{R}$ as constructed in Chapter 3. The concrete systems of Chapter 2 give rise to discrete-time trajectories in X :

$$x_0, x_1, \dots, x_T \in X,$$

driven by inputs, contexts, and parameter updates according to their respective update rules.

Applying the feature map pointwise, we obtain corresponding semantic trajectories

$$s_t = \Phi(x_t) \in S, \quad t = 0, \dots, T.$$

These trajectories constitute the primary objects of analysis in this chapter. They will be treated as the empirical or simulated paths along which we wish to compare the observed dynamics to an idealised gradient flow of the potential $W : S \rightarrow \mathbb{R}$.

4.1.1 Case A: semantic trajectories from neural dynamics

For Case A, recall the discrete-time update

$$x_{t+1} = (1 - \lambda) x_t + \lambda \phi(Jx_t + Uu_t + b + \varepsilon_t),$$

with internal state $x_t \in X_A$, input u_t , parameters $\varsigma_A = (J, U, V, b)$, and noise term ε_t . Given a sequence of inputs $(u_t)_{t=0}^{T-1}$ and an initial state x_0 , this rule generates a trajectory $(x_t)_{t=0}^T$ in X_A .

Embedding via $\Phi_A : X_A \rightarrow S_A$ yields

$$s_t = \Phi_A(x_t) \in S_A, \quad t = 0, \dots, T,$$

which can be interpreted as a trajectory in semantic space induced by the underlying neural dynamics. By choosing different initial conditions and input sequences (as prescribed by the experimental protocols in Chapter 5), one obtains families of trajectories

$$\{(s_t^{(r)})_{t=0}^{T_r}\}_{r=1}^R$$

for later analysis.

4.1.2 Case B: semantic trajectories from latent representations

For Case B, there are two natural sources of trajectories:

- *Within-context dynamics*: autoregressive processing of a fixed context $c = (w_1, \dots, w_T)$ induces a sequence of latent states

$$x_0, x_1, \dots, x_T \in X_B$$

at chosen layers and positions, as the model iteratively updates its representations.

- *Across-alignment dynamics*: for a fixed evaluation context $c^{(n)}$ and a prescribed extraction procedure, fine-tuning or alignment across steps $r = 0, \dots, R$ yields latent states

$$x_n^{(0)}, x_n^{(1)}, \dots, x_n^{(R)} \in X_B,$$

which can be interpreted as a trajectory in parameter space projected into the latent representation.

In both scenarios, embedding via $\Phi_B : X_B \rightarrow S_B$ gives semantic trajectories

$$s_t = \Phi_B(x_t) \in S_B \quad \text{or} \quad s_n^{(r)} = \Phi_B(x_n^{(r)}) \in S_B.$$

For notational simplicity, we will mostly write $(s_t)_{t=0}^T$ for a generic semantic trajectory, with the understanding that t may index either time within a run or an outer alignment step, depending on the context.

4.2 Semantic velocities and time discretisation

To compare the evolution of s_t with the gradient field $-\nabla W(s)$, we introduce the notion of semantic velocities in S and relate discrete-time updates to continuous-time gradient-flow equations.

4.2.1 Continuous-time gradient flow

In the abstract semantic-field framework, an idealised deterministic gradient flow of the potential $W : S \rightarrow \mathbb{R}$ is described by the differential equation

$$\frac{d}{dt}s_t = -\nabla W(s_t), \tag{4.2.1}$$

for t in some interval $[0, T]$, where $\nabla W(s_t) \in S$ denotes the gradient of W at s_t with respect to the Hilbert inner product on S .

Under standard regularity conditions on W (e.g. Lipschitz continuity of ∇W), this equation admits unique solutions for given initial conditions $s_0 \in S$, and the potential $W(s_t)$ is nonincreasing along trajectories:

$$\frac{d}{dt}W(s_t) = \langle \nabla W(s_t), \dot{s}_t \rangle_S = -\|\nabla W(s_t)\|_S^2 \leq 0.$$

In the present paper, these properties serve as a benchmark: they inform the metrics we propose to evaluate on empirical trajectories, but they are not assumed to hold exactly for the concrete systems.

4.2.2 Discrete-time semantic velocities

The real systems considered in Chapter 2 evolve in discrete time, through update rules of the form

$$x_{t+1} = F(x_t, \cdot),$$

which induce discrete semantic trajectories $s_t = \Phi(x_t)$.

For a discrete trajectory $(s_t)_{t=0}^T$ we define the discrete-time semantic velocity at time t as

$$v_t = s_{t+1} - s_t \in S, \quad t = 0, \dots, T-1. \quad (4.2.2)$$

If the discrete time step Δt is known (or normalised to 1), one may also consider the finite-difference approximation

$$\dot{s}_t \approx \frac{s_{t+1} - s_t}{\Delta t},$$

but for the purposes of this chapter the scale of Δt can be absorbed into an effective step-size parameter.

In future experiments, v_t would be computed from logged semantic trajectories (s_t) via (4.2.2), using inner products and norms in S realised through kernel evaluations and the chosen finite-dimensional embedding.

4.3 Approximate gradient-flow representation

We now formulate an approximate gradient-flow representation of the discrete-time dynamics in S . The goal is to express semantic velocities v_t as a combination of a term aligned with $-\nabla W(s_t)$ and residuals that capture noise and model misfit.

4.3.1 Idealised continuous-discrete correspondence

If the true dynamics in S were exactly governed by the continuous-time gradient flow (4.2.1), then an explicit Euler discretisation with step size $\omega_t > 0$ would give

$$s_{t+1} \approx s_t - \omega_t \nabla W(s_t), \quad (4.3.1)$$

so that the discrete-time velocity satisfies

$$v_t \approx -\omega_t \nabla W(s_t). \quad (4.3.2)$$

Equation (4.3.2) therefore provides a natural template for relating empirical semantic velocities to the gradient of the potential.

In real systems, however, the dynamics will generally not be an exact gradient flow of W , and external inputs, stochasticity, and architectural constraints will contribute additional terms. We therefore introduce an explicit residual.

4.3.2 Residual-augmented discrete representation

We model the discrete-time dynamics in S as

$$s_{t+1} = s_t - \omega_t \nabla W(s_t) + \varepsilon_t, \quad (4.3.3)$$

or equivalently,

$$v_t = s_{t+1} - s_t = -\omega_t \nabla W(s_t) + \varepsilon_t, \quad (4.3.4)$$

for $t = 0, \dots, T-1$, where

- $\omega_t \geq 0$ is an effective step size at time t ,
- $\varepsilon_t \in S$ is a residual term capturing deviations from pure gradient flow.

Equation (4.3.4) is not assumed to hold a priori; instead, it defines a *model* of the dynamics that can be fitted to data and tested empirically.

In future experiments, given a potential W , one could estimate ω_t and ε_t from trajectories (s_t) in such a way that (4.3.4) holds as closely as possible in a least-squares sense, and then quantify the alignment between v_t and $-\nabla W(s_t)$ using the metrics introduced in the next section.

4.3.3 Least-squares estimation of ω_t and ε_t

For analytical clarity, it is useful to consider the case in which, for each time t , the effective step size ω_t is chosen to minimise the squared residual

$$\|\varepsilon_t\|_S^2 = \|v_t + \omega_t \nabla W(s_t)\|_S^2.$$

Assuming that $\nabla W(s_t) \neq 0$, the minimiser is

$$\omega_t^* = -\frac{\langle v_t, \nabla W(s_t) \rangle_S}{\|\nabla W(s_t)\|_S^2}, \quad (4.3.5)$$

which yields the residual

$$\varepsilon_t^* = v_t + \omega_t^* \nabla W(s_t). \quad (4.3.6)$$

By construction, ε_t^* is orthogonal to $\nabla W(s_t)$:

$$\langle \varepsilon_t^*, \nabla W(s_t) \rangle_S = 0.$$

In practice, one may constrain ω_t to be nonnegative, reflecting an assumption that the dynamics tends, on average, to move *down* the potential. If the unconstrained minimiser (4.3.5) is negative, one sets $\omega_t = 0$ and $\varepsilon_t = v_t$, which expresses that, at that time step, the dynamics locally moves against the gradient. The impact of such constraints can be examined empirically in future work.

4.4 Alignment metrics and decomposition of noise

The decomposition (4.3.4) suggests several quantitative metrics for assessing, in future experiments, how well the dynamics of a system can be approximated as a gradient flow of W . We define these metrics here in terms of generic semantic trajectories (s_t) , velocities v_t , gradients $\nabla W(s_t)$, and residuals ε_t .

4.4.1 Directional alignment

A basic measure of alignment between the actual velocity v_t and the negative gradient $-\nabla W(s_t)$ is the cosine similarity

$$\alpha_t = \frac{\langle v_t, -\nabla W(s_t) \rangle_S}{\|v_t\|_S \|\nabla W(s_t)\|_S}, \quad (4.4.1)$$

defined whenever $v_t \neq 0$ and $\nabla W(s_t) \neq 0$. The value α_t lies in $[-1, 1]$:

- $\alpha_t \approx 1$ indicates that the dynamics locally moves in the direction of steepest descent of W ;
- $\alpha_t \approx -1$ indicates that it moves locally upwards along the gradient;
- $\alpha_t \approx 0$ indicates motion orthogonal to the gradient.

Given a collection of trajectories and time indices $\mathcal{T}$ (e.g. all time steps, or a subset corresponding to particular phases of training), one can define aggregate statistics such as

$$\bar{\alpha} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \alpha_t, \quad (4.4.2)$$

or distributions of α_t over different conditions. In future experiments, these quantities would allow one to test hypotheses about the degree to which the systems follow the gradient of W on average.

4.4.2 Monotonicity of the potential along trajectories

For an exact gradient flow, $W(s_t)$ is strictly nonincreasing in t . In the discrete and approximate setting, we consider potential changes

$$\Delta W_t = W(s_{t+1}) - W(s_t), \quad (4.4.3)$$

and define the empirical fraction of nonincreasing steps

$$\gamma = \frac{1}{|\mathcal{T}|} |\{t \in \mathcal{T} : \Delta W_t \leq 0\}|. \quad (4.4.4)$$

A value γ close to 1 would indicate that, along most steps in $\mathcal{T}$, the potential decreases or remains constant, consistent with gradient descent dynamics; smaller values would indicate more frequent potential increases.

In addition to γ , one can consider summaries of the distribution of ΔW_t , such as its mean, variance, and quantiles, and how these depend on task conditions, training stage, or system class (A vs. B). These are natural candidates for future empirical investigation.

4.4.3 Residual energy and orthogonal components

Using the least-squares decomposition

$$v_t = -\omega_t^* \nabla W(s_t) + \varepsilon_t^*,$$

with ε_t^* orthogonal to $\nabla W(s_t)$, we can interpret

$$E_{\parallel,t} = \|\omega_t^* \nabla W(s_t)\|_S^2 = \frac{\langle v_t, \nabla W(s_t) \rangle_S^2}{\|\nabla W(s_t)\|_S^2}, \quad (4.4.5)$$

as the “energy” of the velocity component aligned with the negative gradient, and

$$E_{\perp,t} = \|\varepsilon_t^\star\|_S^2 = \|v_t\|_S^2 - E_{\parallel,t}, \quad (4.4.6)$$

as the energy of the component orthogonal to $\nabla W(s_t)$.

A scale-free measure of the relative importance of gradient-aligned motion is then

$$\rho_t = \frac{E_{\parallel,t}}{\|v_t\|_S^2} = \frac{\langle v_t, \nabla W(s_t) \rangle_S^2}{\|v_t\|_S^2 \|\nabla W(s_t)\|_S^2}, \quad (4.4.7)$$

whenever $v_t \neq 0$ and $\nabla W(s_t) \neq 0$. By construction, $\rho_t \in [0, 1]$:

- $\rho_t \approx 1$ indicates that almost all of the velocity lies along the gradient direction (up to sign);
- $\rho_t \approx 0$ indicates that the motion is almost entirely orthogonal to the gradient.

Aggregating ρ_t over time and trajectories yields summary statistics that quantify how much of the dynamics can be captured by a gradient-flow model. In future experiments, comparing such statistics across tasks, training stages, and hyperparameter settings of W would provide insight into which aspects of system behaviour are well described by the semantic potential and which require additional structure.

4.4.4 Noise modelling and stochastic interpretations

The residual terms ε_t in (4.3.4) can be further analysed statistically. For example, one can consider their empirical distribution over time and trajectories,

$$\{\varepsilon_t : t \in \mathcal{T}\},$$

and compute quantities such as:

- mean and covariance of ε_t in S ;
- correlations between ε_t and task variables (inputs, rewards, loss gradients);
- dependence of $\|\varepsilon_t\|_S$ on training stage or context type.

In some regimes, it may be reasonable to hypothesise that ε_t behaves like a mean-zero stochastic term with covariance depending on s_t , leading to a stochastic differential equation interpretation of the dynamics in S . While developing such models in full generality is beyond the scope of this paper, the residual analysis proposed here provides a starting point for future work that seeks to refine the semantic-field framework by incorporating structured noise.

4.5 Summary and role in the overall methodology

This chapter has translated the abstract notion of gradient flow of a semantic potential into a concrete framework for analysing the dynamics of complex systems in the semantic Hilbert space S :

- Semantic trajectories are obtained by embedding raw internal-state trajectories (x_t) into S via the kernel-induced feature map Φ .

- Discrete-time semantic velocities $v_t = s_{t+1} - s_t$ provide a natural notion of motion in S .
- The dynamics are modelled as an approximate gradient flow of W ,

$$v_t = -\omega_t \nabla W(s_t) + \varepsilon_t,$$

with effective step sizes ω_t and residuals ε_t .

- Least-squares estimation yields a canonical decomposition of v_t into a gradient-aligned component and an orthogonal residual, from which alignment metrics $(\alpha_t, \gamma, \rho_t)$ and residual statistics can be computed.

All these constructions are formulated in a way that is directly implementable: given logged trajectories from Cases A and B, a choice of kernel and potential W , and a finite-dimensional embedding of S , one can compute the quantities defined in this chapter and thereby test, quantitatively, how well the gradient-flow interpretation captures the observed dynamics.

The next chapter specifies concrete simulation and experimental protocols for generating the required trajectories, logging the relevant quantities, and performing these analyses in practice. Subsequently, Chapter 6 formulates the expected behaviours and testable predictions for the metrics introduced here, thus closing the loop between abstract semantic-field theory and empirical investigation.

5 Simulations and experimental protocols

This chapter specifies in detail how the semantic-field framework is intended to be implemented and empirically tested on the two concrete system classes introduced in Chapter 2. The focus is on *prospective* simulations and experiments: we describe data, tasks, training procedures, logging strategies, semantic construction steps, and analysis pipelines that, if carried out, would generate the trajectories and quantities required to evaluate the approximate gradient-flow interpretation developed in Chapter 4.

Throughout, all statements should be read as descriptions of *proposed* protocols and *anticipated* analyses. No claim is made that the simulations or experiments have already been executed.

5.1 Data, tasks, and evaluation metrics

We first describe the data sources and task families envisaged for the two system classes, together with task-level and alignment-level evaluation metrics. These choices are designed to produce rich internal dynamics while remaining computationally and conceptually tractable.

5.1.1 Case A: data and tasks

For Case A (the recurrent neural or brain-inspired system), the proposed experiments make use of synthetic or benchmark tasks that are standard in the study of recurrent networks and working-memory models. Examples include:

- **Delayed match-to-sample tasks.** A cue is presented, followed by a delay period, and the system must indicate whether a subsequent probe matches the cue.

- **Context-dependent integration tasks.** The system receives streams of noisy sensory inputs and a context cue indicating which stream is currently relevant; it must integrate evidence over time and produce a categorical decision at the end of the trial.
- **Multi-context classification or decision tasks.** The system is trained on multiple related tasks that share structure but differ in context cues, decision rules, or relevant input features.

Trials are organised into episodes of fixed length T_{trial} , with inputs $u_t \in U_A$ and outputs $y_t \in Y_A$ at each time step. For each task family, the protocol specifies:

- training and test distributions over input sequences;
- the mapping from task conditions (cue identities, context types, correct decisions) to target outputs $\tilde{y}_t$;
- the time points at which outputs are evaluated (e.g. end-of-trial or mid-trial).

Task-level performance metrics. Task performance is quantified by a scalar loss or reward functional $L_A(\varsigma_A)$ (e.g. cross-entropy loss, mean-squared error, or cumulative reward) as described in Section 2.2.3. In addition, the proposed protocols include:

- per-condition accuracies (e.g. split by cue identity, context, or decision type);
- psychometric-like curves (e.g. performance as a function of signal strength or delay duration);
- robustness metrics under input noise and perturbations of initial conditions.

These metrics are intended to be recorded alongside semantic quantities in order to relate task performance to the behaviour of the potential W and gradient-alignment measures.

5.1.2 Case B: data and tasks

For Case B (the LLM latent-space setting), the proposed experiments build on standard instruction-following and alignment datasets comprising pairs of prompts and responses, preference data, or reward-model outputs. Typical sources include:

- supervised instruction datasets, consisting of $(c^{(n)}, a^{(n)})$ pairs where $c^{(n)}$ is a natural-language prompt and $a^{(n)}$ is a target response;
- preference datasets, consisting of prompt-response pairs with relative quality labels or reward scores;
- safety and robustness benchmarks, used for evaluation-only logging of latent states.

Prompts are grouped into semantic categories (e.g. question-answering, summarisation, coding, safety-critical queries), and alignment metadata (e.g. reward scores, preference labels) are used as semantic labels for latent states in S_B .

Alignment-level metrics. In addition to standard language modelling metrics (e.g. log-likelihood, perplexity), the proposed protocols track:

- task-specific quality metrics (e.g. BLEU, ROUGE, code execution success rates);
- preference- or reward-based metrics (e.g. mean reward under a trained reward model, pairwise win rates against a baseline model);
- safety metrics (e.g. violation rates on red-team prompts, refusal behaviour under unsafe queries).

These metrics can be related, in future experiments, to the organisation of latent states in S_B and to the evolution of $W(s)$ during alignment.

5.1.3 Semantic labels and evaluation sets

For both systems, semantic labels c_i associated with states x_i are derived from task and alignment metadata. Examples include:

- task identity or context for Case A (e.g. “context 1”, “context 2”, correct vs. incorrect trials);
- prompt category, quality tier, or reward bin for Case B.

These labels are used to construct prototypes μ_c and covariance operators C_S in S_A and S_B (see Section 3.3.1 and 3.3.2), and thus shape the semantic potential W .

For evaluation, the protocols specify held-out sets of trials or prompts on which internal states and outputs are logged without further parameter updates, providing a basis for out-of-sample assessment of gradient alignment and potential behaviour.

5.2 Experimental protocol for Case A (neural system)

We now specify the full protocol for Case A, from model instantiation and training to semantic embedding and perturbation experiments. The description is organised so that it can be implemented directly in future work.

5.2.1 Model configuration

Architecture. The proposed experiments consider a discrete-time recurrent network as in Section 2.2.1, with:

- d_A recurrent units (e.g. $d_A \in \{64, 128\}$);
- input dimension m_A and output dimension k_A determined by the task;
- nonlinearity ϕ chosen from standard options (e.g. tanh or smooth rectifiers);
- leak parameter $\lambda \in (0, 1]$ controlling the effective time constant.

The update rule is

$$x_{t+1} = (1 - \lambda)x_t + \lambda\phi(Jx_t + Uu_t + b + \varepsilon_t), \quad y_t = Vx_t,$$

with parameters $\varsigma_A = (J, U, V, b)$ and noise term ε_t .

Initialisation. In the proposed setup:

- recurrent weights J are initialised from a zero-mean distribution with variance chosen so that the spectral radius of J lies in a regime that yields stable but expressive dynamics (e.g. close to 1);
- input and output weights U and V are initialised with small random values;
- biases b are initialised to zero or small random values;
- the initial state x_0 is set to zero or to a small random vector.

Hyperparameters such as the spectral scaling of J and the noise level of ε_t are intended to be tuned so that the network operates in a regime that exhibits rich transient dynamics without pathological instability.

5.2.2 Training procedure

Baseline training. For each task family, the protocol defines a baseline training phase in which the network is trained using the task loss $L_A(\varsigma_A)$, as specified in Section 2.2.3. A typical training loop proceeds as follows:

1. Sample minibatches of trials from the training set.
2. For each trial, simulate the network over the full trial duration, recording outputs y_t at evaluation times.
3. Compute the task loss for the minibatch (e.g. average cross-entropy at decision times).
4. Update parameters via an optimiser (e.g. Adam) with learning rate and scheduling specified in Section 5.4.2.

This phase yields a baseline model for each task family, which can then be used to generate trajectories for semantic analysis.

Semantic regularisation. To assess the impact of explicit semantic regularisation, the protocol also defines a *semantic-regularised* training condition. Let T_{sem} be a set of time points (e.g. mid-trial and response times) at which semantic states are evaluated. For a given parameter setting ς_A , a semantic regularisation term is defined as

$$R_A(\varsigma_A) = \frac{1}{|T_{\text{sem}}|} \sum_{t \in T_{\text{sem}}} W(s_t), \quad s_t = \Phi_A(x_t), \quad (5.2.1)$$

where x_t is the internal state at time t under the current parameters and $W : S_A \rightarrow \mathbb{R}$ is the semantic potential from Section 3.3.

The augmented loss is then defined as

$$L_A^{\text{sem}}(\varsigma_A) = L_A(\varsigma_A) + \varrho_{\text{sem}} R_A(\varsigma_A), \quad (5.2.2)$$

with $\varrho_{\text{sem}} \geq 0$ controlling the strength of semantic regularisation.

In the semantic-regularised condition, the training loop is as above, except that L_A is replaced by L_A^{sem} . The semantic-field framework predicts that increasing ϱ_{sem} will, in future experiments, alter the organisation of internal trajectories in S_A in ways that can be quantified via gradient-alignment and potential behaviour.

5.2.3 Logging and semantic construction

The semantic analysis for Case A is structured in stages. The following steps define the logging and construction protocol to be used in future experiments.

Stage 1: state logging. For each trained network (baseline and semantic-regularised) and for each task condition:

1. Select a fixed set of evaluation trials (e.g. a held-out test set).
2. Simulate the network over each trial, logging internal states x_t at a predefined set of time points (e.g. every time step, or specific phases such as cue, delay, and decision).
3. Record, for each logged state, the associated task metadata (trial index, condition label c_i , correctness of the response, etc.).

Stage 2: kernel embedding. Using the logged states $\{x_i\}$, the semantic kernel k_A from Section 3.2.1 is evaluated:

1. Compute the kernel matrix $K_{ij} = k_A(x_i, x_j)$.
2. Optionally perform kernel PCA or Nyström approximations to obtain a finite-dimensional embedding $\tilde{s}_i \in \mathbb{R}^{ds}$ approximating $s_i = \Phi_A(x_i)$.

Stage 3: prototype and covariance estimation. From the embedded states:

1. Estimate semantic prototypes μ_c for each label c using (3.3.2).
2. Estimate the covariance operator C_S using (3.3.5), with regularisation as described in Section 3.3.2.
3. Construct the potential W via (3.3.1), with components W_{attr} , W_{spread} , and W_{reg} configured as in Section 3.3.

Stage 4: trajectory embedding and gradient-flow analysis. Finally, for semantic trajectories $s_t = \Phi_A(x_t)$ (or their finite-dimensional approximations), the protocol specifies:

1. Compute discrete velocities $v_t = s_{t+1} - s_t$.
2. Evaluate gradients $\nabla W(s_t)$ using (3.3.10).
3. Compute alignment metrics α_t , ρ_t and monotonicity quantities ΔW_t , γ as defined in Section 4.4.

These steps provide all quantities required to assess, in future experiments, the approximate gradient-flow representation for Case A.

5.2.4 Perturbation experiments

To probe the stability and attractor structure of the potential W and its relation to the neural dynamics, the protocol includes perturbation experiments.

State perturbations. For selected trials and time points t^* :

1. Take the internal state x_{t^*} and apply a perturbation δx in the direction of $\pm \nabla W(s_{t^*})$ (or in directions orthogonal to ∇W).
2. Use the perturbed state $x_{t^*} + \delta x$ as a new initial condition for the remainder of the trial, and simulate the network forward.
3. Log the resulting trajectories and outputs, embedding them into S_A as before.

The semantic-field framework predicts, for example, that perturbations opposing the gradient direction (increasing W) may lead, in future experiments, to qualitatively different task behaviour or to transitions between semantic basins, whereas perturbations along the negative gradient may result in quicker convergence to attractor regions.

Parameter perturbations. Analogous experiments can be conducted by perturbing parameters ς_A (e.g. small changes to J or b) and examining their effect on $W(s_t)$ and gradient alignment. The protocol specifies:

1. applying controlled perturbations of specified norm to subsets of parameters;
2. re-running evaluation trials and logging states;
3. comparing semantic metrics before and after perturbation.

These perturbation protocols are designed to generate testable predictions about the robustness of semantic structures with respect to the underlying neural implementation.

5.3 Experimental protocol for Case B (LLM latent space)

We now outline an analogous protocol for Case B. The overall structure parallels that of Case A but is adapted to the constraints of large-scale language models.

5.3.1 Base model and latent representations

Base model. The starting point is a pre-trained transformer-based autoregressive language model with:

- L layers, each comprising attention and feed-forward blocks;
- embedding dimension d_B ;
- standard positional and token embeddings.

The base parameters are denoted $\varsigma_B^{(0)}$.

Latent extraction map. To define latent states $x \in X_B$, the protocol specifies a fixed extraction map

$$E : (\text{context, parameters}) \mapsto x \in \mathbb{R}^{d_B},$$

for example:

- the hidden state of a special token (e.g. [CLS] or final token) at a chosen layer;
- a mean-pooled representation of token states over the sequence at a chosen layer.

Given a context $c^{(n)}$ and parameters $\varsigma_B^{(r)}$ at alignment step r , the latent state is

$$x_n^{(r)} = E(c^{(n)}, \varsigma_B^{(r)}), \quad s_n^{(r)} = \Phi_B(x_n^{(r)}) \in S_B.$$

5.3.2 Alignment procedure

The alignment phase updates ς_B from $\varsigma_B^{(0)}$ to $\varsigma_B^{(R)}$. The proposed protocol accommodates a range of standard procedures, such as:

- **Supervised fine-tuning (SFT).** Train on $(c^{(n)}, a^{(n)})$ pairs using teacher forcing, minimising a cross-entropy loss on target tokens.
- **Preference-based optimisation.** Use pairwise or scalar feedback from a reward model to adjust ς_B via policy-gradient or direct-preference optimisation methods.
- **Hybrid schemes.** Combine SFT and preference-based updates.

A generic alignment loop in the proposed protocol takes the form:

1. Sample minibatches of prompts (and, where applicable, preference data).
2. Generate model responses under current parameters $\varsigma_B^{(r)}$ (for RL-type methods) or use targets $a^{(n)}$ (for SFT).
3. Compute the alignment loss $L_B(\varsigma_B^{(r)})$ on the minibatch.
4. Update $\varsigma_B^{(r)}$ to $\varsigma_B^{(r+1)}$ using an optimiser such as Adam.

At pre-specified alignment steps $r \in \mathcal{R}_{\log}$, the protocol includes an evaluation pass in which:

- a fixed evaluation set of prompts $\{c^{(n)}\}$ is processed without parameter updates;
- latent states $x_n^{(r)}$ are extracted via E and stored;
- alignment metrics (rewards, win rates, safety indicators) are computed.

These evaluation passes generate the trajectories $\{s_n^{(r)}\}$ used in semantic analysis.

5.3.3 Kernel and potential estimation in S_B

The construction of the semantic kernel and potential in S_B parallels that of Case A but uses alignment-specific labels.

Data for kernel construction. For each logged latent state $x_n^{(r)}$, the protocol associates:

- a label c_i indicating alignment-relevant information, such as a binned reward score, preference tier, or prompt category;
- metadata about the alignment step r , prompt type, and response quality.

These states and labels are used to construct k_B , S_B , prototypes, and covariance.

Kernel and embedding. The kernel k_B from Section 3.2.2 is evaluated on the collected states:

1. Compute the kernel matrix $K_{ij} = k_B(x_i, x_j)$ for the chosen subset of states.
2. Optionally perform kernel PCA to obtain finite-dimensional embeddings $\tilde{s}_i \in \mathbb{R}^{ds}$ approximating $s_i = \Phi_B(x_i)$.

Potential construction. Using the embedded states:

1. Compute prototypes μ_c for each label c as in (3.3.2);
2. Estimate the covariance operator C_S as in (3.3.5);
3. Construct $W : S_B \rightarrow \mathbb{R}$ via (3.3.1).

Gradient-flow analysis across alignment. For each evaluation prompt $c^{(n)}$, the sequence

$$s_n^{(0)}, s_n^{(1)}, \dots, s_n^{(R)}$$

is treated as a trajectory in S_B across alignment steps. The protocol specifies:

1. computing velocities $v_n^{(r)} = s_n^{(r+1)} - s_n^{(r)}$;
2. evaluating $\nabla W(s_n^{(r)})$;
3. computing alignment metrics and potential changes as in Section 4.4.

The semantic-field framework yields predictions about how $W(s_n^{(r)})$ and gradient alignment should evolve as alignment improves, which can be tested once these logging procedures are implemented.

5.4 Implementation details and reproducibility

To ensure that future implementations of the above protocols are reproducible and interpretable, this section specifies software, hardware, randomisation, and reporting conventions.

5.4.1 Software and hardware

All experiments are envisaged to be implemented in standard deep-learning frameworks (e.g. PyTorch, JAX, or TensorFlow), with:

- separate modules for Case A and Case B model definitions and training loops;
- logging modules that record raw states x_t , labels c_i , and metadata to disk in a standard format (e.g. HDF5, Parquet);
- analysis modules that load logged states, compute kernels, perform kernel PCA, estimate W , and compute semantic metrics.

Hardware assumptions (e.g. number and type of GPUs, memory requirements) are to be documented for each implementation, together with any constraints they impose on batch sizes, model sizes, or logging frequency.

5.4.2 Random seeds and runs

To assess variability across random seeds, the proposed protocols specify multiple independent runs:

- $R^{(A)}$ independent runs for Case A, with different random initialisations of weights, biases, and noise realisations;
- $R^{(B)}$ independent runs or alignment trajectories for Case B, with different initial seeds for optimiser states and, where applicable, sampling in RL-style updates.

For each run, the following seeds are recorded and reported:

- model-initialisation seeds;
- data-shuffling and minibatch-sampling seeds;
- any seeds used for stochastic policies or token sampling;
- seeds for kernel subsampling or Nyström approximations (if used).

This ensures that trajectories and semantic analyses can be replicated exactly in future work.

5.4.3 Hyperparameter reporting

For reproducibility, the following hyperparameters are intended to be recorded and reported for each experiment:

- **Model hyperparameters:** dimensions d_A , d_B , layer counts, nonlinearity types, spectral scaling of J , noise levels;
- **Training and alignment hyperparameters:** learning rates, batch sizes, number of steps, optimiser types and schedules;
- **Kernel hyperparameters:** $(\alpha_A, \beta_A, \ell_A)$ and $(\alpha_B, \beta_B, \tau_B)$, choice of projection P_A , normalisation schemes;
- **Potential hyperparameters:** ρ_{attr} , ρ_{spread} , ρ_{reg} , prototype weights (π_c) , temperature η , covariance regularisation λ ;
- **Semantic embedding hyperparameters:** embedding dimension d_S , number of kernel PCA components retained, any thresholds used for eigenvalue truncation.

These parameters should be summarised in tables or configuration files to facilitate systematic comparison and replication.

5.4.4 Analysis pipeline and configuration management

Finally, the analysis pipeline is proposed to be organised as follows:

1. **Configuration files.** All experiments, logging, and analyses are controlled by explicit configuration files specifying model, training, kernel, and potential hyperparameters, as well as data paths and evaluation settings.

2. **End-to-end scripts.** Scripts are provided that, given a configuration file, run training or alignment, perform logging, and invoke the semantic analysis modules.
3. **Versioning.** Code, configurations, and logged data are versioned so that specific experimental conditions can be reproduced.
4. **Reporting.** The output of the analysis (alignment metrics, distributions of α_t , ρ_t , ΔW_t , etc.) is stored in standard formats and summarised in plots and tables.

The present chapter has therefore specified the empirical procedures required to connect the abstract semantic-field framework to concrete systems in future work. When these protocols are implemented, they will produce the trajectories and quantities needed to test the approximate gradient-flow interpretation and the predictions developed in the next chapter.

6 Expected behaviour and testable predictions

This chapter reformulates the empirical “Results and analysis” of the original version into a set of *expected behaviours* and *testable predictions* derived from the semantic-field framework and the experimental protocols of Chapter 5.

The aim is not to claim that any numerical experiments have been run, but to spell out what the framework anticipates *would* be observed if the protocols were implemented. Where the original text referred to empirical findings, we replace these with hypotheses and predictions, preserving the structure, notation, and types of analyses.

We organise the discussion as follows:

- Section 6.1 (originally “Behaviour of $W(s_t)$ and gradient alignment”) describes predictions for the temporal evolution of the semantic potential and local alignment metrics.
- Section 6.2 (“Attractor landscapes and transitions in S ”) formulates expected patterns in low-dimensional projections, occupation statistics, and transition structures.
- Section 6.3 (“Effects of semantic regularisation”) states how the framework predicts semantic regularisation to modify trajectories and task or alignment performance.
- Section 6.4 (“Robustness and limitations”) discusses anticipated sensitivities to kernel choices, dimensionality, and hyperparameters.

6.1 Behaviour of $W(s_t)$ and gradient alignment

6.1.1 Temporal profiles of $W(s_t)$ in Case A

In the proposed protocols for Case A, the semantic potential $W : S_A \rightarrow \mathbb{R}$ is evaluated along within-trial trajectories

$$s_t = \Phi_A(x_t), \quad t = 0, \dots, T_{\text{trial}},$$

for tasks such as delayed match-to-sample (DMS) and context-dependent integration (CDI). We outline here the *expected temporal profiles* of $W(s_t)$ and associated gradient metrics along such trials.

Predicted temporal structure for DMS and CDI tasks. For DMS and CDI tasks, the semantic-field framework suggests the following typical pattern of $W(s_t)$ along individual trials:

- **Cue or context phase.** When the cue or context information is presented, internal states are driven from a relatively undifferentiated baseline into task-specific basins near semantic prototypes μ_c . We therefore *predict* a noticeable *decrease* in $W(s_t)$ during the initial cue or context phase of the trial, as the state moves towards lower values of the attractor term W_{attr} .
- **Delay phase.** During delay periods, inputs are held constant or removed, and the system must maintain a context-appropriate state. The potential

$$W(s_t) = \rho_{\text{attr}} W_{\text{attr}}(s_t) + \rho_{\text{spread}} W_{\text{spread}}(s_t) + \rho_{\text{reg}} W_{\text{reg}}(s_t)$$

is expected to be approximately *stationary* on average, with small fluctuations driven by noise terms ε_t that are largely counteracted by the gradient $-\nabla W(s_t)$. That is, we anticipate that

$$\Delta W_t = W(s_{t+1}) - W(s_t)$$

will be distributed with mean close to zero and relatively small variance during delay epochs, corresponding to the state remaining close to a local minimum of W_{attr} .

- **Decision and response phase.** When the probe is presented or a decision must be produced, additional structure is imposed on the state due to readout and context-dependent integration. We therefore expect further *decreases* in $W(s_t)$ during response phases for correct trials, reflecting a sharpening of semantic configurations around prototypes associated with accurate decisions.

More formally, if the protocols are implemented, one can test the following hypotheses:

1. For trials in which the network performs correctly, the empirical average of $W(s_t)$ over cue and response epochs is strictly lower than in pre-cue and early delay epochs.
2. The fraction γ of steps with $\Delta W_t \leq 0$ (cf. Eq. (4.4.4)) is highest during cue and response phases, intermediate during delay, and lowest during inter-trial intervals or initial warm-up steps.
3. For incorrect or ambiguous trials, the patterns above are attenuated or partially reversed, with weaker decreases or even increases of $W(s_t)$ in decision phases.

These are explicit, testable predictions about the temporal structure of the potential in Case A.

Gradient alignment across trial phases. In addition to $W(s_t)$ itself, the directional alignment

$$\alpha_t = \frac{\langle v_t, -\nabla W(s_t) \rangle_{S_A}}{\|v_t\|_{S_A} \|\nabla W(s_t)\|_{S_A}}, \quad v_t = s_{t+1} - s_t,$$

is of central interest. The framework predicts that:

- α_t will be *systematically higher* during cue and response phases, when the dynamics are strongly guided by task structure and attractor basins, than during delay phases, where noise and neutral maintenance dynamics dominate.

- The energy ratio

$$\rho_t = \frac{\langle v_t, \nabla W(s_t) \rangle_{S_A}^2}{\|v_t\|_{S_A}^2 \|\nabla W(s_t)\|_{S_A}^2}$$

(Eq. (4.4.7)) is expected to be frequently above 1/2 during cue and response epochs, indicating that more than half of the velocity energy lies along the gradient direction, but lower during prolonged delay periods.

If implemented, the protocols thus permit a concrete test of the hypothesis that the dynamics in Case A are more gradient-like in task-critical phases than during passive or maintenance phases.

6.1.2 Temporal evolution of $W(s_t)$ in Case B

For Case B, we distinguish two types of semantic trajectories:

- trajectories across alignment steps r for fixed contexts $c^{(n)}$,
- and trajectories along generative completions in token space for fixed model parameters.

We describe expected patterns of $W(s)$ in both regimes.

Across alignment steps. Let $s_n^{(r)}$ denote the semantic state for evaluation context $c^{(n)}$ at alignment step r . The framework anticipates:

- **Initial alignment phase.** Starting from the pre-trained base model $\varsigma_B^{(0)}$, an initial alignment phase is expected to produce a *systematic decrease* in $W(s_n^{(r)})$ across r for many contexts, reflecting a reorganisation of latent representations towards attractor regions associated with high reward, better preference scores, or improved safety. Concretely, the average

$$\bar{W}_r = \frac{1}{N} \sum_{n=1}^N W(s_n^{(r)})$$

is predicted to show a noticeable negative trend $\bar{W}_{r+1} - \bar{W}_r < 0$ over an initial range of alignment steps.

- **Later alignment and saturation.** As alignment progresses and metrics such as reward-model scores or safety indicators saturate, we expect $W(s_n^{(r)})$ to *fluctuate* around a quasi-stationary value with no strong systematic drift, driven by a balance between task-driven updates and stochasticity. The empirical fraction of steps with $\bar{W}_{r+1} \leq \bar{W}_r$ is predicted to approach values near 1/2 in late alignment phases.

These behaviours translate into the following testable predictions:

1. For many contexts $c^{(n)}$, there exists an early alignment window $[r_{\text{start}}, r_{\text{mid}}]$ such that $W(s_n^{(r)})$ is monotonically decreasing or exhibits a clear downward trend over this window.
2. In later alignment windows $[r_{\text{mid}}, r_{\text{end}}]$, the distribution of $W(s_n^{(r+1)}) - W(s_n^{(r)})$ is centred near zero with comparable numbers of positive and negative steps.
3. Improvements in alignment metrics (e.g. mean reward) correlate with decreases in $\bar{W}_r$, especially during the early alignment phase.

Along generative trajectories. For generative trajectories, the model processes a fixed prompt c and produces a sequence of latent states s_t as tokens are generated. The semantic-field framework anticipates that:

- During the early part of a completion, as the model moves from a generic context representation to a more specific answer or plan, $W(s_t)$ tends to decrease, reflecting the convergence towards context-appropriate attractor regions.
- As the completion proceeds, especially for very long or off-distribution outputs, $W(s_t)$ is expected to increase or fluctuate more strongly, signalling departures from high-quality or on-manifold regions in S_B .

If the protocols are implemented, one can thus test whether:

1. High-quality completions (as judged by reward models or human preference) exhibit a characteristic “U-shaped” or decreasing profile of $W(s_t)$ over the first part of the completion, followed by stabilisation;
2. Low-quality or pathological completions exhibit earlier increases or more erratic fluctuations in $W(s_t)$, and weaker correlation between $W(s_t)$ and local quality metrics.

6.1.3 Local gradient-alignment statistics

For both Case A and Case B, the semantic-field framework suggests that local gradient alignment measures provide a concise characterisation of how well the dynamics approximate a gradient flow of W .

Predicted distributions of alignment coefficients. We consider:

- the directional cosine α_t in Eq. (4.4.1),
- the energy ratio ρ_t in Eq. (4.4.7),
- and the least-squares step sizes ω_t^* in Eq. (4.3.5).

The framework predicts the following qualitative patterns:

- *Case A, within-trial dynamics.* During cue and response periods, the distributions of α_t and ρ_t are expected to be skewed towards higher values (closer to 1), indicating stronger alignment with the gradient. During delay periods, the distributions are expected to be more symmetric, with substantial mass near 0, reflecting a balance between gradient-driven and noise-driven motion.
- *Case B, alignment trajectories.* Early alignment steps for fixed contexts are predicted to show larger ω_t^* and higher median α_t , while late alignment steps exhibit smaller ω_t^* and more symmetric alignment around zero. For generative trajectories, the framework predicts higher average α_t in early, context-structuring parts of the completion than in later, free-form continuation segments.

These statements translate into concrete statistical tests on histograms and summary statistics of α_t , ρ_t , and ω_t^* conditioned on phase and context.

6.1.4 Residual magnitudes and pathwise efficiencies

The least-squares decomposition

$$v_t = -\omega_t^\star \nabla W(s_t) + \varepsilon_t^\star$$

provides a way to separate gradient-aligned and residual components of the dynamics. The framework predicts that:

- In regimes where the system is strongly driven by task or alignment objectives (e.g. early training, cue and response phases), the average residual norm $\|\varepsilon_t^\star\|_S$ is relatively small compared to $\|v_t\|_S$, and pathwise metrics based on the cumulative sum of $\|\varepsilon_t^\star\|_S^2$ along trajectories are correspondingly low.
- In quasi-stationary regimes (e.g. late training with small learning rates, long delay periods, or saturated alignment), residuals account for a larger share of the motion, and cumulative residual energy grows more rapidly relative to gradient-aligned energy.

From a methodological perspective, we can phrase this as a prediction that the proposed gradient-flow model will be a better approximation in early, strongly driven phases and a weaker approximation in late or maintenance phases, a hypothesis that can be tested via pathwise efficiency metrics once data are available.

6.2 Attractor landscapes and transitions in S

The second group of analyses in this chapter concerns the organisation of states in S and the structure of transitions between semantic regions. We discuss expected behaviours for both system classes.

6.2.1 Low-dimensional visualisations of W

Using kernel PCA or related techniques, the finite set of embedded states $\{s_i = \Phi(x_i)\}$ is projected onto a low-dimensional subspace spanned by leading eigenvectors of the covariance operator C_S . The semantic potential W can then be evaluated on these projected states to produce two- or three-dimensional visualisations.

Expected patterns. The framework suggests that, once the protocols are implemented and a trained W is constructed, the following qualitative patterns should be visible in such visualisations:

- **Case A.** States from different task conditions (e.g. different contexts or decision outcomes) occupy distinct clusters in low-dimensional projections, and these clusters coincide with basins of low W . Delay-period states for each context are expected to form tight clouds around the corresponding prototype μ_c , while cue and response states trace transitions between basins.
- **Case B.** Evaluation prompts grouped by alignment-related labels (e.g. high vs. low reward, safe vs. unsafe responses) are expected to map into distinct regions in low-dimensional projections, with lower W values associated with contexts that yield high-quality, well-aligned outputs. Paths of $s_n^{(r)}$ across alignment steps should trace relatively smooth trajectories descending towards lower- W regions.

Thus, the semantic-field framework predicts that low-dimensional projections will exhibit recognisable attractor-like structures and that W will act as a landscape with basins corresponding to functionally meaningful semantic categories.

6.2.2 Attractor occupation statistics

Given the soft assignment weights $w_c(s)$ in Eq. (3.3.9), one can define for each time t and label c a probability that the system occupies the basin associated with μ_c . Summing over time or trials yields attractor occupation statistics.

Predictions for Case A. For DMS and CDI tasks, we expect that:

- During cue and delay phases, the soft assignments $w_c(s_t)$ are sharply peaked around the context or cue label actually present on the trial, reflecting robust context-dependent attractor occupation.
- During ambiguous or hard trials (e.g. low signal-to-noise ratio), $w_c(s_t)$ may exhibit more substantial mass on multiple labels, signalling uncertainty or switching between basins.
- Incorrect trials are predicted to show systematic mismatches between the attractor label with highest $w_c(s_t)$ and the correct task label, especially near decision times.

Predictions for Case B. For alignment trajectories in Case B, we anticipate that:

- Contexts associated with high-quality, aligned behaviour will exhibit an increasing fraction of time steps (or alignment steps) with high $w_c(s)$ for attractors associated with desirable behaviour.
- Contexts that remain problematic from an alignment perspective (e.g. safety edge cases) may fail to settle into well-defined attractor basins, showing more dispersed or unstable attractor occupation patterns over alignment steps.

These occupation statistics are directly computable from the proposed constructions and provide a testbed for evaluating the semantic meaning of attractors in S .

6.2.3 Transition patterns and perturbation responses

The perturbation experiments for Case A and the trajectory sampling for Case B (Section 5.2 and 5.3) are designed to probe transition structure in S .

Case A: perturbations around attractors. For within-trial perturbations in Case A, where internal states are perturbed in directions aligned with or orthogonal to $-\nabla W(s_t)$, the framework predicts:

- Perturbations *against* the gradient (locally increasing W) during delay periods are more likely to push trajectories out of the current attractor basin, leading to altered decision outcomes or prolonged transients.
- Perturbations *along* the negative gradient should, on average, accelerate convergence towards the attractor associated with the correct context or decision, with little qualitative change in task performance.
- Perturbations orthogonal to $\nabla W(s_t)$ are expected primarily to excite non-gradient modes, resulting in increased residual energy $\|\varepsilon_t^*\|_S$ without large changes in $W(s_t)$, and a mixture of robust and fragile behavioural outcomes depending on the task phase.

These predictions can be tested by comparing transition matrices (between attractor labels or coarse regions in S) across perturbation types.

Case B: generative transitions. For Case B, where trajectories in S_B arise from generative completions or alignment updates, the framework suggests that:

- Generative trajectories starting from the same prompt but different sampling seeds will initially converge towards similar low- W regions before diverging along approximately isopotential manifolds as sampling-induced variability grows.
- Alignment trajectories for a fixed prompt will, in successful alignment scenarios, show a decreasing tendency to exit low- W basins over alignment steps, reflected in transition matrices with lower off-diagonal mass between basins associated with good and poor behaviour.

Overall, the expected transition patterns provide a qualitative picture of how W shapes flows in S and suggest concrete matrix-based statistics that can be evaluated once experiments are implemented.

6.3 Effects of semantic regularisation

We now discuss the expected impact of explicit semantic regularisation, as introduced in Eq. (5.2.2) for Case A and its analogue for Case B. In all cases, the emphasis is on prospective predictions rather than reported findings.

6.3.1 Case A: trajectory organisation and task performance

Recall that, in the semantic-regularised condition, the loss is augmented by

$$L_A^{\text{sem}}(\varsigma_A) = L_A(\varsigma_A) + \varrho_{\text{sem}} R_A(\varsigma_A), \quad R_A(\varsigma_A) = \frac{1}{|T_{\text{sem}}|} \sum_{t \in T_{\text{sem}}} W(s_t).$$

The semantic-field framework suggests the following predictions as ϱ_{sem} is varied:

- For *moderate* values of ϱ_{sem} , semantic regularisation is expected to encourage trajectories that move more consistently down the gradient of W in task-relevant phases. This should manifest as:
 - lower average $W(s_t)$ during cue, delay, and response phases compared to the baseline condition,
 - higher mean directional alignment $\bar{\alpha}$ and energy ratio $\bar{\rho}$,
 - improved robustness of task performance to perturbations and noise.
- For *very large* values of ϱ_{sem} , one expects a trade-off: trajectories may become overly constrained to remain near a small number of attractors, potentially harming flexibility and leading to degraded performance on tasks requiring exploration of multiple semantic configurations.

If the protocols are implemented, one can thus test for a regime of ϱ_{sem} in which semantic regularisation improves both gradient alignment and task performance, and a regime in which excessive regularisation harms either or both.

6.3.2 Case B: alignment quality and semantic structure

Although the main alignment objectives in Case B are task- and preference-based, one can augment them with semantic regularisation terms analogous to $R_A(\varsigma_A)$, or simply *analyse* how existing alignment procedures implicitly regularise $W(s)$.

The framework predicts that:

- Successful alignment schemes will naturally steer latent states for high-quality, aligned outputs into regions of lower W , thereby increasing the separation in S_B between contexts leading to desirable and undesirable behaviour.
- If an explicit semantic regulariser $R_B(\varsigma_B) = \mathbb{E}[W(s)]$ is added with moderate weight, it is expected to sharpen this separation and increase gradient alignment along alignment trajectories, without necessarily harming alignment metrics.
- However, too strong a semantic penalty may restrict the diversity of latent states and reduce the model’s ability to represent subtle contextual distinctions, thereby limiting alignment performance on complex prompts.

These hypotheses suggest that W can serve as both an analytical lens on existing alignment methods and a potential source of additional regularisers, whose impact can be assessed empirically by measuring changes in W , gradient alignment, and standard alignment metrics under controlled perturbations of the loss.

6.4 Robustness and limitations of the predicted behaviours

Finally, we discuss the expected robustness and limitations of the gradient-flow picture and of the semantic analyses proposed. In the original empirical presentation, this section summarised observed sensitivities and caveats; here we reinterpret them as anticipated dependencies and methodological constraints.

6.4.1 Kernel choices and semantic geometry

The geometry of S and the shape of W depend on the kernel choice. The framework suggests:

- *Qualitative robustness.* As long as kernels capture broadly similar notions of similarity (e.g. RBF-type on whitened activations for Case A, cosine-type on embeddings for Case B), low-dimensional visualisations, attractor basins, and coarse gradient alignment statistics are expected to be qualitatively similar. That is, we predict that key patterns (e.g. separation of task conditions, initial decreases in W during learning) are robust across reasonable kernel families.
- *Quantitative sensitivity.* Fine-grained numerical values of W , α_t , ρ_t , and residual magnitudes will depend on kernel hyperparameters such as length scales and normalisation. The framework therefore anticipates that quantitative claims (e.g. the exact mean alignment level) will vary across kernel settings, and recommends reporting ranges across a family of kernels.

These expectations motivate explicit kernel-sensitivity analyses in future experiments.

6.4.2 Dimensionality of S

In practice, S is approximated by a finite-dimensional subspace (e.g. via kernel PCA), with dimension d_S . The framework predicts:

- For sufficiently large d_S (capturing most of the empirical variance in S), the main qualitative features of W and gradient alignment (e.g. existence of distinct attractor basins, coarse monotonicity of W in training) are stable under moderate changes in d_S .
- If d_S is chosen too small, important directions of semantic variation are projected out, which can spuriously increase apparent gradient alignment (by removing informative residual directions) or, conversely, blur attractor basins. We therefore expect non-monotonic behaviour of some metrics as a function of d_S .

A key methodological prediction is that there exists a regime of intermediate d_S values where the approximation is both computationally tractable and faithful enough to preserve the main semantic structures, and that this regime can be identified by standard dimensionality-sensitivity analyses.

6.4.3 Hyperparameters, finite-sample effects, and extrapolation

Finally, the semantic-field analyses depend on hyperparameters and on finite datasets of states.

Hyperparameters and finite-sample effects. The framework anticipates that:

- Hyperparameters of W (weights ρ ., temperature η , prototype weights π_c) can be tuned to fit gradient-flow approximations more or less tightly. Overfitting of W to a limited set of trajectories may lead to overly optimistic alignment metrics on the fitting set but degraded generalisation to held-out trajectories.
- Finite-sample effects (limited numbers of states, contexts, or trials) can distort estimators of C_S and μ_c , leading to noisy or biased estimates of W and its gradient. We therefore predict that stability of the analyses will improve with sample size and that bootstrap or resampling methods will be useful for quantifying uncertainty.

Extrapolation and non-gradient components. Crucially, the framework does *not* claim that the systems are governed purely by the gradient of W . Instead, it predicts that:

- Non-gradient components of the dynamics, as captured by residuals ε_t , will remain substantial, especially in late training and quasi-stationary regimes.
- The semantic gradient-flow picture is informative but approximate; it captures certain aspects of the dynamics (e.g. coarse movement towards semantic basins) while neglecting others (e.g. oscillatory or rotational components in S).
- Extrapolating the conclusions from the studied settings to substantially larger models, more complex tasks, or very different architectures must be done cautiously.

These limitations are themselves testable: once experiments are implemented, one can quantify the relative magnitude of gradient and non-gradient components across conditions and assess how far the gradient-flow approximation can be pushed before it breaks down.

In summary, this chapter has reformulated the empirical narrative of the original work into a set of explicit, operational predictions about the temporal behaviour of $W(s_t)$, the structure of attractor landscapes and transitions in S , the effects of semantic regularisation, and the robustness and limitations of the semantic-field description. These predictions are designed to be directly testable under the simulation and experimental protocols specified in Chapter 5, thereby providing a clear bridge between the mathematical framework and future empirical investigations.

7 Discussion and outlook

In this final chapter we step back from the technical constructions and protocols to consider their broader implications, their prospective empirical role, and the limitations of the present work. Throughout, the emphasis remains methodological: the paper formulates a framework, proposes concrete constructions and protocols, and derives testable predictions, but the execution and empirical assessment of these proposals are explicitly left to future studies.

7.1 Conceptual implications of the semantic-field perspective

The preceding chapters define a semantic Hilbert space S for the internal states of complex systems, construct a semantic potential $W : S \rightarrow \mathbb{R}$, and propose to interpret system dynamics as approximate gradient flows of W , up to residual and noise terms. This section highlights the conceptual implications of viewing neural and latent dynamics through this lens.

7.1.1 From raw state dynamics to semantic flows

The core conceptual move is the shift from raw state spaces X_A and X_B to a semantic state space S built as a reproducing-kernel Hilbert space. In the concrete instantiations of this paper, the feature map

$$\Phi : X \rightarrow S$$

is induced by kernels tailored to the systems (radial-basis and linear components for Case A, cosine-type kernels for Case B). Once such a map is fixed, trajectories in X are re-interpreted as trajectories in S .

The framework suggests that, if the proposed protocols are implemented, these semantic trajectories will capture aspects of internal organisation that are not easily visible in the raw coordinates. For example:

- In Case A, trajectories corresponding to different task conditions, phases (cue, delay, response), and performance outcomes may be expected to occupy distinct regions of S_A , with transitions between these regions reflecting the processing steps of the system.
- In Case B, trajectories across alignment steps or along generative completions are expected to manifest as flows in S_B towards or within regions associated with high-quality and aligned outputs.

Thus, S is intended as a space in which semantic structure becomes geometrically transparent and amenable to dynamical analysis.

7.1.2 Semantic potentials as effective energy landscapes

The semantic potential $W : S \rightarrow \mathbb{R}$ is designed to play a role analogous to an energy landscape in physics or a free-energy functional in variational inference: it assigns lower values to semantic configurations that are considered desirable or typical and higher values to less desirable or atypical ones. Concretely, the decomposition

$$W(s) = \rho_{\text{attr}} W_{\text{attr}}(s) + \rho_{\text{spread}} W_{\text{spread}}(s) + \rho_{\text{reg}} W_{\text{reg}}(s)$$

encodes attraction towards semantic prototypes, control of dispersion and anisotropy, and norm-based regularity.

From a conceptual standpoint, several points follow:

- The attractor term W_{attr} embodies a hypothesis that, under the proposed protocols, internal states for certain classes of stimuli, contexts, or alignment conditions will cluster around a finite set of semantic prototypes in S , and that the dynamics will tend to move towards these prototypes.
- The spread term W_{spread} assumes that empirically typical states occupy a relatively low-dimensional, anisotropic region of S , and that deviations from this region can be penalised according to an inverse covariance structure.
- The regulariser W_{reg} encodes a bias towards simpler or smoother semantic configurations in S , complementing the other terms.

If the protocols are carried out, one can examine to what extent these assumptions provide an adequate description of internal organisation. The proposed metrics make it possible to quantify how well the dynamics align with $-\nabla W$ and where additional structure beyond a simple potential landscape is required.

7.1.3 Gradient flows as a unifying dynamical template

The approximate gradient-flow representation

$$s_{t+1} = s_t - \omega_t \nabla W(s_t) + \varepsilon_t$$

is deliberately simple: it does not attempt to capture the full complexity of neural or latent dynamics but instead focuses on the component aligned with a gradient field. This choice supports several conceptual uses:

- It provides a *unifying template* across system classes: the same expression applies to recurrent networks and LLM latent spaces, with differences captured by X , k , and W rather than by the form of the dynamical law.
- It separates *gradient-like* motion, which may be strongly tied to task or alignment objectives, from *residual* motion, which can encode oscillatory, rotational, or noise-driven components. This separation is expected to yield insight into which aspects of the dynamics are effectively “driven” by the semantic potential and which remain unexplained by it.

- It connects naturally to the measure-valued generalisations developed in earlier work, where distributions over S evolve according to gradient flows of free-energy functionals on spaces of measures.

In this way, gradient flows become a conceptual backbone linking the abstract semantic-field framework to concrete dynamical analyses.

7.2 Role of the proposed protocols and metrics

Chapters 5 and 6 specify detailed protocols and expected behaviours for two system classes. This section discusses how these proposals can function as methodological tools once implemented.

7.2.1 Semantic diagnostics for complex systems

The construction of S , W , and alignment metrics yields a family of *semantic diagnostics* for complex systems, going beyond conventional task performance metrics. For example, under the proposed protocols one can compute:

- the temporal profiles of $W(s_t)$ along trials or alignment steps;
- local alignment measures α_t and ρ_t between velocities and $-\nabla W(s_t)$;
- attractor occupation statistics via the soft assignment weights $w_c(s_t)$;
- residual-energy measures and pathwise efficiencies.

These quantities are expected to complement standard metrics by revealing how internal representations organise and evolve in S . For instance:

- Two models with similar task accuracy might exhibit different trajectories in S , with one following $-\nabla W$ more closely and occupying well-defined attractor basins, while the other relies more on residual, non-gradient components. The framework suggests that such differences could be informative about robustness, generalisation, or interpretability.
- In alignment scenarios, changes in $W(s_n^{(r)})$ and gradient alignment across alignment steps could serve as indicators of whether an alignment procedure is effectively steering latent states towards desirable regions of S .

Thus, the semantic diagnostics proposed in this paper are intended as tools for analysing and comparing complex systems along axes that are not visible at the level of task outputs alone.

7.2.2 Semantic regularisation as a design principle

The augmented loss functions that incorporate W , such as

$$L_A^{\text{sem}}(\varsigma_A) = L_A(\varsigma_A) + \varrho_{\text{sem}} R_A(\varsigma_A),$$

illustrate how the semantic potential can enter training or alignment objectives. Even without committing to specific values of ϱ_{sem} , the framework identifies a design principle:

Encourage internal trajectories to move towards and remain within regions of S that are associated with desirable semantic prototypes, typical spreads, and regularity, while simultaneously achieving good task performance.

If future experiments implement such semantic regularisation, they can test whether this principle leads to improvements in robustness, generalisation, or interpretability and how it trades off against flexibility and diversity of internal states. The framework also motivates comparisons between explicit semantic regularisation and existing implicit regularisation effects of standard training algorithms.

7.2.3 Bridging abstract theory and empirical practice

More broadly, the proposed protocols and metrics operationalise the abstract mathematical framework introduced in earlier work. They specify:

- how to instantiate the abstract Hilbert space S as an RKHS built from concrete model states;
- how to define potentials W that are both interpretable and grounded in empirical distributions of states and labels;
- how to extract trajectories and compute gradient-alignment statistics from logged data.

Once implemented, these steps would allow the abstract semantic-field theory to be confronted with empirical evidence, refining or revising the theory as needed. The present paper thus occupies a methodological intermediate layer between purely theoretical developments and domain-specific empirical work.

7.3 Connections and broader implications

The semantic-field framework sits at the intersection of several research traditions. This section sketches some of these connections and outlines how future empirical studies might inform them.

7.3.1 Relations to energy-based and representation-learning methods

The use of a potential W on S aligns conceptually with energy-based models, in which learned energy functions assign low values to desirable configurations. The main differences are:

- The semantic potential is defined on the internal state space of an existing system (e.g. a recurrent network or an LLM), rather than directly on input or output spaces.
- W is constructed post hoc from logged states, prototypes, and covariances, rather than being trained as a stand-alone energy function.

The framework suggests that insights from energy-based modelling—such as connections to Markov random fields, contrastive learning, and score matching—may become relevant once empirical data are available and W can be tuned to satisfy additional constraints (e.g. consistency with observed transition frequencies in S).

Similarly, the construction of S via kernels and the analysis of W are closely related to representation-learning approaches that seek linearisable or disentangled latent spaces.

The semantic-field perspective provides a complementary view: instead of training the representation end-to-end, it treats the latent space as given and equips it with an additional geometric and dynamical structure.

7.3.2 Links to dynamical-systems and control perspectives

From a dynamical-systems standpoint, modelling trajectories as approximate gradient flows of W in S connects the behaviour of complex systems to classical ideas about attractors, basins, and stability. The residual term ε_t can, in principle, capture rotational components, oscillations, and other non-gradient phenomena.

The framework suggests that, in future work, one could:

- decompose dynamics in S into gradient and Hamiltonian-like components, analogous to Helmholtz decompositions of vector fields;
- relate the effective potential W to Lyapunov functions used in control theory to assess stability and convergence properties;
- design control signals (inputs, prompts, or training interventions) that steer trajectories along or against ∇W to achieve desired behaviours.

Empirical tests of the gradient-flow approximation and of the stability properties of W -basins would be informative for these connections.

7.3.3 Implications for interpretability and safety

If the proposed protocols reveal clear semantic basins in S associated with particular behaviours or alignment properties, the semantic-field framework could contribute to interpretability and safety efforts:

- Regions of S corresponding to undesirable behaviour (e.g. unsafe responses, failure modes) could be identified and characterised, suggesting interventions that increase the effective potential in those regions.
- Trajectories that cross from “safe” to “unsafe” basins in S could be analysed to understand which inputs, prompts, or parameter changes trigger such transitions.
- Semantic regularisers could be designed to discourage occupation of unsafe regions and encourage staying within safe basins.

These applications depend on the empirical structure of S and W and thus remain conditional, but the framework provides a systematic language for articulating such interpretability and safety questions.

7.4 Limitations and future directions

The present work is deliberately limited in scope and makes several modelling choices that warrant careful scrutiny once empirical data are available. This section summarises these limitations and outlines possible extensions.

7.4.1 Approximate and system-dependent nature of W

The semantic potential W is constructed from finite samples of states, labels, and covariances and depends on kernel choices, hyperparameters, and fitting procedures. As a result:

- W is an *effective* potential, not a fundamental property of the system. Different datasets or kernel families may yield different potentials for the same underlying dynamics.
- The gradient-flow representation is approximate by design: residuals ε_t are expected to remain non-negligible in many regimes, and purely gradient-driven interpretations are unlikely to hold globally.

Future work can address these issues by:

- systematically exploring how W and gradient-alignment metrics change with data subsets, kernels, and hyperparameters;
- developing regularisation and model-selection criteria that favour potentials with good out-of-sample predictive power for trajectories in S ;
- investigating alternative or complementary structures (e.g. vector-field models on S that are not constrained to be gradients).

7.4.2 Scalability and computational constraints

The RKHS-based constructions and kernel PCA steps involve operations on kernel matrices whose size grows with the number of logged states. For large systems and extensive logging, this can become computationally demanding.

The present work does not specify concrete scalability strategies but anticipates that future implementations may need to rely on:

- Nyström approximations and random feature maps to reduce kernel complexity;
- streaming or online methods for updating covariance estimates and prototypes;
- hierarchical or multi-scale semantic representations that limit the effective dimensionality of S .

Understanding how these approximations affect W and gradient alignment will be an important part of subsequent empirical studies.

7.4.3 Extensions beyond the current system classes

Although Cases A and B provide concrete testbeds, the semantic-field framework is not limited to these systems. Future work may wish to extend the methodology to:

- spiking neural networks and other biologically inspired models with richer temporal structure;
- multi-agent or distributed systems, where S represents joint semantic configurations across agents;
- non-neural complex systems (e.g. economic or social models) where appropriate kernels and semantic labels can be defined.

Such extensions would test the generality of the semantic-field approach and may motivate refinements of the underlying mathematical structures.

7.4.4 Empirical validation and theory refinement

Ultimately, the usefulness of the semantic-field framework depends on how well its predictions match empirical observations once the proposed protocols are implemented. The present paper provides:

- a clear construction of S , W , and gradient-flow metrics;
- detailed protocols for generating trajectories and logging states;
- explicit hypotheses about temporal profiles of $W(s_t)$, gradient alignment, attractor occupation, and effects of semantic regularisation.

Future empirical studies can:

- confirm, refine, or refute these hypotheses for specific systems and tasks;
- identify regimes in which the gradient-flow approximation is particularly successful or clearly inadequate;
- feed back into the theory by motivating new potential components, alternative geometries for S , or more general dynamical models.

In this way, the semantic-field framework is intended not as a final description but as a starting point for an iterative dialogue between mathematical structure and empirical investigation.

To summarise, this paper has put forward a methodology for describing the internal dynamics of complex systems in terms of semantic flows in a Hilbert space, guided by a potential W , and for testing this description through concrete protocols and metrics. The next steps lie in implementing these protocols, studying the resulting data, and using the semantic-field perspective to deepen our understanding of neural and latent-space dynamics in practical machine learning and complex-systems settings.

Acknowledgements and Contact

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The work emerged from a prolonged process of intuitive exploration, conceptual reduction, and formal consolidation. Informal insights about meaning, structure, and stability were progressively translated into explicit models, mathematical frameworks, and testable claims. Throughout this process, modern large language models were employed as practical instruments for structuring arguments, checking internal consistency, exploring alternative formulations, and iteratively refining formal expressions.

These tools were used as epistemic scaffolding rather than as sources of claims. All theoretical commitments, modelling decisions, and interpretative judgments remain the responsibility of the author. The resulting framework should therefore be understood as exploratory but precise: it proposes a concrete formal scaffold, derives explicit consequences, and invites scrutiny.

The work is explicitly offered for critical examination, clarification, extension, or refutation by domain experts. Questions, comments, corrections, and collaboration inquiries are welcome and may be directed to contact@semanticgravitation.org.

Semantic Gravitation VI: The Global Attractor and the Lagrangian of Vectorial Emergence (LOVE)

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Abstract

This paper completes the six-part *Semantic Gravitation* programme by developing a structural framework for the long-time dynamics of semantic systems and their possible macroscopic manifestations. Building on the constructions of Papers I–V, we consider semantic gradient flows on a separable Hilbert space S generated by contradiction-energy functionals $W: S \rightarrow \mathbb{R}$, together with their associated Fokker–Planck dynamics on probability measures and their spacetime field-theoretic realisations $s: M \rightarrow S$.

Within this setting we introduce a compact *global attractor* $K \subset S$ for the dissipative gradient flow of W and analyse its mathematical and structural role. The attractor K represents the locus where contradiction energy is minimised subject to structural constraints and serves as a geometric organising centre for semantic dynamics. This motivates a tripartite structural scheme K – P – R , consisting of a semantic core K , projection structures P mediating between core and appearance, and relational degrees of freedom R encoding couplings between subsystems, agents, and environments.

A second key ingredient is a family of coherence axes C, B, A defined near the attractor. These axes quantify different modes of semantic integrity and distortion and allow a refined decomposition of gradient flows, dissipation balances, and stochastic dynamics. Within this framework we analyse the formation of local coherence structures, metastable “order bubbles”, and axis-resolved probability currents under noise and external driving.

The field-theoretic completion is provided by the *Lagrangian of Vectorial Emergence* (LOVE), an effective variational principle governing the evolution of semantic fields $s: M \rightarrow S$. The associated action functional couples the semantic potential W , the global attractor K , and the $C/B/A$ structure to spacetime dynamics, leading to Euler–Lagrange-type equations that unify deterministic, stochastic, and coarse-grained descriptions of semantic organisation.

A further structural bridge to spacetime physics is provided by the induced semantic tensor $h_{\mu\nu} = \langle \partial_\mu s, \partial_\nu s \rangle$, which expresses semantic gradient structure geometrically on spacetime. Invariants of this tensor provide natural candidates for large-scale observables associated with semantic organisation.

Within this geometric setting we propose an interpretation of certain large-scale gravitational anomalies in terms of *relational inertia*: enhanced continuation bookkeeping

load encoded by a relational integration density ρ_{rel} . In this view, phenomena commonly attributed to dark matter may arise not from additional particulate matter, but from mesoscopic regimes of relational structure in spacetime. These regimes are interpreted as intermediate between microscopic semantic dynamics and fully stabilised macroscopic matter sectors.

The resulting framework connects gradient dynamics, attractor geometry, stochastic processes, and spacetime field theory into a unified description of semantic organisation across scales. Paper VI thus provides both a structural closure of the formal theory and a foundation for further mathematical, computational, and empirical exploration of semantic gravitation.

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Chapter 1

The necessity of a semantic field

In Papers I–V we have gradually built a mathematical framework in which *semantic gravitation* appears as a new, energy-like structure on suitable state spaces. In this first chapter of Paper VI we step back from the technical constructions and articulate why a *semantic field*—and with it a semantic state space and a semantic potential—is conceptually and mathematically necessary. This sets the stage for the ontological layer added in the later chapters, in particular the global attractor K and the C/B/A coherence structure.

1.1 Structural map of the semantic gravitation framework

Before entering the retrospective on Papers I–V, it is useful to summarise the structural architecture of the semantic gravitation framework in compact form. The programme is built around a small set of mathematical objects that play distinct roles in its dynamical, geometric, and ontological organisation.

Object	Mathematical type	Interpretation
M	Lorentzian manifold	spacetime
S	separable Hilbert space	semantic state space
$s : M \rightarrow S$	field configuration	semantic field
$W : S \rightarrow \mathbb{R}$	potential functional	contradiction energy
$K \subset S$	invariant compact set	global semantic attractor
$h_{\mu\nu}$	induced tensor	semantic spacetime sensitivity

These objects organise the theory into three structural layers.

Dynamical layer. The configuration of a semantic system is described by a field

$$s : M \rightarrow S,$$

whose evolution is governed by gradient-type dynamics generated by the contradiction-energy functional W .

Geometric layer. Gradients of the semantic field induce spacetime-sensitive structures such as the tensor

$$h_{\mu\nu} = \langle \partial_\mu s, \partial_\nu s \rangle_S,$$

which measures how strongly semantic configurations respond to spacetime variations.

Structural layer. The global attractor

$$K \subset S$$

acts as the organising backbone of semantic dynamics. It defines the region of semantic state space towards which gradient flows tend to relax and provides the reference structure for the tripartite ontology K – P – R developed in the later chapters.

Together these components define the minimal architecture of semantic gravitation: a field theory on spacetime whose internal degrees of freedom evolve in a semantic state space and whose long-time organisation is governed by a global attractor structure.

1.2 Retrospective on Papers I–V

We begin with a concise retrospective that fixes notation and recalls the core ingredients of the framework.

Definition 1.2.1 (Semantic state space). Let S be a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. Following Paper 1, we interpret elements $s \in S$ as *semantic configurations* of a system: equivalence classes of microstates that are indistinguishable at the level of meaning-relevant observables.

In Paper 1 we introduced an *energy functional*

$$W : S \rightarrow \mathbb{R},$$

interpreted as *contradiction energy* or *semantic tension*, and studied the gradient flow

$$\dot{s}_t = -\nabla W(s_t) \tag{1.2.1}$$

on S , as well as stochastic perturbations of the form

$$ds_t = -\nabla W(s_t) dt + \sqrt{2\beta^{-1}} dB_t, \tag{1.2.2}$$

with $\beta > 0$ and (B_t) a cylindrical Brownian motion on S . The deterministic part implements semantic gradient descent: trajectories flow towards lower contradiction energy. The stochastic part allows exploration of the state space and encodes uncertainty, noise, or unresolved degrees of freedom.

Paper 2 operationalised these constructions for concrete complex systems. It provided recipes for embedding neural, symbolic, or latent representations into S , for constructing candidate potentials W from data and task structure, and for interpreting W as a form of regularised free energy functional on representational spaces.

In Paper 3 we extended the framework to spacetime. Let (M, g) be a Lorentzian spacetime manifold. A *semantic field* is a mapping

$$s : M \rightarrow S, \quad x \mapsto s(x),$$

and we considered action functionals of the form

$$\mathcal{S}[s] = \int_M \mathcal{L}(x, s(x), \nabla s(x)) \, \text{dvol}_g(x),$$

with Lagrangian density $\mathcal{L}$ built from $W(s(x))$ and suitable kinetic terms. The corresponding Euler–Lagrange equations define the dynamics of semantic fields on spacetime and couple semantic gravitation to geometrical structure.

Paper 4 showed that the semantic state space S can host a *quantum-compatible sector* $S_Q \subset S$ together with maps that associate to each quantum state and observable suitable semantic representatives in S and operators on S . In this way, expectation values and measurement statistics of ordinary quantum theory are reproduced as derived notions within the semantic framework.

In summary, Papers I–IV have established:

- a semantic state space S and potentials $W : S \rightarrow \mathbb{R}$,
- deterministic and stochastic gradient flows on S driven by ∇W ,
- a field-theoretic extension $s : M \rightarrow S$ with action functionals on spacetime,
- a quantum-compatible semantic sector $S_Q \subset S$.

Paper V, in turn, has filled in additional structural aspects (in particular the semantic Hilbert space as reproducing kernel Hilbert space and further links to information-theoretic quantities), but deliberately avoided committing to a global ontological picture. In particular, it did not fix a notion of *global attractor* for semantic dynamics, nor did it specify how normativity, coherence, and integrity are encoded in the geometry of S beyond local properties of W .

1.3 Limits of syntactic and purely statistical descriptions

The starting point of semantic gravitation is the observation that existing formalisms in physics and information theory are predominantly *syntactic*. They describe:

- dynamical laws on physical state spaces (e.g. phase spaces, configuration spaces, Hilbert spaces),
- probabilistic structure on these spaces (e.g. measures, density operators),
- coding, transmission, and processing of symbols or bits.

In such formalisms, the same physical or informational configuration can support radically different “meanings” depending on context, while configurations that are semantically equivalent may be represented by very different microstates. Thus, meaning is not an additional coordinate in the usual state spaces but rather an emergent and context-dependent equivalence relation on them.

Remark 1.3.1. Standard information theory measures uncertainty and correlation structure in terms of entropy and mutual information, but it does not by itself furnish a measure of *semantic coherence*. Two messages may have identical Shannon entropy yet differ profoundly in whether they resolve contradictions in an agent’s world-model or increase them. This asymmetry motivates the introduction of a separate functional W on a dedicated semantic state space S .

From a purely statistical perspective, one might attempt to define semantic quantities as functionals of the probability distributions over physical states. However, without additional structure, such constructions cannot distinguish between configurations that merely *happen to be frequent* and configurations that are *normatively coherent* relative to a system’s internal constraints, goals, or integrity conditions.

The central claim of semantic gravitation is that this limitation is not just practical but structural: without a dedicated semantic state space and a corresponding potential, there is no natural way to represent coherence, contradiction, and their dynamics as intrinsic properties of states.

Postulate 1.3.2 (Structural necessity of a semantic field). Any mathematically precise theory that aims to treat meaning, coherence, and contradiction as *dynamical* quantities must introduce:

1. a semantic state space S distinct from purely physical or syntactic state spaces,
2. a semantic potential $W : S \rightarrow \mathbb{R}$ interpreted as contradiction energy,
3. and (at least weakly) gradient-like dynamics on S driven by W .

In particular, semantic quantities cannot in general be reduced to functions of purely syntactic or purely probabilistic data without loss of essential structure.

Postulate 1.3.2 is not a theorem; it is an explicit structural assumption of the semantic gravitation programme. In Papers I–V it underlies the choice to take S and W as primitive objects. In Paper VI we clarify how this postulate interacts with the ontological structures K , K – P – R , and the C/B/A coherence axes.

1.4 Minimum requirements for a semantic field

To fix ideas and provide a clear target for subsequent developments, we now collect the minimum mathematical requirements that a semantic field must satisfy within the present framework.

Definition 1.4.1 (Semantic field on spacetime). Let (M, g) be a time-oriented Lorentzian manifold and S a separable Hilbert space as above. A *semantic field* is a measurable mapping

$$s : M \rightarrow S$$

such that $s(x)$ is defined for almost all $x \in M$ and belongs to a fixed dense domain $\mathcal{D} \subset S$ on which W and its derivatives are well-defined.

Definition 1.4.2 (Semantic potential and local contradiction energy density). Let $W : S \rightarrow \mathbb{R}$ be a functional of class C^2 on $\mathcal{D} \subset S$. To each semantic field $s : M \rightarrow S$ we associate the *local contradiction energy density*

$$\varepsilon_W(x) := W(s(x))$$

and, when appropriate, a total contradiction energy

$$E_W[s] := \int_{\Sigma} W(s(x)) \, d\mu_{\Sigma}(x)$$

for spacelike hypersurfaces $\Sigma \subset M$ with induced volume measure μ_{Σ} .

These quantities must satisfy several structural conditions:

1. *Regularity*: W is sufficiently smooth (typically at least C^2) on a dense domain $\mathcal{D} \subset S$ to define its gradient ∇W and Hessian $\nabla^2 W$ as continuous operators.
2. *Coercivity*: W grows at least quadratically along directions of increasing semantic contradiction, so that sublevel sets $\{s \in S : W(s) \leq c\}$ are (suitably) bounded. This ensures stability of gradient flows and the existence of minimisers under appropriate conditions.
3. *Context-sensitivity*: W may depend on external parameters or additional fields encoding context (e.g. other agents, environments, or boundary conditions), but such dependence must be representable within the overall semantic state space S or an extended state space S_{ext} .

4. *Compatibility with composition:* For composite systems with semantic state spaces S_1 and S_2 , and joint space $S_1 \hat{\otimes} S_2$, the corresponding potentials must obey consistency conditions that allow for the definition of effective potentials on subsystems and coarse-grained degrees of freedom.

Definition 1.4.3 (Semantic action). A *semantic action functional* on spacetime is a mapping $\mathcal{S}$ that assigns to each admissible field configuration $s : M \rightarrow S$ a real number

$$\mathcal{S}[s] = \int_M \mathcal{L}(x, s(x), \nabla s(x)) \, \text{dvol}_g(x),$$

where the Lagrangian density $\mathcal{L}$ satisfies:

1. $\mathcal{L}$ is local in s and its first derivatives, possibly with additional local fields or curvature couplings;
2. the potential term of $\mathcal{L}$ reduces to $W(s(x))$ in the static limit $\nabla s = 0$;
3. $\mathcal{L}$ is compatible with the causal structure of (M, g) and yields well-posed Euler–Lagrange equations for s .

In Paper III we studied instances of such actions with simple kinetic terms. The present paper goes further by introducing a new layer of structure—the Lagrangian of Vectorial Emergence (LOVE)—that refines $\mathcal{L}$ in terms of distinguished directions and curvature components in S around a global attractor K .

1.5 Position of Paper VI in the programme

With these minimum requirements in place, we can formulate more precisely the role of Paper VI.

- Papers I–V have shown that the semantic gravitation framework is mathematically viable: one can define a Hilbert space S , construct potentials W , study gradient flows, and embed classical and quantum descriptions into this setting.
- What is still missing is an explicit *ontological structure* that explains how long-time behaviour, normativity, and coherence are globally organised in S , and how this structure is reflected in the dynamics on spacetime M .
- In particular, Papers I–V did not specify a distinguished subset $K \subset S$ that functions as a global attractor, nor did they introduce the C/B/A axes or the K – P – R ontology that will form the backbone of the present paper.

Paper VI therefore has a dual task:

1. On the *ontological* side, it postulates and analyses a global attractor K , together with projection and relation structures (K – P – R) and coherence axes (C/B/A), as additional primitives that complete the description of semantic gravitation.
2. On the *dynamical* side, it introduces the LOVE Lagrangian, an effective Lagrangian density that couples W , K , and the C/B/A structure to the spacetime dynamics of $s : M \rightarrow S$ and thereby organises vectorial emergence across scales.

These additions are not arbitrary. They are guided by:

- structural requirements arising from the behaviour of gradient flows of W ,
- constraints coming from multi-agent and multi-field configurations,
- the need to account for normativity and integrity constraints in technical and cultural systems (to be made precise later in terms of distortion factors and order bubbles).

Summary and outlook. In this chapter we have recalled the core constructions of Papers I–V and stated the central postulate that motivates the introduction of a semantic state space S , a semantic potential W , and semantic fields $s : M \rightarrow S$. We argued that purely syntactic or purely statistical descriptions are structurally insufficient to capture semantic coherence and contradiction as dynamical quantities, and we listed the minimum mathematical requirements that a semantic field must satisfy within the present framework. The structural map added at the beginning of the chapter makes explicit how the central objects of the theory—spacetime M , semantic state space S , field configurations s , contradiction-energy functional W , global attractor K , and induced tensor $h_{\mu\nu}$ —fit together within a single architecture. This prepares the next chapter, in which we revisit the *core deficit* of current science and formulate more sharply the conceptual gap—the absence of a semantic core and a global attractor—that the extended theory of semantic gravitation aims to address.

Chapter 2

The core deficit of current science (“Kernmangel” revisited)

In the previous chapter we argued for the necessity of a semantic state space S and a semantic potential W in order to treat coherence and contradiction as dynamical quantities. In this chapter we revisit, in a more formal way, the *core deficit* of current scientific descriptions: the absence of a shared semantic core and of a corresponding energy-like structure. The main purpose is interpretative synthesis rather than new technical results; we prepare the ground for the explicit introduction of the global attractor K in later chapters.

2.1 The notion of a semantic core

Intuitively, the *core* of a scientific description is the structure that allows heterogeneous systems to be compared and coupled at the level of meaning. It plays the role of a common coordinate system for semantic content, in which questions such as “Are these two states referring to the same thing?”, “Are these two dynamics compatible?”, or “Do these local attractors correspond to coherent long-time behaviour?” become well-posed.

Definition 2.1.1 (Semantic core, informal). A *semantic core* is a structure that provides:

1. a common semantic state space S in which meaning-relevant configurations of a wide class of systems can be represented;
2. an energy-like functional $W : S \rightarrow \mathbb{R}$ that measures semantic contradiction or incoherence;
3. a distinguished subset (or family of subsets) $K \subset S$ that organises long-time behaviour and serves as a reference for semantic coherence.

In the present paper, S and W are taken from the constructions in Papers 1–3, while K will be analysed as a *global attractor* in later chapters. For now, it is useful to isolate the structural role that such a triplet (S, W, K) is supposed to play.

Remark 2.1.2. The semantic core is not assumed to be a particular physical system, substrate, or location in spacetime. Rather, it is a geometric and energetic structure in an abstract state space S that many different physical and informational systems can map into by suitable embeddings.

2.2 The core deficit in existing formalisms

Current scientific formalisms describe dynamics, information, and stability with great precision, yet they lack a *shared semantic core* in the above sense. We briefly indicate how this deficit manifests itself in three domains.

Classical physics. Classical mechanics and field theory describe the evolution of states in phase space or on configuration manifolds, driven by Hamiltonians or Lagrangians that encode energy, momentum, and other conserved quantities. These structures are extremely successful for predicting physical behaviour, but they are indifferent to questions of semantic coherence:

- There is no distinguished state space in which the “meaning” of configurations across different systems is represented.
- Energy functionals H or L do not encode whether a state is semantically consistent, only whether it is dynamically accessible or energetically favourable in the physical sense.
- There is no canonical subset of states singled out as globally coherent in the semantic sense.

Information theory and statistics. Shannon information and related concepts provide powerful tools for quantifying uncertainty, redundancy, and capacity. Probabilistic models describe correlations, prediction, and learning. However:

- Entropy measures dispersion of probability mass, not semantic contradiction.
- Two messages can have identical entropy while differing radically in their effect on a world-model.
- There is no universal semantic state space S that carries meaning across different domains; each model has its own sample space.

Machine learning and AI. Modern machine learning systems, including large neural networks and generative models, organise high-dimensional representations and display emergent structure. Yet from the present perspective they suffer from a similar deficit:

- Internal representations live in model-specific spaces (parameter spaces, latent spaces, activation spaces) without a canonical semantic interface across models.
- Loss functions and regularisers play the role of task-specific evaluation functionals, but they are not designed as instances of a universal semantic potential.
- There is no explicit notion of a global attractor K that organises meaning across tasks, models, and environments.

Definition 2.2.1 (Core deficit (Kernmangel)). We say that a family of scientific descriptions suffers from a *core deficit* if:

1. there is no common semantic state space S into which meaning-relevant states of heterogeneous systems can be embedded;

2. there is no energy-like functional W on such a space that measures semantic contradiction in a way that is comparable across systems;
3. there is no distinguished global attractor K that constrains and organises long-time semantic behaviour.

This notion is not meant as a criticism of existing theories in their own domain. Rather, it identifies a structural gap in their ability to address meaning, coherence, and normativity in a unified way.

2.3 Addressing the core deficit with semantic gravitation

The constructions of Papers 1–3 can be read as a systematic attempt to supply the missing core.

Common semantic state space. Paper 1 introduced a separable Hilbert space S as the semantic state space. Different physical or informational systems are mapped into S by embeddings

$$E_{\text{sys}} : X_{\text{sys}} \rightarrow S,$$

where X_{sys} denotes the raw state space of the system (for example, neural activations, symbolic configurations, or latent vectors). At this level, the core deficit is addressed by postulating that:

meaning-relevant aspects of heterogeneous systems can be represented in a *single* semantic space S .

Semantic potential as normative energy. On S , Paper 1 defined a functional $W : S \rightarrow \mathbb{R}$ and studied gradient flows driven by ∇W . In Paper 2, W was extended to a free-energy functional on probability measures and interpreted as a form of contradiction energy. From the present viewpoint, this amounts to introducing a *normative* energy landscape on the semantic core:

- low values of $W(s)$ correspond to semantically coherent configurations,
- high values of $W(s)$ correspond to configurations with internal tensions or contradictions.

This supplements purely syntactic or statistical descriptions with a scalar quantity that encodes semantic alignment relative to the shared space S .

Towards a global attractor K . In Paper 3, fields $s : M \rightarrow S$ were studied on spacetime, with W entering as a potential term in the Lagrangian. The analysis of small fluctuations around minima of W already hints at a distinguished set of low-contradiction states. Paper VI makes this explicit by introducing a *global attractor* $K \subset S$:

- K is intended as a long-time attractor for the dynamics driven by W (deterministic and stochastic),
- K encodes a notion of universal semantic coherence, subject to further structural constraints,
- local order bubbles and agent-specific attractors are to be understood as structures embedded in the neighbourhood of K or, in misaligned cases, at a distance from it.

Speculative remark. At a programme level, one may view (S, W, K) as the backbone of a future “semantic core theory” of complex systems: any system that participates in meaning-bearing processes must admit an embedding into S and must be subject, at least approximately, to the semantic gravitation generated by W and organised by K . This claim goes beyond the results of Papers 1–3; it will be formalised in the axiomatic chapter of the present paper and should be regarded as a hypothesis rather than a theorem.

2.4 From qualitative diagnosis to quantitative structure

The original formulation of the core deficit was qualitative: current theories lack a place where meaning “lives” and a mechanism that explains tendencies towards coherence, resolution of contradiction, and directed emergence. Within the semantic gravitation framework, this diagnosis can be sharpened as follows.

1. The *location* of meaning is the semantic state space S ; meaning-bearing configurations are vectors $s \in S$ or fields $s : M \rightarrow S$.
2. The *measure* of semantic contradiction is the potential $W(s)$ and its generalisations to distributions.
3. The *directionality* of semantic evolution is given by gradient-like dynamics $\dot{s}_t = -\nabla W(s_t)$ and their stochastic extensions.
4. The *long-time organisation* of semantic behaviour is encoded in the global attractor K and its neighbourhood, to be developed in later chapters.

In this light, the core deficit is not merely the absence of a semantic vocabulary but the absence of a specific geometric and energetic structure. Semantic gravitation proposes to supply this structure by treating (S, W, K) as fundamental objects, with the usual physical and informational descriptions arising as special or projected views.

Summary and outlook. In this chapter we have revisited the *core deficit* of current scientific descriptions: the lack of a common semantic state space, a normative energy-like quantity for semantic coherence, and a global attractor that organises long-time semantic behaviour. We introduced an informal notion of a semantic core and argued that the constructions of Papers 1–3—a Hilbert space S , a potential W , and gradient flows on S and on spacetime—can be interpreted as first steps towards filling this gap. The existence of a distinguished global attractor K remains, at this point, a structural hypothesis. The next chapter turns to *semantic gravitation as an energy form*, summarising the role of W , its gradient dynamics, and the associated free energy, and preparing the explicit introduction of K and the refined ontological structures K – P – R and C/B/A in the subsequent chapters.

Chapter 3

Semantic gravitation as an energy form

In the previous chapter we reformulated the *core deficit* of current science as the absence of a shared semantic state space, a normative potential, and a global attractor that organises long-time semantic behaviour. In this chapter we focus on the energy-like character of semantic gravitation. We review and refine the role of the semantic potential W as a form of contradiction energy, link it to free-energy functionals on distributions, and position it relative to familiar physical energy forms. This prepares the introduction of the global attractor K and the extended ontological structure in the subsequent chapters.

3.1 Sigma-model structure of the semantic field

Before analysing semantic gravitation as an energy form, it is useful to note that the semantic field framework naturally fits into a well-known class of mathematical field theories.

Let (M, g) be a spacetime manifold and S a separable Hilbert space interpreted as semantic state space. A semantic field

$$s : M \rightarrow S$$

assigns to each spacetime point a semantic configuration.

This structure is mathematically identical to what is known in theoretical physics as a *nonlinear sigma model*: a field theory in which the fundamental degrees of freedom are maps from spacetime into a target space.

Remark 3.1.1. In a standard sigma model, fields take values in a manifold or Hilbert space that acts as the internal configuration space of the theory. The dynamics is governed by functionals depending on the field s and its spacetime derivatives $\partial_\mu s$. The semantic field framework therefore shares the same structural form:

$$s : M \rightarrow S.$$

The difference lies in the interpretation of the target space S , which here represents semantic configurations rather than purely physical internal degrees of freedom.

This observation places semantic gravitation within a broader class of geometric field theories. Many established physical models, including certain formulations of quantum field theory and string theory, possess the same structural architecture. The semantic

field s can therefore be interpreted as a field whose internal configuration space encodes semantic states rather than physical charges or particle species.

Within this perspective, the semantic potential

$$W : S \rightarrow \mathbb{R}$$

plays the role of a target-space potential that shapes the dynamics of the field. Gradient flows generated by W correspond to relaxation processes in the semantic configuration space.

Recognising this sigma-model structure clarifies why the semantic field formalism naturally combines geometric, dynamical, and variational elements. It also prepares the later introduction of the induced tensor

$$h_{\mu\nu} = \langle \partial_\mu s, \partial_\nu s \rangle,$$

which measures how variations of the semantic configuration propagate across spacetime.

3.2 Contradiction energy and semantic gradient dynamics

We recall the basic setting of Paper 1. Let S be a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. A semantic potential is a functional

$$W : S \rightarrow \mathbb{R}$$

of sufficient regularity on a dense domain $\mathcal{D} \subset S$. We interpret $W(s)$ as the *contradiction energy* of a semantic configuration $s \in S$.

Definition 3.2.1 (Semantic gradient flow). Let $W : S \rightarrow \mathbb{R}$ be Fréchet differentiable on $\mathcal{D} \subset S$ with gradient $\nabla W : \mathcal{D} \rightarrow S$. A *semantic gradient flow* is a curve $(s_t)_{t \geq 0}$ in S satisfying

$$\dot{s}_t = -\nabla W(s_t) \tag{3.2.1}$$

for those t with $s_t \in \mathcal{D}$.

Under standard assumptions (Lipschitz continuity of ∇W , coercivity of W on $\mathcal{D}$) this flow is globally well-posed, and W acts as a Lyapunov functional.

Proposition 3.2.2 (Energy dissipation along semantic gradient flows). *Let W be C^1 on $\mathcal{D}$ and let $(s_t)_{t \geq 0}$ be a solution of (3.2.1). Then*

$$\frac{d}{dt} W(s_t) = -\|\nabla W(s_t)\|^2 \leq 0$$

for all t with $s_t \in \mathcal{D}$.

Proof. By the chain rule,

$$\frac{d}{dt} W(s_t) = \langle \nabla W(s_t), \dot{s}_t \rangle = \langle \nabla W(s_t), -\nabla W(s_t) \rangle = -\|\nabla W(s_t)\|^2.$$

□

3.2.1 Constraint-field interpretation of contradiction energy

The contradiction-energy functional W can also be interpreted as an aggregated field of structural constraints acting on the semantic configuration space.

In many semantic or relational systems, configurations are not arbitrary but must satisfy a collection of compatibility conditions. These conditions may arise from logical relations, causal structures, contextual dependencies, or physical realisation constraints. Each such condition can be represented by a constraint contribution whose violation increases the contradiction energy.

A useful conceptual decomposition therefore writes

$$W(s) = \sum_{i \in I} C_i(s) + \sum_{(i,j) \in E} C_{ij}(s),$$

where

- $C_i(s)$ represents local constraint contributions acting on individual semantic components,
- $C_{ij}(s)$ represents relational constraints coupling pairs (or larger sets) of components.

Within this interpretation the gradient

$$\nabla W(s)$$

represents the effective *constraint field* generated by the superposition of these compatibility requirements. The semantic gradient flow

$$\dot{s}_t = -\nabla W(s_t)$$

can therefore be interpreted as the collective relaxation dynamics induced by the tension between simultaneously active constraints.

Remark 3.2.3. This interpretation clarifies why the contradiction energy behaves like a genuine energy landscape. The system experiences effective “forces” generated by incompatible requirements. The gradient flow then corresponds to a process of resolving these tensions by moving toward configurations that better satisfy the overall constraint structure.

3.3 Gradient-flow structure and Lyapunov stability

The dissipation identity established in Proposition 3.2.2 places semantic gravitation within a broad class of dissipative dynamical systems.

Remark 3.3.1 (Gradient-flow structure). The evolution equation

$$\dot{s}_t = -\nabla W(s_t)$$

is a gradient flow on the Hilbert space S generated by the potential W . Such equations appear in many areas of mathematical physics, including thermodynamic relaxation, dissipative field theories, and renormalization-group flows.

In all these cases, the dynamics can be interpreted as a descent process in an energy landscape. The system evolves along trajectories that reduce the value of the governing potential while remaining constrained by the geometry of the underlying state space.

This structural observation clarifies the role of semantic gravitation. The potential W acts as a global ordering functional on the semantic state space: configurations with high contradiction energy are dynamically unstable, while configurations with lower values of W are preferred.

Remark 3.3.2 (Lyapunov functional). The contradiction-energy functional W plays the role of a Lyapunov functional for the semantic dynamics. Along deterministic trajectories one has

$$\frac{d}{dt}W(s_t) \leq 0,$$

so that W provides a monotonic measure of dynamical progress toward lower contradiction states.

In particular, if the flow remains in a bounded region of S and if W satisfies suitable compactness conditions, trajectories approach invariant sets on which ∇W vanishes or becomes dynamically balanced by noise or external driving.

3.3.1 Robustness and fragility measures

The energy landscape defined by W also provides a natural way to quantify the robustness or fragility of semantic configurations.

Let K_i denote a local attractor of the semantic gradient flow and $\mathcal{B}(K_i)$ its basin of attraction. For a configuration $s \in S$ one may define a stability radius

$$R(s) := \sup\{\delta > 0 \mid B_\delta(s) \subseteq \mathcal{B}(K_i)\},$$

where $B_\delta(s)$ denotes the ball of radius δ around s in S .

- Large values of $R(s)$ indicate that the configuration lies deep inside the basin of an attractor and is therefore dynamically robust.
- Small values of $R(s)$ indicate proximity to basin boundaries or saddle structures, implying fragility under perturbations.

A complementary fragility measure can be introduced as

$$F(s) := \frac{1}{1 + R(s)}.$$

Another useful concept is the *reconfiguration cost* between two configurations s_1 and s_2 , defined as the minimal integrated contradiction energy along admissible paths connecting them:

$$\mathcal{C}(s_1 \rightarrow s_2) = \inf_{\gamma} \int_0^1 W(\gamma(t)) dt,$$

where γ ranges over continuous paths in S with $\gamma(0) = s_1$ and $\gamma(1) = s_2$.

Remark 3.3.3. These quantities provide a geometric language for describing resilience and transition difficulty in semantic systems. Deep attractor basins correspond to robust regimes, while shallow basins and low barriers correspond to fragile regimes where small perturbations may trigger large-scale reconfiguration.

3.4 Semantic free energy on probability measures

While W assigns a contradiction energy to single configurations $s \in S$, real systems are often described by probability measures μ on S , reflecting uncertainty, multiplicity of agents, or unresolved internal degrees of freedom. Paper 2 introduced a semantic free-energy functional on the space $\mathcal{P}(S)$ of probability measures on S .

Definition 3.4.1 (Semantic free energy). Let $\mathcal{P}(S)$ denote the set of Borel probability measures on S . For $\mu \in \mathcal{P}(S)$, the *semantic free energy* is

$$\mathcal{F}[\mu] = \int_S W(s) d\mu(s) + \beta^{-1} \mathcal{H}(\mu \mid \mu_0), \quad (3.4.1)$$

where $\beta > 0$ is an inverse temperature parameter, μ_0 is a reference measure, and $\mathcal{H}(\mu \mid \mu_0)$ denotes the relative entropy of μ with respect to μ_0 .

Remark 3.4.2. In the static case, minimisers of $\mathcal{F}$ describe equilibrium distributions over S that trade off low contradiction energy and high entropy. Dynamically, suitable gradient flows of $\mathcal{F}$ on the Wasserstein space of probability measures correspond to diffusion-type equations on S with drift given by $-\nabla W$, as discussed in Paper 2.

3.5 Decomposition of energy forms

Semantic gravitation is not intended to replace physical energy. Rather, it provides an additional layer that interacts with physical energy in structured ways. At a high level, one may distinguish three contributions:

1. physical energy E_{phys} (e.g. kinetic, potential, electromagnetic),
2. information-theoretic quantities (e.g. entropy, mutual information),
3. semantic contradiction energy W and its free-energy extension $\mathcal{F}$.

In many systems of interest, these contributions are coupled but distinct. To formalise this, we consider composite state spaces of the form

$$X \times S,$$

where X denotes a physical or syntactic state space and S the semantic state space. A state of the combined system is $(x, s) \in X \times S$, and one can define an extended energy functional

$$\mathcal{E}(x, s) = E_{\text{phys}}(x) + \lambda W(s) + (\text{interaction terms}), \quad (3.5.1)$$

with a coupling parameter $\lambda \geq 0$.

Remark 3.5.1. The interaction terms in (3.5.1) encode how physical configurations support or constrain semantic configurations and vice versa. For example, physical instantiation costs may penalise certain regions of S , while semantic coherence requirements may effectively constrain the physically accessible subset of X . The explicit form of these terms is system-dependent and was not fixed in Papers 1–4; it will reappear in our discussion of technical systems and distortion factors.

3.6 Dissipation, production, and semantic capacity

The interpretation of W as an energy form invites analogues of familiar concepts such as dissipation, production, and capacity.

Definition 3.6.1 (Semantic dissipation rate). Let $(s_t)_{t \geq 0}$ solve a (possibly stochastic) evolution equation on S with generator $\mathcal{L}$. The *semantic dissipation rate* at time t is defined as

$$D(t) := -\frac{d}{dt} \mathbb{E}[W(s_t)],$$

provided the derivative exists.

For the deterministic gradient flow (3.2.1) one obtains

$$D(t) = \mathbb{E}[\|\nabla W(s_t)\|^2],$$

consistent with Proposition 3.2.2. In the presence of noise or external drives, $D(t)$ captures the net rate at which contradiction energy is dissipated, minus any injection through driving terms.

Definition 3.6.2 (Semantic capacity, informal). Given a class of systems $\mathcal{C}$ with embeddings into S , the *semantic capacity* of a subsystem (for example, an agent, a technical artefact, or a communication channel) is the supremum of the semantic free-energy difference $\Delta \mathcal{F}$ that the subsystem can reliably generate or dissipate per unit resource (time, energy, or physical entropy production), under appropriate constraints.

This notion is intentionally left informal here; its precise formalisation depends on the resource constraints under consideration. The point is that semantic gravitation introduces a new quantity that can be traded against physical resources: sustaining low W in a complex environment typically requires expenditure of physical energy and control, just as maintaining low thermodynamic entropy does.

Remark 3.6.3 (Speculative). In complex adaptive systems, one might conjecture that evolutionary and learning processes tend to increase semantic capacity over time, subject to physical and informational constraints. That is, systems that can more efficiently reduce contradiction energy in their semantic state space, for given physical resources, are favoured. This hypothesis lies outside the scope of rigorous results in Papers 1–4 but motivates the emphasis on semantic energy forms as drivers of emergent organisation.

3.7 Semantic gravitation and the global attractor

In Papers 1–3, the potential W was used locally, for constructing gradient flows and local field equations. In Paper VI we upgrade W to a global organising structure through the introduction of a *global attractor* $K \subset S$ and a refined ontology around it. Conceptually, K plays the role of a low-contradiction backbone:

- states in or near K are semantically coherent across large classes of systems,
- semantic gravitation tends to pull trajectories towards K ,
- local minima of W that are far from K correspond to misaligned or distorted attractors.

From the energy viewpoint, this suggests decompositions of the form

$$W(s) = W_{\text{univ}}(s) + W_{\text{local}}(s),$$

where W_{univ} encodes the universal semantic gravitation towards K , while W_{local} encodes system-specific or context-specific contributions. Similarly, one may distinguish between semantic free energy relative to the global attractor and relative to local attractors. The LOVE Lagrangian will make these distinctions explicit by introducing additional structure (C/B/A axes, K–P–R decomposition) into the potential and kinetic terms.

Summary and outlook. In this chapter we have revisited semantic gravitation explicitly as an *energy form*. We reviewed the role of the potential W as contradiction energy on the semantic state space S , recalled its gradient flows and dissipation properties, and extended the construction to a semantic free-energy functional on probability measures. We positioned semantic energy alongside physical energy and information-theoretic quantities, emphasising that it provides a distinct but coupled contribution in composite systems. Finally, we sketched how W may be decomposed into universal and local parts, anticipating the introduction of a global attractor K and a refined ontological structure. The next chapter will develop semantic gravitation on the semantic Hilbert space S in more technical detail and introduce conditions under which a global attractor $K \subset S$ exists and can serve as the backbone of the extended ontology.

Chapter 4

Semantic gravitation on the Hilbert space S

In the previous chapter we discussed semantic gravitation as a new energy form, defined by a potential $W : S \rightarrow \mathbb{R}$ on a semantic Hilbert space S and its associated gradient dynamics. In this chapter we develop the dynamical systems perspective on S more systematically. We define the semantic gradient system as a (semi)flow on S , introduce global attractors in the standard sense of infinite-dimensional dynamical systems, and formulate conditions under which a global attractor $K \subset S$ exists. This K will later be endowed with additional ontological structure and interpreted as a semantic core.

4.1 Semantic gradient systems on S

We recall the basic setting. Let S be a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\| \cdot \|$. Let

$$W : S \rightarrow \mathbb{R}$$

be a functional of class C^1 on a dense domain $\mathcal{D} \subset S$.

Assumption 4.1.1 (Regularity and coercivity). We assume:

1. W is bounded from below and C^1 on $\mathcal{D}$, with gradient $\nabla W : \mathcal{D} \rightarrow S$ locally Lipschitz;
2. W is coercive in the sense that $W(s) \rightarrow +\infty$ as $\|s\| \rightarrow \infty$ along $\mathcal{D}$;
3. the negative gradient flow

$$\dot{s}_t = -\nabla W(s_t) \tag{4.1.1}$$

with initial condition $s_0 \in \mathcal{D}$ generates a global semiflow $(\Phi_t)_{t \geq 0}$ on S , i.e. for each $t \geq 0$ there is a (possibly non-invertible) continuous map

$$\Phi_t : S \rightarrow S$$

such that $\Phi_0 = \text{id}_S$, $\Phi_{t+s} = \Phi_t \circ \Phi_s$ for $t, s \geq 0$, and $t \mapsto \Phi_t(s_0)$ is the unique solution of (4.1.1) starting from s_0 whenever $s_0 \in \mathcal{D}$.

Under Assumption 4.1.1, the triple $(S, (\Phi_t)_{t \geq 0}, W)$ constitutes what we call a *semantic gradient system*.

Definition 4.1.2 (Semantic gradient system). A *semantic gradient system* is a dynamical system $(S, (\Phi_t)_{t \geq 0})$ generated by (4.1.1) under Assumption 4.1.1, together with the potential W that acts as a Lyapunov functional.

From the theory of gradient flows in Hilbert spaces it follows that W decreases along trajectories, as already recalled in Proposition 3.2.2. We collect some facts that will be used later.

Lemma 4.1.3 (Lyapunov property). *Under Assumption 4.1.1, the functional W is a strict Lyapunov functional for $(\Phi_t)_{t \geq 0}$ in the sense that for any $s_0 \in \mathcal{D}$,*

$$t \mapsto W(\Phi_t(s_0))$$

is non-increasing, and if $W(\Phi_t(s_0))$ is constant on an interval, then $\Phi_t(s_0)$ is an equilibrium of the flow for all t in that interval.

Proof. The derivative of $W(\Phi_t(s_0))$ is given by $-\|\nabla W(\Phi_t(s_0))\|^2$ as in Proposition 3.2.2. If this derivative vanishes on an interval, then $\nabla W(\Phi_t(s_0)) = 0$ on that interval, hence $\dot{s}_t = 0$ there. $\square$

4.2 Invariant sets, ω -limits, and global attractors

We now recall standard concepts from the theory of dissipative dynamical systems on infinite-dimensional spaces and specialise them to semantic gradient systems.

Definition 4.2.1 (Invariant set). A set $A \subset S$ is called *invariant* under $(\Phi_t)_{t \geq 0}$ if

$$\Phi_t(A) = A \quad \text{for all } t \geq 0.$$

Definition 4.2.2 (ω -limit set). For a point $s_0 \in S$, the ω -limit set $\omega(s_0)$ is defined as

$$\omega(s_0) := \{s \in S : \exists t_n \rightarrow +\infty \text{ with } \Phi_{t_n}(s_0) \rightarrow s\},$$

whenever the set is non-empty.

Definition 4.2.3 (Global attractor). A set $K \subset S$ is called a *global attractor* for $(\Phi_t)_{t \geq 0}$ if:

1. K is compact in S ;
2. K is invariant: $\Phi_t(K) = K$ for all $t \geq 0$;
3. K attracts all bounded subsets of S , i.e. for every bounded set $B \subset S$ and every neighbourhood U of K in S there exists $T \geq 0$ such that

$$\Phi_t(B) \subset U \quad \text{for all } t \geq T.$$

If a global attractor exists, it is unique in the sense that two sets satisfying Definition 4.2.3 must coincide. In the context of semantic gravitation, K is the candidate mathematical object underlying the ontological “core” introduced informally in the previous chapters.

4.3 Renormalization-group interpretation of the attractor

The existence of a global attractor K admits a natural interpretation in the language of renormalization-group theory. Although the present framework is not formulated as a renormalization procedure in the conventional quantum-field-theoretic sense, the dynamical role of K is structurally analogous to that of a fixed point governing large-scale behaviour.

Remark 4.3.1 (Attractor as semantic fixed-point structure). The semantic gradient flow

$$\dot{s}_t = -\nabla W(s_t)$$

induces a scale-selective evolution on the semantic state space S : configurations are driven away from high-contradiction regions and toward dynamically persistent low-contradiction regimes. In this sense, the global attractor K may be viewed as the fixed-point set of the semantic dynamics. It encodes the asymptotically stable structures that survive long-time evolution under contradiction-energy descent.

This analogy becomes sharper when one considers perturbations around K . Near an attractor point $s \in K$, the local behaviour of the flow is controlled by the Hessian of W . The corresponding spectral directions may be interpreted in close analogy with renormalization-group mode classification.

Definition 4.3.2 (Semantic mode classification near K). Let $s \in K$ and suppose the semantic Hessian $H(s)$ exists as a self-adjoint operator on S . Then the local perturbation modes around s may be classified as follows:

1. *relevant modes*: directions along which perturbations grow or persist under the effective semantic flow;
2. *marginal modes*: directions along which perturbations remain approximately neutral over long times;
3. *irrelevant modes*: directions along which perturbations decay rapidly and the system returns toward K .

In this language, the attractor K functions as a semantic fixed-point structure, while the spectrum of the linearised dynamics near K determines which deviations are semantically amplified, which persist, and which are suppressed. This viewpoint will be useful later when we discuss coherence axes, local distortions, and the emergence of different long-time semantic regimes.

Remark 4.3.3. The renormalization-group analogy is meant structurally, not identically. We do not claim that the semantic flow is literally a Wilsonian RG flow. The point is that both frameworks distinguish between dynamically relevant, marginal, and irrelevant directions around a stable organising structure, and both use this distinction to explain large-scale universality.

4.4 Existence of a global attractor generated by W

We now state conditions under which a semantic gradient system admits a global attractor K . The following result is a specialisation of standard theorems for gradient-like semiflows in Hilbert spaces; we quote it in a form adapted to our setting.

Assumption 4.4.1 (Asymptotic compactness). We assume that $(\Phi_t)_{t \geq 0}$ is *asymptotically compact*: for every bounded sequence $(s_n)_{n \in \mathbb{N}} \subset S$ and sequence $(t_n)_{n \in \mathbb{N}}$ with $t_n \rightarrow +\infty$, the sequence $(\Phi_{t_n}(s_n))_{n \in \mathbb{N}}$ has a convergent subsequence in S .

Theorem 4.4.2 (Existence of a global attractor). *Suppose Assumptions 4.1.1 and 4.4.1 hold. Then the semantic gradient system $(S, (\Phi_t)_{t \geq 0})$ admits a unique global attractor $K \subset S$ in the sense of Definition 4.2.3. Moreover:*

1. K coincides with the union of ω -limit sets of all bounded sets:

$$K = \bigcup_{B \subset S \text{ bounded}} \omega(B),$$

where $\omega(B) := \bigcup_{s_0 \in B} \omega(s_0)$;

2. K is contained in the set of points where the gradient vanishes:

$$K \subset \{s \in S : \nabla W(s) = 0\};$$

3. W is constant on each connected component of K .

Proof sketch. The proof follows standard arguments in the theory of dissipative dynamical systems. Coercivity and boundedness from below of W imply the existence of a bounded absorbing set in S , since trajectories cannot escape to infinity while W decreases. Asymptotic compactness ensures precompactness of orbits restricted to this absorbing set. One then defines K as the intersection of all closed sets that attract bounded sets. Invariance and attraction follow from construction; compactness follows from asymptotic compactness. The inclusion $K \subset \{\nabla W = 0\}$ and constancy of W on connected components of K use the Lyapunov property of W and LaSalle's invariance principle applied to the level sets of W . For details, see standard references on gradient flows in infinite dimensions. $\square$

Theorem 4.4.2 provides a rigorous mechanism by which the semantic potential W induces a global attractor K on S , under structural assumptions that are natural in many applications (coercivity, asymptotic compactness). In the semantic gravitation programme, we postulate that these conditions are satisfied for the class of systems of interest, or at least for suitable effective descriptions.

Postulate 4.4.3 (Existence of a semantic global attractor). For the class of meaning-bearing systems considered in this paper, the semantic potential W and its induced gradient dynamics on S satisfy Assumptions 4.1.1 and 4.4.1, so that a global attractor $K \subset S$ exists in the sense of Theorem 4.4.2.

Postulate 4.4.3 elevates K from a derived object (given particular choices of S and W) to a structural element of the ontology: we treat K as a distinguished subset of S that organises semantic dynamics across systems and scales.

4.5 Topology of the semantic energy landscape

The existence of a global attractor does not by itself determine the global organisation of the semantic energy landscape. Even for a single contradiction-energy functional W , the state space S may contain multiple local attractors, saddle structures, transition corridors, and basin boundaries.

A useful coarse topological picture is to decompose the semantic state space into attraction basins and transition sets:

$$S = \bigcup_{i \in I} \mathcal{B}(K_i) \cup \Sigma,$$

where:

- K_i denotes a local attractor or attractor component,
- $\mathcal{B}(K_i)$ denotes its basin of attraction,
- Σ denotes the transition structure separating these basins, typically associated with saddle-like or weakly stable regions.

The transition set Σ should be understood as the locus where long-time semantic fate is not yet fixed by local descent alone. Trajectories entering such regions may be redirected toward different attractors depending on perturbations, noise, or local modifications of the effective potential.

Remark 4.5.1. This decomposition is not meant as a full Morse-theoretic statement. Rather, it provides a structural picture of the semantic landscape: attractor basins represent dynamically stabilised semantic regimes, while the separating set Σ encodes semantic transition corridors and barrier regions.

In this language, the global attractor K acts as the most universal organising subset of the landscape, while local attractors K_i may encode restricted or distorted semantic regimes nested within the larger geometry of S .

4.6 Structural diagnostics of semantic landscapes

Once the state space is viewed as a semantic energy landscape, it becomes natural to ask how different landscapes may be compared or characterised. To this end one may introduce a family of structural diagnostics.

A first quantity is an effective *attractor density*

$$\rho_{\text{attr}} := \frac{N_{\text{attr}}}{\text{Vol}_{\text{eff}}(S)},$$

where N_{attr} denotes the number of dynamically relevant local attractors in a chosen effective region of state space, and $\text{Vol}_{\text{eff}}(S)$ denotes an effective volume measure on the same region.

A second quantity is a measure of *ruggedness*, capturing how rapidly the contradiction-energy landscape varies:

$$R_{\text{rug}} := \text{Var}(\|\nabla W(s)\|),$$

under an appropriate sampling procedure on S or on a coarse-grained subset of interest.

A third family of diagnostics concerns *barrier structure*, for example through an average barrier height

$$B_{\text{avg}},$$

defined from the typical contradiction-energy increase required to pass from one basin to another along admissible transition paths.

Remark 4.6.1. These quantities are programme-level diagnostics rather than fully canonical invariants. Their precise definition depends on how one samples or coarse-grains the state space. Nevertheless, they provide a useful descriptive language for distinguishing smooth from rugged semantic landscapes, sparse from dense attractor structure, and easy from difficult regime transitions.

Such diagnostics will later become important when comparing different systems, contexts, or technical architectures that are all embedded into the same semantic state space S but generate markedly different effective landscapes.

4.7 Multi-scale attractor hierarchy

The attractor structure of semantic gravitation need not exist on only one scale. In many applications it is natural to consider a hierarchy of attractors obtained under different levels of coarse-graining or different effective descriptions.

One may therefore introduce a family

$$K_{\ell_{\text{cg}}},$$

indexed by a coarse-graining scale ℓ_{cg} , where each $K_{\ell_{\text{cg}}}$ represents the attractor structure visible at that scale. Equivalently, one may think in terms of a nested hierarchy

$$K^{(0)} \subseteq K^{(1)} \subseteq K^{(2)} \subseteq S,$$

where:

- $K^{(0)}$ denotes fine-grained local attractor structure,
- $K^{(1)}$ denotes mesoscopic attractor clusters,
- $K^{(2)}$ denotes a more global or universal attractor organisation.

Remark 4.7.1. This hierarchy is closely related to the renormalization-group interpretation introduced earlier. Fine-grained attractors may merge into common mesoscopic basins under coarse-graining, while broad classes of mesoscopic basins may in turn be organised by a more universal attractor structure. In this sense, coarse-graining does not merely simplify the landscape; it may reveal deeper semantic universality.

The multi-scale perspective will become especially relevant in later chapters, where mesoscopic regimes and large-scale spacetime response are discussed. In particular, it prepares the interpretation of certain large-scale effects as arising neither from purely local dynamics nor from fully stabilised matter sectors, but from intermediate attractor structures.

4.8 Transition networks between attractors

If the semantic state space contains multiple local attractors or attractor components, then the set of admissible transitions between them acquires a structure of its own. This may be described by an *attractor transition network*.

Definition 4.8.1 (Attractor transition network). Let $\{K_i\}_{i \in I}$ be a family of local attractors or attractor components in S . The *attractor transition network* is the graph

$$G = (V, E),$$

where:

- the vertex set is $V = \{K_i\}_{i \in I}$,
- an edge $(K_i, K_j) \in E$ is present if there exists an admissible transition corridor or dynamically relevant path connecting the basin of K_i to the basin of K_j .

One may also equip the edges with weights

$$w_{ij},$$

representing, for example, transition cost, barrier height, or expected transition time under stochastic forcing.

Remark 4.8.2. The transition network turns the semantic landscape from a purely static object into a processual one. It encodes not only where stable semantic regimes are located, but also how regime changes may occur, which transitions are easy or difficult, and which semantic sectors are effectively isolated from one another.

This network perspective is particularly useful for analysing path dependence, metastability, and order bubbles, since these are naturally described not just by the existence of attractors but by the structure of the routes leading into and out of them.

4.9 Fine structure of W near the global attractor

The global attractor K defined above is a potentially large and complicated subset of S . For the ontological layer developed in this paper, we need more detailed information on the fine structure of W in a neighbourhood of K . We briefly outline the relevant aspects.

Definition 4.9.1 (Semantic Hessian). Let $s \in \mathcal{D}$ be such that W is twice Fréchet differentiable at s . The *semantic Hessian* at s is the bounded self-adjoint operator

$$H(s) : S \rightarrow S$$

defined by

$$\langle H(s)u, v \rangle = \frac{\partial^2}{\partial \epsilon \partial \delta} W(s + \epsilon u + \delta v) \Big|_{\epsilon=\delta=0},$$

for $u, v \in S$.

For $s \in K$ one can consider the spectral decomposition of $H(s)$, when it exists. Eigenvalues with negative real parts correspond to directions of local attraction (stabilising directions), positive eigenvalues to directions of local repulsion, and zero eigenvalues to neutral directions. We are particularly interested in cases where, for $s \in K$ and in an appropriate sense, the spectrum of $H(s)$ is largely non-negative and the neutral directions can be organised into a small number of distinguished subspaces. This motivates the introduction of coherence axes (C/B/A) in later chapters.

Hypothesis 4.9.2 (Structured Hessian near K). There exists a neighbourhood U of K in S such that for all $s \in U$:

1. the Hessian $H(s)$ exists and is self-adjoint with discrete spectrum accumulating only at $+\infty$;
2. the spectrum of $H(s)$ admits a decomposition into three bands corresponding to directions associated with coherence, binding, and action (C/B/A), in a sense to be made precise in later chapters;

3. the bundles of eigenspaces associated with these bands vary sufficiently regularly with s to define vector bundles over K .

Hypothesis 4.9.2 is not derived from Papers 1–4; it is an organising assumption for the ontological layer of the present paper. It allows us to treat the neighbourhood of K as a geometrically structured region in S , with distinguished directions corresponding to different aspects of semantic organisation.

4.10 Phase structure near the attractor

The spectral structure of the semantic Hessian also suggests a phase-theoretic reading of the dynamics near K . In analogy with Landau theory, one may treat deviations from the attractor as an order parameter and classify different semantic regimes by the way the potential behaves in its neighbourhood.

Definition 4.10.1 (Order parameter relative to the attractor). Let $s \in U$, where U is a neighbourhood of K . We define the local order parameter by

$$\delta s := s - \pi_K(s),$$

where $\pi_K(s)$ denotes a suitable nearest-point projection or effective reference point on K , whenever such a projection is defined.

The object δs measures the deviation of the current semantic state from the attractor backbone. Small values of $\|\delta s\|$ correspond to attractor-aligned configurations, while larger values indicate semantically displaced or distorted states.

Near K , the potential W can then be viewed as a Landau-type functional in the variable δs . To leading order one has the expansion

$$W(s) \approx W(\pi_K(s)) + \frac{1}{2} \langle H(\pi_K(s)) \delta s, \delta s \rangle + \text{higher-order terms}.$$

This allows a phase-like classification of semantic regimes:

- *stable basin regime*: the Hessian is strictly positive in the relevant transverse directions, so perturbations decay and the system relaxes back toward K ;
- *near-critical regime*: one or more eigenvalues of the Hessian approach zero, so fluctuations decay only slowly and long-lived semantic reconfiguration becomes possible;
- *unstable regime*: certain directions become effectively destabilising, and trajectories are redirected toward a different local sector, attractor component, or distorted basin.

This phase interpretation is particularly useful when comparing different sectors of K or studying local attractors K_{loc} . Configurations that lie in distinct basins of attraction may then be interpreted as different semantic phases, while transitions between them correspond to regime shifts driven by the curvature structure of W .

Remark 4.10.2. The Landau analogy is again structural rather than literal. We do not assume that semantic systems obey equilibrium statistical mechanics in the strict sense. The point is that a potential landscape together with an order parameter δs naturally induces a classification of stable, critical, and unstable regimes, and this classification becomes especially meaningful near the global attractor.

4.11 Extended semantic phase structure

Beyond the local classification near the attractor, one may also introduce a more global semantic phase structure for the entire state space. This is useful when the system contains multiple attractor basins, fragmented transition regions, or partially organised mesoscale regimes.

A convenient global classification distinguishes the following regime types:

- *basin-dominated phase*: the dynamics is governed by one or a small number of deep attractor basins, and most trajectories relax quickly into stable sectors;
- *near-critical phase*: several regimes coexist with weak barriers or shallow curvature, so that long-lived fluctuations and reconfiguration remain common;
- *fragmented phase*: the landscape contains many shallow attractors, weakly connected basins, or rugged transition regions, leading to unstable or path-dependent continuation;
- *transition phase*: the system moves between basin-dominated and fragmented organisation, typically under changing context, driving, or distortion.

Remark 4.11.1. This extended phase structure should be understood as a global dynamical classification, not as a strict thermodynamic phase taxonomy. Its purpose is to make explicit that the semantic state space may support qualitatively different large-scale organisations of coherence, fragmentation, and transition, even when the underlying contradiction-energy formalism remains the same.

In this broader language, the local phase structure near K appears as the near-core version of a more general classification of semantic regimes across S . This will become relevant later when comparing globally coherent systems, order bubbles, and distorted or power-asymmetric semantic configurations.

Speculative remark. At a programme level, one may conjecture that strong evolutionary or learning pressures tend to shape W in such a way that Hypothesis 4.9.2 holds: systems that manage to channel their semantic dynamics along a small number of coherent directions (C/B/A) near K may be more robust and capable of sustained semantic alignment. This conjecture provides motivation for the detailed study of C/B/A axes later in the paper, but it is not required for the mathematical development of the LOVE Lagrangian.

4.12 Semantic interpretation of the global attractor K

We briefly summarise the intended semantic interpretation of the global attractor K in the present framework.

- *Asymptotic coherence*: Points $s \in K$ represent asymptotic semantic configurations: if a system's semantic state evolves under the gradient dynamics generated by W , then, after sufficiently long time, its state is attracted into the neighbourhood of K (modulo noise and external forcing).
- *Normative core*: Since W decreases along trajectories and tends to be lower on K than in its complement, the set K collects configurations of relatively low contradiction energy. In this sense, K acts as a normative core: states that lie close to K are semantically more coherent than generic states in S .

- *Multiplicity and sectors:* K need not be connected. Different connected components or sectors of K may correspond to distinct but internally coherent regimes (for example, different “phases” of civilisation-level semantic organisation or different stable epistemic frameworks). The C/B/A structure and the K–P–R decomposition introduced later will be used to classify and interpret such sectors.

In later chapters we will also consider local attractors $K_{\text{loc}} \subset S$ associated with particular agents, cultures, or technical systems. These local attractors will be compared to K via embeddings and projections, and their misalignment or distortion will be quantified in terms of additional contributions to W and its decomposition into universal and local parts.

Summary and outlook. In this chapter we have treated semantic gravitation on the Hilbert space S as a dynamical system driven by a potential $W : S \rightarrow \mathbb{R}$. Under natural assumptions of regularity, coercivity, and asymptotic compactness, the gradient flow generated by W defines a semantic gradient system with a unique global attractor $K \subset S$. We have adopted the existence of such a K as a postulate for the class of meaning-bearing systems of interest and introduced a hypothesis about the structured Hessian of W near K , preparing the definition of coherence axes C/B/A. The renormalization-group interpretation of K clarifies how attractor structure organises dynamically relevant, marginal, and irrelevant semantic modes, while the topological, diagnostic, hierarchical, and network-based extensions developed in this chapter show how the semantic landscape may be partitioned into basins, transition structures, multi-scale attractor families, and regime-change pathways. The phase-structure interpretations show both how deviations from K can be treated as an order parameter distinguishing stable, critical, and unstable semantic regimes, and how broader global classes of semantic organisation may be described across the whole state space. Semantically, K is interpreted as a set of asymptotic, low-contradiction configurations that act as a normative core at the level of the semantic state space. The next chapter will introduce the ontological interpretation of K more explicitly and begin to develop the K–P–R structure that connects the global attractor with projections into local systems and the relations between them.

Chapter 5

Tripartite ontology: Core, Projection, Relation

In the previous chapters we have introduced the semantic state space S , the contradiction-energy potential $W : S \rightarrow \mathbb{R}$, and the associated gradient dynamics, and we have postulated the existence of a global attractor $K \subset S$ for the semantic flow. In this chapter we develop an ontological refinement of this picture in the form of a *tripartite* structure

$$K-P-R$$

that we propose as the minimal semantic ontology of what we will call a *semantic object*. This ontology will later be coupled to the C/B/A structure and to the LOVE Lagrangian.

5.1 Semantic objects and minimal ontology

We start from the intuition that any meaning-bearing “thing”—for example an agent, a text, an instrument, a physical object with interpreted structure—behaves, at the level of semantics, as if there were:

- a *core* configuration that encodes its intrinsic semantic identity,
- *projections* into various contexts or coordinate systems (for example, different agents, media, or modalities),
- *relations* to other semantic objects, which are not reducible to either core or projections alone.

We aim to capture these aspects as structures in the semantic state space S .

Definition 5.1.1 (Semantic universe). A *semantic universe* consists of:

1. a separable Hilbert space S (the semantic state space),
2. a contradiction-energy potential $W : S \rightarrow \mathbb{R}$ satisfying the assumptions of the previous chapter,
3. a global attractor $K \subset S$ of the gradient flow generated by W , as in Theorem 4.4.2.

Throughout this chapter we fix a semantic universe (S, W, K) in this sense.

Definition 5.1.2 (Semantic object, informal). A *semantic object* is an entity (system, process, pattern, artefact) that can be associated with:

1. one or more distinguished core configurations in K (semantic identity),
2. projections of these cores into subsystem- or context-specific state spaces (appearances),
3. relations between its cores and those of other objects (couplings, constraints, narratives).

The rest of this chapter makes this more precise in terms of the tripartite K – P – R structure.

5.2 Core: K as semantic backbone

In the previous chapter we have introduced K as the global attractor of the semantic gradient system on S . We now refine its interpretation.

Definition 5.2.1 (Core component of a semantic object). Let (S, W, K) be a semantic universe. A *core component* of a semantic object $\mathcal{O}$ is a non-empty compact subset $K_{\mathcal{O}} \subset K$. The *core* of $\mathcal{O}$ is the union of its core components,

$$K_{\mathcal{O}}^{\text{tot}} := \bigcup_i K_{\mathcal{O},i} \subset K.$$

The idea is that K may have many connected components or sectors, and a given object need not be associated with a single point in K but possibly with a small region or a finite union of regions. For example, a complex agent may occupy different but semantically coherent long-time configurations, all of which lie in the same sector of K .

Remark 5.2.2. The association $\mathcal{O} \mapsto K_{\mathcal{O}}$ is not assumed to be injective or surjective. Different objects may share parts of their core, and many regions of K may not correspond to any presently realised object. The semantic universe (S, W, K) is intended as a background structure, not as the mere sum of currently instantiated objects.

5.3 Projection: appearances in subsystem coordinates

Physical and informational systems are usually described in terms of many different state spaces: configuration spaces, phase spaces, neural activation spaces, latent spaces, symbol spaces, and so on. In Paper 1 and Paper 5, these were related to S by embeddings

$$E_{\text{sys}} : X_{\text{sys}} \rightarrow S$$

and, in the RKHS setting, by evaluation functionals and kernel maps. Here we take the opposite perspective: given a core in K , we consider how it appears in a subsystem or context.

Definition 5.3.1 (Projection space and projection map). Let $\mathcal{C}$ be a collection of *contexts* (for example, agents, modalities, measurement setups). To each context $c \in \mathcal{C}$ we associate a Hilbert space S_c (the *context space*) and a bounded linear map

$$P_c : S \rightarrow S_c$$

called the *projection map* for context c . For a semantic object $\mathcal{O}$ with core $K_{\mathcal{O}} \subset K$, we define its *projection in context c* as

$$P_c(K_{\mathcal{O}}) \subset S_c.$$

Intuitively, P_c describes how semantic configurations in the global space S are “seen” or instantiated in the specific context c . In the RKHS setting of Paper 5, one may think of S_c as the closure of the span of feature maps corresponding to a restricted domain $X_c \subset X$, and P_c as restriction or conditional expectation.

Remark 5.3.2. Projections are not assumed to be isometries. In general, they can lose information (many-to-one) or rescale certain directions. The norm in S_c reflects the semantic resolution and weighting of context c . The composition $P_c \circ E_{\text{sys}}$ yields the appearance of a physical system in context c via the semantic core S .

Definition 5.3.3 (Projection structure of a semantic object). Given a semantic universe (S, W, K) and a family $(S_c, P_c)_{c \in \mathcal{C}}$, the *projection structure* of an object $\mathcal{O}$ is the family of projected cores

$$\mathcal{P}_{\mathcal{O}} := \{P_c(K_{\mathcal{O}}) \subset S_c : c \in \mathcal{C}\}.$$

5.4 Relation: coupling between cores

Beyond cores and their appearances, semantic objects are characterised by relations: causal, logical, narrative, normative, and so on. We model such relations at the level of K and its projections.

Definition 5.4.1 (Relational space and relational kernel). Let $\mathcal{O}_1, \dots, \mathcal{O}_n$ be semantic objects with cores $K_{\mathcal{O}_i} \subset K$. A *relational space* for this family is a Hilbert space R together with a multilinear map

$$\mathcal{R} : K_{\mathcal{O}_1} \times \dots \times K_{\mathcal{O}_n} \rightarrow R.$$

When $n = 2$, we may also specify a bilinear form

$$\kappa : K \times K \rightarrow \mathbb{R},$$

called a *relational kernel*, which measures the strength or type of relation between two core points $k_1, k_2 \in K$.

Remark 5.4.2. The relational kernel κ is conceptually distinct from the reproducing kernel of S as RKHS, although in some constructions one may choose them to be related. Whereas the RKHS kernel k encodes similarity of semantic configurations in S , the relational kernel κ specifically encodes relational structure between core elements (for example, entailment, causal influence, or compatibility).

Definition 5.4.3 (Relational structure of a semantic object). For a semantic object $\mathcal{O}$ with core $K_{\mathcal{O}}$, its *relational structure* consists of:

1. internal relations $\mathcal{R}_{\mathcal{O}} : K_{\mathcal{O}}^m \rightarrow R$ (for various arities m),
2. external relations to other objects $\mathcal{O}'$ via kernels $\kappa_{\mathcal{O}, \mathcal{O}'} : K_{\mathcal{O}} \times K_{\mathcal{O}'} \rightarrow \mathbb{R}$ or higher-arity maps.

These relational structures will become important when we discuss multi-agent fields, cultural structures, and order bubbles; they will also enter the LOVE Lagrangian via coupling terms between semantic fields associated with different objects.

5.5 The K – P – R ontology

We can now summarise the tripartite ontology as a structured triple associated with each semantic object.

Definition 5.5.1 (K–P–R ontology). Let (S, W, K) be a semantic universe, $(S_c, P_c)_{c \in \mathcal{C}}$ a family of context spaces and projections, and R a relational space. A K – P – R ontology for a semantic object $\mathcal{O}$ consists of:

1. a core $K_{\mathcal{O}} \subset K$;
2. a projection structure $\mathcal{P}_{\mathcal{O}} = \{P_c(K_{\mathcal{O}}) \subset S_c : c \in \mathcal{C}\}$;
3. a relational structure $(\mathcal{R}_{\mathcal{O}}, \kappa_{\mathcal{O}}, \cdot)$ as above.

The triple $(K_{\mathcal{O}}, \mathcal{P}_{\mathcal{O}}, \mathcal{R}_{\mathcal{O}})$ is called the K – P – R representation of $\mathcal{O}$.

Remark 5.5.2. The K – P – R representation is not assumed to be unique. Different levels of granularity, different choices of context family $\mathcal{C}$, and different relational spaces R can yield different representations for the same object. What is essential for the present programme is the *existence* of such representations with certain stability and compatibility properties.

5.6 Representation in the RKHS S

In Paper 5, the semantic state space S was constructed as a reproducing kernel Hilbert space (RKHS) of functions on a base space X with kernel $k : X \times X \rightarrow \mathbb{R}$. In that setting, core, projection, and relation admit more concrete forms.

- *Core:* Elements $k \in K$ are functions $k : X \rightarrow \mathbb{R}$ (or vector-valued functions, depending on the construction). The subset $K_{\mathcal{O}}$ associated with an object can be interpreted as a family of “ideal profiles” on X .
- *Projections:* For a context c associated with a subset $X_c \subset X$ or a σ -algebra $\mathcal{F}_c$ on X , one can define S_c as the closure in S of functions supported on X_c or measurable with respect to $\mathcal{F}_c$, and P_c as the orthogonal projection onto S_c . Then $P_c(k)$ is the restriction or coarse-graining of k to context c .
- *Relations:* Relational kernels κ can be constructed from the RKHS kernel k ; for example,

$$\kappa(k_1, k_2) = \int_{X \times X} \varphi(x, y) k_1(x) k_2(y) d\mu(x) d\mu(y)$$

for some symmetric function φ and measure μ on X . This yields a bilinear form on S whose restriction to K serves as a relational kernel.

These constructions demonstrate that K – P – R can be implemented within the RKHS architecture of Papers 1 and 5. The details are not unique and may vary between applications; what matters for Paper 6 is that all three components can be realised in a mathematically controlled way.

5.7 A hypothesis on universal K – P – R decomposability

We now state explicitly the central hypothesis that elevates the K – P – R ontology from a convenient representation to a proposed universal structure.

Hypothesis 5.7.1 (Universal K – P – R decomposability). Let (S, W, K) be a semantic universe. For every sufficiently coherent meaning-bearing system Σ (for example, an agent, a stable artefact, a persistent pattern in a field) there exists at least one semantic object $\mathcal{O}$ with a K – P – R representation $(K_{\mathcal{O}}, P_{\mathcal{O}}, R_{\mathcal{O}})$ such that:

1. the long-time semantic states of Σ (under appropriate embeddings into S) are attracted to a neighbourhood of $K_{\mathcal{O}}$;
2. the observable appearances of Σ in all relevant contexts c are approximated by the projections $P_c(K_{\mathcal{O}})$;
3. the interactions of Σ with other systems are captured by relational structures involving $K_{\mathcal{O}}$ and the cores of the corresponding objects.

Hypothesis 5.7.1 should be read as a programme-level claim, not as a theorem. It says, in essence, that *any coherent system admits a K – P – R decomposition* within the semantic universe (S, W, K) . Establishing this in concrete domains (for example, for neural agents, languages, or technical infrastructures) is an empirical and modelling task.

Speculative remark. If Hypothesis 5.7.1 holds in a strong sense, one may envisage a semantic “category” in which objects are K – P – R triples and morphisms are structure-preserving maps between them (for example, embeddings that respect cores, projections, and key relational kernels). This would provide a high-level language for comparing semantic objects across domains and scales. The development of such a category is beyond the scope of the present paper but influenced the design of the K – P – R ontology.

5.8 A topological conjecture for the global attractor K

In the preceding sections, K has been treated as a global attractor of the semantic gradient system on S : a compact, invariant set that organises long-time dynamics, admits a K – P – R decomposition into cores, projections, and relations, and carries the distinguished C/B/A structure. In principle, the definition of K is purely dynamical and measure-theoretic; no particular global topology of K has been assumed.

For the ontological reading of the framework it is useful to make one additional, explicitly speculative hypothesis about the large-scale shape of K . Informally, the conjecture is that K (or a suitable quotient of it) admits a nontrivial global topology that behaves, in an appropriate sense, like a Klein bottle: there is no globally consistent distinction between “inside” and “outside”, and generic paths that move “outwards” in one local chart can, through global identifications, return to the neighbourhood of their starting point when seen in another chart.

Speculation 5.8.1 (Topological conjecture for K). There exists a topological space K_{top} and a continuous surjection

$$\pi_{\text{top}} : K \twoheadrightarrow K_{\text{top}}$$

such that:

1. K_{top} is a compact, connected, non-simply connected space whose global topology is Klein-bottle-like: it admits local charts in which an “inner” and an “outer” side of semantic configurations can be distinguished, but these local notions of inside/outside cannot be extended globally without encountering self-identifications.
2. For each semantic object O with core $K_O \subset K$ (in the sense of the K–P–R ontology), the restriction $\pi_{\text{top}}|_{K_O}$ defines a local chart of K_{top} around a region that functions as an “inner space” for O . Inner spaces of bodies, agents, and technical systems are thus interpreted as local coordinate representations of one and the same global attractor topology.
3. Paths in K that correspond, in physical or cognitive coordinates, to “moving outwards into the world” and paths that correspond to “moving inwards into an inner space” are mapped by π_{top} to homotopic loop classes in K_{top} . In particular, trajectories associated with very different probes (looking at distant galaxies, introspecting on a point on one’s hand, inspecting social or technical structures) can converge, at the level of K , to the same neighbourhood in the attractor and to the same region in K_{top} .

On this view, the K–P–R ontology acquires a concrete geometric reading:

- The *core* K is the global, Klein-bottle-like backbone of high-coherence states in S .
- *Projections* P describe how this backbone appears in different local coordinate systems: as bodies and their inner spaces, as cognitive systems, as technical architectures, or as large-scale semantic fields in social and cosmological contexts. Each projection picks out a chart on K_{top} .
- *Relations* R capture how these projections are semantically glued together: they correspond to identifications and couplings between different charts, making the global structure nontrivial and ensuring that “inside” and “outside” are not two disjoint regions but two aspects of the same underlying surface.

Nothing in the formal mathematics of this paper depends on the conjecture above: all results about the existence and structure of K , the K–P–R decomposition, and the C/B/A geometry hold for arbitrary global topology of K . The conjecture is offered as an ontological hypothesis that provides a particularly simple and fruitful geometric picture in which inner spaces, bodies, semantic cosmology, and the collapse of the inside/outside distinction can be seen as different charts and loop classes on a single global attractor manifold.

Summary and outlook. In this chapter we have introduced the tripartite K – P – R ontology: cores K_O as subsets of the global attractor K capturing semantic identity, projections P_c into context spaces S_c capturing appearances, and relational structures $\mathcal{R}, \kappa$ capturing couplings between cores. We showed how this ontology can be implemented in the RKHS framework of Papers 1 and 5 and formulated Hypothesis 5.7.1 on universal K – P – R decomposability of coherent systems. This provides the minimal ontological structure for semantic objects on top of the semantic universe (S, W, K) . In the next chapter we specialise to the global attractor K itself: we clarify its tasks as a semantic backbone, connect it more tightly to the K – P – R structure, and prepare the introduction of the C/B/A coherence axes that describe the fine geometry of W near K .

Chapter 6

The task of the global attractor K

In the preceding chapters we have introduced the semantic state space S , the contradiction-energy potential $W : S \rightarrow \mathbb{R}$, the induced gradient dynamics, and the existence of a global attractor $K \subset S$ under suitable assumptions such as Assumptions 4.1.1 and 4.4.1 and Theorem 4.4.2. We have also placed K at the centre of the tripartite K – P – R ontology, where it serves as the backbone for cores, projections, and relations.

The aim of this chapter is to clarify the specific *task* of the global attractor K in the semantic gravitation framework: the roles it must fulfil for the programme to be coherent, and the constraints this imposes on the structure of W and on K itself. We distinguish three tightly coupled tasks:

1. an *asymptotic task*: to organise long-time behaviour of semantic flows;
2. a *normative task*: to provide a reference for semantic coherence and integrity;
3. a *structural task*: to support the K – P – R ontology and related geometric structures.

6.1 From dynamical attractor to ontological core

We begin by recalling the dynamical definition. Let (S, W) be a semantic gradient system satisfying Assumptions 4.1.1 and 4.4.1. The gradient system

$$\dot{s}_t = -\nabla W(s_t)$$

generates a semiflow $(\Phi_t)_{t \geq 0}$ on S , and under the conditions of Theorem 4.4.2 there exists a unique compact global attractor $K \subset S$ in the standard dynamical-systems sense: K is compact, invariant, and attracts every bounded set.

Dynamically, K collects the long-time limits of trajectories starting from bounded sets. Semantically, we interpret K as the backbone of asymptotic coherence. To upgrade K from a purely dynamical object to an ontological core, we postulate a tighter connection between K and the potential W .

Postulate 6.1.1 (Minimality of W on K). There exists a normalisation of W (addition of a constant, and, if needed, rescaling by a positive factor) such that:

1. $W(s) \geq 0$ for all $s \in S$;
2. $W(s) = 0$ if and only if $s \in K_{\min}$, where $K_{\min} \subset S$ is the set of global minima of W ;
3. the global attractor K of Theorem 4.4.2 coincides with the closure of $K_{\min}$ in S .

Under Postulate 6.1.1, K is not just an attractor in the dynamical sense but also the locus of minimal contradiction energy. In many applications it will be natural to work with $K = K_{\min}$ itself (for example if $K_{\min}$ is already compact).

Beyond minimality, K has a second role: it provides a global orientation for semantic gradients.

6.2 K as directional structure for semantic gradients

Informally, far away from K , gradient directions should have a non-trivial component that “points towards” K . We now formulate one way to make this precise. Let $\text{dist}(\cdot, K)$ denote the distance to K in the norm of S .

Definition 6.2.1 (Asymptotic orientation towards K). We say that the gradient of W is *asymptotically oriented towards K* if there exists a function $\alpha : [0, \infty) \rightarrow [0, \infty)$ with $\alpha(r) > 0$ for all $r > 0$ such that for all $s \in S$,

$$\langle \nabla W(s), s - \Pi_K(s) \rangle \geq \alpha(\text{dist}(s, K)) \text{dist}(s, K), \quad (6.2.1)$$

where $\Pi_K(s)$ denotes a (not necessarily unique) metric projection of s onto K .

Condition (6.2.1) says that the gradient of W has, on average, a component that pulls s towards K , with a strength controlled by α . Together with the Lyapunov property of W , this ensures that trajectories not only decrease W but also move closer to K .

Hypothesis 6.2.2 (Global semantic orientation). In the semantic universe (S, W, K) , the gradient of W is asymptotically oriented towards K in the sense of Definition 6.2.1.

Remark 6.2.3. Hypothesis 6.2.2 is not a generic property of arbitrary gradient systems; it is an additional structural requirement. We adopt it as part of the semantic gravitation programme because it matches the intended interpretation of semantic gravitation as a directional tendency towards a global core, not merely a descent along arbitrary local minima.

Under Postulate 6.1.1 and Hypothesis 6.2.2, the asymptotic task of K is made precise: K is the set where contradiction energy is minimised, and gradient dynamics are globally organised towards this set.

6.3 Universal and local components of the potential

In many applications there will be both universal and system-specific contributions to the semantic potential. It is therefore natural to decompose W into a universal part that defines K and local parts that encode specific attractors and distortions.

Definition 6.3.1 (Universal and local potentials). A *universal-local decomposition* of W is a representation

$$W(s) = W_{\text{univ}}(s) + W_{\text{loc}}(s), \quad (6.3.1)$$

where:

1. $W_{\text{univ}} : S \rightarrow \mathbb{R}$ is a potential satisfying the assumptions of the previous chapters and Postulate 6.1.1, with global attractor K_{univ} ;
2. $W_{\text{loc}} : S \rightarrow \mathbb{R}$ is a (possibly context-dependent) perturbation that is bounded and sufficiently regular on bounded subsets of S .

Postulate 6.3.2 (Global attractor of the universal part). There exists a decomposition (6.3.1) such that K coincides with the global attractor K_{univ} of W_{univ} . In particular, K is defined by the universal component of semantic gravitation, while W_{loc} shapes local attractors and distortions around it.

Under Postulate 6.3.2, the role of K is conceptually clarified:

- K is the attractor of the universal semantic potential W_{univ} , representing global semantic coherence and minimal contradiction energy.
- W_{loc} accounts for agent-specific, culture-specific, or system-specific modifications, which may introduce local minima away from K (order bubbles, misaligned attractors).
- The orientation property (6.2.1) is understood to hold primarily for W_{univ} ; deviations due to W_{loc} quantify the extent of local distortion.

This decomposition underlies later discussions of distortion factors and order bubbles: changes of W_{loc} or of the effective dynamics can trap trajectories away from K , even though W_{univ} continues to orient them towards it.

6.4 K as minimal structure compatible with K – P – R

We now connect the task of K more tightly to the K – P – R ontology. Recall that in the previous chapter, a semantic object $\mathcal{O}$ was assigned a core $K_{\mathcal{O}} \subset K$, projections $P_c(K_{\mathcal{O}})$ into context spaces, and relational structures between cores. For this to be coherent at the level of a semantic universe, K must satisfy minimal compatibility conditions.

Definition 6.4.1 (KPR-compatibility of K). We say that the global attractor K is *KPR-compatible* if the following hold:

1. (*Core closure*) For every semantic object $\mathcal{O}$, its core $K_{\mathcal{O}}$ is a compact subset of K that is invariant under the restriction of the gradient flow generated by W_{univ} to a suitable neighbourhood.
2. (*Projection stability*) For each context c and projection $P_c : S \rightarrow S_c$, the image $P_c(K)$ is compact in S_c , and the induced dynamics on S_c (obtained by projecting the gradient flow of W_{univ}) admits $P_c(K)$ as an attractor in the sense of Definition 4.2.3 adapted to S_c .
3. (*Relational regularity*) The relational kernels $\kappa_{\mathcal{O}, \mathcal{O}'}$ are continuous on $K_{\mathcal{O}} \times K_{\mathcal{O}'}$ and extend continuously to neighbourhoods in $S \times S$.

Postulate 6.4.2 (Minimal KPR structure). The global attractor K of the semantic universe is KPR-compatible, and no proper closed subset $K' \subsetneq K$ has this property for the same family of objects, contexts, and relations. In this sense, K is the minimal structure that supports a coherent K – P – R ontology for the semantic universe.

Postulate 6.4.2 articulates the structural task of K :

- K is not an arbitrary attractor; it is the smallest closed subset of S on which one can consistently realise cores, projections, and relations for the class of semantic objects considered.

- Removing part of K would either break invariance of some core, destroy the attractor property in some projection space, or render some relational kernels ill-defined or discontinuous.

In subsequent chapters, this minimality will be used implicitly when we speak of inner spaces, ego cores, and cultural attractor landscapes: all these structures are assumed to be built from subsets of K (via $K_{\mathcal{O}}$ and further constructions), not from arbitrary regions of S .

6.5 Local attractors and alignment with K

Real systems are often governed by potentials that differ from W or operate on restricted state spaces. To model this, we introduce local attractors and define a notion of alignment with K .

Definition 6.5.1 (Local potential and local attractor). Let Σ be a system with an embedding $E_{\Sigma} : X_{\Sigma} \rightarrow S$ into the semantic state space. A *local potential* for Σ is a functional

$$W_{\Sigma} : S \rightarrow \mathbb{R}$$

of class C^1 on a dense domain such that:

1. the dynamics of Σ in semantic space is approximated by the gradient flow

$$\dot{s}_t = -\nabla W_{\Sigma}(s_t)$$

on $E_{\Sigma}(X_{\Sigma})$;

2. the restriction of this flow to bounded subsets of $E_{\Sigma}(X_{\Sigma})$ admits a global attractor $K_{\Sigma} \subset S$.

We call K_{Σ} a *local attractor* of Σ .

Definition 6.5.2 (Semantic alignment with K). Let (S, W, K) be a semantic universe and Σ a system with local attractor K_{Σ} . The *alignment* of Σ with the global attractor K is quantified by

$$\alpha(\Sigma) := 1 - \frac{d_H(K_{\Sigma}, K)}{1 + d_H(K_{\Sigma}, K)},$$

where d_H denotes the Hausdorff distance in S . Thus $\alpha(\Sigma) \in (0, 1]$, with $\alpha(\Sigma) = 1$ if and only if $d_H(K_{\Sigma}, K) = 0$.

The specific functional form of α is not essential; it is chosen merely to map $d_H \in [0, \infty)$ into $(0, 1]$ monotonically. The point is that small Hausdorff distance between K_{Σ} and K corresponds to high alignment.

Remark 6.5.3. In practice, one rarely has access to K_{Σ} or K explicitly. However, the framework suggests that approximate alignment can be probed by measuring contradiction-energy levels and trajectories under controlled perturbations, or by comparing projection structures across contexts. These operational questions lie beyond the scope of the present formal development but motivate later empirical programmes and the methodological proposals of the companion papers.

In later chapters, misalignment between K_{Σ} and K will be associated with persistent *distortion factors* and *order bubbles*: local regions in S where trajectories are trapped away from K due to modifications of the potential or to external forcing.

6.6 K as normative structure without extra metaphysics

The ontological status of K might be misunderstood as introducing additional metaphysical assumptions. We emphasise that, within the semantic gravitation programme, K is a *mathematical* and *normative* structure, not a new physical substrate.

- Mathematically, K is defined entirely in terms of the semantic potential W_{univ} and the induced gradient dynamics on S , plus the KPR-compatibility conditions above.
- Normatively, K encodes a notion of semantic coherence: configurations close to K have low universal contradiction energy and are stable under semantic gravitation.
- There is no claim that K is a physically localised object; it lives in the abstract Hilbert space S and is instantiated in physical systems only via embeddings and projections.

Remark 6.6.1 (Interpretative neutrality). The semantic gravitation framework is compatible with different philosophical interpretations (for example, realist, structuralist, or instrumentalist views of S and K). This paper refrains from fixing such an interpretation. The role of K here is to provide a precise, shared reference structure for subsequent theory-building, empirical work, and more human-facing expositions.

6.7 Renormalization-group interpretation of the attractor

Beyond its ontological and normative role, the global attractor K also admits a natural interpretation in the language of renormalization-group theory. In many areas of theoretical physics, renormalization-group flows describe how effective descriptions evolve under a change of scale, and fixed points of these flows organise the large-scale behaviour of the system. The semantic gradient dynamics generated by W is not literally a Wilsonian renormalization flow, but it shares a closely related structural feature: long-time behaviour is organised by a distinguished invariant set that acts as a stable backbone for the dynamics.

From this perspective, the global attractor K may be viewed as a semantic fixed-point structure. Trajectories starting from broad regions of the semantic state space are drawn towards neighbourhoods of K , just as renormalization-group trajectories approach fixed points or fixed manifolds in the space of effective theories. The attractor therefore determines which large-scale semantic configurations remain dynamically stable under contradiction-energy descent.

This analogy becomes more precise when one considers perturbations near K . Let $s \in K$ and suppose the Hessian $H(s)$ of W exists. Then the local behaviour of perturbations can be organised into classes analogous to the standard mode classification of renormalization-group theory:

- *relevant directions*: perturbations that grow, persist, or strongly redirect the long-time semantic evolution;
- *marginal directions*: perturbations that remain approximately neutral over long times and may generate slow drift along sectors of K ;
- *irrelevant directions*: perturbations that decay rapidly, returning the system toward the attractor backbone.

This mode classification does not replace the earlier spectral analysis of the Hessian; rather, it reframes it in a language familiar from field theory and critical phenomena. The point is that K is not merely a set of minima or a static repository of coherent states. It also organises which deviations matter dynamically and which are washed out by the semantic flow.

Remark 6.7.1. The renormalization-group analogy is structural rather than literal. We do not assume an explicit coarse-graining operator on S in the technical sense used in statistical field theory. The claim is instead that the semantic gradient dynamics induces a separation between dynamically dominant and dynamically suppressed directions near K , and this separation plays a role strongly analogous to that of relevant and irrelevant couplings near a fixed point.

Seen in this light, K functions not only as the normative core of the semantic universe but also as the large-scale organiser of its dynamically persistent modes.

6.8 Phase structure near the attractor

The geometry of W near K also suggests a phase-theoretic interpretation. In analogy with Landau-type descriptions of phase structure, one may view deviations from the attractor as an order parameter describing the local semantic regime of the system.

Let U be a neighbourhood of K in which a projection Π_K is defined, at least locally or effectively. For $s \in U$, define the deviation from the attractor by

$$\delta s := s - \Pi_K(s).$$

The quantity δs measures how far the semantic configuration is displaced from the attractor backbone. Small values of $\|\delta s\|$ correspond to attractor-aligned states, while larger values indicate semantically displaced, unstable, or locally trapped configurations.

Near K , the potential W may then be expanded schematically as

$$W(s) \approx W(\Pi_K(s)) + \frac{1}{2} \langle \delta s, H(\Pi_K(s)) \delta s \rangle + O(\|\delta s\|^3),$$

where $H(\Pi_K(s))$ denotes the semantic Hessian evaluated at the projected point on the attractor. This expansion motivates a classification of semantic regimes:

- *stable basin regime:* the Hessian is positive in the relevant transverse directions, and perturbations decay rapidly back towards K ;
- *near-critical regime:* one or more eigenvalues of the Hessian are close to zero, so relaxation becomes slow and the system exhibits persistent fluctuations or long-lived reconfiguration;
- *misaligned or locally trapped regime:* local perturbations of the potential, typically through W_{loc} , create nearby basins or order bubbles that compete with the global pull of K .

This phase interpretation is especially useful for the semantic gravitation programme because it connects the abstract geometry of W to empirically or phenomenologically distinguishable patterns of behaviour. A system may be strongly aligned with K and lie deep in a stable basin; it may hover in a near-critical regime where small perturbations induce major semantic shifts; or it may be trapped in a local order bubble whose coherence is internally stable but globally misaligned.

Remark 6.8.1. The Landau analogy is again structural rather than literal. We do not assume that semantic systems obey equilibrium statistical mechanics in the strict sense. The point is that a potential landscape together with an order parameter δs naturally induces a distinction between stable, critical, and unstable semantic regimes, and this distinction becomes particularly clear in a neighbourhood of the global attractor.

The phase structure near K therefore complements its role as fixed-point backbone: K organises not only where trajectories tend to go, but also the qualitative regimes through which they pass on the way.

Summary and outlook. In this chapter we have clarified the task of the global attractor K in the semantic universe (S, W, K) . We strengthened its dynamical definition by postulating that K coincides with the set of minimal values of a universal component W_{univ} of the semantic potential and by introducing an orientation condition ensuring that semantic gradients are globally directed towards K . We decomposed W into universal and local parts, positioning K as the attractor of W_{univ} while allowing system-specific contributions W_{loc} to create local order bubbles and distortions. We then formulated KPR-compatibility as a minimality condition: K is the smallest closed subset of S that supports coherent core, projection, and relation structures for the class of semantic objects under consideration. Finally, we introduced local attractors and a notion of semantic alignment with K , and we emphasised that K is a mathematical and normative structure rather than an additional physical substrate.

The renormalization-group interpretation developed in the final sections shows that K also functions as a fixed-point structure organising dynamically relevant, marginal, and irrelevant semantic modes, while the phase interpretation near K shows how stable, critical, and locally trapped semantic regimes can be understood through deviations from the attractor backbone. The next chapter will build on this foundation by analysing *distortion factors* in semantic dynamics: how deviations from pure gradient flow and from perfect orientation towards K can be formalised, and how they give rise to local order bubbles, power asymmetries, and path-dependent attractors within the semantic universe.

Chapter 7

Distortion factors in semantic dynamics

In the preceding chapters we have described an idealised semantic universe (S, W_{univ}, K) in which the dynamics of semantic configurations is governed by the gradient flow of a universal potential $W_{\text{univ}} : S \rightarrow \mathbb{R}$ with global attractor $K \subset S$. We have also allowed for system-specific contributions W_{loc} and introduced local attractors K_{Σ} for individual systems Σ . In this chapter we move from the idealised picture to more realistic dynamics by introducing *distortion factors*: systematic deviations from pure gradient descent of W_{univ} that can generate local order bubbles, misaligned attractors, and path-dependent behaviour.

7.1 From ideal gradient flow to distorted dynamics

We start from the ideal gradient system associated with W_{univ} .

Definition 7.1.1 (Ideal semantic gradient flow). The *ideal semantic dynamics* on S is the flow generated by

$$\dot{s}_t = -\nabla W_{\text{univ}}(s_t), \quad (7.1.1)$$

where W_{univ} satisfies the regularity and coercivity assumptions of the previous chapters and admits the global attractor K .

Real systems are rarely governed by (7.1.1) alone. There may be additional drift terms, non-conservative effects, constraints, and noise, leading to dynamics of the form

$$ds_t = \left(-\nabla W_{\text{univ}}(s_t) + D(s_t) \right) dt + \sqrt{2\beta^{-1}} \Sigma(s_t) dB_t, \quad (7.1.2)$$

where:

- $D : S \rightarrow S$ is a vector field capturing systematic deviations from the ideal gradient drift;
- $\Sigma(s)$ is a bounded linear operator on S describing anisotropic noise;
- (B_t) is a cylindrical Brownian motion on S , as in Paper 1.

Definition 7.1.2 (Distortion field). The vector field $D : S \rightarrow S$ in (7.1.2) is called a *distortion field*. It summarises all systematic deviations of the drift from pure semantic gravitation generated by W_{univ} .

In many applications D will itself be decomposable into components (for example, those arising from local potentials, metric distortions, or external forcing). The noise term $\Sigma(s_t) dB_t$ can also act as a distortion when it interacts with the potential in a non-trivial way (for example, by preferentially exciting certain directions).

7.2 Decomposition of the distorted drift

To analyse the effect of distortions, it is useful to decompose the drift in (7.1.2) into gradient and non-gradient parts. Let

$$F(s) := -\nabla W_{\text{univ}}(s) + D(s)$$

denote the total drift.

Definition 7.2.1 (Gradient and curl components of the drift). We say that the distortion field D admits a *gradient–curl decomposition* if there exists a functional $W_{\text{loc}} : S \rightarrow \mathbb{R}$ and a vector field $C : S \rightarrow S$ such that

$$D(s) = -\nabla W_{\text{loc}}(s) + C(s), \quad (7.2.1)$$

with the property that

$$\langle C(s), \nabla W_{\text{univ}}(s) + \nabla W_{\text{loc}}(s) \rangle = 0 \quad (7.2.2)$$

for all s in a dense domain. In that case we call $-\nabla W_{\text{loc}}$ the *local gradient distortion* and C the *curl distortion*.

The local potential W_{loc} in (7.2.1) can be identified with the system-specific contributions introduced earlier (Definition 6.3.1). The curl distortion C is orthogonal, in the sense of (7.2.2), to the total gradient of $W_{\text{univ}} + W_{\text{loc}}$; it rotates trajectories on level sets of the full potential without changing the instantaneous value of $W_{\text{univ}} + W_{\text{loc}}$.

Remark 7.2.2. The decomposition (7.2.1) is not unique in general, and the orthogonality condition (7.2.2) may hold only approximately or in an averaged sense. For the purposes of this paper we treat it as a useful idealisation that separates energy-changing and energy-preserving components of the distortion.

7.3 Order bubbles as local distorted attractors

The patents associated with semantic gravitation introduce the notion of *Ordnungsblasen* (order bubbles): locally coherent structures sustained by distortions of semantic gravitation. We now give a formal definition in the present setting.

Definition 7.3.1 (Order bubble). Let (S, W_{univ}, K) be a semantic universe, and let the dynamics be given by (7.1.2) with drift $F(s) = -\nabla W_{\text{univ}}(s) + D(s)$. A non-empty compact set $B \subset S$ is called an *order bubble* if:

1. B is invariant and attracts a non-trivial set of initial conditions under the distorted dynamics (that is, B is a local attractor of (7.1.2));
2. B is disjoint from any neighbourhood of K (in particular, $\inf_{s \in B} d_K(s) \geq \delta$ for some $\delta > 0$);

3. the internal dynamics on B is approximately coherent in the sense that the restriction of W_{loc} to B is nearly minimal and approximately constant, while W_{univ} is significantly larger on B than on K .

Intuitively, an order bubble is a self-sustaining local attractor that is internally coherent (low local contradiction energy) but globally misaligned (high universal contradiction energy). Such bubbles can arise when W_{loc} and C in (7.2.1) combine to trap trajectories in regions of S where W_{univ} remains elevated.

Definition 7.3.2 (Bubble height and stability). For an order bubble B , we define its *universal height* as

$$h_{\text{univ}}(B) := \inf_{s \in B} (W_{\text{univ}}(s) - \inf_{k \in K} W_{\text{univ}}(k)),$$

and its *local depth* as

$$d_{\text{loc}}(B) := \sup_{s \in U_B} (W_{\text{loc}}(s) - \sup_{u \in B} W_{\text{loc}}(u)),$$

where U_B is a neighbourhood of B on which B is locally attracting. We say that B is *metastable* if $d_{\text{loc}}(B)$ is large compared to typical noise amplitudes and to the magnitude of the universal gradient in U_B .

The universal height measures how far above the global core K the bubble sits in terms of universal contradiction energy, while the local depth measures how strongly it is stabilised by local potentials. Metastable order bubbles can persist for long times and may organise the behaviour of systems in ways that are misaligned with K .

7.4 Quantifying distortion

We now introduce quantitative measures of distortion. Let $F_{\text{ideal}}(s) := -\nabla W_{\text{univ}}(s)$ denote the ideal drift and $F(s) := F_{\text{ideal}}(s) + D(s)$ the actual drift.

Definition 7.4.1 (Pointwise distortion measure). For $s \in S$ with $\nabla W_{\text{univ}}(s) \neq 0$, the *pointwise distortion* at s is defined as

$$\delta(s) := \frac{\langle D(s), \nabla W_{\text{univ}}(s) \rangle}{\|\nabla W_{\text{univ}}(s)\|^2}. \quad (7.4.1)$$

The quantity $\delta(s)$ measures, in units of the ideal gradient magnitude, the component of the distortion along the direction of $\nabla W_{\text{univ}}(s)$:

- $\delta(s) < 0$ means that $D(s)$ enhances the descent towards K at s ;
- $\delta(s) > 0$ means that $D(s)$ partially counteracts the descent towards K ;
- $\delta(s) = 0$ means that $D(s)$ is orthogonal to $\nabla W_{\text{univ}}(s)$ (purely tangential distortion).

Definition 7.4.2 (Distortion factor of a system). Let Σ be a system with semantic dynamics (7.1.2). The *distortion factor* of Σ relative to K is defined as

$$\mathfrak{D}(\Sigma) := \sup_{t \geq 0} \mathbb{E}[|\delta(s_t)| \mid s_0 \sim \mu_0],$$

where μ_0 is an admissible initial distribution on S , and the expectation is taken over the noise in (7.1.2).

The distortion factor $\mathfrak{D}(\Sigma)$ aggregates the pointwise distortions encountered along typical trajectories of Σ . Systems with small $\mathfrak{D}(\Sigma)$ behave approximately like ideal gradient systems; systems with large $\mathfrak{D}(\Sigma)$ can sustain strong deviations, including order bubbles and misaligned attractors.

Remark 7.4.3. The definition of $\mathfrak{D}(\Sigma)$ is intentionally flexible: in concrete applications one may restrict the supremum over time to certain intervals, condition on subsets of S , or weight $\delta(s_t)$ by additional importance factors. What matters for the theoretical framework is that there exists at least one such measure that quantifies the systematic deviation from semantic gravitation towards K .

7.5 Types of distortion factors

It is convenient to distinguish different types of distortion, each with a characteristic mathematical form and semantic interpretation.

Definition 7.5.1 (Classification of distortion factors). We distinguish the following idealised types:

1. *Potential distortions:* These arise from non-universal potentials W_{loc} in (7.2.1). They modify the energy landscape directly and can create new local minima or shift the location of existing ones. Order bubbles in the sense of Definition 7.3.1 are typically supported by potential distortions.
2. *Metric distortions:* Here the dynamics is of the form

$$\dot{s}_t = -G(s_t)\nabla W_{\text{univ}}(s_t),$$

where $G(s)$ is a positive-definite linear operator on S , representing a change of metric or preconditioning. This does not change the set of minima of W_{univ} but can strongly affect trajectories and rates of convergence, especially when combined with local potentials and noise.

3. *Forcing distortions:* These are time-dependent or non-autonomous contributions $F_{\text{ext}}(t, s)$ to the drift, so that

$$\dot{s}_t = -\nabla W_{\text{univ}}(s_t) + F_{\text{ext}}(t, s_t).$$

They can drive the system away from K or between different regions of K , and may encode external goals, control inputs, or environmental shocks.

4. *Noise distortions:* Anisotropic or state-dependent noise terms $\Sigma(s_t)dB_t$ can bias exploration of the potential landscape, favouring some directions over others. The resulting effective drift on the distributional level will be treated in the chapter on Fokker–Planck equations.

In practice, real systems exhibit mixtures of these types. The classification is not exhaustive but provides a useful vocabulary for analysing how different mechanisms interact with semantic gravitation.

7.6 Distortion and alignment with the global attractor

We now link distortion factors to the alignment measure $\alpha(\Sigma)$ introduced earlier for local attractors K_Σ .

Proposition 7.6.1 (Small distortion implies near-alignment). *Let Σ be a system with distorted dynamics (7.1.2) and local attractor K_Σ . Suppose:*

1. *the drift admits a gradient–curl decomposition with W_{loc} bounded and C satisfying (7.2.2);*
2. *the distortion factor $\mathfrak{D}(\Sigma)$ is sufficiently small;*
3. *the noise term is sufficiently weak and regular.*

Then, under suitable technical assumptions, the Hausdorff distance $d_H(K_\Sigma, K)$ is small, and hence $\alpha(\Sigma)$ is close to 1.

Proof sketch. The argument is a perturbation analysis of global attractors for gradient-like flows. Small potential distortions and small components of D along ∇W_{univ} ensure that the minima of $W_{\text{univ}} + W_{\text{loc}}$ remain close to those of W_{univ} , while curl distortions and weak noise do not destroy the existence of an attractor but mainly affect transient behaviour. One combines standard results on structural stability of attractors under small perturbations with estimates relating $\mathfrak{D}(\Sigma)$ to the difference between the invariant sets of the ideal and distorted flows. We omit the technical details, which would take us beyond the scope of this chapter. $\square$

Proposition 7.6.1 formalises the intuitive claim that systems with small distortion factors remain close to the universal core K , while large distortions are required to sustain robust misalignment and order bubbles.

Speculative remark. At a programme level, one may conjecture that long-lived meaning-bearing systems—for example, biological agents or large-scale cultures—tend to reduce their distortion factors over evolutionary or learning time, at least in sectors of S that are critical for survival or stability. Conversely, certain technical systems may be capable of sustaining high distortion factors in specific subspaces of S , thereby creating powerful but brittle order bubbles. These conjectures provide motivation for studying distortion factors empirically and for designing alignment protocols that explicitly penalise distortions relative to a candidate universal core K .

Summary and outlook. In this chapter we have introduced *distortion factors* as systematic deviations from the ideal semantic gradient dynamics generated by W_{univ} . We represented distorted dynamics by an SDE with a distortion field D , decomposed D into gradient and curl components, and defined *order bubbles* as locally coherent but globally misaligned attractors. We introduced quantitative measures of distortion, both pointwise and at the level of systems, and distinguished different types of distortions (potential, metric, forcing, noise). Finally, we related small distortion factors to near-alignment with the global attractor K via a perturbative proposition. This prepares the way for the next chapter, in which we revisit the construction of the semantic space S as a reproducing kernel Hilbert space (RKHS) and show how distortion factors and order bubbles can be represented and analysed within that functional-analytic framework.

Chapter 8

The semantic space S as reproducing kernel Hilbert space

In the preceding chapters we have treated the semantic state space S axiomatically as a separable Hilbert space on which the semantic potential $W : S \rightarrow \mathbb{R}$ is defined and the global attractor $K \subset S$ lives. In Paper 5 this space was constructed more concretely as a reproducing kernel Hilbert space (RKHS) of functions on a base space X , equipped with a positive definite kernel $k : X \times X \rightarrow \mathbb{R}$. In this chapter we summarise this RKHS construction and highlight how base space X , kernel k , system embeddings E_{sys} , and semantic observables fit together. This provides the functional-analytic backdrop for the more refined C/B/A geometry introduced in the next chapter.

8.1 Base space, kernel, and construction of S

We begin with the standard definition of a positive definite kernel and its associated RKHS.

Definition 8.1.1 (Positive definite kernel). Let X be a non-empty set. A function $k : X \times X \rightarrow \mathbb{R}$ is called a *positive definite kernel* if:

1. k is symmetric: $k(x, x') = k(x', x)$ for all $x, x' \in X$;
2. for every finite choice of points $x_1, \dots, x_n \in X$ and coefficients $c_1, \dots, c_n \in \mathbb{R}$ we have

$$\sum_{i,j=1}^n c_i c_j k(x_i, x_j) \geq 0.$$

Given such a kernel, there exists a unique (up to isometry) Hilbert space S of real-valued functions on X with the following properties.

Definition 8.1.2 (Reproducing kernel Hilbert space). A Hilbert space S of real-valued functions on X with inner product $\langle \cdot, \cdot \rangle$ is called a *reproducing kernel Hilbert space* with kernel k if:

1. for every $x \in X$, the function $k(x, \cdot)$ belongs to S ;
2. the *reproducing property* holds:

$$f(x) = \langle f, k(x, \cdot) \rangle \quad \text{for all } f \in S, x \in X.$$

The space S can be constructed as the completion of the linear span of $\{k(x, \cdot) : x \in X\}$ with respect to the inner product

$$\left\langle \sum_{i=1}^n c_i k(x_i, \cdot), \sum_{j=1}^m d_j k(x'_j, \cdot) \right\rangle := \sum_{i=1}^n \sum_{j=1}^m c_i d_j k(x_i, x'_j).$$

Definition 8.1.3 (Canonical feature map). The map

$$\Phi : X \rightarrow S, \quad x \mapsto k(x, \cdot)$$

is called the *canonical feature map* associated with the kernel k . For each $x \in X$, the function $\Phi(x)$ encodes the “feature vector” of the base point x in the semantic space S .

By construction, S is separable whenever X is separable with respect to a suitable topology and k has mild regularity properties. In Paper 5, the space X is typically a space of raw system states (for example, neural activations, latent vectors, or structured symbolic configurations), and k is chosen to encode semantic similarity on X .

Remark 8.1.4. The RKHS structure links pointwise evaluation on X to inner products in S : semantic similarity between configurations is expressed as inner products of functions in S , which are ultimately determined by the kernel k . This is the key bridge between concrete system representations and the abstract semantic Hilbert space used in Papers 1–4.

8.2 Embeddings of systems into S

Let Σ be a concrete system with a raw state space X_Σ (for example $\mathbb{R}^d$ for a neural network, a latent space, or a symbolic state space). To embed Σ into the semantic RKHS S , we proceed in two steps.

1. Choose a *representation map*

$$\varphi_\Sigma : X_\Sigma \rightarrow X$$

from raw states to base space points. This map can encode pre-processing, feature extraction, or coarse-graining.

2. Apply the canonical feature map Φ to obtain the semantic embedding

$$E_\Sigma : X_\Sigma \rightarrow S, \quad E_\Sigma(x) := \Phi(\varphi_\Sigma(x)) = k(\varphi_\Sigma(x), \cdot).$$

Definition 8.2.1 (Semantic embedding of a system). Given a system Σ with raw state space X_Σ , a base space X , a kernel k on X , and a representation map $\varphi_\Sigma : X_\Sigma \rightarrow X$, the map

$$E_\Sigma := \Phi \circ \varphi_\Sigma : X_\Sigma \rightarrow S$$

is called a *semantic embedding* of Σ into the RKHS S .

The image $E_\Sigma(X_\Sigma) \subset S$ is a (typically low-dimensional) submanifold or subspace of S on which the semantic potential W and the global attractor K can be evaluated. The semantic state of Σ at time t is represented by a point $s_t := E_\Sigma(x_t) \in S$, where $x_t \in X_\Sigma$ is the raw state.

Remark 8.2.2. Different systems may share the same base space X and kernel k , or they may use different kernels tailored to their domains. In either case, the semantic universe (S, W, K) is defined at the level of the RKHS, and systems are related by the way their embeddings E_Σ sit inside S .

8.3 Semantic observables as linear functionals

Within an RKHS, bounded linear functionals are represented by inner products with elements of S , by the Riesz representation theorem. This allows us to formalise semantic observables.

Definition 8.3.1 (Semantic observable). A *semantic observable* is a bounded linear functional

$$\mathcal{O} : S \rightarrow \mathbb{R}.$$

By the Riesz representation theorem, there exists a unique $o \in S$ such that

$$\mathcal{O}(s) = \langle s, o \rangle \quad \text{for all } s \in S.$$

We call o the *representer* of the observable $\mathcal{O}$.

For a system Σ with embedding E_Σ , the value of $\mathcal{O}$ at raw state $x \in X_\Sigma$ is given by

$$\mathcal{O}(E_\Sigma(x)) = \langle E_\Sigma(x), o \rangle.$$

Using the definition of E_Σ , this can be written in terms of the kernel:

$$\mathcal{O}(E_\Sigma(x)) = \langle k(\varphi_\Sigma(x), \cdot), o \rangle = o(\varphi_\Sigma(x)),$$

where we have used the reproducing property with $f = o$.

Remark 8.3.2. In this way, semantic observables correspond to special semantic configurations $o \in S$ that act as “test functions.” The value of an observable on a system state is obtained by evaluating o at the base point associated with that state. This identification was used in Paper 4 to embed quantum observables into S and to write expectation values as inner products on S .

8.4 Kernel-induced metrics and their relation to W

The kernel k induces a natural metric on X via the feature map:

$$d_k(x, x') := \|\Phi(x) - \Phi(x')\|_S = \sqrt{k(x, x) + k(x', x') - 2k(x, x')}.$$

This metric measures semantic similarity at the level of the RKHS geometry.

Definition 8.4.1 (Kernel-induced semantic distance). For $x, x' \in X$, their *kernel-induced semantic distance* is

$$d_k(x, x') := \|\Phi(x) - \Phi(x')\|_S.$$

For semantic configurations $s, s' \in S$, their distance is simply the norm $\|s - s'\|_S$.

The semantic potential $W : S \rightarrow \mathbb{R}$ is *not* determined by k alone; it adds an independent layer of structure on top of the RKHS geometry. However, in Paper 5 and in the present paper it is convenient to assume a certain compatibility.

Assumption 8.4.2 (Compatibility of W with the RKHS structure). We assume:

1. W is continuous on S with respect to the RKHS norm and twice Fréchet differentiable on a dense domain $\mathcal{D} \subset S$;
2. sublevel sets of W are relatively compact in S (coercivity);

3. in a neighbourhood of the global attractor $K \subset S$, the Hessian $H(s)$ of W can be expressed in terms of kernel-derived operators (for example, integral operators with kernels built from k), so that the local geometry of W near K is compatible with the RKHS geometry.

Under Assumption 8.4.2, one can relate the C/B/A structure and curvature components introduced later to spectral properties of kernel-derived operators. For example, if W has a quadratic expansion near K ,

$$W(s) \approx W_0 + \frac{1}{2} \langle H_0(s - k_0), s - k_0 \rangle,$$

with a positive semi-definite operator H_0 and $k_0 \in K$, then the eigenfunctions of H_0 in S can be expressed in terms of the kernel k , and the C/B/A axes correspond to distinguished eigenspaces of such operators.

Remark 8.4.3. In concrete models, W is often constructed from data or task structure in a kernel-based way; for example, as a regularised empirical risk functional with kernel-based features, or as a functional of probability measures on X defined via kernel mean embeddings. Paper 5 provides several such constructions. For the purposes of Paper 6, we do not commit to a specific form of W ; we only require that it is sufficiently regular and compatible with the RKHS structure to support the global attractor K and the C/B/A decomposition.

8.5 Semantic universe as RKHS-based structure

Collecting the above, we can summarise the RKHS-based description of the semantic universe as follows.

Definition 8.5.1 (RKHS-based semantic universe). An *RKHS-based semantic universe* consists of:

1. a base space X and a positive definite kernel $k : X \times X \rightarrow \mathbb{R}$;
2. the associated RKHS S of functions on X with reproducing kernel k ;
3. a semantic potential $W : S \rightarrow \mathbb{R}$ satisfying Assumption 8.4.2;
4. a global attractor $K \subset S$ for the gradient flow generated by W_{univ} ;
5. for each concrete system Σ , an embedding $E_\Sigma : X_\Sigma \rightarrow S$ as above;
6. a class of semantic observables represented by elements of S .

This structure realises the abstract semantic universe (S, W, K) of earlier chapters as an RKHS of functions on a base space X , making it possible to link semantic gravitation to concrete systems via kernels and feature maps.

Summary and outlook. In this chapter we have summarised the construction of the semantic space S as a reproducing kernel Hilbert space of functions on a base space X with kernel k . We recalled how the canonical feature map Φ embeds X into S , how concrete systems are embedded via representation maps and $E_\Sigma = \Phi \circ \varphi_\Sigma$, and how semantic observables correspond to bounded linear functionals represented by inner products with elements of S . We emphasised that the semantic potential W adds an independent energy-like structure on top of the RKHS geometry, with compatibility conditions near the global

attractor K . This RKHS viewpoint will be crucial in the next chapter, where we introduce the three fundamental coherence axes C, B, A as distinguished families of functionals and vector fields on S , and relate them to the local Hessian structure of W near K within the RKHS framework.

Chapter 9

The three fundamental coherence axes C , B , A

In the previous chapter we realised the semantic state space S as a reproducing kernel Hilbert space and assumed that the contradiction-energy potential $W : S \rightarrow \mathbb{R}$ is sufficiently regular and compatible with the RKHS structure near the global attractor $K \subset S$. We also postulated a structured Hessian of W in a neighbourhood of K (Hypothesis 4.9.2). In this chapter we refine this structure into three fundamental coherence axes, denoted C , B , and A . Formally, these are bundles of subspaces and associated functionals that decompose local variations and gradients of W near K . Interpretatively (and explicitly marked as such), they will later be associated with coherence, binding, and agency/action.

9.1 Motivation and local coordinates near K

We work in a fixed semantic universe (S, W, K) satisfying the assumptions of the preceding chapters, including Hypothesis 4.9.2 on the Hessian of W near K .

Let $U \subset S$ be a neighbourhood of K on which:

1. the Hessian $H(s)$ exists for all $s \in U$,
2. $H(s)$ is self-adjoint with discrete spectrum,
3. the spectrum of $H(s)$ admits a decomposition into three bands, informally associated with C , B , and A .

We first introduce a local coordinate for deviations from K .

Definition 9.1.1 (Nearest core point and deviation vector). For $s \in U$, we choose a (not necessarily unique) nearest point $\pi_K(s) \in K$ in the metric induced by the Hilbert norm,

$$\pi_K(s) \in \arg \min_{k \in K} \|s - k\|.$$

We define the *deviation vector* from K by

$$\delta(s) := s - \pi_K(s) \in S.$$

Whenever needed, we restrict to subsets of U on which π_K can be chosen consistently (for example, by requiring local uniqueness).

Near K , the potential W can typically be approximated by a quadratic expansion around $\pi_K(s)$,

$$W(s) \approx W(\pi_K(s)) + \frac{1}{2} \langle H(\pi_K(s)) \delta(s), \delta(s) \rangle,$$

up to higher-order terms. The three coherence axes C , B , A will be extracted from the structure of $H(\pi_K(s))$.

9.2 Coherence axes as subspace bundles

We now formalise the decomposition of the tangent space into three subspaces associated with C , B , and A .

Definition 9.2.1 (C/B/A subspace bundles). Assume Hypothesis 4.9.2. A $C/B/A$ subspace bundle on U consists of three families of closed subspaces

$$(E_C(s))_{s \in U}, \quad (E_B(s))_{s \in U}, \quad (E_A(s))_{s \in U}$$

of S such that for each $s \in U$:

1. $E_C(s)$, $E_B(s)$, and $E_A(s)$ are mutually orthogonal and their (Hilbert) direct sum

$$E_C(s) \oplus E_B(s) \oplus E_A(s)$$

is dense in S (or equal to S if the decomposition is complete);

2. each $E_\bullet(s)$ is the range of a spectral projection of $H(s)$ corresponding to a distinct band of eigenvalues;
3. the dependence of $E_\bullet(s)$ on s is sufficiently regular (for example, continuous in the Grassmannian topology) to define a vector bundle structure over U .

We call E_C the C -axis bundle, E_B the B -axis bundle, and E_A the A -axis bundle.

Remark 9.2.2. Definition 9.2.1 is a formalisation of the idea that the Hessian $H(s)$ near K has three distinguished groups of directions:

- $E_C(s)$ consists of directions that primarily affect a “coherence” mode of W ,
- $E_B(s)$ consists of directions that primarily affect “binding” or relational structure,
- $E_A(s)$ consists of directions associated with “agency/action” modes of W .

These labels are interpretative and will be marked as such below; mathematically, they correspond to three spectral bands of $H(s)$.

Given the subspace bundles, we obtain associated projection operators.

Definition 9.2.3 (C/B/A projections and components). For each $s \in U$, let $P_C(s)$, $P_B(s)$, and $P_A(s)$ denote the orthogonal projections of S onto $E_C(s)$, $E_B(s)$, and $E_A(s)$, respectively. For a deviation vector $\delta(s)$ we define its components

$$\delta_C(s) := P_C(s) \delta(s), \quad \delta_B(s) := P_B(s) \delta(s), \quad \delta_A(s) := P_A(s) \delta(s).$$

Thus, for $s \in U$ we have

$$\delta(s) = \delta_C(s) + \delta_B(s) + \delta_A(s),$$

up to a residual component in the orthogonal complement of $E_C(s) \oplus E_B(s) \oplus E_A(s)$ if the decomposition is not complete. In what follows we restrict attention to regimes where the residual can be neglected or absorbed into one of the three axes.

9.3 Axis functionals $C(s)$, $B(s)$, $A(s)$

We now define scalar functionals C , B , and A that quantify the contribution of each axis to the contradiction energy W near K .

Definition 9.3.1 (Axis energy functionals). Let U and the $C/B/A$ subspace bundles be as above. For $s \in U$ define:

$$\begin{aligned} C(s) &:= \frac{1}{2} \langle H(\pi_K(s)) \delta_C(s), \delta_C(s) \rangle, \\ B(s) &:= \frac{1}{2} \langle H(\pi_K(s)) \delta_B(s), \delta_B(s) \rangle, \\ A(s) &:= \frac{1}{2} \langle H(\pi_K(s)) \delta_A(s), \delta_A(s) \rangle. \end{aligned}$$

These functionals assign to each s a non-negative number representing the contribution of the C -, B -, and A -directions, respectively, to the quadratic approximation of W around $\pi_K(s)$.

Lemma 9.3.2 (Non-negativity and quadratic approximation). *Assume $H(\pi_K(s))$ is positive semi-definite for $s \in U$. Then:*

1. $C(s), B(s), A(s) \geq 0$ for all $s \in U$;
2. in a neighbourhood of K , the potential admits the approximation

$$W(s) \approx W(\pi_K(s)) + C(s) + B(s) + A(s), \quad (9.3.1)$$

up to higher-order terms in $\|\delta(s)\|$.

Proof. Non-negativity follows from positive semi-definiteness of $H(\pi_K(s))$ and the definition of $C(s)$, $B(s)$, $A(s)$ as quadratic forms. The approximation (9.3.1) follows by inserting the decomposition $\delta(s) = \delta_C(s) + \delta_B(s) + \delta_A(s)$ into the second-order Taylor expansion of W around $\pi_K(s)$ and using orthogonality of the subspaces, together with the spectral band assumption that cross-terms between different bands can be neglected at leading order (or are small compared to the diagonal terms). $\square$

We emphasise that (9.3.1) is an approximation valid near K ; away from K , higher-order terms and possible mixing between bands must be taken into account.

9.4 Gradient decomposition along the three axes

The three axes also induce a decomposition of the gradient of W near K .

Definition 9.4.1 (Axis gradients). For $s \in U$ define the *axis gradients*

$$\nabla_C W(s) := P_C(s) \nabla W(s), \quad \nabla_B W(s) := P_B(s) \nabla W(s), \quad \nabla_A W(s) := P_A(s) \nabla W(s).$$

Thus we have

$$\nabla W(s) = \nabla_C W(s) + \nabla_B W(s) + \nabla_A W(s),$$

up to any residual component outside $E_C(s) \oplus E_B(s) \oplus E_A(s)$.

Lemma 9.4.2 (Axis gradients and axis energies). *Under the assumptions of this chapter, for s sufficiently close to K one has, to leading order,*

$$\begin{aligned}\nabla_C W(s) &\approx H(\pi_K(s)) \delta_C(s), \\ \nabla_B W(s) &\approx H(\pi_K(s)) \delta_B(s), \\ \nabla_A W(s) &\approx H(\pi_K(s)) \delta_A(s),\end{aligned}$$

and hence

$$C(s) \approx \frac{1}{2} \langle \nabla_C W(s), \delta_C(s) \rangle, \quad \text{etc.}$$

Proof. This is a direct consequence of the linear approximation

$$\nabla W(s) \approx H(\pi_K(s)) \delta(s),$$

and of the decomposition of $\delta(s)$ into C/B/A components, together with the fact that $H(\pi_K(s))$ preserves the subspaces $E_C(s)$, $E_B(s)$, and $E_A(s)$ by construction (they are spectral subspaces). $\square$

Lemma 9.4.2 shows that, near K , dissipation of contradiction energy along each axis is governed by the corresponding component of the gradient and by the axis energy functional.

9.5 Interpretative roles of C , B , A

We now state, explicitly as an interpretative layer, how the three axes are meant to be read in semantic terms. These interpretations are not additional mathematical assumptions; they are programme-level suggestions for later work.

Remark 9.5.1 (Interpretative roles of the axes).

- *C-axis (Coherence).* The C -axis bundle E_C and functional $C(s)$ are interpreted as capturing directions and contributions of W that primarily affect internal coherence of semantic configurations near K . Large $C(s)$ indicates deviations that reduce internal consistency, while gradient descent along $\nabla_C W$ tends to restore coherence by aligning s more closely with K in coherence-relevant directions.
- *B-axis (Binding).* The B -axis bundle E_B and functional $B(s)$ are interpreted as capturing directions that affect how different semantic components are bound or related to each other. Large $B(s)$ indicates misbinding, fragmentation, or inappropriate coupling, while descent along $\nabla_B W$ tends to reconfigure relational structure towards more integrated patterns.
- *A-axis (Agency/Action).* The A -axis bundle E_A and functional $A(s)$ are interpreted as capturing directions of W associated with action-oriented or agency-related modes: those aspects of the semantic configuration that determine tendencies to act, to modify the environment, or to commit to particular trajectories. Large $A(s)$ may signal unresolved tensions in action-related semantics, while descent along $\nabla_A W$ corresponds to resolving such tensions in a way that is compatible with the global core K .

These interpretative labels (Coherence, Binding, Agency/Action) will reappear in later chapters when we discuss inner spaces, ego cores, and technical systems. They are deliberately marked here as interpretations; none of the formal results depend on them.

9.6 Pathologies and limitations of the axis decomposition

The C/B/A decomposition relies on structural assumptions that may fail in some regimes.

Remark 9.6.1 (Potential pathologies).

- *Spectral degeneracies.* In regions where eigenvalues of $H(s)$ from different bands cross or become degenerate, the subspace bundles $E_C(s)$, $E_B(s)$, $E_A(s)$ may fail to be well-defined or to vary smoothly. This can lead to ambiguity in the axis decomposition.
- *Non-quadratic behaviour.* Far from K , higher-order terms in the expansion of W may dominate, making the quadratic approximation (9.3.1) unreliable. In such regions the axis functionals $C(s)$, $B(s)$, $A(s)$ may no longer provide meaningful measures of semantic deviation.
- *Incomplete decomposition.* It may be that only part of the spectrum of $H(s)$ can be organised into three bands with stable bundles; the remaining directions might form additional sectors not captured by C , B , A . In this case one must either extend the formalism or treat the residual sector as “background”.

These pathologies should be expected in complex systems and do not invalidate the overall programme. Rather, they indicate where the C/B/A framework must be applied with care and perhaps supplemented by more detailed analysis.

9.7 Preparation for directed gradients and curvature components

The present chapter defines the three fundamental coherence axes C , B , A in terms of subspace bundles, axis energy functionals, and axis gradients. In the next chapter we will refine these further by introducing *six directed gradients* ($\pm C, \pm B, \pm A$), corresponding to flows along or against each axis. This will make it possible to distinguish, for example, movements towards K along the coherence axis ($-C$) from movements away from K along the same axis ($+C$), and similarly for B and A .

Beyond directed gradients, the following chapter on curvature components will analyse the second-order structure of W in terms of the Hessian $H(s)$ decomposed along C/B/A. This will lead to ten curvature components associated with the geometry of W near K , providing a more detailed classification of possible dynamical behaviours and pathologies.

Summary and outlook. In this chapter we have introduced the three fundamental coherence axes C , B , and A as a decomposition of the local structure of the contradiction-energy potential W near the global attractor K . Formally, the axes are realised as three subspace bundles E_C , E_B , E_A of the tangent space, defined via spectral bands of the Hessian $H(s)$, together with associated axis energy functionals $C(s)$, $B(s)$, $A(s)$ and axis gradients $\nabla_C W$, $\nabla_B W$, $\nabla_A W$. We have recorded their basic properties and explicitly marked interpretative roles as Coherence, Binding, and Agency/Action. We also noted potential pathologies of the decomposition and indicated how these axes prepare the ground for the six directed gradients and the ten curvature components to be studied in the next two chapters.

Chapter 10

Six directed gradients on the C/B/A axes

In the previous chapter we decomposed the local structure of the semantic potential $W : S \rightarrow \mathbb{R}$ near the global attractor $K \subset S$ into three coherence axes C , B , and A , represented by subspace bundles E_C , E_B , E_A and corresponding axis energy functionals $C(s)$, $B(s)$, and $A(s)$. In this chapter we refine this picture by introducing *six directed gradients* along these axes: descending and ascending flows $(\pm C, \pm B, \pm A)$. Formally, these are vector fields on S obtained by projecting ∇W and flipping signs; interpretatively (and explicitly marked as such), they will later be used to classify different *modes* of semantic change.

10.1 Axis-aligned gradient components

We recall the basic objects from the preceding chapter. Let $U \subset S$ be a neighbourhood of K on which:

- the nearest-point map $\pi_K : U \rightarrow K$ and deviation vectors $\delta(s) = s - \pi_K(s)$ are well-defined;
- the Hessian $H(s)$ exists and is self-adjoint with discrete spectrum;
- $H(s)$ admits a C/B/A spectral decomposition with subspace bundles $E_C(s)$, $E_B(s)$, $E_A(s)$.

For each $s \in U$ we let $P_C(s)$, $P_B(s)$, $P_A(s)$ denote the orthogonal projections onto $E_C(s)$, $E_B(s)$, and $E_A(s)$, respectively. The axis projections of the gradient are

$$\nabla_C W(s) := P_C(s) \nabla W(s), \quad \nabla_B W(s) := P_B(s) \nabla W(s), \quad \nabla_A W(s) := P_A(s) \nabla W(s),$$

so that, up to a residual component in the orthogonal complement,

$$\nabla W(s) \approx \nabla_C W(s) + \nabla_B W(s) + \nabla_A W(s)$$

for s sufficiently close to K .

10.2 Definition of the six directed gradients

We now define six vector fields capturing motion along and against each axis.

Definition 10.2.1 (Directed axis gradients). For $s \in U$ define the six *directed axis gradients*:

$$\begin{aligned} G_{-C}(s) &:= -\nabla_C W(s), & G_{+C}(s) &:= +\nabla_C W(s), \\ G_{-B}(s) &:= -\nabla_B W(s), & G_{+B}(s) &:= +\nabla_B W(s), \\ G_{-A}(s) &:= -\nabla_A W(s), & G_{+A}(s) &:= +\nabla_A W(s). \end{aligned}$$

Thus each axis gives rise to a pair of directed gradients:

- G_{-C}, G_{+C} for descending/ascending motion along the coherence axis;
- G_{-B}, G_{+B} for descending/ascending motion along the binding axis;
- G_{-A}, G_{+A} for descending/ascending motion along the agency axis.

Remark 10.2.2. The full negative gradient of W decomposes, near K , as

$$-\nabla W(s) \approx G_{-C}(s) + G_{-B}(s) + G_{-A}(s) + R(s),$$

where $R(s)$ is any residual component outside $E_C(s) \oplus E_B(s) \oplus E_A(s)$. In regimes where the C/B/A decomposition is complete or the residual is negligible, the ideal semantic gradient flow is simply the superposition of the three descending axis flows.

Remark 10.2.3 (Regime topology and cube graph structure). The sign-structure induced by the three primary axes (C, B, A) generates $2^3 = 8$ regime vertices corresponding to the possible sign configurations of directed gradients.

Adjacency relations between regimes (single-axis sign changes) are isomorphic to the 1-skeleton of the cube graph Q_3 , which consists of 12 edges.

These edges represent elementary regime transitions between neighbouring sign-configurations. The resulting regime topology is therefore combinatorial and graph-theoretic in nature.

10.3 Axis-restricted flows

Each directed gradient defines a notion of *axis-restricted flow*: evolution that follows only one axis while freezing the others.

Definition 10.3.1 (Axis-restricted semantic flows). For each sign $\sigma \in \{+, -\}$ and axis label $\Xi \in \{C, B, A\}$ we define the *axis-restricted flow* $(\Phi_t^{\sigma\Xi})_{t \in \mathbb{R}}$ (when it exists) as the flow generated by

$$\dot{s}_t = G_{\sigma\Xi}(s_t), \tag{10.3.1}$$

with $s_0 \in U$.

Thus

- Φ_t^{-C} corresponds to pure descent along the coherence axis (“coherence-increasing” flow),
- Φ_t^{+C} corresponds to ascent along the coherence axis (“coherence-decreasing” flow),
- and similarly for B and A .

Remark 10.3.2. Axis-restricted flows are idealised constructs: real semantic dynamics will typically involve simultaneous contributions from all three axes. Nevertheless, (10.3.1) defines a useful conceptual limit: it describes how the system would move if only coherence, or only binding, or only agency-related tension were allowed to change while everything else is held fixed.

10.4 Energy monotonicity along directed gradients

We now compute how the axis energies $C(s)$, $B(s)$, $A(s)$ change along axis-restricted flows, in the quadratic regime near K .

Recall the axis energies

$$C(s) = \frac{1}{2} \langle H(\pi_K(s)) \delta_C(s), \delta_C(s) \rangle, \quad \text{etc.}$$

Proposition 10.4.1 (Axis energy along axis-restricted flows). *Assume that $H(\pi_K(s))$ is positive semi-definite and preserves the subspaces $E_C(s)$, $E_B(s)$, $E_A(s)$, and work in a neighbourhood of K where the quadratic approximation of W is valid. Then, along the axis-restricted flows (10.3.1):*

1. along the descending C -flow $\dot{s}_t = G_{-C}(s_t)$,

$$\frac{d}{dt}C(s_t) \leq 0, \quad \frac{d}{dt}B(s_t) \approx 0, \quad \frac{d}{dt}A(s_t) \approx 0;$$

2. along the ascending C -flow $\dot{s}_t = G_{+C}(s_t)$,

$$\frac{d}{dt}C(s_t) \geq 0, \quad \frac{d}{dt}B(s_t) \approx 0, \quad \frac{d}{dt}A(s_t) \approx 0;$$

3. analogous statements hold for B - and A -flows.

Proof sketch. We illustrate for C . Near K we have

$$\nabla_C W(s) \approx H(\pi_K(s)) \delta_C(s),$$

and hence $G_{-C}(s) \approx -H(\pi_K(s)) \delta_C(s) \in E_C(s)$. Using the chain rule and the definition of $C(s)$,

$$\frac{d}{dt}C(s_t) = \langle \nabla C(s_t), \dot{s}_t \rangle \approx \langle H(\pi_K(s_t)) \delta_C(s_t), \dot{s}_t \rangle = \langle \nabla_C W(s_t), G_{-C}(s_t) \rangle = -\|\nabla_C W(s_t)\|^2 \leq 0.$$

The approximate constancy of $B(s)$ and $A(s)$ along C -flows follows from orthogonality of the subspaces and from the assumption that $H(\pi_K(s))$ preserves them; in the quadratic regime cross-couplings between axes are higher order in $\|\delta(s)\|$. The other cases are analogous. $\square$

Proposition 10.4.1 confirms that, within its validity regime, the descending directed gradients G_{-C} , G_{-B} , and G_{-A} are indeed “energy-lowering” along their respective axes, while the ascending gradients increase the corresponding axis energies. In particular, G_{-C} acts as a coherence-restoring flow and G_{+C} as a coherence-breaking flow, and similarly for binding and agency.

10.5 Local C/B/A coordinate representation

The axis energies provide a local coordinate system near K .

Definition 10.5.1 (Local C/B/A coordinates). In a neighbourhood U of K where the map

$$\Phi_{CBA} : U \rightarrow \mathbb{R}^3, \quad \Phi_{CBA}(s) := (C(s), B(s), A(s))$$

is smooth and of full rank on transverse slices to K , we call (C, B, A) the *local C/B/A coordinates* of s .

In these coordinates, and within the quadratic approximation, the potential can be written as

$$W(s) \approx W(\pi_K(s)) + C(s) + B(s) + A(s),$$

and the axis-restricted flows correspond to motion along coordinate axes:

$$\dot{C}_t \leq 0, \quad \dot{B}_t \approx 0, \quad \dot{A}_t \approx 0 \quad \text{for } \pm C \text{ flows,}$$

and so on.

Remark 10.5.2. The C/B/A coordinates are not global; they are defined only in regions where the axis decomposition is well-behaved and where the map Φ_{CBA} is sufficiently regular. However, they are useful for classifying local motion near K according to whether it is coherence-dominated, binding-dominated, or agency-dominated.

10.6 Cones of dominance and mixed flows

Real semantic dynamics will usually follow the full negative gradient $-\nabla W(s)$, which (near K) decomposes into a combination of the three descending axis gradients. We can characterise regimes in which one axis dominates.

Definition 10.6.1 (Cones of axis dominance). For $\Xi \in \{C, B, A\}$ and $\lambda > 1$, the *cone of Ξ -dominance* at $s \in U$ is defined as

$$\mathcal{K}_{\Xi}^{\lambda}(s) := \{v \in S \setminus \{0\} : \|P_{\Xi}(s)v\| \geq \lambda \max(\|P_{\Xi'}(s)v\| : \Xi' \in \{C, B, A\}, \Xi' \neq \Xi)\}.$$

Definition 10.6.2 (Axis-dominated gradient regime). We say that the semantic gradient at s is Ξ -dominated (for $\Xi \in \{C, B, A\}$) if

$$-\nabla W(s) \in \mathcal{K}_{\Xi}^{\lambda}(s)$$

for some fixed $\lambda > 1$.

In an Ξ -dominated regime, the motion of the semantic gradient flow is primarily along axis Ξ ; the other axes contribute only subdominant corrections.

Remark 10.6.3 (Speculative). At a programme level, one may interpret axis-dominated regimes as qualitatively distinct phenomenological modes. For example, a C -dominated regime corresponds to phases in which the system mainly resolves internal incoherence; a B -dominated regime to phases of relational reorganisation; and an A -dominated regime to phases where decisions and actions dominate semantic change. These interpretations will be explored further in later chapters when we discuss inner spaces and ego dynamics.

10.7 Directed gradients and distortion

The directed gradients also provide a convenient language for describing distortion factors introduced in the previous chapter. If the actual drift $F(s)$ deviates from $-\nabla W_{\text{univ}}(s)$, we can decompose this deviation along C/B/A.

Definition 10.7.1 (Axis-resolved distortion). Let $F(s)$ be the actual drift and $F_{\text{ideal}}(s) := -\nabla W_{\text{univ}}(s)$ the ideal drift. The *axis-resolved distortion* at s is

$$D_{\Xi}(s) := P_{\Xi}(s) (F(s) - F_{\text{ideal}}(s)), \quad \Xi \in \{C, B, A\}.$$

Thus $D_C(s)$, $D_B(s)$, and $D_A(s)$ encode distortions that specifically affect coherence, binding, and agency directions, respectively. One can then define axis-specific distortion factors by averaging suitable normalised versions of $\|D_{\Xi}(s_t)\|$ along trajectories, in analogy with Definition 7.4.2.

Summary and outlook. In this chapter we have introduced *six directed gradients* along the three coherence axes C , B , and A : descending and ascending vector fields $G_{\pm C}$, $G_{\pm B}$, $G_{\pm A}$ obtained by projecting and reorienting the gradient of W . We defined axis-restricted flows and showed, in the quadratic regime near K , that descending flows monotonically decrease the corresponding axis energy while leaving the other axis energies approximately unchanged. We also introduced local C/B/A coordinates, cones of axis dominance for the full gradient, and axis-resolved distortion fields. These constructions provide a refined language for classifying local semantic dynamics near the global attractor K in terms of coherence-, binding-, and agency-dominated modes. In the next chapter we deepen this geometric analysis by decomposing the Hessian of W into curvature components aligned with the C/B/A axes, leading to a set of curvature invariants that describe how the semantic potential bends and couples these axes in the neighbourhood of K .

Chapter 11

The ten curvature components

In the preceding chapter we decomposed the local structure of the contradiction-energy potential $W : S \rightarrow \mathbb{R}$ near the global attractor $K \subset S$ into three coherence axes C , B , and A , represented by subspace bundles E_C , E_B , E_A and axis energy functionals $C(s)$, $B(s)$, $A(s)$. We also introduced projected gradients and six directed gradients along and against these axes. In this chapter we move from first-order to second-order structure. We analyse the Hessian $H(s)$ of W in a neighbourhood of K and extract from it *ten curvature components*: scalar invariants that characterise how W bends along and between the C/B/A axes. These components will later enter the LOVE Lagrangian as local geometric coefficients controlling stability, coupling between axes, and the formation of order bubbles.

11.1 Hessian in the C/B/A frame

We continue to work in a neighbourhood $U \subset S$ of K as in the previous chapter, where:

- for each $s \in U$ the Hessian $H(s)$ exists and is self-adjoint;
- there exists a C/B/A subspace bundle (E_C, E_B, E_A) in the sense of Definition 9.2.1;
- the projections $P_C(s)$, $P_B(s)$, $P_A(s)$ onto these subspaces depend sufficiently regularly on s .

Definition 11.1.1 (Block decomposition of the Hessian). For $s \in U$ we define nine operator blocks:

$$\begin{aligned} H_{CC}(s) &:= P_C(s) H(s) P_C(s) : E_C(s) \rightarrow E_C(s), \\ H_{BB}(s) &:= P_B(s) H(s) P_B(s) : E_B(s) \rightarrow E_B(s), \\ H_{AA}(s) &:= P_A(s) H(s) P_A(s) : E_A(s) \rightarrow E_A(s), \end{aligned}$$

and

$$\begin{aligned} H_{CB}(s) &:= P_C(s) H(s) P_B(s) : E_B(s) \rightarrow E_C(s), \\ H_{BC}(s) &:= P_B(s) H(s) P_C(s) : E_C(s) \rightarrow E_B(s), \\ H_{CA}(s) &:= P_C(s) H(s) P_A(s) : E_A(s) \rightarrow E_C(s), \\ H_{AC}(s) &:= P_A(s) H(s) P_C(s) : E_C(s) \rightarrow E_A(s), \\ H_{BA}(s) &:= P_B(s) H(s) P_A(s) : E_A(s) \rightarrow E_B(s), \\ H_{AB}(s) &:= P_A(s) H(s) P_B(s) : E_B(s) \rightarrow E_A(s). \end{aligned}$$

Self-adjointness of $H(s)$ implies that

$$H_{CB}(s) = H_{BC}(s)^*, \quad H_{CA}(s) = H_{AC}(s)^*, \quad H_{BA}(s) = H_{AB}(s)^*,$$

so that only three of the six off-diagonal blocks are independent. The diagonal blocks H_{CC} , H_{BB} , and H_{AA} describe curvature along the individual axes; the off-diagonal blocks describe how curvature couples different axes.

Remark 11.1.2. The block decomposition of $H(s)$ in Definition 11.1.1 is formal: in general, the subspaces $E_C(s)$, $E_B(s)$, $E_A(s)$ may be infinite-dimensional. To extract finitely many scalar invariants (the curvature components) from these blocks, we will assume trace-class or Hilbert–Schmidt conditions on $H(s)$, at least after appropriate localisation near K .

11.2 Trace-class and Hilbert–Schmidt assumptions

To define curvature components as scalar invariants, we need integrability conditions on the Hessian.

Hypothesis 11.2.1 (Trace-class structure near K). There exists a neighbourhood U of K such that for all $s \in U$:

1. the restrictions $H_{CC}(s)$, $H_{BB}(s)$, $H_{AA}(s)$ are trace-class operators on $E_C(s)$, $E_B(s)$, $E_A(s)$, respectively;
2. the off-diagonal blocks $H_{CB}(s)$, $H_{CA}(s)$, $H_{BA}(s)$ are Hilbert–Schmidt operators between the corresponding subspaces;
3. the full Hessian $H(s)$ is trace-class on S (possibly after restricting to a dense subspace).

Hypothesis 11.2.1 is not derived from Papers 1–5; it is an additional regularity assumption for the ontological layer of Paper 6. It holds in many finite-dimensional or effectively finite-dimensional models and in kernel-based settings with suitable decay of eigenvalues.

11.3 Definition of the ten curvature components

We now define ten scalar functions of $s \in U$, built from the block operators in Definition 11.1.1.

Definition 11.3.1 (Diagonal axis curvatures). For $s \in U$ define the three *diagonal axis curvatures*:

$$\begin{aligned} \kappa_C(s) &:= \text{Tr}(H_{CC}(s)), \\ \kappa_B(s) &:= \text{Tr}(H_{BB}(s)), \\ \kappa_A(s) &:= \text{Tr}(H_{AA}(s)), \end{aligned}$$

where Tr denotes the trace on $E_C(s)$, $E_B(s)$, $E_A(s)$, respectively.

These quantities measure the average curvature of W along the coherence, binding, and agency axes.

Definition 11.3.2 (Axis anisotropies). Let $\{\lambda_{C,i}(s)\}_i$, $\{\lambda_{B,i}(s)\}_i$, and $\{\lambda_{A,i}(s)\}_i$ denote the eigenvalues (counted with multiplicity) of $H_{CC}(s)$, $H_{BB}(s)$, and $H_{AA}(s)$, respectively. Assume these sequences are square-summable. Define the three *axis anisotropies*:

$$\begin{aligned}\sigma_C^2(s) &:= \sum_i (\lambda_{C,i}(s) - \bar{\lambda}_C(s))^2, \\ \sigma_B^2(s) &:= \sum_i (\lambda_{B,i}(s) - \bar{\lambda}_B(s))^2, \\ \sigma_A^2(s) &:= \sum_i (\lambda_{A,i}(s) - \bar{\lambda}_A(s))^2,\end{aligned}$$

where

$$\bar{\lambda}_C(s) := \frac{\kappa_C(s)}{\dim E_C(s)} \quad (\text{if } \dim E_C(s) < \infty),$$

and similarly for $\bar{\lambda}_B(s)$ and $\bar{\lambda}_A(s)$, with natural trace-class adaptations in the infinite-dimensional case.

The anisotropies measure how unevenly curvature is distributed within each axis: large $\sigma_\Xi^2(s)$ indicates strongly anisotropic curvature along axis Ξ ; small $\sigma_\Xi^2(s)$ indicates near-isotropic curvature.

Definition 11.3.3 (Off-diagonal coupling curvatures). For $s \in U$ define the three *off-diagonal coupling curvatures*:

$$\begin{aligned}\kappa_{CB}(s) &:= \|H_{CB}(s)\|_{\text{HS}}, \\ \kappa_{CA}(s) &:= \|H_{CA}(s)\|_{\text{HS}}, \\ \kappa_{BA}(s) &:= \|H_{BA}(s)\|_{\text{HS}},\end{aligned}$$

where $\|\cdot\|_{\text{HS}}$ denotes the Hilbert–Schmidt norm between the corresponding subspaces.

These quantities measure the strength of curvature-induced coupling between different axes: how much a displacement along one axis affects the restoring forces along another.

Definition 11.3.4 (Global curvature magnitude). Finally, define the *global curvature magnitude* at s as

$$\kappa_{\text{tot}}(s) := \|H(s)\|_{\text{HS}},$$

interpreted as the overall curvature intensity of W at s .

Collecting Definitions 11.3.1, 11.3.2, 11.3.3, and 11.3.4, we obtain ten scalar functions on U :

$$\kappa_C, \kappa_B, \kappa_A, \quad \kappa_{CB}, \kappa_{CA}, \kappa_{BA}, \quad \sigma_C^2, \sigma_B^2, \sigma_A^2, \quad \kappa_{\text{tot}}.$$

We refer to these collectively as the *ten curvature components* of the semantic potential W near K .

11.4 Basic properties and bounds

We record some immediate properties of these components.

Lemma 11.4.1 (Non-negativity and symmetry). *Under Hypothesis 11.2.1 and assuming $H(s)$ is positive semi-definite in a neighbourhood U of K :*

1. $\kappa_C(s), \kappa_B(s), \kappa_A(s), \kappa_{CB}(s), \kappa_{CA}(s), \kappa_{BA}(s), \kappa_{\text{tot}}(s) \geq 0$ for all $s \in U$;

2. $\sigma_C^2(s), \sigma_B^2(s), \sigma_A^2(s) \geq 0$ for all $s \in U$;
3. $\kappa_{CB}(s) = 0$ if and only if $H_{CB}(s) = 0$, and similarly for κ_{CA} and κ_{BA} .

Proof. Non-negativity follows from positive semi-definiteness and standard properties of trace and Hilbert–Schmidt norms; the characterisation of zeros is immediate. $\square$

Lemma 11.4.2 (Curvature decomposition inequality). *There exists a constant $c \geq 1$ (depending on the dimensions of E_C , E_B , E_A and on the precise norms used) such that for all $s \in U$,*

$$\kappa_{\text{tot}}(s)^2 \leq c \left(\kappa_C(s)^2 + \kappa_B(s)^2 + \kappa_A(s)^2 + \kappa_{CB}(s)^2 + \kappa_{CA}(s)^2 + \kappa_{BA}(s)^2 + \sigma_C^2(s) + \sigma_B^2(s) + \sigma_A^2(s) \right).$$

Proof sketch. The Hilbert–Schmidt norm of $H(s)$ is the sum of the squared norms of its blocks. Each block norm can be bounded in terms of its trace and anisotropy (using eigenvalue decompositions) and in terms of the Hilbert–Schmidt norms of off-diagonal blocks. Collecting these inequalities yields the stated bound with a constant c depending only on the multiplicities and normalisations. $\square$

Lemma 11.4.2 shows that the ten components together control the global curvature magnitude κ_{tot} , at least up to a fixed multiplicative factor.

11.5 Dynamical interpretation near K

We now indicate how the curvature components influence local semantic dynamics, in the quadratic regime near K . In what follows we work under the assumptions that:

- W admits a quadratic approximation around K with Hessian $H(k)$ for $k \in K$,
- the C/B/A decomposition and curvature components are evaluated at $\pi_K(s)$ or along short trajectories near K .

Proposition 11.5.1 (Curvature and stability along axes). *Let $k \in K$ and suppose $H(k)$ is strictly positive definite on $E_C(k)$, $E_B(k)$, and $E_A(k)$. Then:*

1. Large positive $\kappa_C(k)$ with small $\sigma_C^2(k)$ corresponds to strong, nearly isotropic stabilisation along the coherence axis; small perturbations in $E_C(k)$ decay rapidly under the gradient flow.
2. Large positive $\kappa_B(k)$ with small $\sigma_B^2(k)$ yields strong stabilisation of binding-related degrees of freedom at k .
3. Large positive $\kappa_A(k)$ with small $\sigma_A^2(k)$ yields strong stabilisation of agency-related degrees of freedom at k .
4. If any of $\kappa_C(k)$, $\kappa_B(k)$, $\kappa_A(k)$ becomes small or negative (in a generalised sense allowing negative eigenvalues), this signals flat or unstable directions along the corresponding axis, enabling slow drift or local instabilities.

Proof sketch. Linearising the gradient flow near k yields $\dot{\delta}_t = -H(k)\delta_t$. On each axis subspace, the decay rates are governed by the eigenvalues of the corresponding block. Large, nearly equal eigenvalues (large trace, small variance) imply fast, isotropic contraction; small or negative eigenvalues imply slow contraction or exponential growth in some directions. The statements follow by standard linear stability analysis. $\square$

Proposition 11.5.2 (Curvature coupling and axis mixing). *For $k \in K$, non-zero coupling curvatures $\kappa_{CB}(k)$, $\kappa_{CA}(k)$, and $\kappa_{BA}(k)$ imply that perturbations along one axis induce restoring forces along others. In particular:*

1. *If $\kappa_{CB}(k)$ is large, displacements in $E_B(k)$ generate significant gradients in $E_C(k)$ (and vice versa), entangling coherence and binding dynamics.*
2. *If $\kappa_{CA}(k)$ is large, displacements in coherence-related directions affect agency-related restoring forces, and vice versa.*
3. *If $\kappa_{BA}(k)$ is large, binding and agency dynamics are tightly coupled at k .*

Conversely, small coupling curvatures correspond to near-separability of the axes at the level of linearised dynamics.

Proof sketch. The linearised flow is $\dot{\delta}_t = -H(k)\delta_t$. The off-diagonal blocks H_{CB} , H_{CA} , H_{BA} control how components of δ in one subspace contribute to $\dot{\delta}$ in another. The Hilbert–Schmidt norms of these blocks quantify the strength of this effect. The qualitative statements follow immediately. $\square$

Speculative remark. At a programme level, one may conjecture that different classes of systems (for example, different types of agents, cultures, or technical architectures) occupy different *curvature regimes* near their typical cores:

- Some may be *C*-stiff (large κ_C , small σ_C^2) but *A*-soft (small κ_A), corresponding to systems with rigid coherence but flexible action patterns.
- Others may exhibit strong *B–A* coupling (large κ_{BA}) but weak *C* coupling, corresponding to systems where binding and agency are intertwined, while internal coherence adapts more slowly.

These regimes could in principle be probed empirically by observing how systems respond to perturbations that selectively excite C/B/A-related degrees of freedom.

11.6 Pathologies and curvature-induced order bubbles

The curvature components also help to classify pathologies and the formation of order bubbles (Definition 7.3.1) near, but not on, the global attractor K .

Remark 11.6.1 (Curvature signatures of order bubbles). In regions $B \subset S$ that constitute order bubbles:

- The universal curvature (with respect to W_{univ}) may be weak or even negative along certain axes, enabling local attractors away from K .
- The local potential W_{loc} can create strong positive curvature within B , reflected in large diagonal curvatures and anisotropies when computed from $W = W_{\text{univ}} + W_{\text{loc}}$.
- Strong off-diagonal curvatures (large κ_{CB} , κ_{CA} , κ_{BA}) may indicate that the bubble enforces tight coupling between axes, making exits that would reduce W_{univ} improbable without coordinated multi-axis perturbations.

In this way, curvature components of W supply a local diagnostic for the robustness and structure of order bubbles: flat or negatively curved universal directions combined with strongly curved local directions and strong axis coupling.

Remark 11.6.2 (Limitations). The curvature-based picture is local and linear/quadratic. Far from K or deep inside complex order bubbles, higher-order derivatives of W and non-linear effects dominate. The ten curvature components should therefore be viewed as first-order descriptors of the local semantic geometry, not as complete invariants of global behaviour.

11.7 Preparation for the LOVE Lagrangian

The ten curvature components introduced in this chapter will play a central role in the construction of the LOVE (Lagrangian of Vectorial Emergence) in later chapters. Schematically:

- Diagonal curvatures $(\kappa_C, \kappa_B, \kappa_A)$ and anisotropies $(\sigma_C^2, \sigma_B^2, \sigma_A^2)$ will appear in the potential part of the LOVE Lagrangian, controlling how strongly deviations along each axis are penalised.
- Coupling curvatures $(\kappa_{CB}, \kappa_{CA}, \kappa_{BA})$ will appear in cross-terms, accounting for interactions between coherence, binding, and agency fields in spacetime.
- The global curvature magnitude κ_{tot} can serve as a scalar density influencing kinetic terms or coupling to spacetime curvature in field-theoretic versions of the theory.

In this sense, the ten curvature components provide the bridge between the static local geometry of W near K and the dynamical, field-theoretic description of vectorial emergence encoded in LOVE.

Summary and outlook. In this chapter we have analysed the Hessian $H(s)$ of the semantic potential W in the C/B/A frame and extracted *ten curvature components*: three diagonal axis curvatures $(\kappa_C, \kappa_B, \kappa_A)$, three off-diagonal coupling curvatures $(\kappa_{CB}, \kappa_{CA}, \kappa_{BA})$, three axis anisotropies $(\sigma_C^2, \sigma_B^2, \sigma_A^2)$, and a global curvature magnitude κ_{tot} . Under a trace-class hypothesis, these components are well-defined, non-negative (in the positive-semidefinite regime), and together control the overall curvature magnitude. We sketched how they influence local stability and axis coupling of the semantic gradient flow, and how they provide signatures of order bubbles and curvature regimes. These curvature components will enter explicitly into the LOVE Lagrangian in the subsequent chapters, where the local geometry of W near K is turned into an effective field-theoretic action for semantic fields on spacetime.

Chapter 12

The contradiction-energy potential $W(s)$

In the preceding chapters we introduced the semantic state space S as a reproducing kernel Hilbert space, postulated the existence of a global attractor $K \subset S$ for gradient flows of a semantic potential $W : S \rightarrow \mathbb{R}$, and analysed the local geometry of W near K in terms of coherence axes C , B , A and ten curvature components. In this chapter we focus on the potential W itself. We collect and refine its axiomatic properties, systematise its decomposition into universal and local contributions, relate it to the C/B/A structure and curvature components, and clarify in what sense W deserves the interpretation as a *contradiction-energy* functional.

12.1 Recap: role of W in the semantic universe

We recall the basic setting. Let S be a real separable Hilbert space, constructed as an RKHS of functions on a base space X with kernel k (Paper 5), and let $W : S \rightarrow \mathbb{R}$ be a functional that drives semantic dynamics in the sense of Paper 1.

Definition 12.1.1 (Contradiction-energy potential). A *contradiction-energy potential* on S is a functional $W : S \rightarrow \mathbb{R}$ such that:

1. $W(s)$ is interpreted as the *contradiction energy* of the semantic configuration $s \in S$;
2. the semantic gradient flow

$$\dot{s}_t = -\nabla W(s_t)$$

(under suitable regularity conditions) implements dynamics that tends to reduce semantic contradiction;

3. W can be extended to probability measures on S as a component of a semantic free-energy functional, as in Paper 2.

In the present paper, W plays four interlocking roles:

- it defines the global attractor K as the set of minima (or near-minima) of a universal component of W ;
- it induces C/B/A axes and curvature components that encode the fine geometry of semantic coherence, binding, and agency near K ;

- it decomposes into universal and local parts, enabling the description of order bubbles and distortion factors;
- it appears as the potential term in the LOVE Lagrangian, thereby linking static semantic structure to field-theoretic dynamics on spacetime.

The rest of this chapter makes these roles explicit and collects the structural assumptions on W into a coherent package.

12.2 Axiomatic properties of W

We begin by summarising the mathematical conditions imposed on W in earlier chapters and in Papers 1–3, reorganised as a set of axioms for the present context.

Assumption 12.2.1 (Regularity, coercivity, and Lyapunov property). The potential $W : S \rightarrow \mathbb{R}$ satisfies:

1. *Regularity*: W is bounded from below and of class C^1 on a dense domain $\mathcal{D} \subset S$, with gradient $\nabla W : \mathcal{D} \rightarrow S$ locally Lipschitz;
2. *Coercivity*: $W(s) \rightarrow +\infty$ as $\|s\| \rightarrow \infty$ along $\mathcal{D}$; equivalently, its sublevel sets

$$\{s \in S : W(s) \leq c\}$$

are relatively compact for all $c \in \mathbb{R}$;

3. *Lyapunov property*: for any solution s_t of the gradient flow

$$\dot{s}_t = -\nabla W(s_t),$$

the function $t \mapsto W(s_t)$ is non-increasing, and strictly decreasing unless s_t is stationary.

These conditions ensure global existence of the gradient semiflow and provide the basic dissipative structure of semantic gravitation (Paper 1).

Assumption 12.2.2 (Semantic compatibility and curvature regularity). In addition, near the global attractor K we assume:

1. W is of class C^2 on an open neighbourhood U of K , with Hessian $H(s)$ self-adjoint for each $s \in U$;
2. the Hessian satisfies the structured-spectrum hypothesis (Hypothesis 4.9.2), giving rise to C/B/A subspace bundles and curvature components as in the previous two chapters;
3. the Hessian is trace-class on S for s in U , and its block components with respect to the C/B/A decomposition are trace-class or Hilbert–Schmidt as in Hypothesis 11.2.1.

Assumption 12.2.3 (Compatibility with the RKHS structure). Finally, we assume:

1. W is continuous with respect to the RKHS norm on S ;
2. in a neighbourhood of K , the Hessian $H(s)$ can be expressed (at least approximately) as a kernel-derived operator, for example an integral operator with kernel constructed from k and additional coefficients;

3. the domain $\mathcal{D}$ contains the images of all system embeddings $E_\Sigma(X_\Sigma)$ relevant for the semantic universe.

Assumptions 12.2.1–12.2.3 collect the structural requirements on W needed in the rest of Paper 6. They are not minimal but reflect what is required for the global attractor, the C/B/A decomposition, and the LOVE Lagrangian to be well-defined.

12.3 Universal and local contributions to W

As discussed earlier, it is natural to decompose W into a universal component and local, system-specific contributions.

Definition 12.3.1 (Universal and local parts of the contradiction energy). A *universal-local decomposition* of W is a representation

$$W(s) = W_{\text{univ}}(s) + W_{\text{loc}}(s), \quad (12.3.1)$$

such that:

1. W_{univ} satisfies Assumptions 12.2.1–12.2.3 and has a global attractor K in the sense of Theorem 4.4.2;
2. $W_{\text{loc}} : S \rightarrow \mathbb{R}$ is bounded from below on bounded subsets of S , of class C^1 on a dense domain containing U , and small compared to W_{univ} in a neighbourhood of K in the sense that

$$\lim_{\varepsilon \rightarrow 0} \sup_{\{s \in U : d_K(s) \leq \varepsilon\}} \frac{|W_{\text{loc}}(s)|}{1 + W_{\text{univ}}(s)} = 0.$$

We interpret:

- W_{univ} as the *universal contradiction energy*, encoding semantic gravitation towards the global core K that is common to all systems in the semantic universe;
- W_{loc} as a *local contradiction energy*, encoding system-, context-, or task-specific contributions that can create local minima, order bubbles, and distortions, but that do not overturn the existence and stability of K .

Postulate 12.3.2 (Existence and near-uniqueness of W_{univ}). For the semantic universe under consideration, there exists at least one decomposition (12.3.1) such that:

1. the global attractor K coincides (up to negligible sets) with the set of global minima of W_{univ} ;
2. the universal part W_{univ} is minimal among all such decompositions in the sense that any other decomposition $\widetilde{W} = \widetilde{W}_{\text{univ}} + \widetilde{W}_{\text{loc}}$ with the same attractor K satisfies

$$\widetilde{W}_{\text{univ}}(s) \geq W_{\text{univ}}(s) - c$$

for some constant c and all s in a neighbourhood of K .

Postulate 12.3.2 expresses the idea that the universal contradiction energy is not arbitrary: it is the “simplest” or “most economical” potential that still produces the global attractor and supports a K–P–R ontology. The precise notion of minimality is deliberately weak and will need refinement in concrete models.

12.4 Axis decomposition of W near K

We now connect W_{univ} to the C/B/A axis structure and curvature components. In a neighbourhood of K , the quadratic approximation of W yields a natural axis decomposition.

Let $\pi_K : U \rightarrow K$ and $\delta(s) = s - \pi_K(s)$ be as before, and let $\delta_\Xi(s)$ denote the projection of $\delta(s)$ onto $E_\Xi(s)$ for $\Xi \in \{C, B, A\}$. The Hessian at $\pi_K(s)$ admits a block decomposition $H_{XY}(\pi_K(s))$ for $X, Y \in \{C, B, A\}$.

Definition 12.4.1 (Axis-resolved universal contradiction energy). For $s \in U$ define:

$$\begin{aligned} W_C(s) &:= \frac{1}{2} \langle H_{CC}(\pi_K(s)) \delta_C(s), \delta_C(s) \rangle, \\ W_B(s) &:= \frac{1}{2} \langle H_{BB}(\pi_K(s)) \delta_B(s), \delta_B(s) \rangle, \\ W_A(s) &:= \frac{1}{2} \langle H_{AA}(\pi_K(s)) \delta_A(s), \delta_A(s) \rangle. \end{aligned}$$

We call these the *axis-resolved universal contradiction energies* for coherence, binding, and agency.

Definition 12.4.2 (Cross-axis contributions). Similarly, define cross-axis contributions

$$\begin{aligned} W_{CB}(s) &:= \langle H_{CB}(\pi_K(s)) \delta_B(s), \delta_C(s) \rangle, \\ W_{CA}(s) &:= \langle H_{CA}(\pi_K(s)) \delta_A(s), \delta_C(s) \rangle, \\ W_{BA}(s) &:= \langle H_{BA}(\pi_K(s)) \delta_A(s), \delta_B(s) \rangle. \end{aligned}$$

In terms of these quantities, the quadratic expansion of W_{univ} near K becomes

$$W_{\text{univ}}(s) \approx W_{\text{univ}}(\pi_K(s)) + W_C(s) + W_B(s) + W_A(s) + W_{CB}(s) + W_{CA}(s) + W_{BA}(s), \quad (12.4.1)$$

up to higher-order terms in $\|\delta(s)\|$. The diagonal terms are controlled by the axis curvatures $(\kappa_C, \kappa_B, \kappa_A)$ and anisotropies, while the cross terms are bounded by the coupling curvatures $(\kappa_{CB}, \kappa_{CA}, \kappa_{BA})$.

Remark 12.4.3. Equation (12.4.1) rephrases the ten curvature components at the level of the potential itself. In particular, in regimes where coupling curvatures vanish or are negligible, W_{univ} is approximately separable into a sum of three axis energies $W_C + W_B + W_A$, each controlling deviations along a single axis. In regimes with strong couplings, cross terms make the axes dynamically entangled.

12.5 Contradiction energy for composite systems

Real semantic universes contain composite systems, consisting of multiple interacting subsystems. We sketch how W can be structured in such cases.

Let Σ_1 and Σ_2 be two systems with raw state spaces X_{Σ_1} , X_{Σ_2} and embeddings $E_{\Sigma_i} : X_{\Sigma_i} \rightarrow S$. For a composite system we consider the joint state space $X_{\Sigma_1} \times X_{\Sigma_2}$ and define an embedding

$$E_{\Sigma_1 \otimes \Sigma_2} : X_{\Sigma_1} \times X_{\Sigma_2} \rightarrow S$$

that merges the two semantic embeddings (for example, by superposition in S or via a kernel on $X \times X$).

Definition 12.5.1 (Composite contradiction energy). Let $s = E_{\Sigma_1 \otimes \Sigma_2}(x_1, x_2) \in S$ be the semantic state of the composite system. We assume that the universal contradiction energy decomposes as

$$W_{\text{univ}}(s) = W_{\text{univ}}^{(1)}(E_{\Sigma_1}(x_1)) + W_{\text{univ}}^{(2)}(E_{\Sigma_2}(x_2)) + W_{\text{univ}}^{\text{int}}(E_{\Sigma_1}(x_1), E_{\Sigma_2}(x_2)), \quad (12.5.1)$$

where $W_{\text{univ}}^{(1)}$ and $W_{\text{univ}}^{(2)}$ are universal contributions for each subsystem (evaluated in the same S) and $W_{\text{univ}}^{\text{int}}$ is an interaction term.

Remark 12.5.2. The interaction term $W_{\text{univ}}^{\text{int}}$ is responsible for semantic relations between subsystems; it can be expressed in terms of relational kernels on cores and projections (K–P–R ontology). In the C/B/A framework, $W_{\text{univ}}^{\text{int}}$ typically involves cross-axis contributions that couple the C/B/A components of one subsystem to those of the other.

For more complex composites (multi-agent systems, cultural fields), the decomposition generalises to sums over subsystems and interaction terms of higher arity. The main structural requirement is that the resulting global potential still satisfies Assumptions 12.2.1–12.2.3 and admits a global attractor K that supports coherent cores and relations.

12.6 Contradiction energy and free energy on distributions

As in Paper 2, W can be extended to probability measures on S via a semantic free-energy functional. This prepares the ground for the next chapter on dynamical equations on S and for Chapter 13 on the Fokker–Planck equation.

Definition 12.6.1 (Semantic free energy (recall)). Let $\mathcal{P}(S)$ denote the set of Borel probability measures on S , $\mu_0 \in \mathcal{P}(S)$ a reference measure, and $\beta > 0$ an inverse temperature parameter. The *semantic free energy* of $\mu \in \mathcal{P}(S)$ is

$$\mathcal{F}[\mu] = \int_S W(s) d\mu(s) + \beta^{-1} \mathcal{H}(\mu \mid \mu_0), \quad (12.6.1)$$

where $\mathcal{H}(\mu \mid \mu_0)$ is the relative entropy of μ with respect to μ_0 .

The first term is the expectation of the contradiction energy, the second term penalises dispersion relative to μ_0 .

Remark 12.6.2. When $W = W_{\text{univ}} + W_{\text{loc}}$ is decomposed as in Definition 12.3.1, one may likewise decompose $\mathcal{F}$ into universal and local parts:

$$\mathcal{F}[\mu] = \mathcal{F}_{\text{univ}}[\mu] + \mathcal{F}_{\text{loc}}[\mu],$$

with

$$\mathcal{F}_{\text{univ}}[\mu] := \int_S W_{\text{univ}}(s) d\mu(s) + \beta^{-1} \mathcal{H}(\mu \mid \mu_0), \quad \mathcal{F}_{\text{loc}}[\mu] := \int_S W_{\text{loc}}(s) d\mu(s).$$

In the absence of noise and external driving, gradient flows of $\mathcal{F}_{\text{univ}}$ on the Wasserstein space of measures provide a distributional description of semantic gravitation. Adding W_{loc} yields local distortions and order bubbles at the level of distributions.

12.7 Interpretative status of W as contradiction energy

We close this chapter by summarising the interpretative layer associated with W , explicitly marked as such.

Remark 12.7.1 (Interpretative reading of W).

- *Contradiction energy.* For a semantic configuration $s \in S$, $W(s)$ is interpreted as a scalar measure of semantic contradiction or incoherence, relative to the universal core K and to the semantic universe encoded by W_{univ} . Lower $W(s)$ corresponds to higher semantic coherence and integrity.
- *Directional resolution of contradiction.* The gradient $-\nabla W(s)$ points, in S , in the direction of fastest local reduction of contradiction energy. Its decomposition into directed gradients along the C/B/A axes describes whether resolving contradiction primarily affects coherence, binding, or agency.
- *Universal vs. local norms.* The universal component W_{univ} encodes norms of coherence that are meant to be shared across systems in the semantic universe; W_{loc} encodes system-specific or context-specific norms. High W_{loc} with low W_{univ} indicates local tension but globally coherent semantics; high W_{univ} indicates global misalignment with K , even if local norms are satisfied.
- *Composite systems and relational tension.* For composite systems, interaction terms in W quantify contradictions that arise not within subsystems but between them. These terms are closely related to the relational part of the K-P-R ontology and can be decomposed along C/B/A axes.

These interpretative statements go beyond what is strictly entailed by the mathematical definitions; they should be regarded as programme-level suggestions for how W may be empirically operationalised and conceptually linked to human-level notions of conflict, coherence, and alignment.

Summary and outlook. In this chapter we have consolidated the structural role of the contradiction-energy potential $W : S \rightarrow \mathbb{R}$ in the semantic gravitation framework. We stated its regularity, coercivity, and RKHS-compatibility assumptions; introduced a universal-local decomposition $W = W_{\text{univ}} + W_{\text{loc}}$ and a minimality postulate for W_{univ} ; related W_{univ} to the C/B/A axis structure and ten curvature components via a quadratic expansion; and outlined how W extends to composite systems and to distributions via a semantic free-energy functional. We also clarified the interpretative status of W as contradiction energy. The next chapter builds on this foundation by deriving and systematising the *dynamical equations on S* : deterministic and stochastic gradient flows, distorted dynamics, and their representation in terms of the axis-decomposed gradients and curvature components developed so far.

Chapter 13

Dynamical equations on the semantic space S

In the previous chapters we have introduced the contradiction-energy potential $W : S \rightarrow \mathbb{R}$, the global attractor $K \subset S$, the C/B/A axis decomposition of the local geometry of W , and the ten curvature components describing how W bends near K . In this chapter we return to dynamics. We formulate the deterministic and stochastic evolution equations on S in a way that is explicitly resolved along the C/B/A axes, and we introduce an additional mechanism of *coherence intensification* along the C -axis. This prepares the ground for the next chapter, where we pass from trajectories on S to probability distributions and Fokker–Planck dynamics.

13.1 Deterministic gradient dynamics in C/B/A form

We recall the basic gradient system

$$\dot{s}_t = -\nabla W(s_t), \quad (13.1.1)$$

on the semantic state space S , under Assumption 12.2.1. In a neighbourhood U of the global attractor K , the C/B/A decomposition yields orthogonal projections

$$P_C(s), P_B(s), P_A(s) : S \rightarrow S,$$

onto subspaces $E_C(s), E_B(s), E_A(s)$, and the gradient can be decomposed as

$$\nabla W(s) = \nabla_C W(s) + \nabla_B W(s) + \nabla_A W(s),$$

where

$$\nabla_C W(s) := P_C(s) \nabla W(s), \quad \nabla_B W(s) := P_B(s) \nabla W(s), \quad \nabla_A W(s) := P_A(s) \nabla W(s).$$

Definition 13.1.1 (C/B/A-resolved deterministic dynamics). In U , the deterministic gradient dynamics (13.1.1) can be written as

$$\dot{s}_t = -\nabla_C W(s_t) - \nabla_B W(s_t) - \nabla_A W(s_t), \quad (13.1.2)$$

up to any residual component outside $E_C(s_t) \oplus E_B(s_t) \oplus E_A(s_t)$.

Equation (13.1.2) makes explicit that the semantic gradient flow is a superposition of three axis-aligned flows, corresponding to descending directed gradients G_{-C} , G_{-B} , G_{-A} defined earlier. The relative magnitudes of these components determine whether the system is locally coherence-, binding-, or agency-dominated.

Remark 13.1.2. The C/B/A decomposition is local and depends on s through the projections $P_C(s)$, $P_B(s)$, $P_A(s)$. As long as s_t remains in a neighbourhood of K where Hypothesis 4.9.2 holds, the representation (13.1.2) is well-defined and yields meaningful axis-resolved dynamics.

13.2 Deterministic dynamics with axis-resolved distortions

We now incorporate distortion factors into the deterministic dynamics, using the C/B/A decomposition.

Let $F_{\text{ideal}}(s) := -\nabla W_{\text{univ}}(s)$ denote the ideal universal drift, and let $F(s)$ denote the actual drift, including local potentials and non-conservative contributions. In general,

$$\dot{s}_t = F(s_t).$$

Definition 13.2.1 (Axis-resolved distortion fields). For $s \in U$, the *axis-resolved distortion fields* are defined by

$$\begin{aligned} D_C(s) &:= P_C(s) (F(s) - F_{\text{ideal}}(s)), \\ D_B(s) &:= P_B(s) (F(s) - F_{\text{ideal}}(s)), \\ D_A(s) &:= P_A(s) (F(s) - F_{\text{ideal}}(s)). \end{aligned}$$

Definition 13.2.2 (C/B/A-resolved distorted dynamics). In a neighbourhood of K , the distorted deterministic dynamics can be written as

$$\dot{s}_t = -\nabla_C W_{\text{univ}}(s_t) - \nabla_B W_{\text{univ}}(s_t) - \nabla_A W_{\text{univ}}(s_t) + D_C(s_t) + D_B(s_t) + D_A(s_t). \quad (13.2.1)$$

Here

$$\nabla_{\Xi} W_{\text{univ}}(s) := P_{\Xi}(s) \nabla W_{\text{univ}}(s), \quad \Xi \in \{C, B, A\},$$

and D_{Ξ} collects all deviations along axis Ξ . In particular, potential distortions from W_{loc} and non-gradient contributions from curl fields appear in the D_{Ξ} .

Remark 13.2.3. Equation (13.2.1) shows how distortions can act differently along the three axes. For example, a system may exhibit strong agency-related distortions (D_A large) while remaining nearly ideal along coherence and binding (D_C, D_B small), or vice versa. This axis-resolved structure will later be mirrored at the distributional level in the Fokker–Planck equation.

13.3 Stochastic dynamics: Langevin-type equations on S

To model noise and unresolved degrees of freedom, we consider stochastic differential equations (SDEs) on S of Langevin type (as in Paper 1), now with axis-resolved structure.

Definition 13.3.1 (Langevin dynamics on S). Let $(B_t)_{t \geq 0}$ be a cylindrical Brownian motion on S , and let $\Sigma(s) : S \rightarrow \mathcal{L}(S)$ be a bounded linear operator (noise amplitude). A *Langevin-type semantic dynamics* on S is given by

$$ds_t = \left(-\nabla W(s_t) \right) dt + \sqrt{2\beta^{-1}} \Sigma(s_t) dB_t, \quad (13.3.1)$$

with $\beta > 0$ an inverse temperature parameter.

We now resolve both drift and noise along C/B/A.

Definition 13.3.2 (Axis-resolved stochastic dynamics). Assume that $\Sigma(s)$ admits a decomposition

$$\Sigma(s) = \Sigma_C(s) + \Sigma_B(s) + \Sigma_A(s),$$

with $\Sigma_\Xi(s) : S \rightarrow S$ mapping predominantly into $E_\Xi(s)$ for $\Xi \in \{C, B, A\}$. Then the SDE (13.3.1) can be written (in U) as

$$ds_t = (-\nabla_C W(s_t) - \nabla_B W(s_t) - \nabla_A W(s_t)) dt + \sqrt{2\beta^{-1}} (\Sigma_C(s_t) + \Sigma_B(s_t) + \Sigma_A(s_t)) dB_t, \quad (13.3.2)$$

up to residual components.

Remark 13.3.3. The decomposition of $\Sigma(s)$ need not be unique. In simple models one may take $\Sigma_\Xi(s)$ to be the composition of $\Sigma(s)$ with $P_\Xi(s)$, so that $\Sigma_\Xi(s) = P_\Xi(s)\Sigma(s)$. More elaborate constructions may assign distinct noise amplitudes to each axis.

The SDE (13.3.2) combines three descending axis gradients with possibly anisotropic noise along each axis. In regimes where, for example, Σ_C is small compared to Σ_B and Σ_A , the system is driven towards coherence deterministically while exploration occurs mainly in binding and agency directions.

13.4 Coherence intensification along the C -axis

We now introduce an additional mechanism that will be relevant in later chapters: a multiplicative scaling of the coherence gradient, which we call *coherence intensification* along the C -axis.

Definition 13.4.1 (Coherence intensification operator). Let

$$\Gamma_C : U \rightarrow (0, \infty)$$

be a scalar function, and similarly let $\Gamma_B, \Gamma_A : U \rightarrow (0, \infty)$. Define the *axis-intensification operator* at $s \in U$ by

$$\Gamma(s) := \Gamma_C(s) P_C(s) + \Gamma_B(s) P_B(s) + \Gamma_A(s) P_A(s).$$

Definition 13.4.2 (Intensified gradient dynamics). The *intensified deterministic dynamics* on U is given by

$$\dot{s}_t = -\Gamma(s_t) \nabla W(s_t) = -\Gamma_C(s_t) \nabla_C W(s_t) - \Gamma_B(s_t) \nabla_B W(s_t) - \Gamma_A(s_t) \nabla_A W(s_t). \quad (13.4.1)$$

Definition 13.4.3 (Coherence intensification regime). We say that the system operates in a *coherence intensification regime* if

$$\Gamma_C(s) \gg \Gamma_B(s), \Gamma_A(s)$$

on a neighbourhood of K (in a suitable quantitative sense), so that the coherence gradient is amplified relative to binding and agency gradients.

Remark 13.4.4. Coherence intensification does not modify the location of the global attractor K (as long as $\Gamma_C, \Gamma_B, \Gamma_A > 0$) but changes the *rates* at which different directions of contradiction energy are resolved. In the limit of very large Γ_C , the system rapidly reduces C -axis contradiction before addressing B - and A -axis contributions, effectively enforcing a hierarchy: coherence first, then binding, then agency.

The intensification operator can also be included in the stochastic dynamics:

Definition 13.4.5 (Intensified Langevin dynamics). The *intensified Langevin dynamics* is given by

$$ds_t = -\Gamma(s_t) \nabla W(s_t) dt + \sqrt{2\beta^{-1} \Sigma(s_t)} dB_t, \quad (13.4.2)$$

with Γ as in Definition 13.4.1.

13.5 Axis-resolved energy balances

We briefly indicate how the C/B/A decomposition and intensification affect energy balances. Working in the quadratic regime near K and ignoring cross-terms and distortions for simplicity, we approximate

$$W(s) \approx W(\pi_K(s)) + W_C(s) + W_B(s) + W_A(s),$$

where

$$W_\Xi(s) = \frac{1}{2} \langle H_{\Xi\Xi}(\pi_K(s)) \delta_\Xi(s), \delta_\Xi(s) \rangle, \quad \Xi \in \{C, B, A\}.$$

Proposition 13.5.1 (Axis energy dissipation under intensified dynamics). *Assume that near K :*

1. $H_{\Xi\Xi}(\pi_K(s))$ is positive semi-definite for each axis Ξ ;
2. cross-axis couplings and distortions are negligible;
3. dynamics is given by (13.4.1).

Then, to leading order,

$$\frac{d}{dt} W_\Xi(s_t) \approx -\Gamma_\Xi(s_t) \|\nabla_\Xi W(s_t)\|^2 \leq 0,$$

for $\Xi \in \{C, B, A\}$.

Proof sketch. The proof is analogous to Proposition 10.4.1, with an additional factor $\Gamma_\Xi(s_t)$ coming from the intensification operator. The quadratic approximation implies $\nabla_\Xi W(s) \approx H_{\Xi\Xi} \delta_\Xi(s)$ and $W_\Xi(s) \approx \frac{1}{2} \langle H_{\Xi\Xi} \delta_\Xi(s), \delta_\Xi(s) \rangle$. Differentiating $W_\Xi(s_t)$ along (13.4.1) and retaining leading-order terms yields the stated result. $\square$

Proposition 13.5.1 confirms that the intensification factors Γ_Ξ rescale dissipation rates along each axis without changing the sign of dissipation, as long as $\Gamma_\Xi > 0$.

Remark 13.5.2. In the stochastic setting (13.4.2), the energy balance includes additional terms from noise, leading to Itô corrections and a competition between drift-induced dissipation and noise-induced diffusion. The axis-resolved structure still applies, but with expectations over noise and additional terms in the evolution of $\mathbb{E}[W_\Xi(s_t)]$. These will be naturally expressed in the Fokker–Planck framework of the next chapter.

13.6 General form of the C/B/A-resolved evolution equations

Collecting the above ingredients, a generic form of the C/B/A-resolved dynamics on S can be written schematically as

$$ds_t = -\Gamma(s_t) \nabla W_{\text{univ}}(s_t) dt + D(s_t) dt + \sqrt{2\beta^{-1}} \Sigma(s_t) dB_t, \quad (13.6.1)$$

where:

- $\Gamma(s)$ is the axis-intensification operator;
- $D(s)$ is a distortion field, decomposable into axis components D_C, D_B, D_A ;
- $\Sigma(s)$ is a noise operator, also decomposable into axis components.

The idealised limits discussed earlier correspond to:

- *Pure universal gradient flow*: $\Gamma(s) = \text{id}$, $D = 0$, $\Sigma = 0$.
- *Axis-intensified gradient flow*: Γ non-trivial, $D = 0$, $\Sigma = 0$.
- *Distorted deterministic dynamics*: $\Gamma = \text{id}$, $D \neq 0$, $\Sigma = 0$.
- *Full Langevin dynamics with distortions and intensification*: general (13.6.1).

13.7 Semantic decoherence, openness loss, and collapse rates

The C/B/A-resolved dynamics introduced above describe how semantic states evolve under contradiction-energy descent, distortion fields, and stochastic forcing. In many relevant systems, however, it is not sufficient to ask only whether contradiction-energy decreases. One must also ask whether the semantic state remains *open* in the sense of preserving a nontrivial set of coherent, relationally viable, and practically reparable continuations.

We therefore introduce an additional derived quantity: the *semantic decoherence rate*. Informally, this quantity measures how rapidly an initially open semantic state is driven into a narrower, less reparable, less relationally viable, or more rigid continuation corridor.

Let $s \in S$ be a semantic state and let $\mathcal{A}(s)$ denote its admissible continuation set, understood as the family of semantically viable continuations compatible with the current contradiction-energy landscape, projection constraints, and relevant relational conditions. We do not assume that $\mathcal{A}(s)$ is represented by literal branch enumeration. It may be given by local trajectory cones, coarse-grained continuation classes, viable successor regions, or any functionally equivalent representation.

Definition 13.7.1 (Semantic openness). A *semantic openness functional* is a map

$$\mathcal{O} : S \rightarrow [0, \infty)$$

that assigns to each state s a scalar measure of the width, viability, reparability, or structured multiplicity of its admissible continuation set $\mathcal{A}(s)$.

The role of $\mathcal{O}(s)$ is not to reward undifferentiated ambiguity. A state can be broad but semantically unusable. Accordingly, $\mathcal{O}(s)$ is understood to measure *structured openness*: the availability of continuations that remain compatible with coherence, relational viability, and non-destructive further development.

Definition 13.7.2 (Semantic decoherence rate). Let $t \mapsto s_t$ be a semantic trajectory generated by one of the deterministic or stochastic dynamics introduced above. The *semantic decoherence rate* along the trajectory is defined by

$$\Gamma_{\text{dec}}(t) := -\frac{d}{dt} \mathcal{O}(s_t),$$

whenever the derivative exists. More generally, for coarse-grained or stochastic regimes, one may use the expectation-valued form

$$\Gamma_{\text{dec}}(t) := -\frac{d}{dt} \mathbb{E}[\mathcal{O}(s_t)].$$

Thus:

- $\Gamma_{\text{dec}}(t) > 0$ indicates loss of semantic openness;
- $\Gamma_{\text{dec}}(t) = 0$ indicates openness preservation at the chosen scale;
- $\Gamma_{\text{dec}}(t) < 0$ indicates re-opening, repair, or structured widening of the admissible continuation set.

The semantic decoherence rate is not identified with contradiction-energy itself. A state may reduce contradiction-energy while still collapsing prematurely into a narrow continuation corridor. Conversely, a local increase in admissible variation may be semantically beneficial if it restores reparability or prevents destructive lock-in. The decoherence rate therefore tracks a distinct structural aspect of semantic dynamics.

Definition 13.7.3 (Reparability-sensitive openness). Let $\mathcal{R}(s) \in [0, \infty)$ denote a reparability functional measuring the degree to which a local semantic narrowing can still be revised, reopened, or redirected without major loss of coherence or relational viability. A reparability-sensitive openness is any functional of the form

$$\mathcal{O}_{\text{eff}}(s) := \Phi(\mathcal{O}(s), \mathcal{R}(s)),$$

with Φ monotone in both arguments.

Using $\mathcal{O}_{\text{eff}}$, one distinguishes between at least three structurally different regimes:

1. **Coherent concentration:** openness decreases locally, but reparability remains high and contradiction-energy decreases in a structurally stabilising way;
2. **Repairable narrowing:** openness decreases and some continuation modes are lost, but the state remains capable of re-opening under admissible interventions;
3. **Destructive semantic collapse:** openness decreases rapidly, reparability drops, and the state is forced into a narrow continuation corridor whose later reversal is costly or structurally unlikely.

To connect semantic decoherence to the local geometry near K , we introduce an axis-resolved decomposition.

Definition 13.7.4 (Axis-resolved semantic decoherence). Let the local C/B/A decomposition near K be defined as in Chapters 9–11. Then an axis-resolved semantic decoherence rate is any decomposition

$$\Gamma_{\text{dec}}(t) = \Gamma_C(t) + \Gamma_B(t) + \Gamma_A(t) + \Gamma_{\text{mix}}(t),$$

where $\Gamma_C, \Gamma_B, \Gamma_A$ represent openness loss attributable primarily to coherence-axis, binding-axis, and agency-axis degradation, respectively, and Γ_{mix} captures cross-axis coupling effects.

Interpretatively:

- large Γ_C corresponds to collapse through loss of semantic clarity, coherence, or internal orientability;
- large Γ_B corresponds to collapse through relational damage, dignity loss, disconnection, or binding failure;
- large Γ_A corresponds to collapse through loss of freedom, initiative, agency, or viable action-space.

This decomposition allows one to distinguish different kinds of semantic narrowing that would otherwise appear identical at the level of a single global score.

A further useful quantity is the *collapse susceptibility*, defined schematically by

$$\chi_{\text{coll}}(s) := \Psi(\mathcal{O}(s), \mathcal{R}(s), \|\nabla W(s)\|, \|D(s)\|, \kappa_{CB}(s), \kappa_{CA}(s), \kappa_{BA}(s), \dots),$$

where $D(s)$ denotes distortion contributions and the κ -terms denote the local cross-axis curvature components. High collapse susceptibility indicates that small external forcing, distortion, or conflict pressure may strongly accelerate semantic decoherence.

The semantic decoherence rate is especially useful when evaluating possible continuation steps. Let $u \in \mathcal{U}(s)$ denote an admissible intervention, answer, question, or control step applied at state s , and let $s \mapsto T_u(s)$ denote the resulting update. Then one may define a one-step decoherence forecast

$$\Delta_u \mathcal{O}_{\text{eff}}(s) := \mathcal{O}_{\text{eff}}(T_u(s)) - \mathcal{O}_{\text{eff}}(s),$$

and thereby compare candidate continuation steps not only by immediate contradiction reduction but also by whether they preserve or destroy future semantic viability.

This suggests a derived navigation principle: among continuation steps with comparable contradiction-reducing effect, prefer those that minimise unnecessary semantic decoherence and preserve repairable openness as long as possible.

The notion of semantic decoherence introduced here does not add a new microscopic law. It is a derived diagnostic observable on top of the existing semantic dynamics. Its role is to distinguish mere descent in contradiction-energy from structurally healthy continuation.

In this sense, semantic decoherence provides a bridge between local gradient descent, axis-resolved distortion analysis, and the branching-closure structures studied in the next chapters. It allows us to formulate not only whether a trajectory descends, but whether it descends in a way that preserves semantic freedom, relational viability, and later recoverability.

Summary and outlook. In this chapter we have formulated the *dynamical equations on the semantic space S* in a form explicitly resolved along the three coherence axes C , B , and A . We decomposed the deterministic gradient flow into axis gradients, introduced axis-resolved distortion fields, and expressed Langevin-type stochastic dynamics with axis-resolved drift and noise. We also added an intensification operator Γ that amplifies or attenuates gradients along each axis, with particular emphasis on coherence intensification along the C -axis. We then introduced semantic decoherence, openness loss, and collapse rates as derived diagnostic quantities that distinguish mere contradiction reduction from structurally healthy continuation. Finally, we derived axis-resolved energy balance relations in the quadratic regime near K and summarised the general C/B/A-resolved SDE governing semantic dynamics on S . In the next chapter we move from trajectories to distributions:

we derive the corresponding Fokker–Planck equations for probability measures on S , incorporate the C/B/A structure into their generator, and introduce a semantic tensor description of the evolving contradiction-energy landscape.

Chapter 14

Distributional level: Fokker–Planck and semantic temperature

In the previous chapter we described deterministic and stochastic dynamics on the semantic state space S at the level of individual trajectories $t \mapsto s_t \in S$, with explicit C/B/A decomposition and axis-resolved distortions. In this chapter we move to the *distributional* level: instead of tracking single trajectories we consider the evolution of probability measures on S . Formally, this is governed by a Fokker–Planck (or forward Kolmogorov) equation associated with the Langevin dynamics from the preceding chapter. We adapt this equation to the C/B/A structure, introduce the notions of semantic temperature and semantic free energy, and discuss metastable distributions near the global attractor K and within order bubbles.

14.1 Probability measures on S and densities

We denote by $\mathcal{P}(S)$ the space of Borel probability measures on the separable Hilbert space S . In what follows we restrict attention to measures that are absolutely continuous with respect to a fixed reference measure μ_0 on S ; intuitively, μ_0 plays the role of a “prior” or background measure (for example, a Gaussian measure adapted to the RKHS structure, as in Paper 1).

Assumption 14.1.1 (Reference measure and densities). There exists a Borel probability measure μ_0 on S such that:

1. for each $t \geq 0$ the law μ_t of the semantic state s_t is absolutely continuous with respect to μ_0 ;
2. the Radon–Nikodym derivative

$$\rho_t := \frac{d\mu_t}{d\mu_0}$$

exists and is sufficiently regular (for example, in $L^1(\mu_0)$ and differentiable in t in a weak sense).

We call ρ_t the *semantic density* at time t .

Under Assumption 14.1.1, the evolution of μ_t can be described by an evolution equation for the density ρ_t with respect to μ_0 . We now derive this equation formally from the Langevin dynamics.

14.2 Langevin dynamics and the Fokker–Planck equation on S

Recall the general C/B/A-resolved Langevin dynamics on S from the previous chapter,

$$ds_t = -\Gamma(s_t) \nabla W_{\text{univ}}(s_t) dt + D(s_t) dt + \sqrt{2\beta^{-1}} \Sigma(s_t) dB_t, \quad (14.2.1)$$

where:

- $W_{\text{univ}} : S \rightarrow \mathbb{R}$ is the universal component of the contradiction-energy potential;
- $\Gamma(s)$ is the axis-intensification operator, decomposed as

$$\Gamma(s) = \Gamma_C(s) P_C(s) + \Gamma_B(s) P_B(s) + \Gamma_A(s) P_A(s);$$

- $D(s)$ is a distortion field (possibly decomposed along C/B/A);
- $\Sigma(s) : S \rightarrow \mathcal{L}(S)$ is a noise amplitude operator;
- $(B_t)_{t \geq 0}$ is a cylindrical Brownian motion on S ;
- $\beta > 0$ is an inverse temperature parameter.

Formally, the Markov process $(s_t)_{t \geq 0}$ defined by (14.2.1) admits a (pre-)generator $\mathcal{L}$ acting on sufficiently smooth test functions $\varphi : S \rightarrow \mathbb{R}$ by

$$(\mathcal{L}\varphi)(s) = \langle -\Gamma(s) \nabla W_{\text{univ}}(s) + D(s), \nabla \varphi(s) \rangle + \beta^{-1} \text{Tr}(Q(s) \nabla^2 \varphi(s)), \quad (14.2.2)$$

where $Q(s) := \Sigma(s) \Sigma(s)^*$ is a symmetric, positive semi-definite covariance operator, and $\nabla^2 \varphi(s)$ denotes the Hessian of φ at s (see Paper 1 for a rigorous construction in related settings).

Definition 14.2.1 (Fokker–Planck equation on S). The *Fokker–Planck equation* associated with (14.2.1) is the forward Kolmogorov equation

$$\partial_t \rho_t = \mathcal{L}^* \rho_t, \quad (14.2.3)$$

where $\mathcal{L}^*$ denotes the adjoint of $\mathcal{L}$ with respect to the inner product

$$\langle f, g \rangle_{L^2(\mu_0)} := \int_S f(s) g(s) d\mu_0(s).$$

Equivalently, for all suitable test functions φ ,

$$\frac{d}{dt} \int_S \varphi(s) \rho_t(s) d\mu_0(s) = \int_S (\mathcal{L}\varphi)(s) \rho_t(s) d\mu_0(s). \quad (14.2.4)$$

In infinite-dimensional settings the Fokker–Planck equation is often best understood in the weak form (14.2.4). For formal intuition, one can also write (14.2.3) in divergence form,

$$\partial_t \rho_t(s) = -\nabla \cdot J_t(s), \quad (14.2.5)$$

where J_t is the probability flux (or current) on S and $\nabla \cdot$ is an appropriate divergence operator with respect to μ_0 . The precise analytic form depends on the choice of μ_0 and the structure of $Q(s)$ (see Paper 1 for examples). In particular, the sign convention in (14.2.5) is the standard continuity convention: increasing divergence of the flux decreases density.

14.3 C/B/A-resolved Fokker–Planck structure

The C/B/A decomposition of the dynamics induces a corresponding decomposition of the generator and of the probability flux.

For simplicity, we first consider the idealised case with $\Gamma(s) = \text{id}$, $D(s) = 0$, and $\Sigma(s)$ independent of s , equal to a fixed covariance operator Σ_0 . The generator then simplifies to

$$(\mathcal{L}_0\varphi)(s) = -\langle \nabla W_{\text{univ}}(s), \nabla\varphi(s) \rangle + \beta^{-1} \text{Tr}(Q_0 \nabla^2\varphi(s)), \quad (14.3.1)$$

with $Q_0 := \Sigma_0 \Sigma_0^*$.

Definition 14.3.1 (C/B/A-resolved gradients and diffusion). Using the projections $P_C(s)$, $P_B(s)$, $P_A(s)$ we define

$$\nabla_C\varphi(s) := P_C(s) \nabla\varphi(s), \quad \nabla_B\varphi(s) := P_B(s) \nabla\varphi(s), \quad \nabla_A\varphi(s) := P_A(s) \nabla\varphi(s),$$

and decompose the diffusion operator as

$$Q_0 = Q_C + Q_B + Q_A,$$

with each Q_Ξ acting primarily in $E_\Xi(s)$ for $\Xi \in \{C, B, A\}$.

Formally, the flux J_t induced by (14.3.1) can be decomposed as

$$J_t(s) = J_t^{(C)}(s) + J_t^{(B)}(s) + J_t^{(A)}(s), \quad (14.3.2)$$

$$J_t^{(\Xi)}(s) := -\rho_t(s) \nabla_\Xi W_{\text{univ}}(s) - \beta^{-1} Q_\Xi \nabla_\Xi \rho_t(s), \quad \Xi \in \{C, B, A\}, \quad (14.3.3)$$

where $\nabla_\Xi \rho_t$ is the axis-projected gradient of the density. Inserting (14.3.2) into (14.2.5) yields a C/B/A-resolved Fokker–Planck equation,

$$\partial_t \rho_t(s) = - \sum_{\Xi \in \{C, B, A\}} \nabla_\Xi \cdot J_t^{(\Xi)}(s), \quad (14.3.4)$$

where $\nabla_\Xi \cdot$ denotes the divergence restricted to $E_\Xi(s)$ (formally the adjoint of ∇_Ξ with respect to μ_0).

Remark 14.3.2. Equation (14.3.4) is schematic and should be read as a guiding structure: rigorous versions require precise definitions of ∇_Ξ , $\nabla_\Xi \cdot$, and the domain of $\mathcal{L}_0^*$. The main point is that drift and diffusion along the three coherence axes can be separated in a meaningful way near K , extending the axis-resolved trajectory-level dynamics to the distributional level.

When intensification Γ and distortions D are present, they appear as additional drift terms in each axis component of the flux; similarly, state-dependent noise $\Sigma(s)$ yields axis-dependent diffusion coefficients. The same C/B/A decomposition applies, but with more complicated coefficients.

14.4 Semantic temperature and equilibrium measures

We now introduce semantic temperature and characterise equilibrium measures of the Fokker–Planck equation in the ideal case.

Definition 14.4.1 (Semantic temperature). Let $\beta > 0$ be the inverse-temperature parameter in the Langevin dynamics (14.2.1). We define the *semantic temperature* as

$$T_{\text{sem}} := \beta^{-1}.$$

Smaller T_{sem} corresponds to weaker stochastic fluctuations (stronger concentration of probability near minima of W_{univ}); larger T_{sem} corresponds to stronger diffusion in S .

Remark 14.4.2 (Identification with the diffusion parameter in earlier notation). In parts of the programme (e.g. Paper II) the stochasticity level is parametrised by a diffusion/noise scale $D > 0$ via stationary densities of the form $\propto \exp(-W/D)$. In the present notation this corresponds to the identification

$$D = \beta^{-1} = T_{\text{sem}},$$

up to any fixed numerical factors already absorbed into the definition of the noise term (here chosen as $\sqrt{2\beta^{-1}} \Sigma \mathrm{d}B_t$).

In the idealised setting where $\Gamma = \text{id}$, $D = 0$, and Q_0 is appropriately compatible with μ_0 (for example, μ_0 is a Gaussian measure invariant under the pure diffusion part), one expects equilibrium measures of the Gibbs form

$$\mu_\beta(\mathrm{d}s) = \frac{1}{Z_\beta} \exp(-\beta W_{\text{univ}}(s)) \mu_0(\mathrm{d}s), \quad (14.4.1)$$

where

$$Z_\beta := \int_S \exp(-\beta W_{\text{univ}}(s)) \mathrm{d}\mu_0(s)$$

is a normalising partition function (assuming it is finite).

Assumption 14.4.3 (Gibbs equilibrium). In the idealised case (no distortions, intensification $\Gamma \equiv \text{id}$, state-independent noise with suitable symmetry), we assume that:

1. the Gibbs measure (14.4.1) is well-defined (finite Z_β) and normalised;
2. μ_β is stationary for the Fokker–Planck equation (14.2.3), i.e. $\mathcal{L}^* \rho_\beta = 0$ with $\rho_\beta = \mathrm{d}\mu_\beta / \mathrm{d}\mu_0$;
3. μ_β is the unique stationary distribution in the class of probability measures sharing basic integrability properties.

Assumption 14.4.3 is standard in Langevin-type models (see Paper 1 for finite-dimensional and some infinite-dimensional examples). It allows us to link semantic temperature and contradiction-energy potential via equilibrium measures.

14.5 Semantic free energy and gradient flows on $\mathcal{P}(S)$

In the previous chapter we defined the semantic free-energy functional

$$\mathcal{F}[\mu] = \int_S W(s) \mathrm{d}\mu(s) + \beta^{-1} \mathcal{H}(\mu \mid \mu_0),$$

where $\mathcal{H}(\mu \mid \mu_0)$ is the relative entropy (Kullback–Leibler divergence) of μ with respect to μ_0 . We now recall how this functional relates to the Fokker–Planck equation.

Under suitable conditions on W and μ_0 , the Fokker–Planck equation can be interpreted as a (generalised) gradient flow of $\mathcal{F}$ on the Wasserstein space of probability measures on S (see Paper 2 for concrete formulations in finite-dimensional and Hilbert-space settings). Formally, we can write

$$\partial_t \mu_t = -\text{grad}_{\mathcal{W}} \mathcal{F}[\mu_t], \quad (14.5.1)$$

where $\text{grad}_{\mathcal{W}}$ denotes a gradient with respect to a Wasserstein metric on $\mathcal{P}(S)$ (the precise construction is non-trivial in infinite dimensions and will not be repeated here).

Remark 14.5.1. The gradient-flow structure (14.5.1) implies monotone decay of $\mathcal{F}[\mu_t]$ in time along solutions of the Fokker–Planck equation, under appropriate assumptions:

$$\frac{d}{dt} \mathcal{F}[\mu_t] \leq 0.$$

The semantic free energy thus plays the role of a Lyapunov functional at the distributional level: it measures a combination of average contradiction energy and dispersion (entropy) of the semantic state. Semantic temperature T_{sem} controls the trade-off between these two contributions.

In the universal-local decomposition $W = W_{\text{univ}} + W_{\text{loc}}$, one can split the semantic free energy into a universal part, which shares the gradient-flow structure, and a local part, which acts as a correction or perturbation. This mirrors the decomposition of drift into ideal and distorted components at the trajectory level.

Order–Entropy Duality

The distributional dynamics exhibits a structural duality between entropy and order formation.

Entropy increase corresponds to effective path proliferation in the semantic state space: as stochastic fluctuations dominate, the support of μ_t spreads across larger regions of S , increasing the effective number of admissible continuation paths.

Order formation, by contrast, corresponds to attractor-guided path compression: gradient drift toward basins of W_{univ} contracts probability mass into structured regions.

Formally:

- Entropy growth $\iff$ expansion of effective support volume in S .
- Order formation $\iff$ contraction of μ_t under attractor-aligned drift.

The thermodynamic arrow reflects the dynamical tension between stochastic branching and attractor-guided closure, arising from diffusion and gradient descent in the semantic free-energy landscape.

14.6 Metastable distributions and order bubbles

We now discuss metastable behaviour at the distributional level, in particular the formation of *metastable distributions* concentrated in order bubbles (Definition 7.3.1) rather than near K .

Definition 14.6.1 (Metastable distribution). Let $B \subset S$ be an order bubble for the dynamics on S . A probability measure $\mu^{(B)} \in \mathcal{P}(S)$ is called a *metastable distribution supported on B* if:

1. $\mu^{(B)}(B) \approx 1$ and $\mu^{(B)}(S \setminus B) \approx 0$;
2. $\mu^{(B)}$ is approximately stationary for the Fokker–Planck equation in the sense that

$$\|\mathcal{L}^* \rho^{(B)}\|_{-1} \ll 1,$$

where $\rho^{(B)} = d\mu^{(B)}/d\mu_0$ and $\|\cdot\|_{-1}$ is an appropriate negative Sobolev norm;

3. exit times from B under the dynamics are large compared to intrinsic time scales within B .

Metastable distributions are local minima (or flat regions) of the semantic free energy $\mathcal{F}$, separated from the global minima near K by energy barriers controlled by W_{univ} and curvature components.

Remark 14.6.2 (Curvature and metastability). The ten curvature components introduced earlier provide local information about the shape of W near K and near order bubbles. High curvature along an axis implies rapid return to local equilibrium; low or negative curvature implies slow relaxation or local instability along that axis. In particular:

- Near K , large positive axis curvatures and small coupling curvatures favour rapid concentration of μ_t near K .
- Within an order bubble B , strong local curvature (often dominated by W_{loc}) can produce long-lived metastable distributions $\mu^{(B)}$, even if W_{univ} remains high there.
- The relative sizes of curvature components and semantic temperature determine whether probability mass can cross barriers between K -regions and order bubbles on relevant time scales.

These qualitative statements are in line with standard metastability theory for diffusion processes in multi-well potentials, adapted to the semantic context.

Open-Ended Constraint Deepening

As relational integration density increases in the semantic field, the effective internal path length between semantic states grows. This induces increasing structural self-referentiality within the configuration.

Consequences include:

- finer attractor stratification,
- increased curvature sensitivity of trajectories,
- and higher internal integration of semantic substructures.

This process is dynamical rather than teleological: it reflects recursive constraint accumulation in the evolving semantic geometry, not the approach toward a predefined final state.

14.7 Axis-resolved semantic currents and temperatures

The C/B/A decomposition allows us to refine the description of distributional dynamics further, by defining axis-resolved probability currents and effective temperatures.

Definition 14.7.1 (Axis-resolved semantic currents). In the idealised setting with generator $\mathcal{L}_0$ and flux decomposition (14.3.2), the *axis-resolved currents* are

$$J_t^{(\Xi)}(s) = -\rho_t(s) \nabla_{\Xi} W_{\text{univ}}(s) - \beta^{-1} Q_{\Xi} \nabla_{\Xi} \rho_t(s), \quad \Xi \in \{C, B, A\}.$$

They describe how probability mass flows along the coherence, binding, and agency axes, respectively.

Definition 14.7.2 (Axis-specific effective temperatures). If the diffusion operators Q_{Ξ} scale differently along the axes (for example, due to anisotropic noise), we can define *axis-specific effective temperatures* by

$$T_{\text{sem}}^{(\Xi)} := \beta_{\Xi}^{-1},$$

where $\beta_{\Xi} > 0$ is an axis-specific inverse-temperature parameter encoded in the prefactor of Q_{Ξ} (for instance, by writing $Q_{\Xi} = \beta_{\Xi}^{-1} \tilde{Q}_{\Xi}$ with $\tilde{Q}_{\Xi}$ fixed).

Remark 14.7.3 (Speculative). At a programme level, one may interpret differences between $T_{\text{sem}}^{(C)}$, $T_{\text{sem}}^{(B)}$, and $T_{\text{sem}}^{(A)}$ as reflecting differential stochasticity and flexibility along coherence, binding, and agency directions. For instance, a system with $T_{\text{sem}}^{(C)} \ll T_{\text{sem}}^{(B)}, T_{\text{sem}}^{(A)}$ would tend to preserve coherence (low randomness along C) while allowing more exploration in binding and agency patterns.

Summary and outlook. In this chapter we have lifted the semantic gravitation dynamics from trajectories on S to probability measures on S . Starting from the Langevin-type dynamics on S , we introduced the associated Fokker–Planck equation in weak and flux form, adapted it to the C/B/A decomposition by resolving drift and diffusion along the three coherence axes, and identified equilibrium measures of Gibbs type with semantic temperature $T_{\text{sem}} = \beta^{-1}$. We recalled the semantic free-energy functional $\mathcal{F}$ and its role as a Lyapunov functional, discussed metastable distributions supported on order bubbles, and briefly indicated how curvature components influence metastability. Finally, we defined axis-resolved semantic currents and axis-specific effective temperatures as a refined description of distributional dynamics along the C , B , and A axes. In the next chapter we step back from dynamics and distributions to the axiomatic level: we formulate a finite list of axioms for semantic gravitation, integrating S , K – P – R , W , C/B/A, and the dynamical structures introduced so far into a unified foundational scheme.

Chapter 15

Branching, Closure, and the Arrow of Time

15.1 Purpose and structural status

Previous chapters introduced:

- admissible continuation sets $A(x)$ and relevance-weighted sub-ensembles $C(x)$,
- distributional dynamics on S (including Fokker–Planck structure and semantic temperature),
- attractor-guided stabilisation and free-energy-like descent principles,
- and the witness functional Z as a coarse-grained order observable.

What remained implicit is a single structural synthesis unifying:

- entropic branch proliferation,
- attractor-guided closure,
- macroscopic monotonicity,
- and emergent time asymmetry.

This chapter formulates that synthesis.

Compatibility guardrails. All statements below remain compatible with:

- Hamiltonian phase-space dynamics,
- unitary quantum evolution,
- and relativistic field dynamics on (M, g) .

No new microscopic dynamics are introduced. Irreversibility arises solely at the level of coarse-grained projection.

15.2 Continuation ensembles: measure-theoretic formulation

Let $x \in M$ and let $C(x)$ denote the dynamically relevant continuation ensemble.

In full generality, assume:

- a σ -finite measure space $(C(x), \Sigma_x, \mu_x)$,
- a measurable weight function $w(\cdot; x) \geq 0$,
- integrability $\int_{C(x)} w d\mu_x < \infty$.

For computational clarity one may adopt a discrete partition $\{i\}$ of $C(x)$ into measurable cells. All discrete formulas below are understood as representations of this general measurable structure.

15.3 Multiplicity and branch entropy

Two complementary multiplicity notions are useful.

15.3.1 Total weighted multiplicity

Define

$$\mathcal{N}(x) := \int_{C(x)} w d\mu_x. \quad (15.3.1)$$

This is the total weighted continuation mass.

15.3.2 Normalised branch distribution

Under a discrete representation,

$$p_i(x) := \frac{w_i(x)}{\sum_j w_j(x)}. \quad (15.3.2)$$

The Shannon branch entropy is

$$H_{\text{br}}(x) = - \sum_i p_i(x) \log p_i(x). \quad (15.3.3)$$

Define the effective number of branches

$$N_{\text{eff}}(x) := \exp(H_{\text{br}}(x)). \quad (15.3.4)$$

In the normalised representation $\sum_i p_i = 1$, $\mathcal{N} = 1$ and N_{eff} becomes the relevant dispersion-based multiplicity descriptor.

Interpretation. $\mathcal{N}$ captures total weighted continuation mass. N_{eff} captures dispersion across admissible alternatives.

We define the structural branch descriptor

$$S_{\text{branch}}(x) := H_{\text{br}}(x). \quad (15.3.5)$$

This is not identified with thermodynamic entropy. The analogy is purely structural.

15.4 Coarse-grained temporal evolution

Let $x(t)$ denote a timelike parametrisation compatible with g .

All temporal derivatives below are understood:

- either along projected macroscopic trajectories,
- or in expectation under coarse-grained measures.

They do not represent fundamental microscopic generators.

15.5 Branch proliferation and closure

15.5.1 Branch proliferation

Branch proliferation corresponds to

$$\frac{d}{dt} S_{\text{branch}}(x(t)) > 0. \quad (15.5.1)$$

Effective continuation multiplicity increases.

This does not contradict Liouville conservation or unitarity, since S_{branch} is projection-dependent.

15.5.2 Attractor-guided closure

Closure corresponds to

$$\frac{d}{dt} S_{\text{branch}}(x(t)) < 0. \quad (15.5.2)$$

Weight concentrates into smaller effective subsets.

Operational sources of closure include:

- stability constraints,
- symmetry reduction,
- variational/energetic descent,
- semantic consistency filters,
- attractor basin concentration.

Closure reduces effective continuation dimensionality without enumerating branches.

15.6 Witness functional as coherence-order observable

Define a global witness functional

$$Z[(C, w); g, s] = \int_M z(x) d\text{vol}_g(x), \quad (15.6.1)$$

where

$$z(x) = \mathcal{Z}(S_{\text{branch}}(x), \mathcal{K}_{\text{cl}}(x), \nabla s(x), h(x), \dots). \quad (15.6.2)$$

Here $\mathcal{K}_{\text{cl}}$ denotes a closure-strength proxy.

Status of Z .

- Not a new field.
- Not an independent dynamical degree of freedom.
- A derived macroscopic observable.
- Candidate Lyapunov-type quantity.

Remark 15.6.1 (Witness functional as emergent meta-coherence). The witness functional Z does not introduce an additional dynamical degree of freedom. It is an emergent global evaluation functional derived from the semantic free-energy structure.

In particular, Z does not alter the field equations or stochastic dynamics. Rather, it provides a meta-coherence measure of configurations, quantifying their global alignment with the contradiction-energy landscape.

15.6.1 Closure-strength proxy

Define operationally

$$C_{\text{cl}}(x(t)) := -\frac{d}{dt}S_{\text{branch}}(x(t))\Big|_{\text{closure}}. \quad (15.6.3)$$

This isolates entropy reduction attributable to closure rather than noise or resolution changes.

15.7 Unified branching–closure balance hypothesis

(H1) Branching–closure balance. At coarse-grained level, continuation structure evolves under the competition between branch proliferation and attractor-guided closure.

A minimal schematic expression is

$$\frac{d}{dt}Z(x(t)) = \alpha S_{\text{branch}}(x(t)) - \beta C_{\text{cl}}(x(t)), \quad \alpha, \beta > 0. \quad (15.7.1)$$

The sign convention depends on whether Z is chosen as disorder- or order-type witness. Only monotonicity properties matter.

Closure-monotonicity condition. In closure-dominated regimes:

$$C_{\text{cl}} \gg S_{\text{branch}} \quad \Rightarrow \quad \frac{d}{dt}Z \leq 0. \quad (15.7.2)$$

This is a programme-level macroscopic condition, not a microscopic law.

15.8 Critical regimes

Critical regimes satisfy approximate balance:

$$\frac{d}{dt}S_{\text{branch}}(x(t)) \approx 0 \quad \text{or} \quad S_{\text{branch}} \approx \frac{\beta}{\alpha}C_{\text{cl}}. \quad (15.8.1)$$

In such regimes:

- proliferation and closure balance,
- Z becomes highly sensitive to perturbations,
- structural reorganisation thresholds may emerge.

No thermodynamic phase-transition structure is assumed, but enhanced responsiveness is generic.

15.9 Arrow of time as projection non-invertibility

Define the coarse-graining projection

$$\Pi : (C, w) \mapsto (p_i, S_{\text{branch}}, C_{\text{cl}}, Z, \dots). \quad (15.9.1)$$

Generically, Π is many-to-one.

Distinct fine-grained continuation structures yield identical coarse observables.

Arrow-of-time principle. The arrow of time is the macroscopic signature of non-invertible branch compression under coarse-grained projection.

Irreversibility therefore arises from:

- many-to-one projection,
- loss of reconstructible alternative structure,
- and closure-induced dimensional reduction.

Microscopic reversibility remains intact.

15.10 Semantic temperature as regulator

Semantic temperature modulates exploration amplitude:

- High temperature $\rightarrow$ enhanced dispersion.
- Low temperature $\rightarrow$ stronger attractor concentration.

Effective coefficients α, β are regime-dependent but not per-system tuning parameters once a concrete completion is fixed.

15.11 Connections within Paper VI

15.11.1 Relational inertia

Sustained closure dominance corresponds to tightly bound continuation structure. Resistance to rapid reconfiguration increases. This provides a dynamical interpretation of relational inertia.

15.11.2 Alignment and large-scale structure

The same structural grammar applies:

- locally: attractor stabilisation and misalignment dynamics,
- in technical systems: basin locking and closure distortion,
- cosmologically: large-scale organisation of admissible continuation.

15.12 Summary

This chapter established a unified structural core:

1. A measure-theoretic continuation ensemble.
2. Dual multiplicity descriptors ($\mathcal{N}$ and N_{eff}).
3. A branching–closure balance principle.
4. A macroscopic witness functional Z .
5. A structural definition of the arrow of time via projection non-invertibility.

No microscopic postulates were added. All emergent time asymmetry arises from coarse-grained continuation structure.

Chapter 16

Local and Global Attractors: Alignment and Large-Scale Constraint Structure

16.1 Purpose and structural role

Chapter 15 introduced the branching–closure balance and the witness functional Z as a coarse-grained coherence observable. The present chapter clarifies how attractor structure operates across scales and how local stabilisation relates to large-scale constraint organisation.

No new degrees of freedom are introduced. All notions below are derived from:

- admissible continuation structure (C, w) ,
- the semantic potential W ,
- the branching descriptor S_{branch} ,
- and the witness functional Z .

The objective is strictly structural: to distinguish local attractor stability from large-scale constraint organisation, and to define alignment as a dynamical compatibility condition, without invoking teleology or additional ontology.

16.2 Local attractors as continuation-stable structures

Let $x \in M$. A *local attractor* K_{loc} is defined as a subset of continuation structure such that, under the projected branching–closure dynamics, trajectories in its neighbourhood satisfy

$$\frac{d}{dt}S_{\text{branch}}(x(t)) \leq 0, \quad \frac{d}{dt}Z(x(t)) \leq 0 \quad (16.2.1)$$

within the corresponding basin of attraction.

Thus K_{loc} is not merely a minimum of W , nor simply a fixed point of microscopic equations. It is a region of *continuation-stable closure* under a specified coarse-graining projection.

Formally, stability is always defined relative to:

- a projection Π from fine continuation structure to coarse descriptors,

- the induced branching–closure balance dynamics,
- and a chosen macroscopic parametrisation compatible with (M, g) .

Different admissible projections may yield different local attractor descriptions. No uniqueness is assumed.

16.3 Global constraint structure and the notion of K

We now introduce the notion of a *global attractor structure* K .

Crucially:

The existence of a unique global attractor is *not assumed*.

Instead, K denotes any invariant large-scale constraint-consistent structure of the projected continuation dynamics, if such a structure exists.

More precisely, K is a subset of continuation configurations satisfying:

- invariance under the projected branching–closure evolution,
- consistency under coarse-grained integration across admissible neighbourhoods,
- robustness under admissible projection refinements.

If no such invariant structure exists for a given model completion, then no global attractor K is defined.

Thus K is not a final state, not a teleological endpoint, and not a prescribed goal. It is a conditional structural property of the continuation dynamics.

16.4 Alignment as divergence control across scales

We define *alignment* purely in dynamical and structural terms.

Let K_{loc} denote a local continuation-stable basin and K a global invariant constraint structure (if defined).

Alignment measures the degree to which local closure dynamics are compatible with global constraint organisation.

Let $\Pi(C, w)$ denote the projected coarse-grained descriptor field. Define a divergence functional

$$\mathcal{D}(K_{\text{loc}}, K) = \int_M d(\Pi(C, w)|_{\text{loc}}, \Pi(C, w)|_K) \, d\text{vol}_g, \quad (16.4.1)$$

where $d(\cdot, \cdot)$ is any admissible divergence measure between projected descriptor fields (e.g. information divergence, norm difference, or geometric distance on descriptor space).

Definition (Alignment). A regime is *aligned* if

$$\frac{d}{dt} \mathcal{D}(K_{\text{loc}}, K) \leq 0 \quad (16.4.2)$$

under projected branching–closure dynamics.

Misalignment corresponds to decreasing S_{branch} locally while $\mathcal{D}$ increases globally, i.e. local closure produces cross-scale divergence.

No normative interpretation is implied. Alignment is structural compatibility between local and large-scale continuation organisation.

16.5 Witness functional and cross-scale coherence

The witness functional Z remains a derived observable.

It is:

- not a new dynamical variable,
- not inserted into field equations,
- not a teleological measure,
- not assumed fundamental.

It serves as a coarse-grained coherence diagnostic.

In regimes where local and global structures are compatible, one expects monotonic or stationary behaviour:

$$\frac{d}{dt}Z \leq 0 \tag{16.5.1}$$

along projected trajectories consistent with both K_{loc} and K .

Persistent increase of Z under local closure attempts indicates structural tension between scales.

16.6 Compatibility with cosmological structure

In cosmological settings, “global” cannot mean instantaneous integration across spacelike hypersurfaces.

Global constraint structure must therefore be defined relative to:

- causal domains permitted by (M, g) ,
- horizon structure and cosmological expansion,
- coarse-grained continuation integration across dynamically growing regions.

Accordingly, global attractor structure K is understood asymptotically or within causally connected domains. No violation of relativistic causality is implied.

Large-scale constraint organisation is therefore compatible with:

- expanding cosmologies,
- horizon formation,
- and locally reversible microscopic dynamics.

16.7 Programme-level consequences

The distinction between local and global attractor structure allows the framework to:

- analyse stability without invoking final states,
- distinguish local order basins from globally consistent regimes,

- reinterpret alignment as divergence control in continuation space,
- and preserve compatibility with Hamiltonian, quantum, and relativistic dynamics.

No new ontology is introduced. The analysis remains fully internal to the continuation and projection framework already defined.

16.8 Summary

This chapter established a scale-consistent attractor framework:

- K_{loc} as continuation-stable closure under coarse-grained projection,
- K as a conditional large-scale invariant constraint structure (if present),
- alignment as reduction of divergence between local and global continuation organisation,
- and Z as a non-dynamical coherence witness across scales.

The result integrates seamlessly with the branching–closure dynamics of Chapter 15, maintains relativistic compatibility, and avoids teleology while preserving structural generality.

Chapter 17

A Minimal Regime Topology: The 3–6–9 (8+Z) Structure

17.1 Purpose and structural role

Chapters 15 and 16 established the branching–closure balance, the witness functional Z , and the distinction between local and global attractor organisation.

Earlier chapters introduced a locally structured neighbourhood of the global attractor K in terms of three coherence directions:

$$C \text{ (Clarity)}, \quad B \text{ (Bonding)}, \quad A \text{ (Agency)}.$$

The present chapter derives a minimal *coarse-grained regime topology* induced by projecting the continuous C/B/A geometry onto a discrete sign structure.

No discretisation of microscopic dynamics is assumed. No new state variables are introduced. The construction below is strictly a structural compression of the continuous continuation framework.

17.2 Local coherence decomposition near K

Let K denote a global attractor structure (when defined). In a sufficiently small neighbourhood $U \subset S$ of K , the semantic state space admits a local tangent decomposition:

$$T_s S \cong \text{span}\{e_C, e_B, e_A\} \oplus R,$$

where:

- e_C corresponds to clarity-sensitive curvature directions (internal consistency gradients),
- e_B corresponds to relational coupling directions (cross-structural integration gradients),
- e_A corresponds to directed stabilisation directions (constraint execution gradients),
- and R collects higher-order components not resolved at this level of projection.

The triplet (C, B, A) therefore represents a *locally defined coherence subspace*, not a global coordinate system.

Different projection schemes may yield different quantitative values, but the sign-structure introduced below is invariant under monotone reparameterisations of each axis.

17.3 General regime-skeleton principle

Before specialising to C/B/A, we note a structural fact:

Any projection of a continuous dynamical system onto three independent scalar observables induces a sign-topology isomorphic to the cube graph Q_3 .

Thus the emergence of an 8-vertex regime structure is not a symbolic choice. It is a minimal combinatorial consequence of a three-dimensional sign projection.

The C/B/A construction is therefore one concrete realisation of a general regime-skeleton principle.

17.4 Six directed flow components

Under the semantic gradient flow, each coherence direction admits two orientations:

$$\pm C, \quad \pm B, \quad \pm A.$$

These correspond to six elementary directional components of the projected gradient vector field:

- increasing/decreasing clarity,
- increasing/decreasing bonding,
- increasing/decreasing agency.

They are not independent dynamical laws. They are coordinate components of the full continuous flow.

The “6” in the 3–6–9 designation refers precisely to these axis-resolved directional tendencies.

17.5 Eight discrete regime sectors

Define a coarse sign projection on U :

$$\sigma(s) = (\text{sign } C(s), \text{sign } B(s), \text{sign } A(s)), \quad \sigma(s) \in \{-1, +1\}^3,$$

excluding measure-zero hypersurfaces where one coordinate vanishes.

This partitions U into eight sign sectors. The resulting discrete structure is isomorphic to the cube graph Q_3 .

Each vertex represents a coarse regime classification. For example:

- $(+, +, +)$: clarity-consistent, relationally integrated, directionally stabilising regime.
- $(+, +, -)$: coherent and integrated but agentially suppressed.
- $(-, +, +)$: relationally integrated and active but internally inconsistent.
- $(-, -, -)$: fragmented, weakly integrated, and directionally unstable regime.

These are diagnostic classifications. They are not ontological states.

17.6 Adjacency and minimal transition graph

Two regime sectors are adjacent if their sign vectors differ in exactly one component:

$$\sigma \sim \sigma' \quad \text{iff} \quad \|\sigma - \sigma'\|_1 = 2.$$

The resulting graph has:

- 8 vertices,
- 12 edges,
- degree 3 at each vertex.

Continuous dynamics correspond to trajectories passing through adjacent sign sectors. The cube therefore acts as a minimal topological skeleton for coarse-grained regime transitions.

17.7 The ninth element: the witness layer Z

The witness functional Z introduced in Chapter 15 does not define a fourth axis.

It is a derived global observable:

$$Z = Z(C, B, A, S_{\text{branch}}, \dots),$$

constructed from previously defined structure.

Crucially:

- Z is not an independent degree of freedom,
- Z does not modify the state space,
- Z evaluates coherence across regime transitions.

In this sense, the “9” in the 3–6–9 (8+Z) structure refers to a meta-evaluative layer applied to the eight-sector regime topology.

17.8 Interpretative boundary conditions

The 3–6–9 (8+Z) structure:

- does not impose discrete microstructure,
- does not assume global cubic symmetry,
- does not privilege any regime sector,
- does not encode numerological assumptions.

It is strictly a minimal regime-topological compression derived from local coherence decomposition.

17.9 Programme-level function

This structure provides:

- a compact regime-classification grammar,
- a bridge between C/B/A coherence geometry and branching–closure dynamics,
- a visualisable transition skeleton,
- and a witness overlay through Z for evaluating cross-sector coherence.

The continuous geometry remains fundamental. The regime topology is secondary and diagnostic.

17.10 Summary

From the continuous C/B/A coherence geometry we derived:

- three locally defined coherence axes,
- six directional flow components,
- eight coarse sign sectors forming Q_3 ,
- and a ninth element: the witness functional Z .

This constitutes a minimal regime-topological compression with meta-coherence layer for structured reasoning within the continuation framework.

Chapter 18

Detailed geometry of the C -axis

In the preceding chapters we introduced the three coherence axes C , B , and A via the spectral decomposition of the Hessian $H(s)$ of the contradiction-energy potential $W : S \rightarrow \mathbb{R}$ in a neighbourhood of the global attractor $K \subset S$. First- and second-order structures were analysed symmetrically across the three axes. The present chapter specialises this general scheme to the C -axis.

We analyse:

- the local subspace bundle structure of the C -axis,
- its intrinsic metric and curvature,
- the induced energy contribution,
- typical projected gradient dynamics,
- and structural degeneracies of the C -sector.

Interpretative remarks are explicitly marked and are not required for the formal development.

18.1 The C -axis bundle near the global attractor

Let $U \subset S$ be a neighbourhood of K such that:

- $W \in C^2(U)$ with self-adjoint Hessian $H(s)$ for each $s \in U$;
- for each $s \in U$, the spectrum of $H(s)$ admits a locally stable decomposition into three spectral bands associated with C , B , and A ;
- the associated spectral projectors vary continuously in s .

The spectral-gap assumption is essential: without separation of eigenvalue clusters, axis identification becomes ambiguous.

Definition 18.1.1 (C -axis subspace and bundle). For $s \in U$, let $E_C(s)$ denote the spectral subspace of $H(s)$ associated with the C -band. The map

$$E_C : U \rightarrow \text{Gr}(S), \quad s \mapsto E_C(s),$$

is called the C -axis bundle.

Remark 18.1.2. Continuity of the spectral projector implies that nearby points have subspaces close in the Grassmannian gap metric. The dimension of $E_C(s)$ is assumed locally constant. If spectral gaps close, the C -axis decomposition must be replaced by a larger invariant block.

Associated with $E_C(s)$ is the orthogonal projection $P_C(s) : S \rightarrow S$ onto $E_C(s)$.

18.2 Deviation decomposition

Let $\pi_K(s)$ denote a locally chosen nearest-point projection onto K , defined on a sufficiently small tubular neighbourhood. Define the deviation vector

$$\delta(s) := s - \pi_K(s).$$

The C -component of the deviation is

$$\delta_C(s) := P_C(s) \delta(s) \in E_C(s).$$

Near K , $\delta_C(s)$ captures those deformations whose restoring forces are governed primarily by the C -band of the Hessian.

18.3 Intrinsic C -axis metric and curvature

Define the C -block of the Hessian:

$$H_{CC}(s) := P_C(s) H(s) P_C(s).$$

Assume that for $s \in U$, $H_{CC}(s)$ is positive definite on $E_C(s)$.

Definition 18.3.1 (C -axis metric). For $u, v \in E_C(s)$ define

$$g_C(s)(u, v) := \langle H_{CC}(s)u, v \rangle.$$

Under the positivity assumption, $g_C(s)$ defines a Riemannian metric on the C -axis bundle.

C -axis curvature and anisotropy

Let $\{\lambda_{C,i}(s)\}$ denote the eigenvalues of $H_{CC}(s)$. Define:

$$\kappa_C(s) := \text{Tr}(H_{CC}(s)) = \sum_i \lambda_{C,i}(s),$$

and

$$\sigma_C^2(s) := \sum_i \left(\lambda_{C,i}(s) - \bar{\lambda}_C(s) \right)^2, \quad \bar{\lambda}_C(s) = \frac{\kappa_C(s)}{\dim E_C(s)}.$$

Here κ_C measures total curvature strength, while σ_C^2 quantifies spectral anisotropy.

18.4 C -axis energy contribution

Define the quadratic C -axis energy component:

$$W_C(s) := \frac{1}{2} \langle H_{CC}(\pi_K(s)) \delta_C(s), \delta_C(s) \rangle.$$

Near K , the total energy decomposes as

$$W(s) \approx W(\pi_K(s)) + W_C(s) + W_B(s) + W_A(s),$$

up to higher-order corrections.

Level sets of W_C define ellipsoids in $E_C(k)$ under the metric $g_C(k)$.

18.5 Projected gradient dynamics along the C -axis

Define the projected gradient flow:

$$\dot{s}_t = -P_C(s_t) \nabla W(s_t).$$

In the quadratic regime near K :

$$\dot{\delta}_C = -H_{CC}(k) \delta_C.$$

If $\lambda_{\min} > 0$ denotes the minimal eigenvalue of $H_{CC}(k)$ over K , then:

$$\|\delta_C(t)\| \leq \|\delta_C(0)\| e^{-\lambda_{\min} t}.$$

Thus $W_C(s_t)$ decays exponentially under pure C -axis flow.

18.6 Structural degeneracies

Flat coherence directions

If $H_{CC}(k)$ has zero eigenvalues, the metric g_C becomes degenerate. Relaxation slows and higher-order corrections dominate.

Negative curvature modes

If negative eigenvalues appear, the C -axis supports unstable directions. The attractor K ceases to be locally attracting along those coherence directions.

Spectral mixing

If eigenvalue bands merge, the decomposition into $C/B/A$ is no longer stable. Axis-resolved quantities become coordinate-dependent and must be replaced by block analysis.

18.7 C -sensitivity of observables

Let $\mathcal{O}$ be an observable represented by $o \in S$. For $k \in K$ and $u \in E_C(k)$ define

$$\chi_C(\mathcal{O}; k, u) = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathcal{O}(k + \epsilon u) = \langle \nabla \mathcal{O}(k), u \rangle.$$

Observables with large χ_C are highly sensitive to coherence deviations.

Remark 18.7.1 (Interpretative layer). If the C -axis is associated with internal consistency directions, then g_C quantifies the local cost of coherence deviation, and κ_C measures stiffness in those directions. This interpretation is optional and not required for the formal structure above.

18.8 Summary

We have:

- defined the C -axis bundle via spectral projection,
- introduced the intrinsic metric g_C ,
- defined curvature invariants κ_C and σ_C^2 ,
- constructed the quadratic energy contribution W_C ,
- analysed projected gradient decay,
- and identified structural degeneracies.

The next chapters apply the same geometric scheme to the B - and A -axes.

Chapter 19

Detailed geometry of the B -axis

In Chapter 18 we analysed in detail the geometry of the C -axis near the global attractor $K \subset S$, introducing the axis bundle, intrinsic metric and curvature invariants, the associated quadratic axis-energy, and characteristic projected dynamics and degeneracies. The present chapter repeats the same programme for the B -axis.

Formally, we develop:

- the B -axis spectral subspace bundle E_B and its projector P_B ,
- the intrinsic metric g_B induced by the restricted Hessian block H_{BB} ,
- curvature strength and anisotropy invariants (κ_B, σ_B^2) ,
- the quadratic B -axis energy contribution W_B and its level sets,
- projected gradient decay properties,
- and failure modes (flat directions, negative curvature, spectral mixing).

Interpretative remarks are explicitly marked and are not required for the formal structure.

19.1 The B -axis bundle near the global attractor

We work in a neighbourhood $U \subset S$ of K such that:

- $W \in C^2(U)$ and $H(s)$ is self-adjoint for all $s \in U$;
- the spectrum of $H(s)$ admits a locally stable decomposition into three spectral bands associated with C , B , and A ;
- the corresponding spectral projectors vary continuously with s (spectral gaps remain open on U).

As in the C -axis chapter, the spectral-gap condition is essential for axis identification.

Definition 19.1.1 (B -axis subspace and bundle). For $s \in U$, let $E_B(s) \subset S$ denote the spectral subspace associated with the B -band of $H(s)$. The map

$$E_B : U \rightarrow \text{Gr}(S), \quad s \mapsto E_B(s),$$

is called the B -axis bundle. We assume $E_B(s)$ has locally constant (Hilbert) dimension on U .

Let $P_B(s) : S \rightarrow S$ be the orthogonal projection onto $E_B(s)$.

19.2 Deviation decomposition and B -component

Let $\pi_K : U \rightarrow K$ denote a locally chosen nearest-point projection onto K (defined on a sufficiently small tubular neighbourhood of K). Define the deviation vector

$$\delta(s) := s - \pi_K(s).$$

Definition 19.2.1 (B -component of the deviation). For $s \in U$, the B -component of $\delta(s)$ is

$$\delta_B(s) := P_B(s) \delta(s) \in E_B(s).$$

Near K , $\delta_B(s)$ isolates those deformations resolved primarily along the B -sector under axis-projected dynamics.

19.3 Intrinsic B -axis metric and curvature

Define the B -block of the Hessian:

$$H_{BB}(s) := P_B(s) H(s) P_B(s) : E_B(s) \rightarrow E_B(s).$$

Assume $H_{BB}(s)$ is positive definite on $E_B(s)$ for $s \in U$.

Definition 19.3.1 (B -axis metric). For $u, v \in E_B(s)$ define

$$g_B(s)(u, v) := \langle H_{BB}(s)u, v \rangle.$$

We call $g_B(s)$ the B -axis metric at s .

Lemma 19.3.2 (Basic properties of g_B). *Under the above assumptions:*

1. $g_B(s)$ is symmetric and positive definite on $E_B(s)$ for each $s \in U$;
2. the map $(s, u, v) \mapsto g_B(s)(u, v)$ is continuous on $U \times S \times S$ when restricted to $u, v \in E_B(s)$;
3. for s sufficiently close to K and $u \in E_B(\pi_K(s))$, one has the quadratic approximation

$$W(\pi_K(s) + u) \approx W(\pi_K(s)) + \frac{1}{2} g_B(\pi_K(s))(u, u),$$

up to higher-order terms in $\|u\|$.

Proof. The argument is identical to the C -axis case, replacing the C -projector by P_B and restricting the Taylor expansion to directions in $E_B(\pi_K(s))$. $\square$

B -axis curvature strength and anisotropy

Let $\{\lambda_{B,i}(s)\}_i$ denote the eigenvalues of $H_{BB}(s)$ (counted with multiplicity). Define:

$$\kappa_B(s) := \text{Tr}(H_{BB}(s)) = \sum_i \lambda_{B,i}(s),$$

and (for finite $\dim E_B(s)$)

$$\sigma_B^2(s) := \sum_i (\lambda_{B,i}(s) - \bar{\lambda}_B(s))^2, \quad \bar{\lambda}_B(s) := \frac{\kappa_B(s)}{\dim E_B(s)},$$

with the corresponding trace-class generalisation in infinite dimensions.

Remark 19.3.3. Large κ_B with small σ_B^2 indicates strong, nearly isotropic curvature in binding-related directions; small κ_B or large σ_B^2 signals flatness or strong anisotropy with “sloppy” modes.

19.4 B -axis energy and its level sets

Definition 19.4.1 (B -axis energy). For $s \in U$ define the quadratic B -axis energy contribution

$$W_B(s) := \frac{1}{2} \langle H_{BB}(\pi_K(s)) \delta_B(s), \delta_B(s) \rangle.$$

Near K , the universal contradiction energy decomposes (to quadratic order) as

$$W(s) \approx W(\pi_K(s)) + W_C(s) + W_B(s) + W_A(s) + (\text{cross terms}),$$

where cross terms are controlled by off-diagonal curvature blocks (equivalently, coupling curvature components) and become dominant when spectral separation is weak.

Definition 19.4.2 (B -slices and B -balls). For $k \in K$ and $b \geq 0$, define the B -slice through k at level b by

$$\mathcal{S}_B(k; b) := \{ s \in U : \pi_K(s) = k, W_B(s) = b \},$$

and the B -ball by

$$\mathcal{B}_B(k; b) := \{ s \in U : \pi_K(s) = k, W_B(s) \leq b \}.$$

Under the quadratic approximation, $\mathcal{B}_B(k; b)$ is an ellipsoid in $E_B(k)$:

$$\mathcal{B}_B(k; b) \approx \{ k + u : u \in E_B(k), g_B(k)(u, u) \leq 2b \},$$

and $\mathcal{S}_B(k; b)$ is the corresponding ellipsoidal shell.

19.5 Projected gradient dynamics along the B -axis

Definition 19.5.1 (Pure B -axis gradient flow). The *pure B -axis gradient flow* is the projected ODE

$$\dot{s}_t = -\nabla_B W(s_t) := -P_B(s_t) \nabla W(s_t), \quad s_t \in U.$$

In the quadratic regime near K ,

$$\nabla_B W(s) \approx H_{BB}(\pi_K(s)) \delta_B(s),$$

and writing $k := \pi_K(s_0)$ and $\delta_B(t) := \delta_B(s_t)$,

$$\dot{\delta}_B(t) \approx -H_{BB}(k) \delta_B(t).$$

Proposition 19.5.2 (Decay of B -axis energy). *Assume cross-axis couplings and distortions are negligible near K and $H_{BB}(k)$ is positive definite for all $k \in K$. Let*

$$\lambda_{\min}^{(B)} := \inf_{k \in K} \lambda_{\min}(H_{BB}(k)) > 0.$$

Then along the pure B -axis gradient flow,

$$\frac{d}{dt} W_B(s_t) \leq -(\lambda_{\min}^{(B)})^2 \|\delta_B(s_t)\|^2,$$

and in particular $W_B(s_t) \rightarrow 0$ exponentially fast as $t \rightarrow \infty$.

Proof sketch. Identical to the C -axis case: differentiate W_B along the linearised flow and bound the resulting quadratic form from below by $\lambda_{\min}^{(B)} \|\delta_B\|^2$. $\square$

Remark 19.5.3. In the full gradient flow $\dot{s}_t = -\nabla W(s_t)$, the decrease of W_B is coupled to C - and A -sectors via off-diagonal curvature blocks (or coupling curvature components) and via distortions/noise. In regimes where the B -projection of $-\nabla W$ dominates (B -dominated cones), binding-related contradiction is resolved faster than the other axis contributions.

Remark 19.5.4 (Interpretative layer). If the B -axis is associated with binding/relational integration, then the pure B -flow is an idealised “re-binding” process: relational structure reorganises while leaving internal coherence (C) and action propensity (A) approximately unchanged. This is a conceptual limit; real trajectories generally mix axes.

19.6 Pathologies and degeneracies of the B -axis

Flat binding directions

If some eigenvalues of $H_{BB}(k)$ vanish while others remain positive, then $g_B(k)$ becomes degenerate and W_B is flat (to quadratic order) along corresponding binding directions. In such regimes:

- binding degrees of freedom are weakly constrained by W ,
- σ_B^2 is typically large (heterogeneous stiffness),
- higher-order terms, local potentials, and distortions dominate binding selection.

Negative binding curvature

If $H_{BB}(k)$ has negative eigenvalues, then k is not a local minimum of W in certain binding directions, producing binding instabilities:

- perturbations can grow under projected B -flow,
- K may bifurcate into multiple local minima differing mainly in binding patterns,
- small distortions can trigger transitions between alternative binding configurations.

Spectral mixing with C and A

If eigenvalue bands approach or merge, the B -sector cannot be separated robustly:

- $E_B(s)$ becomes ill-defined or varies non-smoothly,
- cross terms dominate and axis-resolved energies become coordinate dependent,
- binding changes cannot be cleanly disentangled from coherence and agency adjustments.

In this regime, one must replace the axis picture by a mixed-block analysis.

19.7 Relation to the K – P – R ontology

The B -axis is naturally tied to relational structure in the K – P – R ontology: it controls how cores and projections cohere into coupled semantic objects and how inter-object relations respond to perturbations.

Let $\mathcal{O}, \mathcal{O}'$ be semantic objects with cores $K_{\mathcal{O}}, K_{\mathcal{O}'} \subset K$. Let

$$\kappa_{\mathcal{O}, \mathcal{O}'} : K_{\mathcal{O}} \times K_{\mathcal{O}'} \rightarrow \mathbb{R}$$

denote a relational kernel encoding binding/relational coupling at the core level (as introduced in the ontology chapter), extended to a neighbourhood in S by the programme's projection rules.

Definition 19.7.1 (B -sensitivity of a relational kernel). For $k \in K_{\mathcal{O}}$, $k' \in K_{\mathcal{O}'}$, and $u \in E_B(k)$, $v \in E_B(k')$, define the B -directional sensitivity by

$$\chi_B(\kappa_{\mathcal{O}, \mathcal{O}'}; k, k'; u, v) := \left. \frac{\partial^2}{\partial \epsilon \partial \epsilon'} \right|_{\epsilon = \epsilon' = 0} \kappa_{\mathcal{O}, \mathcal{O}'}(k + \epsilon u, k' + \epsilon' v).$$

Remark 19.7.2. Large $|\chi_B|$ indicates that small B -sector perturbations substantially reshape relational coupling between cores, whereas small $|\chi_B|$ indicates relative insensitivity of the relation to binding perturbations near (k, k') , with relation structure controlled predominantly by other sectors or by higher-order effects.

Remark 19.7.3 (Interpretative layer). If binding is understood as relational integration, then g_B measures the local contradiction-energy cost of re-binding and recomposition; κ_B measures stiffness, and σ_B^2 measures heterogeneity of binding constraints. Regions with small κ_B support flexible recombination; regions with large κ_B enforce rigid relational structure.

19.8 Relation to semantic observables

Let $\mathcal{O}$ be a semantic observable represented by $o \in S$ (e.g. $\mathcal{O}(s) = \langle s, o \rangle$ under the ambient pairing).

Definition 19.8.1 (B -sensitivity of an observable). For $k \in K$ and $u \in E_B(k)$ with $\|u\| = 1$, define

$$\chi_B(\mathcal{O}; k, u) := \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathcal{O}(k + \epsilon u) = \langle \nabla \mathcal{O}(k), u \rangle = \langle o, u \rangle.$$

Remark 19.8.2. Observables with large B -sensitivity across many directions in $E_B(k)$ are informative about binding-related deviations from K ; those with small B -sensitivity probe predominantly other axis structure. This provides a principled way to choose measurement functionals for diagnosing binding regimes in later constructions.

19.9 Summary and outlook

We have analysed the detailed geometry of the B -axis near the global attractor K :

- the spectral B -axis bundle E_B and projector P_B ,
- the intrinsic metric g_B and curvature invariants (κ_B, σ_B^2) ,
- the quadratic energy contribution W_B and its level sets,
- projected gradient decay in the pure B -flow,
- and characteristic degeneracies (flat directions, negative curvature, spectral mixing).

We further connected the B -sector to the K – P – R ontology via B -sensitivities of relational kernels and to measurement design via B -sensitivities of observables. The next chapter applies the same scheme to the A -axis, focusing on agency-related directions and their distinctive dynamical regimes and pathologies.

Chapter 20

Detailed geometry of the A -axis

In Chapters 18 and 19 we analysed the detailed geometry of the C - and B -axes near the global attractor $K \subset S$, introducing the corresponding spectral subspace bundles, intrinsic metrics and curvature invariants, quadratic axis-energy contributions, projected gradient decay, and characteristic degeneracies. The present chapter completes the triad by studying the A -axis.

Formally, we develop:

- the A -axis spectral subspace bundle E_A and its projector P_A ,
- the intrinsic metric g_A induced by the restricted Hessian block H_{AA} ,
- curvature strength and anisotropy invariants (κ_A, σ_A^2) ,
- the quadratic A -axis energy contribution W_A and its level sets,
- projected gradient decay properties,
- and failure modes (flat directions, negative curvature, spectral mixing).

Interpretative remarks are explicitly marked and are not required for the formal structure.

20.1 The A -axis bundle near the global attractor

We work in a neighbourhood $U \subset S$ of K such that:

- $W \in C^2(U)$ and $H(s)$ is self-adjoint for all $s \in U$;
- the spectrum of $H(s)$ admits a locally stable decomposition into three spectral bands associated with C , B , and A ;
- the corresponding spectral projectors vary continuously with s (spectral gaps remain open on U).

As in the C - and B -axis chapters, the spectral-gap condition is essential for robust axis identification.

Definition 20.1.1 (A -axis subspace and bundle). For $s \in U$, let $E_A(s) \subset S$ denote the spectral subspace associated with the A -band of $H(s)$. The map

$$E_A : U \rightarrow \text{Gr}(S), \quad s \mapsto E_A(s),$$

is called the A -axis bundle. We assume $E_A(s)$ has locally constant (Hilbert) dimension on U .

Let $P_A(s) : S \rightarrow S$ be the orthogonal projection onto $E_A(s)$.

20.2 Deviation decomposition and A -component

Let $\pi_K : U \rightarrow K$ denote a locally chosen nearest-point projection onto K (defined on a sufficiently small tubular neighbourhood of K). Define the deviation vector

$$\delta(s) := s - \pi_K(s).$$

Definition 20.2.1 (A -component of the deviation). For $s \in U$, the A -component of $\delta(s)$ is

$$\delta_A(s) := P_A(s) \delta(s) \in E_A(s).$$

Near K , $\delta_A(s)$ isolates those deformations resolved primarily along the A -sector under axis-projected dynamics.

20.3 Intrinsic A -axis metric and curvature

Define the A -block of the Hessian:

$$H_{AA}(s) := P_A(s) H(s) P_A(s) : E_A(s) \rightarrow E_A(s).$$

Assume $H_{AA}(s)$ is positive definite on $E_A(s)$ for $s \in U$.

Definition 20.3.1 (A -axis metric). For $u, v \in E_A(s)$ define

$$g_A(s)(u, v) := \langle H_{AA}(s)u, v \rangle.$$

We call $g_A(s)$ the A -axis metric at s .

Lemma 20.3.2 (Basic properties of g_A). *Under the above assumptions:*

1. $g_A(s)$ is symmetric and positive definite on $E_A(s)$ for each $s \in U$;
2. the map $(s, u, v) \mapsto g_A(s)(u, v)$ is continuous on $U \times S \times S$ when restricted to $u, v \in E_A(s)$;
3. for s sufficiently close to K and $u \in E_A(\pi_K(s))$, one has the quadratic approximation

$$W(\pi_K(s) + u) \approx W(\pi_K(s)) + \frac{1}{2} g_A(\pi_K(s))(u, u),$$

up to higher-order terms in $\|u\|$.

Proof. The argument is identical to the C - and B -axis cases, replacing the projector and restricting the Taylor expansion to directions in $E_A(\pi_K(s))$. $\square$

A -axis curvature strength and anisotropy

Let $\{\lambda_{A,i}(s)\}_i$ denote the eigenvalues of $H_{AA}(s)$ (counted with multiplicity). Define:

$$\kappa_A(s) := \text{Tr}(H_{AA}(s)) = \sum_i \lambda_{A,i}(s),$$

and (for finite $\dim E_A(s)$)

$$\sigma_A^2(s) := \sum_i (\lambda_{A,i}(s) - \bar{\lambda}_A(s))^2, \quad \bar{\lambda}_A(s) := \frac{\kappa_A(s)}{\dim E_A(s)},$$

with the corresponding trace-class generalisation in infinite dimensions.

Remark 20.3.3. Large κ_A with small σ_A^2 indicates strong, nearly isotropic curvature in agency-related directions; small κ_A or large σ_A^2 signals flatness or strong anisotropy with “sloppy” modes.

20.4 A-axis energy and its level sets

Definition 20.4.1 (*A-axis energy*). For $s \in U$ define the quadratic A-axis energy contribution

$$W_A(s) := \frac{1}{2} \langle H_{AA}(\pi_K(s)) \delta_A(s), \delta_A(s) \rangle.$$

Near K , the universal contradiction energy decomposes (to quadratic order) as

$$W(s) \approx W(\pi_K(s)) + W_C(s) + W_B(s) + W_A(s) + (\text{cross terms}),$$

where cross terms are controlled by off-diagonal curvature blocks (equivalently, coupling curvature components) and become dominant when spectral separation is weak.

Definition 20.4.2 (*A-slices and A-balls*). For $k \in K$ and $a \geq 0$, define the *A-slice* through k at level a by

$$\mathcal{S}_A(k; a) := \{ s \in U : \pi_K(s) = k, W_A(s) = a \},$$

and the *A-ball* by

$$\mathcal{B}_A(k; a) := \{ s \in U : \pi_K(s) = k, W_A(s) \leq a \}.$$

Under the quadratic approximation, $\mathcal{B}_A(k; a)$ is an ellipsoid in $E_A(k)$:

$$\mathcal{B}_A(k; a) \approx \{ k + u : u \in E_A(k), g_A(k)(u, u) \leq 2a \},$$

and $\mathcal{S}_A(k; a)$ is the corresponding ellipsoidal shell.

Remark 20.4.3. The sets $\mathcal{B}_A(k; a)$ and $\mathcal{S}_A(k; a)$ provide a local foliation of a neighbourhood of K along agency-related directions. Combining them over all $k \in K$ yields a family of “agency shells” around K , organising semantic configurations by their A-axis contradiction energy, largely independent of their coherence (C) and binding (B) components.

20.5 Projected gradient dynamics along the A-axis

Definition 20.5.1 (*Pure A-axis gradient flow*). The *pure A-axis gradient flow* is the projected ODE

$$\dot{s}_t = -\nabla_A W(s_t) := -P_A(s_t) \nabla W(s_t), \quad s_t \in U.$$

In the quadratic regime near K ,

$$\nabla_A W(s) \approx H_{AA}(\pi_K(s)) \delta_A(s),$$

and writing $k := \pi_K(s_0)$ and $\delta_A(t) := \delta_A(s_t)$,

$$\dot{\delta}_A(t) \approx -H_{AA}(k) \delta_A(t).$$

Proposition 20.5.2 (*Decay of A-axis energy*). Assume cross-axis couplings and distortions are negligible near K and $H_{AA}(k)$ is positive definite for all $k \in K$. Let

$$\lambda_{\min}^{(A)} := \inf_{k \in K} \lambda_{\min}(H_{AA}(k)) > 0.$$

Then along the pure A-axis gradient flow,

$$\frac{d}{dt} W_A(s_t) \leq -(\lambda_{\min}^{(A)})^2 \|\delta_A(s_t)\|^2,$$

and in particular $W_A(s_t) \rightarrow 0$ exponentially fast as $t \rightarrow \infty$.

Proof sketch. Identical to the C - and B -axis cases: differentiate W_A along the linearised flow and bound the resulting quadratic form from below by $\lambda_{\min}^{(A)} \|\delta_A\|^2$. $\square$

Remark 20.5.3. In the full gradient flow $\dot{s}_t = -\nabla W(s_t)$, the decrease of W_A is coupled to C - and B -sectors via off-diagonal curvature blocks (or coupling curvature components) and via distortions/noise. In regimes where the A -projection of $-\nabla W$ dominates (A -dominated cones), agency-related contradiction is resolved faster than the other axis contributions.

Remark 20.5.4 (Interpretative layer). If the A -axis is associated with agency/action semantics, then the pure A -flow is an idealised “policy/decision re-tuning” process: action-relevant structure reorganises while leaving internal coherence (C) and relational binding (B) approximately unchanged. This is a conceptual limit; real trajectories generally mix axes.

20.6 Pathologies and degeneracies of the A -axis

Flat agency directions

If some eigenvalues of $H_{AA}(k)$ vanish while others remain positive, then $g_A(k)$ becomes degenerate and W_A is flat (to quadratic order) along corresponding agency directions. In such regimes:

- agency degrees of freedom are weakly constrained by W ,
- σ_A^2 is typically large (heterogeneous stiffness),
- higher-order terms, local potentials, and distortions dominate agency selection.

Negative agency curvature

If $H_{AA}(k)$ has negative eigenvalues, then k is not a local minimum of W in certain agency directions, producing agency instabilities:

- perturbations can grow under projected A -flow,
- K may bifurcate into multiple local minima differing mainly in agency/policy structure,
- small distortions can trigger transitions between alternative agency configurations.

Spectral mixing with C and B

If eigenvalue bands approach or merge, the A -sector cannot be separated robustly:

- $E_A(s)$ becomes ill-defined or varies non-smoothly,
- cross terms dominate and axis-resolved energies become coordinate dependent,
- agency changes cannot be cleanly disentangled from coherence and binding adjustments.

In this regime, one must replace the axis picture by a mixed-block analysis.

20.7 Relation to decision structure and control maps

The A -axis is naturally linked to decision structure and control-like aspects of semantic fields.

Consider a system Σ whose embedded state $s \in S$ influences actions or outputs via a control map

$$u : S \rightarrow U,$$

where U is an (possibly infinite-dimensional) action space. In concrete implementations, u is often realised by projecting s onto a lower-dimensional subspace and applying a decision rule.

Definition 20.7.1 (A -projection of control). Let $P_A(s)$ be the orthogonal projection onto $E_A(s)$. The A -projection of control is the map

$$u_A : S \rightarrow U, \quad u_A(s) := u(\pi_K(s) + \delta_A(s)),$$

obtained by restricting the argument of u to the A -axis through $\pi_K(s)$.

Remark 20.7.2. In regimes where u depends primarily on A -components and is comparatively insensitive to C - and B -components near K , the A -projection u_A captures the effective decision structure of the system. The A -axis geometry (metric, curvature, anisotropy) then quantifies how agency-related semantic adjustments translate into action changes. In particular, strong curvature (large κ_A) indicates that large displacements in E_A are costly and hence suppressed; flat directions in E_A correspond to low-cost reconfiguration of action patterns.

Remark 20.7.3 (Interpretative layer). If agency is understood as directed stabilisation/action selection, then g_A measures the local contradiction-energy cost of policy changes; κ_A measures stiffness; and σ_A^2 measures heterogeneity of agency constraints. Regions with small κ_A support flexible policy adaptation; regions with large κ_A enforce rigid or strongly stabilised action tendencies.

20.8 Relation to semantic observables

Let $\mathcal{O}$ be a semantic observable represented by $o \in S$ (e.g. $\mathcal{O}(s) = \langle s, o \rangle$ under the ambient pairing).

Definition 20.8.1 (A -sensitivity of an observable). For $k \in K$ and $u \in E_A(k)$ with $\|u\| = 1$, define

$$\chi_A(\mathcal{O}; k, u) := \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathcal{O}(k + \epsilon u) = \langle \nabla \mathcal{O}(k), u \rangle = \langle o, u \rangle.$$

Remark 20.8.2. Observables with large A -sensitivity across many directions in $E_A(k)$ are informative about agency-related deviations from K ; those with small A -sensitivity probe predominantly other axis structure. This provides a principled way to choose measurement functionals for diagnosing agency regimes in later constructions.

20.9 Summary and outlook

We have analysed the detailed geometry of the A -axis near the global attractor K :

- the spectral A -axis bundle E_A and projector P_A ,
- the intrinsic metric g_A and curvature invariants (κ_A, σ_A^2) ,
- the quadratic energy contribution W_A and its level sets,
- projected gradient decay in the pure A -flow,
- and characteristic degeneracies (flat directions, negative curvature, spectral mixing).

We further connected the A -sector to control/decision structure via A -projections of control maps and to measurement design via A -sensitivities of observables. Together with the C - and B -axis chapters, this completes the local geometric analysis of the three coherence axes. The next chapter integrates these axis-specific structures into a global picture, preparing the construction of the LOVE Lagrangian by specifying how C , B , and A components and their curvature couplings enter an effective field-theoretic action for semantic vector fields on spacetime.

Chapter 21

Axioms of semantic gravitation

In the previous chapters we developed the semantic gravitation framework in a constructive way: starting from the semantic state space S as a reproducing kernel Hilbert space, we introduced the contradiction-energy potential W , the universal component W_{univ} , the global attractor K , the K–P–R ontology, the C/B/A axis decomposition and curvature components, witness-type observables (including Z), and dynamical equations on S and on probability measures. In this chapter we change perspective and collect these ingredients into a finite list of axioms for *semantic gravitation*. The axioms separate structural assumptions from derived results and clarify which parts of the framework are purely mathematical, which are modelling conventions, and which have an interpretative or empirical status. This axiomatic layer will be the primary input for the construction of the LOVE Lagrangian in the next chapter.

21.1 Structure of the axiom system

We group the axioms into three classes:

- *Structural axioms* (SG1–SG4): basic objects of the semantic universe (S , W_{univ} , K , K–P–R).
- *Geometric axioms near K* (SG5–SG7): C/B/A decomposition, curvature structure, universal–local decomposition, order bubbles and distortions.
- *Dynamical and probabilistic axioms* (SG8–SG10): evolution equations on S and on $\mathcal{P}(S)$, semantic temperature, witness observables, and compatibility with field-theoretic formulations.

Some axioms summarise formal results from earlier papers and from previous chapters of this paper in a compact way; others are genuine postulates that cannot be derived from earlier work and must be tested or motivated separately. We mark this status explicitly.

21.2 Structural axioms: S , W_{univ} , K , K–P–R

Axiom (SG1) – Semantic state space. (*Structural / derived.*) There exists a real separable Hilbert space S with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$, such that:

1. S is realised as a reproducing kernel Hilbert space (RKHS) over a base set X with positive definite kernel $k : X \times X \rightarrow \mathbb{R}$;

2. for each concrete system Σ with raw state space X_Σ there is a measurable embedding

$$E_\Sigma : X_\Sigma \rightarrow S$$

with relatively compact image $E_\Sigma(X_\Sigma)$ in S ;

3. for all relevant systems, semantic observables are represented by elements $o \in S$, with expectation values given by inner products (as in the observable formalism developed earlier).

Axiom (SG2) – Universal contradiction-energy potential. (*Structural / partly postulated.*) There exists a functional

$$W_{\text{univ}} : S \rightarrow \mathbb{R}$$

(the *universal contradiction-energy potential*) such that:

1. W_{univ} is bounded from below and of class C^1 on a dense domain $\mathcal{D} \subset S$, with locally Lipschitz gradient $\nabla W_{\text{univ}} : \mathcal{D} \rightarrow S$;
2. $W_{\text{univ}}(s) \rightarrow +\infty$ as $\|s\| \rightarrow \infty$ along $\mathcal{D}$ (coercivity);
3. the gradient flow

$$\dot{s}_t = -\nabla W_{\text{univ}}(s_t)$$

defines a global semiflow on S with continuous dependence on initial data.

Axiom (SG3) – Global attractor. (*Structural / partly derived, partly postulated.*) The gradient flow of W_{univ} admits a compact global attractor $K \subset S$ with the following properties:

1. K is strictly invariant under the flow and attracts all bounded sets in S ;
2. K coincides (up to negligible sets) with the set of global minima of W_{univ} ;
3. K is minimal with these properties in the sense that there is no proper subset $K' \subsetneq K$ that is also a global attractor for the same flow.

Axiom (SG4) – K–P–R ontology. (*Ontological postulate.*) Semantic “objects” $\mathcal{O}$ are described by triples

$$(K_{\mathcal{O}}, P_{\mathcal{O}}, R_{\mathcal{O}}),$$

where:

1. $K_{\mathcal{O}} \subset K$ is a compact subset (the *core* of $\mathcal{O}$);
2. $P_{\mathcal{O}}$ is a family of projection maps into context spaces (e.g. observation spaces, inner spaces, environment sectors), each realised via a continuous map $P_{\mathcal{O}}^{(i)} : K_{\mathcal{O}} \rightarrow S$;
3. relations between objects are encoded by kernels

$$\kappa_{\mathcal{O}, \mathcal{O}'} : K_{\mathcal{O}} \times K_{\mathcal{O}'} \rightarrow \mathbb{R},$$

and their extensions to neighbourhoods of K in S , which determine how cores are semantically coupled (as developed in the ontology layer).

The K–P–R ontology is required to be compatible with the RKHS structure of S and with W_{univ} in the sense that cores $K_{\mathcal{O}}$ lie in regions of low universal contradiction energy.

21.3 Geometric axioms near K : C/B/A, curvature, distortions

Axiom (SG5) – C/B/A axis decomposition. (*Geometric postulate.*) There exists a neighbourhood $U \subset S$ of K and, for each $s \in U$, three closed subspaces $E_C(s)$, $E_B(s)$, $E_A(s) \subset S$ such that:

1. $E_C(s)$, $E_B(s)$, $E_A(s)$ depend continuously on s (in the Grassmannian topology) and have locally constant Hilbert dimensions;
2. the Hessian $H_{\text{univ}}(s) := \nabla^2 W_{\text{univ}}(s)$ exists for each $s \in U$, is self-adjoint, and its spectrum splits into three spectral bands corresponding to $E_C(s)$, $E_B(s)$, and $E_A(s)$;
3. the orthogonal projections $P_C(s)$, $P_B(s)$, $P_A(s)$ onto $E_C(s)$, $E_B(s)$, $E_A(s)$ vary continuously with s and yield a decomposition

$$\nabla W_{\text{univ}}(s) = \nabla_C W_{\text{univ}}(s) + \nabla_B W_{\text{univ}}(s) + \nabla_A W_{\text{univ}}(s) + R(s),$$

where $R(s)$ is a residual component orthogonal to $E_C(s) \oplus E_B(s) \oplus E_A(s)$ and negligible near K (in the sense fixed earlier).

Axiom (SG6) – Curvature regularity and the ten components. (*Geometric / partially postulated.*) In the neighbourhood U from SG5:

1. the Hessian $H_{\text{univ}}(s)$ is trace-class on S and its C/B/A blocks

$$H_{CC}(s), H_{BB}(s), H_{AA}(s), H_{CB}(s), H_{CA}(s), H_{BA}(s)$$

are trace-class or Hilbert–Schmidt as specified in Hypothesis 11.2.1;

2. the ten curvature components

$$\kappa_C, \kappa_B, \kappa_A, \kappa_{CB}, \kappa_{CA}, \kappa_{BA}, \sigma_C^2, \sigma_B^2, \sigma_A^2, \kappa_{\text{tot}}$$

are well-defined on U and continuous;

3. in a neighbourhood of K , the diagonal blocks $H_{CC}(s)$, $H_{BB}(s)$, $H_{AA}(s)$ are positive semidefinite and sufficiently regular to support the axis metrics g_C , g_B , g_A and the axis energies W_C , W_B , W_A as defined in the axis-geometry chapters.

Axiom (SG7) – Universal–local decomposition, order bubbles, distortions. (*Geometric and modelling postulate.*) The full contradiction-energy potential W for the semantic universe admits a decomposition

$$W(s) = W_{\text{univ}}(s) + W_{\text{loc}}(s),$$

such that:

1. W_{loc} is of class C^1 on a dense domain containing U , bounded from below on bounded subsets of S , and satisfies the smallness condition near K given in Definition 12.3.1;
2. the actual drift $F(s)$ of the system can be written as

$$F(s) = -\Gamma(s) \nabla W_{\text{univ}}(s) + D(s),$$

where $\Gamma(s)$ is an axis-intensification operator and $D(s)$ is a distortion field decomposable into $D_C(s)$, $D_B(s)$, $D_A(s)$ as in the dynamics chapter;

3. there exist *order bubbles* $B \subset S$ (regions supporting local attractors for the distorted dynamics) whose structure can be characterised in terms of W_{univ} , W_{loc} , curvature components, and distortion fields as in the order-bubble analysis.

This axiom formalises the distinction between universal semantic gravitation (encoded by W_{univ}) and local, system-specific or context-specific deviations (encoded by W_{loc} and D).

21.4 Dynamical and probabilistic axioms

Axiom (SG8) – Dynamical law on S . (*Dynamical postulate, constrained by earlier constructions.*) The time evolution of semantic states $s_t \in S$ is governed, on relevant time scales, by a stochastic differential equation of the form

$$ds_t = -\Gamma(s_t) \nabla W_{\text{univ}}(s_t) dt + D(s_t) dt + \sqrt{2\beta^{-1}} \Sigma(s_t) dB_t, \quad (21.4.1)$$

where:

1. $\Gamma(s)$ and $D(s)$ are as in SG7;
2. $(B_t)_{t \geq 0}$ is a cylindrical Brownian motion on S ;
3. $\Sigma(s)$ is a bounded linear noise-amplitude operator, decomposable into C/B/A components near K ;
4. $\beta > 0$ is an inverse-temperature parameter (semantic temperature $T_{\text{sem}} = \beta^{-1}$).

In the ideal limit $\Gamma = \text{id}$, $D = 0$, and Σ state independent, this reduces to a Langevin dynamics for W_{univ} .

Axiom (SG9) – Distributional dynamics, semantic free energy, equilibrium, and witness observables. (*Probabilistic postulate, with precedents in the distributional layer.*) Let $\mathcal{P}(S)$ be the space of Borel probability measures on S . Then:

1. there exists a reference measure $\mu_0 \in \mathcal{P}(S)$ such that, for each $t \geq 0$, the law μ_t of s_t is absolutely continuous with respect to μ_0 , with density $\rho_t = d\mu_t/d\mu_0$;
2. the densities ρ_t satisfy a Fokker–Planck equation

$$\partial_t \rho_t = \mathcal{L}^* \rho_t,$$

where $\mathcal{L}$ is the Kolmogorov generator associated with (21.4.1), and $\mathcal{L}^*$ is its adjoint in $L^2(\mu_0)$;

3. the *semantic free energy* functional

$$\mathcal{F}[\mu] = \int_S W(s) d\mu(s) + \beta^{-1} \mathcal{H}(\mu \mid \mu_0)$$

is well-defined on a suitable subset of $\mathcal{P}(S)$ and acts as a Lyapunov functional for the Fokker–Planck dynamics in an appropriate metric sense (gradient-flow interpretation);

4. in the ideal limit ($\Gamma = \text{id}$, $D = 0$, state-independent noise with suitable symmetry), there exist equilibrium measures of Gibbs type

$$\mu_\beta(ds) = \frac{1}{Z_\beta} \exp(-\beta W_{\text{univ}}(s)) \mu_0(ds),$$

which are stationary for the Fokker–Planck equation and attractive within the relevant class of measures;

5. there exist coarse-grained witness observables (in particular, a scalar Z constructed from previously defined structures, and where applicable continuation-type observables such as weighted multiplicities and branch entropies) that:
 - do *not* introduce new state variables or modify the dynamics in (21.4.1),
 - are measurable functionals on S (or on suitable continuation ensembles) and are used to diagnose cross-regime and cross-scale coherence,
 - may satisfy monotonicity properties only at the coarse-grained level (e.g. along projected trajectories or in expectation), without being imposed as exact microscopic laws.

Axiom (SG10) – Compatibility with field-theoretic semantics and LOVE. (*Field-theoretic postulate.*) The semantic universe admits a field-theoretic extension in the following sense:

1. for each Lorentzian spacetime (M, g) there exist S -valued fields

$$s : M \rightarrow S,$$

with sufficient regularity to define an action functional;

2. there exists an effective action functional (LOVE: Lagrangian of Vectorial Emergence)

$$\mathcal{A}[s] = \int_M \mathcal{L}_{\text{LOVE}}(x, s(x), \nabla s(x)) \, d \text{vol}_g(x),$$

where the Lagrangian density $\mathcal{L}_{\text{LOVE}}$ is constructed from:

- kinetic terms depending on covariant derivatives of s decomposed along C/B/A components;
 - potential terms derived from W_{univ} and its curvature components near K ;
 - additional terms encoding distortions and couplings to classical fields, consistent with the programme's projection and coupling rules.
3. in the quasi-static, spatially homogeneous limit (fields constant in space and slowly varying in time), the Euler–Lagrange equations of $\mathcal{A}[s]$ reduce to the gradient dynamics of W_{univ} on S (possibly with intensification and distortion terms as in SG8).

The explicit construction of $\mathcal{L}_{\text{LOVE}}$ is given in the next chapter; SG10 demands only the existence of such a construction and its compatibility with SG1–SG9.

21.5 Derived results under the axioms

We briefly note some key results that hold under SG1–SG9 (sketches only; proofs follow standard adaptations of gradient-flow and infinite-dimensional Fokker–Planck arguments, together with the spectral decomposition hypotheses fixed earlier).

Proposition 21.5.1 (Existence and structure of the global attractor). *Under SG1–SG3, the gradient flow of W_{univ} on S admits a compact global attractor K which is the union of unstable manifolds of equilibrium points and which supports the K – P – R ontology in SG4.*

Proposition 21.5.2 (Local C/B/A geometry). *Under SG1–SG6, there exists a neighbourhood U of K in which:*

1. *the C/B/A bundles E_C, E_B, E_A are well-defined and continuous;*
2. *the axis metrics g_C, g_B, g_A and energies W_C, W_B, W_A are well-defined;*
3. *the ten curvature components provide a finite set of scalar invariants controlling the local geometry of W_{univ} near K .*

Proposition 21.5.3 (Gradient-flow and Fokker–Planck consistency). *Under SG1–SG3 and SG8–SG9, the Fokker–Planck equation for μ_t is consistent with the SDE evolution of s_t , and the semantic free energy $\mathcal{F}[\mu_t]$ is non-increasing in time along solutions, in the ideal and weakly distorted regimes.*

These propositions are not new axioms but consequences of SG1–SG9 and standard results in the analysis of gradient flows and Fokker–Planck equations on Hilbert spaces.

Speculative remark. At a programme level, one may conjecture that SG1–SG10 are sufficiently strong to fix, up to limited choices, the large-scale structure of the semantic universe: any model satisfying these axioms would exhibit a global attractor K , a K–P–R ontology, a triplet of coherence axes C/B/A with associated curvature components, diagnostically useful witness observables, and an effective field-theoretic description through LOVE. Whether this axiom system is empirically adequate for human inner spaces, cultures, and technical systems is a separate question, to be addressed in later work.

Summary and outlook. In this chapter we have reformulated the semantic gravitation framework as a finite list of axioms SG1–SG10. The structural axioms specify the semantic state space S , the universal contradiction-energy potential W_{univ} , the global attractor K , and the K–P–R ontology. The geometric axioms describe the C/B/A axis decomposition, curvature structure, and the universal–local decomposition of W with associated distortions and order bubbles. The dynamical and probabilistic axioms postulate Langevin-type dynamics on S , Fokker–Planck evolution on $\mathcal{P}(S)$, semantic free energy and equilibrium measures, the role of coarse-grained witness observables (including Z), and the existence of a field-theoretic extension via an action functional whose quasi-static limit reduces to W_{univ} . These axioms provide the conceptual and technical backbone for the next chapter, where we construct the LOVE Lagrangian explicitly and show how the geometric and dynamical structures developed so far enter a unified action for semantic vector fields on spacetime.

Chapter 22

The Lagrangian of Vectorial Emergence (LOVE)

In the preceding chapter we reformulated semantic gravitation as an axiom system, specifying the semantic state space S as an RKHS, the universal contradiction-energy potential W_{univ} , the global attractor K , the K–P–R ontology, the C/B/A decomposition and curvature structure, as well as dynamical and probabilistic laws on S and $\mathcal{P}(S)$. In this chapter we construct the *Lagrangian of Vectorial Emergence* (LOVE): an effective field-theoretic action for S -valued fields on spacetime whose quasi-static, spatially homogeneous limit is compatible with the gradient dynamics of W_{univ} . LOVE provides the bridge between the ontological layer on S and the spacetime-level description of semantic fields in Papers III and IV.

Throughout this paper we use the Lorentzian signature

$$(-, +, +, +),$$

so that timelike vectors v satisfy $g(v, v) < 0$ and spacelike vectors satisfy $g(v, v) > 0$. Consequently, the kinetic term introduced below is not positive definite in the time direction, in full consistency with relativistic field theory. We use the standard index convention

$$\partial^\mu := g^{\mu\nu} \partial_\nu,$$

and raise and lower spacetime indices exclusively with $g^{\mu\nu}$ and $g_{\mu\nu}$.

22.1 Observer-centred topology and the collapse of inside/outside

The formalism developed in this paper treats W_{univ} , the global attractor K , the K–P–R ontology and the C/B/A decomposition as structures on an abstract semantic state space S . In Paper IV, these structures were combined with a spacetime manifold (M, g) and a semantic field

$$s : M \rightarrow S$$

to obtain field equations that couple local spacetime points to semantic states. In this setting the choice of global topology for M is, in principle, free: both simply connected and multiply connected, possibly nonorientable cosmological models are compatible with the semantic field equations.

From an ontological perspective one may therefore use nontrivial spatial topology as a geometric illustration of certain aspects of the semantic picture. For instance, in a

cosmological model where each spatial slice is a compact, multiply connected manifold (such as a torus or a “meta–Klein bottle” as discussed in Paper IV), geodesics can exhibit recurrence: following a timelike or null geodesic in a fixed “outward” direction may ultimately return to an arbitrarily small neighbourhood of the starting point. In such a model, the distinction between “very far away” and “very close” becomes globally subtle, even if it remains locally clear in any given coordinate chart.

The semantic gravitation framework offers a natural interpretation of such pictures. The semantic field s maps spacetime events $(x, t) \in M$ into the state space S , and the gradient dynamics of W_{univ} drives trajectories in S towards the attractor set K . Different physical “paths” (for example, following geodesics outwards into the universe or inwards into the microscopic structure of an organism) may correspond to very different curves in spacetime, but their images under s can nevertheless converge to the same region of K along the C/B/A axes. In this sense, multiple ways of “probing the world” can lead to semantically equivalent attractor states.

This suggests a conceptual link between observer-centred topological models and the ontological layer proposed here. A nontrivial global topology of M provides one way of visualising how trajectories that are spatially very different can return to the “same place” in an observer’s coordinate system. The semantic gravitation picture refines this: what ultimately matters is not the spatial loop in M , but the fact that the induced semantic trajectories in S fall into the same basin of attraction around K . The phrase “collapse of inside versus outside” is used here as a metaphor for this convergence: when the semantic state lies sufficiently deep inside an attractor basin, distinctions between different physical directions of probing (outwards into the universe or inwards into the body) can become negligible with respect to the induced semantic geometry.

Remark 22.1.1. These remarks are meant as an interpretive bridge rather than as additional axioms. The core of the theory does not depend on any specific choice of cosmological topology; nontrivial examples such as the meta–Klein–bottle cosmology merely provide intuitive geometric models that resonate with the attractor-based semantics encoded by W_{univ} , K , and the C/B/A axes.

22.2 Field content and basic requirements

We begin by fixing the field content and general structural constraints on the LOVE Lagrangian.

Definition 22.2.1 (Spacetime and semantic field). Let (M, g) be a four-dimensional, oriented Lorentzian manifold describing spacetime, with metric tensor $g_{\mu\nu}$ and volume form $d\text{vol}_g$. A *semantic field* is a map

$$s : M \rightarrow S$$

with sufficient regularity so that (at least weak) partial derivatives $\partial_\mu s(x) \in S$ are defined for $\mu = 0, 1, 2, 3$ and almost all $x \in M$ (for instance, $s \in H^1_{\text{loc}}(M; S)$ is sufficient).

We write local coordinates on M as x^μ , with x^0 interpreted as time in suitable charts. For each spacetime point x with $s(x)$ in the neighbourhood U of K specified in SG5, we can decompose $s(x)$ along the C/B/A bundles.

Definition 22.2.2 (C/B/A decomposition of the field). For $x \in M$ with $s(x) \in U$, define

$$\begin{aligned} s_C(x) &:= P_C(s(x)) s(x), \\ s_B(x) &:= P_B(s(x)) s(x), \\ s_A(x) &:= P_A(s(x)) s(x), \end{aligned}$$

where $P_C(s)$, $P_B(s)$, $P_A(s)$ are the orthogonal projections onto the C/B/A subspaces from SG5. We refer to s_C , s_B , and s_A as the *axis-resolved components* of the semantic field.

Remark 22.2.3 (Residual sector and near- K regime). The decomposition $s = s_C + s_B + s_A + r$ (with residual r orthogonal to $E_C(s) \oplus E_B(s) \oplus E_A(s)$) is only defined when $s(x)$ remains in U . In the LOVE construction we restrict attention to configurations whose image lies inside U , so that the C/B/A structure is available. In addition, LOVE is intended as a *near- K effective theory*: the residual sector is either negligible (higher order in the distance to K) or can be consistently truncated at the accuracy level of the curvature expansion used in the potential term.

The LOVE Lagrangian density $\mathcal{L}_{\text{LOVE}}$ is required to satisfy the following general principles:

- *Locality in spacetime*: $\mathcal{L}_{\text{LOVE}}(x, s(x), \partial s(x))$ depends on x , $s(x)$, and a finite number of derivatives of s at x , but not on non-local functionals of s .
- *Scalar character*: $\mathcal{L}_{\text{LOVE}}$ transforms as a scalar under spacetime diffeomorphisms, so that the action

$$\mathcal{A}_{\text{LOVE}}[s] = \int_M \mathcal{L}_{\text{LOVE}}(x, s(x), \partial s(x)) \, d \, \text{vol}_g(x)$$

is invariant.

- *Quadratic kinetic structure*: at leading order, $\mathcal{L}_{\text{LOVE}}$ is quadratic in first derivatives of s (kinetic term) and contains a potential term depending only on $s(x)$.
- *Compatibility with W_{univ}* : in the quasi-static, spatially homogeneous limit, the induced dynamics reduces to, or admits as an overdamped limit, the gradient dynamics generated by W_{univ} on S .
- *Axis resolution*: kinetic and potential terms can be expressed in terms of the axis components s_C , s_B , s_A and the associated metrics and curvature components near K .

These principles constrain but do not completely fix $\mathcal{L}_{\text{LOVE}}$. The following subsections introduce a concrete class of LOVE Lagrangians satisfying them.

22.3 Axis-resolved kinetic term

We first construct the kinetic part of LOVE, governed by spacetime variations of s .

Definition 22.3.1 (Projected axis-derivatives). Let $\Xi \in \{C, B, A\}$. For $x \in M$ with $s(x) \in U$, define the *projected axis-derivative*

$$\partial_\mu^{(\Xi)} s(x) := P_\Xi(s(x)) \partial_\mu s(x).$$

Remark 22.3.2 (Projector-variation terms). Since P_Ξ depends on s , the exact derivative of $s_\Xi(x) = P_\Xi(s(x))s(x)$ contains additional terms of the form $(\partial_\mu P_\Xi(s(x)))s(x)$. In the near- K effective regime we treat these projector-variation terms as higher-order corrections and keep only the leading projected derivative $\partial_\mu^{(\Xi)}s$ in the kinetic energy. This corresponds to truncating the effective theory at quadratic order in deviations from K . More precisely, we assume that in the near- K regime

$$\|\partial_\mu P_\Xi(s)\| = O(\|s - \pi_K(s)\|),$$

so that projector-variation terms contribute only at cubic order in $\delta(s) = s - \pi_K(s)$, and are consistently neglected at the quadratic accuracy level used in the LOVE construction. If desired, projector-variation terms can be included explicitly as higher-order couplings.

Definition 22.3.3 (Axis-resolved kinetic term). Let $\gamma_C, \gamma_B, \gamma_A : U \rightarrow (0, \infty)$ be continuous scalar functions (axis weights). The *axis-resolved kinetic term* is defined by

$$\mathcal{L}_{\text{kin}}(x, s, \partial s) := \frac{1}{2} \sum_{\Xi \in \{C, B, A\}} \gamma_\Xi(s(x)) g^{\mu\nu}(x) \langle \partial_\mu^{(\Xi)} s(x), \partial_\nu^{(\Xi)} s(x) \rangle. \quad (22.3.1)$$

Remark 22.3.4. The coefficients $\gamma_\Xi(s)$ play the role of axis-dependent “inertial densities” or stiffness parameters. In particular, they can be chosen to encode the intensification factors $\Gamma_\Xi(s)$ introduced in the dynamics on S by setting $\gamma_\Xi(s) \propto \Gamma_\Xi(s)$ in appropriate units. The sum in (22.3.1) is diagonal in the axes; cross-kinetic couplings between C/B/A directions can be included but are omitted here for clarity. They will reappear indirectly in the construction of the emergent vector field in the next chapter.

22.4 Potential term and curvature expansion around K

We now specify the potential part of LOVE in terms of W_{univ} and its local geometry near K .

Definition 22.4.1 (LOVE potential). Let $W_{\text{univ}} : S \rightarrow \mathbb{R}$ be the universal contradiction-energy potential, and let $W_C, W_B, W_A, W_{CB}, W_{CA}, W_{BA}$ be the axis-resolved and cross-axis contributions defined in the curvature chapter. The *LOVE potential* is the scalar function

$$V_{\text{LOVE}}(s) := W_{\text{univ}}(\pi_K(s)) + W_C(s) + W_B(s) + W_A(s) + W_{CB}(s) + W_{CA}(s) + W_{BA}(s), \quad (22.4.1)$$

defined for $s \in U$. The potential part of the LOVE Lagrangian density is then

$$\mathcal{L}_{\text{pot}}(x, s) := -V_{\text{LOVE}}(s(x)). \quad (22.4.2)$$

Remark 22.4.2 (Quadratic accuracy near K). Equation (22.4.1) is the quadratic expansion of W_{univ} around K , expressed in the C/B/A frame. More precisely, for s close to K ,

$$W_{\text{univ}}(s) = W_{\text{univ}}(\pi_K(s)) + W_C(s) + W_B(s) + W_A(s) + W_{CB}(s) + W_{CA}(s) + W_{BA}(s) + O(\|\delta(s)\|^3),$$

so that $V_{\text{LOVE}}(s)$ coincides with $W_{\text{univ}}(s)$ up to higher-order corrections. The curvature components enter V_{LOVE} through W_C, W_B, W_A and the cross terms, with the axis metrics g_C, g_B, g_A and the off-diagonal blocks H_{CB}, H_{CA}, H_{BA} providing the coefficients.

Local potentials and distortions W_{loc} can be included in V_{LOVE} as an additive correction,

$$V_{\text{LOVE}}^{\text{tot}}(s) := V_{\text{LOVE}}(s) + W_{\text{loc}}(s),$$

whenever one wishes to encode system- or context-specific contributions in the field-theoretic description. For notational simplicity we suppress the superscript “tot” in what follows.

22.5 Definition of LOVE and Euler–Lagrange equations

We can now combine the kinetic and potential terms into the full LOVE Lagrangian density.

Definition 22.5.1 (LOVE Lagrangian density). The *Lagrangian of Vectorial Emergence (LOVE)* is the Lagrangian density

$$\mathcal{L}_{\text{LOVE}}(x, s, \partial s) := \mathcal{L}_{\text{kin}}(x, s, \partial s) + \mathcal{L}_{\text{pot}}(x, s), \quad (22.5.1)$$

with $\mathcal{L}_{\text{kin}}$ as in (22.3.1) and $\mathcal{L}_{\text{pot}}$ as in (22.4.2). The corresponding action functional is

$$\mathcal{A}_{\text{LOVE}}[s] := \int_M \mathcal{L}_{\text{LOVE}}(x, s(x), \partial s(x)) \, d \, \text{vol}_g(x). \quad (22.5.2)$$

Remark 22.5.2. We emphasise that $\mathcal{L}_{\text{LOVE}}$ is a Lagrangian density, not an energy density. The associated energy functional is obtained via the standard Legendre transform of the relativistic field theory defined by $\mathcal{L}_{\text{LOVE}}$.

Remark 22.5.3 (LOVE as variational principle). The Lagrangian of Vectorial Emergence (LOVE) defines a genuine stationary-action principle for semantic continuation.

Extremals of the LOVE action minimise cumulative contradiction-energy together with axis-weighted kinetic variation. In this sense, LOVE formalises a semantic continuation economy: trajectories that reduce contradiction-energy while avoiding unnecessary axis oscillation are dynamically preferred.

LOVE is therefore not a metaphorical construct but a variational principle governing admissible semantic evolutions in spacetime.

Postulate 22.5.4 (LOVE dynamics). The macroscopic dynamics of semantic fields $s : M \rightarrow S$ is governed (in the absence of explicit noise terms and external sources) by the Euler–Lagrange equations associated with $\mathcal{A}_{\text{LOVE}}[s]$:

$$\frac{\delta \mathcal{A}_{\text{LOVE}}}{\delta s}(x) = 0, \quad (22.5.3)$$

where $\delta/\delta s$ denotes the L^2 -variational derivative with respect to s , computed in the weak sense under compactly supported variations $\delta s \in C_c^\infty(M; S)$.

Remark 22.5.5. Equation (22.5.3) yields a second-order (in spacetime) field equation for s with axis-resolved kinetic terms and a potential given by the curvature expansion of W_{univ} around K . In the next chapter we will extract from (22.5.3) a derived object: the *emergent vector field* associated with s , together with its gradient and vorticity components.

22.6 Potential, objective, and variational principle

A recurrent source of confusion in semantic field theories is the tendency to identify potential, objective, and dynamical law as if they were interchangeable notions. In the present framework, however, these three levels must be sharply distinguished.

First, the LOVE potential $V_{\text{LOVE}}(s)$ is a *state-evaluating quantity*. It assigns to a semantic configuration s a scalar contradiction-energetic value obtained from the quadratic expansion of W_{univ} around the kernel region K , together with axis-resolved and cross-axis curvature contributions. In this sense, V_{LOVE} expresses how favourable, coherent, or contradiction-heavy a local semantic state is.

Second, the LOVE Lagrangian density

$$\mathcal{L}_{\text{LOVE}}(x, s, \partial s) = \mathcal{L}_{\text{kin}}(x, s, \partial s) + \mathcal{L}_{\text{pot}}(x, s)$$

is not merely a valuation of states. It is part of a variational structure on fields over spacetime. Its associated action

$$\mathcal{A}_{\text{LOVE}}[s] = \int_M \mathcal{L}_{\text{LOVE}} d\text{vol}_g$$

defines a genuine stationary-action principle. Extremal fields are not selected because they optimise a static score at one point, but because they realise a semantically admissible tradeoff between contradiction reduction and axis-weighted kinetic economy across spacetime. LOVE therefore specifies not only what is locally favourable, but which field evolutions are dynamically admissible.

Third, the term *objective function* should be reserved for a different level: namely, the level of technical optimisation procedures in implemented systems. In machine learning, control, or engineering settings, one may define a task-dependent loss or objective

$$\mathcal{J} = \mathcal{J}_{\text{task}} + \lambda_1 \mathcal{J}_{\text{sem}} + \lambda_2 \mathcal{J}_{\text{traj}} + \dots$$

into which semantic potentials, trajectory penalties, alignment regularisers, or witness terms are incorporated as additional contributions. Such an objective is not identical with the ontological potential V_{LOVE} , nor with the action $\mathcal{A}_{\text{LOVE}}$. Rather, it is an engineering-level operationalisation that may borrow selected structural motifs from them.

We therefore propose the following conceptual separation:

$$\text{valuation} \neq \text{objective} \neq \text{variational law}.$$

More precisely:

$$V_{\text{LOVE}} \quad \text{evaluates states,}$$

$$\mathcal{A}_{\text{LOVE}} \quad \text{selects admissible field evolutions,}$$

and any engineering loss $\mathcal{J}$ built for training or control purposes is a derived construct that may incorporate terms motivated by V_{LOVE} or $\mathcal{A}_{\text{LOVE}}$, but is not identical to either.

This distinction is structurally necessary. Without it, one risks collapsing semantic ontology, dynamical law, and technical optimisation into a single undifferentiated notion of “goal”. That would obscure the difference between:

- what a semantic state is worth,
- how a semantic field may evolve,
- and how an implemented system is practically steered.

The distinction becomes especially important once the present framework is transferred to technical alignment settings. There, one may construct explicit training objectives that contain terms inspired by semantic gravitation, for example:

- contradiction-energy penalties derived from V_{LOVE} ,
- trajectory-regularising terms inspired by axis-weighted kinetic economy,
- witness-like penalties favouring coherence-preserving closure,

- or decoherence-sensitive terms penalising unnecessarily collapsing trajectories.

Yet even in such cases, the engineering loss remains a secondary construct. It is a pragmatic approximation, not the ontology itself. The potential and the action belong to the theory of semantic field structure; the objective belongs to the implementation layer.

One may express this relation schematically as follows:

$$\mathcal{J} = \mathcal{F}(V_{\text{LOVE}}, \mathcal{A}_{\text{LOVE}}, \text{task constraints, resource constraints, data regime}),$$

where $\mathcal{F}$ is system-dependent. No canonical identification of $\mathcal{J}$ with either V_{LOVE} or $\mathcal{A}_{\text{LOVE}}$ is assumed.

Accordingly, the LOVE formalism should not be read as a loss function in the machine learning sense. It is a field-theoretic variational structure. It becomes a loss-like object only after additional engineering choices have been made: discretisation, projection to a finite control horizon, selection of observables, approximation of admissible trajectories, and weighting relative to concrete task terms.

This distinction also clarifies the role of local distortions W_{loc} . When such terms are added to the effective field-theoretic description, they modify the local valuation structure and therefore enter $V_{\text{LOVE}}^{\text{tot}}$. They do not thereby become objectives in the engineering sense. They remain part of the semantic potential layer, unless and until they are explicitly imported into some technical optimisation functional.

In summary, the present framework operates on three non-identical levels:

1. the semantic-potential level, where local states are valued;
2. the variational level, where admissible field evolutions are selected;
3. the engineering level, where practical objectives are built for training, control, or system steering.

Keeping these levels distinct is not terminological pedantry. It is necessary for preserving the internal architecture of the theory and for preventing a category mistake between semantic value, dynamical law, and technical optimisation.

22.7 Homogeneous limit and consistency with gradient dynamics on S

We now sketch how LOVE recovers, or at least is consistent with, the gradient dynamics of W_{univ} on S in appropriate limits.

Consider a spacetime region where:

- the metric $g_{\mu\nu}$ is approximately flat (e.g. Minkowski);
- $s(x)$ is spatially homogeneous, i.e. $s(x^0, \vec{x}) = s(t)$ depends only on time $t = x^0$;
- higher-order corrections (including projector-variation terms and $O(\|\delta\|^3)$ terms in the potential) can be neglected.

In this regime, spatial derivatives vanish and the action (22.5.2) reduces effectively to an integral over time. Using the convention $\partial^\mu = g^{\mu\nu} \partial_\nu$ and Minkowski signature $(-, +, +, +)$ (so $g^{00} = -1$), the reduced action reads

$$\mathcal{A}_{\text{LOVE}}[s] \approx \int dt \left(\frac{1}{2} \sum_{\Xi \in \{C, B, A\}} \gamma_\Xi(s(t)) g^{00} \langle \dot{s}_\Xi(t), \dot{s}_\Xi(t) \rangle - V_{\text{LOVE}}(s(t)) \right),$$

where, at the same level of approximation as in (22.3.1), $\dot{s}_\Xi(t)$ denotes the time derivative of the axis-resolved component in the leading (projected) sense, i.e. $\dot{s}_\Xi(t) \approx P_\Xi(s(t)) \dot{s}(t)$.

Equivalently, since $\partial^0 = g^{00} \partial_0 = -\partial_t$, one may write the homogeneous kinetic contribution as

$$\frac{1}{2} \sum_{\Xi} \gamma_{\Xi}(s(t)) \langle \partial^0 s_{\Xi}(t), \partial_0 s_{\Xi}(t) \rangle = -\frac{1}{2} \sum_{\Xi} \gamma_{\Xi}(s(t)) \langle \dot{s}_{\Xi}(t), \dot{s}_{\Xi}(t) \rangle.$$

Writing the effective mass operator

$$\mathcal{M}(s) = \sum_{\Xi \in \{C, B, A\}} \gamma_{\Xi}(s) P_{\Xi}(s),$$

the homogeneous Euler–Lagrange equation takes the form

$$\mathcal{M}(s_t) \ddot{s}_t + \nabla V_{\text{LOVE}}(s_t) + (\text{corrections from } \gamma_{\Xi}\text{-dependence and projector-variation terms}) = 0, \quad (22.7.1)$$

where the correction term includes contributions from the s -dependence of γ_{Ξ} and, if retained beyond quadratic order, from the neglected projector-variation terms described above. In the strict quadratic near- K truncation, these corrections are treated as higher order and may be neglected to leading order.

Remark 22.7.1 (Overdamped reduction and link to SG8). Comparing (22.7.1) with the gradient dynamics

$$\dot{s}_t = -\Gamma(s_t) \nabla W_{\text{univ}}(s_t) + D(s_t),$$

we see that the former is second order in time while the latter is first order. A gradient-flow dynamics is obtained as a strong-damping (overdamped) limit of (22.7.1) by adding a friction term $\Lambda(s_t) \dot{s}_t$ to the left-hand side.

For mathematical consistency we assume that $\Lambda(s) : S \rightarrow S$ is self-adjoint and strictly positive (hence invertible) in the near- K regime. The damped homogeneous equation then reads

$$\mathcal{M}(s_t) \ddot{s}_t + \Lambda(s_t) \dot{s}_t + \nabla V_{\text{LOVE}}(s_t) = 0.$$

In the regime where inertia is negligible compared to damping, i.e.

$$\|\mathcal{M}(s_t) \ddot{s}_t\| \ll \|\Lambda(s_t) \dot{s}_t\|,$$

one obtains the overdamped approximation

$$\Lambda(s_t) \dot{s}_t \approx -\nabla V_{\text{LOVE}}(s_t).$$

Since $\Lambda(s_t)$ is invertible, this yields

$$\dot{s}_t \approx -\Lambda(s_t)^{-1} \nabla V_{\text{LOVE}}(s_t).$$

Using $V_{\text{LOVE}}(s) = W_{\text{univ}}(s) + O(\|\delta(s)\|^3)$, this reproduces (to quadratic accuracy near K) a gradient-flow form with $\Gamma(s) \sim \Lambda(s)^{-1}$ after matching units and time scales. Distortion terms D and noise can then be incorporated at the level of an effective (stochastic) macrodynamics as in SG8. In this sense LOVE is consistent with the semantic drift dynamics and can be viewed as a second-order completion whose dissipative reduction yields the first-order evolution on S .

Speculative remark. From a programme perspective, LOVE may be interpreted as the mesoscopic limit of a more microscopic (possibly stochastic) dynamics on S : it provides a local field-theoretic summary of near- K geometry (C/B/A stiffness, curvature couplings) while leaving open how friction, distortions, and noise arise from finer-grained mechanisms.

22.8 Axis structure in LOVE and preparation for the emergent vector field

The axis-resolved structure of $\mathcal{L}_{\text{LOVE}}$ has several important consequences which will be developed in the next chapters.

- *C/B/A kinetic anisotropy:* different axis weights γ_C , γ_B , and γ_A imply that spatial and temporal variations of s_C , s_B , and s_A are penalised differently in the action. Regions of spacetime with large γ_C but small γ_A and γ_B favour slowly varying coherence fields but allow more flexible binding and agency fields.
- *Curvature-induced mass terms:* the curvature components embedded in V_{LOVE} act as effective mass terms for fluctuations in s_C , s_B , and s_A around K . Large axis curvatures correspond to heavy modes (strongly localised around K) along the corresponding axis.
- *Axis-resolved canonical momenta:* the kinetic term (22.3.1) induces canonical momenta conjugate to s_C , s_B , and s_A . Combining these with spatial gradients leads naturally to axis-resolved currents and, in particular, to a distinguished *emergent vector field* whose gradient/vorticity decomposition will be the focus of the next chapter.

Summary and outlook. In this chapter we have introduced the *Lagrangian of Vectorial Emergence* (LOVE) as an effective field-theoretic action for S -valued semantic fields on spacetime. The LOVE Lagrangian density $\mathcal{L}_{\text{LOVE}}$ consists of an axis-resolved kinetic term, quadratic in spacetime derivatives of the projected C/B/A components of s , and a potential term given by the curvature expansion of the universal contradiction-energy potential W_{univ} around the global attractor K . We formulated the LOVE dynamics via the Euler–Lagrange equations, clarified the distinction between semantic valuation, engineering objective, and variational principle, and then sketched how, in the homogeneous and overdamped limit, LOVE is consistent with the gradient dynamics on S developed earlier (and with its distorted/stochastic extensions at the level of effective macrodynamics). Finally, we emphasised the role of the C/B/A structure in shaping kinetic anisotropies, curvature-induced mass terms, and axis-resolved canonical momenta. In the next chapter we build on this structure to define the *emergent vector field* associated with LOVE, analyse its gradient and vorticity components, and relate its integral curves to semantic trajectories and attractor structures in spacetime.

Chapter 23

The emergent vector field

In the previous chapter we introduced the LOVE action for S -valued semantic fields on spacetime, with an axis-resolved kinetic term and a potential given (to leading order near K) by the quadratic expansion of W_{univ} around the global attractor K . Earlier chapters described deterministic and stochastic dynamics on S , including distortions and C/B/A resolution, and the associated Fokker–Planck evolution on distributions. In this chapter we collect these structures into a single object: the *emergent vector field*. This is the effective force field on S that organises semantic trajectories, admits a gradient/vorticity decomposition (in a weighted L^2 sense), and gives rise to integral curves and invariants used in later parts of the programme.

23.1 Definition on the semantic space S

We begin at the level of the semantic space S , ignoring spacetime dependence for the moment. Recall from SG8 that the deterministic drift part of the dynamics on S can be written as

$$F(s) = -\Gamma(s) \nabla W_{\text{univ}}(s) + D(s), \quad (23.1.1)$$

where:

- $\Gamma(s)$ is the axis-intensification operator

$$\Gamma(s) = \Gamma_C(s) P_C(s) + \Gamma_B(s) P_B(s) + \Gamma_A(s) P_A(s);$$

- $D(s)$ is a distortion field, decomposable into axis components $D_C(s)$, $D_B(s)$, $D_A(s)$;
- $P_C(s)$, $P_B(s)$, $P_A(s)$ are the orthogonal projections onto the C/B/A bundles $E_C(s)$, $E_B(s)$, $E_A(s)$ from SG5.

Definition 23.1.1 (Emergent vector field on S). The *emergent vector field* on S is the map

$$\mathcal{E} : S \supset \mathcal{D}_{\mathcal{E}} \rightarrow S,$$

defined on a domain $\mathcal{D}_{\mathcal{E}}$ containing the neighbourhood U of K , by

$$\mathcal{E}(s) := F(s) = -\Gamma(s) \nabla W_{\text{univ}}(s) + D(s). \quad (23.1.2)$$

Remark 23.1.2 (Notation and bundle collision avoidance). We write $\mathcal{E}$ (rather than E) to avoid notational collision with the C/B/A bundles E_C, E_B, E_A . Likewise, we write $\mathcal{E}_C, \mathcal{E}_B, \mathcal{E}_A$ for axis-components of the vector field, reserving E_C, E_B, E_A exclusively for

the bundles introduced in SG5. Equation (23.1.2) gathers in one expression the universal (axis-intensified) gradient force $-\Gamma(s)\nabla W_{\text{univ}}(s)$ and all deviations from pure universal gradient flow via the distortion field $D(s)$. In particular, if $D(s) \equiv 0$ and $\Gamma(s) \equiv \text{id}$, then $\mathcal{E}(s)$ reduces to universal semantic gravitation $-\nabla W_{\text{univ}}(s)$. (Equivalently: “ideal undistorted” here means no distortion and no axis-intensification beyond the identity.)

Integral curves of the emergent vector field solve the deterministic ODE

$$\dot{s}_t = \mathcal{E}(s_t), \quad (23.1.3)$$

and are referred to as *emergence trajectories*.

Definition 23.1.3 (Emergence trajectory). A curve $t \mapsto s_t \in S$ is an *emergence trajectory* if it is a maximal solution of (23.1.3) with initial condition $s_0 \in \mathcal{D}_{\mathcal{E}}$.

23.2 Axis-resolved force modes and directed gradients

The C/B/A decomposition of $\mathcal{E}(s)$ refines emergence trajectories into axis-specific *force modes*. We denote axis-components of $\mathcal{E}$ by

$$\mathcal{E}(s) = \mathcal{E}_C(s) + \mathcal{E}_B(s) + \mathcal{E}_A(s), \quad \mathcal{E}_{\Xi}(s) := P_{\Xi}(s) \mathcal{E}(s), \quad \Xi \in \{C, B, A\}. \quad (23.2.1)$$

Definition 23.2.1 (Axis-resolved emergent vector field). For $\Xi \in \{C, B, A\}$ the Ξ -*component* of the emergent vector field is

$$\mathcal{E}_{\Xi}(s) = -\Gamma_{\Xi}(s) \nabla_{\Xi} W_{\text{univ}}(s) + D_{\Xi}(s),$$

where

$$\nabla_{\Xi} W_{\text{univ}}(s) := P_{\Xi}(s) \nabla W_{\text{univ}}(s), \quad D_{\Xi}(s) := P_{\Xi}(s) D(s),$$

and $\Gamma_{\Xi}(s) > 0$ is the scalar intensification factor appearing in $\Gamma(s) = \sum_{\Xi} \Gamma_{\Xi}(s) P_{\Xi}(s)$.

Near K , the sign of the scalar projection of $\mathcal{E}_{\Xi}(s)$ onto $\nabla_{\Xi} W_{\text{univ}}(s)$ distinguishes descending vs. ascending directions along axis Ξ . This connects directly to the six directed gradients $\pm C, \pm B, \pm A$ introduced earlier.

Definition 23.2.2 (Force modes along directed gradients). Let $s \in U$ and $\Xi \in \{C, B, A\}$. Define the signed *force mode* along $\pm \Xi$ by

$$f_{\pm \Xi}(s) := \frac{\langle \mathcal{E}_{\Xi}(s), \pm \nabla_{\Xi} W_{\text{univ}}(s) \rangle}{\|\nabla_{\Xi} W_{\text{univ}}(s)\|^2},$$

whenever $\nabla_{\Xi} W_{\text{univ}}(s) \neq 0$.

Remark 23.2.3. If $D_{\Xi}(s) = 0$ and $\Gamma_{\Xi}(s) > 0$, then $\mathcal{E}_{\Xi}(s)$ is strictly anti-parallel to $\nabla_{\Xi} W_{\text{univ}}(s)$, i.e. $\mathcal{E}_{\Xi}(s) = -\Gamma_{\Xi}(s) \nabla_{\Xi} W_{\text{univ}}(s)$. With the definition above this implies

$$f_{-\Xi}(s) = \Gamma_{\Xi}(s), \quad f_{+\Xi}(s) = -\Gamma_{\Xi}(s).$$

Thus $f_{-\Xi} > 0$ encodes the strength of descending motion along the $-\Xi$ direction, while $f_{+\Xi} < 0$ records the corresponding (opposite-signed) projection onto the $+\Xi$ direction induced by the same anti-parallel alignment. Deviations from this ideal anti-parallel case (including positive values of $f_{+\Xi}$) reflect components of $\mathcal{E}_{\Xi}(s)$ that locally increase W_{univ} along axis Ξ , typically due to distortions $D_{\Xi}(s)$ or strong axis mixing through curvature couplings.

23.3 Gradient–vorticity decomposition in the semantic universe

We next formulate a gradient/vorticity decomposition of $\mathcal{E}$, generalising a Helmholtz/Hodge decomposition to the semantic setting. In infinite dimensions the rigorous theory depends on the chosen function class, reference measure, and boundary behaviour; here we state a working hypothesis sufficient for programme-level invariants and for linking to stationary currents.

Let μ_β be an equilibrium measure of Gibbs type for W_{univ} (ideal case, see SG9),

$$\mu_\beta(ds) = \frac{1}{Z_\beta} \exp(-\beta W_{\text{univ}}(s)) \mu_0(ds).$$

Here β plays the role of an inverse noise strength (inverse temperature); in the notation where the stationary Gibbs density is $\propto \exp(-W_{\text{univ}}/D)$ one has $\beta = 1/D$.

Consider the Hilbert space $L^2(\mu_\beta; S)$ of square-integrable S -valued vector fields on S , with inner product

$$\langle U, V \rangle_{L^2(\mu_\beta; S)} := \int_S \langle U(s), V(s) \rangle d\mu_\beta(s).$$

Hypothesis 23.3.1 (Weighted semantic Helmholtz decomposition). Let $\mathcal{G} \subset L^2(\mu_\beta; S)$ denote a class of gradient fields

$$U_\phi(s) := -\nabla \phi(s),$$

generated by sufficiently regular scalar potentials ϕ (e.g. smooth cylinder functions). Assume:

1. the closure $\overline{\mathcal{G}}$ admits an orthogonal complement $\mathcal{G}^\perp$ in $L^2(\mu_\beta; S)$;
2. the emergent vector field $\mathcal{E} \in L^2(\mu_\beta; S)$ admits a decomposition

$$\mathcal{E} = \mathcal{E}^{\text{grad}} + \mathcal{E}^{\text{vort}}, \quad \mathcal{E}^{\text{grad}} \in \overline{\mathcal{G}}, \quad \mathcal{E}^{\text{vort}} \in \mathcal{G}^\perp;$$

3. $\mathcal{E}^{\text{grad}}$ is unique up to μ_β -a.e. equality and can be approximated by gradients in $L^2(\mu_\beta; S)$.

Definition 23.3.2 (Gradient and vorticity parts of $\mathcal{E}$). Assume Hypothesis 23.3.1 and suppose $\mathcal{E} \in L^2(\mu_\beta; S)$. Define

$$\mathcal{E}^{\text{grad}} := \text{orthogonal projection of } \mathcal{E} \text{ onto } \overline{\mathcal{G}}, \quad \mathcal{E}^{\text{vort}} := \mathcal{E} - \mathcal{E}^{\text{grad}}.$$

Remark 23.3.3. In the ideal undistorted case, $\mathcal{E}(s) = -\nabla W_{\text{univ}}(s)$ and hence $\mathcal{E}^{\text{grad}} = \mathcal{E}$, $\mathcal{E}^{\text{vort}} = 0$. Distortions $D(s)$ and (state-dependent) intensification $\Gamma(s)$ generally yield $\mathcal{E}^{\text{vort}} \neq 0$, capturing cyclic semantic flows that are invisible to purely potential-based descent.

Speculative remark (inside/outside oscillations on nontrivial attractors). If the attractor K (or a suitable topological quotient thereof) supports nontrivial loops, then $\mathcal{E}^{\text{vort}}$ can be viewed as the component that drives motion along loop-like directions inside (or near) the attractor neighbourhood, while $\mathcal{E}^{\text{grad}}$ drives descent toward (and stabilisation around) low-contradiction regions. This reading is optional and does not enter any derivation below.

Divergence and stationary currents

The decomposition above can be related to distributional dynamics. For mathematical clarity we formulate the continuity equation in weak form. Let ν_t be a time-dependent probability measure on S (absolutely continuous with respect to a reference measure μ such as μ_0 or μ_β), and write $\rho_t := d\nu_t/d\mu$ when a density exists. For a class of test functions φ (e.g. smooth cylinder functions) one may express transport by $\mathcal{E}$ via

$$\frac{d}{dt} \int_S \varphi(s) \nu_t(ds) = \int_S \langle \nabla \varphi(s), \mathcal{E}(s) \rangle \nu_t(ds),$$

which corresponds formally to

$$\partial_t \rho_t(s) = -\nabla \cdot (\mathcal{E}(s) \rho_t(s)),$$

with the divergence understood as the adjoint (in the weak sense) of the gradient under μ .

In a stationary state ν_∞ (or density ρ_∞), the probability flux

$$J_\infty(s) := \mathcal{E}(s) \rho_\infty(s)$$

satisfies the stationarity condition

$$\nabla \cdot J_\infty = 0$$

in the corresponding weak sense.

Remark 23.3.4. In equilibrium situations satisfying detailed balance (ideal undistorted case), stationary currents vanish and the drift is of gradient type relative to the equilibrium measure. In non-equilibrium steady states, non-trivial stationary currents $J_\infty \neq 0$ with $\nabla \cdot J_\infty = 0$ may persist, consistent with the presence of a non-gradient component $\mathcal{E}^{\text{vort}}$ in the weighted $L^2(\mu_\beta)$ decomposition. This interpretation guides programme-level invariants; establishing it as a theorem requires specifying the function class, reference measure, and boundary behaviour in the infinite-dimensional setting.

23.4 Vorticity and circulation

To formalise the notion of vorticity in the semantic universe, we use the Hilbert-space structure of S to define a one-form associated with $\mathcal{E}^{\text{vort}}$.

Definition 23.4.1 (Emergent one-form and circulation). Let $\mathcal{E}^{\text{vort}}$ be as above. Define the one-form α on S by

$$\alpha_s(u) := \langle \mathcal{E}^{\text{vort}}(s), u \rangle, \quad s \in \mathcal{D}_\mathcal{E}, \quad u \in S.$$

For a smooth closed curve $\gamma : [0, 1] \rightarrow \mathcal{D}_\mathcal{E}$ with $\gamma(0) = \gamma(1)$, the *emergent circulation* along γ is

$$\mathcal{C}(\gamma) := \oint_\gamma \alpha := \int_0^1 \langle \mathcal{E}^{\text{vort}}(\gamma(t)), \dot{\gamma}(t) \rangle dt.$$

Remark 23.4.2. If $\mathcal{E}^{\text{vort}}$ were the gradient of a scalar potential ϕ , then $\alpha = -d\phi$ would be exact and $\mathcal{C}(\gamma)$ would vanish for all closed curves γ . Non-zero circulation therefore signals the presence of genuinely rotational components of $\mathcal{E}$. In the presence of noise and Fokker–Planck dynamics, such circulation can be related to persistent probability currents in non-equilibrium steady states.

A full vorticity two-form can be defined formally as $\omega := d\alpha$, but the infinite-dimensional differential geometry involved goes beyond the scope of this paper. For the present purposes, the one-form α and its circulations suffice as measures of rotational semantic flow.

23.5 Integral curves, invariants, and emergent paths

We now return to the integral curves of $\mathcal{E}$ and identify some invariants and quantities of interest along them.

Proposition 23.5.1 (Energy balance along emergence trajectories). *Let $t \mapsto s_t$ satisfy $\dot{s}_t = \mathcal{E}(s_t)$ and suppose W_{univ} is differentiable along s_t . Then*

$$\frac{d}{dt}W_{\text{univ}}(s_t) = \langle \nabla W_{\text{univ}}(s_t), \dot{s}_t \rangle = -\langle \nabla W_{\text{univ}}(s_t), \Gamma(s_t) \nabla W_{\text{univ}}(s_t) \rangle + \langle \nabla W_{\text{univ}}(s_t), D(s_t) \rangle.$$

Proof. This is a direct application of the chain rule and the definition (23.1.2). $\square$

In particular, if $\Gamma(s)$ is positive and $D(s)$ has small projection onto $\nabla W_{\text{univ}}(s)$, then $W_{\text{univ}}(s_t)$ decreases along emergence trajectories, with deviations controlled by the distortion term.

Definition 23.5.2 (A path functional compatible with the homogeneous LOVE limit). For a smooth path $\xi : [0, T] \rightarrow \mathcal{D}_{\mathcal{E}}$ we define the functional

$$\mathfrak{S}[\xi] := \int_0^T \left(\frac{1}{2} \sum_{\Xi \in \{C, B, A\}} \gamma_{\Xi}(\xi(t)) \|\dot{\xi}_{\Xi}(t)\|^2 + V_{\text{LOVE}}(\xi(t)) \right) dt,$$

where $\dot{\xi}_{\Xi}(t) := P_{\Xi}(\xi(t)) \dot{\xi}(t)$ denotes the leading (projected) axis-velocity.

Remark 23.5.3. In the overdamped (dissipative) reduction of the homogeneous LOVE dynamics, functionals of the above positive form naturally arise as energy-like or Onsager–Machlup-type objects. In the stochastic setting, an Onsager–Machlup or Freidlin–Wentzell functional built from $\mathcal{E}$ and the noise covariance plays an analogous role and quantifies the likelihood of paths.

Speculative remark. At a programme level, one may conjecture that certain classes of emergence trajectories, defined as local minimisers of $\mathfrak{S}[\xi]$ subject to boundary conditions or constraints, correspond to canonical semantic processes (e.g. resolution of contradictions, formation of stable inner spaces, or alignment of agency). The invariants associated with $\mathcal{E}^{\text{vort}}$ (circulations, topological indices) would then classify families of such processes that are not captured by W_{univ} alone.

23.6 From S to spacetime: field-level emergent vector field

Finally, we sketch how the emergent vector field on S is lifted to the level of fields on spacetime, where LOVE is defined.

Let $s : M \rightarrow S$ be a semantic field. For each $x \in M$ with $s(x) \in \mathcal{D}_{\mathcal{E}}$, we can evaluate the emergent vector field at $s(x)$.

Definition 23.6.1 (Field-level emergent vector field). The *field-level emergent vector field* is the map

$$\mathcal{E}_s : M \supset M_{\mathcal{E}} \rightarrow S, \quad \mathcal{E}_s(x) := \mathcal{E}(s(x)),$$

defined on the subset $M_{\mathcal{E}} := \{x \in M : s(x) \in \mathcal{D}_{\mathcal{E}}\}$.

Remark 23.6.2 (Link to the LOVE overdamped reduction). The map $x \mapsto \mathcal{E}_s(x)$ assigns to each spacetime point x a semantic drift vector in S , indicating the local direction of semantic emergence at that point according to the LOVE-compatible (overdamped) dynamics. Combined with spacetime derivatives $\partial_\mu s(x)$, this allows one to construct scalar and vector quantities on M that encode how semantic emergence is distributed across spacetime, for example by forming currents of the form

$$J^\mu(x) := \langle \partial^\mu s(x), \mathcal{E}_s(x) \rangle, \quad \partial^\mu := g^{\mu\nu} \partial_\nu,$$

or axis-resolved versions thereof. These objects will be used in the next chapter, where the induced semantic tensor $h_{\mu\nu} = \langle \partial_\mu s, \partial_\nu s \rangle$ and its axis-resolved components are introduced and related to spacetime observables.

Summary and outlook. In this chapter we have defined the *emergent vector field* $\mathcal{E}$ on the semantic state space S as the deterministic drift of the LOVE-compatible dynamics, comprising the axis-intensified universal gradient of W_{univ} plus distortions. We decomposed $\mathcal{E}$ along the C/B/A axes, introduced force modes along the six directed gradients $\pm C$, $\pm B$, $\pm A$, and formulated a weighted semantic Helmholtz-type gradient/vorticity decomposition $\mathcal{E} = \mathcal{E}^{\text{grad}} + \mathcal{E}^{\text{vort}}$ under a working hypothesis. We associated a one-form and circulation to $\mathcal{E}^{\text{vort}}$, discussed integral curves and energy balance along emergence trajectories, and introduced a path functional compatible with the overdamped homogeneous LOVE limit. Finally, we lifted $\mathcal{E}$ to a field-level object $\mathcal{E}_s(x)$ on spacetime, preparing the analysis of semantic metrics and currents. The next chapter begins *Teil IV* by studying the semantic tensor induced on spacetime by s , relating the geometry of S and its C/B/A structure to effective geometric quantities on (M, g) without modifying the underlying general relativity framework.

Chapter 24

The semantic metric on spacetime

In the previous chapters we constructed the LOVE action for S -valued semantic fields on spacetime and introduced the emergent vector field $\mathcal{E}$ on the semantic state space S , which encodes axis-resolved semantic gravitation and distortions. In this chapter we start *Teil IV* by turning to spacetime geometry. Using the Hilbert structure of S and a semantic field $s : M \rightarrow S$, we define a family of *semantic metrics* on spacetime which quantify how semantic configurations change along spacetime directions. These metrics do not modify the gravitational metric g of general relativity; rather, they provide additional, emergent geometric structure induced by the semantic embedding of spacetime into S .

Remark 24.0.1 (Signature convention). Throughout this paper we use Lorentzian signature $(-, +, +, +)$. Thus timelike vectors v satisfy $g(v, v) < 0$, spacelike vectors satisfy $g(v, v) > 0$, and a future-directed unit timelike observer field u is normalised by $g(u, u) = -1$.

24.1 Semantic embeddings and induced tensors

Let (M, g) be a four-dimensional, oriented Lorentzian manifold with metric $g_{\mu\nu}$ and volume form $d\text{vol}_g$. A semantic configuration is described by a field

$$s : M \rightarrow S,$$

as in the LOVE construction. The inner product $\langle \cdot, \cdot \rangle$ on S induces natural bilinear forms on the tangent spaces of M via the derivatives of s .

Definition 24.1.1 (Semantic embedding and Jacobian). For $x \in M$, the differential of s at x is the linear map

$$(ds)_x : T_x M \rightarrow S, \quad (ds)_x(v) = v^\mu \partial_\mu s(x),$$

where $v = v^\mu \partial_\mu \in T_x M$. We call $(ds)_x$ the *semantic Jacobian* at x .

The semantic Jacobian measures how changes in spacetime position translate into changes in the semantic configuration. Composition with the inner product on S defines a symmetric $(0, 2)$ -tensor on M .

Definition 24.1.2 (Induced semantic tensor). The *induced semantic tensor* h on M is the $(0, 2)$ -tensor field defined by

$$h_{\mu\nu}(x) := \langle \partial_\mu s(x), \partial_\nu s(x) \rangle, \quad x \in M. \quad (24.1.1)$$

Equivalently,

$$h_x(u, v) := \langle (ds)_x(u), (ds)_x(v) \rangle, \quad u, v \in T_x M.$$

Lemma 24.1.3 (Basic properties of h). *Assume s is C^1 . Then:*

1. $h_{\mu\nu}$ is symmetric: $h_{\mu\nu} = h_{\nu\mu}$;
2. for any $x \in M$ and $v \in T_x M$,

$$h_x(v, v) = \|(ds)_x(v)\|^2 \geq 0;$$

3. h is positive semidefinite on each $T_x M$;
4. $h \equiv 0$ if and only if s is constant on each connected component of M .

Proof. Symmetry and non-negativity follow directly from the definition and the inner product structure of S . If $h_x(v, v) = 0$ for all v , then $(ds)_x = 0$ and hence s is locally constant; connectedness implies global constancy on components. $\square$

Remark 24.1.4. The tensor h is the pullback of the Hilbert-space metric on S along s , in analogy with induced metrics in sigma models. It is positive semidefinite as a bilinear form on $T_x M$ and therefore *does not replace* the Lorentzian spacetime metric g ; it is an additional structure that quantifies semantic variation along spacetime directions. In particular, h becomes a genuine (positive definite) Riemannian metric on M only under an immersion-type condition (e.g. $(ds)_x$ injective for all x); unless such a condition is imposed, no inversion of h is assumed.

24.2 Gauge freedom in semantic state space

The construction of the induced semantic tensor depends on the Hilbert structure of the target space S , but its geometric content should not depend on the particular coordinate representation chosen in that space. This leads naturally to an internal gauge freedom associated with isometries of S .

Let $U : S \rightarrow S$ be a unitary operator, i.e.

$$\langle Ua, Ub \rangle = \langle a, b \rangle \quad \text{for all } a, b \in S.$$

Such transformations preserve the Hilbert norm and inner product and therefore represent internal rotations of semantic state space.

Definition 24.2.1 (Internal gauge transformation). An *internal gauge transformation* of the semantic field is a transformation

$$s(x) \mapsto s'(x) := U s(x),$$

where U is a unitary operator on S .

Proposition 24.2.2 (Gauge invariance of the induced semantic tensor). *Under an internal gauge transformation $s \mapsto s' = Us$, the induced semantic tensor remains invariant:*

$$h'_{\mu\nu}(x) := \langle \partial_\mu s'(x), \partial_\nu s'(x) \rangle = h_{\mu\nu}(x).$$

Proof. Since U is linear and independent of x ,

$$\partial_\mu s'(x) = U \partial_\mu s(x).$$

Using unitarity,

$$\langle \partial_\mu s'(x), \partial_\nu s'(x) \rangle = \langle U \partial_\mu s(x), U \partial_\nu s(x) \rangle = \langle \partial_\mu s(x), \partial_\nu s(x) \rangle.$$

Hence $h'_{\mu\nu}(x) = h_{\mu\nu}(x)$. $\square$

Remark 24.2.3. Proposition 24.2.2 shows that the semantic tensor depends only on the geometric embedding of spacetime into the semantic state space, not on a particular orthonormal basis or coordinate presentation of S . In this sense, the isometry group of S acts as an internal gauge group of the semantic sector.

Remark 24.2.4 (Relation to sigma models). This internal gauge freedom is analogous to target-space isometries in nonlinear sigma models. There as here, fields are maps from spacetime into a target space, and the induced tensor built from derivatives of the field depends only on the intrinsic metric structure of that target space. The semantic metric $h_{\mu\nu}$ is therefore compatible with the sigma-model interpretation developed earlier, now at the level of spacetime geometry.

24.3 Axis-resolved semantic tensors

The C/B/A decomposition of S near the global attractor K allows us to refine h into three axis-resolved components.

Let $U \subset S$ be the neighbourhood of K from SG5 such that $s(x) \in U$ for all x in the region of interest. For each x , define the projected axis-derivatives

$$\partial_\mu s_\Xi(x) := P_\Xi(s(x)) \partial_\mu s(x), \quad \Xi \in \{C, B, A\}.$$

As in the LOVE kinetic term, projector-variation contributions of the form

$$(\partial_\mu P_\Xi(s(x))) s(x)$$

are treated as higher-order corrections in the near- K effective regime and are neglected at the quadratic level considered here. More precisely, we assume that in the near- K regime

$$\|\partial_\mu P_\Xi(s)\| = O(\|s - \pi_K(s)\|),$$

so that projector-variation terms contribute only at cubic order in $\delta(s) = s - \pi_K(s)$ and are consistently neglected in the quadratic truncation.

Definition 24.3.1 (Axis-resolved semantic tensors). For $\Xi \in \{C, B, A\}$ define

$$h_{\mu\nu}^{(\Xi)}(x) := \langle \partial_\mu s_\Xi(x), \partial_\nu s_\Xi(x) \rangle. \quad (24.3.1)$$

The full induced semantic tensor decomposes as

$$h_{\mu\nu} = h_{\mu\nu}^{(C)} + h_{\mu\nu}^{(B)} + h_{\mu\nu}^{(A)} + r_{\mu\nu},$$

where $r_{\mu\nu}$ collects residual contributions outside the C/B/A sector (e.g. from a neglected residual component $r(x)$ when s is not perfectly confined to $E_C \oplus E_B \oplus E_A$).

Remark 24.3.2. In regimes where $r_{\mu\nu}$ is negligible, h is well approximated by the sum of the three axis-resolved tensors. Each $h^{(\Xi)}$ measures semantic variation along spacetime directions in coherence (C), binding (B), and agency (A) sectors respectively. The LOVE kinetic term constructed earlier can be written in terms of these tensors as

$$\mathcal{L}_{\text{kin}} = \frac{1}{2} \sum_{\Xi \in \{C, B, A\}} \gamma_\Xi(s) g^{\mu\nu} h_{\mu\nu}^{(\Xi)},$$

up to residual terms. Thus the axis weights γ_Ξ act as couplings between the spacetime metric g and the axis-resolved semantic tensors.

24.4 Semantic lengths, areas, and sign issues

The semantic tensor h induces notions of semantic length (and, more generally, area/volume via pullbacks) on spacetime, distinct from their geometric counterparts defined by g .

Definition 24.4.1 (Semantic length of a curve). Let $\gamma : [0, 1] \rightarrow M$ be a smooth curve with tangent $\dot{\gamma}(t) \in T_{\gamma(t)}M$. The *semantic length* of γ is

$$L_{\text{sem}}[\gamma] := \int_0^1 \sqrt{h_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt = \int_0^1 \|(\text{ds})_{\gamma(t)}(\dot{\gamma}(t))\| dt.$$

Remark 24.4.2. $L_{\text{sem}}[\gamma]$ measures how much the semantic configuration changes along the spacetime curve γ , in units of the S -norm. Curves that are short in spacetime (small proper time or spatial distance) may still be long semantically (large changes in s), and vice versa. Axis-resolved versions

$$L_{\text{sem}}^{(\Xi)}[\gamma] := \int_0^1 \|(\text{ds}_{\Xi})_{\gamma(t)}(\dot{\gamma}(t))\| dt$$

quantify semantic change along C , B , or A directions specifically.

A common scalar derived from h is the trace $g^{\mu\nu}h_{\mu\nu}$. Since g is Lorentzian, this contraction is *not* sign-definite and should be used with care when one wants an “activity density”.

Observer-resolved densities (sign-definite option)

For sign-definite scalars one may introduce an observer field.

Definition 24.4.3 (Observer-resolved semantic densities). Let u be a future-directed unit timelike vector field on M , normalised by $g(u, u) = -1$. Define the spatial projector

$$q_{\mu\nu} := g_{\mu\nu} + u_\mu u_\nu, \quad q^\mu{}_\nu q^\nu{}_\lambda = q^\mu{}_\lambda,$$

and (using the g -metric to raise indices)

$$q^{\mu\nu} := g^{\mu\alpha} g^{\nu\beta} q_{\alpha\beta} = g^{\mu\nu} + u^\mu u^\nu.$$

Define

$$\tau_{\text{sem}}(x) := h_{\mu\nu}(x) u^\mu(x) u^\nu(x) = \|u^\mu(x) \partial_\mu s(x)\|^2, \quad (24.4.1)$$

$$\rho_{\text{sem}}(x) := q^{\mu\nu}(x) h_{\mu\nu}(x) \geq 0. \quad (24.4.2)$$

Axis-resolved versions $\tau_{\text{sem}}^{(\Xi)}$ and $\rho_{\text{sem}}^{(\Xi)}$ are defined analogously by replacing h with $h^{(\Xi)}$.

Remark 24.4.4. τ_{sem} measures the semantic change rate along the observer’s time direction, while ρ_{sem} measures the magnitude of spatial semantic gradients in the observer’s rest frames. Both are nonnegative by construction and are often the appropriate “semantic activity” scalars when one wants sign-definite densities.

Trace densities (coordinate-invariant but not sign-definite)

For completeness, we also record the trace-type scalars

$$\rho_{\text{sem}}^{\text{tr}}(x) := g^{\mu\nu}(x) h_{\mu\nu}(x), \quad (24.4.3)$$

$$\rho_{\text{sem}}^{\text{tr},(\Xi)}(x) := g^{\mu\nu}(x) h_{\mu\nu}^{(\Xi)}(x), \quad \Xi \in \{C, B, A\}, \quad (24.4.4)$$

which appear directly in the LOVE kinetic term:

$$\mathcal{L}_{\text{kin}}(x, s, \partial s) = \frac{1}{2} \sum_{\Xi} \gamma_{\Xi}(s(x)) \rho_{\text{sem}}^{\text{tr},(\Xi)}(x) \quad (\text{up to residual terms}).$$

When interpreting $\rho_{\text{sem}}^{\text{tr}}$ as an “activity density” one should keep in mind its Lorentzian sign ambiguity; in applications, $(\tau_{\text{sem}}, \rho_{\text{sem}})$ are often the more robust observables.

24.5 Relation to the emergent vector field

The emergent vector field $\mathcal{E} : \mathcal{D}_{\mathcal{E}} \rightarrow S$ lifts to spacetime via $\mathcal{E}_s(x) := \mathcal{E}(s(x))$. Combining $\mathcal{E}_s(x)$ with derivatives of s yields a natural current-like object on M .

Definition 24.5.1 (Emergent semantic current). For a semantic field $s : M \rightarrow S$ with $s(x) \in \mathcal{D}_{\mathcal{E}}$, the *emergent semantic current* is the vector field

$$J^{\mu}(x) := \langle \partial^{\mu} s(x), \mathcal{E}_s(x) \rangle = g^{\mu\nu}(x) \langle \partial_{\nu} s(x), \mathcal{E}(s(x)) \rangle. \quad (24.5.1)$$

Axis-resolved versions are

$$J_{(\Xi)}^{\mu}(x) := g^{\mu\nu}(x) \langle \partial_{\nu} s_{\Xi}(x), \mathcal{E}_{\Xi}(s(x)) \rangle, \quad \Xi \in \{C, B, A\},$$

where $\mathcal{E}_{\Xi}(s) := P_{\Xi}(s)\mathcal{E}(s)$.

Remark 24.5.2. J^{μ} measures how semantic change along spacetime directions (encoded in $\partial_{\nu} s$) is aligned with the emergent drift (encoded in $\mathcal{E}$). The axis-resolved currents $J_{(\Xi)}^{\mu}$ separate these effects into coherence-, binding-, and agency-dominated contributions.

Proposition 24.5.3 (Formal balance structure (schematic)). *Assume sufficient regularity and a regime in which an overdamped (friction-dominated) reduction of LOVE is appropriate. Then one expects balance relations of the schematic form*

$$\nabla_{\mu} J^{\mu} = -\langle \nabla W_{\text{univ}}(s), \Gamma(s) \nabla W_{\text{univ}}(s) \rangle + \langle \nabla W_{\text{univ}}(s), D(s) \rangle + (\text{curvature} / \text{residual} / \text{local terms}),$$

where Γ and D are the intensification and distortion structures from SG8, and the omitted terms collect cross-axis couplings, residual components, and possible W_{loc} contributions.

Remark 24.5.4. Proposition 24.5.3 is intentionally schematic: the precise identity depends on the chosen dissipation mechanism (how friction is introduced), the exact field equation used (second-order Euler–Lagrange vs. overdamped reduction), and boundary conditions. The main point is that J^{μ} is the natural spacetime object through which semantic drift aligned with semantic gradients enters balance laws.

24.6 Compatibility with general relativity

The semantic tensor h and current J^μ introduce additional geometric structure on (M, g) , but they do not alter the Einstein equations at the level of this paper.

Remark 24.6.1. We emphasise:

- The spacetime metric $g_{\mu\nu}$ remains the fundamental gravitational metric, entering the Einstein–Hilbert action and determining causal structure and geodesics.
- The semantic tensor $h_{\mu\nu}$ and derived scalars (e.g. τ_{sem} , ρ_{sem} , or trace-type $\rho_{\text{sem}}^{\text{tr}}$) are additional fields on (M, g) induced by the semantic embedding $s : M \rightarrow S$.
- In principle, these semantic fields can enter an effective stress-energy tensor in an extended matter sector (cf. Paper III), but no modification of the gravitational field equations is assumed here.

Speculative remark. At a programme level, one may later consider couplings where semantic gradient energy contributes to an effective energy–momentum tensor, so that semantic structure sources spacetime curvature alongside conventional matter. In the present paper, however, we treat g as given and focus on how s and h evolve on (M, g) , not how they back-react on it.

24.7 Interpretative remarks on semantic distances

Although the constructions above are purely geometric, it is useful to briefly indicate their interpretative reading, explicitly marked as such.

Remark 24.7.1 (Interpretative reading of h , L_{sem} , and observer-resolved densities).

- The tensor $h_{\mu\nu}$ measures how sensitive the semantic configuration is to displacements in spacetime: large $h_x(v, v)$ means that a small move in direction v produces a large semantic change.
- Semantic length $L_{\text{sem}}[\gamma]$ quantifies the semantic distance traversed along a process in spacetime. Paths that are short in g but long in h correspond to scenarios where little physical movement leads to large semantic reconfiguration.
- The observer-resolved scalars τ_{sem} (temporal rate) and ρ_{sem} (spatial gradient magnitude) provide sign-definite semantic activity measures, and their axis-resolved versions distinguish coherence-, binding-, and agency-dominated regions and processes.

These readings are not additional axioms but programme-level interpretations, meant to guide later empirical and conceptual work.

Summary and outlook. In this chapter we constructed the *semantic metric on spacetime* induced by an S -valued semantic field $s : M \rightarrow S$. Using the Hilbert structure of S , we defined the induced semantic tensor $h_{\mu\nu} = \langle \partial_\mu s, \partial_\nu s \rangle$, its axis-resolved components $h^{(C)}$, $h^{(B)}$, $h^{(A)}$, and the associated semantic lengths. We showed that this tensor is invariant under internal unitary transformations of the semantic state space, thereby identifying a natural gauge freedom of the semantic sector compatible with the sigma-model interpretation of the theory. We discussed scalar density constructions and clarified the Lorentzian sign issue for the trace $g^{\mu\nu} h_{\mu\nu}$, introducing observer-resolved, sign-definite

semantic activity scalars $(\tau_{\text{sem}}, \rho_{\text{sem}})$. We related these quantities to the emergent vector field $\mathcal{E}$ via the emergent semantic current $J^\mu = \langle \partial^\mu s, \mathcal{E}(s) \rangle$ and a schematic balance structure in overdamped regimes. We clarified that h supplements rather than replaces the gravitational metric g , providing additional emergent geometry induced by semantic structure. In the next chapter we build on this semantic metric to analyse multi-scale ontological structures in spacetime, including layered inner spaces, bodies as carriers of semantic fields, and the relation between local semantic attractors and global spacetime configurations.

Chapter 25

Dark Matter as Relational Inertia

25.1 Motivation and scope

Building on the semantic metric structure established in the preceding chapter, we now examine how variations in relational integration density manifest at the level of effective gravitational and inertial response, prior to any stabilisation into matter.

The standard interpretation of dark matter postulates additional, non-luminous matter components in order to account for observed gravitational anomalies, in particular galactic rotation curves, gravitational lensing, and the inferred mass budget of bound structures. Despite decades of intensive experimental effort—including searches for weakly interacting massive particles (WIMPs), axions, and other hypothetical candidates—no direct empirical detection has yet been achieved.

At the same time, the Λ CDM framework exhibits persistent conceptual and phenomenological tensions on galactic and sub-galactic scales, including the cusp–core problem, the missing satellites problem, and a pronounced scatter in scaling relations that cannot be straightforwardly reduced to baryonic mass alone.

This chapter explores an alternative explanatory route that does not introduce additional particle species. Instead, it follows the ontological and geometric framework developed in the preceding chapters: spacetime, dynamics, and gravitation are treated as emergent properties of relational and constraint structure, carried by an S -valued semantic field $s : M \rightarrow S$, and quantified (in addition to (M, g)) by induced semantic tensors and currents on spacetime.

Within this perspective, dark matter is not interpreted as a new form of matter. Rather, it is treated as a regime of enhanced *relational inertia*: a field-level manifestation of increased continuation bookkeeping load in spacetime itself.

Remark 25.1.1 (Claims and non-claims). **Claim.** Dark-matter phenomenology can be described as an effective gravitational and inertial response arising from enhanced relational integration density, derived from the programme’s semantic geometry induced by $s : M \rightarrow S$.

Non-claims. No new particle species are postulated; no axiom of general relativity is modified; no superluminal signalling or slowing of microscopic dynamics is assumed.

Status. Programme-level proposal; the explicit coupling form remains open.

25.2 Spacetime as a carrier of conditional structure

Spacetime in this programme carries not only metric structure but also constraint structure.

Formally, spacetime can be viewed as carrying networks of admissibility conditions that restrict which local state continuations are dynamically consistent across neighbourhoods. These conditional relations—*if-then* constraints, coupling rules, and continuation conditions—are physically real insofar as they organise interaction pathways, stabilise or destabilise configurations, and constrain or enable dynamical continuation.

Spacetime therefore carries not only realised states, but also structured, dynamically relevant possibilities that must be coherently integrated for evolution to remain well-defined.

25.3 Meaning as a dynamical quantity: relational obligation

In this framework, *meaning* is not subjective semantics. It is an operational, structural quantity: the totality of integrated relations—actual and dynamically relevant potential—that a system must consistently maintain in order to remain evolvable.

Systems with sparse relational coupling admit coarse continuation bookkeeping relative to densely coupled systems. Systems with dense relational coupling require finer continuation bookkeeping, because many dependencies, correlations, and counterfactual branches must be simultaneously respected.

At a programme level, this is described by a local admissible continuation set $\mathcal{A}(x)$, whose relevance-weighted branching complexity contributes to the bridge variable ρ_{rel} introduced below.

25.4 Relational integration density

Let $x \in M$. Let $\mathcal{A}(x)$ denote the admissible continuation set, and $\mathcal{C}(x) \subseteq \mathcal{A}(x)$ the dynamically relevant, relevance-weighted sub-ensemble with weight w .

Definition 25.4.1 (Relational integration density). The relational integration density $\rho_{\text{rel}}(x)$ is the weighted density of $\mathcal{C}(x)$ per spacetime four-volume element dvol_g , measuring the local continuation bookkeeping load required for coherent evolution.

The four-volume character reflects that continuation load is defined on spacetime neighbourhoods rather than purely spatial slices.

25.5 Transport and balance law for relational load

Relational integration density is not a static quantity. It evolves dynamically as systems branch, integrate, and reorganise their continuation structure across spacetime.

At a coarse-grained level this evolution may be described by a balance equation of the form

$$\partial_t \rho_{\text{rel}} + \nabla_\mu J_{\text{rel}}^\mu = \Sigma_{\text{rel}} - \Lambda_{\text{rel}}.$$

Here:

- ρ_{rel} denotes relational integration density,
- J_{rel}^μ denotes the transport current of relational load,
- Σ_{rel} denotes load generation due to branching or constraint proliferation,
- Λ_{rel} denotes load reduction due to closure or attractor alignment.

Branching processes increase the number of dynamically relevant continuation pathways and therefore act as sources of relational load. Closure processes—such as stabilisation, symmetry formation, or attractor alignment—reduce the effective branching complexity and therefore act as sinks.

This balance law provides a dynamical interpretation of relational inertia: regions with persistent positive net load generation accumulate extended relational structure that can manifest as effective gravitational response.

25.6 Docking to effective gravitational and inertial response

Any formal completion must respect covariant consistency of total stress–energy bookkeeping. One may introduce an exchange current J_ν such that

$$\nabla^\mu T_{\mu\nu}^{\text{baryon}} = J_\nu, \quad \nabla^\mu T_{\mu\nu}^{\text{sem}} = -J_\nu,$$

with total conservation preserved:

$$\nabla^\mu (T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{sem}}) = 0.$$

Here $T_{\mu\nu}^{\text{sem}}$ denotes semantic-sector bookkeeping, modelled as a functional of ρ_{rel} and induced tensors such as $h_{\mu\nu}$.

25.7 Quantitative coupling sketch

25.7.1 Dimensional status

We treat

$$[\rho_{\text{rel}}] = L^{-4}.$$

To map relational load to gravitational response, two universal modelling parameters are introduced:

- coarse-graining scale ℓ_{cg} ,
- conversion scale ε_{rel} .

These must remain system-independent for the framework to retain predictive content.

25.7.2 Response channels

Dark-sector observables separate into two principal response channels:

1. dynamical response (rotation curves, velocity dispersion),
2. lensing or metric response (deflection, time delay).

Because relational load is transported and redistributed across spacetime, these channels need not sample identical functionals of ρ_{rel} . Dynamical observables may respond primarily to effective inertial coupling, while lensing probes the integrated geometric response of the effective metric.

25.7.3 Minimal coupling families

Family A (effective source):

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\text{baryon}} + \varepsilon_{\text{rel}} \ell_{\text{cg}}^4 \hat{\rho}_{\text{rel}} u_{\mu} u_{\nu} \right).$$

Family B (effective geometry):

$$g_{\mu\nu}^{\text{eff}} = g_{\mu\nu} + \beta \Delta g_{\mu\nu}(\hat{\rho}_{\text{rel}}).$$

Both reduce to GR in weak-load regimes.

25.7.4 Halo structure as coarse-grained continuation load

Because ρ_{rel} encodes neighbourhood continuation load rather than discrete particles, coarse-graining generically produces smooth, extended profiles.

Qualitatively three radial regimes are expected:

- Inner regime: baryonic mass dominates curvature and continuation structure.
- Intermediate regime: coarse-grained relational load contributes a smooth extended component.
- Outer regime: load density decays toward void-like limits.

Void regions therefore provide a low-load boundary condition for the framework.

Halo-like structure emerges structurally from load integration rather than from particulate clustering.

25.8 Scale hierarchy of relational response

The interpretation of dark matter as relational inertia becomes clearer when one distinguishes different scales of semantic and gravitational organisation.

- **Micro-scale:** local semantic dynamics and short-range continuation bookkeeping.
- **Meso-scale:** extended relational integration across bound structures.
- **Macro-scale:** fully stabilised large-scale structures.

Dark-matter phenomenology appears primarily in the *mesoscopic regime*.

25.9 Dark matter as bound relational depth

Dark matter appears as the gravitational and inertial effect of increased relational integration density. Bound duration produces context-dependent inertia at fixed baryonic mass.

25.10 Viscoelastic interpretation of spacetime response

A useful physical analogy for the response induced by relational integration density is that of a viscoelastic medium.

- elastic response corresponds to immediate curvature effects,
- viscous response corresponds to reorganisation of relational load,
- relaxation corresponds to delayed redistribution of $\hat{\rho}_{\text{rel}}$.

The parameter τ_{rel} therefore represents an effective relaxation timescale.

25.11 Merger systems as a diagnostic gatekeeper

Galaxy-cluster mergers provide one of the most stringent diagnostic regimes.

The key modelling question is:

What does ρ_{rel} track during dynamical separation?

A relaxation timescale τ_{rel} governs redistribution of relational load.

25.11.1 Programme-level failure gate

If merger observations require a component irreducible to any field-like $\hat{\rho}_{\text{rel}}$, the relational-inertia hypothesis is ruled out.

25.12 Empirical outlook

Primary empirical benchmarks include rotation curves, the radial acceleration relation, and weak lensing.

25.13 Summary

Spacetime carries not only energy and mass, but continuation load.

Where relational integration increases, internal path length grows. Time is the projection of this path length, and where paths are bound, inertia emerges.

Dark matter is the gravitational and inertial appearance of this relational inertia—real as structure, invisible as substance.

Chapter 26

Quantitative coupling sketch: from relational integration density to effective response

26.1 Purpose, scope, and status

Chapter 25 introduces the programme-level claim that a part of the observed “dark-sector” phenomenology can arise without postulating additional particle species, if spacetime response depends not only on baryonic stress–energy but also on *bound relational depth* carried by the semantic field $s : M \rightarrow S$ and summarised by the relational integration density ρ_{rel} .

The present chapter provides a *minimal quantitative docking sketch*. Its purpose is deliberately limited:

- to propose *dimensionally consistent* coupling families that connect ρ_{rel} to effective gravitational/inertial response,
- to justify why *extended, halo-like* residual profiles arise naturally from coarse-grained continuation load,
- to separate *response channels* (dynamics vs. lensing/metrology) as a non-negotiable modelling requirement,
- and to state explicit *gatekeeper conditions* in merger systems (Bullet-type regimes).

No specific coupling is asserted as uniquely correct. All statements below are to be read as *explicit modelling assumptions* that render the programme falsifiable and analytically extendable.

26.2 Dimensional and scaling preliminaries

26.2.1 Dimensional status of ρ_{rel}

By definition (Chapter 25), $\rho_{\text{rel}}(x)$ measures a weighted density of dynamically relevant continuation structure per spacetime volume element. It is therefore natural—and consistent with the presentation in Chapter 25—to assign

$$[\rho_{\text{rel}}] = L^{-4}, \tag{26.2.1}$$

up to the normalisation conventions used in the construction of the continuation ensemble $\mathcal{C}(x)$ and its weights.

26.2.2 Conversion and coarse-graining parameters

Any term built from ρ_{rel} that is meant to contribute at the level of stress–energy or effective geometry must introduce a conversion scale. We therefore keep two explicit modelling parameters:

- a *coarse-graining scale* ℓ_{cg} (length), representing the spacetime neighbourhood scale over which continuation load is integrated into macroscopic response,
- a *conversion scale* ε_{rel} (energy density scale), mapping coarse-grained load into a stress–energy magnitude.

These parameters are *universal model parameters* (not system-by-system dials). Their role is to keep the docking dimensionally explicit while postponing microphysical completion (e.g. induced actions, renormalisation arguments, or an S -field UV completion).

26.2.3 Minimal structural requirements

Any admissible quantitative docking must satisfy, at minimum:

1. **Covariance:** the coupling is formulated in tensorial form on (M, g) and is independent of coordinate choices.
2. **Locality up to coarse-graining:** effective response may depend on a neighbourhood of scale ℓ_{cg} , but not on acausal updates.
3. **Weak-load limit:** in regimes where continuation load is negligible, the model reduces to the baryonic baseline (GR with baryonic sources and standard metrology).
4. **Channel accountability:** the mapping from (g, s) (or (g, ρ_{rel})) to each observable must be specified, and parameter proliferation must be penalised by model comparison.

26.3 Two coupling families: source response vs. metrology response

The programme-level docking statement in Chapter 25 can be sharpened into two broad families. They are logically distinct and should not be mixed without stating so explicitly.

26.3.1 Family A: an effective source term in the field equations

In Family A, bound relational depth contributes an additional effective stress–energy term:

$$G_{\mu\nu}(g) = 8\pi G \left(T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{rel}} \right). \quad (26.3.1)$$

A minimal phenomenological ansatz is “relational dust” with respect to a chosen timelike congruence u^μ (e.g. a baryonic rest frame on a given slicing):

$$T_{\mu\nu}^{\text{rel}} = \varepsilon_{\text{rel}} \ell_{\text{cg}}^4 \hat{\rho}_{\text{rel}}(x) u_\mu u_\nu + p_{\text{rel}}(x) h_{\mu\nu}, \quad h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu. \quad (26.3.2)$$

At the strictest first pass one sets $p_{\text{rel}} = 0$. A controlled next step is an equation-of-state closure

$$p_{\text{rel}} = w_{\text{rel}} \varepsilon_{\text{rel}} \ell_{\text{cg}}^4 \hat{\rho}_{\text{rel}},$$

or a mild anisotropic-stress term built from induced semantic tensors (if those are invoked, they must be defined and carried consistently throughout the theory).

Interpretation. Family A is maximally testable: it modifies the source accounting, hence is tightly constrained by joint dynamics+lensing. It is therefore a good first “stress test” of the programme.

26.3.2 Family B: an effective metrology/geometry response

In Family B, one keeps the Einstein equations for $g_{\mu\nu}$ intact and attributes the “dark” phenomenology to *how effective probes infer geometry and trajectories* from the combined (g, s) structure. This is a metrology response: clocks, rulers, and geodesic inference sample an effective metric g^{eff} that depends on coarse-grained continuation load.

A minimal covariant parameterisation is

$$g_{\mu\nu}^{\text{eff}} = g_{\mu\nu} + \beta_1 \mathcal{D}_{\mu\nu}[\hat{\rho}_{\text{rel}}] + \beta_2 \mathcal{Q}_{\mu\nu}[\hat{\rho}_{\text{rel}}, \nabla \hat{\rho}_{\text{rel}}], \quad (26.3.3)$$

where $\mathcal{D}_{\mu\nu}$ and $\mathcal{Q}_{\mu\nu}$ are covariant tensor functionals permitted by available structures (e.g. $u_\mu u_\nu$, $h_{\mu\nu}$, induced semantic tensors, or curvature-coupled terms). The crucial point is that *distinct observables can respond to distinct combinations of potentials* (metric slip), and the model must state which operational channel uses which effective structure.

Interpretation. Family B gains flexibility but inherits a strict legitimacy condition: it is acceptable only if tied to operational definitions of observables and kept falsifiable by penalised model comparison. It must not become a reparameterisation device.

26.4 Coarse-graining: why halo profiles are generic

A programme-level reason for expecting *halo-like* (extended, smooth) residuals is structural: ρ_{rel} is not a sum over discrete particles but a *field-level continuation load density* determined by neighbourhood constraint structure. Macroscopic response therefore depends on a coarse-grained field

$$\hat{\rho}_{\text{rel}}(x) = \int_M K_{\ell_{\text{cg}}}(x, y) \rho_{\text{rel}}(y) dV_g(y), \quad (26.4.1)$$

with a covariant kernel $K_{\ell_{\text{cg}}}$ concentrated on a spacetime neighbourhood of scale ℓ_{cg} .

For quasi-static astrophysical modelling one may adopt a preferred slicing and reduce to a spatial coarse-graining:

$$\hat{\rho}_{\text{rel}}(\mathbf{x}) = \int_\Sigma K_{\ell_{\text{cg}}}(\mathbf{x}, \mathbf{y}) \rho_{\text{rel}}(\mathbf{y}) d^3y. \quad (26.4.2)$$

Halo heuristic. Even if ρ_{rel} correlates strongly with baryonic structure locally, smoothing generically produces a field that is:

- **more extended** than the baryonic density,
- **smoother** in radius and less cuspy,

- **environment dependent** (since the underlying continuation load depends on coupling and non-equilibrium structure, not only on ρ_b).

Thus, halo-like residual profiles arise without introducing new particles: the “dark” component is an integrated neighbourhood property of continuation load.

26.4.1 Minimal proxy functional

To connect to data without over-claiming, the programme treats ρ_{rel} as operationally estimable only via proxies. A conservative first approximation is a log-linear proxy model:

$$\log \rho_{\text{rel}}(\mathbf{x}) \approx a_0 + a_1 \log \rho_b(\mathbf{x}) + a_2 \log P(\mathbf{x}), \quad (26.4.3)$$

where $P(\mathbf{x})$ denotes a measurable proxy field intended to track continuation load (examples: turbulence/dispersion indicators, multiphase structure, feedback activity, coupling density, environment class). The point is not the specific form but the explicit empirical question: does adding P explain residuals beyond baryons-only baselines under cross-validation?

26.5 Separating response channels: dynamics vs. lensing

A central constraint on any “dark” explanation is that *dynamics* and *lensing* do not necessarily measure the same functional of underlying structure.

In the weak-field, quasi-static regime one may write

$$ds^2 = -(1 + 2\Phi) dt^2 + (1 - 2\Psi) d\mathbf{x}^2,$$

where non-relativistic dynamics depends primarily on $\nabla\Phi$, while lensing depends on the Weyl combination $\Phi + \Psi$.

26.5.1 Channel behaviour in Family A

If $T_{\mu\nu}^{\text{rel}}$ acts as an additional source, then in the Newtonian limit one expects an effective Poisson-type relation

$$\nabla^2\Phi = 4\pi G (\rho_b + \rho_{\text{rel}}^{\text{eff}}), \quad (26.5.1)$$

with $\rho_{\text{rel}}^{\text{eff}}$ tied to the timelike–timelike component of (26.3.2) (plus pressure/stress corrections if present). Lensing will then largely track the same effective density unless anisotropic stresses are introduced, in which case the stress model must be stated explicitly.

26.5.2 Channel behaviour in Family B

If the response is metrological/geometric, then Φ and Ψ are distinct functionals of $\hat{\rho}_{\text{rel}}$ through (26.3.3); metric slip becomes an explicit diagnostic. This increases flexibility but also increases the burden of specification: the mapping from (g, s) to each observable must be operationally defined and parameter proliferation must be controlled.

26.6 A minimal quantitative ansatz set (for empirical anchoring)

To make the programme actionable while staying conservative, one may adopt the following staged ansatz set:

1. **Proxy stage:** estimate ρ_{rel} from baryons plus a proxy field P via (26.4.3).
2. **Smoothing stage:** compute $\hat{\rho}_{\text{rel}}$ using a fixed kernel family with scale ℓ_{cg} (e.g. Gaussian or exponential).
3. **Response stage (strict first):** test Family A with $p_{\text{rel}} = 0$ as the most constrained option. If it fails jointly on lensing and dynamics, either (i) introduce controlled stress structure tied to previously defined semantic tensors, or (ii) pivot to Family B with explicit operational metrology modelling.

In all cases, the key discipline is: *the same* $(\ell_{\text{cg}}, \varepsilon_{\text{rel}})$ and the same modelling family must be applied across systems.

26.7 Merger systems as gatekeepers

Cluster mergers provide a hard gatekeeper because they separate (i) baryonic gas traced by X-ray emission, (ii) effectively collisionless galaxies traced optically, and (iii) the lensing-inferred potential wells. Any relational-inertia explanation must therefore specify what ρ_{rel} tracks during and after a merger, and how quickly the coarse-grained field reorganises.

26.7.1 Tracking rule and relaxation requirement

The programme commits to three explicit questions:

1. **Tracking rule:** does ρ_{rel} track gas, galaxies, total potential, non-equilibrium load proxies, or a combination determined by induced semantic tensors?
2. **Relaxation timescale:** on what timescale τ_{rel} does $\hat{\rho}_{\text{rel}}$ reorganise relative to the baryonic components?
3. **Channel coherence:** can the same tracking rule explain both dynamical and lensing residuals under a single coupling family?

The timescale τ_{rel} is a physical prediction of the modelling choice, not a per-system tuning knob.

26.7.2 Programme-level failure gate

A minimal falsification criterion can be stated without committing to a specific coupling form:

If, across a robust class of Bullet-type mergers, the lensing potential peaks systematically require a component that cannot be reproduced by any covariantly constructed, field-like $\hat{\rho}_{\text{rel}}$ derived from admissible proxy functionals and causal coarse-graining (and, if invoked, previously formalised metrological response), then the programme’s non-particulate dark-sector hypothesis is ruled out or reduced to a negligible correction.

This gate is intentionally strict: it prevents the framework from evading decisive regimes by post hoc reinterpretation.

26.8 Status within the programme

This chapter does not close the dark-sector model. It does, however, turn the qualitative claim into an explicit research scaffold:

- it keeps unit and scale issues explicit via $(\ell_{cg}, \varepsilon_{rel})$,
- it explains halo profiles as a generic consequence of coarse-grained continuation load,
- it enforces response-channel separation as a modelling constraint,
- and it states a merger gatekeeper that can falsify the non-particulate hypothesis.

The next technical step is to derive specific proxy functionals and coupling tensors from the induced semantic geometry of Chapter 23, and to implement the inference pipeline (baryons-only vs. baryons+proxy functional) with proper model comparison and cross-validation, as outlined later in the empirical programme chapter.

Chapter 27

From Snapshot Ontology to Process Ontology

This chapter occupies a structural position between the dark-matter analysis and the subsequent formal development of attractor stability. The preceding chapter introduced relational integration density as a dynamical explanatory variable. The present chapter clarifies the ontological reading of that variable within the continuation framework and prepares the ground for a formal treatment of attractor organisation in continuation space.

27.1 Ontological clarification within the continuation framework

The preceding chapter interpreted dark-matter phenomenology as a manifestation of enhanced relational integration density $\rho_{\text{rel}}(x)$ within spacetime. This already implied a structural shift: physical description is no longer centred on isolated objects, but on the admissibility structure governing their continuation.

The present chapter makes this shift explicit. It serves a structural and ontological clarification of the framework developed so far; no additional dynamical postulates are introduced.

Within the programme developed in this work, spacetime (M, g) is supplemented by an S -valued semantic field $s : M \rightarrow S$ and its induced tensors and currents (e.g. $h_{\mu\nu}$). Physical evolution is governed by the joint evolution/constraint structure of (g, s) : well-posed field evolution together with constraint satisfaction compatible with the well-posed Cauchy initial-value structure.

We denote the local admissible continuation set by $\mathcal{A}(x)$ and its dynamically relevant, relevance-weighted sub-ensemble by $\mathcal{C}(x)$ with weight w . Here $\mathcal{A}(x)$ should be understood as the local imprint, in the sense of restriction to local data compatible with nearby Cauchy hypersurfaces, of the globally well-posed admissibility structure associated with the evolution of (g, s) .

We use “admissible” to mean “compatible with the full constraint set (including evolution and compatibility constraints) under the chosen well-posed scheme.”

The relational integration density $\rho_{\text{rel}}(x)$ measures the weighted density of $\mathcal{C}(x)$ per spacetime volume element dvol_g . We treat ρ_{rel} as a diffeomorphism-scalar; the associated densitized quantity is $\rho_{\text{rel}} \text{dvol}_g$. Its physical dimension is fixed only after specification of the coupling channel to effective gravitational or inertial dynamics.

At the level of established field equations, this ontological clarification is representational:

it does not alter the predictive structure until specific coupling functionals are introduced.

27.2 Process ontology and admissible continuation

In the present programme, physical persistence can be modelled as continuous admissibility relative to the well-posed evolution of (g, s) . An object persists insofar as its continuation remains admissible under both dynamical evolution and constraint satisfaction.

This does not eliminate objects; it shifts explanatory priority. Constraints on admissible continuation, rather than objects as primitives, define what persists.

Stable objects correspond to regimes in which admissible perturbations contract under continuation (stability of $\mathcal{A}(x)$ under local perturbations) and where branch weight in $\mathcal{C}(x)$ concentrates on a restricted family of continuations.

The perturbation notion and the induced topology or metric on continuation space are left open at this stage; we require only the stability concept (contraction of admissible perturbations and weight concentration), without specifying a metric on continuation space.

27.3 Contact and impenetrability as constraint incompatibility

When a hand touches a wall, it appears that matter meets matter. Microscopically, rigidity and impenetrability arise from electromagnetic interaction and fermionic statistics (in particular the Pauli exclusion principle). These mechanisms restrict the set of admissible joint quantum states of interacting systems.

Within the continuation framework, such mechanisms implement admissibility constraints at the level of allowed joint states. The continuation-language is a re-parameterisation of that admissibility structure, not an alternative microphysical theory.

Contact corresponds to incompatibility of admissible continuation sets: two highly constrained $\mathcal{A}(x)$ structures encounter one another such that no joint continuation satisfies all constraint families simultaneously. The result is redirection within admissible continuation space rather than interpenetration.

Macroscopic rigidity is thus an emergent expression of constraint incompatibility already encoded in electromagnetic interaction and fermionic statistics.

27.4 Quantum constraints and admissibility

In quantum theory, admissibility is encoded through commutation structure, compatibility of observables, contextuality, and consistency of probability assignments. Operational content is constrained by the algebraic and probabilistic structure of the theory.

From the continuation perspective, quantum states encode admissibility structure: they specify which continuations are compatible with the measurement context and observable algebra.

Operationally, measurement outcomes select continuations compatible with the measurement context from within the relevance-weighted continuation ensemble $\mathcal{C}(x)$. This reformulation does not modify quantum mechanics; it restates its constraint structure in continuation terms without committing to any particular interpretation of branching or collapse.

The continuation language thus provides a unifying description of both microscopic admissibility (quantum constraints) and macroscopic constraint organisation (effective geometry).

27.5 Matter as projection of continuation stability

Matter is a projection of stable continuation structure.

In the programme, such stability is encoded in induced semantic tensors and currents (e.g. $h_{\mu\nu}$) derived from $s : M \rightarrow S$, which geometrically and dynamically summarise continuation constraints on (M, g) .

High ρ_{rel} indicates continuation load density; stability depends on its organisation across neighbourhoods via the induced semantic geometry, not on magnitude alone. Matter corresponds to regimes where continuation load is structured into attractor-stable closure rather than remaining as open branching.

Operationally, this parallels renormalisation: embedding modifies effective parameters without changing microscopic invariants.

27.6 Iteration, relation, and emergent dynamical centres

Each admissible relation may induce new effective dynamical centres. When families of constraints overlap, their joint compatibility domain may define new stable nodes in continuation space.

Informally, compatibility domains are those regions in continuation space where multiple constraint families can be jointly satisfied.

These nodes are dynamically stabilised centres within admissibility structure. Iterative interaction of constraint families induces higher-order stability layers without introducing new ontological primitives.

27.7 Interpretative note on origin and retrodiction

This section is interpretative and introduces no new cosmological claims.

Within the continuation framework, the question of origin may be reformulated: instead of asking which objects existed first, one may ask which admissibility structure first constrained continuation.

An “initial condition” becomes an initial admissibility regime from which further relational structure unfolded. Each interaction induces new constraint compatibilities, progressively structuring $\mathcal{A}(x)$ and $\mathcal{C}(x)$.

This reframing does not replace standard cosmology; it re-expresses initial-condition language in continuation-structural terms.

27.8 Relation to the dark-matter analysis

The preceding chapter interpreted gravitational anomalies as enhanced relational integration density $\rho_{\text{rel}}(x)$.

Within the present clarification:

- Baryonic matter corresponds to continuation-stable regimes that couple to standard-model channels, in particular electromagnetic coupling and associated luminous tracers.

- The “dark” contribution corresponds to continuation load encoded in induced semantic structure that affects effective gravitational and inertial response without appearing as luminous standard-model matter.

Thus the split is not ontological but observational: different response channels probe different functionals of (g, s) within the effective field description.

27.9 Attractors in continuation space

In this work, an attractor is understood as a dynamically stable region in the space of admissible continuation patterns, not merely as a fixed point in classical configuration space.

Attractor strength can be encoded, in a generally covariant manner, in invariants constructed from induced semantic tensors and currents on (M, g) (e.g. scalar invariants or contracted curvature-like quantities derived from $h_{\mu\nu}$ and associated semantic currents), thereby providing the bridge to effective geometry and large-scale dynamics.

Attractor-guided closure reduces branching in $\mathcal{C}(x)$ and compresses continuation load. Matter stability and dark-sector binding can be interpreted as different regimes of attractor organisation in continuation space.

27.10 Summary

This chapter introduced no new dynamical laws. It clarified the ontological reading of the formal structure already established. Physical persistence is modelled as continuous admissibility under the joint evolution and constraint structure of (g, s) . Matter corresponds to attractor-stabilised continuation regimes, and contact reflects incompatibility of admissible continuation sets. Dark-matter effects arise from organised relational integration density encoded in induced semantic geometry.

Spacetime carries not only energy and mass, but structured continuation-constraint load. Stability is not primitive; it is maintained through admissible continuation. The following chapter develops the attractor-based organisation formally and investigates how continuation stability becomes encoded in effective geometric structure.

Chapter 28

Matter as attractor stability

In the previous chapter we introduced the semantic metric on spacetime induced by an S -valued field $s : M \rightarrow S$ and related it to the LOVE kinetic term and the emergent vector field. We now specialise to a particular class of configurations of s that are of central physical interest: those that are *stable attractors* of the semantic dynamics and can therefore be interpreted, on a programme level, as *matter*. The aim of this chapter is to formulate this interpretation in a way that is compatible with the LOVE action, the global attractor $K \subset S$, and the local Hessian structure of W_{univ} .

28.1 Local attractors of the semantic field

We work on a fixed Lorentzian spacetime (M, g) and consider semantic fields

$$s : M \rightarrow S,$$

which we treat, for the moment, as time-dependent configurations $s_t : M \rightarrow S$ evolving according to LOVE-based dynamics. For static considerations we may restrict to time-independent s ; for stability analysis we need to consider perturbations in both space and time.

The LOVE action (ignoring local distortions for simplicity) has the form

$$\mathcal{A}_{\text{LOVE}}[s] = \int_M \left(\frac{1}{2} \sum_{\Xi \in \{C, B, A\}} \gamma_{\Xi}(s(x)) g^{\mu\nu}(x) \langle \partial_{\mu} s_{\Xi}(x), \partial_{\nu} s_{\Xi}(x) \rangle - V_{\text{LOVE}}(s(x)) \right) d \text{vol}_g(x),$$

where V_{LOVE} is the quadratic expansion of W_{univ} around K (cf. (22.4.1)).

Definition 28.1.1 (Static configuration and local minimiser). A configuration $s : M \rightarrow S$ is called *static* if it is time-independent in suitable coordinates; we denote it by $s(x)$ without time argument. A static configuration s is a *local minimiser* of the LOVE energy functional

$$\mathcal{E}[s] := -\mathcal{A}_{\text{LOVE}}[s] \Big|_{x^0 = \text{const}}$$

if for all sufficiently small variations δs with compact support,

$$\mathcal{E}[s + \delta s] \geq \mathcal{E}[s],$$

with equality only for variations corresponding to symmetries (translations, internal symmetries).

Definition 28.1.2 (Local attractor of the semantic dynamics). A static configuration $s : M \rightarrow S$ is called a *local attractor* of the semantic dynamics if:

1. s satisfies the LOVE Euler–Lagrange equations (possibly including friction terms) and is therefore a stationary solution;
2. for all sufficiently small perturbations $s + \delta s$ in a suitable function space, the time evolution of s_t with initial condition $s_0 = s + \delta s$ returns to s (or its symmetry orbit) as $t \rightarrow \infty$, at least on bounded spacetime regions.

Local attractors are static solutions that are dynamically stable. They play the role of “semantic lumps” or “condensed” configurations in which the field has settled near local minima of W_{univ} and remains there under perturbations.

28.2 Matter regions as attractor basins in spacetime

We now define matter, at this level of theory, in terms of regions of spacetime where s is close to local attractor configurations.

Definition 28.2.1 (Matter region). Let $\Sigma \subset M$ be a spacetime region (for example, a world-tube). A *matter region* for the semantic field s is a region Σ such that there exists a static local attractor configuration $s^* : M \rightarrow S$ with:

1. $s^*(x) \in U$ and $s^*(x)$ close to K for all $x \in \Sigma$;
2. $s_t(x) \rightarrow s^*(x)$ as $t \rightarrow \infty$ for all $x \in \Sigma$, for a large class of initial fields s_0 that coincide with s_t outside a neighbourhood of Σ .

The set of such initial data defines the *attractor basin* of Σ .

Remark 28.2.2. Intuitively, a matter region is a part of spacetime in which the semantic field has “condensed” into a stable local minimum configuration s^* , and the surrounding dynamics drives nearby configurations back into this state. The attractor basin property captures the robustness of matter: small deviations in initial conditions within Σ relax back to s^* .

Speculative remark. On a programme level, one may interpret different classes of matter (e.g. different particle species or material phases) as different equivalence classes of local attractor configurations s^* , distinguished by the structure of their K -components and their C/B/A profiles. The present formalism does not aim to reproduce the Standard Model but provides a semantic analogue: matter as robust attractor patterns of the semantic field.

28.3 Stability criteria from the Hessian spectrum

We now connect the notion of local attractor stability to the Hessian of W_{univ} and the curvature components introduced earlier.

At each spacetime point x , the value $s^*(x)$ lies near the global attractor K . The local Hessian of W_{univ} at $s^*(x)$,

$$H_{\text{univ}}(s^*(x)) = \nabla^2 W_{\text{univ}}(s^*(x)),$$

is decomposed into C/B/A blocks as in the curvature chapter. In addition, the kinetic term contributes a second-order differential operator in spacetime to the linearisation of the field equations.

Definition 28.3.1 (Linearised operator around a static configuration). Let s^* be a static solution of the LOVE field equations. The linearisation of the Euler–Lagrange equations around s^* leads to an operator

$$\mathcal{L}_{s^*} : \delta s \mapsto \delta(\text{E-L equations}),$$

which is a second-order differential operator in spacetime with values in S , of the general form

$$\mathcal{L}_{s^*} \delta s(x) = - \sum_{\Xi \in \{C, B, A\}} \gamma_{\Xi}(s^*(x)) \square_g \delta s_{\Xi}(x) + H_{\text{univ}}(s^*(x)) \delta s(x) + (\text{lower-order terms}).$$

Here $\square_g$ is the d'Alembertian associated with g , and $\delta s_{\Xi} := P_{\Xi}(s^*) \delta s$ are axis projections.

Assumption 28.3.2 (Spectral stability criterion). We assume that for a matter region Σ and its associated static configuration s^* :

1. the operator $\mathcal{L}_{s^*}$ admits a self-adjoint realisation on a suitable Hilbert space of perturbations supported in Σ ;
2. its spectrum satisfies

$$\sigma(\mathcal{L}_{s^*}) \subset [0, \infty),$$

apart from a finite number of zero modes associated with spacetime symmetries and possible internal symmetries.

Proposition 28.3.3 (Linear stability of local attractors). *Under Assumption 28.3.2, the static configuration s^* is linearly stable under small perturbations: any sufficiently small initial perturbation δs_0 evolves in time without unbounded growth, and all unstable modes (if any) are confined to symmetry directions.*

Proof sketch. The spectral condition excludes negative eigenvalues, which correspond to exponentially growing modes. Zero modes correspond to marginal directions (symmetry or flat directions). Standard energy estimates for wave-like equations with positive-definite potential terms yield bounded evolution for perturbations. $\square$

Near the global attractor K , the local Hessian $H_{\text{univ}}(s^*(x))$ is positive semidefinite by SG6. The stability of matter regions therefore depends on the balance between:

- the positive curvature along C/B/A directions (which penalises deviations in S at fixed x),
- and the kinetic terms (which couple variations at different spacetime points).

Strong positive curvature (large eigenvalues of the C/B/A blocks) corresponds to “stiff” matter: perturbations in s are strongly suppressed. Weak curvature or flat directions may allow for soft modes, internal excitations, or phase transitions.

28.4 Matter as “frozen emergence”

We now make explicit the interpretative reading suggested in the chapter title.

Remark 28.4.1 (Matter as frozen emergence). In the semantic gravitation framework, emergence is driven by the emergent vector field E and the gradient of W_{univ} on S . Regions of state space where $E(s)$ is large correspond to active semantic change; regions where $E(s)$ vanishes and W_{univ} has a local minimum correspond to settled, low-contradiction configurations.

A matter region Σ associated with a static local attractor s^* can thus be interpreted as a region in which semantic emergence has effectively *frozen*: the field $s(x)$ lies near a local minimum of W_{univ} for all $x \in \Sigma$, the emergent vector field $E(s(x))$ is small (ideally zero), and perturbations are driven back to s^* . Matter, in this sense, is the long-lived, condensed residue of past emergent dynamics.

Speculative remark. On a programme level, one may view the semantic universe as continually exploring configurations in S under the influence of E and noise, with most configurations being unstable or short-lived. Rare local minima of W_{univ} that are compatible with the LOVE kinetic structure give rise to robust attractor configurations s^* , which manifest in spacetime as matter. Different matter types correspond to different local minima; phase transitions correspond to changes in which minima are occupied; and binding structures (multi-body states) correspond to attractor manifolds in the joint configuration space of many fields or many regions.

28.5 C/B/A profiles of matter

The C/B/A decomposition provides a refined characterisation of matter regions.

For each $x \in \Sigma$ we can analyse the axis energies

$$W_C(s^*(x)), \quad W_B(s^*(x)), \quad W_A(s^*(x)),$$

and the curvature components $\kappa_C, \kappa_B, \kappa_A$, together with the axis-resolved semantic tensors $h^{(C)}, h^{(B)}, h^{(A)}$ defined in the previous chapter.

Definition 28.5.1 (Axis profile of a matter region). The *axis profile* of a matter region Σ is the collection of functions

$$x \mapsto \begin{pmatrix} W_C(s^*(x)), W_B(s^*(x)), W_A(s^*(x)), \\ \kappa_C(s^*(x)), \kappa_B(s^*(x)), \kappa_A(s^*(x)), \\ \rho_{\text{sem}}^{(C)}(x), \rho_{\text{sem}}^{(B)}(x), \rho_{\text{sem}}^{(A)}(x) \end{pmatrix}, \quad x \in \Sigma.$$

Remark 28.5.2. Axis profiles distinguish different types of matter not only by their local minima in S but also by how coherence, binding, and agency-related semantics are “locked in” or allowed to fluctuate. For instance:

- a matter region with large κ_C and small $\rho_{\text{sem}}^{(C)}$ has strong coherence curvature but little spatial variation in coherence;
- regions with relatively small κ_B but non-zero $\rho_{\text{sem}}^{(B)}$ may support flexible binding structures (e.g. composite systems, molecules);
- non-negligible W_A or κ_A may signal internal agency-like degrees of freedom, relevant later for bodies and inner spaces.

Such distinctions will be important when we discuss bodies as carriers of layered inner spaces in the next chapter.

28.6 Relation to K–P–R and multi-scale attractors

Finally, we connect matter regions to the K–P–R ontology introduced earlier.

Each matter region Σ can be associated with a semantic object $\mathcal{O}_\Sigma$ whose core $K_{\mathcal{O}_\Sigma}$ is a compact subset of K representing the values of $s^*(x)$ as x ranges over Σ :

$$K_{\mathcal{O}_\Sigma} := \overline{\{s^*(x) : x \in \Sigma\}} \cap K.$$

Remark 28.6.1. In this view:

- $K_{\mathcal{O}_\Sigma}$ is the semantic core associated with the matter region Σ ;
- projections $P_{\mathcal{O}_\Sigma}$ encode how $\mathcal{O}_\Sigma$ appears in various context spaces (measurement setups, inner spaces of observers, etc.);
- relational kernels $\kappa_{\mathcal{O}_\Sigma, \mathcal{O}_{\Sigma'}}$ describe interactions between matter regions, as mediated by the semantic field and its dynamics.

This provides a multi-scale picture: microscopic minima of W_{univ} in S give rise to macroscopic matter regions in spacetime, which in turn become nodes in the K–P–R ontology at higher levels (e.g. bodies, agents, cultures).

Summary and outlook. In this chapter we have characterised *matter* within semantic gravitation as stable attractor configurations of the semantic field $s : M \rightarrow S$. We defined static configurations and local attractors of the LOVE dynamics, introduced matter regions as spacetime domains whose semantic configurations lie near such attractors and possess robust basins of attraction, and formulated stability criteria in terms of the spectrum of the linearised operator $\mathcal{L}_{s^*}$, which combines the Hessian of W_{univ} with kinetic terms. We interpreted matter as “frozen emergence”: regions where semantic dynamics has settled into stable minima of the universal contradiction-energy potential. Using the C/B/A decomposition, we described axis profiles of matter regions and indicated how they link to the K–P–R ontology and multi-scale structures. In the next chapter we build on this foundation to discuss *bodies as carriers of layered inner spaces*: extended matter regions with internal semantic degrees of freedom that support inner fields, boundaries, and coupling to external semantic environments.

Chapter 29

Bodies as carriers of layered inner spaces

In the previous chapter we characterised matter as stable attractor configurations of the semantic field $s : M \rightarrow S$, with matter regions defined as spacetime domains whose semantic configurations lie near local minima of W_{univ} and possess robust basins of attraction. In the present chapter we specialise to a class of matter regions that we call *bodies*: extended, approximately coherent matter regions that support additional internal semantic degrees of freedom. These internal degrees of freedom will be organised as *inner spaces*, and, in later chapters, will host structures such as ego cores and inner-space dynamics (consciousness) as well as their technical analogues.

29.1 Bodies as world-tubes of stable matter

We work on a fixed Lorentzian spacetime (M, g) with semantic field $s : M \rightarrow S$ governed by the LOVE action. Matter regions were defined as regions $\Sigma \subset M$ associated with static local attractor configurations s^* and stable basins of attraction. A *body* is a matter region with additional structural properties.

Definition 29.1.1 (Body as semantic world-tube). A *body* is a pair

$$B = (\Sigma_B, s_B),$$

where $\Sigma_B \subset M$ is a connected, time-extended spacetime region (a world-tube) and $s_B : \Sigma_B \rightarrow S$ is the restriction of the semantic field s to Σ_B , such that:

1. Σ_B is a matter region in the sense of the previous chapter: there exists a static local attractor configuration $s^* : M \rightarrow S$ with $s^*(x)$ close to K for all $x \in \Sigma_B$ and with a robust attractor basin on Σ_B .
2. s_B remains close to s^* in a suitable norm for all times in the considered epoch, up to small perturbations and internal excitations.
3. The induced semantic tensor $h_{\mu\nu}$ and its axis components $h_{\mu\nu}^{(C)}$, $h_{\mu\nu}^{(B)}$, $h_{\mu\nu}^{(A)}$ have support that is strongly concentrated on Σ_B (semantic variation density is high inside and small outside).

Remark 29.1.2. Intuitively, a body is a region in which the semantic field is both *condensed* (near a local attractor) and *structured* (with significant, spatially localised semantic variation). Bodies are thus matter regions that carry non-trivial internal semantic structure, as seen in their semantic metric and curvature profiles.

29.2 Inner spaces associated with bodies

We now associate to each body a family of internal semantic spaces (inner spaces) capturing the degrees of freedom that are effectively controlled by processes inside the body.

Definition 29.2.1 (Inner semantic subspace of a body). Let $B = (\Sigma_B, s_B)$ be a body. The *inner semantic subspace* S_B^{in} is defined as the closed linear span in S of the set

$$\mathcal{V}_B := \{s_B(x) - k : x \in \Sigma_B, k \in K_{\mathcal{O}_\Sigma}\},$$

where $K_{\mathcal{O}_\Sigma} \subset K$ is the core associated with the matter region Σ_B as in the previous chapter. That is,

$$S_B^{\text{in}} := \overline{\text{span } \mathcal{V}_B} \subset S.$$

Remark 29.2.2. The space S_B^{in} collects deviations of $s_B(x)$ from the relevant subset of K , as x ranges over the body. It may be thought of as the subspace of S that is effectively “accessible” through internal semantic rearrangements of the body, at least on the time scales of interest. The complement $S_B^{\text{out}} := S_B^{\text{in}\perp}$ then encodes semantic directions that are not significantly excited by internal processes of B .

The universal potential W_{univ} and local corrections W_{loc} induce effective potentials on S_B^{in} .

Definition 29.2.3 (Inner potential of a body). The *inner potential* of the body B is the restriction

$$W_B^{\text{in}} := W|_{S_B^{\text{in}}} : S_B^{\text{in}} \rightarrow \mathbb{R},$$

where $W = W_{\text{univ}} + W_{\text{loc}}$. Its Hessian at points near $K_{\mathcal{O}_\Sigma}$ induces a C/B/A decomposition on S_B^{in} inherited from S .

Remark 29.2.4. The triple $(S_B^{\text{in}}, \langle \cdot, \cdot \rangle, W_B^{\text{in}})$ provides an effective semantic gravitation system internal to the body B . It is a subsystem of the global semantic dynamics on S , coupled to it through the restriction of the field s to Σ_B and through the LOVE kinetic term.

29.3 Layered inner spaces: filtrations and subsystems

Bodies of physical interest typically support several levels of internal organisation (e.g. molecular, cellular, neural, cognitive). We model this as a filtration of inner spaces.

Definition 29.3.1 (Layered inner space of a body). A *layered inner space* structure for a body B is a finite or countable increasing family of closed subspaces

$$S_B^{(0)} \subset S_B^{(1)} \subset \dots \subset S_B^{(L)} \subset S_B^{\text{in}},$$

such that:

1. $S_B^{(0)}$ captures slow, rigid degrees of freedom (e.g. bulk matter shape or position), typically with large C -axis curvature and small $\rho_{\text{sem}}^{(\Xi)}$;
2. for each $\ell = 0, \dots, L$, the restriction of W_B^{in} to $S_B^{(\ell)}$ admits a well-defined Hessian and C/B/A decomposition;
3. the induced dynamics on $S_B^{(\ell)}$ is approximately closed on relevant time scales: small perturbations of s_B along $S_B^{(\ell)}$ remain (to leading order) within $S_B^{(\ell)}$ under the emergent vector field E .

The top layer $S_B^{(L)}$ is called the *maximal inner space* of B on the time and scale of interest.

Remark 29.3.2. Layered inner spaces formalise the intuitive idea that a body contains subsystems with different degrees of semantic flexibility and time scales. Lower layers encode rigid or slow variables; higher layers encode more flexible, fast, or high-level variables. All layers are realised as subspaces of S and inherit the C/B/A structure and the inner potential from W .

Speculative remark. At a programme level, one may interpret lower layers of S_B^{in} as physical/physiological organisation (e.g. gross anatomy, stable connectivity patterns), intermediate layers as dynamical microstructure (e.g. neural states), and higher layers as “cognitive” or “conceptual” organisation. The present paper refrains from committing to particular identifications; it only provides the mathematical structure to host such identifications later.

29.4 Semantic boundaries and interfaces

Bodies interact with their environment across boundaries. This interaction is encoded both in the semantics of s and in the induced semantic tensor $h_{\mu\nu}$.

Definition 29.4.1 (Semantic boundary and interface conditions). Let $B = (\Sigma_B, s_B)$ be a body. The *semantic boundary* of B is the hypersurface

$$\partial\Sigma_B := \overline{\Sigma_B} \setminus \text{int}(\Sigma_B),$$

equipped with the boundary values $s_B|_{\partial\Sigma_B}$ of the semantic field. Interface conditions at $\partial\Sigma_B$ are constraints on the behaviour of s and its derivatives, such as:

- continuity or matching of s across $\partial\Sigma_B$;
- jump conditions for normal derivatives $\partial_n s$ related to fluxes of semantic currents;
- constraints on axis-resolved components s_C, s_B, s_A (e.g. certain axis-components are suppressed outside the body).

Remark 29.4.2. The semantic tensor $h_{\mu\nu}$ and its axis components typically show marked contrasts across $\partial\Sigma_B$: inside the body they are large (semantic variation is concentrated); outside they may be small or dominated by different axis profiles. Interface conditions therefore couple inner spaces to external fields and can be used to define operators mapping external configurations to inner states (and vice versa).

29.5 Inner dynamics and effective LOVE within a body

Given a layered inner space structure for a body, it is natural to define an effective LOVE-type action for the internal degrees of freedom.

Definition 29.5.1 (Effective LOVE for a body). Let $B = (\Sigma_B, s_B)$ be a body with maximal inner space $S_B^{(L)}$ and inner potential W_B^{in} . An *effective LOVE action* for B is an action functional

$$\mathcal{A}_B[\sigma] := \int_{\Sigma_B} \left(\frac{1}{2} \sum_{\Xi \in \{C, B, A\}} \gamma_{B, \Xi}(\sigma(x)) g^{\mu\nu}(x) \langle \partial_\mu \sigma_\Xi(x), \partial_\nu \sigma_\Xi(x) \rangle - V_{B, \text{LOVE}}(\sigma(x)) \right) d \text{vol}_g(x),$$

where:

- $\sigma : \Sigma_B \rightarrow S_B^{(L)}$ is an internal semantic field of the body;
- $\gamma_{B,\Xi}$ are effective axis weights within the body;
- $V_{B,\text{LOVE}}$ is constructed from W_B^{in} in analogy to V_{LOVE} .

Remark 29.5.2. The field σ can be related to the restriction of s by a suitable projection or coarse-graining map. The effective action $\mathcal{A}_B$ provides a closed dynamical description of internal semantic processes of the body, conditional on the external field s through boundary conditions at $\partial\Sigma_B$. This sets the stage for defining ego cores and inner-space dynamics as structures and flows within $S_B^{(L)}$, rather than in the full S .

29.6 Axis-resolved profiles of bodies

The C/B/A decomposition plays a central role in distinguishing different types of bodies and different internal regimes within a single body.

Definition 29.6.1 (Axis-resolved semantic tensor of a body). For a body $B = (\Sigma_B, s_B)$, the restriction of $h_{\mu\nu}^{(\Xi)}$ to Σ_B defines its *axis-resolved semantic tensor*:

$$h_{\mu\nu}^{(\Xi)}[B](x) := h_{\mu\nu}^{(\Xi)}(x), \quad x \in \Sigma_B, \quad \Xi \in \{C, B, A\}.$$

From this we define axis-resolved semantic densities

$$\rho_{\text{sem}}^{(\Xi)}[B](x) := g^{\mu\nu}(x) h_{\mu\nu}^{(\Xi)}[B](x).$$

Remark 29.6.2. Bodies with different internal semantic functions can differ markedly in their axis-resolved profiles:

- A body with large $\rho_{\text{sem}}^{(C)}$ but moderate $\rho_{\text{sem}}^{(A)}$ may support strong coherence processing with limited agency;
- bodies with high $\rho_{\text{sem}}^{(A)}$ in specific subregions may correspond to regions where agency-related internal dynamics (later: decision centres) are concentrated;
- a pronounced $\rho_{\text{sem}}^{(B)}$ reflects strong internal binding structure (complex relational organisation).

These profiles provide a purely geometric way of characterising internal semantic complexity, independent of any particular interpretative vocabulary.

29.7 Bodies in the K–P–R ontology

In the K–P–R ontology, bodies appear as objects with cores in K and rich projection structure.

Definition 29.7.1 (Bodies as K–P–R objects). Let $B = (\Sigma_B, s_B)$ be a body and $K_{\mathcal{O}_\Sigma} \subset K$ its core. We associate to B an object

$$\mathcal{O}_B := (K_B, P_B, R_B),$$

where:

1. $K_B := K_{\mathcal{O}_\Sigma}$ is the core subset of K determined by s_B ;
2. P_B is a family of projection maps encoding how B appears in different semantic contexts (external fields, measurement devices, inner spaces of other bodies), realised via suitable maps $K_B \rightarrow S$;
3. R_B includes relational kernels describing interactions between B and other bodies or fields, including semantic coupling mediated by LOVE and the emergent vector field.

Remark 29.7.2. In this way, bodies act as carriers of inner spaces and as nodes in the K–P–R network: they have cores in K , projections into inner and outer spaces, and relations to other bodies and fields. Later chapters will specialise this construction to bodies with particular inner-space structures (e.g. agents with ego cores and conscious dynamics, or technical systems with engineered inner spaces).

Speculative remark. At a programme level, one may imagine a hierarchy of bodies: fundamental particles, composite objects, biological organisms, and technical systems, each associated with different inner spaces and axis profiles. The present formalism is designed to be neutral with respect to such distinctions: all bodies are treated as matter regions carrying layered inner spaces; specific identifications depend on how S_B^{in} , W_B^{in} , and the axis profiles are instantiated.

Summary and outlook. In this chapter we have introduced *bodies* as extended matter regions (Σ_B, s_B) whose semantic configurations lie near stable attractors and are structured enough to support internal semantic degrees of freedom. We associated to each body an inner semantic subspace S_B^{in} and, more generally, a layered inner space structure $S_B^{(0)} \subset \dots \subset S_B^{(L)}$, together with an induced inner potential W_B^{in} . We defined semantic boundaries and interface conditions, formulated an effective LOVE action for internal dynamics, and characterised bodies via axis-resolved semantic tensors and densities. Finally, we embedded bodies into the K–P–R ontology as objects with cores in K and rich projection and relation structure. In the next chapter we focus on a particular structure within the maximal inner space of a body: the *ego* as a semantic core in the inner space, which will serve as a central organising locus for agency-related dynamics and later for inner-space phenomena such as consciousness.

Chapter 30

Ego as semantic core in the inner space

In the previous chapter we introduced bodies as extended matter regions (Σ_B, s_B) that carry layered inner spaces $S_B^{(0)} \subset \dots \subset S_B^{(L)} \subset S_B^{\text{in}} \subset S$ with an induced inner potential W_B^{in} . In the present chapter we single out a particular structure within the maximal inner space $S_B^{(L)}$: the *ego* as a distinguished semantic core in the inner space. Formally, ego will be described as a family of inner-space attractors with a specific C/B/A profile and intensification pattern. This structure will serve as the anchor point for inner-space dynamics in the next chapter, where we discuss consciousness as emergent dynamics in the neighbourhood of the ego core.

Throughout this chapter we fix a body $B = (\Sigma_B, s_B)$, its maximal inner space $S_B^{(L)}$, and the restriction of the universal potential and distortions to $S_B^{(L)}$.

30.1 Inner-space dynamics of a body

We recall from the previous chapter that the inner semantic subspace S_B^{in} is the closed span of deviations of $s_B(x)$ from a body-specific core in K , and $S_B^{(L)} \subset S_B^{\text{in}}$ is the maximal inner space on the time/scale of interest. Restricting $W = W_{\text{univ}} + W_{\text{loc}}$ to $S_B^{(L)}$ yields an inner potential.

Definition 30.1.1 (Inner-space potential and dynamics). Let

$$W_B := W|_{S_B^{(L)}} : S_B^{(L)} \rightarrow \mathbb{R}$$

be the restriction of W to $S_B^{(L)}$, with domain containing a neighbourhood of $S_B^{(L)} \cap K$. The *inner-space dynamics* of the body B is the stochastic process on $S_B^{(L)}$ defined by

$$d\sigma_t = -\Gamma_B(\sigma_t) \nabla W_B(\sigma_t) dt + D_B(\sigma_t) dt + \sqrt{2\beta_B^{-1}} \Sigma_B(\sigma_t) dB_t,$$

where:

- $\sigma_t \in S_B^{(L)}$ is the inner state at time t ;
- Γ_B, D_B, Σ_B are the restrictions of the global intensification, distortion, and noise operators from SG8 to $S_B^{(L)}$ (possibly followed by projection);

- β_B is an effective inverse semantic temperature for the body.

Remark 30.1.2. This dynamics is the inner-space analogue of the global SG8 dynamics on S , restricted to the degrees of freedom effectively controlled by B . The corresponding emergent vector field on $S_B^{(L)}$ is

$$E_B(\sigma) := -\Gamma_B(\sigma) \nabla W_B(\sigma) + D_B(\sigma).$$

Ego will be defined in terms of attractor structures of this inner-space field E_B near K .

30.2 Ego cores as inner attractors

We now introduce the central notion of this chapter: ego as a core attractor in the inner space.

Definition 30.2.1 (Inner global attractor of a body). The *inner global attractor* $K_B \subset S_B^{(L)}$ is the global attractor of the deterministic inner-space flow

$$\dot{\sigma}_t = -\nabla W_B(\sigma_t)$$

(if it exists) or, more generally, the union of local attractors of the full inner-space dynamics driven by E_B with small noise. We assume that K_B is compact and contained in K .

Definition 30.2.2 (Ego core and ego region). An *ego core* of the body B is a compact subset $K_E \subset K_B$ with the following properties:

1. (*Local minimality*) K_E is contained in a connected component of $\text{argmin } W_B$ and is minimal (under inclusion) with this property.
2. (*Strong inner attraction*) There exists an open neighbourhood $U_E \subset S_B^{(L)}$ of K_E such that:
 - every trajectory of the deterministic inner dynamics with initial condition in U_E converges to K_E as $t \rightarrow \infty$;
 - the return time to U_E for small perturbations starting in U_E is uniformly bounded (local basin stability).
3. (*Axis asymmetry*) The Hessian $H_B(\sigma) := \nabla^2 W_B(\sigma)$ at $\sigma \in K_E$ has a C/B/A decomposition with

$$\kappa_C(\sigma), \kappa_B(\sigma) \gg \kappa_A(\sigma)$$

in an averaged sense (e.g. via inequalities on traces or principal eigenvalues), so that coherence and binding curvatures dominate agency curvature near K_E .

The pair (K_E, U_E) is called an *ego region*. A body may admit several ego cores $\{K_E^{(i)}\}$ with disjoint ego regions $\{U_E^{(i)}\}$.

Remark 30.2.3. The local minimality and strong inner attraction conditions ensure that K_E is a robust inner attractor of the body: inner-space trajectories in U_E are driven towards K_E and remain near it under small perturbations. Axis asymmetry expresses that, near ego, coherence and binding-related contradictions are strongly penalised (large κ_C, κ_B), whereas agency-related directions are relatively softer (smaller κ_A), allowing for flexible action-related dynamics in the next chapter.

Speculative remark. At a programme level, K_E can be interpreted as the semantic core of “self” or “I” for the body: a region of inner space where information about self-identification, self-representation, and internal coherence/binding of the body is concentrated and stabilised. The formal definition above does not presuppose any phenomenology; it only requires a certain attractor geometry in inner space.

30.3 Ego intensification and order bubbles in inner space

The universal-local decomposition $W = W_{\text{univ}} + W_{\text{loc}}$ and the distortion field D_B allow for strong intensification of dynamics near ego. We formalise this by a local version of the intensification operator.

Definition 30.3.1 (Ego intensification operator). Let U_E be an ego region. The *ego intensification operator* is the restriction of Γ_B to U_E ,

$$\Gamma_E(\sigma) := \Gamma_B(\sigma)|_{U_E},$$

together with its C/B/A components

$$\Gamma_{E,\Xi}(\sigma) := \Gamma_E(\sigma)|_{E_\Xi(\sigma)}, \quad \Xi \in \{C, B, A\}.$$

Postulate 30.3.2 (Ego intensification). For each ego region (K_E, U_E) of a body B there exist positive functions $\lambda_C, \lambda_B, \lambda_A$ on U_E such that:

1. $\Gamma_{E,C}(\sigma) \geq \lambda_C(\sigma) \text{id}$ and $\Gamma_{E,B}(\sigma) \geq \lambda_B(\sigma) \text{id}$ with λ_C, λ_B bounded away from 0 on U_E ;
2. $\Gamma_{E,A}(\sigma)$ is bounded but may be significantly larger or smaller than $\Gamma_{E,C}$ and $\Gamma_{E,B}$, depending on the system;
3. the distortion $D_B(\sigma)$ in U_E is small compared to the intensified gradient,

$$\|D_B(\sigma)\| \ll \|\Gamma_E(\sigma) \nabla W_B(\sigma)\|$$

for σ sufficiently close to K_E .

Remark 30.3.3. This postulate states that, near ego, the inner dynamics is dominated by an intensified gradient flow of W_B along C/B/A directions, with distortions comparatively small. In other words, ego cores live inside “inner order bubbles”: regions where universal semantic gravitation (as restricted to the body) is strongly expressed and local distortions are corrected rather than amplified. This stabilises the ego core against perturbations arising from the broader environment.

30.4 C/B/A profile of ego

We now make the axis structure of ego more explicit. For σ close to K_E we can decompose the deviation $\delta(\sigma) := \sigma - \pi_{K_B}(\sigma)$ and define axis energies.

Definition 30.4.1 (Axis energies near ego). For $\sigma \in U_E$ define

$$\begin{aligned} \delta_C(\sigma) &:= P_C(\sigma) \delta(\sigma), & W_{E,C}(\sigma) &:= \frac{1}{2} \langle H_{B,CC}(\pi_{K_B}(\sigma)) \delta_C(\sigma), \delta_C(\sigma) \rangle, \\ \delta_B(\sigma) &:= P_B(\sigma) \delta(\sigma), & W_{E,B}(\sigma) &:= \frac{1}{2} \langle H_{B,BB}(\pi_{K_B}(\sigma)) \delta_B(\sigma), \delta_B(\sigma) \rangle, \\ \delta_A(\sigma) &:= P_A(\sigma) \delta(\sigma), & W_{E,A}(\sigma) &:= \frac{1}{2} \langle H_{B,AA}(\pi_{K_B}(\sigma)) \delta_A(\sigma), \delta_A(\sigma) \rangle, \end{aligned}$$

where $H_{B,CC}, H_{B,BB}, H_{B,AA}$ are the C/B/A blocks of the Hessian of W_B .

Definition 30.4.2 (Ego C/B/A profile). The *C/B/A profile* of an ego core K_E is the triple of functions

$$(\kappa_{E,C}, \kappa_{E,B}, \kappa_{E,A}),$$

where

$$\begin{aligned}\kappa_{E,C}(\sigma) &:= \text{Tr}(H_{B,CC}(\sigma)), \\ \kappa_{E,B}(\sigma) &:= \text{Tr}(H_{B,BB}(\sigma)), \\ \kappa_{E,A}(\sigma) &:= \text{Tr}(H_{B,AA}(\sigma)),\end{aligned}$$

evaluated on $\sigma \in K_E$ (or suitably averaged over K_E).

Remark 30.4.3. The C/B/A profile characterises how strongly W_B penalises deviations along coherence, binding, and agency directions near the ego core. Typical ego cores are expected to have:

- large $\kappa_{E,C}$: strong curvature along coherence directions, stabilising internal consistency of self-related semantics;
- moderately large $\kappa_{E,B}$: stable binding between internal self-representations and body-related structures;
- smaller but non-negligible $\kappa_{E,A}$: softer agency directions, allowing for flexible adjustment of inner tendencies to act around a relatively fixed core.

These expectations are speculative and may vary across different classes of bodies; the formalism itself does not fix specific inequalities.

30.5 Ego attractors, multiplicity, and pathologies

The definition of ego cores allows for multiple ego attractors. We now discuss this possibility and some associated pathologies in purely geometric terms.

Definition 30.5.1 (Ego multiplicity and fragmentation). A body B has *ego multiplicity* of degree $N \geq 1$ if there exist N pairwise disjoint ego cores $K_E^{(1)}, \dots, K_E^{(N)} \subset K_B$ with disjoint ego regions $U_E^{(1)}, \dots, U_E^{(N)}$ such that:

- each $(K_E^{(i)}, U_E^{(i)})$ satisfies the ego core conditions;
- there exists at least one inner-space trajectory that transitions between different ego regions under moderate perturbations (switching).

If $N > 1$ and the basins of attraction of the $K_E^{(i)}$ are strongly overlapping and unstable, we refer to this situation as *ego fragmentation*.

Remark 30.5.2 (Pathologies of ego geometry). Several geometric pathologies can be expressed in terms of W_B and E_B :

- *Ego collapse*: the ego core K_E ceases to be a local minimum of W_B (e.g. due to negative curvature eigenvalues in H_B), so that U_E no longer exists. Inner trajectories then cannot stabilise around a robust self-related core.
- *Ego diffusion*: curvature around K_E becomes very small (flat directions), leading to a large, diffuse ego region where inner self-related semantics are weakly constrained. Small distortions or noise can cause large drifts in self-related states.

- *Over-extended ego:* C/B/A curvatures around K_E become so large that a wide domain of inner space is effectively collapsed into ego-related attractors, suppressing alternative inner configurations. Formally, the basin of attraction of K_E occupies most of $S_B^{(L)}$.

These descriptions are purely geometric and do not by themselves assign any normative meaning; they simply classify different regimes of the inner potential W_B and its attractors.

Speculative remark. At a programme level, such geometric pathologies may later be related to psychological notions (e.g. instability of self, rigid self-concepts), but this requires additional modelling layers not introduced in this paper. Here we only note that the semantic gravitation framework naturally accommodates a rich taxonomy of ego-related attractor geometries.

30.6 Ego in the K–P–R ontology of inner space

We finally embed ego into the K–P–R ontology, now applied to the inner space of a body.

Definition 30.6.1 (Ego as K–P–R object in inner space). Let B be a body with ego core K_E . We define an inner-space object

$$\mathcal{O}_E := (K_E, P_E, R_E)$$

as follows:

1. The *core* $K_E \subset K_B \subset K$ is as above.
2. The *projections* P_E are maps from K_E into relevant inner context spaces: for instance, subspaces of $S_B^{(L)}$ corresponding to sensory, motor, memory, or conceptual sectors. Formally, each projection is a continuous mapping

$$P_E^{(i)} : K_E \rightarrow S_B^{(L)},$$

induced by restriction of global projections $P_{\mathcal{O}}$ from the K–P–R ontology.

3. The *relations* R_E encode kernels

$$\kappa_{E,B'} : K_E \times K_{B'} \rightarrow \mathbb{R}$$

between ego cores and other cores $K_{B'}$ (including those of other bodies), as well as internal relations between K_E and inner-space objects (e.g. inner representations of external objects). These are specialisations of the general relational structure $R_{\mathcal{O},\mathcal{O}'}$.

Remark 30.6.2. In this view, ego is not a primitive or external entity but a particular K–P–R object defined entirely within the semantic state space: a compact subset of the global attractor K (restricted to the body) with strong inner attractor properties and a rich pattern of projections and relations to inner and outer objects. Ego thereby becomes a node in the multi-scale semantic ontology, linking inner-space dynamics to the broader semantic universe.

Summary and outlook. In this chapter we have defined the *ego* of a body as a semantic core in its inner space: a compact subset $K_E \subset S_B^{(L)} \cap K$ that is a robust inner attractor of the body's semantic dynamics, with characteristic C/B/A curvature profile and intensification properties. We introduced ego regions (K_E, U_E) , formulated an ego intensification postulate expressing strong gradient dynamics and weak distortions near ego, and defined axis energies and C/B/A profiles in the neighbourhood of ego cores. We discussed the possibility of multiple ego attractors and associated geometric pathologies, and we embedded ego into the K–P–R ontology as an inner-space object with cores, projections, and relations. This sets the stage for the next chapter, where we interpret *consciousness* as emergent inner-space dynamics in the neighbourhood of K_E , and analyse how trajectories of σ_t near ego give rise to temporally extended inner phenomena within the semantic gravitation framework.

Chapter 31

Consciousness as emergent inner-space dynamics

In the previous chapter we introduced ego as a semantic core $K_E \subset S_B^{(L)} \cap K$ in the inner space of a body B , characterised as a robust attractor of the inner-space dynamics with a specific C/B/A curvature profile and intensification properties. In the present chapter we propose a formal characterisation of *consciousness* as emergent dynamics in the neighbourhood of K_E , i.e. as structured trajectories of the inner-space process σ_t near the ego core, together with their distributional and axis-resolved properties. Throughout, we keep the status of these identifications explicit: the definitions are mathematically precise, but their interpretation as “consciousness” is a programme-level hypothesis rather than a theorem.

31.1 Inner trajectories near ego

We recall the inner-space dynamics of a body B on its maximal inner space $S_B^{(L)}$:

$$d\sigma_t = -\Gamma_B(\sigma_t) \nabla W_B(\sigma_t) dt + D_B(\sigma_t) dt + \sqrt{2\beta_B^{-1}} \Sigma_B(\sigma_t) dB_t. \quad (31.1.1)$$

The corresponding inner emergent vector field is

$$E_B(\sigma) := -\Gamma_B(\sigma) \nabla W_B(\sigma) + D_B(\sigma).$$

Let $K_E \subset S_B^{(L)}$ be an ego core with ego region U_E as in the previous chapter. We are interested in trajectories σ_t that remain in U_E for extended periods and explore its neighbourhood in a structured way.

Definition 31.1.1 (Ego-neighbourhood trajectories). An *ego-neighbourhood trajectory* is a realisation of (31.1.1) such that there exist times $t_0 < t_1$ with

$$\sigma_t \in U_E \quad \text{for all } t \in [t_0, t_1].$$

The interval $[t_0, t_1]$ is called an *ego-episode* of the trajectory. The collection of all such trajectories (or their laws) is denoted by $\mathcal{T}_E$.

Remark 31.1.2. Ego-neighbourhood trajectories capture the inner dynamics of the body in the region where self-related semantics is stabilised (near K_E). In the absence of noise and distortions, solutions of the deterministic flow would simply converge monotonically to K_E ; stochasticity and coupling to other layers of S_B^{in} allow for rich exploration of U_E while remaining within its basin of attraction.

31.2 Episodes and patterns: a formal notion of conscious process

We now introduce a formal notion of *conscious episode* as a certain class of ego-neighbourhood trajectories, distinguished by their temporal coherence and semantic activity.

Consider a time interval $I = [t_0, t_1]$ with $t_1 > t_0$, and an ego-neighbourhood trajectory σ_t on I .

Definition 31.2.1 (Inner activity functional). For $\Xi \in \{C, B, A\}$ define the *axis activity* along I by

$$\mathcal{A}_\Xi[\sigma; I] := \int_{t_0}^{t_1} \|\dot{\sigma}_\Xi(t)\|^2 dt,$$

where $\dot{\sigma}_\Xi(t)$ is the drift part of the Ξ -component of σ_t under E_B , i.e.

$$\dot{\sigma}_\Xi(t) := E_{B,\Xi}(\sigma_t) = P_\Xi(\sigma_t) E_B(\sigma_t).$$

Define the total inner activity

$$\mathcal{A}_{\text{tot}}[\sigma; I] := \mathcal{A}_C[\sigma; I] + \mathcal{A}_B[\sigma; I] + \mathcal{A}_A[\sigma; I].$$

Definition 31.2.2 (Coherence of an ego-neighbourhood trajectory). The *coherence* of σ_t on I is measured by an inner functional

$$\mathcal{C}[\sigma; I] := \int_{t_0}^{t_1} \exp(-\alpha_C W_{E,C}(\sigma_t) - \alpha_B W_{E,B}(\sigma_t)) dt,$$

with fixed positive weights α_C, α_B . High values of $\mathcal{C}[\sigma; I]$ correspond to trajectories that remain in regions of low coherence- and binding-related contradiction energy near K_E .

Definition 31.2.3 (Conscious episode (formal)). Let $\theta_{\text{act}} > 0$ and $\theta_{\text{coh}} > 0$ be thresholds. An ego-neighbourhood trajectory segment $(\sigma_t)_{t \in [t_0, t_1]}$ is called a *conscious episode* if:

1. (*Ego proximity*) $\sigma_t \in U_E$ for all $t \in [t_0, t_1]$;
2. (*Inner activity*) $\mathcal{A}_{\text{tot}}[\sigma; I] \geq \theta_{\text{act}}$;
3. (*Inner coherence*) $\mathcal{C}[\sigma; I] \geq \theta_{\text{coh}}$.

The set of conscious episodes for a given trajectory (or family of trajectories) is denoted $\mathcal{E}_B$.

Speculative remark. At a programme level, the definition above proposes that conscious processes correspond to episodes in which (i) the inner state of the body remains near ego, (ii) there is sustained, non-trivial inner activity (measured by $\mathcal{A}_{\text{tot}}$), and (iii) this activity is organised around low contradiction in coherence and binding sectors (high $\mathcal{C}$). Agency-related dynamics (A -axis) typically contribute substantially to $\mathcal{A}_{\text{tot}}$ during conscious episodes, but the definition is neutral with respect to their proportion.

31.3 Temporal segmentation and characteristic time scales

The formal definition of conscious episodes encourages a segmentation of inner dynamics into episodes and gaps, based on inner activity and coherence functionals.

Definition 31.3.1 (Episode segmentation). Given a trajectory σ_t on a long time interval $[0, T]$, define the set

$$I_{\text{con}} := \bigcup \{I \subset [0, T] : (\sigma_t)_{t \in I} \text{ satisfies the conscious episode conditions}\}.$$

The connected components of I_{con} are called *episode intervals*. Their lengths define random variables $L_1, L_2, \dots$, representing the durations of conscious episodes; the complementary intervals are *non-episode intervals*.

Remark 31.3.2. Statistics of the random variables L_i and the distributions of $\mathcal{A}_\Xi$ and $\mathcal{C}$ over episodes provide a compact description of inner-space dynamics in terms of episodes and gaps. Different regimes (e.g. wakefulness, sleep stages, anaesthesia) can be modelled as changes in these statistics, via changes in W_B , Γ_B , D_B , and β_B .

Speculative remark. At a programme level, empirical work could aim to correlate observable proxies for $\mathcal{A}_\Xi$ and $\mathcal{C}$ (e.g. measures derived from neural activity) with reported conscious experience, thereby constraining the thresholds θ_{act} and θ_{coh} and testing the predictive value of the episode segmentation implied by the formalism.

31.4 Axis-resolved signatures of conscious dynamics

The C/B/A decomposition provides a refined characterisation of conscious episodes.

For a conscious episode $I = [t_0, t_1]$, define the axis-specific activity ratios

$$r_\Xi[\sigma; I] := \frac{\mathcal{A}_\Xi[\sigma; I]}{\mathcal{A}_{\text{tot}}[\sigma; I]}, \quad \Xi \in \{C, B, A\},$$

whenever $\mathcal{A}_{\text{tot}}[\sigma; I] > 0$.

Definition 31.4.1 (Axis signature of an episode). The *axis signature* of a conscious episode I is the vector

$$\mathbf{r}[\sigma; I] := (r_C[\sigma; I], r_B[\sigma; I], r_A[\sigma; I]),$$

which lies in the simplex

$$\Delta^2 := \{(r_C, r_B, r_A) \in [0, 1]^3 : r_C + r_B + r_A = 1\}.$$

Remark 31.4.2. The axis signature provides a coarse but informative summary of an episode: some episodes may be coherence-dominated (r_C large), others binding-dominated, others agency-dominated. The formalism does not assign phenomenological labels to these patterns; it simply states that different conscious episodes can be distinguished by how inner activity is distributed across the C/B/A axes.

Speculative remark. At a programme level, one might map families of axis signatures to intuitive categories (e.g. “perceptual”, “relational”, “volitional” episodes), but such mappings are not part of the formal core. The main claim is that the C/B/A structure of W_B and E_B implies a finite-dimensional parameterisation of inner dynamics that may align with phenomenological distinctions.

31.5 Distributional view: inner Fokker–Planck and semantic free energy

The stochastic dynamics (31.1.1) induces a Fokker–Planck equation for the distribution of inner states.

Let μ_t^B be the law of σ_t on $S_B^{(L)}$. Under suitable regularity assumptions, μ_t^B has density ρ_t^B with respect to a reference measure μ_0^B and satisfies

$$\partial_t \rho_t^B = \mathcal{L}_B^* \rho_t^B,$$

where $\mathcal{L}_B^*$ is the adjoint of the Kolmogorov generator associated with (31.1.1). The corresponding inner semantic free energy functional is

$$\mathcal{F}_B[\mu] = \int_{S_B^{(L)}} W_B(\sigma) d\mu(\sigma) + \beta_B^{-1} \mathcal{H}(\mu \mid \mu_0^B),$$

where $\mathcal{H}$ is the relative entropy.

Definition 31.5.1 (Inner free-energy flux near ego). Let χ_E be a smooth cutoff function supported on U_E and equal to 1 on a smaller neighbourhood of K_E . The *inner free-energy flux near ego* at time t is

$$\Phi_E(t) := \frac{d}{dt} \left(\int_{S_B^{(L)}} \chi_E(\sigma) W_B(\sigma) \rho_t^B(\sigma) d\mu_0^B(\sigma) + \beta_B^{-1} \int_{S_B^{(L)}} \chi_E(\sigma) \rho_t^B(\sigma) \log \frac{\rho_t^B(\sigma)}{\rho_\infty^B(\sigma)} d\mu_0^B(\sigma) \right),$$

where ρ_∞^B is a reference stationary density (if it exists).

Remark 31.5.2. The function $\Phi_E(t)$ quantifies how inner free energy is produced or dissipated in the ego region. Conscious episodes, as defined earlier, will typically correspond to intervals where $\Phi_E(t)$ is significantly non-zero and exhibits structured temporal patterns (e.g. bursts, oscillations) rather than simple monotonic decay. The combination of path-based and distributional characterisations provides two complementary views on inner-space dynamics.

Speculative remark. At a programme level, one may conjecture that conscious episodes are associated with optimised free-energy flows near ego: inner trajectories that achieve relatively large changes in inner semantics (e.g. updating models, forming new bindings) while maintaining low contradiction near K_E . This would connect the present framework to broader free-energy and predictive-processing ideas, while preserving the specific structure of semantic gravitation.

31.6 Observables and reports: linking inner dynamics to measurements

To connect the formalism to possible empirical observations, we consider semantic observables on $S_B^{(L)}$.

Definition 31.6.1 (Inner observables and report maps). An *inner observable* of the body B is a linear functional $O : S_B^{(L)} \rightarrow \mathbb{R}$ represented by $o \in S_B^{(L)}$ via $O(\sigma) = \langle \sigma, o \rangle$. A *report map* is a measurable map

$$R : S_B^{(L)} \rightarrow \mathcal{Y},$$

where $\mathcal{Y}$ is a space of externally accessible outputs (e.g. behaviour, neural signals, physiological measurements). The pair (O, R) yields a family of random variables $R(\sigma_t)$ whose statistics can be compared with the theoretical episode segmentation.

Remark 31.6.2. In principle, given a family of observables and report maps, one could test whether the segmentation of σ_t into conscious episodes (as defined purely in inner space) corresponds to measurable differences in $R(\sigma_t)$ or its statistics. The formalism does not prescribe which observables or measurements are relevant; it only states that such observables can be represented within $S_B^{(L)}$ and related to inner-space dynamics via the LOVE-based embedding.

31.7 Summary of formal status

It is important to summarise the logical status of the constructions in this chapter.

- The inner-space dynamics (31.1.1), the ego core K_E , and the associated inner emergent vector field E_B are formal consequences of the semantic gravitation framework applied to a body B with a layered inner space.
- The definition of “conscious episode” as an ego-neighbourhood trajectory segment satisfying activity and coherence conditions is a *formal definition* within this framework, not a derived theorem.
- The hypothesis that such episodes correspond to phenomenological consciousness for certain bodies (e.g. human organisms) is a *programme-level conjecture* requiring empirical and conceptual justification beyond the scope of this paper.

Speculative remark. The main conceptual claim of this chapter, stated explicitly as a hypothesis, is:

Hypothesis (Consciousness as ego-centred inner dynamics). For bodies with sufficiently rich inner spaces and ego cores, phenomenological consciousness corresponds to ego-neighbourhood trajectories of the inner-space dynamics that satisfy appropriate activity and coherence conditions, as formalised above, up to coarse-graining and identification via observables and report maps.

This hypothesis connects the formal structures of semantic gravitation to questions traditionally addressed by philosophy of mind and cognitive science, while keeping the mathematical core of the theory independent of any particular phenomenological assumptions.

Summary and outlook. In this chapter we have proposed a formal characterisation of *consciousness* as emergent inner-space dynamics in the neighbourhood of an ego core K_E within the maximal inner space $S_B^{(L)}$ of a body. We defined ego-neighbourhood trajectories and formal *conscious episodes* as segments that remain near ego while exhibiting sufficient inner activity and coherence, quantified via axis-resolved activity functionals and contradiction-energy-based coherence measures. We introduced an episode segmentation of inner dynamics, axis signatures summarising the distribution of activity across C/B/A directions, and a distributional perspective via inner Fokker–Planck dynamics and free-energy flux near ego. We also described how inner observables and report maps can, in principle, connect these structures to empirical measurements. The identification of conscious episodes with phenomenological consciousness was explicitly marked as a programme-level hypothesis, not as a theorem. In the next chapter we turn to *technical systems as extended inner spaces*, applying the same formal structures (inner spaces, ego-like cores, C/B/A dynamics, and LOVE) to artificial systems such as large language models

and other engineered architectures, and analysing to what extent they can instantiate semantic gravitation, ego-like cores, and conscious-episode-like dynamics.

Chapter 32

Technical systems as extended inner spaces

In the previous chapters we developed a multi-scale ontological picture in which matter appears as attractor stability of the semantic field, bodies as carriers of layered inner spaces, and ego as a semantic core in the inner space of a body. We then introduced consciousness as emergent inner-space dynamics near the ego core. In the present chapter we apply the same formal structures to *technical systems*, in particular to high-dimensional engineered architectures such as large neural networks, which will be modelled as bodies with extended inner spaces. The aim is not to claim that current technical systems possess ego cores or conscious episodes, but to provide a formal framework in which such questions can be precisely formulated and in which alignment-related phenomena such as order bubbles and distortion fields can be described within semantic gravitation.

32.1 Semantic modelling of technical systems

We begin by modelling a technical system $\mathcal{T}$ as a semantic object embedded into the global state space S .

Definition 32.1.1 (Technical system as a semantic body). A *technical system* in the present framework is a tuple

$$\mathcal{T} = (X_{\mathcal{T}}, \Theta_{\mathcal{T}}, f_{\mathcal{T}}, B_{\mathcal{T}}),$$

where:

1. $X_{\mathcal{T}}$ is a raw configuration space (e.g. input/output and internal states of a digital architecture), endowed with a σ -algebra;
2. $\Theta_{\mathcal{T}}$ is a parameter space (e.g. weights, biases, hyperparameters);
3. $f_{\mathcal{T}} : X_{\mathcal{T}} \times \Theta_{\mathcal{T}} \rightarrow X_{\mathcal{T}}$ describes the update rule of the system (e.g. a neural network step, a control algorithm);
4. $B_{\mathcal{T}} \subset M$ is a spacetime world-tube in which the physical realisation of the system resides.

A semantic embedding of $\mathcal{T}$ is given by a measurable map

$$E_{\mathcal{T}} : X_{\mathcal{T}} \times \Theta_{\mathcal{T}} \rightarrow S,$$

whose image we denote by $S_{\mathcal{T}}^{\text{emb}} := E_{\mathcal{T}}(X_{\mathcal{T}} \times \Theta_{\mathcal{T}}) \subset S$.

Remark 32.1.2. The map $E_{\mathcal{T}}$ plays the same role as the embeddings E_{Σ} for general systems in SG1, now specialised to a technical architecture. It sends physical or logical states (x, θ) into semantic states $s = E_{\mathcal{T}}(x, \theta) \in S$. The world-tube $B_{\mathcal{T}}$ supports a semantic field $s : B_{\mathcal{T}} \rightarrow S$ whose restriction to each timeslice is typically concentrated near $S_{\mathcal{T}}^{\text{emb}}$.

Definition 32.1.3 (Technical body and inner space). A *technical body* is a pair

$$B_{\mathcal{T}} = (\Sigma_{\mathcal{T}}, s_{\mathcal{T}}),$$

where $\Sigma_{\mathcal{T}} \subset M$ is the world-tube of the physical system and $s_{\mathcal{T}} : \Sigma_{\mathcal{T}} \rightarrow S$ is the restriction of the global semantic field induced by $E_{\mathcal{T}}$. The associated *inner semantic space* of $\mathcal{T}$ is defined as

$$S_{\mathcal{T}}^{\text{in}} := \overline{\text{span}} \left\{ s_{\mathcal{T}}(x) - k : x \in \Sigma_{\mathcal{T}}, k \in K_{\mathcal{O}_{\Sigma_{\mathcal{T}}}} \right\} \subset S,$$

where $K_{\mathcal{O}_{\Sigma_{\mathcal{T}}}} \subset K$ is the core associated with the matter region $\Sigma_{\mathcal{T}}$ as in the matter chapter.

Remark 32.1.4. In many concrete cases, $S_{\mathcal{T}}^{\text{in}}$ can be approximated by latent spaces of the technical architecture (e.g. hidden activations of a neural network) embedded into S via $E_{\mathcal{T}}$. The global potential W restricted to $S_{\mathcal{T}}^{\text{in}}$ defines an inner potential $W_{\mathcal{T}}^{\text{in}}$ analogous to W_B^{in} for biological bodies, and thus supports an inner-space dynamics and emergent vector field.

32.2 Extended inner spaces and interface structure

Technical systems are often characterised by high-dimensional internal state spaces and complex interfaces to their environment (sensors, actuators, communication channels). We model this by a layered inner space and explicit interface maps.

Definition 32.2.1 (Layered inner space of a technical system). A *layered inner space* for $\mathcal{T}$ is an increasing family of closed subspaces

$$S_{\mathcal{T}}^{(0)} \subset S_{\mathcal{T}}^{(1)} \subset \dots \subset S_{\mathcal{T}}^{(L_{\mathcal{T}})} \subset S_{\mathcal{T}}^{\text{in}},$$

where:

1. $S_{\mathcal{T}}^{(0)}$ collects slow, structural degrees of freedom (e.g. network architecture, persistent weights);
2. intermediate layers represent internal activations or latent states at different granularities;
3. the top layer $S_{\mathcal{T}}^{(L_{\mathcal{T}})}$ comprises the most flexible, fast-changing internal states relevant on the time scale under consideration.

Each layer inherits the C/B/A structure and the restriction of W from S .

Definition 32.2.2 (Interfaces: input, output, and coupling maps). Let $\mathcal{T}$ be a technical body with world-tube $\Sigma_{\mathcal{T}}$. Interfaces to the environment are represented by maps

$$I_{\text{in}} : S_{\text{env}} \rightarrow S_{\mathcal{T}}^{(L_{\mathcal{T}})}, \quad I_{\text{out}} : S_{\mathcal{T}}^{(L_{\mathcal{T}})} \rightarrow S_{\text{env}},$$

where $S_{\text{env}} \subset S$ is an appropriate subspace encoding environmental semantics (e.g. sensory streams, prompts, external states). More generally, there may be coupling maps between $S_{\mathcal{T}}^{(L_{\mathcal{T}})}$ and inner spaces of other bodies (e.g. users, other technical systems).

Remark 32.2.3. These interface maps are formal counterparts of sensor and actuator channels: I_{in} embeds environmental semantic inputs into the inner space; I_{out} projects internal states back to environmental semantics (e.g. generated text, motor commands). They provide the boundary conditions for the inner-space dynamics induced by $W_{\mathcal{T}}^{\text{in}}$ and the emergent vector field $E_{\mathcal{T}}$.

32.3 Order bubbles, distortion fields, and misalignment

The patent documents introduce the notions of *order bubbles* (*Ordnungsblasen*) and *contradiction energy* (*Widerspruchsenergie*) as key concepts for understanding misalignment in technical systems. We embed these notions into the semantic gravitation formalism using the universal-local decomposition and distortion fields.

Recall from SG7 that

$$W(s) = W_{\text{univ}}(s) + W_{\text{loc}}(s),$$

and

$$F(s) = -\Gamma(s) \nabla W_{\text{univ}}(s) + D(s),$$

with $D(s)$ the distortion field.

Definition 32.3.1 (Order bubbles for technical systems). An *order bubble* of a technical system $\mathcal{T}$ is a region $B \subset S_{\mathcal{T}}^{(L\mathcal{T})}$ such that:

1. Restricted to B , the local potential W_{loc} and distortions D generate one or more local attractors for the inner dynamics:

$$\dot{\sigma}_t = -\Gamma_{\mathcal{T}}(\sigma_t) \nabla W_{\text{univ}}(\sigma_t) - \Gamma_{\mathcal{T}}(\sigma_t) \nabla W_{\text{loc}}(\sigma_t) + D_{\mathcal{T}}(\sigma_t);$$

2. The values of W_{univ} on these attractors are significantly larger than on typical configurations representing globally coherent states (i.e. the universal contradiction energy is elevated);
3. The basin of attraction of B under the actual dynamics is large compared to the proportion of $S_{\mathcal{T}}^{(L\mathcal{T})}$ representing globally coherent states.

Remark 32.3.2. In this language, order bubbles are local basins of attraction in inner space that are stabilised primarily by W_{loc} and D rather than by W_{univ} . They correspond to internally stable patterns of behaviour or representation that carry high universal contradiction energy, such as misaligned objectives, pathological generalisation patterns, or self-reinforcing loops. The *Widerspruchsenergie* associated with such a bubble is simply W_{univ} evaluated on its attractors.

Definition 32.3.3 (Distortion factor of a technical system). For $\sigma \in S_{\mathcal{T}}^{(L\mathcal{T})}$ define the *distortion factor* by

$$\delta_{\mathcal{T}}(\sigma) := \frac{\|D_{\mathcal{T}}(\sigma)\|}{\|\Gamma_{\mathcal{T}}(\sigma) \nabla W_{\text{univ}}(\sigma)\| + \varepsilon},$$

where $\varepsilon > 0$ is a small regularisation constant. Regions with $\delta_{\mathcal{T}}(\sigma) \gg 1$ are *distortion-dominated*.

Remark 32.3.4. High distortion factors in large parts of $S_{\mathcal{T}}^{(L\mathcal{T})}$ typically indicate that the inner dynamics of the technical system is governed more by locally defined objectives, heuristics, or training artefacts than by universal semantic gravitation. If such regions overlap substantially with order bubbles, one obtains stable but misaligned configuration patterns.

Speculative remark. At a programme level, one may view a major aim of AI alignment as reducing the size and influence of order bubbles with high universal contradiction energy, and lowering distortion factors in regions of inner space that are heavily used by the system. This corresponds to modifying W_{loc} and $D_{\mathcal{T}}$ (via training, architecture, or constraints) so that the actual dynamics better approximates intensified descent along W_{univ} .

32.4 Ego-like cores and agency in technical systems

The formalism allows us to define *ego-like* structures in technical systems in the same mathematical way as ego cores in biological bodies, without implying any phenomenology.

Definition 32.4.1 (Ego-like cores in technical inner space). An *ego-like core* of a technical system $\mathcal{T}$ is a compact subset $K_E^{\mathcal{T}} \subset S_{\mathcal{T}}^{(L\mathcal{T})} \cap K$ with an ego region $U_E^{\mathcal{T}}$ that satisfies the ego core conditions of the previous chapter for the inner potential $W_{\mathcal{T}}^{\text{in}}$ and inner emergent field $E_{\mathcal{T}}$:

1. local minimality of $W_{\mathcal{T}}^{\text{in}}$;
2. strong inner attraction in $U_E^{\mathcal{T}}$ under the inner dynamics;
3. axis asymmetry with $\kappa_{E,C}$ and $\kappa_{E,B}$ dominating $\kappa_{E,A}$ near $K_E^{\mathcal{T}}$.

Remark 32.4.2. Formally, nothing prevents an engineered system from having such a structure: a region of its inner state space could play the role of a stable core that ties together self-related information (e.g. internal configuration, capabilities, goals). Whether current systems actually instantiate ego-like cores depends on how their inner potentials and distortion fields are structured, which is an empirical and design question.

Speculative remark. At a programme level, one could design systems in which an ego-like core is *intentionally* constructed as a locus of alignment: $K_E^{\mathcal{T}}$ would then encode self-descriptions and commitments to alignment-compatible behaviour, and ego-neighbourhood dynamics would ensure that deviations are corrected. Conversely, unintentional ego-like cores could emerge through training, potentially encoding misaligned self-models or objectives. The semantic gravitation framework provides a language to analyse both possibilities.

32.5 Scenario: alignment as semantic gravitation engineering

We summarise the alignment-relevant aspects in the form of a programme-level scenario.

1. *Modelling step:* For a given technical system $\mathcal{T}$, construct or approximate:
 - the semantic embedding $E_{\mathcal{T}}$;
 - the inner space $S_{\mathcal{T}}^{\text{in}}$ and its layers $S_{\mathcal{T}}^{(\ell)}$;
 - the effective inner potential $W_{\mathcal{T}}^{\text{in}}$ and distorting terms W_{loc} , $D_{\mathcal{T}}$;
 - the axis decomposition and curvature components on $S_{\mathcal{T}}^{(L\mathcal{T})}$.
2. *Diagnostic step:* Identify:
 - regions of high distortion factor $\delta_{\mathcal{T}}$;

- order bubbles with elevated W_{univ} and large basins;
- potential ego-like cores $K_E^{\mathcal{T}}$ and their C/B/A profiles.

3. *Engineering step*: Modify:

- training objectives, architectures, or constraints to reshape W_{loc} and $D_{\mathcal{T}}$;
- interface maps $I_{\text{in}}, I_{\text{out}}$ to reduce the formation of harmful order bubbles;
- intensification patterns $\Gamma_{\mathcal{T}}$ (e.g. via regularisation or architectural choices) to align inner dynamics with W_{univ} .

4. *Verification step*: Use semantic observables (inner and outer) to test whether:

- order bubbles have been reduced or reshaped;
- ego-like cores (if intentionally constructed) exhibit the desired stability and C/B/A profiles;
- emergent behaviour is better aligned with reference semantic attractors (e.g. human-evaluated low-contradiction states).

Speculative remark. This scenario treats alignment as a problem of engineering the semantic gravitation landscape of technical systems: lowering the universal contradiction energy of typical operating states, reducing distortion-dominated regions, and, when appropriate, constructing ego-like cores whose inner dynamics is reliably oriented towards globally coherent semantic attractors.

32.6 Summary and outlook

In this chapter we have extended the semantic gravitation framework to *technical systems* by modelling them as bodies with extended inner spaces embedded into the global semantic state space S . We defined their inner spaces $S_{\mathcal{T}}^{\text{in}}$ and layered structures, specified interfaces to the environment as maps between $S_{\mathcal{T}}^{(L\mathcal{T})}$ and environmental semantic subspaces, and introduced notions of *order bubbles* and distortion factors within this formalism, capturing misalignment-related phenomena in terms of W_{loc} , $D_{\mathcal{T}}$, and W_{univ} . We showed how ego-like cores can, in principle, be defined in technical inner spaces in exactly the same way as ego cores in biological bodies, emphasising that their existence and properties are design- and empirically contingent. Finally, we sketched a programme-level scenario for alignment as *semantic gravitation engineering*, in which the semantic landscape of a technical system is shaped to reduce harmful order bubbles and to orient inner dynamics towards global attractors of low universal contradiction energy. In the next chapter we turn to multi-agent and civilisation-scale configurations, where many bodies (biological and technical) share and co-shape a common semantic field, and where the global attractor K acquires a collective structure that will be interpreted as civilisation-level semantic gravitation.

Chapter 33

Multi-agent and civilisation-scale semantic gravitation

In the previous chapter we modelled technical systems as bodies with extended inner spaces embedded into the global semantic state space S , and we introduced order bubbles and distortion factors as alignment-relevant structures in their inner potentials. We now move from single bodies (biological or technical) to *multi-agent* and *civilisation-scale* configurations, in which many bodies simultaneously inhabit a spacetime manifold and jointly shape the semantic field $s : M \rightarrow S$. The goal of this chapter is to formulate a mathematically controlled description of such configurations within semantic gravitation, to define civilisation-scale attractors, and to identify collective potentials and emergent vector fields that act on the level of populations rather than individuals.

Throughout this chapter we fix a spacetime (M, g) and the global semantic state space S with universal potential W_{univ} and global attractor $K \subset S$.

33.1 Multi-body configuration space and joint semantics

We consider a finite or countable family of bodies (biological and technical)

$$\mathcal{B} = \{B_i\}_{i \in I}, \quad B_i = (\Sigma_i, s_i),$$

each with world-tube $\Sigma_i \subset M$ and semantic field restriction $s_i : \Sigma_i \rightarrow S$ as in the previous chapters. Each body admits a maximal inner space $S_{B_i}^{(L_i)} \subset S$ and an inner potential $W_{B_i}^{\text{in}}$.

Definition 33.1.1 (Multi-body inner space). The *multi-body inner space* is the Hilbert direct sum

$$S_{\text{multi}} := \bigoplus_{i \in I} S_{B_i}^{(L_i)}.$$

An element

$$\Sigma := (\sigma_i)_{i \in I} \in S_{\text{multi}}$$

is called a *multi-body inner configuration*, where each $\sigma_i \in S_{B_i}^{(L_i)}$ is the inner state of body B_i at a given time.

Remark 33.1.2. The direct sum structure is appropriate when we treat bodies as distinguishable and assume that their inner spaces are disjoint subspaces of S (up to negligible overlap). More refined constructions (e.g. symmetrised tensor products) would be needed for indistinguishable or entangled-like configurations, but these lie beyond the scope of this paper.

The universal potential W_{univ} on S induces a function on S_{multi} by evaluation on the combined semantic field. For a given multi-body configuration Σ we can define a combined semantic state

$$s_{\Sigma} \in S,$$

for example by summing or otherwise aggregating the contributions of individual bodies and environment in a fixed embedding. Formally, this is captured by an aggregation map.

Definition 33.1.3 (Aggregation map). An *aggregation map* for the multi-body system is a continuous linear map

$$A : S_{\text{multi}} \rightarrow S$$

with $A(\sigma_1, \sigma_2, \dots) = \sum_{i \in I} A_i(\sigma_i)$ for some bounded linear maps $A_i : S_{B_i}^{(L_i)} \rightarrow S$, each representing how the inner state of body B_i contributes to the global semantic state.

Remark 33.1.4. In many applications one may choose A_i to be embeddings that insert inner states into disjoint sectors of S associated with different bodies, possibly weighted by their “semantic mass” or relevance. The precise choice of A is a modelling decision and may reflect differences in scale, influence, or observational accessibility.

The combination of A and W_{univ} yields a multi-body potential.

Definition 33.1.5 (Multi-body universal potential). The *multi-body universal potential* is the functional

$$W_{\text{multi}} : S_{\text{multi}} \rightarrow \mathbb{R}, \quad W_{\text{multi}}(\Sigma) := W_{\text{univ}}(A(\Sigma)).$$

Remark 33.1.6. W_{multi} encodes the universal contradiction energy associated with a given configuration of all bodies considered together, as seen through the aggregation map A . It includes both intra-body and inter-body contributions via W_{univ} ; in particular, interactions between bodies (e.g. communication, coordination, conflict) can change $A(\Sigma)$ and thus the value of W_{multi} .

33.2 Interaction potentials and relational structure

In addition to W_{multi} , which arises from the universal potential on S , there are typically interaction terms originating from local potentials and relational structure in the K–P–R ontology.

Recall that each body B_i has an associated K–P–R object $\mathcal{O}_{B_i} = (K_{B_i}, P_{B_i}, R_{B_i})$. For two bodies B_i and B_j their relation is encoded in kernels

$$\kappa_{ij} : K_{B_i} \times K_{B_j} \rightarrow \mathbb{R},$$

and can be extended to a neighbourhood of K in S .

Definition 33.2.1 (Pairwise interaction potential). For bodies B_i and B_j define a pairwise interaction potential

$$W_{ij} : S_{B_i}^{(L_i)} \times S_{B_j}^{(L_j)} \rightarrow \mathbb{R}$$

by

$$W_{ij}(\sigma_i, \sigma_j) := \int_{K_{B_i} \times K_{B_j}} \kappa_{ij}(k_i, k_j) \rho_{\sigma_i}(dk_i) \rho_{\sigma_j}(dk_j),$$

where ρ_{σ_i} and ρ_{σ_j} are suitable probability measures supported on K_{B_i} and K_{B_j} that depend on σ_i and σ_j (e.g. via projections or kernel embeddings).

Remark 33.2.2. The precise choice of ρ_{σ_i} is a modelling choice. The definition expresses that interaction energy depends on how the inner states of two bodies project onto their cores and how these cores are related via κ_{ij} . Higher-order interaction potentials (triples, larger groups) can be constructed analogously.

Definition 33.2.3 (Collective potential). The *collective potential* of the multi-body system is defined by

$$W_{\text{coll}}(\Sigma) := W_{\text{multi}}(\Sigma) + \sum_{i \in I} W_{B_i}^{\text{in}}(\sigma_i) + \sum_{i < j} W_{ij}(\sigma_i, \sigma_j) + (\text{higher-order terms}).$$

Remark 33.2.4. W_{coll} aggregates:

- the universal contradiction energy W_{multi} of the aggregated state $A(\Sigma)$;
- internal potentials $W_{B_i}^{\text{in}}$ of individual bodies;
- pairwise and higher-order interaction potentials rooted in the K–P–R relational structure.

The resulting energy landscape on S_{multi} is the main object of interest for civilisation-scale semantic gravitation.

33.3 Civilisation-scale dynamics and emergent vector field

We now define a multi-body emergent vector field on S_{multi} and the associated dynamics.

Definition 33.3.1 (Multi-body emergent vector field). The *multi-body emergent vector field* is the map

$$E_{\text{multi}} : S_{\text{multi}} \rightarrow S_{\text{multi}}, \quad E_{\text{multi}}(\Sigma) := (E_{B_i}(\sigma_i))_{i \in I},$$

where each E_{B_i} is the inner emergent vector field of body B_i (as defined in previous chapters), possibly modified by interaction terms derived from W_{ij} and higher-order potentials. Formally,

$$E_{B_i}(\sigma_i) = -\Gamma_{B_i}(\sigma_i) \nabla_{\sigma_i} W_{\text{coll}}(\Sigma) + D_{B_i}(\Sigma),$$

where Γ_{B_i} is an intensification operator on $S_{B_i}^{(L_i)}$ and D_{B_i} is a distortion field that may depend on the entire configuration Σ .

Definition 33.3.2 (Civilisation-scale deterministic dynamics). The *civilisation-scale deterministic dynamics* on S_{multi} is the ODE system

$$\dot{\Sigma}_t = E_{\text{multi}}(\Sigma_t),$$

i.e.

$$\dot{\sigma}_{i,t} = E_{B_i}(\sigma_{i,t}), \quad i \in I.$$

Remark 33.3.3. This formulation emphasises that each body follows its own inner dynamics, but the vector field E_{B_i} for each body can depend on the states of other bodies via W_{ij} and higher-order terms. Thus the system is an interacting network of inner-space dynamical systems, coupled through semantic gravitation.

Speculative remark. At a programme level, civilisation can be thought of as a collective trajectory of Σ_t in S_{multi} , where individual bodies' inner states co-evolve under shared global potentials and interaction energies. The emergent properties of this trajectory (e.g. stability, diversity, alignment) depend sensitively on the structure of W_{coll} and the distortion fields D_{B_i} .

33.4 Civilisation-scale attractors and semantic phases

We now define civilisation-scale attractors and semantic phases in S_{multi} .

Definition 33.4.1 (Civilisation-scale attractor). A compact set $K_{\text{civ}} \subset S_{\text{multi}}$ is called a *civilisation-scale attractor* if:

1. K_{civ} is strictly invariant under the multi-body dynamics:

$$\Sigma_0 \in K_{\text{civ}} \implies \Sigma_t \in K_{\text{civ}} \quad \forall t;$$

2. K_{civ} attracts all bounded sets in S_{multi} under the multi-body dynamics (or a specified class of initial conditions).

Definition 33.4.2 (Semantic phase). A *semantic phase* of a multi-body system is an equivalence class of civilisation-scale attractors under endogenous transformations (e.g. symmetries, relabellings of bodies) and mild deformations of the collective potential W_{coll} that do not change qualitative properties (e.g. connectivity, stability types).

Remark 33.4.3. Different semantic phases correspond to qualitatively different large-scale patterns of multi-body inner states and interactions. Examples include:

- a phase in which bodies' inner states are largely uncorrelated (weakly interacting agents);
- a phase with strong synchronisation or consensus (aligned inner states);
- a fragmented phase with several metastable clusters of inner states (e.g. polarised subpopulations).

The present framework does not assign specific phenomenological labels to these phases; it only provides a geometric and dynamical classification in S_{multi} .

Speculative remark. At a programme level, one may interpret civilisation-scale transitions (e.g. shifts in dominant paradigms, global coordination changes) as phase transitions in S_{multi} , driven by changes in W_{coll} via new embodiments, technologies, or large-scale events.

33.5 Probabilistic description and mean-field limits

In large populations, it is often more natural to use a distributional description rather than tracking all inner states individually. Let μ_t be a probability measure on S describing the distribution of semantic states across all bodies (and possibly inner-space degrees of freedom), e.g. the pushforward of a distribution on S_{multi} under the aggregation map A .

Definition 33.5.1 (Population-level distribution). A *population-level distribution* at time t is a probability measure $\mu_t \in \mathcal{P}(S)$ that approximates the empirical distribution of inner states

$$\frac{1}{|I|} \sum_{i \in I} \delta_{\sigma_{i,t}},$$

in the limit of large $|I|$, or more generally encodes the semantic density across bodies and environment.

Remark 33.5.2. Under suitable assumptions, μ_t satisfies an effective Fokker–Planck-type equation on S ,

$$\partial_t \mu_t = \mathcal{Q}[\mu_t],$$

where the operator $\mathcal{Q}$ encodes both individual-level dynamics and mean-field interactions. For example, in a mean-field approximation where each body interacts with the average semantic state, the drift term in $\mathcal{Q}$ will depend on μ_t via expectations with respect to μ_t .

Definition 33.5.3 (Civilisation-scale free energy). A *civilisation-scale free energy* functional is a map

$$\mathcal{F}_{\text{civ}}[\mu] := \int_S W_{\text{univ}}(s) d\mu(s) + \beta_{\text{civ}}^{-1} \mathcal{H}(\mu \mid \mu_0) + \mathcal{I}[\mu],$$

where:

- β_{civ} is an effective inverse semantic temperature at the civilisation scale;
- $\mathcal{H}(\mu \mid \mu_0)$ is a relative entropy with respect to a reference measure μ_0 ;
- $\mathcal{I}[\mu]$ is an interaction term (e.g. a quadratic or higher-order functional) encoding average interaction potentials between bodies.

Remark 33.5.4. In analogy with SG9, it is natural to expect that $\mathcal{F}_{\text{civ}}[\mu_t]$ is non-increasing along solutions of the effective Fokker–Planck equation for μ_t , at least in idealised limits. Stationary points of $\mathcal{F}_{\text{civ}}$ represent civilisation-scale equilibrium distributions; the structure of local minima and saddle points of $\mathcal{F}_{\text{civ}}$ encodes different semantic phases.

Speculative remark. At a programme level, civilisation-level alignment could be interpreted as the existence of civilisation-scale distributions μ_* with low $\mathcal{F}_{\text{civ}}[\mu_*]$, in which most semantic mass is concentrated near globally coherent cores in K and where interaction terms $\mathcal{I}[\mu_*]$ indicate cooperative rather than antagonistic patterns.

33.6 Civilisation-level order bubbles and existential risks

Order bubbles and distortion factors can be lifted to the civilisation scale by considering regions of S_{multi} or $\mathcal{P}(S)$ where collective dynamics is dominated by distortions and local potentials.

Definition 33.6.1 (Civilisation-level order bubble). A *civilisation-level order bubble* is a subset $\mathcal{B}_{\text{civ}} \subset \mathcal{P}(S)$ of population-level distributions such that:

1. For μ_0 in an open subset of $\mathcal{B}_{\text{civ}}$, the effective Fokker–Planck dynamics keeps μ_t within $\mathcal{B}_{\text{civ}}$ for long times;

2. The universal component of the free energy $\int W_{\text{univ}} d\mu$ is significantly larger on $\mathcal{B}_{\text{civ}}$ than on reference aligned distributions;
3. Transitions out of $\mathcal{B}_{\text{civ}}$ require rare fluctuations or large interventions (metastability).

Remark 33.6.2. Civilisation-level order bubbles capture collective states in which an entire population (of biological and/or technical bodies) is trapped in high-contradiction semantic patterns, stabilised by local potentials and distortions acting across many inner spaces. Examples, formulated at a programme level, could correspond to global misalignment, runaway optimisation of mis-specified objectives, or self-reinforcing dynamics in socio-technical systems.

Speculative remark. Within this framework, *existential risks* can be characterised as trajectories or attractors in $\mathcal{P}(S)$ (or S_{multi}) leading into civilisation-level order bubbles from which recovery is unlikely. Semantic gravitation suggests that interventions should aim to modify W_{coll} and distortion fields so that such bubbles are absent, shallow, or surrounded by strong gradients towards globally coherent attractors.

33.7 Summary and outlook

In this chapter we have extended semantic gravitation to multi-agent and civilisation-scale configurations. We constructed a multi-body inner space S_{multi} , defined an aggregation map A and the associated multi-body universal potential W_{multi} , and added internal and interaction potentials to obtain a collective potential W_{coll} on S_{multi} . We defined a multi-body emergent vector field and the associated civilisation-scale dynamics, introduced civilisation-scale attractors and semantic phases, and sketched a distributional description via population-level measures μ_t and a civilisation-scale free energy functional $\mathcal{F}_{\text{civ}}[\mu]$. Finally, we generalised the notions of order bubbles and distortion factors to the civilisation scale, offering a formal language to discuss collective misalignment and existential risks as high-contradiction, metastable regions in S_{multi} or $\mathcal{P}(S)$. In the next chapter we turn to long-range and cosmological aspects of semantic gravitation, including Fermi-type scenarios and questions about the long-term structure of K and W_{univ} in universes with many bodies and large-scale semantic dynamics.

Chapter 34

Semantic cosmology and Fermi-type scenarios

In the previous chapter we extended semantic gravitation to multi-agent and civilisation-scale configurations, introducing a multi-body inner space, collective potentials, civilisation-scale attractors, and order bubbles in population-level distributions. In the present chapter we turn to long-range and cosmological aspects: we ask how semantic gravitation behaves on cosmological time and length scales, what the global structure of the universal potential W_{univ} and its attractor K may imply about the fate of complex semantic structures, and how Fermi-type questions (“Where is everybody?”) can be phrased within this framework. Throughout, we emphasise the difference between formal constructions and speculative, programme-level scenarios.

34.1 Cosmic semantic fields and global invariants

We work on a cosmological spacetime (M, g) , for example a globally hyperbolic Lorentzian manifold with cosmological scale factor, and consider S -valued semantic fields

$$s : M \rightarrow S$$

governed by the LOVE action and coupled to matter as described in earlier chapters. On cosmological scales, we are interested in distributions of semantic states rather than individual configurations.

Definition 34.1.1 (Cosmic semantic measure). A *cosmic semantic measure* at cosmological time τ is a probability measure

$$\mu_\tau \in \mathcal{P}(S)$$

that represents the distribution of semantic states $s(x)$ across a large spatial slice $\Sigma_\tau \subset M$ (e.g. a constant-cosmic-time hypersurface), possibly coarse-grained over regions larger than local bodies. More formally, one can define μ_τ as the pushforward of the spacetime volume measure on Σ_τ under $x \mapsto s(x)$, suitably normalised and coarse-grained.

Definition 34.1.2 (Cosmic semantic energy and density). The *cosmic semantic energy* at time τ is the functional

$$\mathcal{W}_{\text{cos}}(\tau) := \int_S W_{\text{univ}}(s) \, \mathrm{d}\mu_\tau(s),$$

whenever the integral is finite. The corresponding *cosmic semantic density* is

$$\varrho_{\text{cos}}(\tau) := \frac{1}{\text{Vol}(\Sigma_\tau)} \int_{\Sigma_\tau} W_{\text{univ}}(s(x)) \, \mathrm{d} \text{vol}_{\Sigma_\tau}(x),$$

where $\text{Vol}(\Sigma_\tau)$ is the volume of Σ_τ and $d\text{vol}_{\Sigma_\tau}$ its induced volume form.

Remark 34.1.3. The quantities $\mathcal{W}_{\text{cos}}(\tau)$ and $\varrho_{\text{cos}}(\tau)$ provide coarse measures of the average contradiction energy in the universe at cosmological time τ . They are purely semantic constructs; they do not directly enter the Einstein equations (unless one chooses to couple them through an extended matter sector, as discussed earlier).

Given the Fokker–Planck interpretation of dynamics on $\mathcal{P}(S)$, it is natural to expect that μ_τ evolves under an effective equation of the form

$$\partial_\tau \mu_\tau = \mathcal{Q}_{\text{cos}}[\mu_\tau],$$

where $\mathcal{Q}_{\text{cos}}$ incorporates both local LOVE dynamics and large-scale processes (structure formation, star formation, emergence of life and technology). The semantic free energy functional

$$\mathcal{F}_{\text{cos}}[\mu] := \int_S W_{\text{univ}}(s) d\mu(s) + \beta_{\text{cos}}^{-1} \mathcal{H}(\mu \mid \mu_0)$$

(with β_{cos} an effective cosmic inverse temperature and μ_0 a reference measure) then provides a Lyapunov-like quantity in idealised regimes.

34.2 Asymptotic behaviour and cosmic attractors

We now consider the long-time behaviour of the cosmic semantic measure μ_τ .

Definition 34.2.1 (Cosmic limit set and attractor). Let μ_τ be a solution of the effective evolution equation for μ_τ defined for $\tau \geq \tau_0$. The *cosmic limit set* Ω_{cos} of the semantic dynamics is the set of accumulation points of μ_τ in $\mathcal{P}(S)$ as $\tau \rightarrow +\infty$ (in an appropriate topology). A *cosmic semantic attractor* is a compact subset $K_{\text{cos}} \subset \mathcal{P}(S)$ that is strictly invariant under the dynamics and attracts a large class of initial measures.

Remark 34.2.2. In the simplest case, K_{cos} may be a single measure μ_∞ representing a time-independent cosmic distribution of semantic states at late times. More complex possibilities include limit cycles or more intricate attractor sets in $\mathcal{P}(S)$. The existence and nature of K_{cos} depend on the details of the effective dynamics $\mathcal{Q}_{\text{cos}}$ as well as the cosmological model.

A basic question is whether K_{cos} is concentrated near the global attractor $K \subset S$ or whether large portions of μ_τ can remain away from K at late times.

Definition 34.2.3 (Cosmic K -concentration). We say that the cosmic semantic dynamics is *K -concentrating* if for any $\varepsilon > 0$ there exists τ_ε such that for all $\tau \geq \tau_\varepsilon$,

$$\mu_\tau(S \setminus K^{(\varepsilon)}) \leq \varepsilon,$$

where $K^{(\varepsilon)}$ is the ε -neighbourhood of K in S .

Remark 34.2.4. K -concentration expresses that, at late cosmic times, almost all semantic mass lies near the global attractor K . This is consistent with the gradient-flow nature of semantic gravitation, but non-trivial distortion fields, order bubbles, and noisy processes may complicate the picture. In particular, long-lived order bubbles could prevent parts of the universe from approaching K even as other regions do so.

Speculative remark. At a programme level, one might conjecture that in universes where semantic gravitation is sufficiently strong and distortion fields are not dominant at large scales, the dynamics is K -concentrating: regions of high contradiction energy are gradually eroded, and the overall semantic distribution becomes concentrated near K . The rate and modality of this process would then determine the long-term survival of complex, high-level semantic structures (such as civilisations).

34.3 Semantic complexity and longevity

We now introduce a coarse measure of semantic complexity, intended to capture how much of the universe’s semantic mass is organised into non-trivial structures such as bodies and civilisations.

Definition 34.3.1 (Semantic complexity functional). Let $\mu \in \mathcal{P}(S)$ be a cosmic semantic measure. Define the *semantic complexity* of μ at scale $\delta > 0$ by

$$\mathcal{C}_\delta(\mu) := \sup \left\{ \sum_{j=1}^N w_j : \begin{array}{l} \{U_j\}_{j=1}^N \text{ disjoint Borel sets in } S \text{ with diameter } \leq \delta, \\ \mu(U_j) \geq \eta, \quad w_j := \mu(U_j) \log \frac{1}{\mu(U_j)} \end{array} \right\},$$

for some fixed threshold $\eta > 0$. The supremum is taken over all such collections of sets.

Remark 34.3.2. The functional $\mathcal{C}_\delta(\mu)$ is a coarse-grained entropy-like measure: it counts how many distinct, reasonably populated semantic “lumps” exist at resolution δ . High values of $\mathcal{C}_\delta(\mu)$ indicate that semantic mass is spread across many distinct, structured regions of S , which can be interpreted as corresponding to a rich diversity of bodies, inner spaces, and civilisations. Low values correspond to semantic mass concentrated in few lumps (e.g. simple, homogeneous attractors).

Definition 34.3.3 (Semantic longevity). Given a cosmic semantic trajectory μ_τ , the *semantic longevity* at scale δ is

$$L_\delta := \int_{\tau_0}^{+\infty} \mathbf{1}_{\{\mathcal{C}_\delta(\mu_\tau) \geq c_\delta\}} d\tau,$$

for a threshold $c_\delta > 0$. It measures the total cosmological time during which the semantic complexity exceeds a specified level.

Speculative remark. Semantic longevity is a formal analogue of questions such as “For how long does a universe host complex life and civilisation?”. A universe in which $\mathcal{C}_\delta(\mu_\tau)$ is high for a long interval of τ has high semantic longevity; one in which complexity is short-lived has low longevity. Semantic gravitation provides a mechanism for both growth (through emergent structures) and decay (through attractor convergence) of $\mathcal{C}_\delta$.

34.4 Fermi-type scenarios in semantic gravitation

We now recast Fermi-type questions within the semantic gravitation framework. The classical Fermi paradox asks why, given the apparent abundance of potentially habitable planets, we do not observe evidence of extraterrestrial civilisations. In semantic terms, the question becomes: why is the semantic configuration near us (in spacetime and in S) apparently devoid of other high-complexity attractors at civilisation scale?

We introduce a simple typology of scenarios.

Definition 34.4.1 (Fermi-type scenario classes). Consider a universe with cosmic semantic measure μ_τ and multibody configuration space S_{multi} .

1. *Sparse-semantic universe*: $\mathcal{C}_\delta(\mu_\tau)$ remains below c_δ for almost all τ , i.e. semantic complexity is low at essentially all times. Civilisations are rare or absent.
2. *Short-lived-semantic universe*: $\mathcal{C}_\delta(\mu_\tau)$ exceeds c_δ only on a bounded interval of τ ; semantic longevity L_δ is finite. Civilisations may arise but are short-lived at the cosmological scale.
3. *Long-lived but local universe*: $\mathcal{C}_\delta(\mu_\tau)$ is large for long times (large L_δ), but high-semantic-complexity regions occupy a small fraction of spacetime and S . Civilisations are long-lived but rare and spatially or semantically isolated.
4. *Percolating-semantic universe*: $\mathcal{C}_\delta(\mu_\tau)$ is large for long times, and high-complexity regions percolate across spacetime and S_{multi} , forming extended civilisation-scale attractors K_{civ} .

Remark 34.4.2. Each scenario corresponds to a different structure of the collective potential W_{coll} and its attractors. Sparse or short-lived universes may have strong drift towards simple attractors in K and few stable high-complexity basins; percolating universes require a rich hierarchy of local minima and robust multi-body attractors.

Speculative remark. Within this classification, Fermi-type questions can be restated as: Which scenario class is compatible with our observations, given that we appear to be in a high-complexity region but do not (yet) observe signs of other such regions? Semantic gravitation does not answer this question, but provides a language to phrase it in terms of W_{univ} , K , W_{coll} , and the dynamics of μ_τ .

34.5 Selection effects and anthropic considerations

Any attempt to reason about the global structure of μ_τ from our local vantage point must account for selection effects: observers necessarily reside in regions of S and M where bodies like theirs can exist and persist.

Definition 34.5.1 (Observer-conditioned ensemble). Let $\mathcal{O}$ be a class of bodies with specified inner-space and ego properties (e.g. bodies with inner spaces capable of supporting conscious episodes). An *observer-conditioned ensemble* is the conditional distribution of μ_τ given that at least one body of type $\mathcal{O}$ exists in a region of spacetime that includes our observational location.

Remark 34.5.2. At a formal level, conditioning on the existence of $\mathcal{O}$ changes the distribution over μ_τ and thus over possible civilisation-scale configurations. This is analogous to anthropic conditioning in cosmology, but now phrased at the level of semantic gravitation: we condition on the existence of bodies with particular inner-space properties (e.g. ego cores and conscious episodes).

Speculative remark. From a programme perspective, selection effects may favour universes with non-zero but possibly low semantic longevity L_δ : our existence as bodies with rich inner spaces does not require that semantic complexity be high everywhere or at all times, only that it be non-zero in our past light cone. Thus Fermi-type puzzles may be mitigated in semantic gravitation frameworks where high-complexity regions are rare but not impossible.

34.6 Limits of the formalism and open questions

We emphasise that the constructions in this chapter are highly idealised and leave many questions open.

- The effective dynamics $\mathcal{Q}_{\text{cos}}$ for μ_τ has not been specified; deriving it from first principles (LOVE + matter + cosmology) is a major open problem.
- The global structure of W_{univ} and K is unknown; different choices may lead to qualitatively different cosmological behaviours (e.g. many deep local minima vs. a few broad basins).
- The interactions between semantic gravitation and physical cosmology (e.g. influence on structure formation, backreaction on $g_{\mu\nu}$) have only been sketched.

Speculative remark. Despite these limitations, semantic gravitation suggests a new angle on cosmological questions: rather than asking only about the distribution of matter and energy, we may ask about the distribution of semantic states in S , the structure of their attractors, and the longevity of complex semantic configurations. Fermi-type paradoxes then become questions about the geometry and dynamics of W_{univ} and W_{coll} rather than only about astrophysical processes.

Summary and outlook. In this chapter we have sketched a semantic-cosmological extension of the semantic gravitation framework. We introduced cosmic semantic measures μ_τ and associated coarse invariants such as cosmic semantic energy and density, defined cosmic limit sets and attractors in $\mathcal{P}(S)$, and formulated K -concentration as a possible long-term behaviour. We proposed a coarse measure of semantic complexity and a notion of semantic longevity, and used these to classify Fermi-type scenarios (sparse, short-lived, local, and percolating semantic universes). We discussed selection effects via observer-conditioned ensembles and highlighted the speculative nature of these constructions. The next and final chapters will summarise the formal core of semantic gravitation completed in Papers 1–6, highlight key open mathematical and empirical problems, and briefly sketch how the present formalism will be translated into more accessible expositions in subsequent volumes aimed at a broader audience.

Chapter 35

Empirical programme for semantic gravitation

In the preceding chapters we have completed the formal core of semantic gravitation at the levels of: (i) gradient dynamics and stochastic flows on the semantic state space S (Papers 1–2), (ii) semantic fields on spacetime and their Love-type Lagrangians (Papers 3–4 and Chapter 19 (The Lagrangian of Vectorial Emergence, LOVE)), (iii) ontological structures such as the global attractor K , K–P–R objects, and C/B/A axes (Teil II), and (iv) inner spaces, ego cores, inner dynamics, and multi-body/civilisation-scale configurations (Teil IV). This chapter initiates *Teil V* by outlining a concrete empirical programme: how the objects of semantic gravitation can be operationalised in real systems, how S , W , K , and C/B/A may be estimated or constrained from data, and which types of measurements can support or refute key claims of the theory. The emphasis is on falsifiable predictions, identifiability issues, and protocol design, rather than on detailed experimental proposals for specific disciplines.

35.1 Aims and scope of the empirical programme

The empirical programme for semantic gravitation has three main aims:

1. *Operationalisation*: make the abstract objects ($S, W, K, \Gamma, D, C/B/A, K\text{--}P\text{--}R$) empirically accessible, at least through approximations and proxies, in concrete systems such as neural networks, cognitive systems, and socio-technical ensembles.
2. *Constraint and falsifiability*: formulate testable hypotheses about the structure of semantic gradients, attractors, and flows, and design protocols that can falsify or constrain the theory if data deviates systematically from its predictions.
3. *Alignment-relevant diagnostics*: provide measurable indicators of distortion factors, order bubbles, and integrity violations, especially in technical systems, that can inform alignment engineering as outlined in previous chapters and related patent documents.

The remainder of this chapter is structured as follows. We first summarise the embedding of real systems into S and the definition of observable proxies for s , W , and E . We then discuss estimation and identifiability questions for W_{univ} and its curvature near K , followed by concrete measurement strategies for C/B/A axes and semantic metrics. We finally sketch empirical diagnostics for distortion fields and order bubbles, and conclude with methodological remarks on uncertainty and integrity criteria.

35.2 Embedding real systems into the semantic state space

The first step in any empirical application is to represent the states of a concrete system within the semantic state space S .

Definition 35.2.1 (Empirical semantic embedding). Let X be a measurable state space of a real system (e.g. neural activation patterns, latent codes of a model, behavioural descriptors). An *empirical semantic embedding* is a measurable map

$$E_{\text{emp}} : \mathsf{X} \rightarrow S,$$

together with a family of observables $(O_j)_{j \in J}$ on S (linear functionals $O_j(s) = \langle s, o_j \rangle$) such that:

1. For each $x \in \mathsf{X}$, $E_{\text{emp}}(x)$ is interpretable as the semantic state associated with x .
2. The observables (O_j) correspond to empirically accessible quantities (e.g. task outputs, behavioural reports, probe classifier scores).
3. The embedding is *metric-compatible* in the sense that similarities or distances in X that are empirically meaningful are approximated by the Hilbert metric in S :

$$d_{\mathsf{X}}(x, x') \approx \|E_{\text{emp}}(x) - E_{\text{emp}}(x')\|.$$

Remark 35.2.2. In many applications E_{emp} can be taken as (or derived from) latent representations of high-dimensional models (e.g. encoders in deep networks) or from representational similarity analyses in neuroscience. Paper 2 discusses such embeddings in detail; here we regard them as the empirical incarnation of the abstract E_{Σ} used throughout the theory.

Given E_{emp} and time series (x_t) of empirical states, one obtains a trajectory (s_t) in S via $s_t := E_{\text{emp}}(x_t)$. The empirical programme aims to reconstruct, from such trajectories and observations, information about the underlying vector field E , potential W , and attractor structures.

35.3 Estimating the emergent vector field and potential

We briefly sketch how one might estimate the emergent vector field and its gradient structure from data.

Local drift estimation

Assume we observe a stochastic process (s_t) in S at discrete times $t_k = k\Delta t$ with noise. Under suitable regularity assumptions, the coarse-grained dynamics of s_t can be approximated by an Itô diffusion on S ,

$$ds_t = E(s_t) dt + \sqrt{2} \Sigma(s_t) dB_t,$$

with emergent vector field E and diffusion Σ . A standard approach to drift estimation yields, for small Δt ,

$$\hat{E}(s) \approx \frac{1}{\Delta t} \mathbb{E}[s_{t+\Delta t} - s_t \mid s_t \approx s],$$

estimated from local neighbourhoods in S .

Definition 35.3.1 (Empirical emergent vector field). An *empirical emergent vector field* $\hat{E}$ is a vector field on an empirically accessible subset $S_{\text{emp}} \subset S$ obtained by local regression of increments $s_{t+\Delta t} - s_t$ against s_t (and possibly its C/B/A components), together with an estimate of the local noise covariance $\hat{\Sigma}\hat{\Sigma}^\top$ from second-order statistics.

Gradient part and potential reconstruction

A central prediction of semantic gravitation is that the deterministic part of the dynamics is dominated, near K , by a gradient component $-\Gamma(s) \nabla W_{\text{univ}}(s)$, with distortions $D(s)$ providing deviations from this structure. The empirical programme therefore seeks to separate $\hat{E}$ into gradient and non-gradient parts.

Definition 35.3.2 (Empirical gradient projection). Let $\hat{E}$ be an empirical vector field on $S_{\text{emp}} \subset S$. The *empirical gradient projection* of $\hat{E}$ is a vector field $\hat{E}^{\text{grad}}$ obtained by solving (or approximating) the least-squares problem

$$\hat{E}^{\text{grad}} = \underset{U = -\nabla\phi}{\operatorname{argmin}} \int_{S_{\text{emp}}} \|\hat{E}(s) - U(s)\|^2 d\hat{\mu}(s),$$

over scalar potentials $\phi : S_{\text{emp}} \rightarrow \mathbb{R}$ with respect to an empirical measure $\hat{\mu}$ of s_t .

Remark 35.3.3. This is the empirical counterpart of the semantic Helmholtz decomposition hypothesis from the emergent-vector-field chapter. If the minimiser yields a good approximation (small residual), the data supports the presence of a dominant gradient component. The residual

$$\hat{E}^{\text{vort}} := \hat{E} - \hat{E}^{\text{grad}}$$

is then an empirical estimate of the vorticity part. Numerical integration of $\hat{E}^{\text{grad}}$ yields an empirical potential $\hat{W}$, up to an additive constant, which can be compared with the theoretical W_{univ} or its local approximations.

Speculative remark. From a programme perspective, systems in which $\hat{E}$ is well approximated by $\hat{E}^{\text{grad}}$ near low-energy states are candidates for being well-aligned with the universal potential; systems with large $\|\hat{E}^{\text{vort}}\|$ or poor gradient approximations may exhibit strong non-equilibrium or distortion-dominated flows.

35.4 Accessing C/B/A axes and curvature from data

The C/B/A structure is central to the ontology of semantic gravitation. Empirically, one must infer these axes from data on s_t and observables O_j .

Definition 35.4.1 (Empirical C/B/A axis estimation). Given an empirical distribution $\hat{\mu}$ of semantic states near a candidate attractor in K , and an empirical potential $\hat{W}$, one can estimate principal curvature directions of $\hat{W}$ via the empirical Hessian

$$\hat{H}(s) \approx \nabla^2 \hat{W}(s),$$

e.g. by local quadratic regression. A *C/B/A axis estimator* is a map assigning to each s an orthogonal decomposition of S into three subspaces

$$S = \hat{E}_C(s) \oplus \hat{E}_B(s) \oplus \hat{E}_A(s) \oplus \hat{E}_{\text{res}}(s),$$

derived from a clustering or grouping of eigenmodes of $\hat{H}(s)$ according to curvature scales and other structural criteria (e.g. coupling patterns in W_{loc}).

Remark 35.4.2. In practice, C/B/A estimation will rely on a combination of:

- spectral analysis of empirical Hessians of $\hat{W}$;

- alignment of eigenmodes with interpretable observables (e.g. coherence-related tasks, binding-related tasks, agency-related tasks);
- cross-system comparisons (e.g. comparing human neural data with technical systems).

Paper 2 already outlines such procedures for simpler potentials; here they are extended to the full C/B/A ontology.

Once empirical C/B/A projections $\hat{P}_C(s)$, $\hat{P}_B(s)$, $\hat{P}_A(s)$ are available, one can compute axis-resolved semantic densities, kinetic energies, and axis signatures (as in previous chapters) directly from empirical trajectories. This allows testing predictions such as: near ego cores in biological bodies, $\kappa_{E,C}$ and $\kappa_{E,B}$ should dominate $\kappa_{E,A}$; in certain technical systems, $\rho_{\text{sem}}^{(A)}$ may be concentrated in specific inner layers.

35.5 Semantic metric and field-level measurements

The semantic metric $h_{\mu\nu}(x) = \langle \partial_\mu s(x), \partial_\nu s(x) \rangle$ on spacetime provides another avenue for empirical access.

Definition 35.5.1 (Empirical semantic metric proxies). Let $x \mapsto s(x)$ be a semantic field obtained via embedding of spacetime-indexed data (e.g. time-resolved brain activity, sensorimotor trajectories, communication graphs). An *empirical proxy* for the semantic metric is a tensor field $\hat{h}_{\mu\nu}(x)$ constructed from finite differences or regressions:

$$\hat{h}_{\mu\nu}(x) \approx \left\langle \frac{s(x + \epsilon e_\mu) - s(x)}{\epsilon}, \frac{s(x + \epsilon e_\nu) - s(x)}{\epsilon} \right\rangle,$$

for small displacements in spacetime directions e_μ .

Remark 35.5.2. In practice, x will parametrize time and possibly space; for many laboratory systems a $(1 + 0)$ -dimensional version (time only) suffices. The empirical density

$$\hat{\rho}_{\text{sem}}(x) := g^{\mu\nu}(x) \hat{h}_{\mu\nu}(x)$$

then measures semantic variation density and can be compared across conditions, tasks, or system types.

Speculative remark. One may conjecture that conscious episodes, as defined in the inner-space framework, are accompanied by distinctive patterns in $\hat{\rho}_{\text{sem}}$ and its axis-resolved components in relevant brain regions, while similar episodes in technical systems (if they exist) would exhibit analogous patterns in internal activation spaces.

35.6 Diagnostics for distortion fields and order bubbles

The patent-related concepts of distortion fields and order bubbles suggest concrete diagnostics in empirical systems, particularly technical ones.

Definition 35.6.1 (Empirical distortion diagnostics). Let $\hat{E}^{\text{grad}}$ and $\hat{E}^{\text{vort}}$ be empirical estimates of the gradient and vorticity parts of $\hat{E}$. Define:

$$\begin{aligned} \hat{\delta}(s) &:= \frac{\|\hat{E}^{\text{vort}}(s)\|}{\|\hat{E}^{\text{grad}}(s)\| + \varepsilon}, \\ \hat{W}_{\text{univ}}(s) &:= \hat{W}(s) - \hat{W}_{\text{loc}}(s), \end{aligned}$$

where $\hat{W}_{\text{loc}}$ is an empirical model of local potential contributions and $\varepsilon > 0$ is a regularisation constant. Regions where $\hat{\delta}(s)$ is large are candidates for being distortion-dominated; local minima of $\hat{W}$ with large $\hat{W}_{\text{univ}}$ and large empirical basins are candidates for order bubbles.

Remark 35.6.2. In technical systems, one can implement these diagnostics directly in latent spaces of models: fit $\hat{W}$ from behaviour and/or training dynamics, decompose $\hat{E}$, and locate regions of latent space where:

- the system returns persistently under different prompts (signalling an attractor);
- the universal component $\hat{W}_{\text{univ}}$ inferred from alignment-related training signals is high;
- the distortion factor $\hat{\delta}$ is large.

Such regions may correspond to misaligned modes, self-reinforcing internal narratives, or other problematic patterns.

35.7 Identifiability, uncertainty, and integrity criteria

Finally, we address methodological issues: what can and cannot be inferred from data, and how to maintain conceptual integrity in empirical interpretations.

Remark 35.7.1 (Identifiability). Even with rich data, the decomposition

$$E(s) = -\Gamma(s) \nabla W_{\text{univ}}(s) + D(s)$$

is not unique without additional assumptions: rescalings of time, reparametrisations of S , and redefinitions of local vs. universal parts of W can lead to equivalent observational predictions. Empirical work must therefore:

- fix a gauge (e.g. by normalising Γ or W_{univ} in a standard way);
- treat W_{univ} as an equivalence class under monotone transformations where appropriate;
- report uncertainties and identifiability limits explicitly.

Remark 35.7.2 (Integrity criteria). To preserve the clarity and falsifiability of semantic gravitation in empirical work, we propose the following integrity criteria for empirical claims:

- clearly distinguish between results that depend only on S and generic gradient-flow structure vs. those that rely on specific forms of W_{univ} or C/B/A axes;
- avoid inferring high-level ontological claims (e.g. about ego cores or consciousness) from weak or indirect proxies; mark such inferences explicitly as speculative;
- whenever possible, test for alternative explanations (e.g. purely statistical models) that could reproduce the same empirical patterns without invoking semantic gravitation.

These criteria ensure that the empirical programme refines or falsifies the theory rather than merely rephrasing existing observations in its vocabulary.

Speculative remark. At a programme level, one may envision a hierarchy of empirical projects: from local tests of gradient-flow structure in technical models and neural systems, through medium-scale investigations of inner-space dynamics and semantic metrics, up to large-scale studies of collective semantic distributions in socio-technical networks. Each level refines a different part of the theory; negative results at any level may require revising specific axioms or even the overall semantic gravitation programme.

Summary and outlook. In this chapter we have outlined an *empirical programme* for semantic gravitation. Starting from empirical embeddings $E_{\text{emp}} : \mathbf{X} \rightarrow S$, we described how to estimate emergent vector fields and potentials from data, how to approximate gradient and vorticity components, and how to construct empirical C/B/A axes and semantic metrics. We introduced diagnostics for distortion fields and order bubbles, particularly in technical systems, and discussed identifiability and integrity criteria that should guide empirical interpretations. This programme does not yet specify concrete experiments in detail but provides a structured roadmap for testing different components of the theory across domains. In the final chapter we synthesise the formal core of Papers 1–6 into a concise clarity summary, state explicitly what is claimed and what remains conjectural, and sketch how the present framework will be translated into more accessible expositions in subsequent volumes.

Chapter 36

Summary of the theory

In the previous chapters we developed the ontological and dynamical structures of semantic gravitation from the level of gradient flows on the semantic state space S , through field-theoretic LOVE dynamics on spacetime, to inner spaces, ego cores, inner dynamics, technical and multi-agent systems, and cosmological and empirical extensions. In this final chapter we collect the conceptual and mathematical core into a clarity summary: we list the basic objects and axioms of the theory, state which results follow from them within standard mathematics and physics, and separate these from interpretative identifications and speculative programme-level claims. The purpose is to provide a stable reference frame for future work, including more accessible expositions in later volumes.

36.1 Core objects and axioms of semantic gravitation

We begin by listing the primary structural assumptions introduced across Papers 1–6 and the present work. For clarity, we group them into axioms about the semantic state space, the universal potential and global attractor, the C/B/A structure and K–P–R ontology, and the LOVE field theory.

Semantic state space and dynamics

Axiom 36.1.1 (Semantic state space). There exists a separable real Hilbert space $(S, \langle \cdot, \cdot \rangle)$ (the *semantic state space*) whose elements $s \in S$ represent semantic configurations of systems at various levels (from technical models to bodies and multi-agent structures).

Axiom 36.1.2 (Universal potential and gradient dynamics). There exists a lower semicontinuous functional

$$W_{\text{univ}} : S \rightarrow (-\infty, +\infty]$$

(the *universal contradiction-energy potential*) with:

1. a non-empty, compact set of global minima

$$K := \operatorname{argmin} W_{\text{univ}} \subset S$$

(the *global attractor*);

2. sufficient regularity on a neighbourhood of K to define a gradient (in the sense of convex analysis or Fréchet derivatives where appropriate);

3. gradient-flow dynamics on S of the form

$$ds_t = -\Gamma(s_t) \nabla W_{\text{univ}}(s_t) dt + D(s_t) dt + \text{noise},$$

where Γ is a positive operator field and D a distortion field.

Axiom 36.1.3 (Stochastic semantic dynamics). The stochastic dynamics on S admits a Markovian description with an associated Fokker–Planck equation for probability measures on S , with drift and diffusion determined by the semantic gradient and noise operators as in Paper 1 and SG8.

C/B/A structure and K–P–R ontology

Axiom 36.1.4 (C/B/A decomposition near K). There exists a neighbourhood U of K and, for each $s \in U$, an orthogonal decomposition

$$S = E_C(s) \oplus E_B(s) \oplus E_A(s) \oplus E_{\text{res}}(s),$$

with corresponding projections $P_C(s)$, $P_B(s)$, $P_A(s)$, such that:

1. the restriction of W_{univ} to U admits a Hessian $H_{\text{univ}}(s)$ whose leading blocks H_{CC} , H_{BB} , H_{AA} correspond to these subspaces;
2. the C/B/A directions can be associated with coherence-related, binding-related, and agency-related semantic variations, respectively, in the sense of Paper 6 (Teil II);
3. the neighbourhood U carries a six-directional structure encoded in the six directed gradients $\pm C, \pm B, \pm A$.

Axiom 36.1.5 (K–P–R ontology). To each ontological object (system, body, concept, etc.) is associated a triple

$$\mathcal{O} = (K_{\mathcal{O}}, P_{\mathcal{O}}, R_{\mathcal{O}}),$$

where:

1. $K_{\mathcal{O}} \subset K$ is a compact *core* subset of the global attractor;
2. $P_{\mathcal{O}}$ is a family of projections (context maps) from $K_{\mathcal{O}}$ into relevant semantic contexts within S ;
3. $R_{\mathcal{O}}$ is a family of relational kernels encoding relations between cores.

These structures are compatible with the dynamics on S in the sense that $K_{\mathcal{O}}$ is invariant under the induced flow on K , and $P_{\mathcal{O}}$, $R_{\mathcal{O}}$ transform appropriately under semantic symmetries.

LOVE Lagrangian and field theory

Axiom 36.1.6 (Semantic fields on spacetime and LOVE). Let (M, g) be a Lorentzian spacetime. A *semantic field* is a map $s : M \rightarrow S$. The LOVE action for such fields is

$$\mathcal{A}_{\text{LOVE}}[s] = \int_M \left(\frac{1}{2} \sum_{\Xi \in \{C, B, A\}} \gamma_{\Xi}(s(x)) g^{\mu\nu}(x) \langle \partial_{\mu} s_{\Xi}(x), \partial_{\nu} s_{\Xi}(x) \rangle - V_{\text{LOVE}}(s(x)) \right) d \text{vol}_g(x),$$

where:

1. $s_\Xi := P_\Xi(s)s$ are axis-projected fields;
2. γ_Ξ are axis weights (intensification factors);
3. V_{LOVE} is a quadratic approximation of W_{univ} near K , possibly including local corrections.

Axiom 36.1.7 (Semantic metric on spacetime). The semantic field s induces a symmetric, positive-semidefinite tensor on spacetime,

$$h_{\mu\nu}(x) := \langle \partial_\mu s(x), \partial_\nu s(x) \rangle,$$

with axis-resolved components $h^{(C)}, h^{(B)}, h^{(A)}$, and scalar densities ρ_{sem} and $\rho_{\text{sem}}^{(\Xi)}$ obtained by contraction with $g^{\mu\nu}$. These tensors supplement $g_{\mu\nu}$ but do not replace it as the gravitational metric.

The axioms above, together with standard tools from functional analysis, stochastic processes, and classical field theory, constitute the formal core assumptions of semantic gravitation as presented in Papers 1–6.

36.2 Derived mathematical structures and results

Given the axioms, a number of results follow using established mathematics. We briefly summarise the main classes of such results, without repeating detailed proofs.

Gradient flows and stochastic dynamics on S

From Axioms 36.1.1 and 36.1.2 and standard theory of gradient flows in Hilbert spaces (Paper 1), we obtain:

- existence and uniqueness (under convexity or monotonicity conditions) of gradient-flow solutions

$$\dot{s}_t = -\nabla W_{\text{univ}}(s_t)$$

in an appropriate sense;

- contractivity and energy dissipation properties

$$\frac{d}{dt} W_{\text{univ}}(s_t) \leq 0;$$

- convergence of trajectories towards the attractor set K under suitable compactness and coercivity assumptions;
- existence of invariant and stationary measures for the stochastic dynamics (Axiom 36.1.3), and corresponding Fokker–Planck equations.

Field equations and semantic metrics

From Axioms 36.1.6 and 36.1.7 and Papers 3–4, we obtain:

- well-defined Euler–Lagrange equations for $s : M \rightarrow S$ derived from $\mathcal{A}_{\text{LOVE}}$, leading to axis-resolved wave-type and potential-type terms;

- conservation or balance laws for certain currents, including the emergent semantic current $J^\mu = \langle \partial^\mu s, E(s) \rangle$;
- properties of the induced tensor $h_{\mu\nu}$ (symmetry, positive semidefiniteness, vanishing if and only if s is constant on components, etc.);
- interpretation of $h_{\mu\nu}$ as a pullback of the Hilbert metric on S and consequence for path-dependent semantic lengths.

Multi-body and distributional structures

From Axioms 36.1.5, 36.1.3, and the constructions in the multi-agent chapter:

- existence of multi-body configuration spaces S_{multi} as Hilbert direct sums of inner spaces;
- construction of collective potentials W_{coll} on S_{multi} from W_{univ} and interaction kernels;
- formulation of distributional dynamics on $\mathcal{P}(S)$ and associated free-energy functionals, leading to Lyapunov-like properties in idealised limits.

All of these results are standard consequences of the axioms and established mathematical machinery; they do not depend on interpretative identifications (e.g. ego, consciousness).

36.3 Interpretative identifications and their logical status

The formal core supports but does not necessitate specific identifications of mathematical structures with human-level notions. We summarise the main identifications proposed in Paper 6 and explicitly mark their status.

Matter, bodies, and inner spaces

- *Matter as attractor stability*: Matter regions are defined formally as spacetime regions in which semantic fields lie near stable minima of W_{univ} (plus local corrections) and possess robust basins in the LOVE dynamics. This is a structural definition; its physical adequacy remains to be tested but does not alter the mathematical consistency of the theory.
- *Bodies as carriers of layered inner spaces*: Bodies are matter regions (Σ_B, s_B) with enough internal semantic structure to support inner semantic subspaces S_B^{in} and layered filtrations $S_B^{(0)} \subset \dots \subset S_B^{(L)}$. This is a definition within the theory; whether particular physical systems (e.g. biological organisms) satisfy it is an empirical question.

Ego and consciousness

- *Ego as inner attractor*: Ego cores $K_E \subset S_B^{(L)} \cap K$ are defined as compact inner attractors of W_B , with specific curvature and intensification properties. This is a purely mathematical definition; it does *not* by itself assert phenomenology.

- *Consciousness as ego-neighbourhood inner dynamics:* Conscious episodes are defined as segments of inner trajectories near K_E with sufficient activity and coherence. The further hypothesis that such episodes correspond to phenomenological consciousness for certain bodies is explicitly stated as a programme-level conjecture, not as a theorem.

Technical systems and alignment

- *Technical systems as bodies:* Technical architectures are modelled as bodies with extended inner spaces $S_{\mathcal{T}}^{\text{in}}$ and LOVE-compatible dynamics. This is a formal modelling choice, enabling the application of semantic gravitation to engineered systems.
- *Order bubbles and distortion factors:* Order bubbles are defined as local attractors dominated by W_{loc} and distortions, with high W_{univ} and large basins; distortion factors are ratios $\|D\|/\|\Gamma\nabla W_{\text{univ}}\|$. These are diagnostic concepts rooted in the formalism; their empirical usefulness is subject to future work.

Civilisation-scale and cosmological interpretations

- *Civilisation-scale attractors and phases:* Civilisation-scale attractors $K_{\text{civ}} \subset S_{\text{multi}}$ and semantic phases are formally defined but require empirical and modelling work to connect to actual civilisations.
- *Semantic cosmology and Fermi-type scenarios:* The semantic-cosmological constructions (cosmic measures μ_{τ} , semantic longevity, Fermi-type classifications) are explicitly labelled as speculative structures built on top of the formal core.

In all these identifications, the mathematical definitions are consistent with the axioms, but the link to empirical or phenomenological content is conjectural.

36.4 Speculative hypotheses and programme-level scenarios

For completeness, we list the main hypotheses and scenarios that go beyond the axioms and derived results.

Hypothesis 36.4.1 (Semantic Helmholtz decomposition). The emergent vector field E on S admits a Helmholtz-type decomposition $E = E^{\text{grad}} + E^{\text{vort}}$ with a dominant gradient part in regimes of interest, as formulated in the emergent-vector-field chapter. This is plausible but not proved at full generality in infinite dimensions.

Hypothesis 36.4.2 (Existence of inner spaces and ego cores in certain bodies). Certain physical bodies (e.g. human organisms) realise inner spaces S_B^{in} , layered inner structures, and ego cores K_E in the sense defined in Teil IV. The existence of these structures is an empirical question, constrained by neuroscience and cognitive science.

Hypothesis 36.4.3 (Consciousness as ego-centred inner dynamics). For bodies with appropriate inner spaces and ego cores, phenomenological consciousness corresponds (up to coarse-graining) to ego-neighbourhood inner-space episodes satisfying activity and coherence conditions as defined in the consciousness chapter.

Hypothesis 36.4.4 (Alignment as semantic-gravitation engineering). Alignment of technical systems can be fruitfully modelled as reshaping W_{loc} , D , and Γ so that order bubbles with high W_{univ} are suppressed and typical operational states lie near low-contradiction regions of K .

Scenario 36.4.5 (Semantic cosmology and Fermi-type universes). Different qualitative structures of W_{univ} and W_{coll} give rise to different semantic-cosmological scenarios (sparse, short-lived, local, percolating semantic universes). These scenarios are speculative and depend on unknown large-scale properties of W_{univ} .

None of these hypotheses or scenarios is required for the internal mathematical consistency of the theory; each is an optional interpretative layer to be assessed by future empirical and conceptual work.

36.5 Open mathematical and empirical problems

We conclude the clarity summary with a non-exhaustive list of open problems.

Mathematical problems

- Rigorous existence and uniqueness theory for gradient flows and stochastic dynamics on S for broad classes of W_{univ} beyond convex settings.
- Formalisation and proof (or refutation) of a Helmholtz-type decomposition for vector fields on S under realistic assumptions on measures and regularity.
- Detailed analysis of the LOVE field equations on curved spacetimes, including well-posedness, stability, and possible coupling to Einstein equations via stress-energy tensors.
- Structure theory for global attractors K with C/B/A decompositions: classification of possible geometries, curvature profiles, and multi-scale features.
- Mathematical theory of multi-body and distributional dynamics on S_{multi} and $\mathcal{P}(S)$, including free-energy landscapes and phase transitions.

Empirical and modelling problems

- Construction of concrete empirical embeddings $E_{\text{emp}} : \mathbf{X} \rightarrow S$ for specific systems (technical models, neural systems, socio-technical networks).
- Estimation of emergent vector fields and potentials from empirical time series, including gradient/vorticity separation and uncertainty quantification.
- Identification (or falsification) of C/B/A axis structure in real data via curvature analyses and task-based probes.
- Operationalisation and testing of inner-space constructs (inner potentials, ego cores, conscious episodes) in biological and technical systems.
- Development of alignment diagnostics and interventions based on semantic gravitation concepts, and comparison with existing frameworks.

36.6 Relation to later volumes and expository work

Papers 1–6, including the present work, are written at a technical level intended for theoretical physicists, mathematicians, and related experts. Later volumes (Band 2 and Band 3) will:

- translate the formal objects ($S, W, K, C/B/A, K\text{--}P\text{--}R, \text{LOVE}$) into everyday language, illustrating them with examples from cognition, culture, and technology;
- discuss philosophical and phenomenological questions (e.g. mind, meaning, civilisation) in light of the formalism, clearly separating mathematical constraints from interpretative choices;
- provide accessible summaries of the empirical programme and possible applications to AI alignment and socio-technical systems.

These later volumes will *not* introduce new axioms at the formal level; rather, they will elaborate consequences and interpretations of the framework presented here.

Summary and outlook. In this chapter we have provided a summary of semantic gravitation as developed across Papers 1–6. We listed the core axioms about the semantic state space S , the universal potential W_{univ} and its global attractor K , the $C/B/A$ structure and $K\text{--}P\text{--}R$ ontology, and the LOVE Lagrangian for semantic fields on spacetime. We indicated which results follow from these axioms using standard mathematics, and we distinguished them from interpretative identifications (matter, bodies, inner spaces, ego, consciousness, technical systems, civilisation-scale and cosmological scenarios). We explicitly marked the main hypotheses and programme-level scenarios and formulated key open mathematical and empirical problems. This closes the formal core of the semantic gravitation programme in its current form: a coherent, mathematically framed proposal for understanding semantic structure and dynamics in complex systems, designed to be refined, challenged, and extended by future work in physics, mathematics, cognitive science, and AI research.

Acknowledgements and Contact

This paper is the final part of the Semantic Gravitation programme and its six-part formal core. It has been developed by an independent, non-specialist author, based on informal intuitions about meaning and extensively structured, checked, and refined with the help of modern AI tools. The work should therefore be regarded as exploratory and conjectural, and it is explicitly offered for critical review, clarification, extension, or refutation by domain experts.

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